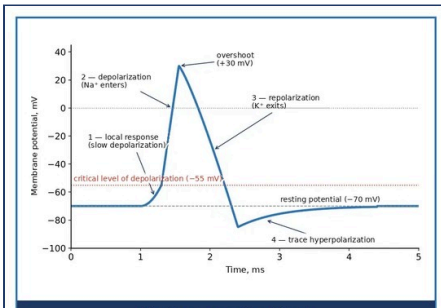
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Department of Normal and Pathological Physiology

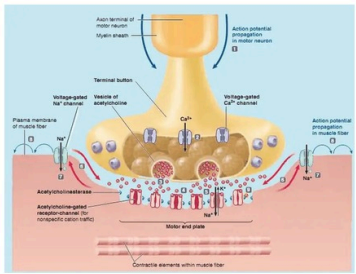
NORMAL PHYSIOLOGY

complete study manual

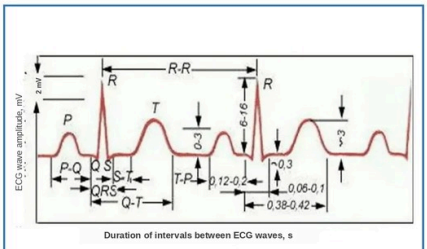
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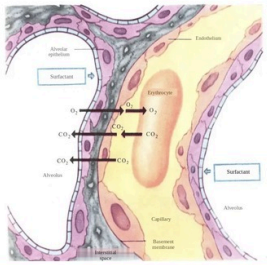
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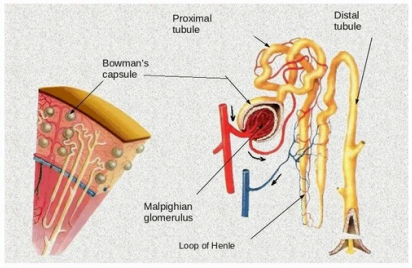
Neuromuscular synapse



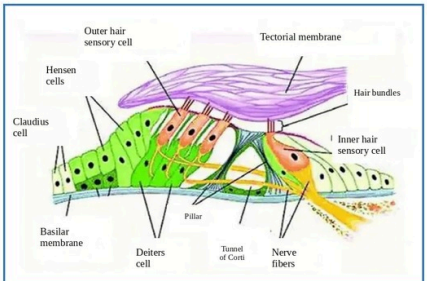
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Alveolar-capillary membrane



Nephron structure



Organ of Corti

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Tashkent — 2026

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Normal Physiology: Complete Study Manual. A study manual for practical classes. — Tashkent, 2026.

This manual is based on the 30 practical-class topics of the calendar-thematic plan of the subject “Normal Physiology” (2nd year, semesters III–IV). It covers the physiology of excitable tissues, the blood system, the cardiovascular system, metabolism and energy, thermoregulation, the endocrine glands, the respiratory, digestive and excretory systems, the central and autonomic nervous system, the analyzers (sensory systems) and higher nervous activity.

Each chapter consists of the learning objectives, theory (with “Remember”, “Clinical significance”, “Memory aid” and “Caution” boxes), a laboratory (practical) work, algorithms of practical skills, review questions, 15 test questions, 3 clinical cases and a quick review table. The answers to the tests are given at the end of the book.

The manual is based on the textbooks “Normal fiziologiya” by O.T. Alavia (2006, 2018), “Normal fiziologiya” by Sh.Q. Qodirov and “Odam fiziologiyasi” (Human Physiology) edited by G.F. Babsky, supplemented by L.S. Costanzo’s “Physiology” and other international sources. The figures are taken from these sources or drawn on their basis, and the source is given under each of them.

The text and structure of the manual were prepared with the help of the artificial intelligence Claude (Anthropic) on the basis of the textbooks listed. The compiler selected the material, defined its structure and checked it. Before applying the clinical information in practice, it is recommended to compare it with the main textbooks.

The manual is intended for 2nd-year students of the general medicine, pediatrics and preventive medicine faculties of higher medical education institutions.

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Preface

Dear student! Normal physiology is the science that studies the vital activity of a healthy organism, its systems, organs, tissues and cells, the mechanisms that regulate this activity and its adaptation to the external environment. Together with anatomy, histology and biochemistry it is the foundation of medicine: in pathological physiology, pharmacology and all clinical subjects, to answer the question “what is disturbed in disease?” one must first know the answer to the question “how does it work normally?”.

The manual follows the 30 practical-class topics of the department’s calendar-thematic plan: in semester III (topics 1–15) — excitable tissues, the blood system, the cardiovascular system, thermoregulation and the endocrine glands; in semester IV (topics 16–30) — the physiology of respiration, digestion, metabolism, excretion, the central nervous system, the autonomic nervous system, the analyzers and higher nervous activity. Every chapter is built in the same order:

- **Learning objectives** — what you should know and be able to do by the end of the class;
- **theory** — the main concepts of the topic, tables, diagrams and figures;
- **laboratory work** — the experiments, examination methods and analysis of the results performed in class;
- **practical skills** — step-by-step algorithms;
- **review questions, test questions and clinical cases** — for self-assessment;
- **quick review** — a table for revising at a glance before the exam.

The colored boxes in the text mean the following:

REMEMBER
The most important definitions, numbers and classifications — they must be memorized.

CLINICAL SIGNIFICANCE
The significance of the topic in clinical practice, examples related to patients.

MEMORY AID
Techniques and abbreviations that make memorizing easier.

CAUTION
Common mistakes, dangerous situations and discrepancies between textbooks.

The animal experiments in the laboratory works are classic models of physiology; today they are often replaced by videos, computer models or non-invasive examinations performed on the students themselves, and they are carried out in compliance with the requirements of bioethics and under the teacher’s supervision. The numbers (reference values) in the manual may differ slightly between textbooks — in such cases, follow the values accepted at the department. It is recommended to read each topic together with the main textbook.

The compiler

TOPIC 1

Introduction. Physiology as a science and the regulation of functions. Bioelectric phenomena in excitable tissues

Learning objectives

To introduce the student to the subject, tasks, research methods and history of normal physiology; to master the concepts of nervous and humoral regulation of functions in the body, homeostasis and the integrity of the organism; to explain the mechanisms by which the resting and action potentials arise in excitable tissues.

By the end of the class the student should know:

- the subject of physiology, its branches and its links with other sciences;
- the methods of physiological research (observation, acute and chronic experiments, modern instrumental methods);
- the main stages in the development of physiology and its scientists (W. Harvey, R. Descartes, I.M. Sechenov, I.P. Pavlov, Avicenna (Ibn Sina));
- the concepts of homeostasis, functional system and feedback;
- the nervous, humoral and local mechanisms of the regulation of functions;
- excitable tissues and their general properties;
- the structure of the cell membrane, ion channels and the function of the Na^+/K^+ pump;
- the mechanisms and phases of the resting potential and the action potential.

The student should be able to:

- perform and explain Galvani's first and second experiments (Matteucci's experiment);
- point out the phases on a graph of the action potential and explain the ionic mechanism of each;
- draw a regulatory loop (receptor — center — effector — feedback) with examples.

1. Physiology as a science: its tasks and methods

Physiology (Greek physis — nature, logos — study) is the science that studies the vital activity (functions) of the living organism, its organs and systems, tissues and cells, the mechanisms that regulate these functions and the interaction of the organism with the external environment. **Normal physiology** studies the functions of the healthy human body and is the theoretical foundation of medicine: to understand disease (pathological physiology), one must first know the norm.

Table 1.1. *The main branches of physiology*

Branch	What it studies
General physiology	Processes common to all tissues: excitability, excitation, inhibition, principles of regulation
Special physiology	Individual systems: blood, cardiovascular, respiratory, digestive, excretory, endocrine and nervous systems, sensory systems (analyzers)
Age physiology	Changes in functions during ontogenesis (childhood, old age)
Occupational and sports physiology	Adaptation of the organism to physical and mental load

Branch	What it studies
Physiology of higher nervous activity	Behavior, conditioned reflexes, memory, speech
Ecological physiology	Adaptation to climate, altitude and underwater conditions

Methods of physiological research

1. **Observation** — the oldest method; recording a process without interfering with it.
2. **Acute experiment (vivisection)** — under anesthesia an organ is exposed and acted on directly; the experiment is short-term. Drawback: anesthesia and surgical trauma affect function.
3. **Chronic experiment** — long-term observation under natural conditions in an animal prepared in advance by surgery (a fistula, organ transplantation). I.P. Pavlov studied the physiology of digestion with this method and was awarded the Nobel Prize in 1904.
4. **Method of isolated organs and tissues** — the frog nerve-muscle preparation, the isolated heart (Langendorff method).
5. **Modern instrumental methods** — electrophysiological (ECG, EEG, EMG), the microelectrode technique, patch clamp, telemetry, tomography, molecular genetic methods, computer modeling.

> **REMEMBER**
> The main method of physiology is **experiment**. An acute experiment gives quick results, but the integrity of the organism is disrupted; a chronic experiment makes it possible to study the organism in its natural state. In humans, mainly **noninvasive** methods are used (ECG, spirography, blood pressure measurement).

A brief history of physiology

- **Hippocrates** (5th–4th century BC) — the doctrine of temperaments.
- **Galen** (2nd century) — vivisection experiments on animals.
- **Avicenna (Abu Ali ibn Sina)** (980–1037) — in “The Canon of Medicine” he described in detail the heartbeat, the types of pulse, and questions of nutrition and hygiene.
- **William Harvey** (1628) — discovered the circulation of the blood; scientific physiology dates from this year.
- **René Descartes** (17th century) — founder of the concept of the reflex.
- **Luigi Galvani** (1791) — discovered “animal electricity”.
- **I.M. Sechenov** (1863) — “Reflexes of the Brain”; discovered central inhibition.
- **I.P. Pavlov** — the physiology of digestion, conditioned reflexes and the doctrine of higher nervous activity.
- **P.K. Anokhin** — the theory of functional systems.
- **Claude Bernard** (19th century) — the idea of the “constancy of the internal environment”; **W. Cannon** (1929) — the term “homeostasis”.

2. Regulation of functions in the body

Homeostasis

Homeostasis is the relative constancy of the composition and properties of the internal environment of the body (blood, lymph, tissue fluid) and the stability of the basic physiological functions. Homeostatic parameters may be **rigid** (pH, osmotic pressure, body temperature — kept within very narrow limits) or **plastic** (blood pressure, heart rate, blood glucose — vary within wider limits).

Table 1.2. *The main homeostatic parameters (in adults)*

Parameter	Normal value
Arterial blood pH	7,35–7,45
Osmotic pressure of blood plasma	7.6 atm (285–295 mOsm/l)
Body temperature (axillary)	36.4–36.9 °C
Blood glucose (fasting, capillary blood)	3.3–5.5 mmol/l
pO_2 / pCO_2 in arterial blood	95–100 / 40 mm Hg
Plasma Na^+ / K^+	135–145 / 3.5–5.0 mmol/l

Regulatory mechanisms

Table 1.3. *Mechanisms of the regulation of functions*

Mechanism	Signal carrier	Speed and characteristics	Example
Nervous (reflex)	Nerve impulse, neurotransmitter	Fast (milliseconds), precisely targeted, short-lasting	Withdrawing the hand on touching something hot
Humoral	Hormones, metabolites, ions — via the blood	Slow, widespread, long-lasting	Lowering of blood glucose by insulin
Local (autocrine, paracrine)	Tissue hormones, prostaglandins, NO	Only on nearby cells	Dilation of vessels in a working muscle
Neurohumoral	Joint action of the nervous and endocrine systems	Hypothalamus — pituitary — peripheral glands	Release of epinephrine (adrenaline) and cortisol in stress

Nervous and humoral regulation do not work separately but as a single **neurohumoral regulation**: the nervous system controls the endocrine glands, and hormones influence the activity of nerve cells.

A **reflex** is the response of the organism to stimulation of receptors, carried out with the participation of the central nervous system. A reflex is carried out through a **reflex arc**: receptor → afferent (sensory) pathway → nerve center → efferent (motor) pathway → effector (working organ). Information about the result of the response returns to the center — this is called **reverse afferentation (feedback)**; therefore modern physiology uses the concept of a **reflex ring** instead of a "reflex arc".



Figure 1.1. *The regulatory loop of physiological functions: negative feedback*

REMEMBER

Negative feedback returns a deviated parameter to normal (the main mechanism of homeostasis): if blood thyroxine rises, the release of thyroid-stimulating hormone decreases. **Positive feedback** amplifies a process and is rare: oxytocin release during labor, the blood coagulation cascade, the opening of Na^+ channels in the rising phase of the action potential.

Functional system (P.K. Anokhin)

A **functional system** is a dynamic association of various organs and tissues directed toward achieving a result useful for the adaptation of the organism. Its components are the **useful result** (the system-

forming factor), the **receptors** that evaluate the result, **reverse afferentation**, the **central architecture** (afferent synthesis, decision making, the acceptor of the action result) and the **executive mechanisms** (autonomic, hormonal, behavioral). For example, the functional system that maintains blood pO_2 includes the respiratory, circulatory and blood systems.

The integrity of the organism

The organism is not a sum of separate organs but a single system. Its integrity is ensured by the nervous system (especially its higher divisions), the endocrine system and the circulation of blood and lymph. A change in the function of one organ leads to changes in other systems as well (for example, during physical work breathing, heart rate and sweating change together with muscle activity).

3. Excitable tissues and their properties

Tissues that can pass from a state of rest to an active state in response to a stimulus and generate an **action potential** in doing so are called **excitable tissues**. They include **nerve**, **muscle** (skeletal, cardiac, smooth) and **glandular** tissue.

Table 1.4. *The main properties of excitable tissues*

Property	Definition	Measure
Excitability	The ability to pass into a state of excitation in response to a stimulus	Stimulation threshold (the lower it is, the higher the excitability), chronaxie
Conductivity	The ability to spread excitation along the membrane	Conduction velocity, m/s
Lability (functional mobility)	The number of action potentials that can be generated per unit of time	Impulses/s (500–1000 in nerve, 200–250 in muscle, 100–125 in a synapse)
Contractility	Contraction on excitation (muscle only)	Force, amplitude of contraction
Secretion	Release of a secretion on excitation (glands)	Volume of secretion

By their nature, **stimuli** are **physical** (mechanical, thermal, electrical, light, sound), **chemical** (acids, alkalis, hormones, neurotransmitters) and **physicochemical** (changes in osmotic pressure or pH). By biological significance, **adequate** stimuli (to which a receptor is specially adapted — light for the eye) and **inadequate** stimuli ("seeing sparks" after a blow to the eye) are distinguished. In experiments **electric current** is used most often: it can be dosed precisely, does not damage the tissue and is close to natural biocurrents.

4. The cell membrane and ion channels

The cell membrane, 7–10 nm thick, consists of a **phospholipid bilayer** with proteins embedded in it (the fluid mosaic model). The lipid layer is almost impermeable to ions; ions pass through **ion channels** and **carrier proteins**.

Table 1.5. *Transport of substances across the membrane*

Type	Energy	Examples
Simple diffusion	Not required, down the concentration gradient	O_2, CO_2, lipid-soluble substances

Type	Energy	Examples
Diffusion through channels	Not required	Na^+, K^+, Ca^{2+}, Cl^- ion channels
Facilitated diffusion	Not required, with a carrier protein	Passage of glucose through GLUT
Primary active transport	Energy of ATP	Na^+/K^+-ATPase, Ca^{2+}-ATPase
Secondary active transport	Energy of the Na^+ gradient	Absorption of glucose and amino acids together with Na^+ (SGLT)
Endocytosis, exocytosis	ATP	Release of neurotransmitters, phagocytosis

Ion channels are protein molecules that span the membrane; they have a **selectivity filter** (letting through only a particular ion) and a **gating mechanism**. By the way they open, channels are **voltage-gated** (open when the membrane potential changes — fast Na^+ channels), **ligand-gated** (open when a neurotransmitter binds — the N-cholinergic receptor for acetylcholine), **mechanosensitive** (open on stretch) and **"leak" channels** (always open — the K^+ channels at rest). The fast Na^+ channel has two gates: the outer **activation (m)** gate and the inner **inactivation (h)** gate.

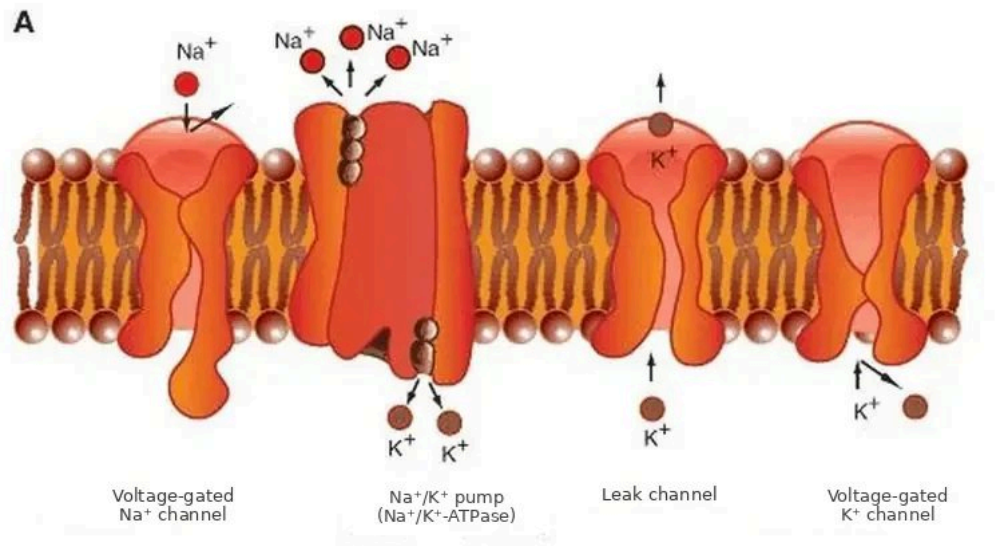


Figure 1.2. *Ion channels and the Na^+/K^+ pump (Alavia O.T., 2018)*

The Na^+/K^+ pump (Na^+/K^+-ATPase), using the energy of one ATP molecule, moves **3 Na^+** ions out of the cell and **2 K^+** ions into it. As a result: 1) the ion concentration gradients are maintained; 2) the pump is **electrogenic** — one extra positive charge is moved out in each cycle, contributing to the membrane potential (−5…−10 mV); 3) cell volume is regulated. The pump accounts for 30–40% of the energy expenditure of a tissue (up to 70% in nervous tissue).

Table 1.6. *Intracellular and extracellular ion concentrations (mammalian skeletal muscle, mmol/l)*

Ion	Inside the cell	Outside the cell	Ratio
K^+	140–155	4–5	≈ 35 : 1 (more inside)
Na^+	10–14	140–145	≈ 1 : 12 (more outside)
Cl^-	4–9	110–120	≈ 1 : 20
Ca^{2+}	0,0001	2–2,5	≈ 1 : 20 000
Organic anions (proteins)	Many	Very few	—

5. The resting membrane potential

The **resting potential (membrane potential)** is the potential difference between the inner and outer surfaces of the membrane of a cell at rest. The inner surface of the membrane is negatively charged relative to the outer surface. Its magnitude is −60…−70 mV in a nerve cell, −80…−90 mV in skeletal muscle and −50…−60 mV in smooth muscle.

The mechanism of the resting potential (membrane-ion theory: J. Bernstein, A. Hodgkin, A. Huxley)

1. **Unequal distribution** of ions inside and outside the cell (the result of the work of the Na^+/K^+ pump).
2. **Different permeability** of the resting membrane to different ions: the permeability to K^+ is 25–100 times higher than to Na^+ (K^+ "leak" channels are always open).
3. K^+ ions **leave the cell** down their concentration gradient, while the large organic anions cannot cross the membrane and stay inside — the inner surface of the membrane becomes negative and the outer surface positive.
4. The electric field that arises opposes further exit of K^+; the point at which the concentration and electrical forces balance is called the **potassium equilibrium potential** (≈ −90 mV). It can be calculated with the **Nernst equation**.
5. A small leak of Na^+ into the cell makes the potential somewhat less negative than the K^+ equilibrium potential (−70 mV); the Na^+/K^+ pump adds an electrogenic contribution.

REMEMBER

The main cause of the resting potential is the **diffusion of K^+ ions out of the cell**. Therefore, if the K^+ concentration in the extracellular fluid rises (hyperkalemia), the gradient decreases and the membrane **depolarizes**.

CLINICAL SIGNIFICANCE

Changes in blood potassium are dangerous for the heart and muscles. In **hyperkalemia** (renal failure, extensive tissue damage) the resting potential decreases and Na^+ channels become inactivated — cardiac conduction is disturbed and cardiac arrest is possible. In **hypokalemia** (vomiting, diuretics) the membrane becomes hyperpolarized — muscle weakness and arrhythmias appear.

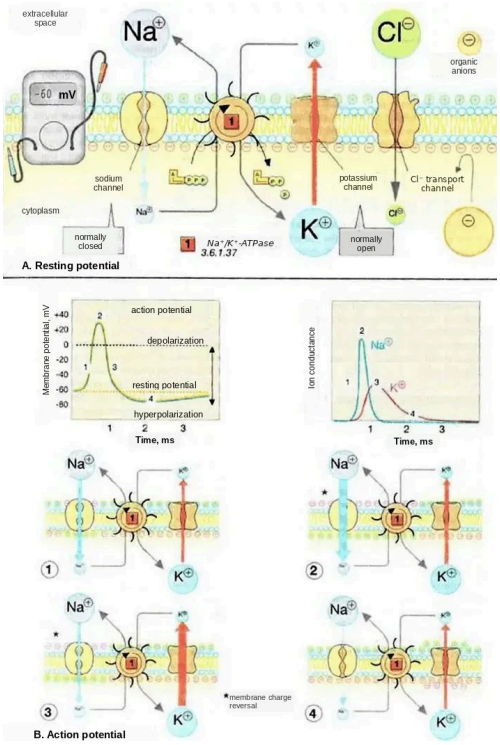


Figure 1.3. *Ionic mechanisms of the resting and action potentials: A — resting potential; B — phases of the action potential and ion permeability (Alavia O.T., 2018)*

6. The action potential

The **action potential** is a rapid, short-lasting change in the membrane potential during excitation, in which the membrane charge is reversed (the inner surface becomes positive). In a nerve fiber it lasts 0.5–2 ms and its amplitude is 100–120 mV.

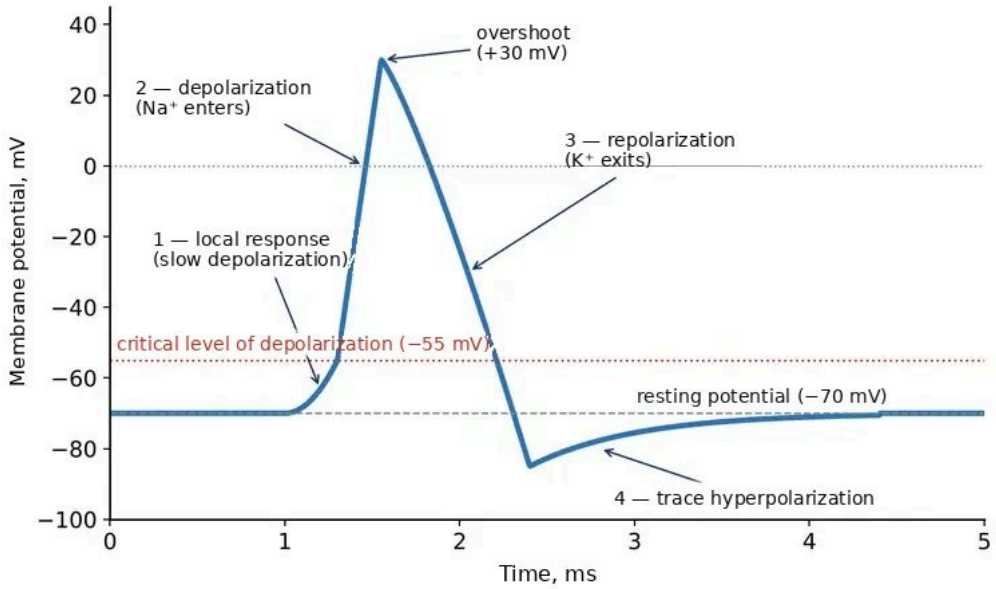


Figure 1.4. *Phases of the action potential (drawn after Alavia O.T., 2018; Costanzo L.S., 2018)*

Table 1.7. *Phases of the action potential and their ionic mechanisms*

Phase	Membrane potential	Ionic mechanism
Local response (slow depolarization)	From −70 to −55 mV	The stimulus opens some of the Na^+ channels; if the critical level is not reached, the potential dies away

Phase	Membrane potential	Ionic mechanism
Rapid depolarization and reversal (overshoot)	From −55 to +30 mV	At the critical level of depolarization the voltage-gated fast Na^+ channels open abruptly (positive feedback — a regenerative process), and Na^+ flows into the cell
Repolarization	From +30 to −70 mV	Na^+ channels are inactivated (the h gate closes), voltage-gated K^+ channels open and K^+ leaves the cell
After-hyperpolarization	Below −70 (−80…−90 mV)	K^+ channels close slowly, K^+ continues to leave
After-depolarization (in some tissues)	Slightly depolarized	Slowing of repolarization

REMEMBER

The **critical level of depolarization (CLD)** is the membrane potential at which an action potential arises (≈ −50…−55 mV in a nerve). The difference between the resting potential and the CLD is called the **threshold depolarization** (≈ 15–20 mV): the smaller it is, the higher the excitability. After an action potential the ion concentrations hardly change (very few ions cross in a single impulse); they are restored by the Na^+/K^+ pump.

The difference between the local response and the action potential

Table 1.8. *Local response and action potential*

Feature	Local response	Action potential
Stimulus strength	Below threshold	Threshold or above
The “all-or-none” law	Not obeyed — the amplitude depends on stimulus strength (graded)	Obeyed — the amplitude is constant
Summation	Summate	Do not summate
Spread	With decrement, over 1–2 mm	Without decrement, along the whole membrane
Refractoriness	Absent	Present
Excitability	Increased	Falls to zero, then recovers

MEMORY AID

Remember the order of the action potential as **“NaK”**: first **Na** enters (depolarization), then **K** leaves (repolarization). At rest, “potassium escapes, the anions stay” — that is why the inside is negative.

Changes in excitability during the action potential

- During the **local response** excitability increases.
- **Absolute refractory period** (depolarization and the beginning of repolarization) — even a very strong stimulus cannot evoke a new action potential: the Na^+ channels are open or inactivated.
- **Relative refractory period** (the end of repolarization) — only a suprathreshold stimulus evokes a response.
- **Supernormal period (exaltation)** — during after-depolarization excitability is above normal.
- **Subnormal period** — during after-hyperpolarization excitability is reduced. These periods are discussed in detail in the next topic (laws of stimulation).

CAUTION

Local anesthetics (lidocaine, procaine (novocaine)) **block voltage-gated Na^+ channels**: no action potential arises and the pain impulse is not conducted. Tetrodotoxin (puffer fish poison) also blocks Na^+ channels and causes paralysis of the respiratory muscles. Tetraethylammonium, on the other hand, blocks K^+ channels and slows repolarization.

7. Laboratory work

Experiment 1: preparing the frog nerve-muscle preparation

1. Immobilize the frog as instructed by the teacher (destruction of the spinal cord — a “spinal frog”).
2. Cut the body just below the forelimbs and remove the internal organs; leave the hind limbs and the lumbar part of the vertebral column.
3. Pull the skin downward like a stocking.
4. Divide the vertebral column in the middle and separate the two legs into individual preparations.
5. Carefully isolate the sciatic nerve in the thigh region, and dissect out the calf muscle (m. gastrocnemius) together with the Achilles tendon.
6. Keep the preparation moist with Ringer's solution (if it dries, excitability is lost).

Experiment 2: Galvani's first experiment

1. Take bimetallic forceps made of copper and zinc.
2. Touch the tips of the forceps simultaneously to the nerves of the preparation (in front of the vertebral column).
3. Watch the muscle contract at each touch. **Results and analysis:** L. Galvani explained the contraction by “animal electricity”. A. Volta, however, proved that the cause of the contraction is the **current of a galvanic cell** formed by two different metals and the tissue fluid (an electrolyte). Thus the first experiment is not yet proof of animal electricity.

Experiment 3: Galvani's second experiment (without metal)

1. Lift the sciatic nerve of a nerve-muscle preparation with a glass hook.
2. Lower the nerve simultaneously onto the **injured** (cut) surface of the muscle and onto its uninjured surface.
3. Watch the muscle contract. **Results and analysis:** the muscle contracts even without metal. The injured surface is negatively charged relative to the uninjured one (the **injury current**, demarcation potential); the nerve acts as a bridge between the two surfaces and the current stimulates it. This experiment was the first proof of the **existence of bioelectric potentials** in tissues.

Experiment 4: Matteucci's secondary contraction experiment

1. Prepare two nerve-muscle preparations.
2. Place the nerve of the second preparation on the muscle of the first preparation.
3. Stimulate the nerve of the first preparation with an electric current. **Results and analysis:** both muscles contract. The **action potentials** generated when the first muscle contracts stimulate the second nerve lying on it. The experiment proves that bioelectric potentials are generated during excitation.

8. Practical skills

Skill: analyzing a graph of the action potential

1. Mark the abscissa (time, ms) and ordinate (potential, mV) axes.
2. Draw the resting potential level (−70 mV) and the critical level of depolarization (−55 mV).
3. Show the phases of the local response, rapid depolarization, overshoot, repolarization and after-hyperpolarization.
4. Under each phase write its ionic mechanism (Na^+ entry, inactivation of Na^+ channels, K^+ exit).
5. Below the graph mark the periods of excitability (absolute and relative refractoriness, supernormal, subnormal).

Skill: constructing a regulatory loop

1. Choose a regulated parameter (for example, blood pressure).
2. Indicate the receptors that sense it (baroreceptors of the aortic arch and carotid artery).
3. Mark the afferent pathway, the center (the circulatory center in the medulla oblongata), the efferent pathway and the effectors (heart, vessels).
4. Determine the type of feedback (negative) and explain the direction of the response when the parameter changes.

Review questions

1. What are the subject and tasks of physiology?
2. What are the differences, advantages and drawbacks of acute and chronic experiments?
3. Describe the contributions of W. Harvey, I.M. Sechenov and I.P. Pavlov to the development of physiology.
4. What is homeostasis? Give examples of rigid and plastic parameters.
5. Explain the differences between nervous and humoral regulation.
6. Explain the concepts of the reflex arc and the reflex ring.
7. Give examples of negative and positive feedback.
8. What are the components of P.K. Anokhin's functional system?
9. List the excitable tissues and their main properties.
10. Explain the types and structure of ion channels.
11. What is the function of the Na^+/K^+ pump, and what does its electrogenicity consist of?
12. How does the resting potential arise?
13. Explain the phases of the action potential and their ionic mechanisms.
14. How does the local response differ from the action potential?
15. What do Galvani's first and second experiments prove?

Test questions

1. With which discovery is the beginning of scientific physiology associated?
   A) Introduction of the concept of the reflex
   B) The discovery of the circulation of the blood by W. Harvey
   C) The discovery of animal electricity
   D) The discovery of central inhibition
   E) The doctrine of conditioned reflexes

2. What is the main advantage of a chronic experiment?
   A) The possibility of studying the organism in its natural, intact state for a long time
   B) It is performed under anesthesia
   C) It gives quick results
   D) It is performed on an isolated organ
   E) The animal dies after the experiment

3. Which is a “rigid” homeostatic parameter?
   A) Heart rate
   B) Blood pressure
   C) Blood glucose level
   D) Respiratory rate
   E) Arterial blood pH

4. Which is an example of positive feedback?
   A) Decreased TSH release when thyroxine rises
   B) Slowing of the heart rate when blood pressure rises
   C) Increased oxytocin release during labor
   D) Insulin release when glucose rises
   E) Sweating when body temperature rises

5. How many ions does the Na^+/K^+ pump transport using one ATP?
   A) 2 Na^+ in, 3 K^+ out
   B) 1 Na^+ out, 1 K^+ in
   C) 3 K^+ out, 2 Na^+ in
   D) 3 Na^+ out, 2 K^+ in
   E) Only Na^+ out

6. What is the main cause of the resting potential?
   A) Diffusion of K^+ ions out of the cell
   B) Entry of Na^+ ions into the cell
   C) Entry of Ca^{2+} ions into the cell
   D) Exit of Cl^- ions from the cell
   E) Exit of organic anions from the cell

7. With which process is the depolarization phase of the action potential associated?
   A) Opening of K^+ channels
   B) Opening of voltage-gated fast Na^+ channels and entry of Na^+
   C) Activation of the Na^+/K^+ pump
   D) Entry of Cl^- into the cell
   E) Inactivation of Na^+ channels

8. What happens in the repolarization phase of the action potential?
   A) Na^+ channels open
   B) Ca^{2+} enters the cell
   C) Na^+ channels are inactivated and K^+ leaves the cell
   D) The Na^+/K^+ pump stops
   E) Membrane permeability does not change

9. Which is a property of the local response?
   A) It spreads without decrement
   B) It obeys the “all-or-none” law
   C) It has a refractory period
   D) It summates and its amplitude depends on stimulus strength
   E) Its amplitude is constant

10. How does the membrane potential change when the K^+ concentration in the extracellular fluid rises?
    A) The membrane depolarizes
    B) Hyperpolarization develops
    C) It does not change
    D) The amplitude of the action potential increases
    E) The membrane charge is reversed

11. What did Galvani's second experiment prove?
    A) That two different metals generate a current
    B) The existence of bioelectric potentials in tissues (contraction without metal)
    C) The speed of excitation in a nerve
    D) The development of tetanus in a muscle
    E) The harmful effect of electric current

12. What is the mechanism of action of local anesthetics (lidocaine)?
    A) They block K^+ channels
    B) They activate the Na^+/K^+ pump
    C) They enhance acetylcholine release
    D) They open Ca^{2+} channels
    E) They block voltage-gated Na^+ channels

13. Which tissue is not an excitable tissue?
    A) Nervous tissue
    B) Skeletal muscle
    C) Bone tissue
    D) Glandular tissue
    E) Cardiac muscle

14. What is excitability during the absolute refractory period?
    A) Doubled
    B) Above normal
    C) Normal
    D) Slightly reduced; a strong stimulus evokes a response
    E) Zero — no stimulus evokes a response

15. Who is the author of the theory of functional systems?
    A) I.P. Pavlov
    B) I.M. Sechenov
    C) W. Cannon
    D) P.K. Anokhin
    E) C. Bernard

Clinical cases

Case 1

In a patient with chronic renal failure, blood potassium reached 7.2 mmol/l. The patient developed muscle weakness and a disturbance of the heart rhythm.

Questions: How does a rise in blood potassium affect the resting potential? Why is the activity of excitable tissues disturbed?

Answer: The resting potential is generated mainly by the diffusion of K^+ out of the cell. When extracellular K^+ increases, the concentration gradient decreases and less K^+ leaves — the membrane depolarizes. At first excitability may increase, but with prolonged depolarization the voltage-gated Na^+ channels are inactivated and a full action potential cannot arise. The result is muscle weakness, disturbed cardiac conduction and, in severe cases, cardiac arrest.

Case 2

Before dental treatment, a dentist injected lidocaine. A few minutes later the patient no longer felt pain in that area, while sensation was preserved in other areas.

Questions: By what mechanism does lidocaine abolish pain? Which phase of the action potential is disturbed?

Answer: Lidocaine blocks the voltage-gated Na^+ channels in the membrane of nerve fibers. The (painful) stimulus cannot open the Na^+ channels, depolarization does not reach the critical level, and the rapid depolarization phase of the action potential does not develop. Excitation does not spread along the fiber and the pain signal does not reach the center. The effect is limited to the injected area, so sensation is preserved elsewhere.

Case 3

A student performed Galvani's first experiment: when the copper-zinc forceps touched the nerve, the muscle contracted. He explained this as “the muscle has its own electricity”.

Questions: Is the student's conclusion correct? Which experiment actually proves the presence of bioelectric potentials in tissue?

Answer: The conclusion is wrong. As A. Volta showed, in the first experiment the two different metals and the tissue fluid form a galvanic cell, and its current stimulates the nerve. The presence of bioelectric potentials in tissue is proved by Galvani's second experiment (without metal): when the nerve touches the injured and uninjured surfaces of the muscle, the muscle contracts under the action of the injury current. Matteucci's secondary contraction experiment shows that action potentials are generated during excitation.

Quick review

Table 1.9. *Key concepts of topic 1*

Concept	Brief definition
Physiology	The science that studies the functions of the organism and their regulation
Chronic experiment	Long-term observation under natural conditions in an animal operated on in advance (Pavlov)
Homeostasis	Constancy of the internal environment (Bernard, Cannon)
Negative feedback	Returns a deviation to normal
Functional system	A dynamic association directed toward a useful result (Anokhin)
Excitable tissues	Nerve, muscle, gland
Na^+/K^+ pump	3 Na^+ out, 2 K^+ in, 1 ATP; electrogenic
Resting potential	−60…−90 mV; mainly due to the exit of K^+
Critical level of depolarization	The level at which the action potential begins (≈ −55 mV)
Depolarization	Fast Na^+ channels open and Na^+ enters
Repolarization	Inactivation of Na^+ channels, exit of K^+
Local response	Subthreshold; graded, summates, spreads with decrement
Absolute refractoriness	Excitability equals zero
Galvani's 2nd experiment	Contraction without metal — proof of bioelectric potentials
Matteucci's experiment	Secondary contraction — proof of the action potential

TOPIC 2

Laws of stimulation

Learning objectives

To teach the student the laws by which excitable tissues respond to stimuli — the laws of strength, duration and rate of rise of stimulus strength (the gradient law), the “all-or-none” law, the polar law of direct current, the changes in excitability during excitation and the concept of lability; to teach the determination of threshold strength and chronaxie.

By the end of the class the student should know:

- the relationship between the stimulation threshold (threshold strength) and excitability;
- the law of strength and the “all-or-none” law, and the limits of their application;
- the strength–duration curve (the Hoorweg–Weiss–Lapicque curve) and the concepts of rheobase, utilization time and chronaxie;
- the phenomenon of accommodation and the gradient law;
- Pflüger's polar law, catelectrotonus and anelectrotonus, and Verigo's cathodal depression;
- the phasic changes in excitability during excitation;
- the concept of lability (N.E. Vvedensky) and its value in different tissues.

The student should be able to:

- determine the stimulation threshold of a nerve and a muscle in a nerve-muscle preparation;
- demonstrate the law of strength experimentally (threshold, submaximal and maximal contraction);
- demonstrate the polar law with direct current;
- explain the clinical significance of chronaxie.

1. Stimulation threshold and excitability

To evoke excitation, a stimulus must meet three conditions: **sufficient strength**, **sufficient duration** and **a sufficiently fast rise in strength**. Each has its own law.

The **stimulation threshold (threshold strength)** is the smallest strength of a stimulus that can evoke excitation. The threshold is the main measure of excitability, and the two are **inversely proportional**: the lower the threshold, the higher the excitability.

Table 2.1. *Types of stimuli by strength*

Stimulus	Description	Response
Subthreshold (subliminal)	Weaker than threshold	Only a local response, no action potential
Threshold (liminal)	The smallest effective strength	The most excitable fibers are excited
Suprathreshold (submaximal)	Stronger than threshold, weaker than maximal	The number of excited fibers increases
Maximal	The smallest strength that excites all fibers	The largest response
Supramaximal	Stronger than maximal	The response does not increase

REMEMBER

Excitability is **inversely proportional** to the threshold. The threshold of a nerve is lower than that of a muscle — the nerve is more excitable; this is why, in a preparation, it is easier to stimulate a muscle through its nerve (indirect stimulation).

2. The law of strength and the “all-or-none” law

The law of strength: a tissue of complex structure (a whole skeletal muscle, a nerve trunk) **increases its response** as the stimulus strength rises from threshold to maximal; beyond the maximal strength the response no longer changes. The reason is that a muscle consists of thousands of fibers with different excitability: as the strength rises, more and more fibers are recruited into excitation. With a very strong stimulus the response may decrease — the phenomenon of **pessimum**.

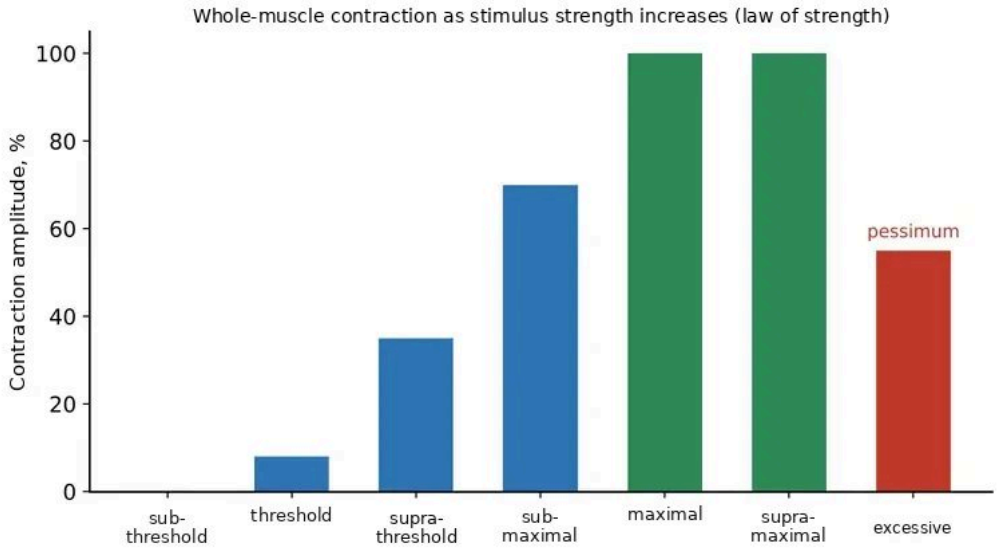


Figure 2.1. *The law of strength: change in the contraction of a whole muscle as stimulus strength increases (drawn after data of Alavia O.T., 2018)*

The “all-or-none” law (H. Bowditch, 1871): a single excitable structure (one nerve or muscle fiber) or a functional syncytium (cardiac muscle) does not respond to a subthreshold stimulus (“none”), while to a threshold or stronger stimulus it **always gives the same, maximal** response (“all”).

CAUTION

The “all-or-none” law is **relative**: 1) a subthreshold stimulus is not “nothing” either — it evokes a local response and increases excitability; 2) the size of the response depends not on the stimulus but on the state of the tissue at that moment (fatigue, temperature, ions). A whole skeletal muscle obeys not this law but the law of strength, while the heart, as a syncytium, obeys the “all-or-none” law.

3. The law of duration: the strength–duration curve

To evoke excitation, a stimulus must last for a certain time. There is an **inverse relationship** between the strength of a stimulus and its duration: the shorter the stimulus, the greater the strength required. This relationship is expressed by the **strength–duration curve** (the Hoorweg–Weiss–Lapicque hyperbola).

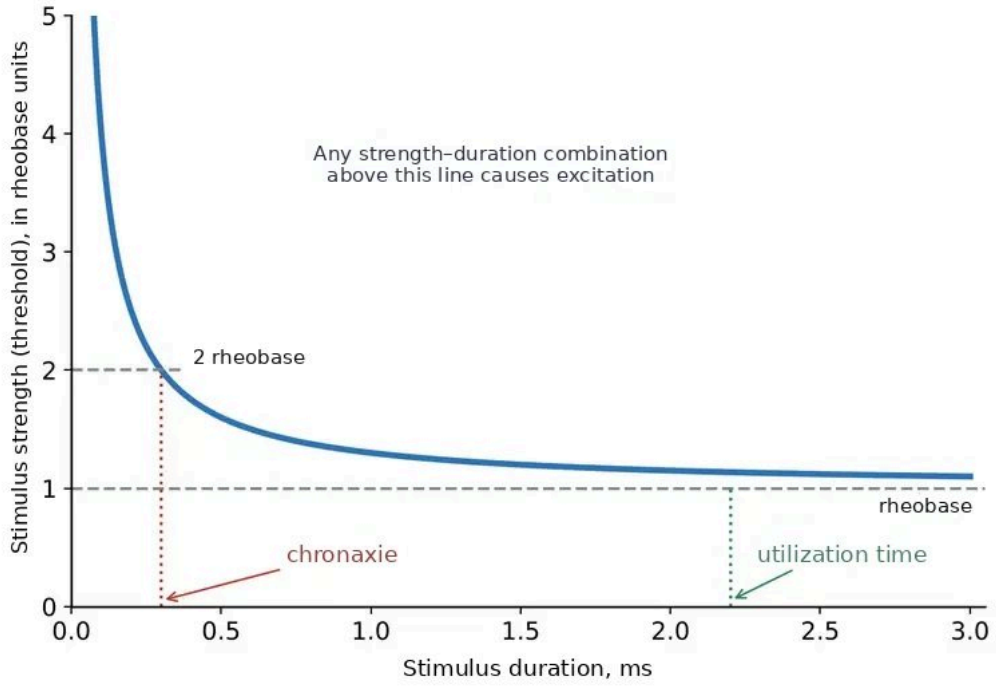


Figure 2.2. *The strength–duration curve: rheobase, utilization time and chronaxie (drawn after Alavia O.T., 2018)*

Table 2.2. *Parameters of the strength–duration curve*

Parameter	Definition	Meaning
Rheobase	The smallest strength of direct current that evokes excitation when applied for a long time	Threshold strength; a measure of excitability in terms of strength
Utilization time	The shortest time for which a current equal to the rheobase must act to evoke excitation	Difficult to measure precisely (the curve is flat there)
Chronaxie (L. Lapicque)	The shortest time for which a current equal to twice the rheobase must act to evoke excitation	A measure of excitability in terms of time; measured precisely

On the left part of the curve, very short stimuli (shorter than 0.01 ms) do not evoke excitation at any strength — the Na^+ channels do not have time to open. On the right part, a current weaker than the rheobase is ineffective even if it acts for a very long time.

Table 2.3. *Chronaxie of different tissues (approximate)*

Tissue	Chronaxie, ms
Fast-conducting (motor) nerve fibers in humans	0,08–0,16
Human skeletal muscle (flexors)	0,08–0,16
Human skeletal muscle (extensors)	0,16–0,32
Cardiac muscle	1–3
Smooth muscles	5–100 and more

CLINICAL SIGNIFICANCE

Chronaximetry is a method of electrodiagnosis in neurology. When a nerve is damaged, the chronaxie of the muscle increases (excitability decreases) and it becomes difficult to stimulate the muscle through its nerve; when the nerve recovers, chronaxie returns to normal. It is used to follow peripheral nerve injuries and to choose the parameters of electrical stimulation in physiotherapy.

4. The gradient (accommodation) law

The gradient law (du Bois-Reymond): excitation depends not only on the absolute strength of a stimulus but also on its **rate of rise per unit of time**. If the strength rises steeply (quickly), excitation occurs; if it rises slowly, excitation may not occur even when the strength exceeds the threshold.

Accommodation is the adaptation of a tissue to a slowly rising stimulus; the critical level of depolarization rises (the threshold increases). Mechanism: during slow depolarization the Na^+ channels are gradually **inactivated** and K^+ permeability increases; as a result the regenerative Na^+ current does not arise.

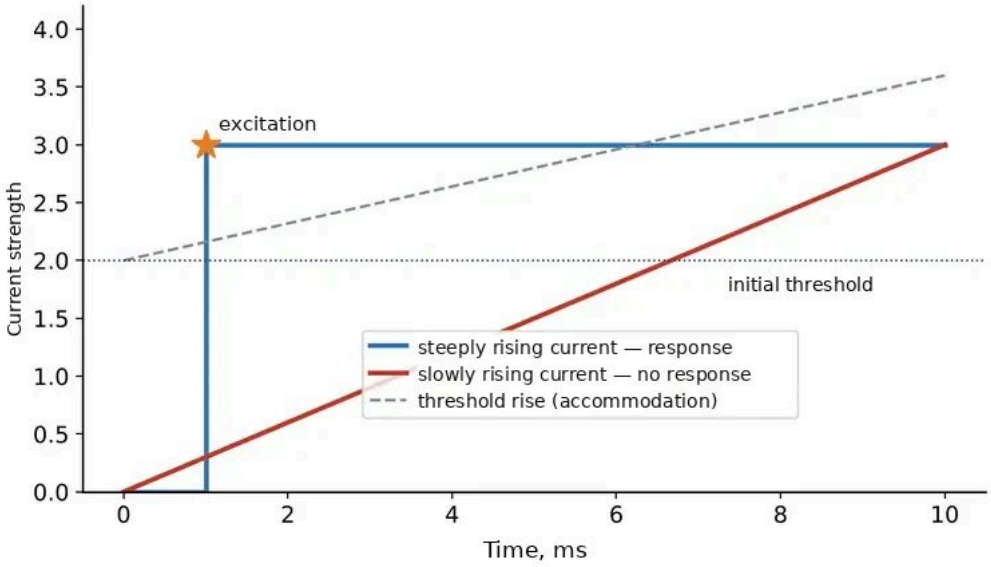


Figure 2.3. *Accommodation: rate of rise of current strength and the threshold (diagram)*

> **REMEMBER**
> The rate of accommodation differs between tissues: it is high in motor nerve fibers and low in smooth muscles and sensory fibers. Tissues with low accommodation respond even to slowly rising stimuli.

5. The polar law of direct current

Pflüger's polar law (1859): direct current stimulates a tissue only when the circuit is **made (closed)** and **broken (opened)**; while the current is flowing there is no excitation. When the circuit is **closed, excitation arises under the cathode**, and when it is **opened — under the anode**. The cathodal effect on closing is stronger than the anodal effect on opening.

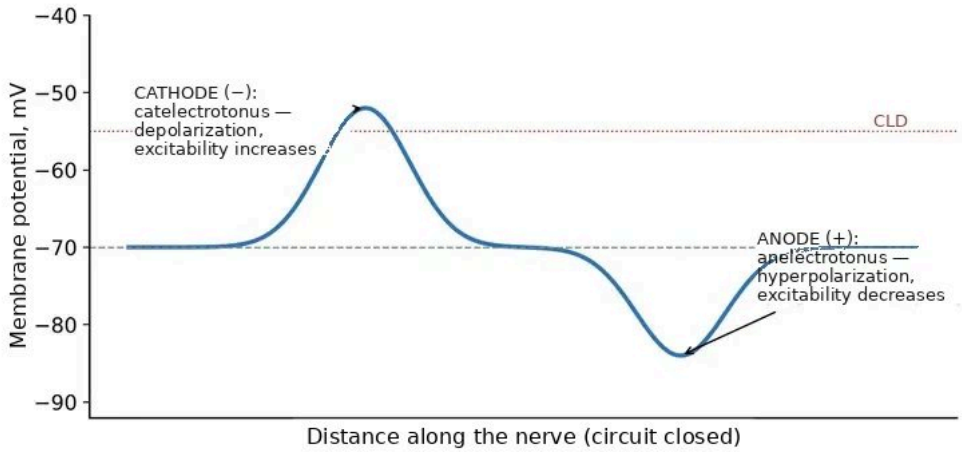


Figure 2.4. *The polar law: change in the membrane potential under the cathode and anode when direct current is switched on (diagram)*

Table 2.4. *Electrotonic phenomena*

Phenomenon	Where	Effect on the membrane	Excitability
Catelectrotonus	Under the cathode (−)	Depolarization (approaches the CLD)	Increased
Anelectrotonus	Under the anode (+)	Hyperpolarization	Decreases
Cathodal depression (B.F. Verigo)	Under the cathode, with prolonged current	Prolonged depolarization — inactivation of Na^+ channels	Decreases, may even disappear
Anodal break effect	Under the anode, when the current is switched off	After hyperpolarization the CLD is lowered	Excitation arises

The appearance of excitation under the anode **when the current is switched off** is explained as follows: while the current flows, the membrane is hyperpolarized, the inactivation of Na^+ channels is removed and the CLD falls. When the current is switched off, the membrane potential is quickly restored, reaches the new, lowered CLD and evokes excitation.

MEMORY AID

“Cathode on make, anode on break”: when the circuit is closed (current switched on) excitation arises at the cathode, and when it is opened (current switched off) — at the anode. The cathode “warms up” the membrane (depolarization), the anode “cools it down” (hyperpolarization).

CLINICAL SIGNIFICANCE

In physiotherapy, **galvanization** and the introduction of drugs by current (**electrophoresis**) are based on the polar law. Because excitability is reduced under the anode, the anode is placed over a painful area to relieve pain, while the cathode is used to increase excitability.

6. Changes in excitability during excitation

When an action potential arises, the excitability of the tissue changes in phases. These phases explain the response to rhythmic stimuli, the development of tetanus and the features of the heart's work.

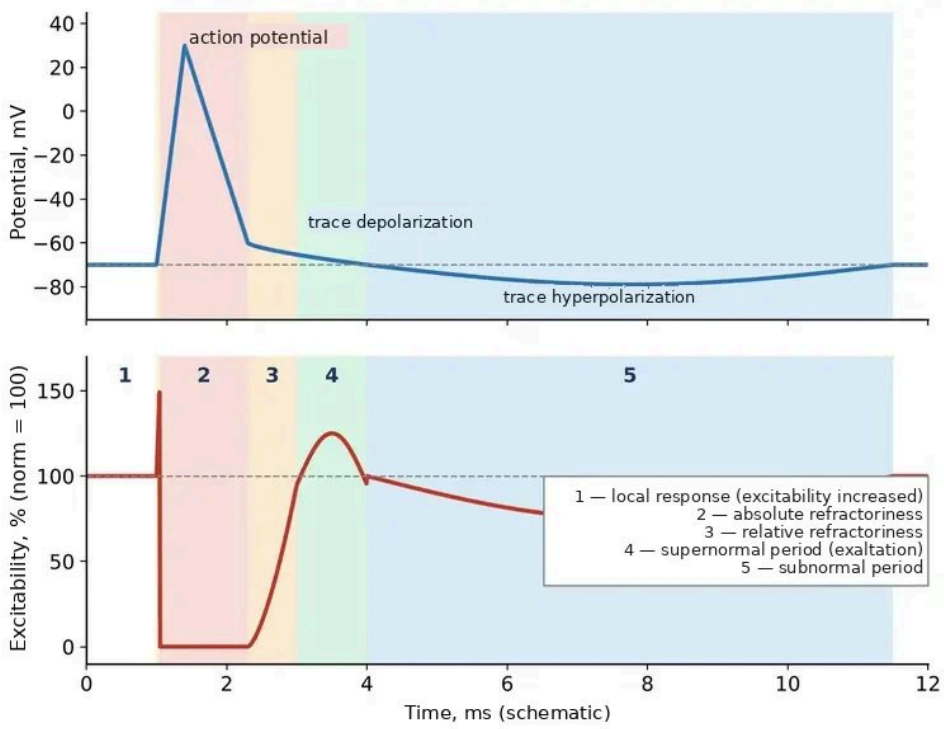


Figure 2.5. *The action potential (top) and the phases of excitability (bottom) (drawn after a figure by Alavia O.T., 2018)*

Table 2.5. *Phases of excitability*

Phase	Corresponding part of the action potential	Excitability	Cause
Primary exaltation	Local response	Increased	The membrane is close to the CLD
Absolute refractoriness	Depolarization and the beginning of repolarization	Zero	Na^+ channels are open or inactivated
Relative refractoriness	The end of repolarization	Reduced, recovering	Some Na^+ channels are still inactivated, K^+ permeability is high
Supernormal (exaltation)	After-depolarization	Above normal	The membrane potential is close to the CLD
Subnormal	After-hyperpolarization	Below normal	The membrane potential is farther from the CLD

Duration of the absolute refractory period: 0.5–1 ms in a nerve fiber, 2–3 ms in skeletal muscle, 250–300 ms in cardiac muscle (almost the entire duration of the action potential).

> **REMEMBER**
> Absolute refractoriness ensures that excitation spreads in **one direction** and sets the upper limit of the lability of a tissue. Because the refractory period of the heart is long, cardiac muscle **cannot contract tetanically**.

7. Lability (functional mobility)

Lability (N.E. Vvedensky, 1886) is the maximum number of action potentials that an excitable tissue can generate per unit of time (per 1 s) in step with the rhythm of stimulation, without transforming it.

Lability depends primarily on the **duration of the absolute refractory period**: the shorter it is, the higher the lability.

Table 2.6. *Lability of different structures*

Structure	Lability, impulses/s
Myelinated nerve fiber	500–1000
Skeletal muscle	200–250
Neuromuscular synapse	100–150
Cardiac muscle	3–5
Smooth muscle	1 and less

Lability is not a constant value: training, temperature and rhythmic stimulation change it. The increase in lability under rhythmic stimulation is called **assimilation of rhythm** (A.A. Ukhtomsky). The structure with the lowest lability determines the operating frequency of the whole system (for example, the synapse in a nerve-muscle preparation).

8. Laboratory work

Experiment 1: determining the stimulation threshold of a nerve and a muscle

1. Prepare a frog nerve-muscle preparation and attach the muscle to the lever of a myograph (or kymograph).
2. Place the electrodes of the electrical stimulator on the sciatic nerve; increase the voltage gradually, starting from zero.
3. Record the voltage at which the muscle first contracts noticeably — this is the **threshold of the nerve** (indirect stimulation).
4. Place the electrodes directly on the muscle and repeat the experiment — the **threshold of the muscle** (direct stimulation).
5. Compare the two values. **Expected results and analysis:** the threshold of the nerve is lower than that of the muscle, i.e., the excitability of the nerve is higher. This is related to the membrane properties of nerve fibers (closer to the CLD).

Experiment 2: studying the law of strength

1. Starting from the threshold strength, increase the stimulus strength step by step (for example, by 0.1 V each time).
2. At each strength record the amplitude of the single contraction on the kymograph tape.
3. Mark the strength at which the amplitude stops increasing — the **maximal strength**.
4. Increase the strength further and show that the amplitude does not change (supramaximal strength). **Expected results and analysis:** from threshold to maximal strength the contraction amplitude increases, because the muscle consists of fibers of different excitability and, as the strength rises, more fibers are excited (the law of strength). At maximal strength all fibers are excited, so a further increase does not enhance the response. Each individual fiber, however, obeys the “all-or-none” law.

Experiment 3: demonstrating Pflüger's polar law

1. Place the nerve of a nerve-muscle preparation on two nonpolarizable electrodes; connect a direct current source through a switch.

2. Place the cathode closer to the muscle and the anode farther away; close and open the circuit and record the contractions.
3. Reverse the poles (anode closer to the muscle) and repeat the experiment.
4. Vary the current strength between weak, moderate and strong and record the results in a table. **Expected results and analysis:** with a weak current, contraction is seen only on closing the circuit (excitation under the cathode); with a moderate current — both on closing and on opening. With a strong current, if the anode is close to the muscle, the cathodal excitation on closing cannot pass through the hyperpolarized region under the anode (anodal block) — contraction occurs only on opening.

9. Practical skills

Skill: determining chronaxie from the strength–duration curve

1. Mark the stimulus strength on the ordinate and the duration on the abscissa.
2. Find the rheobase level from the flat part of the curve.
3. Multiply the rheobase by two and mark this point on the ordinate.
4. From this point draw a horizontal line until it crosses the curve.
5. From the point of intersection drop a perpendicular onto the abscissa — this is the **chronaxie**.
6. Compare the result with the type of tissue and draw a conclusion about its excitability.

Skill: relating the phases of excitability to the action potential

1. Draw a graph of the action potential.
2. Below it, draw the corresponding excitability graph in time (normal = 100%).
3. Mark the periods: local response — exaltation; depolarization and the beginning of repolarization — absolute refractoriness; the end of repolarization — relative refractoriness; after-potentials — the supernormal and subnormal periods.
4. Briefly write the ionic mechanism of each phase.

Review questions

1. What conditions must a stimulus meet to evoke excitation?
2. What is the stimulation threshold, and how is it related to excitability?
3. State the law of strength and explain its mechanism.
4. To which structures does the “all-or-none” law apply, and why is it relative?
5. Draw and explain the strength–duration curve.
6. Define rheobase, utilization time and chronaxie.
7. What is the clinical significance of chronaximetry?
8. Explain the gradient law and the mechanism of accommodation.
9. State Pflüger's polar law.
10. What are catelectrotonus and anelectrotonus?
11. How does Verigo's cathodal depression arise?
12. Why does excitation under the anode arise when the current is switched off?
13. Through what phases does excitability pass during excitation?
14. What is lability, and what does it depend on?

15. Why can cardiac muscle not contract tetanically?

Test questions

1. What is the relationship between excitability and the stimulation threshold?
   A) Directly proportional
   B) There is no relationship
   C) Inversely proportional
   D) Related only in muscle
   E) Related only in nerve

2. Which law does a whole skeletal muscle obey?
   A) The “all-or-none” law
   B) The law of strength
   C) The polar law
   D) Bowditch's law
   E) It obeys no law

3. Which structure obeys the “all-or-none” law?
   A) A whole skeletal muscle
   B) A nerve trunk
   C) A layer of smooth muscle
   D) Cardiac muscle
   E) Whole glands

4. What is rheobase?
   A) The smallest current strength that evokes excitation when applied for a long time
   B) The duration of action of a current equal to twice the rheobase
   C) The strength that evokes the maximal response
   D) The absolute refractory period
   E) The number of impulses per 1 s

5. Chronaxie is —
   A) the duration of action of a current equal to the rheobase
   B) the threshold strength
   C) the shortest time for which a current equal to twice the rheobase must act to evoke excitation
   D) the maximal strength
   E) the duration of the refractory period

6. What is the mechanism of accommodation?
   A) Activation of Na^+ channels
   B) A fall in the critical level of depolarization
   C) Arrest of the Na^+/K^+ pump
   D) An increase in the resting potential
   E) Inactivation of Na^+ channels during slow depolarization and a rise in the CLD

7. Where does excitation arise when a direct current circuit is closed?
   A) Under the anode
   B) Under the cathode
   C) Between the two poles
   D) Under both poles at the same time
   E) It does not arise

8. How does excitability change in anelectrotonus?
   A) It decreases (hyperpolarization)
   B) Increased
   C) It does not change
   D) It first decreases and then increases
   E) It becomes zero

9. What is Verigo's cathodal depression associated with?
   A) Hyperpolarization of the membrane
   B) Closure of K^+ channels
   C) Inactivation of Na^+ channels during prolonged depolarization
   D) Activation of the Na^+/K^+ pump
   E) Opening of calcium channels

10. Which part of the action potential corresponds to the absolute refractory period?
   A) After-hyperpolarization
   B) After-depolarization
   C) Local response
   D) Depolarization and the beginning of repolarization
   E) The resting state

11. What is excitability like in the supernormal period?
   A) Zero
   B) Above normal
   C) Below normal
   D) Normal
   E) It responds only to a strong stimulus

12. Which structure has the highest lability?
   A) Smooth muscle
   B) Cardiac muscle
   C) Neuromuscular synapse
   D) Skeletal muscle
   E) Myelinated nerve fiber

13. What mainly determines lability?
   A) The duration of the absolute refractory period
   B) The magnitude of the resting potential
   C) Muscle mass
   D) Stimulus strength
   E) The distance between the electrodes

14. Why does cardiac muscle not contract tetanically?
    A) Its high excitability
    B) It obeys the law of strength
    C) Its short chronaxie
    D) The absence of automaticity
    E) Its long absolute refractory period

15. How does muscle chronaxie change when a nerve is damaged?
    A) It decreases
    B) It becomes zero
    C) It does not change
    D) It increases (excitability decreases)
    E) It becomes equal to the rheobase

Clinical cases

Case 1

A patient injured in the forearm has damage to the ulnar nerve. A neurologist performed electrodiagnosis and found that the chronaxie of the muscles innervated by this nerve was 10 times higher than normal. On repeat examination three months later, the chronaxie had begun to decrease.

Questions: What does chronaxie show? What do its increase and subsequent decrease mean?

Answer: Chronaxie is the shortest time for which a current equal to twice the rheobase must act to evoke excitation; it is a measure of excitability in terms of time. When a nerve is damaged, the muscle loses its innervation, indirect stimulation through nerve fibers becomes impossible, and the muscle's own excitability is low — so chronaxie rises sharply. The beginning of a decrease in chronaxie indicates regeneration of the nerve fibers and restoration of innervation. That is why chronaximetry is used to follow the course of nerve injuries.

Case 2

A student stimulated the nerve of a nerve-muscle preparation with a current of 2 V — the muscle contracted. He then raised the voltage from zero to 4 V very slowly (over several seconds) — the muscle did not contract.

Questions: Why did the stronger current not evoke a response? Which law does this illustrate?

Answer: This is the gradient law and the phenomenon of accommodation. When the current rises slowly, the membrane depolarizes gradually; meanwhile the Na^+ channels are inactivated, K^+ permeability increases and the critical level of depolarization rises. As a result, even when the current exceeds the initial threshold, no regenerative Na^+ current arises and no action potential appears. For excitation, the stimulus strength must rise quickly enough.

Case 3

A physiotherapist placed a direct current electrode over the painful area of a patient. A student asked: “Which pole should be placed over the painful area to reduce pain?”

Questions: Which pole should be placed there, and why? Which law is this based on?

Answer: The anode is placed over the painful area. According to Pflüger's polar law, the membrane under the anode is hyperpolarized (anelectrotonus), its potential moves away from the critical level and excitability decreases — pain receptors and fibers are excited less. Under the cathode, by

contrast, excitability increases because of depolarization (catelectrotonus), so the cathode is used to stimulate muscles and nerves.

Quick review

Table 2.7. *Key concepts of topic 2*

Concept	Brief definition
Threshold strength	The smallest strength that evokes excitation; inversely proportional to excitability
Law of strength	Whole muscle: the response grows as strength increases (up to maximal)
“All-or-none”	A single fiber and the heart: the response is constant, maximal
Pessimum	Decrease of the response to an excessively strong or frequent stimulus
Rheobase	The smallest effective current strength with prolonged action
Utilization time	The shortest time needed for a current of one rheobase
Chronaxie	The shortest time needed for a current of 2 rheobases (Lapicque)
Accommodation	Adaptation to a slowly rising stimulus; the CLD rises
Polar law	Excitation under the cathode on closing, under the anode on opening
Catelectrotonus	Depolarization under the cathode, excitability increases
Anelectrotonus	Hyperpolarization under the anode, excitability decreases
Cathodal depression	Decrease in excitability with prolonged action of the cathode (Verigo)
Absolute refractoriness	Excitability 0; 0.5–1 ms in nerve, 250–300 ms in the heart
Supernormal period	After-depolarization — excitability increased
Lability	Maximum number of impulses per 1 s (Vvedensky); 500–1000 in nerve

TOPIC 3

Physiology of nerve fibers and synapses

Learning objectives

To teach the student the structure and types of nerve fibers, the mechanisms and laws of conduction of excitation along them, the lability of nerve fibers and the phenomenon of parabiosis, and the structure and types of synapses and the mechanism of synaptic transmission of excitation (using the neuromuscular synapse as an example).

By the end of the class the student should know:

- the structure of the neuron and the nerve fiber, and the difference between myelinated and unmyelinated fibers;
- the Erlanger–Gasser classification of nerve fibers (groups A, B, C);
- the mechanism of continuous conduction in an unmyelinated fiber and saltatory conduction in a myelinated fiber;
- the laws of conduction of excitation in a nerve;
- N.E. Vvedensky's doctrine of parabiosis and the phases of parabiosis;
- the classification of synapses and the stages of transmission of excitation in a chemical synapse;
- the action of acetylcholine at the neuromuscular synapse and the end-plate potential;
- the properties of synapses and the effects of drugs and poisons on them.

The student should be able to:

- demonstrate bilateral conduction and isolated conduction in a nerve fiber experimentally;
- identify the phases of parabiosis in a nerve-muscle preparation;
- prove that a synapse fatigues faster than a nerve.

1. Structure of the neuron and nerve fibers

The **neuron** is the structural and functional unit of the nervous system. It consists of a body (soma), short branching processes — **dendrites** (which receive information) — and one long process — the **axon** (which transmits information). The site where the axon leaves the body — the **axon hillock** (initial segment) — has the lowest threshold: this is where the action potential is generated (the trigger zone).

A **nerve fiber** is an axon surrounded by glial cells (**Schwann cells** in the periphery, **oligodendrocytes** in the CNS). In myelinated fibers the Schwann cell wraps the axon in many layers, forming a lipid-rich **myelin sheath**. The myelin is interrupted every 1–2 mm; these sites are called **nodes of Ranvier**, and voltage-gated Na^+ channels are packed very densely in them.

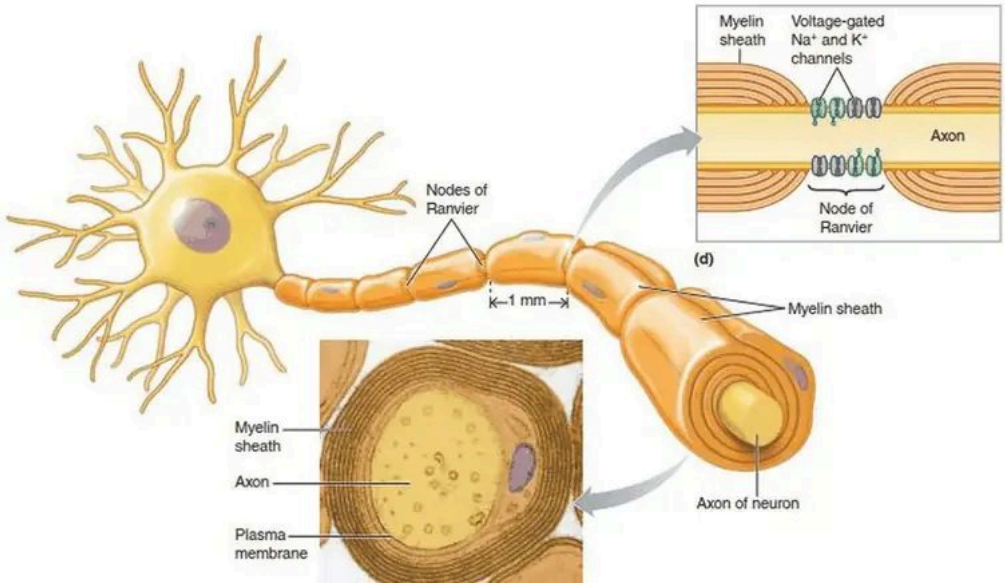


Figure 3.1. *Structure of a myelinated nerve fiber: the myelin sheath and nodes of Ranvier (Alavia O.T., 2018)*

Table 3.1. *Classification of nerve fibers (after Erlanger–Gasser)*

Type	Diameter, µm	Velocity, m/s	Myelin	Function
Aα	12–20	70–120	Present	Motor fibers of skeletal muscles, afferents from muscle spindles
Aβ	5–12	30–70	Present	Afferents from skin mechanoreceptors (touch, pressure)
Aγ	3–6	15–30	Present	Motor fibers to muscle spindles
Aδ	2–5	12–30	Present	Afferents for fast (sharp) pain and temperature
B	1–3	3–15	Thin	Preganglionic fibers of the autonomic nervous system
C	0,4–1,2	0,5–2	Absent	Slow (aching) pain, temperature; postganglionic sympathetic fibers

REMEMBER

Conduction velocity depends on the **diameter** of the fiber and its **myelination**: the thicker the fiber and the thicker its myelin, the higher the velocity. The fastest are Aα fibers (up to 120 m/s), the slowest are C fibers (0.5–2 m/s).

2. Conduction of excitation in a nerve fiber

The inner surface of an excited region is positive, while that of the neighboring unexcited region is negative. **Local (circular) currents** arise between them, depolarize the neighboring region to the critical level and evoke a new action potential in it. In this way excitation spreads **without decrement**.

- **In an unmyelinated fiber** conduction is **continuous**: each neighboring segment of the membrane is excited in turn; the velocity is low.
- **In a myelinated fiber** conduction is **saltatory (jumping)**: myelin acts as an electrical insulator, so the action potential is generated only at the nodes of Ranvier and “jumps” from one node to the next (even skipping 2–3 nodes). This increases the velocity 50-fold and saves energy (the Na^+/K^+ pump works only at the nodes).

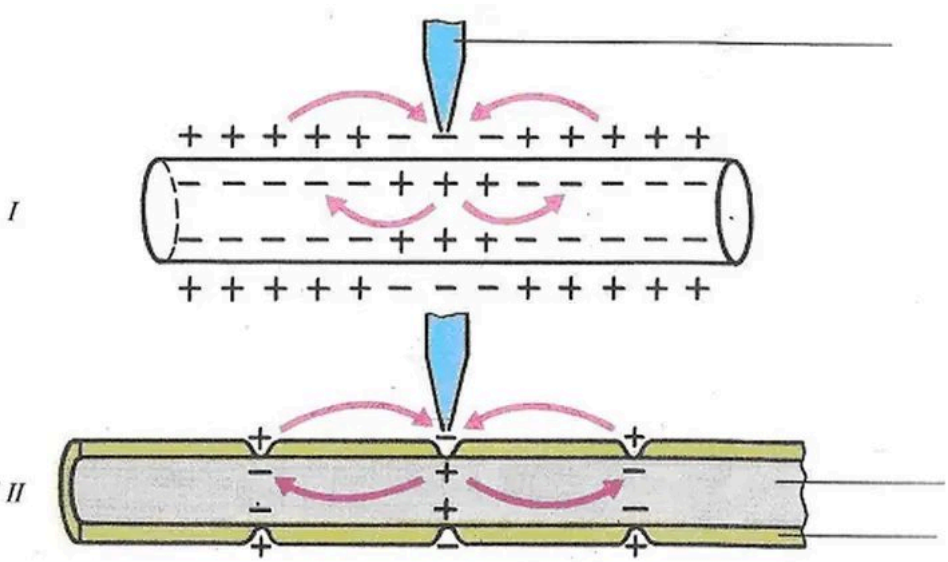


Figure 3.2. *Conduction of excitation: I — continuous in an unmyelinated fiber; II — saltatory in a myelinated fiber (Alavia O.T., 2018)*

CLINICAL SIGNIFICANCE

Multiple sclerosis and **Guillain–Barré syndrome** are demyelinating diseases. When myelin is destroyed, saltatory conduction is disturbed: impulses slow down or stop. The result is paralysis, sensory disturbances and impaired vision. Electroneuromyography reveals a decrease in conduction velocity.

Laws of conduction of excitation in a nerve

Table 3.2. *Laws of conduction of excitation*

Law	Content	Proof / significance
Anatomical and physiological integrity	Conduction stops when the fiber is cut or its physiological state is disturbed (cooling, procaine (novocaine), ligation)	The basis of local anesthesia
Bilateral conduction	If the middle of a fiber is stimulated, excitation spreads in both directions	Biocurrents are recorded on both sides; in the body, synapses ensure one-way direction
Isolated conduction	Each fiber in a nerve conducts excitation separately, without passing it to neighboring fibers	Precise innervation of each muscle and receptor; resistance of myelin and tissue fluid
Relative indefatigability	A nerve hardly fatigues even when stimulated for hours	Little energy is spent per impulse; Vvedensky's experiment
Conduction without decrement	The action potential amplitude does not decrease along the fiber	It is regenerated at every point

3. Lability of the nerve and parabiosis

N.E. Vvedensky (1901) established that when various damaging agents (KCl, procaine (novocaine), ether, cold) act on part of a nerve, its lability decreases and it responds to excitation in a peculiar way. This state was called **parabiosis** (“beside life”). Parabiosis is a reversible process: when the damaging agent is removed, nerve function is restored.

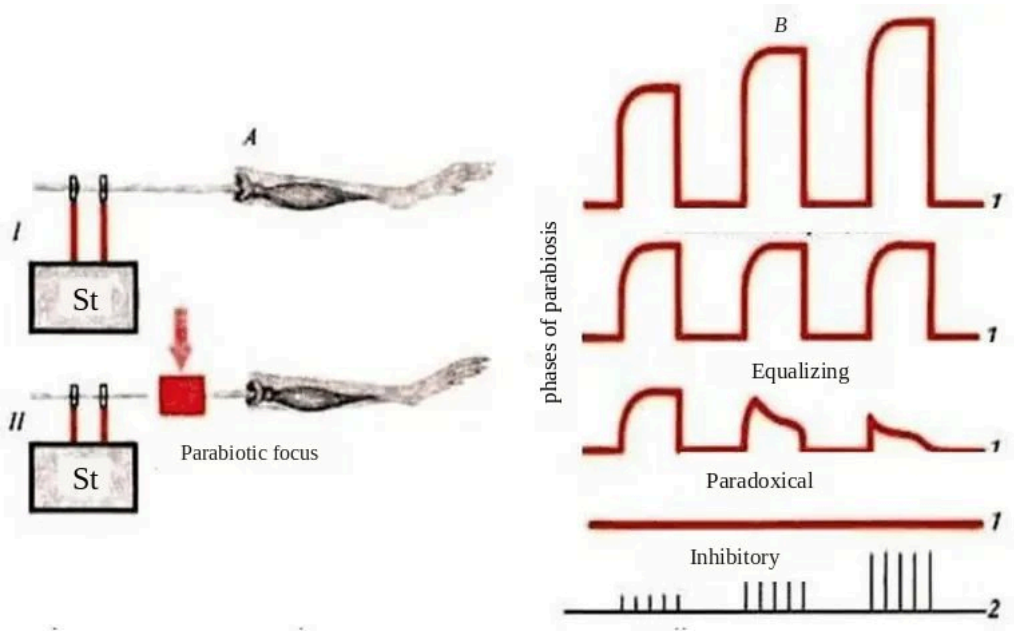


Figure 3.3. *The parabiosis experiment (A) and the phases of parabiosis (B): 1 — muscle contraction; 2 — stimulus strength (Alavia O.T., 2018)*

Table 3.3. *Phases of parabiosis*

Phase	Response to a weak stimulus	Response to a strong stimulus	Mechanism
1. Equalizing (provisional)	Normal	Weakened — the same response to strong and weak stimuli	Lability decreases; frequently arriving impulses fall into the refractory period
2. Paradoxical	A larger response to a weak stimulus	A weak response or none at all to a strong stimulus	Lability is reduced further; strong (frequent) impulses deeply inhibit the region
3. Inhibitory	No response	No response	Persistent depolarization — inactivation of Na^+ channels; excitation is not conducted

REMEMBER

Vvedensky regarded parabiosis as **a peculiar localized, persistent (stationary) state of excitation**. This doctrine shows **the unity of the processes of excitation and inhibition**: inhibition can be a special form of excitation.

CLINICAL SIGNIFICANCE

The phases of parabiosis are seen in local anesthesia: after injection of procaine (novocaine), strong painful stimuli are blocked first (paradoxical phase), then all impulses stop (inhibitory phase). As the anesthetic is absorbed, the phases pass in reverse order and sensation is restored.

4. Synapses: structure and classification

A **synapse** (Greek synapsis — connection; a term of C. Sherrington) is a special structure that transmits excitation from one cell to another. Every synapse consists of a **presynaptic membrane**, a **synaptic cleft** and a **postsynaptic membrane**.

Table 3.4. *Classification of synapses*

Feature	Types
Location	Central (in the CNS) and peripheral (neuromuscular, neuroglandular, in autonomic ganglia)
Between which structures	Axodendritic, axosomatic, axoaxonal, dendrodendritic
Mode of transmission	Chemical (via a neurotransmitter), electrical (via gap junctions — connexons), mixed
Effect	Excitatory and inhibitory
Neurotransmitter	Cholinergic (acetylcholine), adrenergic (norepinephrine), dopaminergic, serotonergic, GABAergic, glycinergic, glutamatergic, peptidergic

In **electrical synapses** the synaptic cleft is very narrow (2–4 nm), and neighboring cells are connected directly through protein channels (connexons); excitation passes without delay and in both directions. They are found in cardiac muscle, smooth muscles and some parts of the CNS.

5. Transmission of excitation in a chemical synapse

Most synapses in the human body are **chemical** synapses. In them the synaptic cleft is 20–50 nm wide; the presynaptic terminal contains **synaptic vesicles** that store the neurotransmitter, and the postsynaptic membrane carries **receptors** sensitive to it.

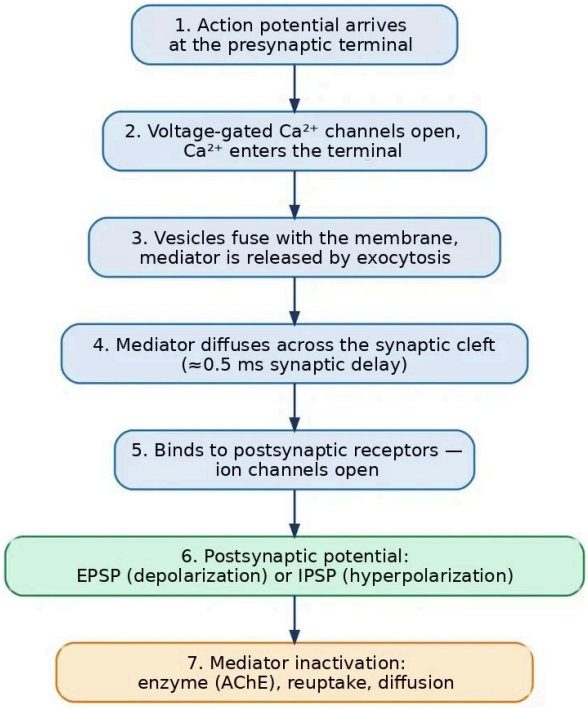


Figure 3.4. *Stages of transmission of excitation in a chemical synapse*

The amount of neurotransmitter in one vesicle is called a **quantum** (one quantum of acetylcholine ≈ 5–10 thousand molecules). Even at rest some vesicles are released spontaneously, producing small **miniature potentials**.

Neuromuscular synapse

The terminal of a motor neuron axon forms a **neuromuscular synapse (end plate)** with a skeletal muscle fiber. The neurotransmitter is **acetylcholine (ACh)**; the postsynaptic membrane carries **N-**

cholinergic receptors (ligand-gated Na^+/K^+ channels).

1. The action potential arrives at the presynaptic terminal, and Ca^{2+} channels open.
2. Under the action of Ca^{2+}, vesicles containing ACh undergo exocytosis (100–300 quanta per impulse).
3. ACh binds to N-cholinergic receptors, cation channels open and Na^+ enters.
4. An **end-plate potential (EPP)** arises on the postsynaptic membrane — like the local response, it is graded and summates.
5. The EPP depolarizes the adjacent muscle membrane to the critical level and an **action potential** arises — the muscle contracts.
6. ACh is broken down within 1–2 ms by the enzyme **acetylcholinesterase (AChE)** into choline and acetic acid; choline returns to the presynaptic terminal and is used to synthesize new ACh.

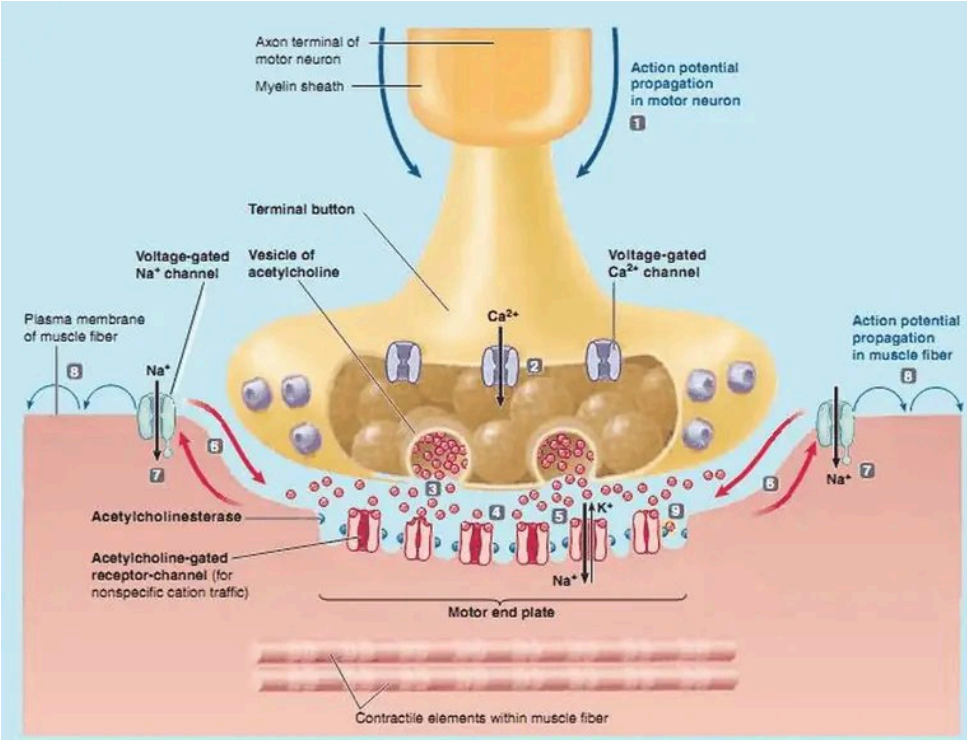


Figure 3.5. *Structure of the neuromuscular synapse and transmission of excitation in it (Alavia O.T., 2018)*

REMEMBER

The neuromuscular synapse has a high **safety factor**: the EPP produced by one impulse is 3–4 times larger than needed for an action potential. Therefore, normally every motor impulse is sure to excite the muscle fiber.

Properties of chemical synapses

1. **One-way conduction** — the neurotransmitter is released only from the presynaptic terminal, and the receptors are only on the postsynaptic membrane.
2. **Synaptic delay** — 0.2–0.5 ms at each synapse (for release of the neurotransmitter, its diffusion and binding to the receptor).
3. **Low lability** — 100–150 impulses/s.
4. **Rapid fatigue** — depletion of the neurotransmitter store, desensitization of receptors.
5. **High sensitivity to chemicals** — drugs and poisons act selectively on synapses.

6. **Summation and facilitation** — with successive impulses, transmitter release increases because Ca^{2+} accumulates.

Table 3.5. *Substances acting on the neuromuscular synapse*

Site of action	Substance	Result
Block of ACh release	Botulinum toxin	Flaccid paralysis (botulism)
Competitive blockade of N-cholinergic receptors	Curare, tubocurarine, atracurium	Muscle relaxation (used in anesthesia)
Depolarizing blockade	Suxamethonium (succinylcholine, Ditilin)	First fibrillation, then relaxation
AChE inhibitors	Neostigmine (Proserin) (reversible); organophosphorus compounds (irreversible)	Accumulation of ACh; in therapeutic doses muscle strength increases, in poisoning — convulsions and paralysis
Autoantibodies to receptors	Myasthenia gravis	Muscle fatigability and weakness

CLINICAL SIGNIFICANCE

Myasthenia gravis is an autoimmune disease in which autoantibodies are formed against N-cholinergic receptors. The number of receptors decreases, the EPP becomes smaller and the safety factor falls: muscles tire quickly (drooping eyelids, double vision, difficulty swallowing). Treatment uses AChE inhibitors (pyridostigmine), which increase the amount of ACh in the synapse.

MEMORY AID

Steps of the synapse — **"Arrives–Enters–Releases–Binds–Opens–Breaks down"**: the impulse **arrives**, Ca^{2+} **enters**, the neurotransmitter is **released**, it **binds** to the receptor, the channel **opens**, the neurotransmitter is **broken down**.

6. Laboratory work

Experiment 1: demonstrating isolated conduction in a nerve

1. Isolate the frog's sciatic nerve together with its branches to both legs (or, in one leg, observe the branches to the gastrocnemius and other muscles).
2. Stimulate part of the nerve trunk and observe which muscles contract.
3. Change the site of stimulation. **Expected results and analysis:** each time only the muscles innervated by the stimulated fibers contract; excitation does not pass to neighboring fibers. This proves the law of isolated conduction.

Experiment 2: Vvedensky's parabiosis experiment

1. Prepare a nerve-muscle preparation. Place a piece of cotton wool soaked in KCl or procaine solution on the middle part of the nerve (the parabiotic region).
2. Place the stimulating electrodes above the parabiotic region (closer to the center).
3. Every 1–2 minutes stimulate alternately with weak and strong stimuli and record the muscle contractions.
4. Note the sequence of phases.
5. Remove the cotton wool, wash the nerve with Ringer's solution and observe the recovery of function.

Expected results and analysis: first the responses to strong and weak stimuli become equal (equalizing phase), then the response to a weak stimulus becomes larger than to a strong one (paradoxical phase), and finally there is no response at all (inhibitory phase). After washing, the phases pass in reverse order and conduction is restored — parabiosis is a reversible process.

Experiment 3: relative indefatigability of the nerve and rapid fatigue of the synapse

1. Stimulate the nerve of a nerve-muscle preparation for a long time at a frequency of 30–50 Hz.
2. Wait until the muscle stops contracting (becomes fatigued).
3. Immediately place the electrodes directly on the muscle and stimulate it. **Expected results and analysis:** the muscle that does not contract when stimulated through the nerve contracts again when stimulated directly. Hence neither the muscle nor the nerve is fatigued — the **neuromuscular synapse** fatigues first (its neurotransmitter store is depleted). This shows that the synapse has the lowest lability.

7. Practical skills

Skill: explaining the structure of a synapse and the stages of transmission with a diagram

1. Draw the presynaptic terminal, vesicles, synaptic cleft and postsynaptic membrane.
2. Mark the Ca^{2+} channels, receptors and the AChE enzyme.
3. Number and write down the stages from the arrival of the impulse to the inactivation of the neurotransmitter.
4. Give an example of a substance acting on each stage (botulinum toxin, curare, neostigmine (Proserin)).

Skill: identifying the type of nerve fiber by its conduction velocity

1. Determine the time (ms) between the action potentials recorded at two points.
2. Measure the distance (mm) between the points.
3. Calculate the velocity: v = distance / time (mm/ms = m/s).
4. Compare the result with the Erlanger–Gasser table and identify the fiber type.

Review questions

1. Explain the structure of the neuron and the function of each part.
2. What is the difference between myelinated and unmyelinated nerve fibers?
3. How are nerve fibers classified according to Erlanger–Gasser?
4. How is excitation conducted in an unmyelinated fiber?
5. What is saltatory conduction, and what are its advantages?
6. List and explain the laws of conduction of excitation in a nerve.
7. What is parabiosis? Explain its phases.
8. What is the significance of Vvedensky's doctrine?
9. How are synapses classified?
10. What are the differences between electrical and chemical synapses?
11. Explain the stages of transmission of excitation in a chemical synapse.
12. What is the end-plate potential, and what are its properties?

13. List the properties of chemical synapses.
14. How do curare, botulinum toxin and neostigmine (Proserin) act on the neuromuscular synapse?
15. What fatigues first in a nerve-muscle preparation, and how is this proved?

Test questions

1. In which part of a neuron is the action potential generated?
   A) In the dendrites
   B) In the axon terminal
   C) In the region of the nucleus
   D) In the axon hillock (initial segment)
   E) In the myelin sheath

2. Which nerve fibers have the highest conduction velocity?
   A) Aα
   B) Aδ
   C) B
   D) C
   E) Aγ

3. To which group do the unmyelinated fibers conducting slow (aching) pain belong?
   A) Aα
   B) C
   C) Aβ
   D) B
   E) Aγ

4. Where is the action potential generated in saltatory conduction?
   A) Under the myelin sheath
   B) Continuously along the whole membrane
   C) Only at the nodes of Ranvier
   D) Only in the axon terminal
   E) In the nucleus of the Schwann cell

5. Which law is proved by the spread of excitation in both directions when the middle of a nerve is stimulated?
   A) Isolated conduction
   B) Bilateral conduction
   C) Physiological integrity
   D) Relative indefatigability
   E) Law of strength

6. Which sign is typical of the paradoxical phase of parabiosis?
   A) The same response to strong and weak stimuli
   B) No response at all
   C) A normal response
   D) A stronger response to a strong stimulus
   E) A larger response to a weak stimulus than to a strong one

7. Who discovered parabiosis?
   A) I.P. Pavlov
   B) I.M. Sechenov
   C) N.E. Vvedensky
   D) A.A. Ukhtomsky
   E) C. Sherrington

8. Which ion must enter the presynaptic terminal for neurotransmitter release in a chemical synapse?
   A) Ca^{2+}
   B) Na^{+}
   C) K^{+}
   D) Cl^{-}
   E) Mg^{2+}

9. What is the neurotransmitter of the neuromuscular synapse?
   A) Norepinephrine
   B) Dopamine
   C) GABA
   D) Acetylcholine
   E) Serotonin

10. How is acetylcholine inactivated at the neuromuscular synapse?
    A) By monoamine oxidase
    B) By reuptake
    C) By catechol-O-methyltransferase
    D) Only by diffusion
    E) By breakdown by acetylcholinesterase

11. What is the mechanism of action of curare?
    A) It stops the release of acetylcholine
    B) It inhibits AChE
    C) They open Ca^{2+} channels
    D) It competitively blocks N-cholinergic receptors
    E) It stops the Na^{+}/K^{+} pump

12. Which feature is not a property of chemical synapses?
    A) One-way conduction
    B) Synaptic delay
    C) Rapid fatigue
    D) Sensitivity to chemicals
    E) Bilateral conduction without delay

13. What causes muscle weakness in myasthenia gravis?
    A) Increased release of acetylcholine
    B) Decreased AChE activity
    C) Autoantibodies to N-cholinergic receptors and a decrease in the number of receptors
    D) Destruction of myelin
    E) Hypertrophy of muscle fibers

14. What fatigues first in a nerve-muscle preparation stimulated for a long time?
    A) Neuromuscular synapse
    B) The nerve fiber
    C) The muscle fiber
    D) All at the same time
    E) None of them fatigues

15. What is a characteristic feature of electrical synapses?
    A) Transmission by means of a neurotransmitter
    B) Transmission of excitation almost without delay and in both directions
    C) A wide synaptic cleft (20–50 nm)
    D) Rapid fatigue
    E) Only one-way transmission

Clinical cases

Case 1

A 30-year-old woman complains that in the evening her eyelids droop, she tires quickly when speaking and swallowing food becomes difficult. After rest in the morning her condition improves. After injection of neostigmine (Proserin), muscle strength was temporarily restored.

Questions: What disease can be suspected? Explain the mechanism of the disease and the action of neostigmine.

Answer: Myasthenia gravis. In this disease autoantibodies are formed against N-cholinergic receptors, and the number of receptors on the postsynaptic membrane decreases. As a result the end-plate potential becomes smaller and the safety factor of the neuromuscular synapse falls; with repeated activity, when transmitter release decreases slightly, the EPP falls below threshold — the muscles tire quickly (worse toward evening). Neostigmine inhibits acetylcholinesterase, acetylcholine persists longer in the synaptic cleft and activates the remaining receptors more — muscle strength is temporarily restored.

Case 2

A patient who had eaten home-canned food developed double vision, drooping eyelids, impaired swallowing and later weakness of the respiratory muscles. Sensation was not affected.

Questions: The action of which poison can be suspected? Which stage of synaptic transmission does it disrupt?

Answer: Botulinum toxin (botulism). The toxin enters the presynaptic terminal and cleaves the proteins that ensure fusion of acetylcholine-containing vesicles with the membrane — exocytosis of the neurotransmitter stops. Even when the action potential reaches the terminal, acetylcholine is not released, no end-plate potential arises and flaccid paralysis develops. Sensation is preserved because sensory pathways do not use cholinergic neuromuscular synapses. Paralysis of the respiratory muscles is life-threatening.

Case 3

A student stimulated the nerve of a nerve-muscle preparation rhythmically for a long time. The muscle stopped contracting. He concluded that “the muscle is fatigued”.

Questions: Is the conclusion correct? How can it be checked? Explain the result.

Answer: The conclusion may be wrong. To check it, the electrodes should be placed directly on the muscle: if the muscle contracts, it is not fatigued. The nerve also practically does not fatigue (the law of relative indefatigability). Hence the neuromuscular synapse fatigues first — it has the lowest lability, and with prolonged stimulation its neurotransmitter store is depleted and its receptors are desensitized.

Quick review

Table 3.6. *Key concepts of topic 3*

Concept	Brief definition
Axon hillock	The most excitable part of a neuron, where the action potential is generated
Node of Ranvier	An unmyelinated segment, densely packed with Na^+ channels
Saltatory conduction	Jumping from node to node; fast and economical
Aα fibers	70–120 m/s; motor and proprioceptive
C fibers	0.5–2 m/s; unmyelinated; slow pain, postganglionic
Laws of conduction	Integrity, bilateral, isolated, indefatigability
Parabiosis	Decreased lability in a damaged region (Vvedensky)
Phases of parabiosis	Equalizing → paradoxical → inhibitory
Chemical synapse	Presynaptic membrane, cleft (20–50 nm), postsynaptic membrane
Ca^{2+}	Required for exocytosis of the neurotransmitter
Acetylcholine	Neurotransmitter of the neuromuscular synapse; N-cholinergic receptor
EPP	End-plate potential — graded, summates
AChE	The enzyme that breaks down acetylcholine
Properties of synapses	One-way conduction, delay, low lability, rapid fatigue
Myasthenia gravis	Autoantibodies to receptors; treatment — AChE inhibitors

TOPIC 4

Methods of studying the properties of muscle. Incomplete (serrated) and complete (smooth) tetanus

Learning objectives

To teach the student the structure, physiological properties and mechanism of contraction of skeletal and smooth muscles, the types and modes of muscle contraction (single twitch, incomplete and complete tetanus), motor units, and the laws of muscle strength, work and fatigue; to master the methods of studying muscle activity in humans (dynamometry, ergography, electromyography).

By the end of the class the student should know:

- the types of muscle and their main physiological properties;
- the structure of the skeletal muscle fiber and the sarcomere;
- excitation–contraction coupling and the sliding filament theory;
- the types of contraction (isotonic, isometric, auxotonic) and its modes (single twitch, incomplete and complete tetanus);
- the phenomena of optimum and pessimum;
- the motor unit and its types;
- muscle strength, work and fatigue, and the law of average loads;
- the features of smooth muscles.

The student should be able to:

- record a single twitch, incomplete and complete tetanus of a nerve-muscle preparation on kymograph tape;
- perform and evaluate hand dynamometry in a human;
- record an ergogram and analyze the dynamics of fatigue.

1. Types of muscle and their properties

There are three kinds of muscle tissue in the human body: **striated skeletal**, **striated cardiac** and **smooth** muscle.

Table 4.1. *Comparative characteristics of the types of muscle*

Feature	Skeletal muscle	Cardiac muscle	Smooth muscle
Structure	Long multinucleated fibers, cross-striated	Cells with one or two nuclei, cross-striated, intercalated discs	Spindle-shaped mononuclear cells, no striation
Control	Voluntary (somatic nervous system)	Involuntary, automaticity	Involuntary (autonomic nervous system, hormones), automaticity in some
Resting potential	−80…−90 mV	−80…−90 mV	−50…−60 mV, unstable
Contraction speed	Fast (single twitch 0.05–0.1 s)	Moderate (0.3 s)	Very slow (seconds)
Source of Ca^{2+}	Sarcoplasmic reticulum	Extracellular + SR	Mainly extracellular

Feature	Skeletal muscle	Cardiac muscle	Smooth muscle
Ca^{2+}-binding protein	Troponin	Troponin	Calmodulin
Tetanus	Present	Absent (long refractoriness)	Tonic contraction
Plasticity	Low	Low	High

The physiological properties of muscle are **excitability**, **conductivity**, **contractility**, **elasticity** (return to the original state after stretching) and **extensibility**; smooth muscles additionally have **plasticity** (keeping their tone unchanged in the stretched state) and **automaticity**.

The main functions of muscles: movement of the body in space and maintenance of posture, respiratory movements, heat production (60–70% of the energy of contraction becomes heat), assisting the movement of blood and lymph (the muscle pump), and movement of internal organs (smooth muscles).

2. Structure of the skeletal muscle fiber

Muscle → muscle bundles → **muscle fibers** (multinucleated cells, up to several cm long) → **myofibrils** → **sarcomeres**. The sarcomere is the part of a myofibril between two **Z lines** (2–2.5 µm long at rest), the functional unit of the contractile apparatus.

- **Thick filaments** — myosin; located in the middle of the sarcomere, they form the **A band** (anisotropic, dark).
- **Thin filaments** consist of actin, tropomyosin and troponin; they are attached to the Z line and form the **I band** (isotropic, light).
- The light part in the middle of the A band containing only thick filaments is the **H zone**, with the **M line** at its center.

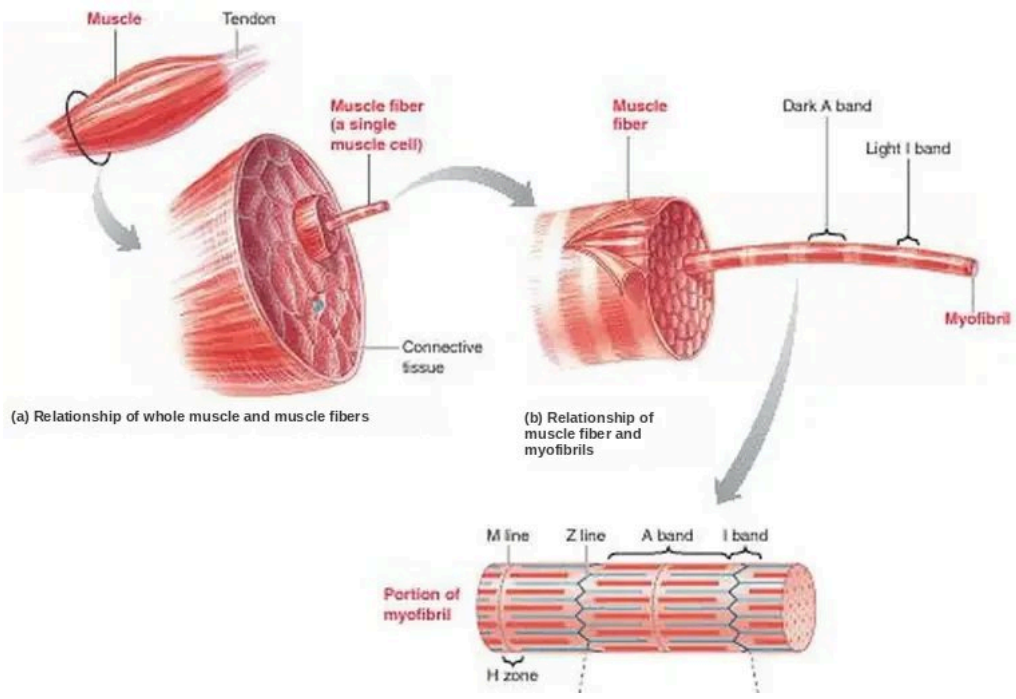


Figure 4.1. *Structure of skeletal muscle: muscle — fiber — myofibril — sarcomere (Alavia O.T., 2018)*

The membrane of the muscle fiber (sarcolemma) dips into the fiber as **transverse tubules (the T system)**. On both sides of them lie the terminal cisterns of the **sarcoplasmic reticulum (SR)**,

forming a **triad**. The SR is the store of Ca^{2+} ions.

3. Mechanism of muscle contraction

Excitation–contraction coupling

1. The action potential generated at the neuromuscular synapse spreads along the sarcolemma and passes into the fiber through the **T tubules**.
2. The voltage-sensitive **dihydropyridine receptors** (L-type Ca^{2+} channels) in the T tubule change their conformation and mechanically open the **ryanodine receptors** (Ca^{2+} channels) of the SR.
3. Ca^{2+} leaves the SR for the sarcoplasm: its concentration rises from 10^{-7} to 10^{-5} mol/l.
4. Ca^{2+} binds to **troponin C**; the troponin–tropomyosin complex shifts and the myosin-binding sites on actin are exposed.

The sliding filament theory (A. Huxley, H. Huxley)

During contraction the filaments themselves do not shorten: **the thin actin filaments slide between the thick myosin filaments**. As a result the Z lines approach each other, **the I band and H zone narrow, while the width of the A band does not change**.

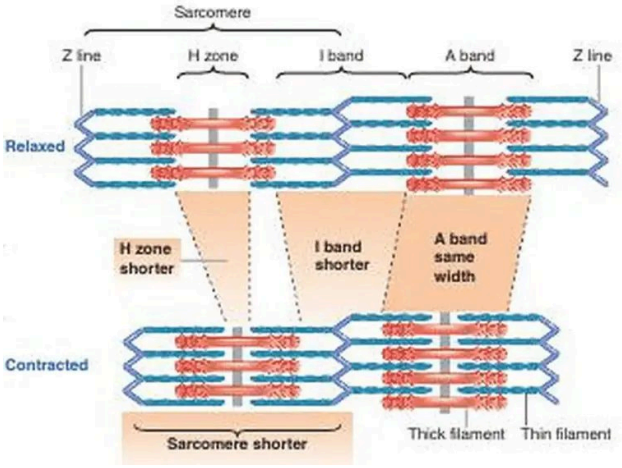


Figure 4.2. *The sarcomere at rest (top) and contracted (bottom) (Alavia O.T., 2018)*

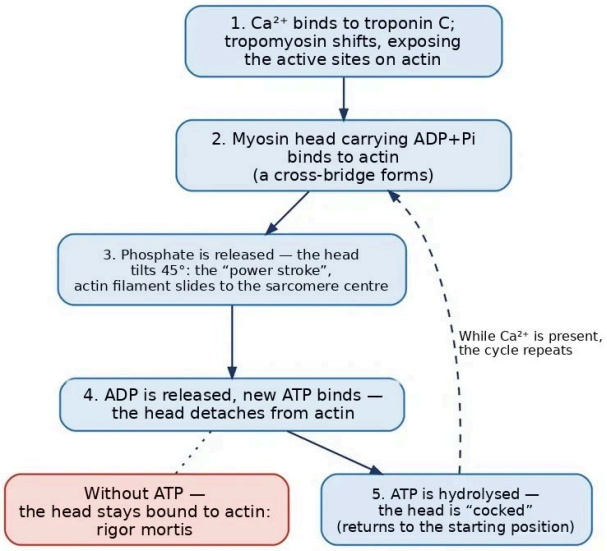


Figure 4.3. *The cross-bridge cycle*

Relaxation

Relaxation is also an **active process**: the **Ca^{2+}-ATPase (calcium pump)** in the SR membrane pumps Ca^{2+} back into the SR using the energy of ATP; once Ca^{2+} falls, troponin–tropomyosin covers the active sites and the muscle relaxes.

REMEMBER

ATP performs three tasks in muscle: 1) energy for the movement of cross-bridges; 2) **detachment** of the myosin head from actin; 3) the work of the Ca^{2+} pump and the Na^{+}/K^{+} pump. After death ATP runs out and the cross-bridges remain attached — **rigor mortis** develops.

MEMORY AID

In sliding, **"A stays Always"** (the A band does not change), **"I and H get thIn"** (the I band and H zone narrow).

4. Types and modes of contraction

Table 4.2. *Types of muscle contraction*

Type	Definition	Example
Isotonic	The muscle shortens while its tension does not change	Flexing the arm without a load
Isometric	The length does not change, the tension increases	Holding a load that cannot be lifted, maintaining posture
Auxotonic (mixed)	Both length and tension change	Most natural movements
Concentric	The muscle shortens while overcoming a load	Lifting a dumbbell
Eccentric	The muscle lengthens while under tension	Lowering a load slowly, walking down stairs

Single twitch

A single threshold or stronger stimulus evokes a **single twitch**. Its periods (in the frog gastrocnemius, ≈ 0.1 s in total) are the **latent period** (from the stimulus to the start of contraction, ≈ 0.01 s — for excitation–contraction coupling), the **contraction phase** (≈ 0.04 s) and the **relaxation phase** (≈ 0.05 s). The action potential (1–3 ms) is much shorter than the contraction — therefore the muscle can be excited again during the contraction.

Summation and tetanus

If a second stimulus arrives before the first contraction has ended, the contractions add up (**summation**). Rhythmic stimuli produce a long-lasting strong contraction — **tetanus**.

Table 4.3. *Types of tetanus*

Type	Stimulus frequency (frog muscle)	Condition	Curve
Incomplete (serrated) tetanus	10–20 Hz	Each subsequent stimulus falls into the relaxation phase of the previous contraction	Serrated (wavy)
Complete (smooth) tetanus	25–50 Hz and higher	Each subsequent stimulus falls into the contraction phase of the previous contraction	A smooth plateau

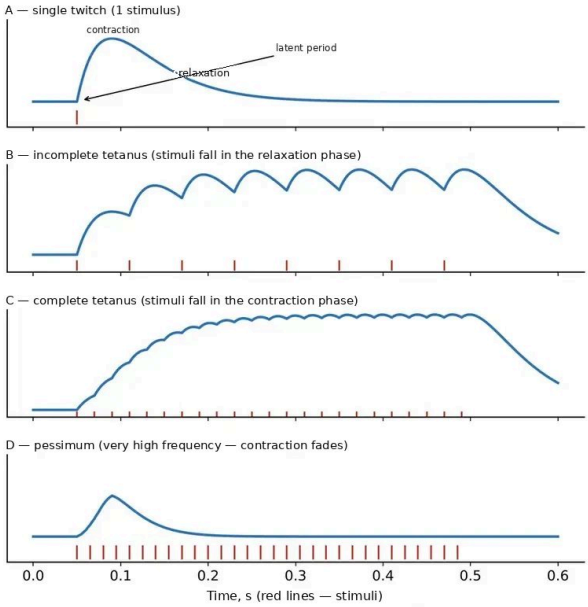


Figure 4.4. *Modes of muscle contraction: A — single twitch; B — incomplete tetanus; C — complete tetanus; D — pessimum (drawn after Alavia O.T., 2018)*

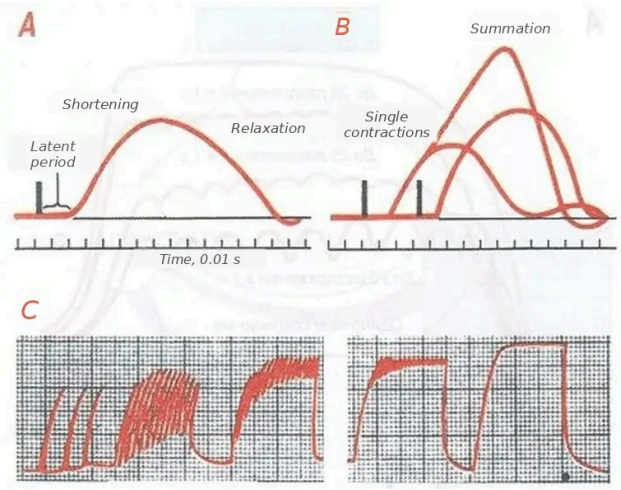


Figure 4.5. *Single twitch and summation (A, B), kymograms of incomplete and complete tetanus (C) (Alavia O.T., 2018)*

In tetanus the contraction amplitude is 2–4 times greater than that of a single twitch, because Ca^{2+} is maintained at a high concentration in the sarcoplasm and all cross-bridges are active. In natural human movements, skeletal muscles contract **tetanically**: motor neurons send impulses at a frequency of 10–50 Hz.

Optimum and pessimum (N.E. Vvedensky)

As the stimulus frequency increases, the amplitude of tetanus grows and becomes greatest at a certain frequency — the **optimum**: each impulse falls into the phase of exaltation after the previous excitation. If the frequency is increased further, the impulses fall into the refractory period and the contraction decreases or stops — the **pessimum**. Pessimum is mainly related to the low lability of the neuromuscular synapse.

REMEMBER

The type of tetanus is determined by the **interval between stimuli**: if the interval is shorter than a single twitch but longer than the contraction phase — incomplete (serrated) tetanus; if it is shorter than the contraction phase — complete (smooth) tetanus.

CLINICAL SIGNIFICANCE

In **tetanus** (the disease), tetanus toxin stops the release of glycine and GABA from inhibitory interneurons in the spinal cord. Motor neurons send impulses continuously at a high frequency, and skeletal muscles contract tetanically and painfully (spasm of the masticatory muscles — “risus sardonicus”, opisthotonus). In **hypocalcemia**, on the other hand, the excitability of nerves and muscles increases and tetanic convulsions (tetany) appear.

5. The motor unit

A **motor unit** is a set consisting of one motor neuron, its axon and the muscle fibers innervated by that axon. It is the functional unit of muscle: when the motor neuron is excited, all its fibers contract simultaneously.

Table 4.4. *Types of motor units*

Feature	Slow (S, type I, “red”)	Fast fatigue-resistant (FR, IIa)	Fast fatigable (FF, IIb, “white”)
Motor neuron	Small, low threshold	Medium	Large, high threshold
Contraction speed	Slow	Fast	Fast
Force	Small	Medium	Large
Fatigue	Very resistant	Resistant	Fatigues quickly
Energy source	Oxidation (many mitochondria, myoglobin, capillaries)	Oxidation and glycolysis	Anaerobic glycolysis
Example	Postural muscles that keep the body upright	Marathon, prolonged work	Jumping, sprinting

In the eye muscles one motor unit consists of 3–10 fibers, in finger muscles 10–25, and in the large calf and thigh muscles up to 1500–2000 fibers: the smaller the motor unit, the more precise the movement. Muscle force is controlled by the **number of active motor units** and their **impulse frequency**; small (slow) units are recruited first, then larger ones (**Henneman's size principle**).

6. Muscle strength, work and fatigue

Muscle strength is the maximum load a muscle can lift or its maximum isometric tension. It depends on the **physiological cross-section** (the sum of the cross-sections of all fibers). **Absolute muscle strength** is the ratio of maximum strength to the physiological cross-section (5–10 kg/cm^2 in humans). Pennate muscles are stronger because their physiological cross-section is large.

Muscle work is the product of the mass of the lifted load and the height to which it is lifted (A = P × h). If the load is zero or maximal, the work is zero; the greatest work is done at an **average load** — **the law of average loads**. Likewise, the most work is done when the rhythm of work is also average (**the law of average rhythm**).

Fatigue is a temporary decrease in working capacity as a result of prolonged or intense work; it is restored after rest. In an isolated preparation, fatigue occurs first at the **neuromuscular synapse**; in the whole organism, mainly in the **nerve centers** (I.M. Sechenov). Sechenov showed the effect of **active rest**: if, while the tired arm is resting, work is done with the other arm, working capacity is restored faster.

> **CAUTION**
> In the past, **accumulation of lactic acid** was considered the main cause of fatigue. According to modern data, muscle fatigue involves accumulation of inorganic phosphate, decreased Ca^{2+} release, exhaustion of glycogen stores, accumulation of K^+ outside the cells and central factors (motivation, CNS fatigue).

7. Smooth muscles

Smooth muscles are found in the walls of internal organs (stomach, intestine, urinary bladder, uterus), blood vessels and bronchi. Their contraction is slow but long-lasting and requires little energy (**tone**).

- **Single-unit (visceral)** smooth muscles — the cells are connected by gap junctions and contract as a functional syncytium; most of them have **automaticity** (pacemaker cells): intestine, stomach, ureter.

- **Multi-unit** smooth muscles — each cell is innervated separately: the muscles of the pupil, the ciliary muscle, large arteries. **Mechanism of contraction:** Ca^{2+} (entering mainly from outside the cell) binds to **calmodulin** → the complex activates **myosin light-chain kinase** → myosin is phosphorylated and its interaction with actin begins. Smooth muscles contract in response to stretch (the **myogenic response**, the Bayliss effect in vessels).

CLINICAL SIGNIFICANCE

Disturbed contraction of smooth muscles underlies many diseases: spasm of bronchial smooth muscle in **bronchial asthma**, increased arteriolar tone in **arterial hypertension**, intestinal spasm in **intestinal colic**. That is why Ca^{2+} channel blockers (nifedipine), β_2-adrenergic agonists (salbutamol (albuterol)) and antispasmodics (drotaverine) are widely used.

8. Laboratory work

Experiment 1: recording a single twitch, incomplete and complete tetanus

1. Prepare a frog nerve-muscle preparation, attach the muscle tendon to the myograph lever and bring the tip of the lever into contact with the kymograph drum.
2. Stimulate the nerve with a single suprathreshold stimulus — record the curve of a single twitch. With the drum rotating fast, mark the latent period and the contraction and relaxation phases.
3. Set the stimulus frequency on the electrical stimulator to 10–15 Hz and stimulate for 2–3 s — record incomplete tetanus.
4. Increase the frequency to 30–50 Hz — record complete tetanus.
5. Increase the frequency further (100 Hz and higher) and observe the pessimum; show that contraction is restored when the frequency is reduced (optimum). **Expected results and analysis:** at low frequency each stimulus falls into the relaxation phase of the previous contraction and incomplete tetanus arises; at high frequency the stimuli fall into the contraction phase, the contractions fuse completely and complete tetanus appears. The amplitude of tetanus is larger than that of a single twitch. At a very high frequency the impulses fall into the refractory period and exceed the lability of the synapse — pessimum.

Experiment 2: hand dynamometry

1. The subject stands upright and extends the arm to the side; the dynamometer is held in the palm with the dial facing inward.
2. It is squeezed with maximum force 2–3 times (with 1 minute of rest in between); the best result is recorded.
3. Both hands are measured.
4. The **strength index** is calculated: hand strength (kg) / body mass (kg) × 100%. **Interpretation:** the normal strength index is 65–80% in men and 48–50% in women. The working hand is usually 5–10% stronger.

Experiment 3: ergography (the Mosso ergograph)

1. The subject's arm is fixed in the ergograph, and a cord with a 2–4 kg load is tied to the middle finger.
2. To the rhythm of a metronome (every 1–2 s) the load is lifted as high as possible; the lifting height is recorded on the kymograph tape.
3. The work continues until the load can no longer be lifted (until fatigue).

4. The experiment is repeated after 5 minutes of passive rest and 5 minutes of active rest (work with the other arm).
5. The work done is calculated: A = mass of the load × sum of the lifting heights. **Expected results and analysis:** the ergogram first shows warming up (the height increases slightly), then stable work and fatigue (the height decreases). After active rest working capacity recovers better than after passive rest — the Sechenov phenomenon.

9. Practical skills

Skill: analyzing a kymogram

1. Find the stimulus mark and the time line.
2. Measure the duration of the latent period and the contraction and relaxation phases of a single twitch.
3. Determine whether the tetanus curve has teeth.
4. Compare the interval between stimuli with the phases of a single twitch and justify the type of tetanus.
5. Calculate the ratio of the tetanus amplitude to the single twitch amplitude.

Skill: evaluating an electromyogram (EMG)

1. Place surface electrodes over the muscle and the ground electrode nearby.
2. Record at rest (bioelectric “silence”) and during weak and maximal voluntary contraction.
3. Note the increase in EMG amplitude and frequency as force increases (recruitment of new motor units and increased firing frequency).
4. Evaluate the change in amplitude and the decrease in frequency during fatigue.

Review questions

1. What are the main differences between skeletal, cardiac and smooth muscles?
2. What physiological properties and functions do muscles have?
3. Explain the structure of the sarcomere.
4. How is excitation–contraction coupling carried out?
5. Explain the sliding filament theory.
6. What is the role of ATP in the cross-bridge cycle?
7. How does muscle relaxation occur?
8. Give examples of isotonic, isometric and auxotonic contractions.
9. What periods does a single twitch consist of?
10. Under what conditions do incomplete and complete tetanus arise?
11. What are optimum and pessimum?
12. What is a motor unit? Describe its types.
13. Explain the law of average loads.
14. Where and how does fatigue occur? What is active rest?
15. What is special about the mechanism of contraction of smooth muscles?

Test questions

1. The sarcomere is the part between which structures?
   A) Between two M lines
   B) Between two H zones
   C) Between two Z lines
   D) Between the A and I bands
   E) Between two triads

2. To which protein does Ca^{2+} bind in skeletal muscle?
   A) Troponin C
   B) Calmodulin
   C) Myosin
   D) Actin
   E) Tropomyosin

3. Which part of the sarcomere does not change when a muscle contracts?
   A) The I band
   B) The A band
   C) The H zone
   D) The length of the sarcomere
   E) The distance between the Z lines

4. What causes rigor mortis?
   A) Lack of Ca^{2+}
   B) Breakdown of actin
   C) Failure of myosin heads to detach from actin because of the absence of ATP
   D) Accumulation of lactic acid
   E) Lack of troponin

5. What is a contraction in which muscle length does not change but tension increases called?
   A) Isotonic
   B) Auxotonic
   C) Concentric
   D) Isometric
   E) Eccentric

6. When does incomplete tetanus arise?
   A) When each stimulus falls into the contraction phase
   B) When each stimulus falls into the relaxation phase of the previous contraction
   C) With a single stimulus
   D) When the stimuli arrive after the single twitch has ended
   E) When the stimulus falls into the refractory period

7. When does complete tetanus arise?
   A) When the stimuli fall into the relaxation phase
   B) When there is complete relaxation between stimuli
   C) At a very low frequency
   D) When each stimulus falls into the contraction phase of the previous contraction
   E) When the stimulus is below threshold

8. What is the main cause of the pessimum phenomenon?
   A) Strong contraction of the muscle
   B) An excess of ATP
   C) An increase in Ca^{2+}
   D) Fatigue of the nerve
   E) Very frequent impulses falling into the refractory period and exceeding the lability of the synapse

9. What is a motor unit?
   A) One motor neuron and the muscle fibers it innervates
   B) One sarcomere
   C) One myofibril
   D) A muscle bundle
   E) One muscle fiber

10. In which muscles is the motor unit smallest (fewest fibers)?
   A) In the thigh muscles
   B) In the calf muscles
   C) In the back muscles
   D) In the gluteal muscles
   E) In the eye muscles

11. At what load does a muscle perform the greatest work?
   A) At an average load
   B) At zero load
   C) At maximal load
   D) At supramaximal load
   E) It does not depend on the load

12. Where does fatigue occur first in an isolated nerve-muscle preparation?
   A) In the nerve fiber
   B) In the muscle fiber
   C) In the sarcoplasmic reticulum
   D) At the neuromuscular synapse
   E) In the myofibrils

13. With which protein does Ca^{2+} bind to start contraction in smooth muscle?
   A) Troponin C
   B) Tropomyosin
   C) Actin
   D) Myoglobin
   E) Calmodulin

14. What ensures muscle relaxation?
   A) Release of Ca^{2+} into the sarcoplasm
   B) Reuptake of Ca^{2+} by the Ca^{2+}-ATPase of the SR
   C) Lack of ATP
   D) Phosphorylation of myosin
   E) Binding of troponin to Ca^{2+}

15. Who demonstrated the effect of active rest?
    A) I.P. Pavlov
    B) N.E. Vvedensky
    C) I.M. Sechenov
    D) A. Mosso
    E) A. Huxley

Clinical cases

Case 1

An unvaccinated patient whose foot had been pierced by a rusty nail developed, a week later, spasm of the masticatory muscles (inability to open the mouth), “risus sardonicus” of the face and later strong spasms of the back muscles.

Questions: What disease is this? Why do the skeletal muscles contract strongly for a long time? Which mode of contraction does this resemble?

Answer: Tetanus. Tetanus toxin stops the release of glycine and GABA from inhibitory interneurons in the spinal cord and brainstem. Motor neurons remain uninhibited and continuously send impulses at a high frequency. Each impulse then falls into the contraction phase of the previous contraction, and the muscles stay contracted for a long time in a state of complete tetanus. The muscles themselves and the neuromuscular synapse are not damaged — the disturbance lies in central inhibition.

Case 2

A 20-year-old athlete performs well in the 100 m sprint but has poor endurance in a marathon. His friend, on the contrary, runs long distances well.

Questions: How can this difference be explained by the types of muscle fibers (motor units)?

Answer: The sprinter's muscles contain many fast fatigable (FF, IIb, “white”) motor units: they contract quickly and strongly and obtain energy from anaerobic glycolysis, but tire quickly. In the marathon runner, slow (S, type I, “red”) and fast fatigue-resistant (FR) units predominate: they have many mitochondria, much myoglobin and many capillaries and obtain energy by oxidation — so they can work for a long time. Although the fiber ratio is mainly inherited, training partly changes their properties.

Case 3

A student stimulated a nerve-muscle preparation on a kymograph at a frequency of 20 Hz and obtained a wavy curve. The duration of a single twitch is 0.1 s, the contraction phase 0.04 s.

Questions: Which type of tetanus did the student obtain? What frequency is needed to obtain complete tetanus?

Answer: At 20 Hz the interval between stimuli is 0.05 s. This is shorter than the duration of a single twitch (0.1 s) but longer than the contraction phase (0.04 s) — each stimulus falls into the relaxation phase of the previous contraction, so incomplete tetanus arose. For complete tetanus the interval must be shorter than the contraction phase: less than 0.04 s, i.e., a frequency above 25 Hz (in practice 30–50 Hz).

Quick review

Table 4.5. *Key concepts of topic 4*

Concept	Brief definition
Sarcomere	The part between two Z lines; the contractile unit
Excitation–contraction coupling	Action potential → T tubule → release of Ca^{2+} from the SR
Troponin C	Binds Ca^{2+} in skeletal muscle
Sliding theory	Actin slides between myosin; the A band does not change
Role of ATP	Cross-bridge movement, cross-bridge detachment, Ca^{2+} pump
Isometric contraction	Length does not change, tension increases
Single twitch	Latent period, contraction, relaxation (≈ 0.1 s)
Incomplete (serrated) tetanus	The stimulus falls into the relaxation phase
Complete (smooth) tetanus	The stimulus falls into the contraction phase
Optimum / pessimum	The largest / reduced tetanus (Vvedensky)
Motor unit	Motor neuron + its muscle fibers
Absolute muscle strength	Maximum strength / physiological cross-section (5–10 kg/cm^2)
Law of average loads	The greatest work at an average load
Active rest	The Sechenov phenomenon
Smooth muscle	Calmodulin, myosin phosphorylation; tone, automaticity, plasticity

TOPIC 5

Biological fluids of the body. Physicochemical properties of blood. ESR

Learning objectives

To teach the student the internal environment of the body and its fluid compartments, the concept of the blood system, the functions, volume and composition of blood, the composition of plasma, the main physicochemical properties of blood (viscosity, specific gravity, osmotic and oncotic pressure, pH and buffer systems), and the erythrocyte sedimentation rate (ESR) and its significance; to master the determination of the hematocrit and ESR.

By the end of the class the student should know:

- the fluid compartments of the body and their volumes;
- the concept of the blood system (G.F. Lang) and the functions of blood;
- the blood volume, hematocrit and composition of plasma;
- the plasma proteins and their functions;
- osmotic and oncotic pressure, and isotonic, hypotonic and hypertonic solutions;
- the active reaction of blood (pH), the buffer systems, acidosis and alkalosis;
- the factors affecting the ESR and its normal values.

The student should be able to:

- determine the hematocrit;
- determine and evaluate the ESR by the Panchenkov method;
- determine the osmotic resistance of erythrocytes.

1. The internal environment of the body and biological fluids

In the adult body water makes up **≈ 60%** of body mass (75–80% in a newborn, less in the elderly and in obese people). Water is located in two main compartments.

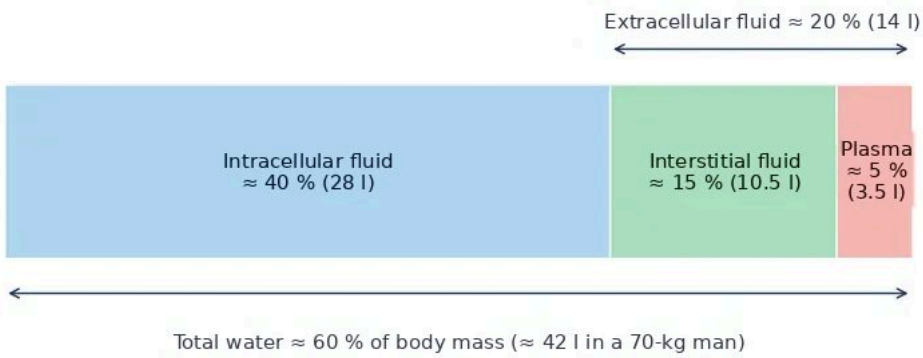


Figure 5.1. *Fluid compartments of the body (for a 70 kg man; drawn after Costanzo L.S., 2018)*

Table 5.1. *Fluid compartments and their main ions*

Compartment	Percentage of body mass	Main cation	Main anion
Intracellular fluid	≈ 40 %	K^+	Phosphates, proteins
Interstitial (tissue) fluid	≈ 15 %	Na^+	Cl^-, HCO_3^-
Blood plasma	≈ 5 %	Na^+	Cl^-, HCO_3^-, proteins
Transcellular fluids (cerebrospinal, synovial, pleural, ocular fluids)	1–2 %	Na^+	Cl^-

The **internal environment** is formed by the fluids surrounding the cells: **blood**, **lymph** and **tissue fluid**. They constantly exchange with one another: blood plasma filters from the capillaries into the tissues and forms tissue fluid, part of which returns to the blood through the lymphatic vessels.

2. The blood system and the functions of blood

The **blood system** (G.F. Lang, 1939) is a single system consisting of the peripheral blood, the hematopoietic organs (red bone marrow, lymph nodes, spleen, thymus), the organs of blood destruction (spleen, liver) and the neurohumoral apparatus that regulates them.

Table 5.2. *Functions of blood*

Function	Content
Transport	Transport of all substances — the general function
Respiratory	Transport of O_2 from the lungs to the tissues and of CO_2 from the tissues to the lungs
Trophic (nutritive)	Delivery of nutrients to the cells
Excretory	Transport of the end products of metabolism (urea, uric acid, creatinine) to the excretory organs
Homeostatic	Maintaining the constancy of pH, osmotic pressure and ion composition
Thermoregulatory	Distribution and redistribution of heat
Protective	Immunity (leukocytes, antibodies), blood coagulation
Regulatory (humoral)	Transport of hormones and other active substances

Blood volume and composition

In adults the blood volume is **6–8%** of body mass, i.e., 4.5–6 l (≈ 70 ml/kg). At rest part of the blood is kept in **reservoirs** (spleen, liver, subcutaneous vessels, lungs) and joins the circulating blood during physical work or blood loss.

Blood consists of **plasma** (55–60%) and **formed elements** (40–45%). The ratio of the volume of formed elements to the total blood volume is called the **hematocrit**.

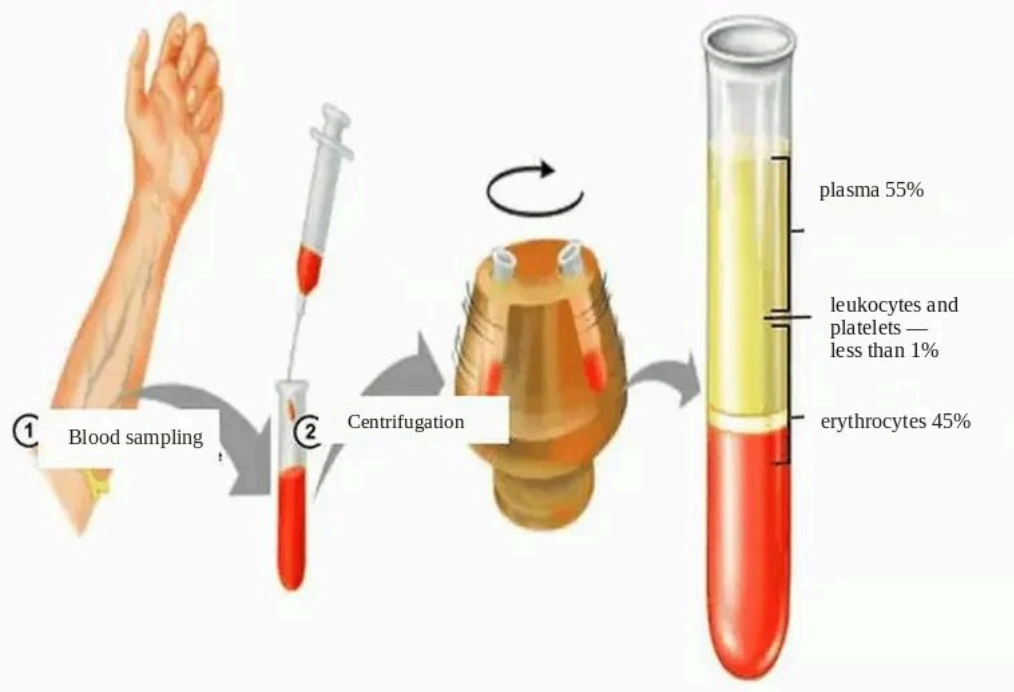


Figure 5.2. *Determination of the hematocrit: blood collection, centrifugation and separation into layers (Alavia O.T., 2018)*

Table 5.3. *The main blood parameters (adults)*

Parameter	Men	Women
Hematocrit	40–48 % (0,40–0,48)	36–42 % (0,36–0,42)
Erythrocytes	4.0–5.0 × 10^{12}/l	3.7–4.7 × 10^{12}/l
Hemoglobin	130–160 g/l	120–140 g/l
Leukocytes	4–9 × 10^{9}/l	4–9 × 10^{9}/l
Platelets	180–320 × 10^{9}/l	180–320 × 10^{9}/l
ESR	1–10 mm/h	2–15 mm/h

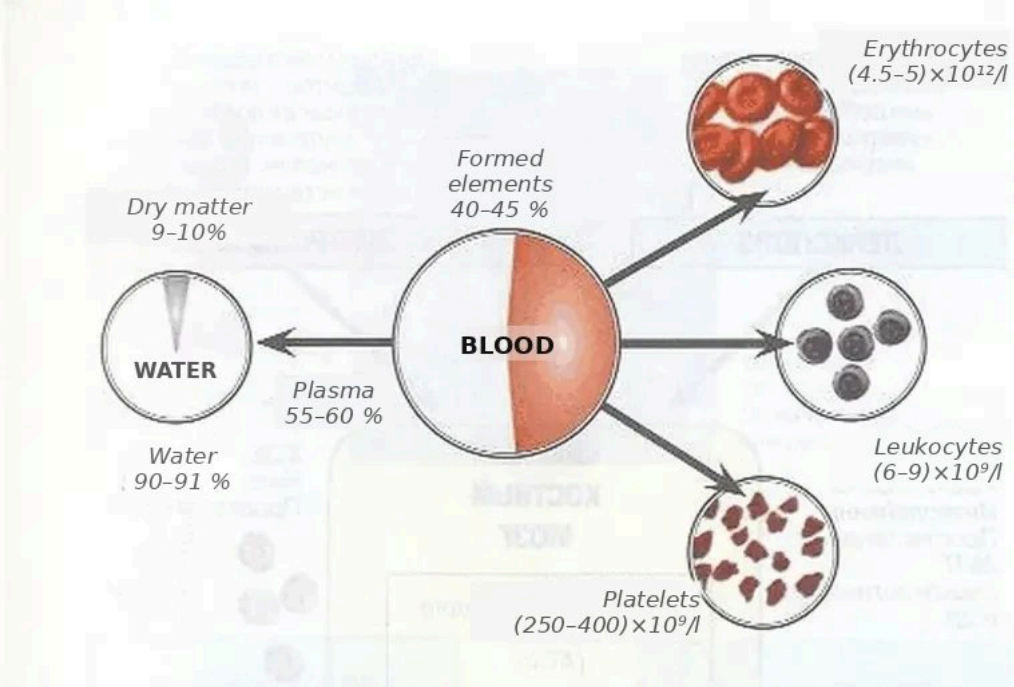


Figure 5.3. *Composition of blood (Alavia O.T., 2018)*

3. Composition of blood plasma

Plasma consists of 90–92% water and 8–10% dry matter. The main part of the dry matter is **proteins** (65–85 g/l).

Table 5.4. *Plasma proteins*

Protein	Amount	Function
Albumins	35–50 g/l (55–60%)	Generate 80% of the oncotic pressure; transport bilirubin, fatty acids, drugs and hormones; a reserve of amino acids
Globulins (α_1, α_2, β, γ)	20–35 g/l	α, β — transport of lipids, iron (transferrin), copper (ceruloplasmin); γ-globulins — antibodies (immunoglobulins)
Fibrinogen	2–4 g/l	Blood coagulation (converted into fibrin)

The **albumin/globulin ratio** is normally 1.2–2.0. If fibrinogen is removed from plasma, **serum** is obtained.

Functions of plasma proteins: generating oncotic pressure (holding water in the vessels), transport, protection (immunity), blood coagulation, buffering, providing viscosity and serving as a protein reserve.

Other substances in plasma: **glucose** (3.3–5.5 mmol/l), lipids, **residual nitrogen** (nonprotein nitrogenous substances: urea 2.5–8.3 mmol/l, creatinine, uric acid, amino acids), mineral substances (≈ 0.9%; Na^+ 135–145, K^+ 3.5–5.0, Ca^{2+} 2.2–2.6, Cl^- 98–107 mmol/l), hormones, enzymes, vitamins.

4. Physicochemical properties of blood

Table 5.5. *Physicochemical parameters of blood*

Parameter	Normal value	What it depends on
Specific gravity — blood	1,050–1,060	Mainly on the number of erythrocytes
Specific gravity — plasma	1,025–1,034	On the protein content
Viscosity — blood (relative to water)	4–5	Number of erythrocytes (hematocrit)
Viscosity — plasma	1,7–2,2	Proteins
Osmotic pressure	7.6 atm (≈ 285–295 mOsm/kg)	Mainly NaCl
Oncotic pressure	25–30 mm Hg (0.03–0.04 atm)	Plasma proteins, mainly albumin
pH — arterial blood	7,40 (7,35–7,45)	Buffer systems, lungs and kidneys
pH — venous blood	7,36	More CO_2
Freezing point (depression)	−0.56…−0.58 °C	Osmotic pressure

Osmotic pressure

Osmotic pressure is the force with which dissolved substances (mainly ions) draw the solvent (water) to themselves through a semipermeable membrane. **NaCl** generates 60% of the osmotic pressure of plasma. Solutions whose osmotic pressure equals that of plasma are called **isotonic** (0.9% NaCl — “physiological saline”, 5% glucose); those with a lower pressure are **hypotonic**, and those with a higher one are **hypertonic**.

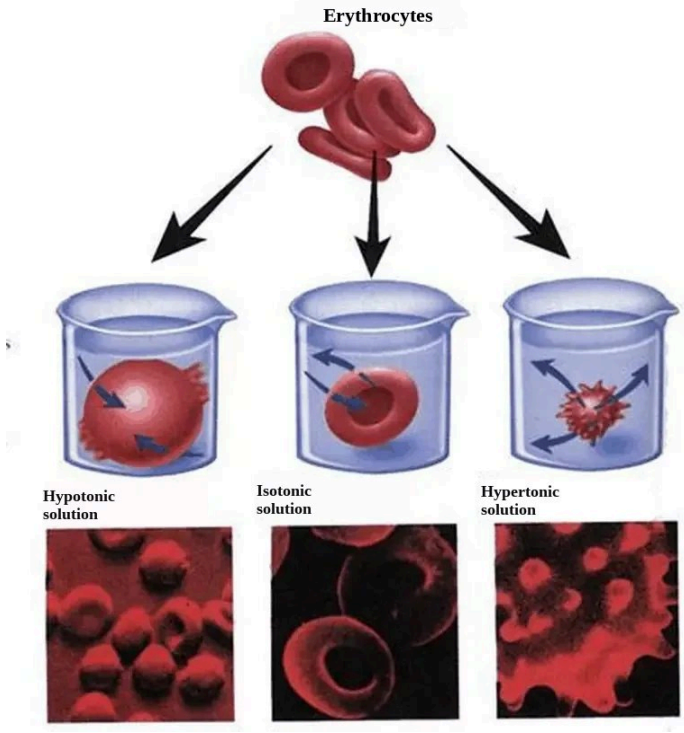


Figure 5.4. *Erythrocytes in hypotonic, isotonic and hypertonic solutions (Alavia O.T., 2018)*

- **In a hypotonic solution** water enters the erythrocyte, which swells and bursts — **osmotic hemolysis** ("lacquered blood").
- **In a hypertonic solution** water leaves the erythrocyte, which shrinks — **plasmolysis** (crenation). **The osmotic resistance of erythrocytes** is their resistance to bursting in hypotonic solutions. Normally hemolysis begins at **0.46–0.48%** NaCl (minimal resistance) and is complete at **0.32–0.34%** NaCl (maximal resistance).

CLINICAL SIGNIFICANCE

Only isotonic solutions are infused intravenously in large volumes. If a hypotonic fluid (for example, distilled water) is infused, hemolysis develops and the kidneys are damaged. In **hereditary microspherocytosis** (Minkowski–Chauffard disease) the erythrocytes are spherical and their osmotic resistance is sharply reduced — hemolysis begins already at 0.6–0.7% NaCl.

Oncotic pressure

Oncotic (colloid osmotic) pressure is the part of osmotic pressure generated by plasma proteins; it is small (25–30 mm Hg) but very important, because proteins hardly pass through the capillary wall. Oncotic pressure **holds water inside the vessels** and returns tissue fluid to the blood in the venous part of the capillaries (the Starling balance).

REMEMBER

Albumin generates **80%** of the oncotic pressure (its molecule is small and there are many of them). When albumin decreases (starvation, liver cirrhosis, nephrotic syndrome), oncotic pressure falls and **oncotic (hypoproteinemic) edema** develops.

The active reaction of blood (pH) and buffer systems

Blood pH is a very rigid homeostatic parameter: the limits compatible with life are 6.8–7.8. The constancy of pH is ensured by the **buffer systems**, the **lungs** (removal of CO_2) and the **kidneys** (excretion of H^+, reabsorption of HCO_3^-).

Table 5.6. *Buffer systems of blood*

Buffer system	Composition	Share of buffer capacity	Features
Hemoglobin	HHb / $KHbO_2$	≈ 75 %	The most powerful; works in erythrocytes
Bicarbonate	H_2CO_3 / $NaHCO_3$ (1:20)	≈ 10% (the most important in plasma)	An open system — CO_2 is removed through the lungs; works quickly
Phosphate	NaH_2PO_4 / Na_2HPO_4	≈ 1 %	Important inside cells and in urine
Plasma proteins	Protein acid / protein salt	≈ 7 %	Amphoteric properties

The **alkaline reserve** is the amount of bicarbonate in plasma; it shows the capacity to neutralize acids (normally 24 ± 2 mmol/l, 50–65 vol% CO_2).

Acidosis is a fall in pH below 7.35, **alkalosis** a rise above 7.45. By origin they are **respiratory** (accumulation or excessive loss of CO_2) and **metabolic** (nonrespiratory); if the pH is kept normal by the buffer systems, they are **compensated**; if the pH changes, they are **decompensated**.

MEMORY AID

For the bicarbonate buffer remember the ratio **“1 to 20”**: H_2CO_3 : $NaHCO_3$ = 1 : 20. If the ratio is maintained, the pH is 7.4.

5. Erythrocyte sedimentation rate (ESR)

When anticoagulated blood stands upright in a tube, the erythrocytes gradually settle and a layer of plasma forms above them. The **ESR** is the height of the plasma layer formed in one hour (mm/h).

Mechanism: erythrocytes carry a negative charge and repel each other. Sedimentation depends on the erythrocytes sticking together into “coin stacks” (rouleaux, aggregates): aggregates are heavier and settle faster. Aggregation is enhanced by **large, positively charged protein molecules — fibrinogen**, **γ-globulins**, paraproteins; **albumin** reduces it.

Table 5.7. *Factors affecting the ESR*

Increase the ESR	Decrease the ESR
Increase in fibrinogen and globulins (inflammation, infection, rheumatic fever)	Increase in albumin
Decrease in erythrocytes (anemia)	Increase in erythrocytes (erythrocytosis)
Pregnancy (especially the second half, up to 40–50 mm/h)	Increase in bile acids (obstructive jaundice)
Paraproteinemia (multiple myeloma — 60–80 mm/h and higher)	Change in erythrocyte shape (sickle cell anemia)
Older age, menstruation	Hemoconcentration

CAUTION

The ESR is a **nonspecific** parameter: it shows whether inflammation is present and how active it is, but does not establish the diagnosis. The ESR changes late (rises 1–2 days after the onset of disease) and may remain high for several weeks after recovery. **C-reactive protein** is more sensitive for early assessment of acute inflammation.

CLINICAL SIGNIFICANCE

A sharp rise in ESR (100 mm/h and higher) occurs in multiple myeloma, sepsis, giant cell arteritis and metastases of malignant tumors. The dynamics of the ESR are useful for monitoring the effectiveness of treatment (rheumatic fever, tuberculosis).

6. Laboratory work

Experiment 1: determining the hematocrit

1. Rinse a hematocrit capillary with an anticoagulant (heparin).
2. Fill the capillary to 7/8 of its length with blood taken from a finger and seal one end with plasticine.
3. Place the capillary in a hematocrit centrifuge and spin for 1.5–3 minutes (or 30 minutes at 3000 rpm in an ordinary centrifuge).
4. Using the scale, determine the ratio of the height of the column of formed elements to the total column height. **Interpretation:** 40–48% in men, 36–42% in women. A decreased hematocrit indicates anemia or hydremia; an increased one — erythrocytosis or dehydration (hemoconcentration).

Experiment 2: determining the ESR by the Panchenkov method

1. Rinse a Panchenkov capillary with 5% sodium citrate solution, draw citrate up to the “P” mark (50 divisions) and expel it onto a watch glass.
2. Draw blood from a finger into the capillary up to the “K” mark twice, add it to the citrate on the watch glass (blood : citrate = 4 : 1) and mix.
3. Draw the mixture into the capillary up to the “K” (0) mark.
4. Place the capillary in the stand strictly upright and note the time.
5. After one hour measure the height of the plasma column in mm. **Interpretation:** 1–10 mm/h in men, 2–15 mm/h in women. A high result suggests an inflammatory process or a change in protein composition.

CAUTION

In modern laboratories the **Westergren method** (a longer tube, 200 mm scale) is the international standard; its normal values are close to those of the Panchenkov method, but at high values the Westergren method is more accurate.

Experiment 3: determining the osmotic resistance of erythrocytes

1. Pour 1 ml of NaCl solutions from 0.7% to 0.2% (in steps of 0.02%) into test tubes.
2. Add 1 drop of blood to each tube, mix and leave for 1 hour (or centrifuge).
3. The tube in which the solution has turned pink but a sediment is still present — **minimal resistance** (onset of hemolysis).
4. The tube in which the solution is completely red and transparent (“lacquered blood”) and there is no sediment — **maximal resistance** (complete hemolysis). **Interpretation:** normally the minimal resistance is 0.46–0.48% and the maximal 0.32–0.34% NaCl.

7. Practical skills

Skill: evaluating the main physicochemical parameters in a complete blood count

1. Compare the hematocrit with the normal range for the person's sex.
2. Compare the ESR with the normal range for sex and age.
3. Evaluate the total protein (65–85 g/l) and albumin (35–50 g/l).
4. If the ESR is increased, suspect conditions associated with increased fibrinogen and globulins (inflammation, pregnancy).
5. If albumin is decreased, point out the risk of oncotic edema.

Skill: a simple assessment of acid–base status

1. Determine the pH: below 7.35 — acidosis, above 7.45 — alkalosis.
2. Evaluate pCO_2 (normal 35–45 mm Hg): if the direction of change matches the pH — a respiratory disorder.
3. Evaluate HCO_3^- (normal 22–26 mmol/l): if it changes in the same direction as the pH — a metabolic disorder.
4. Determine whether compensation is present.

Review questions

1. In what compartments is water distributed in the body?
2. What is the internal environment, and what does it include?
3. Who introduced the concept of the blood system, and which organs does it include?
4. List the main functions of blood.
5. What is the blood volume in adults, and where are the blood reservoirs?
6. What is the hematocrit, and what is its normal value?
7. Explain the plasma proteins and their functions.
8. What is osmotic pressure, and which substance mainly generates it?
9. What changes occur in erythrocytes in isotonic, hypotonic and hypertonic solutions?
10. What is oncotic pressure, and what is its significance?
11. List the buffer systems of blood and indicate the most powerful one.
12. What are acidosis and alkalosis? Name their types.
13. What is the mechanism of the ESR?
14. Which factors increase and decrease the ESR?
15. How is the ESR determined by the Panchenkov method?

Test questions

1. What proportion of body mass does blood volume make up in adults?
   A) 6–8 %
   B) 15–20 %
   C) 1–2 %
   D) 25–30 %
   E) 40 %

2. What is the normal hematocrit in men?
   A) 20–30 %
   B) 40–48 %
   C) 55–60 %
   D) 60–70 %
   E) 10–15 %

3. Which protein generates most of the oncotic pressure of plasma?
   A) Albumin
   B) Fibrinogen
   C) γ-Globulin
   D) α-Globulin
   E) Hemoglobin

4. What is the osmotic pressure of blood plasma?
   A) 25–30 mm Hg
   B) 1 atm
   C) 7.6 atm
   D) 15 atm
   E) 0.03 atm

5. How do erythrocytes change in a hypotonic solution?
   A) They shrink
   B) It does not change
   C) They aggregate
   D) Their sedimentation rate decreases
   E) They swell and burst (osmotic hemolysis)

6. Which solution is isotonic?
   A) 0.45% NaCl
   B) 10% NaCl
   C) Distilled water
   D) 0.9% NaCl
   E) 3% NaCl

7. Which is the most powerful buffer system of blood?
   A) Bicarbonate
   B) Hemoglobin
   C) Phosphate
   D) Plasma proteins
   E) Ammonia

8. What is the normal pH of arterial blood?
   A) 6,8–7,0
   B) 7,0–7,2
   C) 7,50–7,60
   D) 7,8–8,0
   E) 7,35–7,45

9. Removal of which protein from plasma yields serum?
    A) Albumin
    B) γ-Globulin
    C) Fibrinogen
    D) Transferrin
    E) Ceruloplasmin

10. Which factor increases the ESR?
    A) Increase in albumin
    B) An increase in fibrinogen and globulins
    C) Erythrocytosis
    D) An increase in bile acids
    E) Hemoconcentration

11. What is the normal ESR in women?
    A) 2–15 mm/h
    B) 20–30 mm/h
    C) 0–1 mm/h
    D) 40–50 mm/h
    E) 60–80 mm/h

12. What condition develops when albumin decreases?
    A) Hemoconcentration
    B) Osmotic hemolysis
    C) Alkalosis
    D) Oncotic edema
    E) Erythrocytosis

13. At what NaCl concentration does hemolysis of erythrocytes normally begin?
    A) 0,9 %
    B) 0,2 %
    C) 0,46–0,48 %
    D) 0,7 %
    E) 1,5 %

14. What does blood viscosity mainly depend on?
    A) On the glucose content
    B) On mineral salts
    C) On the number of leukocytes
    D) On the number of erythrocytes (hematocrit)
    E) On hormones

15. At what ratio of H_2CO_3 to $NaHCO_3$ in the bicarbonate buffer is the pH 7.4?
    A) 1 : 1
    B) 20 : 1
    C) 1 : 5
    D) 10 : 1
    E) 1 : 20

Clinical cases

Case 1

A patient with liver cirrhosis has fluid accumulation in the legs and the abdominal cavity (edema, ascites). Total blood protein is 52 g/l, albumin 24 g/l.

Questions: Explain the mechanism of the edema. Which parameter has changed?

Answer: The liver is the main organ that synthesizes albumin; in cirrhosis albumin synthesis decreases (24 g/l, normal 35–50 g/l). Albumin generates 80% of the oncotic pressure of plasma, so oncotic pressure falls. The force holding water inside the capillaries decreases: more fluid leaves at the arterial end and less returns at the venous end. As a result fluid accumulates in the tissues — oncotic (hypoproteinemic) edema. In cirrhosis, increased pressure in the portal vein also contributes to ascites.

Case 2

A healthy woman at 32 weeks of pregnancy has an ESR of 42 mm/h. Other parameters are normal, and she has no complaints.

Questions: Is this a pathology? Explain the cause of the increased ESR.

Answer: This is a physiological state. In the second half of pregnancy the blood levels of fibrinogen and globulins rise, albumin decreases relatively, and the erythrocyte concentration falls slightly because of hydremia. Large protein molecules reduce the charge of erythrocytes and enhance their aggregation into “coin stacks” (rouleaux) — the aggregates settle faster. In pregnancy the ESR may rise to 40–50 mm/h.

Case 3

By mistake a nurse infused 500 ml of distilled water intravenously into a patient. A few hours later the urine turned red-brown.

Questions: What condition has developed? Explain the mechanism. What solutions should be infused intravenously in large volumes?

Answer: Osmotic hemolysis. Distilled water is sharply hypotonic relative to blood; water enters the erythrocytes down the osmotic gradient, they swell and burst, and hemoglobin is released into the plasma. Free hemoglobin is excreted through the kidneys, colors the urine red-brown (hemoglobinuria) and may damage the renal tubules. Only isotonic solutions (0.9% NaCl, Ringer's solution, 5% glucose) are infused intravenously in large volumes.

Quick review

Table 5.8. *Key concepts of topic 5*

Concept	Brief definition
Total body water	≈ 60% of body mass; 40% intracellular, 20% extracellular
Blood system	Peripheral blood + hematopoietic and blood-destroying organs + regulation (Lang)
Blood volume	6–8% of body mass (4.5–6 l)
Hematocrit	40–48% in men, 36–42% in women
Plasma proteins	65–85 g/l: albumin, globulins, fibrinogen
Serum	Plasma without fibrinogen

Concept	Brief definition
Osmotic pressure	7.6 atm; mainly NaCl
Oncotic pressure	25–30 mm Hg; 80% albumin
Isotonic solution	0.9% NaCl, 5% glucose
Osmotic resistance	Hemolysis begins at 0.46–0.48% and is complete at 0.32–0.34%
pH	Arterial blood 7.35–7.45
Hemoglobin buffer	The most powerful (≈ 75%)
Bicarbonate buffer	H_2CO_3 : $NaHCO_3$ = 1 : 20; the main buffer in plasma
Normal ESR	Men 1–10, women 2–15 mm/h
Increased ESR	Fibrinogen, globulins, anemia, pregnancy

TOPIC 6

Erythrocytes. Hemoglobin and its types

Learning objectives

To teach the student the structure, number, functions and life cycle of erythrocytes, erythropoiesis and its regulation, the structure, types, compounds and amount of hemoglobin, the color index and the erythrocyte indices; to master counting erythrocytes, determining hemoglobin and calculating the color index.

By the end of the class the student should know:

- the functional significance of the shape, size and structure of erythrocytes;
- the normal erythrocyte count and its physiological changes;
- the functions of erythrocytes;
- the stages of erythropoiesis and its regulation (erythropoietin, vitamins, iron);
- the destruction of erythrocytes and the formation of bilirubin;
- the structure, physiological types and compounds of hemoglobin;
- the color index and the erythrocyte indices (MCV, MCH, MCHC).

The student should be able to:

- count erythrocytes in a Goryaev chamber;
- determine hemoglobin by the Sahli method or the hemiglobincyanide method;
- calculate and evaluate the color index.

1. Structure and properties of erythrocytes

Erythrocytes (red blood cells) in mammals and humans are anucleate cells shaped like biconcave discs. Their diameter is **7.2–7.5 µm**, their thickness 2.5 µm at the edge and 1 µm in the center, and their volume 80–94 fl.

Advantages of the erythrocyte shape:

1. **Large surface area** — a biconcave disc has a surface area 1.7 times greater than a sphere, which speeds up gas diffusion.
2. **Short diffusion distance** — the distance from any point to the surface is small.
3. **Deformability** — the erythrocyte folds to pass through capillaries 3 µm in diameter (thanks to the membrane skeleton formed by spectrin, actin and ankyrin).
4. **Absence of a nucleus and mitochondria** — there is much room for hemoglobin; the erythrocyte does not consume the oxygen it carries (it obtains energy from anaerobic glycolysis). **Hemoglobin** makes up ≈ 34% of the erythrocyte (95% of its dry mass). Blood group antigens (A, B, Rh) are located on its membrane.

Erythrocyte count

4.0–5.0 × 10^{12}/l in men and **3.7–4.7 × 10^{12}/l** in women. Up to 6–7 × 10^{12}/l in newborns.

- **Erythrocytosis** is an increase in the number of erythrocytes. It may be **relative** (hemoconcentration: dehydration, heavy sweating, vomiting) or **absolute** (increased erythropoiesis: high altitude, chronic lung and heart diseases, athletes; pathological — erythremia).

- **Erythropenia** is a decrease: anemias, hydremia (large fluid intake, pregnancy).

Table 6.1. *Functions of erythrocytes*

Function	Content
Respiratory (main)	Transport of O_2 from the lungs to the tissues and of part of the CO_2 from the tissues to the lungs
Buffer	The hemoglobin buffer system — 75% of the buffer capacity of blood
Transport	Carrying amino acids, lipids and drugs on their surface
Immunological	Antigens on the membrane (blood groups), transport of immune complexes
Participation in blood coagulation	Procoagulants are released when they are destroyed
Enzymatic	Carbonic anhydrase — in CO_2 transport

2. Erythropoiesis and its regulation

Erythropoiesis is the formation of erythrocytes; in adults it takes place in the **red bone marrow** (flat bones, vertebrae, sternum, pelvic bones). In the embryo erythropoiesis occurs first in the yolk sac, then in the liver and spleen. About 200 billion erythrocytes (2–3 million per second) are produced per day.

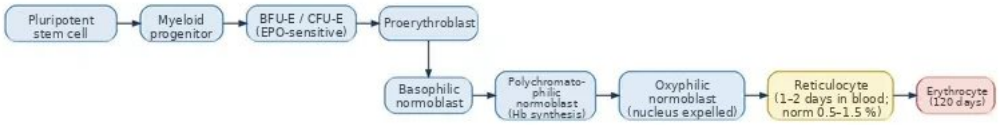


Figure 6.1. *Stages of erythropoiesis*

Reticulocytes are young erythrocytes; residual ribosomes (a reticular substance) become visible in them with special staining. Normally they make up **0.5–1.5%** (5–15‰) of the peripheral blood. The reticulocyte count reflects the erythropoietic activity of the bone marrow: it increases after blood loss or treatment with iron preparations (**reticulocyte crisis**).

Regulation of erythropoiesis

Erythropoietin (EPO) is the main humoral regulator of erythropoiesis, a glycoprotein hormone. It is produced mainly in the **kidneys** (90%) and in small amounts in the liver. The main factor that increases its synthesis is **tissue hypoxia**.

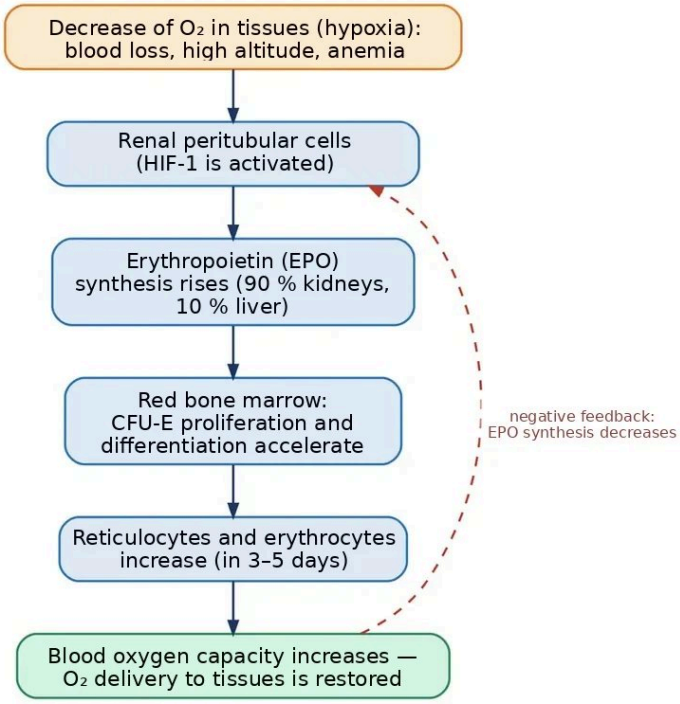


Figure 6.2. *Regulation of erythropoiesis by erythropoietin*

Table 6.2. *Factors affecting erythropoiesis*

Factor	Effect
Erythropoietin	Enhances proliferation and differentiation of erythroid progenitors
Androgens, thyroxine, somatotropin, cortisol	Enhance erythropoiesis (this is why men have more erythrocytes)
Estrogens	Slightly suppress it
Iron	Needed for heme synthesis (20–25 mg is required daily, mostly recycled from destroyed erythrocytes; 1–2 mg is absorbed from food)
Vitamin B_{12} and folic acid	Needed for DNA synthesis and cell division
Vitamin B_6, C, copper, cobalt	For heme synthesis and iron absorption
Castle's intrinsic factor	Produced by the parietal cells of the stomach; needed for the absorption of B_{12} in the small intestine
Sympathetic nervous system	Enhances erythropoiesis; the parasympathetic system suppresses it

CLINICAL SIGNIFICANCE

In chronic renal failure EPO synthesis decreases and severe anemia develops; such patients are treated with recombinant erythropoietin. In sport EPO is used illegally as **doping** — the hematocrit rises, the blood thickens and the risk of thrombosis increases. **After gastric resection** Castle's intrinsic factor is lost and B_{12}-deficiency (megaloblastic) anemia develops.

3. Life cycle and destruction of erythrocytes

Erythrocytes live in the blood for an average of **120 days**. The membrane of aging erythrocytes becomes rigid, and they are phagocytosed by macrophages mainly in the **spleen** (and in the liver and bone marrow) (**intracellular hemolysis**). A small proportion is destroyed inside the blood vessels (**intravascular hemolysis**).

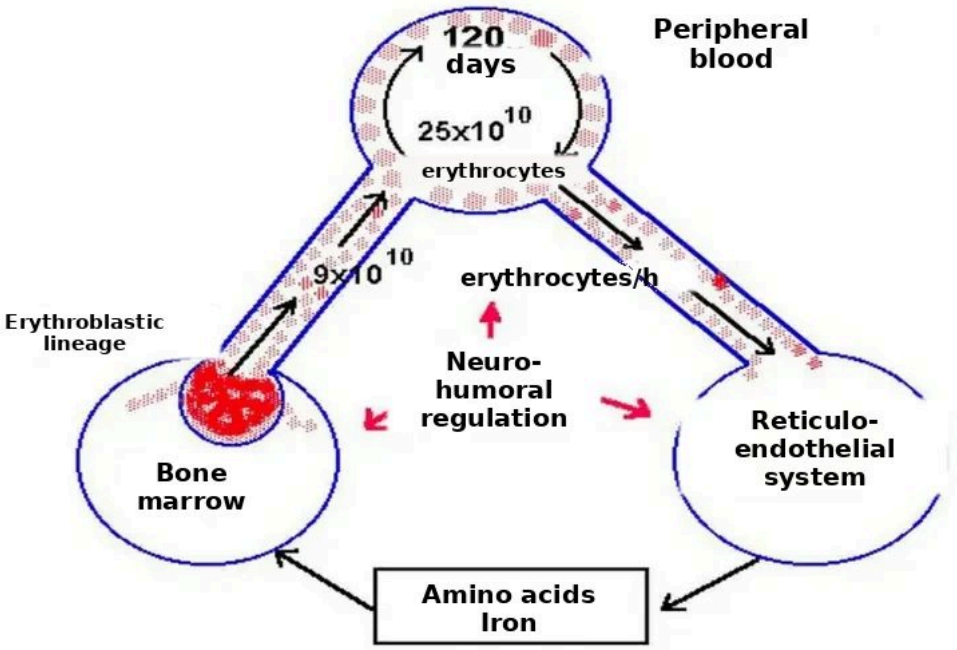


Figure 6.3. *The erythron: formation, circulation and destruction of erythrocytes (Alavia O.T., 2018)*

Breakdown of hemoglobin: **globin** is split into amino acids; **iron** returns to the bone marrow with transferrin or is stored in the liver as ferritin and hemosiderin; **heme** → biliverdin → **bilirubin** (unconjugated, carried by albumin) → conjugated with glucuronic acid in the liver (conjugated bilirubin) → into the intestine with bile → stercobilin (color of stool) and urobilin (color of urine).

> **REMEMBER**
> The **erythron** (Castle) is the set of erythrocytes in the peripheral blood, their precursors in the bone marrow and the erythrocytes at the sites of their destruction. Normally the formation and destruction of erythrocytes are in balance.

4. Hemoglobin

Hemoglobin (Hb) is a complex protein (chromoprotein) with a molecular mass of 64,500 Da. It consists of **4 hemes** and **globin** (4 polypeptide chains). **Heme** is a compound of protoporphyrin IX with **ferrous iron (Fe^{2+})**; oxygen is bound by the iron in heme. One Hb molecule can bind **4 molecules of O_2**.

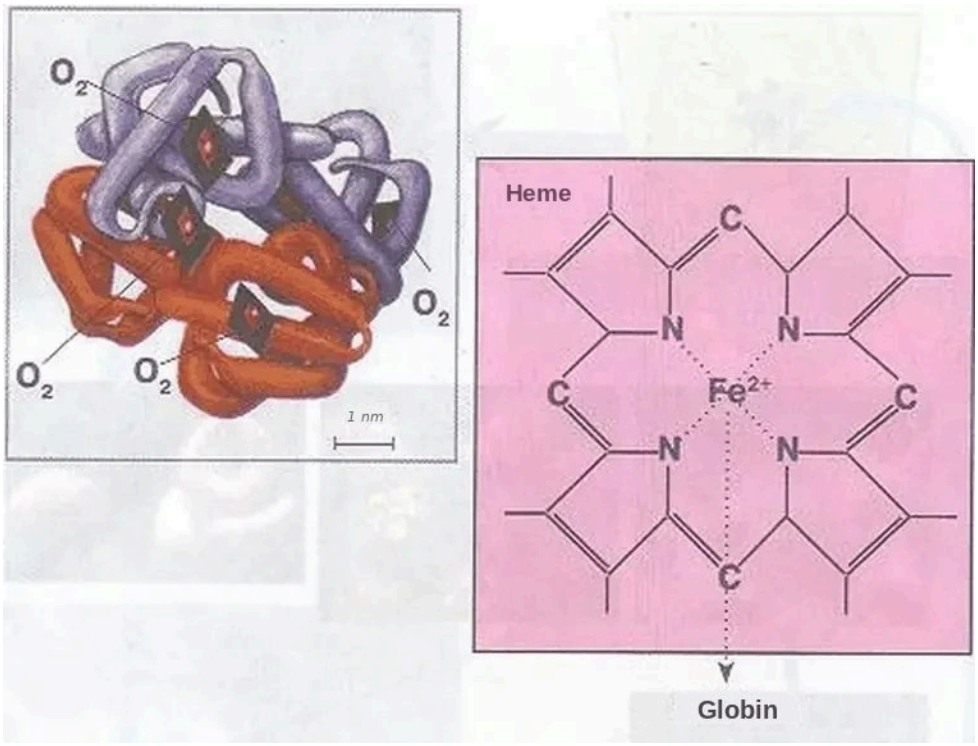


Figure 6.4. *The hemoglobin molecule: four subunits and the structure of heme (Alavia O.T., 2018)*

Hemoglobin content

130–160 g/l in men and **120–140 g/l** in women. 1 g of Hb binds **1.34 ml of O_2** (Hüfner's number). Therefore, with Hb of 150 g/l the **oxygen capacity** of blood is ≈ 200 ml O_2/l (20 vol%).

Physiological types of hemoglobin

Table 6.3. *Types of hemoglobin*

Type	Chains	When	Features
Embryonic (Gower 1, 2, Portland)	ζ, ε chains	In the first 3 months of pregnancy	Formed in the yolk sac
Fetal (HbF)	$\alpha_2\gamma_2$	The main type in fetal life; 70–80% at birth; < 1–2% by 1 year	High affinity for O_2 (binds little 2,3-DPG) — the fetus takes O_2 from the mother's blood
Adult (HbA_1)	$\alpha_2\beta_2$	95–98% in adults	The main hemoglobin
Adult (HbA_2)	$\alpha_2\delta_2$	2–3% in adults	—

Pathological hemoglobins: HbS (sickle cell anemia — glutamic acid replaced by valine in the β chain), **HbH**, **hemoglobin Bart's** (thalassemias) and others (more than 300).

Compounds of hemoglobin

Table 6.4. *Compounds of hemoglobin*

Compound	Formation	Features
Oxyhemoglobin (HbO_2)	Hb + O_2 (in the lungs)	Physiological, reversible; the bright red color of arterial blood
Deoxyhemoglobin (reduced Hb)	Hb that has given up O_2 (in the tissues)	Physiological; the dark red color of venous blood

Compound	Formation	Features
Carbaminohemoglobin ($HbCO_2$)	Hb + CO_2 (bound to amino groups of globin)	Physiological; 10–15% of CO_2 is carried in this form
Carboxyhemoglobin (HbCO)	Hb + carbon monoxide (CO)	Pathological; the affinity of CO for Hb is 200–300 times higher than that of O_2; it dissociates slowly
Methemoglobin (MetHb)	Fe^{2+} is oxidized to Fe^{3+} (nitrates, nitrites, some drugs)	Pathological; cannot carry O_2; normally < 1–2%
Glycated Hb (HbA_{1c})	Hb + glucose (nonenzymatic)	Normally < 6%; reflects the average glycemia over 3 months

CLINICAL SIGNIFICANCE

Carbon monoxide (CO) poisoning — from gas stoves, coal stoves and car exhaust. Carboxyhemoglobin does not carry O_2 and also hinders the release of O_2 by the remaining Hb — severe hypoxia. A "cherry-red" color of the skin is typical. Treatment: fresh air, 100% oxygen and, in severe cases, hyperbaric oxygenation. In **methemoglobinemia** (nitrate-containing water, aniline) the skin is bluish-gray; treatment — methylene blue, ascorbic acid.

MEMORY AID

"CO — the Cherry Con": carbon monoxide "occupies" Hb and the skin becomes cherry-colored, yet the tissues suffocate. A pulse oximeter is "fooled" into reading HbCO as HbO_2 — the saturation may look normal!

5. Color index and erythrocyte indices

The **color index (CI)** is the relative saturation of one erythrocyte with hemoglobin:

CI = (Hb, g/l × 3) / (the first three digits of the erythrocyte count)

For example: Hb 150 g/l, erythrocytes 5.0×10^{12}/l → CI = 150 × 3 / 500 = 0.9.

Table 6.5. *Interpretation of the color index*

CI	Term	Where it occurs
0,85–1,05	Normochromia	Normal; acute blood loss, hemolytic anemias
Less than 0.85	Hypochromia	Iron-deficiency anemia, thalassemia
Above 1.05	Hyperchromia	B_{12}- and folate-deficiency (megaloblastic) anemia

Table 6.6. *Erythrocyte indices (automated hematology analyzers)*

Index	Meaning	Normal value
MCV	Mean corpuscular volume	80–100 fl
MCH	Mean corpuscular hemoglobin	27–33 pg
MCHC	Mean corpuscular hemoglobin concentration	320–360 g/l
RDW	Erythrocyte distribution width by volume (anisocytosis)	11,5–14,5 %

CAUTION

The color index is a traditional parameter widely used in the practice of our country and the CIS; in international practice it has been replaced by the **MCH**. The MCH also shows the amount of hemoglobin in one erythrocyte, but in absolute units (picograms). The **MCV** (microcytic, normocytic, macrocytic) is the most important index for determining the type of anemia.

6. Laboratory work

Experiment 1: counting erythrocytes in a Goryaev chamber

1. Put 4 ml of 3% NaCl solution (or Hayem's fluid) into a test tube.
2. Take 0.02 ml of blood from a finger, add it to the tube (dilution 1 : 200) and mix.
3. Cover the Goryaev chamber with a coverslip (rub it on until “Newton's rings” appear); put a drop of the diluted blood under the coverslip.
4. After 1 minute, count the erythrocytes under the microscope (×8 objective) in 5 large squares along the diagonal (each consisting of 16 small squares — 80 small squares in total). Count cells on the borders by the rule “those on the left and upper lines are counted”.
5. Calculation: X = (A × 4000 × 200) / 80, where A is the number of erythrocytes in 80 small squares; the count per 1 µl; simplified: X = A × 10,000 (per 1 µl) or A / 100 × 10^{12}/l. **Interpretation:** compare the result with the normal range for the person's sex.

Experiment 2: determining hemoglobin by the Sahli method

1. Pour 0.1 N HCl up to the mark into the middle tube of the Sahli hemometer.
2. Take 20 µl (0.02 ml) of blood, add it to the HCl and mix; wait 5 minutes — hemoglobin forms **hematin hydrochloride** (brown) with hydrochloric acid.
3. Add distilled water drop by drop until the color matches the standard tubes.
4. Read the hemoglobin content (g/dl or g/l) from the tube scale.

CAUTION

The Sahli method is subjective and has a large error (± 10%). In modern laboratories the international standard is the **hemiglobincyanide method** (photometric) or automated analyzers. The Sahli method is used for teaching purposes.

Experiment 3: calculating the color index

1. Take the results of the erythrocyte count and hemoglobin content.
2. Calculate by the formula CI = Hb (g/l) × 3 / the first three digits of the erythrocyte count.
3. Evaluate the result: normo-, hypo- or hyperchromia.

7. Practical skills

Skill: suggesting the type of anemia from a blood test

1. Compare Hb and the erythrocyte count with the normal values (whether anemia is present).
2. Calculate the color index (or MCH).
3. Evaluate erythrocyte size from the MCV.
4. Evaluate the reticulocyte count: if increased, the bone marrow is active (blood loss, hemolysis); if decreased, erythropoiesis is suppressed.

5. Hypochromic microcytic — think of iron deficiency; hyperchromic macrocytic — B_{12}/folate deficiency; normochromic — acute blood loss or hemolysis.

Skill: calculating the oxygen capacity from the hemoglobin content

1. Take the Hb content in g/l.
2. Multiply by Hüfner's number (1.34 ml O_2/g).
3. Express the result in ml O_2/l (normally ≈ 190–210 ml/l).

Review questions

1. How are the shape and structure of erythrocytes adapted to their function?
2. What is the normal erythrocyte count, and what are its physiological changes?
3. What is the difference between relative and absolute erythrocytosis?
4. List the functions of erythrocytes.
5. Where and in what stages does erythropoiesis take place?
6. What are reticulocytes, and what is the significance of counting them?
7. Where is erythropoietin produced, and what enhances its synthesis?
8. Which vitamins and trace elements are needed for erythropoiesis?
9. What is the significance of Castle's intrinsic factor?
10. Where and how are erythrocytes destroyed?
11. Explain the structure of hemoglobin.
12. What are the physiological types of hemoglobin, and how do they differ?
13. List the physiological and pathological compounds of hemoglobin.
14. How is the color index calculated and interpreted?
15. What do the MCV, MCH and MCHC indices show?

Test questions

1. What is the diameter of an erythrocyte?
   A) 2–3 µm
   B) 10–12 µm
   C) 15–20 µm
   D) 4–5 µm
   E) 7.2–7.5 µm

2. What is the normal erythrocyte count in men?
   A) $3.0–3.5 \times 10^{12}$/l
   B) $6–7 \times 10^{12}$/l
   C) $4–9 \times 10^{9}$/l
   D) $4.0–5.0 \times 10^{12}$/l
   E) $180–320 \times 10^{9}$/l

3. Where is erythropoietin mainly produced?
   A) In the kidneys
   B) In the red bone marrow
   C) In the spleen
   D) In the pituitary gland
   E) In the stomach

4. What is the main factor that enhances erythropoietin synthesis?
   A) Hyperoxia
   B) Erythrocytosis
   C) Tissue hypoxia
   D) Estrogens
   E) The parasympathetic nervous system

5. For the absorption of which substance is Castle's intrinsic factor needed?
   A) Iron
   B) Folic acid
   C) Vitamin C
   D) Vitamin B_{12}
   E) Copper

6. What is the average life span of erythrocytes?
   A) 7–10 days
   B) 120 days
   C) 1 year
   D) 30 days
   E) 3–5 days

7. What is the normal reticulocyte content of the peripheral blood?
   A) 5–10 %
   B) 0 %
   C) 0,5–1,5 %
   D) 20–25 %
   E) 50 %

8. In what state is the iron in heme?
   A) Fe^{3+}
   B) Fe^{0}
   C) Fe^{4+}
   D) As a free ion
   E) Fe^{2+}

9. What is the composition of fetal hemoglobin (HbF)?
   A) $\alpha_2\beta_2$
   B) $\alpha_2\delta_2$
   C) β_4
   D) $\alpha_2\gamma_2$
   E) $\zeta_2\varepsilon_2$

10. Which is the main hemoglobin in adults?
    A) HbF
    B) HbA_2
    C) HbS
    D) Gower 1
    E) HbA_1

11. Which compound is formed in carbon monoxide poisoning?
    A) Carboxyhemoglobin
    B) Carbaminohemoglobin
    C) Methemoglobin
    D) Oxyhemoglobin
    E) Deoxyhemoglobin

12. In what state is the iron in methemoglobin?
    A) Fe^{2+}
    B) Fe^{3+} (oxidized)
    C) Detached from heme
    D) Bound to CO
    E) Bound to O_2

13. If Hb is 120 g/l and erythrocytes are 4.5×10^{12}/l, what is the color index?
    A) 0,8
    B) 1,0
    C) 1,2
    D) 0,6
    E) 1,5

14. How much oxygen does 1 g of hemoglobin bind?
    A) 0.34 ml
    B) 13.4 ml
    C) 1.34 ml
    D) 20 ml
    E) 3 ml

15. Which anemia is characterized by hyperchromia (CI > 1.05)?
    A) Iron-deficiency anemia
    B) B_{12}-deficiency (megaloblastic) anemia
    C) Thalassemia
    D) Acute blood loss
    E) Chronic blood loss

Clinical cases

Case 1

A 28-year-old woman complains of weakness, fatigability, dizziness and hair loss. For many years her menstrual periods have been heavy. Blood test: Hb 85 g/l, erythrocytes 3.6×10^{12}/l, MCV 72 fl.

Questions: Calculate the color index. What anemia can be suspected? Explain its cause.

Answer: CI = 85 × 3 / 360 = 0.71 — hypochromia. The MCV is reduced (72 fl) — microcytosis. Hypochromic microcytic anemia — iron-deficiency anemia. Chronic blood loss (heavy menstruation)

has depleted the iron stores; iron is needed for heme synthesis, so hemoglobin synthesis decreases and the erythrocytes are small and pale. The weakness and dizziness are due to the reduced oxygen capacity of the blood and tissue hypoxia.

Case 2

In winter, family members who slept in a closed room with a burning coal stove were found in the morning with headache, nausea and weakness. One of them had cherry-red skin and mucous membranes. A pulse oximeter showed a saturation of 98%.

Questions: What kind of poisoning is this? Why is the pulse oximeter reading normal? What is the principle of first aid?

Answer: Carbon monoxide (CO) poisoning. CO binds to hemoglobin 200–300 times more strongly than oxygen, forming carboxyhemoglobin; it does not carry oxygen and also hinders the release of oxygen to the tissues by the remaining oxyhemoglobin — severe tissue hypoxia. Because carboxyhemoglobin has a color similar to oxyhemoglobin, an ordinary pulse oximeter cannot distinguish them and shows a falsely normal saturation. First aid: move the patient to fresh air and give 100% oxygen (which speeds up the elimination of CO); in severe cases, hyperbaric oxygenation.

Case 3

A mountaineer who moved from Tashkent to the Pamir mountains (3500 m) was found a month later to have erythrocytes of 5.8×10^{12}/l, hemoglobin of 175 g/l and reticulocytes of 2.5%.

Questions: Explain the mechanism of these changes. Is this a relative or an absolute erythrocytosis?

Answer: This is an absolute (true) compensatory erythrocytosis. At high altitude the partial pressure of oxygen in the atmospheric air is low, so arterial pO_2 decreases and tissue hypoxia develops. Hypoxia enhances erythropoietin synthesis in the kidneys; erythropoietin accelerates the proliferation of erythroid progenitors in the red bone marrow. The increase in reticulocytes reflects the activity of the bone marrow. The increase in erythrocytes and hemoglobin raises the oxygen capacity of the blood and compensates for the hypoxia.

Quick review

Table 6.7. *Key concepts of topic 6*

Concept	Brief definition
Erythrocyte	Anucleate biconcave disc, 7.2–7.5 µm
Erythrocyte count	Men 4.0–5.0, women 3.7–4.7 × 10^{12}/l
Life span	120 days; destroyed mainly in the spleen
Reticulocytes	0.5–1.5%; an indicator of bone marrow activity
Erythropoietin	In the kidney; enhanced by hypoxia
B_{12} and folate	DNA synthesis; their deficiency causes megaloblastic anemia
Castle's intrinsic factor	In the stomach; for the absorption of B_{12}
Hemoglobin	4 hemes (Fe^{2+}) + globin; men 130–160, women 120–140 g/l
HbA_1	$\alpha_2\beta_2$; the main type in adults
HbF	$\alpha_2\gamma_2$; high affinity for O_2
Hüfner's number	1 g of Hb = 1.34 ml of O_2

Concept	Brief definition
Carboxyhemoglobin	Hb + CO; affinity 200–300 times higher than for O_2
Methemoglobin	Fe^{3+}; does not carry O_2
Color index	Hb × 3 / first 3 digits of the erythrocyte count; normal 0.85–1.05
MCV	Mean volume 80–100 fl

TOPIC 7

Leukocytes

Learning objectives

To teach the student the general properties, types, number and functions of leukocytes, the leukocyte formula (differential count) and its analysis, leukopoiesis and its regulation, and the role of leukocytes in nonspecific and specific (immune) defense; to master counting leukocytes and determining the leukocyte formula in a blood smear.

By the end of the class the student should know:

- the general properties and number of leukocytes;
- the types of physiological and pathological leukocytosis and leukopenia;
- the leukocyte formula and its normal values;
- the functions of neutrophils, eosinophils, basophils, monocytes and lymphocytes;
- the stages of leukopoiesis and its regulation;
- the stages of phagocytosis;
- cellular and humoral immunity and the role of T and B lymphocytes.

The student should be able to:

- count leukocytes in a Goryaev chamber;
- prepare and stain a blood smear and calculate the leukocyte formula;
- detect and evaluate a “shift to the left” in the leukocyte formula.

1. General characteristics of leukocytes

Leukocytes (white blood cells) are nucleated, colorless cells 6–20 µm in size. They are only **transported** in the blood and perform their main function **in the tissues**: they pass through the capillary wall (**diapedesis**) and move toward the focus of inflammation.

General properties: amoeboid movement, diapedesis, **chemotaxis** (movement toward chemical substances), phagocytosis and secretion of biologically active substances.

Number: 4–9 × 10^9/l in adults (4–10 × 10^9/l in some sources). Up to 10–30 × 10^9/l in newborns.

- **Leukocytosis** is an increase above 9 (10) × 10^9/l.
- **Leukopenia** is a decrease below 4 × 10^9/l.

Table 7.1. *Types of leukocytosis*

Type	Cause	Features
Physiological — alimentary (after meals)	Food intake	Leukocytes accumulate mainly in the intestinal submucosa; therefore blood tests are taken on an empty stomach
Physiological — myogenic	Heavy physical work	May increase 3–5 times; release from reservoirs
Physiological — emotional	Stress, pain	Redistribution under the action of epinephrine
Physiological — pregnancy	Late pregnancy, labor	—

Type	Cause	Features
Reactive (pathological)	Bacterial infection, inflammation, necrosis, acute blood loss	Increased leukopoiesis
Leukemias	Blood malignancy	May be very high

Physiological leukocytoses are of a **redistributive** nature (release from reservoirs and from the marginal layer into the circulating blood), whereas pathological leukocytoses are associated with increased leukopoiesis.

Causes of **leukopenia**: viral infections (influenza, measles), ionizing radiation, cytostatics and some other drugs, autoimmune diseases, bone marrow damage.

2. Types of leukocytes and the leukocyte formula

By the presence or absence of specific granules in the cytoplasm, leukocytes are divided into two groups:

- **Granulocytes (granular)** — neutrophils, eosinophils, basophils.
- **Agranulocytes (nongranular)** — monocytes and lymphocytes.

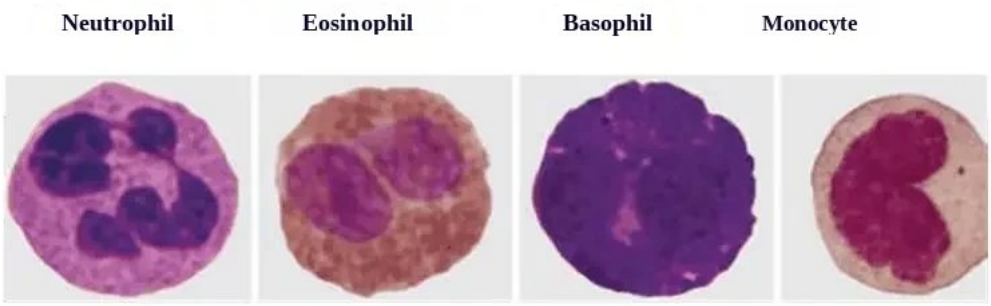


Figure 7.1. *Types of leukocytes: neutrophil, eosinophil, basophil, monocyte (smear stained by Romanowsky–Giemsa; Alavia O.T., 2018)*

The **leukocyte formula (leukogram)** is the percentage ratio of the individual types of leukocytes.

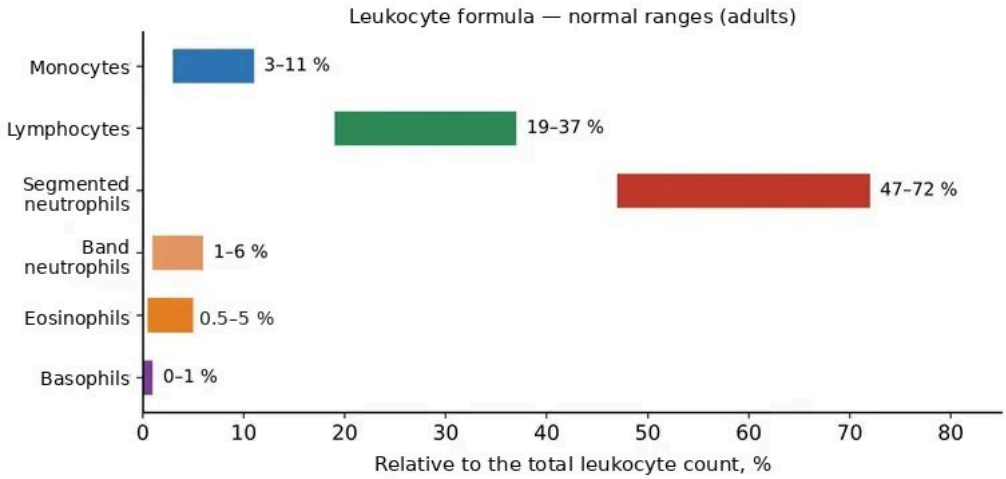


Figure 7.2. *Normal ranges of the leukocyte formula (adults)*

Table 7.2. *The leukocyte formula and the functions of leukocytes*

Type	Normal, %	Absolute count, × 10^9/l	Main function
Neutrophils — band	1–6	0,04–0,3	Young forms; phagocytosis
Neutrophils — segmented	47–72	2,0–5,5	Phagocytosis of bacteria — the first line of defense
Eosinophils	0,5–5	0,02–0,3	Defense against helminths, limiting allergic reactions (release histaminase)
Basophils	0–1	0–0,065	Release of histamine and heparin; participation in immediate-type allergy
Lymphocytes	19–37	1,2–3,0	Specific (immune) defense: T — cellular, B — humoral immunity
Monocytes	3–11	0,09–0,6	Become macrophages in the tissues; phagocytosis, antigen presentation

MEMORY AID

To remember the leukocyte formula in decreasing order: **“Never Let Monkeys Eat Bananas”** — **N**eutrophils > **L**ymphocytes > **M**onocytes > **E**osinophils > **B**asophils (60 – 30 – 6 – 3 – 0.5%).

Shifts of the leukocyte formula

By degree of maturity, neutrophils pass through: myelocyte → metamyelocyte (juvenile) → band → segmented. Normally only band and segmented neutrophils are present in the peripheral blood.

- A **shift to the left** is an increase in young neutrophils (band forms, metamyelocytes and even myelocytes). In acute bacterial infection and inflammation it shows that the bone marrow is working intensively. Nuclear shift index = (myelocytes + metamyelocytes + band forms) / segmented forms; normally 0.05–0.08.
- A **shift to the right** is an increase in segmented neutrophils with 5 or more segments (hypersegmentation); seen in B_{12}-deficiency anemia and radiation sickness.

CLINICAL SIGNIFICANCE

The leukogram aids diagnosis: **neutrophilia** with a shift to the left — bacterial infection (pneumonia, appendicitis); **lymphocytosis** — viral infections (pertussis, mononucleosis), tuberculosis; **eosinophilia** — helminth infestation, allergic diseases (bronchial asthma); **monocytosis** — tuberculosis, infectious mononucleosis, recovery from sepsis; **agranulocytosis** — a sharp decrease in granulocytes, with a risk of severe infections.

3. Functions of the types of leukocytes

Neutrophils

The most numerous (50–70% of leukocytes) and mobile cells; they are the **first** to reach the focus of infection. They circulate in the blood for 6–8 hours and live in the tissues for 1–2 days. Functions: **phagocytosis** of bacteria (“microphages” — I.I. Mechnikov); killing bacteria with lysozyme, lactoferrin, cationic proteins, oxygen radicals and myeloperoxidase; release of pyrogens (fever). Pus consists mainly of dead neutrophils.

Eosinophils

Their granules stain deep pink with eosin. Functions: **defense against helminths** (they damage the parasite with major basic protein and peroxidase); **limiting allergic reactions** — they release histaminase (which breaks down histamine) and arylsulfatase; phagocytosis of antigen–antibody complexes. There are fewer of them in the blood by day and more at night (the effect of cortisol).

Basophils

The least numerous. Their granules contain **histamine** (dilates vessels and increases their permeability) and **heparin** (an anticoagulant). IgE receptors are located on their membrane; when an allergen binds, degranulation occurs — **immediate-type allergy** (anaphylaxis). Similar cells in the tissues are mast cells.

Monocytes

The largest leukocytes (15–20 µm), with a bean-shaped nucleus. After circulating in the blood for 2–3 days they pass into the tissues and become **macrophages** ("macrophages" — Mechnikov): Kupffer cells in the liver, alveolar macrophages in the lungs, osteoclasts in bone, microglia in the CNS. Functions: phagocytosis of large particles and dead cells ("scavengers"), **antigen presentation** (to T lymphocytes) and secretion of cytokines (IL-1, TNF-α).

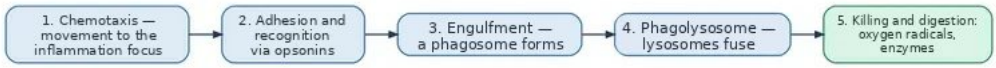


Figure 7.3. *Stages of phagocytosis*

Lymphocytes

The main cells of specific (acquired) immunity. Some of them live for years and even decades (**memory cells**).

Table 7.3. *Types of lymphocytes*

Type	Share in blood	Site of maturation	Function
T lymphocytes	60–80 %	Thymus	Cellular immunity: T killers (CD8) kill virus-infected and tumor cells; T helpers (CD4) regulate the immune response; T regulatory cells limit it
B lymphocytes	10–15 %	Red bone marrow	Humoral immunity: under the action of an antigen they become plasma cells and produce antibodies (immunoglobulins)
NK cells (natural killers)	5–15 %	Bone marrow	Kill tumor and virus-infected cells without prior sensitization

CLINICAL SIGNIFICANCE

HIV infection damages $CD4^+$ T helpers; when their number falls below 200 cells/µl, AIDS develops — the immune response is sharply weakened and opportunistic infections and tumors appear.

4. Leukopoiesis and its regulation

Granulocytes and monocytes are formed in the **red bone marrow**, and lymphocytes in the bone marrow and lymphoid organs (thymus, lymph nodes, spleen, Peyer's patches).

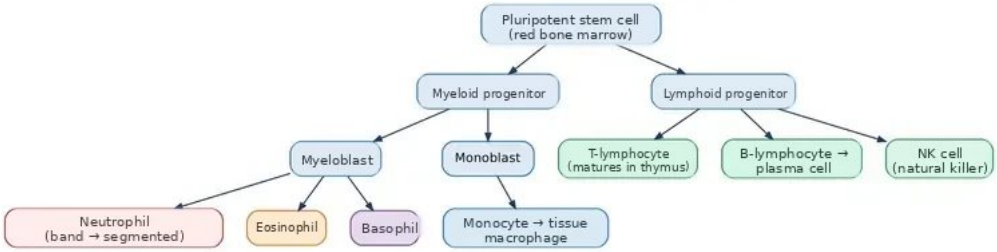


Figure 7.4. *Scheme of leukopoiesis*

Leukopoiesis is regulated by **colony-stimulating factors (CSF)**: **G-CSF** (granulocytes), **M-CSF** (monocytes), **GM-CSF**; and also by **interleukins** (IL-3 — general, IL-5 — eosinophils, IL-7 — lymphocytes). They are produced by macrophages, endothelium, T lymphocytes and fibroblasts; their synthesis increases in infection and inflammation. Substances formed when leukocytes break down also enhance leukopoiesis (feedback).

> **CLINICAL SIGNIFICANCE**
> Recombinant **G-CSF** (filgrastim) is used to treat neutropenia after chemotherapy and to collect stem cells from donors.

5. A brief note on immunity

Immunity is the ability of the organism to defend itself against genetically foreign substances and living organisms.

Table 7.4. *Types of immunity*

Type	Factors	Features
Nonspecific (innate) — cellular	Neutrophils, macrophages, NK cells, eosinophils	Fast, against any foreign agent, no memory
Nonspecific — humoral	Complement, lysozyme, interferons, C-reactive protein	—
Specific (acquired) — cellular	T lymphocytes	Against a particular antigen, immunological memory present
Specific — humoral	B lymphocytes → plasma cells → antibodies (IgG, IgM, IgA, IgE, IgD)	Against a particular antigen, memory present

Specific immunity may be **active** (after disease or through vaccination) or **passive** (maternal antibodies via the placenta and milk, ready-made serum).

6. Laboratory work

Experiment 1: counting leukocytes in a Goryaev chamber

1. Put 0.4 ml of 3–5% acetic acid solution (tinted with methylene blue) into a test tube — it hemolyzes the erythrocytes and stains the leukocyte nuclei.
2. Take 0.02 ml of blood from a finger, add it to the tube (dilution 1 : 20) and mix.
3. Fill the Goryaev chamber and wait 1 minute.
4. At low magnification (×8) count the leukocytes in 100 large squares (1600 small squares).
5. Calculation: X = A × 50 (per 1 µl), where A is the number of leukocytes in 100 large squares; or A / 20 × 10^9/l. **Interpretation:** normal 4–9 × 10^9/l.

Experiment 2: preparing a blood smear and determining the leukocyte formula

1. Place a small drop of blood near the edge of a clean, degreased glass slide.
2. Touch the drop with a ground-edge spreader slide at an angle of 45°; once the blood spreads along its edge, draw the smear with one even movement (the smear should be thin, with a “broom-shaped” end).
3. Air-dry the smear and fix it in methyl alcohol for 3–5 minutes.
4. Stain with Romanowsky–Giemsa stain for 20–30 minutes, rinse with water and dry.
5. Using the immersion objective (×90), move along the edges of the smear in a “meander” pattern and count 100 (or 200) leukocytes, classifying them by type.
6. Calculate the percentage of each type and its absolute count: absolute count = total leukocyte count × percentage / 100. **Interpretation:** compare the result with the normal leukogram and determine whether there is a shift to the left.

> **CAUTION**
> Evaluating the leukocyte formula only in percentages can be misleading. For example, even if lymphocytes are 50%, when the total leukocyte count is 3×10^9/l their absolute number (1.5×10^9/l) is normal — this is a **relative lymphocytosis** (in fact, neutropenia). Therefore **absolute counts** should always be calculated as well.

7. Practical skills

Skill: interpreting a leukogram

1. Evaluate the total leukocyte count (leukocytosis, normal, leukopenia).
2. Compare the percentage of each type with the normal range.
3. Calculate absolute counts and distinguish relative from absolute changes.
4. Evaluate the proportion of band forms among the neutrophils and calculate the nuclear shift index.
5. Name the likely condition from the predominant change: neutrophilia + shift to the left — bacterial inflammation; lymphocytosis — viral infection; eosinophilia — allergy or helminths.

Skill: distinguishing leukocytes in a blood smear

1. Look at the shape of the nucleus: segmented (neutrophil), bilobed (eosinophil), indistinct and covered with granules (basophil), bean-shaped (monocyte), large and round (lymphocyte).
2. Evaluate the color of the cytoplasm and its granules: fine pinkish-violet (neutrophil), large orange-red (eosinophil), large dark violet (basophil), grayish-blue (monocyte), a thin blue rim (lymphocyte).
3. Compare the size of the cell with an erythrocyte.

Review questions

1. What are the general properties of leukocytes?
2. What is the normal leukocyte count?
3. List the types of physiological leukocytosis and explain their mechanism.
4. What causes leukopenia?
5. What is the leukocyte formula, and what are its normal values?

6. What are shifts of the leukocyte formula to the left and to the right?
7. Explain the functions of neutrophils.
8. What are the functions of eosinophils, and when is eosinophilia observed?
9. What substances do basophils and mast cells release?
10. What are the functions of monocytes and macrophages?
11. What is the difference between T and B lymphocytes?
12. Explain the stages of phagocytosis.
13. How is leukopoiesis regulated?
14. How do nonspecific and specific immunity differ?
15. Why is a blood test taken on an empty stomach?

Test questions

1. What is the normal leukocyte count in adults?
   A) 1–2 × 10^9/l
   B) 4.0–5.0 × 10^{12}/l
   C) 4–9 × 10^9/l
   D) 15–20 × 10^9/l
   E) 180–320 × 10^9/l

2. Which cells are the most common in the leukocyte formula?
   A) Segmented neutrophils
   B) Lymphocytes
   C) Monocytes
   D) Eosinophils
   E) Basophils

3. To which type does postprandial (alimentary) leukocytosis belong?
   A) Reactive
   B) Leukemoid
   C) Pathological
   D) Infectious
   E) Physiological (redistributive)

4. Which leukocytes increase in helminth infestation?
   A) Neutrophils
   B) Basophils
   C) Monocytes
   D) Eosinophils
   E) Lymphocytes

5. Which leukocytes contain histamine and heparin?
   A) Eosinophils
   B) Basophils
   C) Neutrophils
   D) Lymphocytes
   E) Monocytes

6. Which cells become macrophages in the tissues?
   A) Neutrophils
   B) Lymphocytes
   C) Eosinophils
   D) Monocytes
   E) Basophils

7. From which cells are antibody-producing cells derived?
   A) From B lymphocytes
   B) From T killers
   C) From neutrophils
   D) From basophils
   E) From NK cells

8. What does a shift of the leukocyte formula to the left mean?
   A) An increase in lymphocytes
   B) An increase in eosinophils
   C) Hypersegmentation of segmented neutrophils
   D) An increase in young (band and younger) neutrophils
   E) An increase in monocytes

9. Which cells mainly provide cellular immunity?
   A) B lymphocytes
   B) Erythrocytes
   C) Basophils
   D) Platelets
   E) T lymphocytes

10. Through which enzyme do eosinophils limit an allergic reaction?
    A) Histaminase
    B) Myeloperoxidase
    C) Lysozyme
    D) Carbonic anhydrase
    E) Acetylcholinesterase

11. Which change is typical of viral infections?
    A) Eosinophilia
    B) Neutrophilia with a shift to the left
    C) Lymphocytosis
    D) Basophilia
    E) Thrombocytosis

12. Where do T lymphocytes mature?
    A) In the spleen
    B) In the thymus
    C) In the liver
    D) In the lymph nodes
    E) In the red bone marrow

13. What is the main factor stimulating the formation of granulocytes?
    A) Erythropoietin
    B) Thrombopoietin
    C) Insulin
    D) Aldosterone
    E) G-CSF (granulocyte colony-stimulating factor)

14. What did I.I. Mechnikov call neutrophils?
    A) Macrophages
    B) Microphages
    C) Killers
    D) Helpers
    E) Plasma cells

15. What is the passage of leukocytes through the vessel wall into the tissue called?
    A) Phagocytosis
    B) Chemotaxis
    C) Diapedesis
    D) Degranulation
    E) Opsonization

Clinical cases

Case 1

A 19-year-old man came to the hospital with pain in the right lower abdomen, nausea and a body temperature of 38 °C. Blood test: leukocytes 15.8 × 10^9/l; band neutrophils 14%, segmented 70%, lymphocytes 12%, monocytes 4%.

Questions: Analyze the leukogram. What do these changes indicate?

Answer: Leukocytosis (15.8 × 10^9/l), neutrophilia (84%) and an increase in band neutrophils (14%, normal 1–6%) — a shift of the leukocyte formula to the left. Lymphocytes are relatively decreased. Such a change is typical of acute bacterial inflammation (the clinical picture matches acute appendicitis). Cytokines formed in the focus of inflammation (IL-1, TNF-α) and G-CSF release neutrophils, including young forms, from the bone marrow into the blood and enhance granulopoiesis.

Case 2

A student gave blood after breakfast. The leukocyte count was 10.5 × 10^9/l. When retested on an empty stomach the next day, it was 6.2 × 10^9/l.

Questions: What caused the first result? Why is a blood test taken on an empty stomach?

Answer: This is alimentary (postprandial) physiological leukocytosis. After food intake leukocytes accumulate in the intestinal submucosa and partly enter the circulating blood; this is redistributive in nature and passes within a few hours. To avoid misinterpreting the result (mistaking it for inflammation), the blood test is taken in the morning on an empty stomach, without physical exertion or emotional stress.

Case 3

A patient seen in spring with nasal congestion, sneezing and itchy eyes had a blood test showing leukocytes of 7.0×10^9/l and eosinophils of 12%.

Questions: What change has been found? Explain its mechanism and the role of eosinophils in this condition.

Answer: Eosinophilia (12%, normal 0.5–5%). The clinical picture matches allergic rhinitis (pollinosis). The allergen degranulates mast cells and basophils that carry bound IgE — histamine and other mediators are released. Mast cells release a chemotactic factor for eosinophils, and Th2 lymphocytes enhance eosinophil production through IL-5. Eosinophils release histaminase and arylsulfatase, breaking down histamine and leukotrienes, and so limit the allergic reaction. That is why the eosinophil count rises in allergy.

Quick review

Table 7.5. *Key concepts of topic 7*

Concept	Brief definition
Leukocyte count	$4–9 \times 10^9$/l
Diapedesis	Passage through the capillary wall into the tissue
Physiological leukocytosis	Alimentary, myogenic, emotional, pregnancy — redistributive
Neutrophils	47–72% segmented, 1–6% band; phagocytosis of bacteria (microphages)
Eosinophils	0.5–5%; helminths, limiting allergy (histaminase)
Basophils	0–1%; histamine, heparin, IgE
Monocytes	3–11%; become macrophages, antigen presentation
Lymphocytes	19–37%; specific immunity
T lymphocytes	In the thymus; cellular immunity (killers, helpers)
B lymphocytes	In the bone marrow; plasma cell → antibodies
Shift to the left	Increase in young neutrophils — acute bacterial infection
Shift to the right	Hypersegmentation — B_{12} deficiency
Phagocytosis	Chemotaxis → adhesion → engulfment → phagolysosome → killing
G-CSF	Regulator of granulopoiesis
Mechnikov	Founder of the theory of phagocytosis

TOPIC 8

Platelets

Learning objectives

To teach the student the structure, number, formation and life span of platelets, their role in hemostasis and their other functions (angiotrophic, adhesion, aggregation, secretion, retraction), the platelet factors and the methods for evaluating platelet number and function; to master determining the bleeding time and testing capillary resistance.

By the end of the class the student should know:

- the structure, size and number of platelets;
- thrombocytopoiesis and its regulation (thrombopoietin);
- the receptors of the platelet membrane and the contents of its granules;
- the functions of platelets;
- the stages of vascular-platelet hemostasis;
- the platelet factors and their significance;
- the causes and consequences of thrombocytopenia and thrombocytosis.

The student should be able to:

- count platelets by the Fonio method;
- determine the bleeding time by the Duke method;
- test capillary resistance (tourniquet and pinch tests);
- evaluate clot retraction.

1. Structure and number of platelets

Platelets (thrombocytes) are anucleate, disc-shaped cell fragments **2–4 µm** in diameter. They are shed from the cytoplasm of the giant bone marrow cells — **megakaryocytes**.

Number: 180–320 × 10^9/l in adults (150–400 × 10^9/l in international practice). Their life span is **7–10 days**; aged platelets are destroyed in the spleen and liver. About 1/3 of platelets are stored **in the spleen**.

- **Thrombocytopenia** is a decrease below 150 (180) × 10^9/l. Below 50 × 10^9/l bleeding after injury increases, and below 20 × 10^9/l spontaneous hemorrhages appear (petechiae, nosebleeds).
- **Thrombocytosis** is an increase above 400 × 10^9/l: physiological (physical work, after meals, pregnancy — redistributive), reactive (after blood loss, inflammation, after splenectomy) or neoplastic (essential thrombocythemia). The risk of thrombosis increases.

Structure

A platelet has a peripheral **hyalomere** (a light part containing microtubules and the contractile protein **thrombosthenin** — actomyosin) and a central **granulomere** (granules). The membrane has an **open canalicular system**, through which the contents of the granules are released.

Table 8.1. *Glycoproteins (receptors) of the platelet membrane*

Receptor	Ligand	Function
GPIb-IX-V	Von Willebrand factor	Adhesion — sticking to the damaged subendothelium

Receptor	Ligand	Function
GPIa-IIa, GPVI	Collagen	Adhesion and activation
GPIIb-IIIa	Fibrinogen, von Willebrand factor	Aggregation — linking platelets to one another
$P2Y_{12}$	ADP	Activation and aggregation
PAR-1	Thrombin	Strong activation

Table 8.2. *Platelet granules*

Granules	Composition
α granules	Fibrinogen, von Willebrand factor, factor V, platelet-derived growth factor (PDGF), PF4 (antiheparin factor), β-thromboglobulin, fibronectin
Dense (δ) granules	ADP, ATP, serotonin, Ca^{2+}, epinephrine, histamine
Lysosomes	Hydrolytic enzymes

2. Thrombocytopoiesis and its regulation

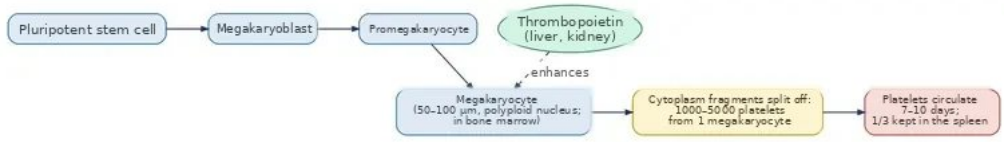


Figure 8.1. *Scheme of thrombocytopoiesis*

The main regulator of thrombocytopoiesis is **thrombopoietin**; it is synthesized mainly in the **liver** at a constant rate. Thrombopoietin binds to receptors on the membranes of platelets and megakaryocytes and is "captured" and destroyed by them. Therefore, when platelets decrease, free thrombopoietin increases and thrombocytopoiesis intensifies; when they increase, it weakens (feedback). IL-3, IL-6 and IL-11 also stimulate thrombocytopoiesis.

> **CLINICAL SIGNIFICANCE**
> **In liver cirrhosis** thrombopoietin synthesis decreases and the enlarged spleen (hypersplenism) "traps" many platelets — thrombocytopenia develops and a bleeding tendency appears. Thrombopoietin receptor agonists (eltrombopag, romiplostim) are used to treat immune thrombocytopenia.

3. Functions of platelets

Table 8.3. *Functions of platelets*

Function	Content
Angiotrophic	They nourish endothelial cells and maintain the integrity of the vessel wall; every day ≈ 15% of platelets merge with the endothelium for this purpose. In thrombocytopenia the capillaries become fragile
Adhesion	Sticking to the damaged vessel wall
Aggregation	Sticking to one another to form a platelet plug
Secretion	Release of the active substances in the granules
Coagulation	Concentrating plasma clotting factors on their surface (the phospholipid matrix — PF3) and speeding up coagulation

Function	Content
Retraction	Compacting the clot and drawing the vessel edges together through contraction of thrombosthenin
Vascular tone	Release of serotonin and thromboxane A_2 to constrict the damaged vessel
Regeneration	PDGF and other growth factors speed up wound healing
Transport and protection	Transport of serotonin and histamine; binding of immune complexes

4. Vascular-platelet (primary) hemostasis

Vascular-platelet hemostasis stops bleeding from microcirculatory vessels (diameter up to 100 μm, low pressure) within 1–3 minutes. It consists of the following stages:

1. **Reflex spasm** — contraction of the damaged vessel (a pain reflex, then the action of serotonin, thromboxane A_2 and epinephrine released from platelets and endothelin from the endothelium).
2. **Adhesion** — subendothelial collagen is exposed; von Willebrand factor binds to collagen and links to the **GPIb** receptor of platelets.
3. **Activation and secretion** — platelets change shape (put out processes), release ADP, serotonin and Ca^{2+} and synthesize **thromboxane A_2** (TxA_2) from arachidonic acid.
4. **Reversible aggregation** — under the action of ADP and TxA_2 the **GPIIb-IIIa** receptors are activated, and platelets stick to one another through fibrinogen “bridges”.
5. **Irreversible aggregation** — under the action of thrombin the platelet membranes break down and the platelets turn into a solid mass; a white (platelet) thrombus is formed.
6. **Retraction** — thrombosthenin contracts and compacts the thrombus.

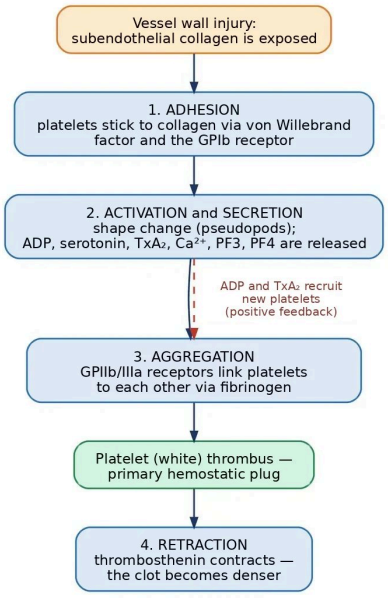


Figure 8.2. *Functions of platelets in hemostasis*

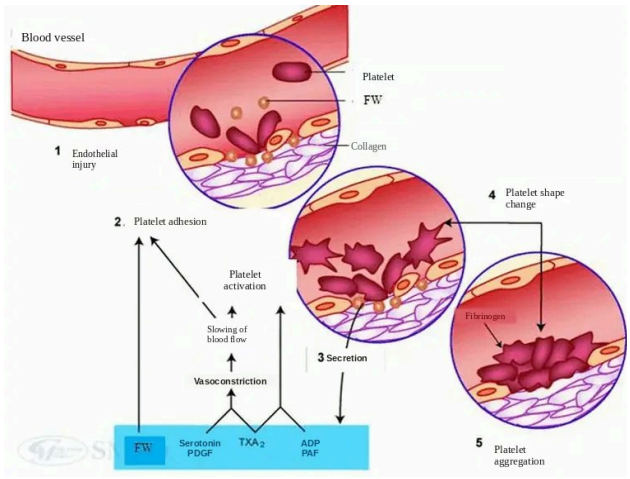


Figure 8.3. *Stages of vascular-platelet hemostasis: vascular spasm, platelet adhesion and aggregation (Alavia O.T., 2018)*

REMEMBER

Normally the endothelium **prevents** platelets from sticking: it produces **prostacyclin (PGI_2)** and **nitric oxide (NO)**, which inhibit aggregation and dilate the vessel. Platelet **thromboxane A_2**, on the contrary, enhances aggregation and vasoconstriction. The balance of these two substances determines the state of the platelets.

CLINICAL SIGNIFICANCE

Aspirin (acetylsalicylic acid) in a low dose (75–100 mg) irreversibly inhibits cyclooxygenase in platelets and stops the synthesis of thromboxane A_2. Because platelets have no nucleus, the enzyme is not resynthesized — the effect lasts for the rest of the platelet's life (7–10 days). That is why aspirin is used to prevent myocardial infarction and stroke and is stopped 5–7 days before elective surgery. **Clopidogrel** blocks the $P2Y_{12}$ (ADP) receptors.

Platelet factors

Table 8.4. *The main platelet factors*

Factor	Function
PF3 (phospholipid factor)	Phospholipid matrix for complexes of clotting factors (formation of prothrombinase)
PF4 (antiheparin factor)	Neutralizes heparin
PF5 (fibrinogen)	Aggregation and clot formation
PF6 (thrombosthenin)	Clot retraction
PF10 (serotonin)	Constricts the vessel
PF11 (thromboxane A_2)	Aggregation and vasoconstriction
ADP	Enhances aggregation

MEMORY AID

The sequence of platelet action — **“Stick–Activate–Aggregate–Retract”**: **S**ticking (adhesion) → **A**ctivation → **A**ggregation → **R**etraction.

5. Evaluating platelet number and function

Table 8.5. *Tests for evaluating vascular-platelet hemostasis*

Test	Normal value	What it shows
Platelet count	180–320 × 10^9/l	Thrombocytopenia or thrombocytosis
Bleeding time (Duke method)	2–4 minutes	Platelet number and function, state of the vessel wall
Bleeding time (Ivy method)	4–8 minutes	The same, standardized
Capillary resistance (tourniquet test — Rumpel–Leede–Konchalovsky)	≤ 10 petechiae in a circle 2.5 cm in diameter	Fragility of the vessel wall, thrombocytopenia
Pinch test	No hemorrhage at the pinched site	Capillary fragility
Clot retraction	44–65% in 1 hour (retraction index)	Platelet count and thrombosthenin activity
Platelet aggregation (aggregometry)	Response to an inducer (ADP, collagen, epinephrine)	Disorders of platelet function (thrombocytopathies)

CAUTION

Clotting time and **bleeding time** are two different parameters. The bleeding time evaluates **vascular-platelet** hemostasis, while the clotting time (5–10 minutes by Lee–White) evaluates **coagulation** hemostasis. In hemophilia the bleeding time is normal but the clotting time is prolonged; in thrombocytopenia, the reverse — the bleeding time is prolonged and the clotting time is normal.

6. Laboratory work

Experiment 1: counting platelets by the Fonio method

1. Wipe the skin of a finger with alcohol and place a drop of 14% magnesium sulfate solution on it (to prevent platelet aggregation).
2. Puncture the finger through this drop; the blood that emerges mixes with the magnesium sulfate.
3. Make a smear from the mixture, fix it and stain with Romanowsky–Giemsa stain.
4. Using the immersion objective, count 1000 erythrocytes and the platelets in the same fields of view.
5. Calculation: platelet count ($\times 10^9$/l) = erythrocyte count ($\times 10^{12}$/l) × the number of platelets per 1000 erythrocytes. **Example:** erythrocytes 4.5×10^{12}/l, 55 platelets per 1000 erythrocytes → $4.5 \times 55 = 247.5 \times 10^9$/l.

CAUTION

Today platelets are counted in automated hematology analyzers. In blood collected with EDTA, platelets sometimes aggregate and the analyzer shows them as low — **pseudothrombocytopenia**. If the result is doubtful, a smear should be examined or the sample recollected with citrate.

Experiment 2: determining the bleeding time by the Duke method

1. Wipe the earlobe with alcohol and let it dry.
2. Puncture it with a scarifier to a depth of 3–4 mm and start a stopwatch.
3. Every 30 seconds blot the emerging drop of blood with filter paper (without touching the skin).
4. Note the time when no new spot is left on the paper — the time when bleeding stopped. **Interpretation:** normally 2–4 minutes. Prolongation may be due to thrombocytopenia, thrombocytopathy, von Willebrand disease or aspirin intake.

Experiment 3: testing capillary resistance (tourniquet test)

1. Put a blood pressure cuff on the upper arm and keep the pressure midway between systolic and diastolic pressure (≈ 90–100 mm Hg) for 5 minutes.
2. Remove the cuff and, after 5 minutes, draw a circle 2.5 cm in diameter below the elbow crease.
3. Count the petechiae inside the circle. **Interpretation:** up to 10 petechiae — normal; 10–20 — doubtful; more than 20 — capillary fragility (thrombocytopenia, vasculitis, vitamin C deficiency).

Experiment 4: evaluating clot retraction

1. Take 5 ml of venous blood into a graduated test tube and place it in a thermostat at 37 °C.
2. After 1 hour separate the clot and measure the volume of the expressed serum.
3. Retraction index = serum volume / blood volume × 100%. **Interpretation:** normally 44–65%. Retraction is impaired in thrombocytopenia and Glanzmann thrombasthenia.

7. Practical skills

Skill: determining the type of hemostatic disorder in a hemorrhagic syndrome

1. Evaluate the type of bleeding: petechiae, ecchymoses, bleeding from mucous membranes (platelet type) or hemorrhages into joints and muscles (hematoma type — coagulation).
2. Evaluate the platelet count.
3. Compare the bleeding time and the clotting time.
4. Add the result of the tourniquet test.
5. If the bleeding time is prolonged and the clotting time is normal, conclude that vascular-platelet hemostasis is disturbed; if the reverse, that coagulation hemostasis is disturbed.

Skill: evaluating antiplatelet therapy and surgical risk

1. Find out whether the patient takes aspirin or clopidogrel.
2. Take into account that the drug effect persists for the life span of the platelets (7–10 days).
3. Agree with the physician on when to stop the drug before elective surgery.
4. Check the bleeding time.

Review questions

1. What are the structure and size of platelets?
2. What are the normal platelet count and life span?
3. How does thrombocytopoiesis proceed, and how is it regulated?
4. List the main receptors of the platelet membrane and their functions.
5. What substances do the platelet granules contain?
6. What is the angiotrophic function of platelets?
7. Explain the stages of vascular-platelet hemostasis.
8. What is the role of von Willebrand factor in adhesion?
9. Which substances enhance aggregation?
10. Explain the opposite effects of prostacyclin and thromboxane A_2.
11. How does aspirin act on platelets?
12. List the platelet factors.
13. What is clot retraction, and which protein provides it?
14. How is the bleeding time determined, and what does it show?
15. What causes thrombocytopenia and thrombocytosis?

Test questions

1. What is the normal platelet count in adults?
   A) $4–9 \times 10^9$/l
   B) $4.0–5.0 \times 10^{12}$/l
   C) $50–100 \times 10^9$/l
   D) $180–320 \times 10^9$/l
   E) $500–800 \times 10^9$/l

2. From which cells are platelets formed?
   A) From monocytes
   B) From lymphocytes
   C) From erythroblasts
   D) From plasma cells
   E) From megakaryocytes

3. What is the life span of platelets?
   A) 120 days
   B) 1–2 days
   C) 7–10 days
   D) 30 days
   E) 1 year

4. What is the main regulator of thrombocytopoiesis?
   A) Thrombopoietin
   B) Erythropoietin
   C) G-CSF
   D) Insulin
   E) Aldosterone

5. What is the sticking of platelets to a damaged vessel wall called?
   A) Aggregation
   B) Adhesion
   C) Retraction
   D) Secretion
   E) Diapedesis

6. Which factor links the platelet to the subendothelium in adhesion?
   A) Fibrinogen
   B) Thrombin
   C) Prostacyclin
   D) Heparin
   E) Von Willebrand factor

7. Which receptor links platelets to one another in aggregation?
   A) GPIb
   B) GPIIb-IIIa
   C) PAR-1
   D) GPVI
   E) IgE receptor

8. Which endothelial substance inhibits platelet aggregation?
   A) Thromboxane A_2
   B) ADP
   C) Serotonin
   D) Endothelin
   E) Prostacyclin (PGI_2)

9. The antiplatelet effect of aspirin is based on stopping the synthesis of which substance?
    A) Thromboxane A_2
    B) Prostacyclin
    C) ADP
    D) Serotonin
    E) Fibrinogen

10. Which contractile protein provides clot retraction?
    A) Fibrin
    B) Actin
    C) Thrombosthenin
    D) Calmodulin
    E) Myosin

11. What is the normal bleeding time by the Duke method?
    A) 10–15 minutes
    B) 30 seconds
    C) 20–30 minutes
    D) 2–4 minutes
    E) 1 hour

12. What is the angiotrophic function of platelets?
    A) Phagocytosis
    B) Nourishing the endothelium and maintaining the integrity of the vessel wall
    C) Producing antibodies
    D) Carrying oxygen
    E) Releasing histamine

13. Which platelet factor neutralizes heparin?
    A) PF3
    B) PF6
    C) PF4
    D) PF10
    E) PF11

14. If the bleeding time is prolonged and the clotting time is normal, which hemostasis is disturbed?
    A) Coagulation hemostasis
    B) Fibrinolysis
    C) The anticoagulant system
    D) Vascular-platelet hemostasis
    E) None

15. What proportion of platelets is stored in the spleen?
    A) ≈ 1/3
    B) Almost all
    C) More than half
    D) They are not stored
    E) ≈ 1/10

Clinical cases

Case 1

Two weeks after influenza an 8-year-old child developed small pinpoint hemorrhages in the skin (petechiae), bruises and nosebleeds. Platelets 18 × 10^9/l, bleeding time 12 minutes, clotting time 7 minutes.

Questions: Which link of hemostasis is disturbed? Why did the bleeding time and clotting time change differently?

Answer: Vascular-platelet (primary) hemostasis is disturbed — severe thrombocytopenia (18 × 10^9/l), probably immune thrombocytopenia (antibodies to platelets formed after a viral infection). Because there are few platelets, adhesion and aggregation are insufficient and no platelet plug forms — the bleeding time is prolonged. The angiotrophic function is also impaired and the capillaries become fragile — petechiae appear. The clotting time depends on the plasma clotting factors; since they are normal, the clotting time has not changed.

Case 2

A patient with coronary heart disease takes 100 mg of aspirin daily. An elective tooth extraction has been scheduled. The dentist advised stopping aspirin 7 days beforehand.

Questions: How does aspirin act on platelets? Why should it be stopped a week before the operation?

Answer: Aspirin irreversibly acetylates cyclooxygenase in platelets and stops the formation of thromboxane A_2 from arachidonic acid. Thromboxane A_2 is a powerful inducer of platelet aggregation and vasoconstriction; without it the platelet plug forms slowly and the bleeding time is prolonged. Because platelets have no nucleus, they cannot synthesize new enzyme — the effect lasts for the whole life of each platelet (7–10 days). After aspirin is stopped, most platelets are renewed within 5–7 days and hemostasis is restored. (In patients at high cardiac risk, the decision to stop aspirin is made by a cardiologist.)

Case 3

A patient with liver cirrhosis has an enlarged spleen and platelets of 65 × 10^9/l. He complains of bleeding gums.

Questions: Explain the mechanisms of the thrombocytopenia.

Answer: There are two main mechanisms. First, thrombopoietin is synthesized mainly in the liver; in cirrhosis its production decreases and the formation of platelets from megakaryocytes weakens. Second, the pressure in the portal vein rises and the spleen enlarges (splenomegaly, hypersplenism); normally 1/3 of the platelets are in the spleen, but in an enlarged spleen up to 80–90% are trapped and destroyed. In addition, in cirrhosis the liver also synthesizes fewer clotting factors, which further increases bleeding.

Quick review

Table 8.6. *Key concepts of topic 8*

Concept	Brief definition
Platelets	Anucleate, 2–4 µm; 180–320 × 10^9/l; live 7–10 days
Megakaryocyte	A giant bone marrow cell; the source of platelets

Concept	Brief definition
Thrombopoietin	Synthesized in the liver; regulator of thrombocytopoiesis
GPIb	Adhesion via von Willebrand factor
GPIIb-IIIa	Aggregation via fibrinogen
α granules	Fibrinogen, vWF, PDGF, PF4
Dense granules	ADP, serotonin, Ca^{2+}
Thromboxane A_2	Aggregation and vasoconstriction; stopped by aspirin
Prostacyclin, NO	In the endothelium; inhibits aggregation
Angiotrophic function	Nourishing the endothelium, capillary integrity
Retraction	Thrombosthenin; compacting the clot
PF3	Phospholipid matrix for coagulation
PF4	Antiheparin factor
Duke method	Bleeding time 2–4 minutes
Tourniquet test	≤ 10 petechiae — normal

TOPIC 9

Hemostasis. Fibrinolysis

Learning objectives

To teach the student the hemostatic system and its components, the plasma clotting factors, the stages and mechanisms of coagulation hemostasis (blood coagulation), the anticoagulant system, and fibrinolysis and its regulation; to master determining the clotting time and evaluating the parameters of the coagulogram.

By the end of the class the student should know:

- the components of the hemostatic system and the types of hemostasis;
- the plasma clotting factors and their numbering;
- the extrinsic, intrinsic and common pathways and the stages of blood coagulation;
- the significance of vitamin K;
- the primary and secondary anticoagulants;
- the stages of fibrinolysis, its activators and inhibitors;
- the role of nervous and humoral factors in the regulation of hemostasis;
- the main parameters of the coagulogram.

The student should be able to:

- determine the clotting time by the Lee–White method;
- determine the clotting time of capillary blood by the Sukharev method;
- interpret the results of the prothrombin time, INR and APTT.

1. The hemostatic system

Hemostasis is the system that stops bleeding when blood vessels are damaged and at the same time keeps the blood in a liquid state inside the vessels. The hemostatic system includes the **vessel wall** (endothelium), **blood cells** (mainly platelets), **plasma clotting factors**, **anticoagulants** and the **fibrinolytic system**.

Table 9.1. *Types of hemostasis*

Type	Where it predominates	Result	Duration
Vascular-platelet (primary)	Small vessels (capillaries, venules, arterioles)	Platelet (white) thrombus	1–3 minutes
Coagulation (secondary)	Large vessels; all vessels for a firm thrombus	Fibrin clot (red thrombus)	5–10 minutes

The two mechanisms are closely interrelated: platelets concentrate clotting factors on their surface (PF3), and thrombin activates platelets.

2. Plasma clotting factors

According to the international nomenclature, clotting factors are designated by **Roman numerals** (in the order of their discovery). The active form is indicated by the letter “a” (for example, Xa).

Table 9.2. *Plasma clotting factors*

Factor	Name	Features
I	Fibrinogen	In the liver; converted into fibrin
II	Prothrombin	In the liver, vitamin K-dependent; converted into thrombin
III	Tissue factor (thromboplastin)	In damaged tissue and endothelium; initiates the extrinsic pathway
IV	Calcium ions (Ca^{2+})	Required at almost all stages
V	Proaccelerin	Cofactor of the prothrombinase complex
VII	Proconvertin	Vitamin K-dependent; extrinsic pathway
VIII	Antihemophilic globulin A	Cofactor of the tenase complex; its deficiency — hemophilia A
IX	Christmas factor	Vitamin K-dependent; its deficiency — hemophilia B
X	Stuart–Prower factor	Vitamin K-dependent; the start of the common pathway
XI	Plasma thromboplastin antecedent	Intrinsic pathway
XII	Hageman factor	Contact activation, initiates the intrinsic pathway
XIII	Fibrin-stabilizing factor (fibrinase)	Makes fibrin insoluble

(Factor VI was removed from the list because it turned out to be activated factor V.) In addition, **prekallikrein** (Fletcher factor) and **high-molecular-weight kininogen** (Fitzgerald factor) also take part in contact activation.

REMEMBER

Vitamin K-dependent factors: II, VII, IX, X (and also the anticoagulants protein C and S). In the liver vitamin K γ-carboxylates these factors, giving them the ability to bind Ca^{2+}. If vitamin K is deficient or its antagonists (warfarin) are taken, the factors are inactive and blood coagulation slows down.

MEMORY AID

Vitamin K-dependent factors: **“1972” — 10, 9, 7, 2** (X, IX, VII, II).

3. Coagulation hemostasis (blood coagulation)

Blood coagulation is an enzymatic **cascade**: each activated factor activates the next, and the signal is amplified at every step. In the classical scheme blood coagulation proceeds in three stages:

1. **Formation of prothrombinase** (the longest stage, 5–7 minutes) — through the extrinsic and intrinsic pathways.
2. **Formation of thrombin** — under the action of prothrombinase, prothrombin (II) is converted into thrombin (IIa) (2–5 seconds).
3. **Formation of fibrin** — thrombin cleaves peptides from fibrinogen and the fibrin monomers polymerize; factor XIIIa links them covalently, turning them into insoluble fibrin (2–5 seconds).

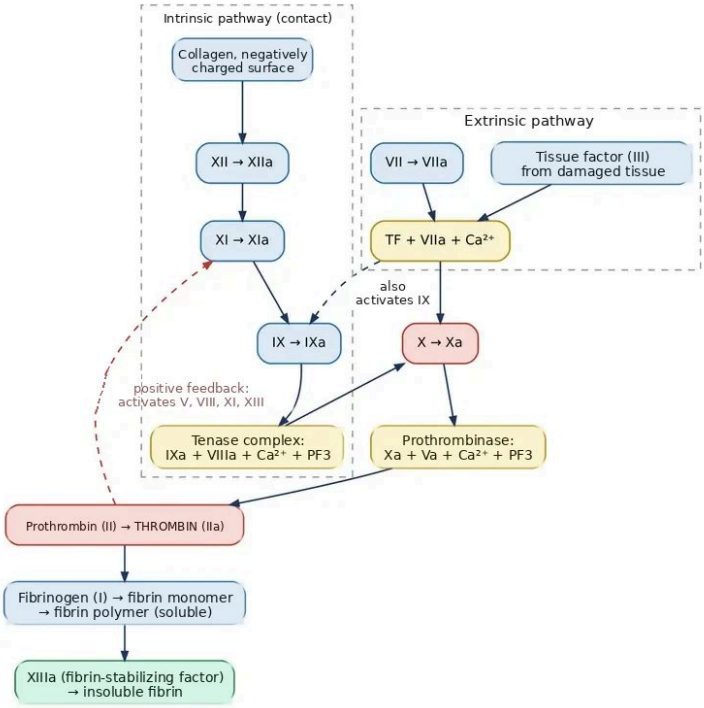


Figure 9.1. *The coagulation cascade: extrinsic, intrinsic and common pathways*

Table 9.3. *Extrinsic and intrinsic pathways of blood coagulation*

Feature	Extrinsic pathway	Intrinsic pathway
Initiating factor	Tissue factor (III) — from damaged tissue	Contact of factor XII with a negatively charged surface (collagen, glass)
Factors involved	III, VII, Ca^{2+}	XII, XI, IX, VIII, Ca^{2+}, PF3
Speed	Fast (10–15 seconds)	Slow (5–10 minutes)
Laboratory test	Prothrombin time (PT), INR	Activated partial thromboplastin time (APTT)

CAUTION

The classical scheme of “extrinsic and intrinsic pathways” is convenient for understanding laboratory tests, but in the body they do not work separately. According to the modern **cell-based model**, coagulation **is initiated** by the tissue factor–VIIa complex (a small amount of thrombin), then thrombin **amplifies** the process by activating factors V, VIII and XI and the platelets, and a large amount of thrombin is formed on the surface of activated platelets (**propagation**). That is why factor XII deficiency does not cause bleeding, whereas deficiency of factors VIII and IX causes severe bleeding (hemophilia).

Functions of thrombin

Thrombin is the central enzyme of hemostasis: it converts fibrinogen into fibrin; activates factors V, VIII, XI and XIII (positive feedback); strongly activates platelets; and, by binding to thrombomodulin on the endothelium, activates **protein C** (an anticoagulant effect — negative feedback).

Retraction

After the clot has formed, it is compacted by the contraction of platelet thrombosthenin and serum is squeezed out of it (**retraction**). This draws the edges of the vessel together and makes the thrombus firmer.

CLINICAL SIGNIFICANCE

Hemophilia A (factor VIII deficiency, 80–85%) and **hemophilia B** (factor IX deficiency) are X-linked recessive hereditary diseases; they occur mainly in boys, with mothers as carriers. Hemorrhages into the joints (hemarthrosis), hematomas in the muscles and prolonged bleeding after injury are typical. The APTT is prolonged, while the prothrombin time and bleeding time are normal. Treatment: concentrates of the missing factor.

4. The anticoagulant system

Factors that prevent blood from clotting in the vessels: **the smooth, negatively charged surface of the endothelium**, with its glycocalyx and heparan sulfate; prostacyclin and NO; the speed of blood flow (which dilutes and carries away active factors); the liver, which clears activated factors; and **anticoagulants**.

Table 9.4. *The main anticoagulants*

Group	Substance	Mechanism of action
Primary (always present in plasma)	Antithrombin III	Inactivates thrombin, Xa, IXa, XIa and XIIa; heparin enhances its action 1000-fold
Primary	Heparin	In mast cells and basophils; a cofactor of antithrombin III
Primary	Protein C (activated) + protein S	Breaks down Va and VIIIa
Primary	Tissue factor pathway inhibitor (TFPI)	Blocks the TF–VIIa–Xa complex
Primary	α_2-Macroglobulin	Binds thrombin and other proteases
Secondary (formed during coagulation and fibrinolysis)	Fibrin (antithrombin I)	Adsorbs thrombin
Secondary	Fibrin(ogen) degradation products	Inhibit fibrin polymerization and platelet aggregation

CLINICAL SIGNIFICANCE

The Leiden mutation (factor V Leiden) is the most common hereditary thrombophilia: the altered factor V is not broken down by activated protein C and the risk of venous thrombosis increases. Deficiency of **antithrombin III**, protein C and protein S also predisposes to thrombosis.

REMEMBER

Anticoagulants used in clinical practice: **heparin** (acts through antithrombin III; fast-acting, given intravenously or subcutaneously; monitored by the APTT), **warfarin** (a vitamin K antagonist; reduces the synthesis of II, VII, IX and X; its effect begins in 3–5 days; monitored by the INR), and new oral anticoagulants (rivaroxaban, apixaban — Xa inhibitors; dabigatran — a thrombin inhibitor). To prevent blood from clotting in a test tube, **sodium citrate** or EDTA (which bind Ca^{2+}) are used.

5. Fibrinolysis

Fibrinolysis is the enzymatic dissolution of a fibrin clot; it restores the patency of the vessel. The main enzyme is **plasmin**, which is formed from inactive **plasminogen**.

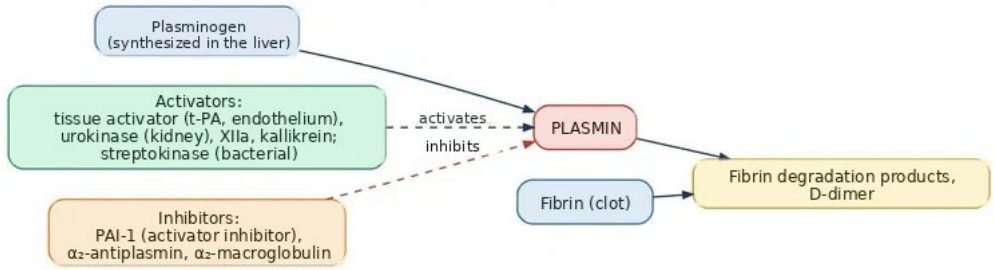


Figure 9.2. *The fibrinolytic system*

Table 9.5. *Stages of fibrinolysis*

Stage	Process
1. Formation of activators	Tissue plasminogen activator (t-PA) is released from the endothelium; urokinase is formed in the kidney; intrinsic route: XIIa, kallikrein
2. Formation of plasmin	Plasminogen binds t-PA on the fibrin surface and is converted into plasmin (in the presence of fibrin the activity of t-PA increases 100-fold)
3. Breakdown of fibrin	Plasmin splits fibrin into soluble degradation products (including D-dimer)

Inhibitors of fibrinolysis: PAI-1 (plasminogen activator inhibitor — from the endothelium and platelets), α_2-antiplasmin (rapidly inactivates free plasmin), α_2-macroglobulin.

CLINICAL SIGNIFICANCE

Thrombolytic therapy — in acute myocardial infarction, ischemic stroke and massive pulmonary embolism, t-PA (alteplase) or streptokinase is given to dissolve the thrombi; to be effective it must be used as soon as possible (within hours) after the onset of disease. An increase of **D-dimer** in the blood shows that fibrin is being formed and broken down in the body — it is used to rule out deep vein thrombosis and pulmonary embolism. Aminocaproic and tranexamic acids inhibit fibrinolysis and reduce bleeding.

6. Regulation of hemostasis

- The **sympathetic nervous system** and **adrenaline** (stress, pain, fear) accelerate blood coagulation (**stress enhancement of hemocoagulation**) — an evolutionary defense that reduces blood loss from wounds. At the same time fibrinolysis is also enhanced.
- The **parasympathetic nervous system** slightly slows coagulation.
- **Cortisol and vasopressin** accelerate coagulation; at the end of pregnancy the clotting factors increase (preparation for blood loss in childbirth).
- The cerebral cortex influences the rate of coagulation through conditioned reflexes.

7. Laboratory work

Exercise 1: determining the venous blood clotting time by the Lee–White method

1. Take 1 ml of venous blood into a clean, dry glass test tube and start the stopwatch.
2. Place the tube in a 37 °C water bath.
3. Tilt the tube by 45° every 30 seconds.
4. Note the time when the blood stops flowing (the clot does not fall out when the tube is inverted). **Interpretation:** normally 5–10 minutes. Prolongation — clotting factor deficiency (hemophilia), heparin therapy; shortening — hypercoagulation.

Exercise 2: determining the capillary blood clotting time by the Sukharev method

1. Wipe away the first drop from the finger with cotton wool.
2. Draw blood to a height of 25–30 mm into a Panchenkov capillary (or Sukharev capillary) and start the stopwatch.
3. Tilt the capillary right and left every 30 seconds.
4. The time when the blood column stops moving is the start of clotting; the appearance of fibrin threads in the blood is the end of clotting. **Interpretation:** start of clotting 30 seconds – 2 minutes, end 3–5 minutes.

Exercise 3: interpreting coagulogram results (case-based)

Compare the coagulogram results given by the instructor with the table below.

Table 9.6. *Main coagulogram parameters*

Parameter	Normal value	What it evaluates
Prothrombin time (PT)	11–15 s	Extrinsic and common pathways (VII, X, V, II, I)
INR (international normalized ratio)	0.8–1.2 (target 2.0–3.0 during warfarin therapy)	Standardized PT
Prothrombin index	80–105 %	A parameter used in CIS practice
APTT	25–38 s	Intrinsic and common pathways (XII, XI, IX, VIII, X, V, II, I)
Thrombin time	14–20 s	Conversion of fibrinogen into fibrin
Fibrinogen	2–4 g/l	Clotting substrate; acute-phase protein
D-dimer	< 0.5 µg/ml	Fibrinolytic activity

8. Practical skills

Skill: determining the level of the disorder from the coagulogram

1. Evaluate the platelet count and bleeding time (primary hemostasis).
2. Evaluate the PT (INR): only PT prolonged — factor VII deficiency, warfarin effect, early vitamin K deficiency.
3. Evaluate the APTT: only APTT prolonged — deficiency of factors VIII, IX, XI (hemophilia), heparin effect.
4. If both are prolonged — deficiency of common pathway factors (X, V, II, fibrinogen), liver diseases, DIC syndrome.
5. Additionally evaluate the state of fibrinolysis by D-dimer and fibrinogen.

Skill: monitoring anticoagulant therapy

1. Find out which drug the patient is taking.
2. For warfarin, monitor the INR (target usually 2.0–3.0).
3. For (unfractionated) heparin, monitor the APTT (it should be prolonged 1.5–2.5 times above normal).
4. Watch for signs of bleeding; in warfarin overdose the antidote is vitamin K, for heparin — protamine sulfate.

Review questions

1. What parts make up the hemostatic system?
2. What is the difference between vascular-platelet and coagulation hemostasis?
3. List the plasma clotting factors.
4. Which factors depend on vitamin K, and why?
5. Explain the stages of blood coagulation.
6. How do the extrinsic and intrinsic pathways differ?
7. List the functions of thrombin.
8. What does the cell-based model of blood coagulation explain?
9. What are the cause and laboratory signs of hemophilia A and B?
10. Which factors prevent blood from clotting inside the vessels?
11. List the primary and secondary anticoagulants.
12. What is the mechanism of action of heparin and warfarin?
13. Explain the stages of fibrinolysis.
14. What is the clinical significance of D-dimer?
15. How does stress affect blood coagulation?

Test questions

1. Which factor initiates the extrinsic pathway?
   A) Factor XII
   B) Tissue factor (III)
   C) Factor VIII
   D) Fibrinogen
   E) Factor XIII

2. Which clotting factors are vitamin K-dependent?
   A) I, V, VIII, XIII
   B) III, IV, XI, XII
   C) V, VIII, XI, XII
   D) I, II, III, IV
   E) II, VII, IX, X

3. Hemophilia A is associated with deficiency of which factor?
   A) Factor VIII
   B) Factor IX
   C) Factor XI
   D) Factor VII
   E) Factor XII

4. Which factor makes fibrin insoluble?
   A) Thrombin
   B) Plasmin
   C) Factor V
   D) Factor XIII
   E) Factor X

5. What is the composition of the prothrombinase complex?
   A) IXa + VIIIa + Ca^{2+}
   B) TF + VIIa + Ca^{2+}
   C) Xa + Va + Ca^{2+} + phospholipid
   D) XIIa + XIa
   E) Thrombin + thrombomodulin

6. Heparin acts through which anticoagulant?
   A) Antithrombin III
   B) Protein C
   C) Fibrin
   D) Plasmin
   E) α_2-antiplasmin

7. What is the main enzyme of fibrinolysis?
   A) Thrombin
   B) Urokinase
   C) Kallikrein
   D) Antithrombin III
   E) Plasmin

8. Which parameter is used to monitor warfarin therapy?
   A) APTT
   B) INR
   C) Bleeding time
   D) Platelet count
   E) D-dimer

9. How does sodium citrate prevent blood from clotting in a test tube?
   A) It blocks thrombin
   B) It breaks down fibrinogen
   C) It binds Ca^{2+} ions
   D) It activates antithrombin III
   E) It destroys platelets

10. Which factors does activated protein C break down?
    A) II and X
    B) VII and IX
    C) I and XIII
    D) Va and VIIIa
    E) XI and XII

11. What is the normal venous blood clotting time by the Lee–White method?
    A) 30 seconds
    B) 5–10 minutes
    C) 20–30 minutes
    D) 1 hour
    E) 2–4 minutes

12. Which pathway does the APTT evaluate?
    A) Only the extrinsic pathway
    B) Fibrinolysis
    C) The intrinsic and common pathways
    D) Vascular-platelet hemostasis
    E) Only fibrinogen

13. Which plasminogen activator is released from the endothelium?
    A) Streptokinase
    B) Urokinase
    C) PAI-1
    D) α_2-antiplasmin
    E) Tissue plasminogen activator (t-PA)

14. Which parameter is prolonged in a patient with hemophilia?
    A) APTT
    B) Bleeding time
    C) Prothrombin time
    D) Reduced platelet count
    E) INR

15. How do stress and adrenaline affect blood coagulation?
    A) They stop coagulation
    B) They have no effect
    C) They only weaken fibrinolysis
    D) They accelerate coagulation
    E) They reduce platelets

Clinical cases

Case 1

After a fall, a 6-year-old boy has a swollen, painful knee joint (bleeding into the joint). Earlier he had prolonged bleeding when his teeth erupted. His mother's brother has the same disease. Platelets 250 × 10^9/l, bleeding time 3 minutes, prothrombin time 13 s, APTT 78 s.

Questions: What disease can be suspected? Explain the laboratory results. How is the disease inherited?

Answer: Hemophilia (probably A — factor VIII deficiency). The platelet count and bleeding time are normal — vascular-platelet hemostasis is not disturbed. The prothrombin time is normal — the extrinsic pathway is preserved. The APTT is sharply prolonged — deficiency of intrinsic pathway factors (VIII or IX). Bleeding into joints and muscles (hematoma type) is typical of a coagulation hemostasis disorder. The disease is inherited in an X-linked recessive manner: the mother is a carrier and boys are affected (the mother's brother is also ill). The exact type is determined by measuring factor VIII and IX activity.

Case 2

A patient with a mechanical heart valve prosthesis takes warfarin. After starting an antibiotic his gums began to bleed and the INR rose to 5.8.

Questions: Explain the mechanism of action of warfarin. Why did the INR rise, and what help is given?

Answer: Warfarin is a vitamin K antagonist: in the liver it stops the γ-carboxylation of the vitamin K-dependent factors (II, VII, IX, X); they cannot bind Ca^{2+} and become inactive — blood coagulation slows down. The intestinal microflora synthesizes part of vitamin K_2; because the antibiotic suppressed the microflora (and affected warfarin metabolism), vitamin K decreased and the effect of warfarin increased — the INR exceeded the therapeutic range (2.5–3.5) and bleeding appeared. Help: stop or reduce the warfarin dose, give vitamin K, and in severe bleeding — fresh frozen plasma or prothrombin complex concentrate.

Case 3

After a long bus journey a 45-year-old patient's left calf became swollen and painful. Then sudden shortness of breath and chest pain appeared. Blood D-dimer 3.2 µg/ml.

Questions: What complication can be suspected? Explain the mechanism of the rise in D-dimer. What are the principles of treatment?

Answer: Deep vein thrombosis and pulmonary embolism. During prolonged immobile sitting the blood flow in the leg veins slows (part of Virchow's triad) and a thrombus forms; a fragment of it breaks off and lodges in the pulmonary artery. The body's fibrinolytic system tries to break down the fibrin of the formed thrombus: plasmin splits covalently cross-linked fibrin, forming D-dimer — its rise shows that fibrin is being formed and broken down in the body. Treatment: anticoagulants (heparin, then oral anticoagulants) — to stop new thrombus formation; in massive embolism, thrombolytics (t-PA) — to dissolve the thrombus.

Quick review

Table 9.7. *Key concepts of topic 9*

Concept	Brief definition
Hemostasis	The system that stops bleeding and keeps blood liquid
Clotting factors	I–XIII (no VI); IV — Ca^{2+}
Vitamin K-dependent	II, VII, IX, X (“1972”)
Extrinsic pathway	Tissue factor + VIIa; PT, INR
Intrinsic pathway	XII → XI → IX + VIII; APTT
Prothrombinase	Xa + Va + Ca^{2+} + phospholipid
Thrombin	Fibrinogen → fibrin; activates V, VIII, XI, XIII
Factor XIII	Stabilizes fibrin
Hemophilia A / B	Factor VIII / IX deficiency; APTT prolonged
Antithrombin III	Main anticoagulant; heparin cofactor
Protein C and S	Breaks down Va and VIIIa
Warfarin	Vitamin K antagonist; INR monitoring
Plasmin	Breaks down fibrin; formed from plasminogen
t-PA	Endothelial plasminogen activator; thrombolytic
D-dimer	Fibrin degradation product; marker of thrombosis

TOPIC 10

Blood groups. Rh factor. Principles of blood transfusion

Learning objectives

To teach the student the doctrine of erythrocyte antigen systems, the ABO blood groups, agglutinogens and agglutinins, the Rh system and Rh incompatibility, the rules of blood transfusion and compatibility tests; to master determining the blood group and Rh factor with standard sera and monoclonal reagents (tsoliclones).

By the end of the class the student should know:

- the history of the discovery of blood groups (K. Landsteiner, J. Jansky);
- the agglutinogens and agglutinins of the ABO system and the description of the four blood groups;
- the mechanism of the agglutination reaction;
- the inheritance of the ABO system;
- the Rh system, Rh-positive and Rh-negative blood;
- Rh conflict and the mechanism of hemolytic disease of the newborn;
- the rules for transfusion of blood and its components and the compatibility tests;
- transfusion complications.

The student should be able to:

- determine the blood group with standard sera and tsoliclones;
- determine the Rh factor;
- explain the procedure for the individual (ABO) and Rh compatibility tests and the biological test.

1. The doctrine of blood groups

A **blood group** is the set of hereditarily determined antigens (**agglutinogens**) on the erythrocyte membrane. Blood groups were discovered in 1901 by the Austrian scientist **K. Landsteiner** (three groups; Nobel Prize in 1930); the fourth group was described in 1907 by the Czech scientist **J. Jansky**. Today more than 400 antigens and more than 40 systems are known on erythrocytes (ABO, Rh, Kell, Duffy, Kidd, MNSs and others), but in clinical practice the most important are the **ABO** and **Rhesus (Rh)** systems.

2. The ABO system

In the ABO system erythrocytes may carry two kinds of agglutinogen — **A** and **B**; the plasma contains natural antibodies against them — **agglutinins α (anti-A)** and **β (anti-B)**. When an agglutinogen and agglutinin of the same name (A and α, B and β) meet, the erythrocytes stick together — the **agglutination** reaction occurs, followed by hemolysis. Therefore one person's blood never contains an agglutinogen and agglutinin of the same name.

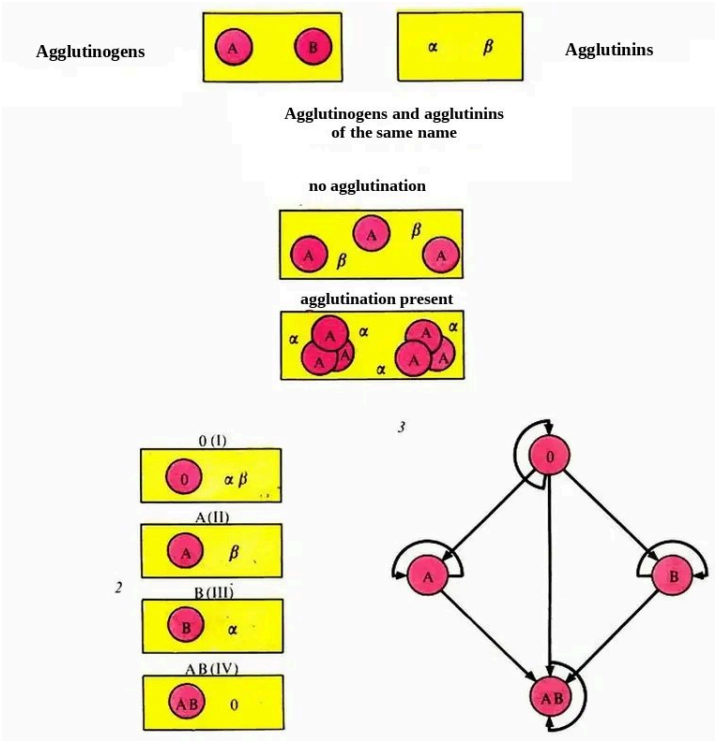


Figure 10.1. *The ABO system: agglutinogens and agglutinins, the agglutination reaction and a scheme of blood group compatibility (Alavia O.T., 2018)*

Table 10.1. *Blood groups of the ABO system*

Blood group	On erythrocytes (agglutinogen)	In plasma (agglutinin)	Frequency in the population (Europe and Central Asia, approx.)
0 (I)	None (H antigen)	α and β	30–40 %
A (II)	A	β	30–40 %
B (III)	B	α	15–25 %
AB (IV)	A and B	Absent	5–10 %

Agglutinins (IgM class) appear 2–8 months after birth and reach their highest titer at 5–10 years of age; they form in response to similar antigens of intestinal bacteria and food (which is why they are called “natural” antibodies). Agglutinogen A has subtypes **A_1** and **A_2** (A_2 is weaker).

Inheritance of the ABO groups

The ABO groups are inherited through a single gene on chromosome 9 with three alleles (**I^A, I^B, i**). I^A and I^B are **codominant**, i is recessive.

Table 10.2. *Genotypes of the blood groups*

Phenotype	Genotype
0 (I)	ii
A (II)	I^AI^A or I^Ai
B (III)	I^BI^B or I^Bi
AB (IV)	I^AI^B

MEMORY AID

The ABO rule: **“Whatever is on the erythrocyte, its enemy is absent from the plasma”**. Group A — A antigen and anti-B; group B — B antigen and anti-A; group 0 — no antigen, both antibodies; group AB — both antigens, no antibodies.

3. The Rh system

The **Rh factor (Rh)** was identified in 1940 by K. Landsteiner and A. Wiener on the erythrocytes of rhesus monkeys. The Rh system has several antigens (D, C, c, E, e); the most immunogenic is the **D antigen**. People who carry it are **Rh-positive (Rh^+)** — ≈ 85% of Europeans; the rest are **Rh-negative (Rh^-)** — ≈ 15% (Rh^- is less common among Asian peoples).

Difference from the ABO system: the Rh system has **no natural antibodies**. Anti-D antibodies (IgG class) in an Rh^- person form only after **sensitization** by Rh^+ erythrocytes: when Rh^+ blood is transfused or when an Rh^- woman is pregnant with an Rh^+ fetus.

Rh conflict

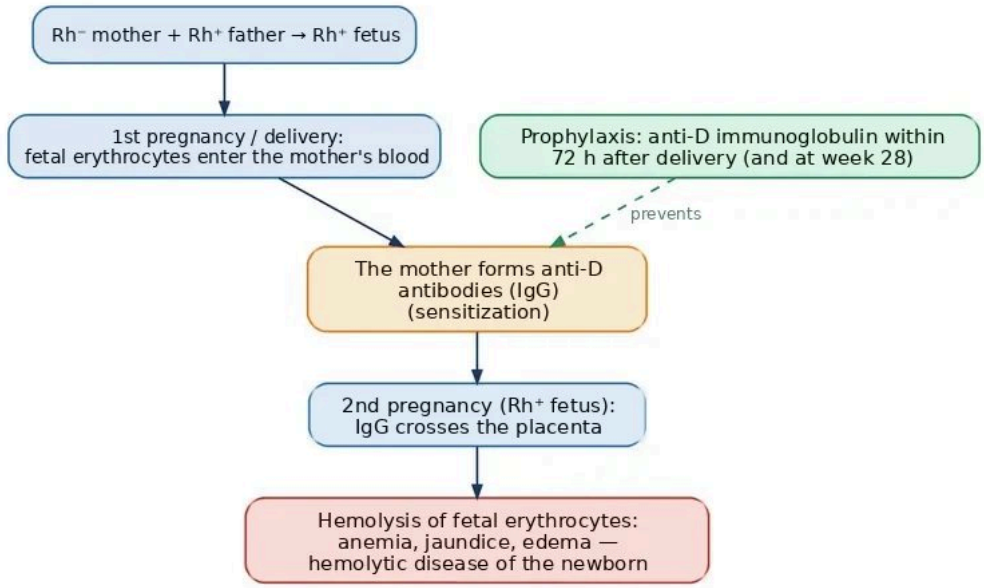


Figure 10.2. *Rh conflict and the development of hemolytic disease of the newborn*

CLINICAL SIGNIFICANCE

Hemolytic disease of the newborn — anti-D IgG antibodies formed in an Rh^- mother cross the placenta and destroy the erythrocytes of the Rh^+ fetus: anemia, jaundice (bilirubin is toxic to the brain — “kernicterus”), and in severe cases hydrops fetalis. Usually the first pregnancy passes without conflict (antibodies form during childbirth); the risk increases in subsequent pregnancies. **Prevention:** an Rh^- woman is given **anti-D immunoglobulin** at 28 weeks of pregnancy and within 72 hours after the birth of an Rh^+ child — it removes the fetal erythrocytes that entered the mother's blood before the immune system “notices” them. Conflict by the ABO system is also possible (most often the mother is group 0 and the child A or B), but it is usually milder.

4. Principles of blood transfusion

Formerly **Ottenberg's rule** was applied in blood transfusion: “the donor's erythrocytes must not be agglutinated by the recipient's plasma” (the donor plasma was assumed to be diluted in the recipient's

blood). Accordingly, group 0 (I) was called the “universal donor” and AB (IV) the “universal recipient”.

CAUTION

Today **only group-specific blood (ABO- and Rh-compatible) is transfused**. The concept of the “universal donor” is outdated: group 0 (I) blood contains high-titer α and β agglutinins and, when transfused in large amounts, can hemolyze the recipient's erythrocytes. Only in emergencies, when group-specific blood is unavailable, may up to 500 ml of 0 (I) Rh^- **packed red cells** (without plasma) be transfused. In addition, whole blood is rarely transfused: the patient receives only the needed **component** (packed red cells, platelet concentrate, fresh frozen plasma).

Table 10.3. *ABO compatibility in transfusion of blood components (basic principle)*

Recipient group	Packed red cells	Fresh frozen plasma
0 (I)	0	0, A, B, AB
A (II)	A (in emergency — 0)	A, AB
B (III)	B (in emergency — 0)	B, AB
AB (IV)	AB (in emergency — A, B, 0)	AB

For plasma the compatibility is reversed: plasma contains agglutinins, so AB (IV) plasma is “universal” (it has no agglutinins).

Mandatory tests before blood transfusion

1. Determine (confirm) the ABO and Rh group of the recipient's and donor's blood.
2. **Individual ABO compatibility test**: recipient serum + donor erythrocytes (10 : 1) at room temperature for 5 minutes; if there is no agglutination — compatible.
3. **Rh compatibility test** (with polyglucin or gelatin, at 37 °C).
4. **Biological test**: 10–15 ml of blood is given as a stream and the patient is observed for 3 minutes; this is repeated three times. If signs of incompatibility appear (pain in the lower back and chest, facial flushing, shortness of breath, tachycardia, a fall in blood pressure), the transfusion is stopped immediately.

CLINICAL SIGNIFICANCE

Hemotransfusion shock — when incompatible blood is transfused, acute intravascular hemolysis develops: pain in the lower back, chills, a fall in blood pressure, hemoglobinuria (“black urine”), acute kidney failure and DIC syndrome. The main cause of these complications is organizational errors (mixing up the patient or the bag), so before every transfusion the documents and the blood group are checked again.

5. Laboratory work

Exercise 1: determining the blood group with standard sera

1. Mark three places on a clean white plate: 0 (I), A (II), B (III).
2. Place 1 drop (≈ 0.1 ml) of the corresponding standard serum in each place (it is recommended to use two batches of serum for each group).
3. Next to each serum place a drop of finger-prick blood 10 times smaller (≈ 0.01 ml) (serum : blood = 10 : 1).

4. Mix each drop with a separate clean glass rod.
5. Rock the plate gently and observe for 5 minutes; add a drop of saline to the drops in which agglutination appears (to exclude pseudo-agglutination).
6. Evaluate the result using the table.

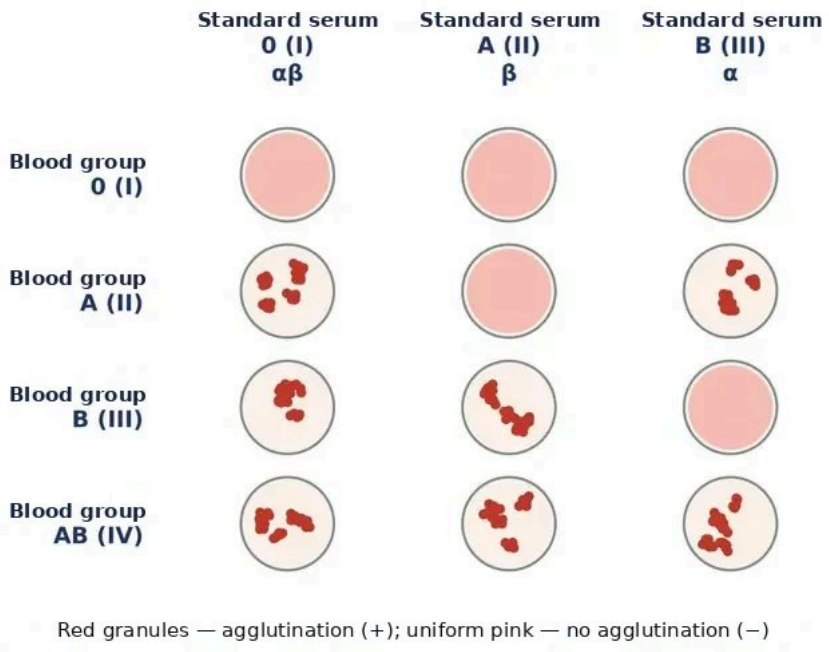


Figure 10.3. *Results of determining the blood group with standard sera (scheme)*

Table 10.4. *Interpreting the result with standard sera*

0 (I) serum (αβ)	A (II) serum (β)	B (III) serum (α)	Blood group
−	−	−	0 (I)
+	−	+	A (II)
+	+	−	B (III)
+	+	+	AB (IV) — a control test is required

If agglutination occurs in all three sera, an additional control is done with standard serum of group AB (IV): there should be no agglutination in it.

Exercise 2: determining the blood group with tsoliclones (anti-A, anti-B)

1. Place 1 drop (0.1 ml) each of the anti-A (pink) and anti-B (blue) tsoliclones on the plate.
2. Put a drop of blood 10 times smaller next to each and mix.
3. Observe for 2.5 minutes.

Table 10.5. *Interpreting the result with tsoliclones*

Anti-A	Anti-B	Blood group
−	−	0 (I)
+	−	A (II)
−	+	B (III)
+	+	AB (IV) (control with anti-AB or saline)

REMEMBER

Tsoliclones (monoclonal antibodies) are the main reagents in current practice: compared with standard sera they are more accurate, have a high titer and do not depend on donor blood. Tsoliclones directly detect the **agglutinogen** (the antigen on the erythrocyte).

Exercise 3: determining the Rh factor with the anti-D tsoliclone

1. Place 1 drop (0.1 ml) of the anti-D tsoliclone on the plate.
2. Add 1 drop (0.02–0.05 ml) of blood next to it and mix.
3. Rock and observe for 3 minutes. **Interpretation:** agglutination present — Rh^+ (Rh-positive); absent — Rh^- (Rh-negative).

6. Practical skills

Skill: algorithm for preparing a blood transfusion

1. Establish the indication for transfusion and decide which component is needed.
2. Re-determine the patient's ABO and Rh group.
3. Check the expiry date, integrity and macroscopic appearance (absence of hemolysis and clots) of the blood in the bag (container).
4. Confirm the donor blood group with tsoliclones.
5. Perform the individual ABO and Rh compatibility tests.
6. Carry out the biological test (3 times 10–15 ml, at 3-minute intervals).
7. During and after the transfusion observe the patient's condition, blood pressure, pulse, temperature and urine color; record all data in the documents.

Skill: solving a blood group inheritance problem

1. Write the possible genotypes from the parents' phenotypes.
2. Determine the gametes.
3. Construct a Punnett square.
4. Determine the possible and impossible groups in the children.

Review questions

1. Who discovered blood groups, and when?
2. What are an agglutinogen and an agglutinin?
3. Describe the four blood groups of the ABO system.
4. When does the agglutination reaction occur?
5. How are the ABO groups inherited?
6. What is the Rh factor, and when was it discovered?
7. What is the main difference between the Rh system and the ABO system?
8. How does Rh conflict develop?
9. How is hemolytic disease of the newborn prevented?
10. What is Ottenberg's rule, and why is it no longer applied?
11. What tests are carried out before blood transfusion?
12. How is the biological test performed?

13. What are the signs and mechanism of hemotransfusion shock?
14. How is the blood group determined with standard sera?
15. What are the advantages of tsoliclones?

Test questions

1. Who discovered the blood groups of the ABO system?
   A) I.I. Mechnikov
   B) J. Jansky
   C) A. Wiener
   D) K. Landsteiner
   E) I.P. Pavlov

2. Which agglutinin is present in the plasma of group A (II) blood?
   A) α
   B) α and β
   C) β
   D) No agglutinin
   E) Anti-D

3. Which blood group has no agglutinogen on its erythrocytes?
   A) AB (IV)
   B) A (II)
   C) B (III)
   D) Rh^+
   E) 0 (I)

4. When does agglutination occur?
   A) When agglutinogen A meets agglutinin β
   B) When an agglutinogen and agglutinin of the same name meet
   C) When agglutinogen B is present without agglutinin β
   D) Only in group AB
   E) When Rh^- blood is mixed with Rh^- blood

5. In typing with standard sera, agglutination occurred in the 0 (I) and B (III) sera, but not in the A (II) serum. What is the blood group?
   A) 0 (I)
   B) B (III)
   C) AB (IV)
   D) A (II)
   E) Cannot be determined

6. Which is the most immunogenic antigen of the Rh system?
   A) C
   B) c
   C) E
   D) e
   E) D

7. When do anti-D antibodies form in an Rh^- person?
   A) Naturally, 2–8 months after birth
   B) They never form
   C) Only after sensitization by Rh^+ erythrocytes
   D) In response to food antigens
   E) Only in ABO incompatibility

8. What is given to an Rh^- mother after childbirth to prevent Rh conflict?
   A) Heparin
   B) Packed red cells
   C) Vitamin K
   D) Rh^+ blood
   E) Anti-D immunoglobulin

9. If both parents are group 0 (I), what group will the child have?
   A) Only 0 (I)
   B) A (II) or 0 (I)
   C) AB (IV)
   D) B (III)
   E) Any group

10. In typing with tsoliclones there is agglutination with anti-A but not with anti-B. What is the blood group?
    A) 0 (I)
    B) A (II)
    C) B (III)
    D) AB (IV)
    E) Rh^-

11. According to current rules, what blood is transfused?
    A) Group 0 (I) to any group
    B) Group AB (IV) to everyone
    C) Blood compatible only by Rh
    D) Group-specific blood (ABO- and Rh-compatible) or its components
    E) Only the parents' blood

12. How is the biological test performed?
    A) 10–15 ml of blood is given as a stream and the patient is observed for 3 minutes; this is repeated three times
    B) Serum and erythrocytes are mixed in a test tube
    C) Blood is injected into the skin
    D) 500 ml of blood is transfused at once
    E) It is performed only in the laboratory

13. To which class of immunoglobulins do the agglutinins of the ABO system belong?
    A) IgG
    B) IgM
    C) IgE
    D) IgA
    E) IgD

14. What is the cause of hemotransfusion shock?
    A) Acute intravascular hemolysis when incompatible blood is transfused
    B) Cold blood
    C) A large volume of saline
    D) Accelerated blood coagulation
    E) An increase in leukocytes

15. Which group is considered “universal” for transfusion of fresh frozen plasma?
    A) 0 (I)
    B) A (II)
    C) AB (IV)
    D) B (III)
    E) Rh^- 0 (I)

Clinical cases

Case 1

An Rh^- woman is in her third pregnancy. Her first child was born healthy; the second pregnancy ended in abortion, and no prophylaxis was given. Her husband is Rh^+. At 30 weeks of pregnancy a high titer of anti-D antibodies was found in her blood.

Questions: What condition is developing? Explain its mechanism and the danger to the fetus. What prophylaxis should have been given?

Answer: Rh conflict (Rh sensitization). During the first birth and the abortion, Rh^+ fetal erythrocytes entered the mother's blood; the mother has no natural anti-D antibodies, but after sensitization IgG-class anti-D antibodies were formed. IgG can cross the placenta and hemolyzes the Rh^+ fetal erythrocytes — hemolytic disease of the newborn: anemia, jaundice, toxic effect of bilirubin on the brain, and in severe cases hydrops fetalis and fetal death. Prophylaxis: anti-D immunoglobulin should have been given within 72 hours after each birth (of an Rh^+ child), abortion or miscarriage, and at 28 weeks of pregnancy.

Case 2

After an accident, a patient who has lost a lot of blood needs emergency surgery. The patient's blood group is A (II) Rh^+. The blood bank has run out of group A (II) blood, but 0 (I) Rh^- packed red cells are available.

Questions: Can group 0 (I) packed red cells be transfused in this situation? Under what conditions? Why are packed red cells chosen rather than whole blood?

Answer: In an emergency, when group-specific blood is unavailable, up to 500 ml of 0 (I) Rh^- packed red cells may be transfused. Group 0 (I) erythrocytes have no A or B agglutinogens, so the recipient's β agglutinins do not agglutinate them. Packed red cells rather than whole blood are chosen because 0 (I) plasma contains high-titer α and β agglutinins — in large volumes they can hemolyze the recipient's A erythrocytes. In any case, individual compatibility and biological tests are performed; group-specific blood is used as soon as possible.

Case 3

A student determined a blood group with standard sera: agglutination appeared in all three sera — 0 (I), A (II), B (III).

Questions: What is the blood group? How should the result be confirmed?

Answer: If agglutination occurs in all three sera, the erythrocytes carry A and B agglutinogens — group AB (IV). However, this result may also be due to nonspecific (pseudo-) agglutination (for example, rouleaux formation or serum contamination). To confirm it, a control test with standard serum of group AB (IV) is performed: it has no agglutinins, so there should be no agglutination. Saline is also added to the agglutinated drops, and the result is confirmed with tsoliclones.

Quick review

Table 10.6. *Key concepts of topic 10*

Concept	Brief definition
Landsteiner	Discovered the ABO (1901) and Rh (1940, with Wiener) systems
Agglutinogen	Antigen on the erythrocyte (A, B, D)
Agglutinin	Antibody in the plasma (α, β)
0 (I)	No antigen; α, β present
A (II)	A antigen; β
B (III)	B antigen; α
AB (IV)	A and B antigens; no agglutinin
Agglutination	When A + α or B + β meet
Inheritance	I^A, I^B codominant, i recessive
Rh^+	D antigen present (≈ 85%)
Anti-D	Not natural; after sensitization, IgG
Rh conflict	Rh^- mother, Rh^+ fetus — hemolytic disease
Prophylaxis	Anti-D immunoglobulin (within 72 hours)
Transfusion rule	Only group-specific, compatible components
Compatibility tests	Individual ABO and Rh tests and the biological test

TOPIC 11

The hemodynamic function of the heart. Physiological properties of cardiac muscle. Methods of studying cardiac activity

Learning objectives

To teach the student the pumping function of the heart, the physiological properties of cardiac muscle (automaticity, excitability, conductivity, contractility) and their specific features, the phases of the cardiac cycle, the main parameters of cardiac activity, the electrical, mechanical and acoustic phenomena of the heart and the methods of studying them (ECG, auscultation, phonocardiography, echocardiography).

By the end of the class the student should know:

- the structure and hemodynamic function of the heart;
- the conduction system of the heart and the gradient of automaticity;
- the ionic mechanisms of the action potential in typical and atypical cardiomyocytes;
- the excitability and refractoriness of cardiac muscle and the mechanism of the extrasystole and compensatory pause;
- the phases of the cardiac cycle and their duration;
- parameters such as stroke and minute volume, ejection fraction and cardiac index;
- the ECG leads, waves and intervals and their meaning;
- the heart sounds and their origin.

The student should be able to:

- perform the Stannius ligature experiment on the frog heart and explain the gradient of automaticity;
- record an extrasystole and compensatory pause in the frog heart;
- record an ECG in a human and determine its main parameters;
- auscultate the heart sounds.

1. The hemodynamic function of the heart

The heart is the central organ of the circulatory system, a rhythmically working **pump**. By ejecting blood into the vessels it creates a pressure gradient in them and ensures the continuous movement of blood through the systemic and pulmonary circulations. The heart consists of two pumps: the **left heart** (systemic circulation, high pressure) and the **right heart** (pulmonary circulation, low pressure). Both ventricles contract simultaneously and eject the same volume of blood.

The heart **valves** ensure one-way movement of blood: the **atrioventricular (cuspid) valves** (bicuspid — mitral — on the left, tricuspid on the right) between the atria and ventricles; the **semilunar valves** (aortic and pulmonary) between the ventricles and arteries. The valves open and close passively under the action of the pressure difference.

2. Physiological properties of cardiac muscle

Cardiac muscle consists of two kinds of cells: **working (typical) cardiomyocytes**, which provide contraction, and **atypical cells of the conduction system**, which generate and conduct impulses.

Automaticity

Automaticity is the ability of the heart to be rhythmically excited and to contract under the influence of impulses generated within the heart itself. Automaticity is **myogenic** in nature: an isolated heart works for hours in a nutrient solution (the Langendorff method). The source of automaticity is the **conduction system**.

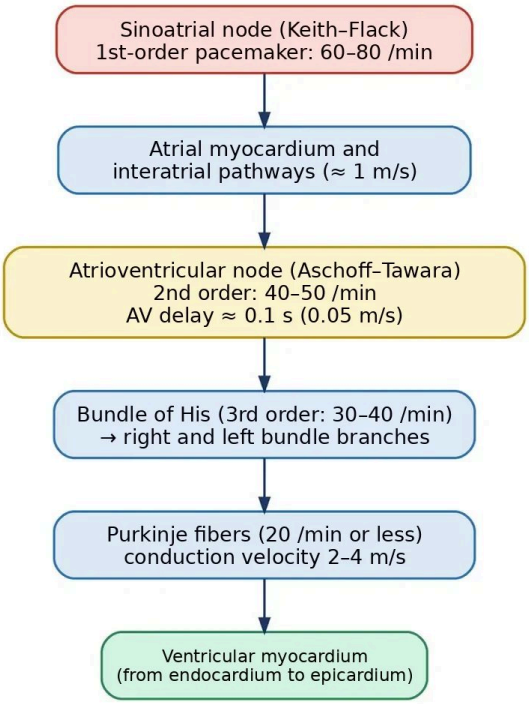


Figure 11.1. *The conduction system of the heart and the gradient of automaticity*

Gradient of automaticity (W. Gaskell): the level of automaticity is high in the upper parts of the conduction system and decreases further down. Normally the rhythm is set by the fastest impulse generator, the **sinoatrial node** (**sinus rhythm**); the other centers are subordinate to its rhythm (latent pacemakers).

Action potential in pacemaker cells: the resting potential is not stable (maximum diastolic potential −60 mV). In phase 4 **slow diastolic depolarization** occurs, due to the "funny" **I_f current** (Na^+ entry), T-type Ca^{2+} channels and a decrease in K^+ efflux. On reaching threshold (−40 mV), depolarization proceeds slowly through **L-type Ca^{2+} channels**; then repolarization occurs with K^+ efflux.

Phases of the action potential in **working cardiomyocytes** (duration 250–300 ms): **0** — rapid depolarization (fast Na^+ channels); **1** — early rapid repolarization (K^+ and Cl^-); **2** — **plateau** (Ca^{2+} enters through L-channels, balanced by K^+ efflux); **3** — rapid final repolarization (K^+ leaves); **4** — rest (−90 mV).

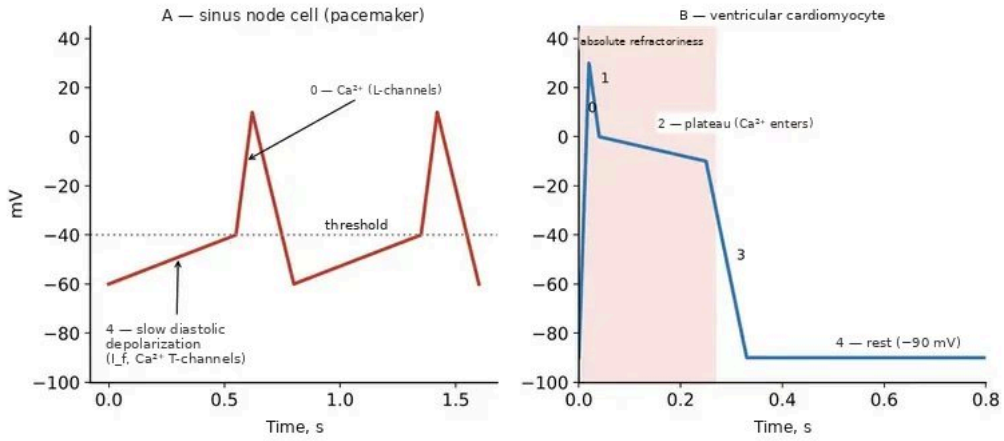


Figure 11.2. *Action potentials of a pacemaker cell (A) and a ventricular cardiomyocyte (B) (drawn after Costanzo L.S., 2018)*

Excitability and refractoriness

Because of the plateau phase, the **absolute refractory period** of cardiomyocytes is very long (≈ 0.27 s) — it covers almost all of systole. Therefore:

1. cardiac muscle **cannot contract tetanically** — relaxation and filling with blood are ensured;
2. excitation does not circulate in a ring (re-entry) and return.

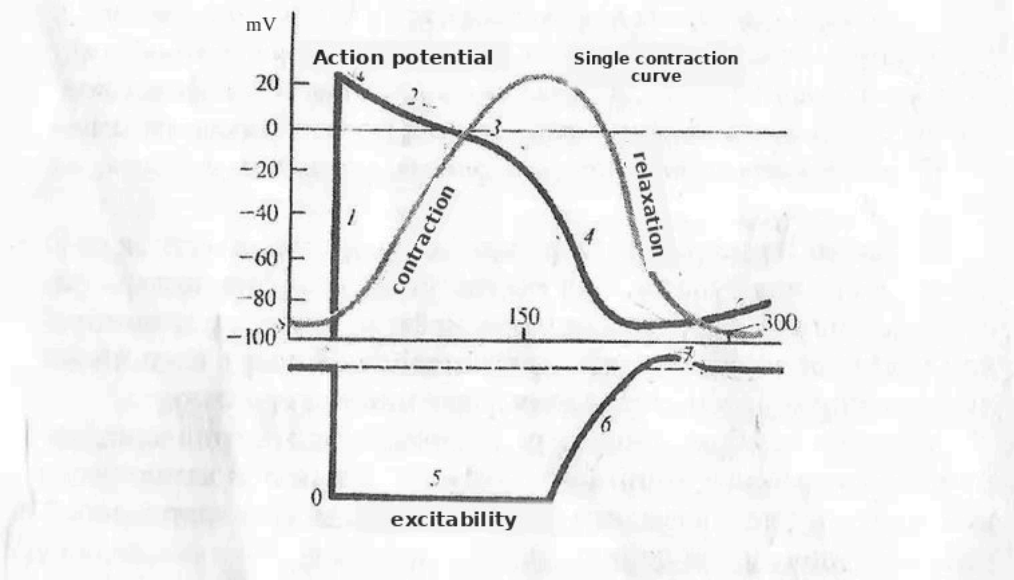


Figure 11.3. *Cardiomyocyte action potential, contraction curve and phases of excitability (Alavia O.T., 2018)*

An **extrasystole** is a premature contraction. If an extra stimulus arrives in **diastole** (after relative refractoriness has ended), the heart contracts prematurely. After a ventricular extrasystole a **compensatory pause** is observed: the next impulse from the sinus node falls into the refractory period of the ventricle and gets no response, and the ventricle waits for the following impulse.

Conductivity

Excitation spreads in the heart through **gap junctions between cardiomyocytes** (nexuses, intercalated discs) — the heart works as a **functional syncytium** and obeys the “all or nothing” law. Conduction velocity: in the atria ≈ 1 m/s, in the AV node 0.02–0.05 m/s, in the bundle of His and Purkinje fibers 2–4 m/s, in the ventricular myocardium 0.3–1 m/s. The **AV delay** (≈ 0.1 s) ensures that the atria finish contracting and the ventricles fill with blood.

Contractility

For contraction of cardiac muscle part of the Ca^{2+} enters **from outside the cell** (in the plateau phase) and triggers the release of additional Ca^{2+} from the SR (**calcium-induced calcium release**). Therefore the force of contraction depends on extracellular Ca^{2+}. The force of contraction is regulated by the **Frank–Starling law** (the more the myocardium is stretched, the stronger it contracts) and by neurohumoral factors (topic 12).

REMEMBER

The 4 properties of cardiac muscle: **automaticity, excitability, conductivity, contractility**. Its main differences from skeletal muscle: automaticity, a functional syncytium ("all or nothing"), a long refractory period (no tetanus), and the need for extracellular Ca^{2+} for contraction.

3. The cardiac cycle

The **cardiac cycle** is the period consisting of one contraction (systole) and relaxation (diastole). At a heart rate of 75 /min the cycle lasts **0.8 s**: atrial systole 0.1 s, atrial diastole 0.7 s; ventricular systole 0.33 s, ventricular diastole 0.47 s. The period when the atria and ventricles are relaxed at the same time is the **general pause** (≈ 0.4 s).

Table 11.1. *Phases of ventricular systole and diastole (75 /min)*

Period and phase	Duration, s	Content
Systole	0,33	—
Tension period: asynchronous contraction	0,05	Excitation spreads through the myocardium; pressure hardly changes
Tension period: isovolumetric contraction	0,03	All valves closed; pressure rises rapidly (to 80 mm Hg in the left ventricle)
Ejection period: rapid ejection	0,12	The semilunar valves open; pressure reaches 120 mm Hg on the left and 25 mm Hg on the right
Ejection period: slow ejection	0,13	Pressure begins to fall
Diastole	0,47	—
Protodiastolic period	0,04	Relaxation begins; the semilunar valves close
Isovolumetric relaxation	0,08	All valves closed; pressure falls rapidly
Rapid filling	0,08	The atrioventricular valves open and blood flows into the ventricles
Slow filling (diastasis)	0,17	—
Atrial systole (presystole)	0,1	The ventricles fill with an additional 10–30% of blood

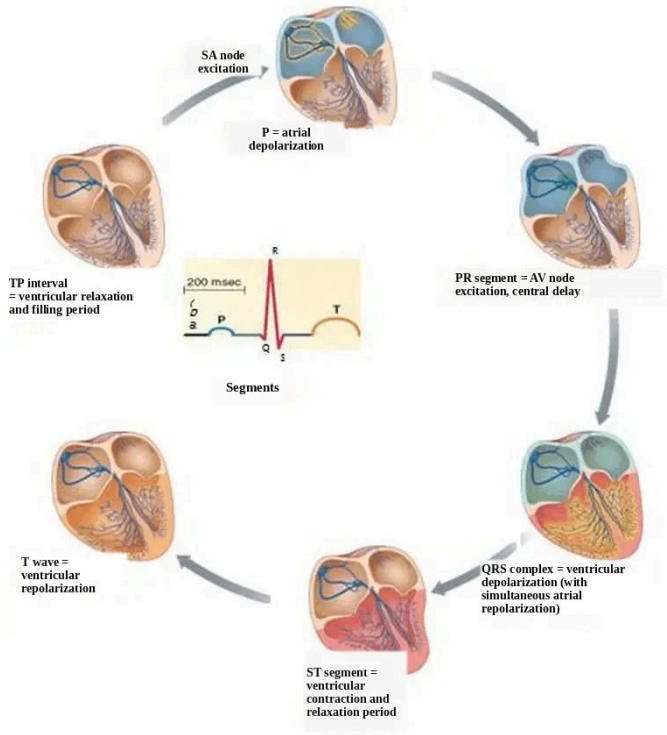


Figure 11.4. *Spread of excitation in the cardiac cycle and its correspondence to the ECG elements (Alavia O.T., 2018)*

CAUTION

When the heart rate increases, it is mainly **diastole that shortens**; systole changes less. At a very high heart rate (more than 180 /min) the ventricles do not fill adequately and the coronary vessels (which receive blood mainly in diastole) are poorly supplied with blood — the efficiency of the heart decreases.

4. Main parameters of cardiac activity

Table 11.2. *Parameters of cardiac activity (at rest, adults)*

Parameter	Definition	Normal value
Heart rate (HR)	Contractions per minute	60–80 /min
End-diastolic volume	Blood in the ventricle at the end of diastole	120–140 ml
Systolic (stroke) volume	Blood ejected in one contraction	60–80 ml
Residual (end-systolic) volume	Blood remaining in the ventricle after systole	50–60 ml
Ejection fraction	Stroke volume / end-diastolic volume	55–70 %
Minute volume (cardiac output)	Stroke volume × HR	4.5–5 l/min (up to 25–30 l/min during physical work)
Cardiac index	Minute volume / body surface area	2.5–4 l/min/m^2

CLINICAL SIGNIFICANCE

The **ejection fraction** is determined by echocardiography and is the main clinical indicator of cardiac contractility. When it falls below 40%, **heart failure with reduced ejection fraction** is diagnosed (after myocardial infarction, in cardiomyopathies).

5. Methods of studying cardiac activity

Electrocardiography (ECG)

The **ECG** is the curve of electrical potentials generated during cardiac excitation, recorded from the body surface. It was first recorded by W. Einthoven (1903).

Table 11.3. *ECG leads*

Leads	Electrode placement
Standard (I, II, III) — Einthoven's triangle	I: right arm (−) — left arm (+); II: right arm (−) — left leg (+); III: left arm (−) — left leg (+)
Augmented unipolar (aVR, aVL, aVF)	The active electrode is on the right arm, left arm and left leg, respectively
Chest (V_1–V_6)	V_1 — 4th intercostal space at the right sternal border; V_2 — at the left sternal border; V_4 — 5th intercostal space, midclavicular line; V_3 — midway between V_2 and V_4; V_5 — anterior axillary line; V_6 — midaxillary line (at the level of V_4)

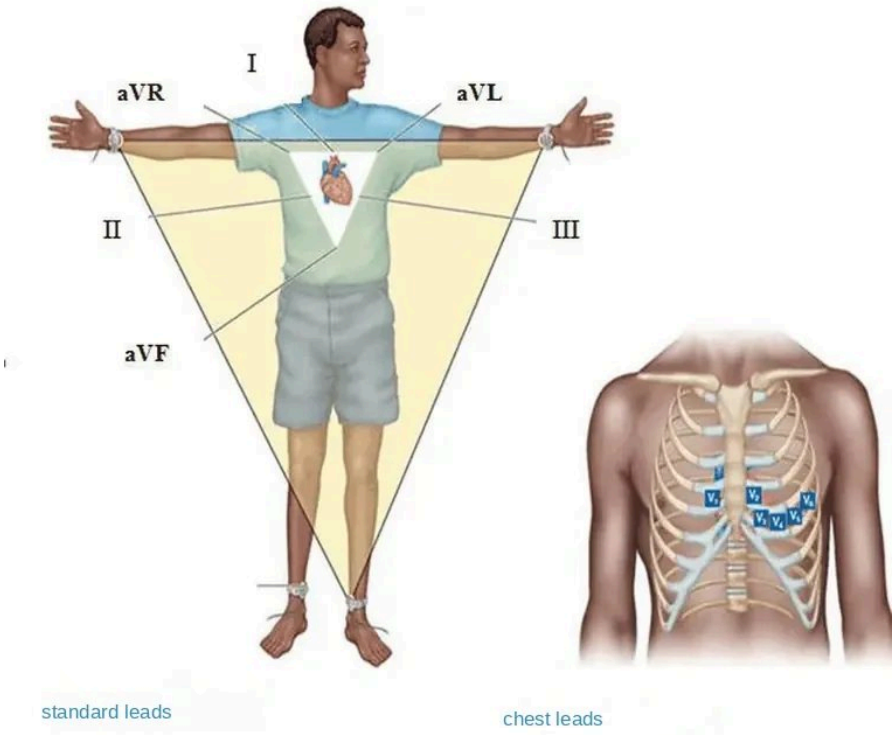


Figure 11.5. *Standard, augmented and chest ECG leads (Alavia O.T., 2018)*

Table 11.4. *ECG elements (standard lead II)*

Element	What it reflects	Duration, s
P wave	Atrial depolarization	0,08–0,10
PQ (PR) interval	Conduction of excitation from the atria to the ventricles (AV conduction)	0,12–0,20
QRS complex	Ventricular depolarization	0,06–0,10
ST segment	The fully excited state of the ventricles (isoelectric)	—
T wave	Ventricular repolarization	0,16–0,24
QT interval	Electrical systole of the ventricles	0.36–0.44 (depends on HR)

Element	What it reflects	Duration, s
RR interval	Duration of the cardiac cycle	0.75–1.0 (at HR 60–80)

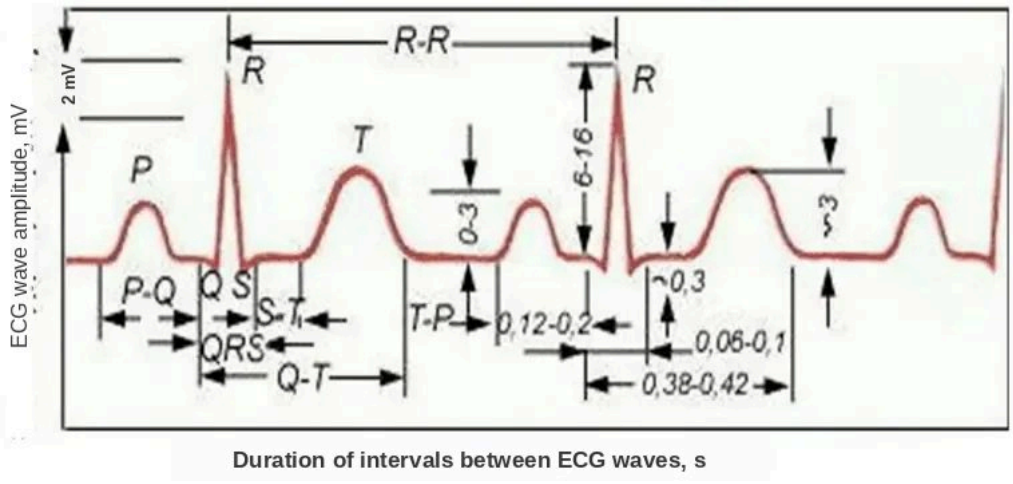


Figure 11.6. *ECG waves and intervals (Alavia O.T., 2018)*

Atrial repolarization is not visible on the ECG because it coincides with the QRS complex.

MEMORY AID

Remember the ECG as "**P — the atria (they fire first); QRS — the ventricles; T — recovery (repolarization)**". HR = 60 / RR (s).

Heart sounds

Auscultation is listening to the heart sounds; **phonocardiography** is recording them graphically.

Table 11.5. *Heart sounds*

Sound	Origin	Features	Where best heard
I (systolic)	Closure of the atrioventricular valves, tension of the ventricular myocardium	Low-pitched, longer (0.1–0.17 s); simultaneous with the apex beat and the carotid pulse	Apex of the heart (mitral), lower end of the sternum (tricuspid)
II (diastolic)	Closure of the semilunar valves	High-pitched, short (0.06–0.08 s)	2nd intercostal space at the right (aorta) and left (pulmonary artery) sternal border
III	Rapid ventricular filling (wall vibration)	Faint; normal in children and athletes	Apex
IV	Atrial systole	Usually not heard	—

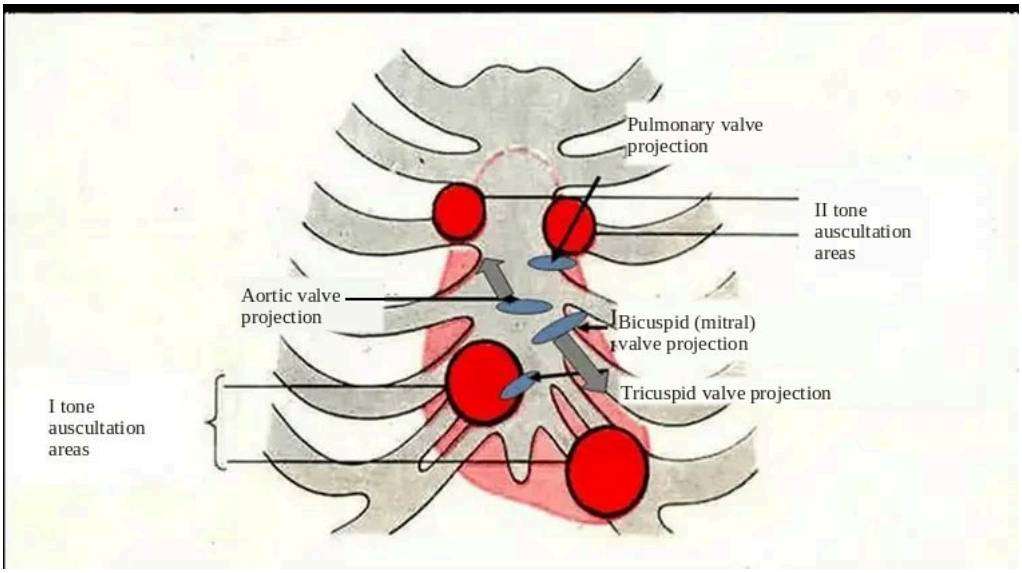


Figure 11.7. *Projection of the heart valves and the points for listening to the heart sounds (Alavia O.T., 2018)*

Order of auscultation: 1) apex (mitral valve); 2) 2nd intercostal space on the right (aorta); 3) 2nd intercostal space on the left (pulmonary artery); 4) base of the xiphoid process (tricuspid valve); 5) Botkin–Erb point — 3rd–4th intercostal space at the left sternal border (aorta).

Other methods

- **Echocardiography (ultrasound)** — evaluation of the heart chambers, wall thickness, valves and ejection fraction; Doppler — speed and direction of blood flow.
- **Palpation of the apex beat** — normally in the 5th intercostal space, 1–1.5 cm medial to the left midclavicular line.
- **Sphygmography, polycardiography** — recording of the arterial pulse and the phase structure of the cycle.
- **Holter monitoring** — recording the ECG over 24 hours.

6. Laboratory work

Exercise 1: Stannius ligatures (frog heart)

1. Immobilize the frog, open the chest and expose the heart. Count the heart rate for 1 minute.
2. **Ligature I** — tie a thread at the boundary between the sinus venosus and the right atrium. The sinus venosus continues to contract at its previous rhythm; the atria and ventricle stop temporarily and then begin to contract at a slower rhythm.
3. **Ligature II** — tie it at the boundary between the atria and the ventricle (the AV node region). The ventricle contracts at an even slower rhythm.
4. **Ligature III** — tie it at the lower third of the ventricle. The apex of the heart does not contract (it has no center of automaticity). **Result and analysis:** the experiment proves the **gradient of automaticity**: the main center of automaticity is in the sinus venosus (in humans, in the sinoatrial node), the lower centers have slower rhythms, and the apex of the heart has no automaticity.

Exercise 2: extrasystole and compensatory pause (frog heart)

1. Connect the apex of the heart to a kymograph lever with a serrefine and record the contractions.

2. Stimulate the ventricle with a single electrical stimulus in different phases of systole and then in diastole.
3. Compare the results on the kymogram.

Result and analysis: a stimulus given in systole causes no response (absolute refractoriness). A stimulus given in diastole causes a premature contraction — an **extrasystole**; it is followed by a prolonged **compensatory pause**, because the next impulse from the sinus venosus falls into the refractory period of the ventricle after the extrasystole.

Exercise 3: recording an ECG in a human

1. Lay the subject on a couch; wipe the electrode sites with alcohol and apply contact gel.
2. Place the limb electrodes: red — right arm, yellow — left arm, green — left leg, black (ground) — right leg.
3. Place the chest electrodes at points V_1–V_6.
4. Check the calibration signal (1 mV = 10 mm) and the paper speed (50 or 25 mm/s); record the ECG in 12 leads.
5. In standard lead II measure the amplitude of the waves and the duration of the intervals.
6. Determine the rhythm (sinus rhythm — a P wave before every QRS, equal RR intervals) and the HR (60 / RR).

Exercise 4: auscultation of the heart sounds

1. Listen to the subject in a sitting or lying position.
2. Place the stethoscope successively at the auscultation points.
3. At the same time palpate the carotid pulse and identify the first sound (simultaneous with the pulse).
4. Compare the first and second sounds by pitch, duration and pauses: the first sound is louder at the apex, the second at the base of the heart.

7. Practical skills

Skill: determining the HR and main intervals on the ECG

1. Determine the paper speed (at 50 mm/s 1 mm = 0.02 s; at 25 mm/s 1 mm = 0.04 s).
2. Measure the distance between two adjacent R waves and calculate RR (s).
3. HR = 60 / RR.
4. Measure PQ, QRS and QT and compare them with the norm.
5. Check the criteria of sinus rhythm: the P wave is positive (in lead II) and precedes each QRS, PQ is constant, RR intervals are equal.

Skill: calculating the minute volume and ejection fraction

1. Stroke volume (SV) = end-diastolic volume − end-systolic volume.
2. Ejection fraction = SV / end-diastolic volume × 100%.
3. Minute volume = SV × HR.
4. Compare the results with the norm and evaluate cardiac activity.

Review questions

1. What is the hemodynamic function of the heart?
2. What physiological properties does cardiac muscle have?
3. Describe the conduction system of the heart.
4. What is the gradient of automaticity, and how is it proved?
5. How is the action potential generated in pacemaker cells?
6. Explain the phases of the action potential of a working cardiomyocyte.
7. Why can cardiac muscle not contract tetanically?
8. What is the mechanism of the extrasystole and compensatory pause?
9. What is the significance of the AV delay?
10. Name the phases of the cardiac cycle and their duration.
11. How are the stroke and minute volume and the ejection fraction determined?
12. List the ECG leads.
13. What do the ECG waves and intervals reflect?
14. How do the heart sounds arise, and where are they heard?
15. What modern methods of studying cardiac activity are there?

Test questions

1. Which structure normally sets the heart rhythm?
   A) The atrioventricular node
   B) The sinoatrial node
   C) The bundle of His
   D) Purkinje fibers
   E) The ventricular myocardium

2. Which current is phase 4 (slow diastolic depolarization) in pacemaker cells associated with?
   A) The fast Na^+ current
   B) The Cl^- current
   C) The I_f (“funny”) current and T-type Ca^{2+} channels
   D) Large K^+ efflux
   E) Na^+/K^+ pump

3. What is the plateau phase of the action potential of a working cardiomyocyte associated with?
   A) Rapid entry of Na^+
   B) Rapid efflux of K^+
   C) Entry of Cl^-
   D) Stopping of the Na^+/K^+ pump
   E) Entry of Ca^{2+} through L-type channels

4. Why does cardiac muscle not contract tetanically?
   A) Because of its automaticity
   B) Because of low contractility
   C) Because it is a functional syncytium
   D) Because of the long absolute refractory period
   E) Because of fast conduction

5. What is observed after Stannius ligature I?
   A) The whole heart stops
   B) Only the ventricle contracts
   C) The sinus venosus contracts at its previous rhythm; the atria and ventricle stop temporarily
   D) The heart contracts faster
   E) The apex of the heart contracts

6. Where is conduction of excitation slowest?
   A) The AV node
   B) Purkinje fibers
   C) The bundle of His
   D) The atrial myocardium
   E) The ventricular myocardium

7. How long does the cardiac cycle last at a HR of 75 /min?
   A) 0.4 s
   B) 0.8 s
   C) 1.2 s
   D) 0.33 s
   E) 1.0 s

8. What is the state of the valves in the isovolumetric contraction phase?
   A) The atrioventricular valves are open
   B) The semilunar valves are open
   C) All valves are closed
   D) All valves are open
   E) Only the aortic valve is open

9. What is the normal ejection fraction?
   A) 20–30 %
   B) 55–70 %
   C) 85–95 %
   D) 35–40 %
   E) 100 %

10. What does the QRS complex on the ECG reflect?
    A) Atrial depolarization
    B) Ventricular repolarization
    C) AV conduction
    D) Ventricular depolarization
    E) Atrial repolarization

11. What is the normal duration of the PQ interval?
    A) 0.12–0.20 s
    B) 0.06–0.10 s
    C) 0.36–0.44 s
    D) 0.30–0.40 s
    E) 0.02–0.04 s

12. What is the first heart sound associated with?
    A) Closure of the semilunar valves
    B) Rapid ventricular filling
    C) Atrial systole
    D) Closure of the atrioventricular valves and ventricular tension
    E) Opening of the semilunar valves

13. What is the minute volume at rest?
    A) 60–80 ml
    B) 1–2 l/min
    C) 10–15 l/min
    D) 25–30 l/min
    E) 4.5–5 l/min

14. What is the cause of the compensatory pause?
    A) The next impulse from the sinus node falls into the refractory period of the ventricle
    B) Arrest of the sinus node
    C) Block of the AV node
    D) Fatigue of the ventricle
    E) Lack of Ca^{2+}

15. Where is the V_4 electrode placed?
    A) 4th intercostal space, at the right sternal border
    B) 4th intercostal space, at the left sternal border
    C) On the anterior axillary line
    D) On the midaxillary line
    E) 5th intercostal space, on the left midclavicular line

Clinical cases

Case 1

A 70-year-old patient has attacks of dizziness and fainting. On the ECG the P waves are regular (70 /min), while the QRS complexes are independent of them, widened, at a rate of 35 /min.

Questions: What is the condition? Under the influence of which pacemaker are the ventricles contracting? Explain from the standpoint of the gradient of automaticity.

Answer: Complete (third-degree) atrioventricular block. The sinus node excites the atria at a normal rhythm (70 /min), but the impulses do not pass through the AV node to the ventricles. The ventricles contract under the influence of lower pacemakers — the bundle of His or Purkinje fibers; according to the gradient of automaticity their rate is low (30–40 /min or less), and because excitation does not spread through the conduction system in the usual way, the QRS is widened. The slow rhythm reduces the minute volume — the brain receives too little blood and fainting occurs (Morgagni–Adams–Stokes attacks). Treatment — an artificial pacemaker (electrocardiostimulator).

Case 2

An athlete at rest has a HR of 50 /min and a stroke volume of 100 ml. His non-athlete peer has a HR of 75 /min and a stroke volume of 66 ml.

Questions: Calculate the minute volume of both. Explain the difference.

Answer: Athlete: 100 ml × 50 = 5000 ml/min (5 l/min). Non-athlete: 66 ml × 75 ≈ 4950 ml/min (≈ 5 l/min). At rest the minute volume is almost the same, because the body's needs are the same. In the athlete regular training has enlarged the left ventricular cavity and strengthened the myocardium (physiological hypertrophy) — the stroke volume is large; therefore the same minute volume is achieved with fewer contractions (bradycardia; vagal tone is also high). This makes the work of the heart economical and provides a large reserve during physical load.

Case 3

A student stimulated a frog heart with an electric current during the diastole phase. A premature contraction followed by a long pause appeared on the kymogram. A stimulus given during systole caused no response.

Questions: Explain the results.

Answer: During systole cardiac muscle is in the absolute refractory period (long because of the plateau phase), so the stimulus caused no response. In diastole excitability had been restored, and the stimulus caused a premature excitation and contraction — an extrasystole. After it, the next impulse from the sinus venosus coincided with the refractory period of the ventricle following the extrasystole and got no response; the ventricle waited for the next sinus impulse — a compensatory pause arose.

Quick review

Table 11.6. *Key concepts of topic 11*

Concept	Brief definition
Automaticity	Rhythmic contraction due to impulses generated in the heart; myogenic
Sinus node	First-order pacemaker, 60–80 /min
The AV node	Second order, 40–50 /min; AV delay ≈ 0.1 s
Gradient of automaticity	Decreases from top to bottom (Gaskell, Stannius)
I_f current	Slow diastolic depolarization of the pacemaker
Plateau	Ca^{2+} enters through L-channels; long refractoriness
Extrasystole	Premature contraction in response to a stimulus in diastole
Compensatory pause	The next impulse falls into the refractory period
Cardiac cycle	0.8 s (75 /min): systole 0.33 s, diastole 0.47 s
Stroke volume	60–80 ml
Minute volume	4.5–5 l/min
Ejection fraction	55–70 %
P / QRS / T	Atrial depolarization / ventricular depolarization / ventricular repolarization
PQ	0.12–0.20 s — AV conduction
First and second sounds	Closure of the atrioventricular / semilunar valves

TOPIC 12

Regulation of cardiac activity

Learning objectives

To teach the student the intracardiac (myogenic and intracardiac nervous system) and extracardiac (nervous and humoral) mechanisms of regulating cardiac activity, the types of influence of the sympathetic and parasympathetic nerves on the heart, cardiac reflexes, and the effects of hormones and ions; to master studying reflex changes in cardiac activity in humans.

By the end of the class the student should know:

- the levels of regulation of cardiac activity;
- the heterometric (Frank–Starling) and homeometric (Anrep, Bowditch) regulatory mechanisms;
- the intracardiac peripheral reflexes;
- the types (chrono-, ino-, dromo-, bathmo-, tonotropic) and mechanisms of the influence of the vagus and sympathetic nerves on the heart;
- the tone of the cardiac nerve centers;
- the intrinsic and conjugate reflexes of the heart (Goltz, Dagnini–Aschner, Bainbridge, baroreceptor reflexes);
- the effects of hormones, mediators and ions on cardiac activity.

The student should be able to:

- stimulate the vagus nerve in the frog and demonstrate its effect on the heart;
- perform the Dagnini–Aschner oculocardiac reflex and the orthostatic test in humans;
- detect respiratory sinus arrhythmia.

1. Levels of regulation of cardiac activity

Cardiac activity constantly changes according to the body's needs: if the minute volume is 5 l at rest, during heavy physical work it reaches 25–30 l/min. This is achieved by changing the **heart rate** and the **stroke volume**.

Table 12.1. *Mechanisms of regulation of cardiac activity*

Level	Mechanism	Example
Intracardiac — myogenic (intracellular)	Heterometric and homeometric regulation	Frank–Starling law, Anrep phenomenon, Bowditch “staircase”
Intracardiac — nervous	Reflexes of the intracardiac (metasympathetic) nervous system	Peripheral intracardiac reflexes
Extracardiac — nervous	Autonomic nervous system (vagus and sympathetic nerves), central reflexes	Baroreceptor reflex, Goltz reflex
Extracardiac — humoral	Hormones, mediators, ions, metabolites	Effects of adrenaline, thyroxine, K^+, Ca^{2+}

2. Intracardiac regulatory mechanisms

Heterometric regulation (Frank–Starling law)

Frank–Starling law: the more the myocardium is stretched in diastole (the more blood enters the ventricle), the more strongly it contracts in the next systole. Mechanism — the sarcomere approaches its optimal length (2.2 μm), the area of interaction between actin and myosin filaments increases, and the Ca^{2+} sensitivity of troponin C rises.

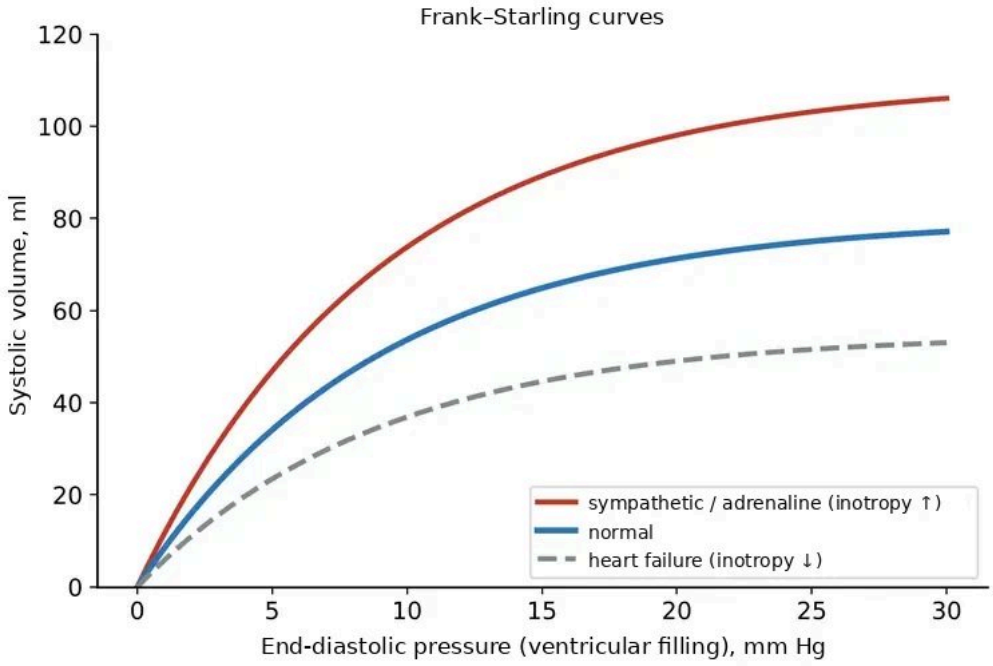


Figure 12.1. *Frank–Starling curves and their shift under inotropic influences (drawn after Costanzo L.S., 2018)*

> **REMEMBER**
> The Frank–Starling law **equalizes the work of the two ventricles**: if the right ventricle ejects more blood, more blood comes from the lungs to the left ventricle, and it too contracts more strongly. As a result the blood volumes in the systemic and pulmonary circulations remain balanced. **"The more the heart receives, the more it ejects."**

Homeometric regulation

Change in the force of contraction without a change in the initial length of the myocardial fiber:

- **Anrep phenomenon** — when the pressure (resistance) in the aorta rises, the force of ventricular contraction increases gradually and the stroke volume is maintained.
- **Bowditch "staircase"** (rate dependence) — when the contraction frequency increases, the force of each contraction increases step by step (accumulation of Ca^{2+} inside the cell).

Peripheral intracardiac reflexes

Inside the heart there is an **intracardiac (metasympathetic) nervous system** consisting of afferent, intercalary and efferent neurons; it carries out reflexes even without the central nervous system. For example, when the right atrium is stretched, contraction of the left ventricle increases (when the atrium is underfilled — the reverse). These reflexes are preserved even after heart transplantation.

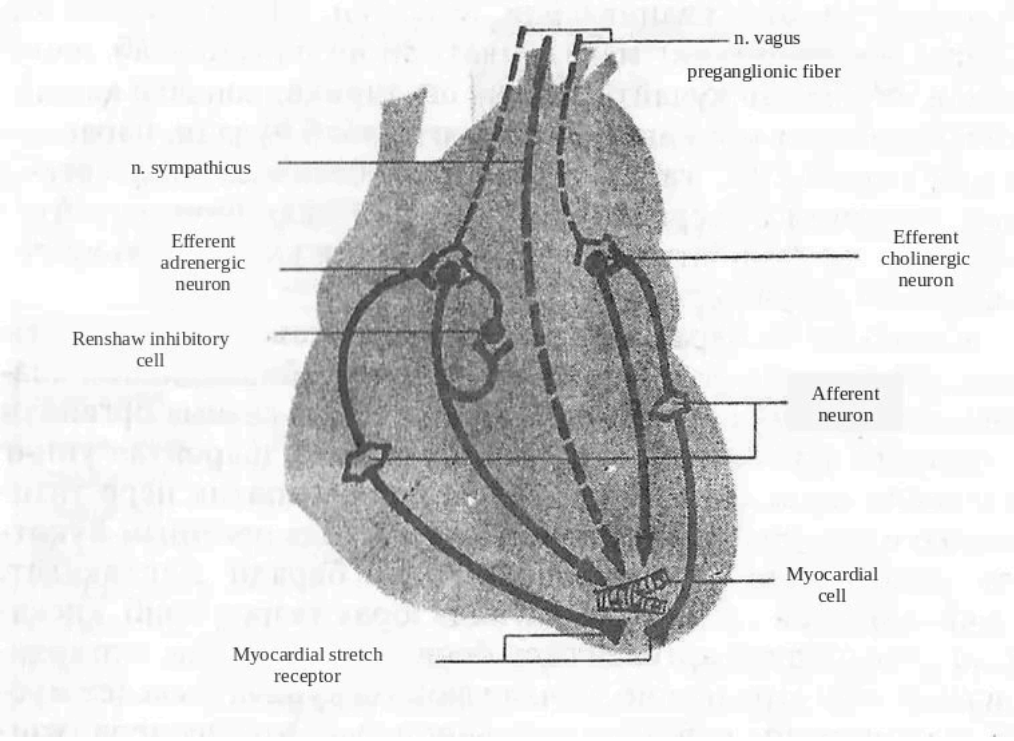


Figure 12.2. *The intracardiac nervous system: adrenergic and cholinergic efferent neurons, an afferent neuron and myocardial stretch receptors (Alavia O.T., 2018)*

3. Nervous regulation of the heart

The heart has **dual autonomic innervation**.

Parasympathetic (vagus nerve) influence

The preganglionic neurons are located in the **medulla oblongata** (the dorsal nucleus of the vagus and the nucleus ambiguus); the postganglionic neurons are inside the heart. The mediator is **acetylcholine**, the receptors are **M_2-cholinergic receptors**. The right vagus acts mainly on the sinus node, the left on the AV node.

Experiment of I. and E. Cyon (1866): stimulation of the vagus nerve in the frog slows and stops the heart. With prolonged stimulation the heart begins to contract again — **"escape" from the vagal effect** (lower pacemakers become active).

Mechanism: acetylcholine activates a G protein via M_2 receptors and **opens K^+ channels ($I_{K\text{-}ACh}$)** — the membrane hyperpolarizes; it inhibits adenylate cyclase and lowers cAMP — the I_f and Ca^{2+} currents decrease; slow diastolic depolarization slows down.

Sympathetic influence

The preganglionic neurons are in the lateral horns of the upper thoracic segments of the spinal cord (Th_1–Th_5); the postganglionic neurons are in the cervical and upper thoracic sympathetic ganglia (stellate ganglion). The mediator is **noradrenaline**, the receptors are **β_1-adrenergic receptors**. Sympathetic nerves go to all parts of the heart, including the ventricles.

Mechanism: β_1 receptors → G_s protein → adenylate cyclase → **cAMP** ↑ → protein kinase A → L-type Ca^{2+} channels and I_f channels are phosphorylated, and the return of Ca^{2+} to the SR is accelerated (phospholamban).

Table 12.2. *Effects of the autonomic nerves on the heart*

Type of effect	Meaning	Vagus nerve	Sympathetic nerve
Chronotropic	Heart rate	Negative (decreases)	Positive (increases)
Inotropic	Force of contraction	Negative (mainly in the atria)	Positive
Dromotropic	Conduction (especially in the AV node)	Negative	Positive
Bathmotropic	Excitability	Negative	Positive
Tonotropic	Tone of cardiac muscle	Negative	Positive
Lusitropic	Rate of relaxation	—	Positive (relaxation accelerates)

MEMORY AID
Five effects — **“CIDBT”**: **C**hronotropic (rate), **I**notropic (force), **D**romotropic (conduction), **B**athmotropic (excitability), **T**onotropic (tone). Sympathetic — all “+”, vagus — all “−”.

Tone of the cardiac centers

At rest the vagal centers have constant (tonic) activity — **vagal tone**. Therefore the intrinsic rate of the sinus node (≈ 100 /min — its “own rhythm”) is normally slowed to 60–80 /min. When the vagus is cut or atropine (an M-cholinergic blocker) is given, the heart rate increases. Athletes have high vagal tone — bradycardia at rest. In newborns vagal tone is low — HR 120–140 /min.

Respiratory sinus arrhythmia

HR increases on inspiration and decreases on expiration. The reason is that during inspiration the lung stretch receptors and the activity of the respiratory center reduce the tone of the vagal center. This is a **physiological** condition and is well expressed in young people.

4. Cardiac reflexes

Table 12.3. *Reflexes affecting cardiac activity*

Reflex	Receptors and stimulus	Response	Type
Baroreceptor (depressor) reflex	Aortic arch and carotid sinus; rise in pressure	HR and force of contraction decrease, vessels dilate	Intrinsic, negative feedback
Bainbridge reflex	Right atrium and orifices of the venae cavae; increased venous return	HR increases	Intrinsic
Goltz reflex	Abdominal organs; a blow to the abdomen or pinching of the intestine (in the frog)	The heart slows or stops	Conjugate (via the vagus)
Dagnini–Aschner (oculocardiac) reflex	Pressure on the eyeballs (trigeminal nerve)	HR decreases by 4–12 /min	Conjugate (via the vagus)
Chemoreceptor reflex	Carotid and aortic bodies; hypoxia, hypercapnia	HR and pressure increase	Conjugate
Pain and cold reflex	Skin receptors	Increase in HR (sometimes slowing)	Conjugate

Intrinsic reflexes start from the receptors of the cardiovascular system itself; **conjugate reflexes** from the receptors of other systems.

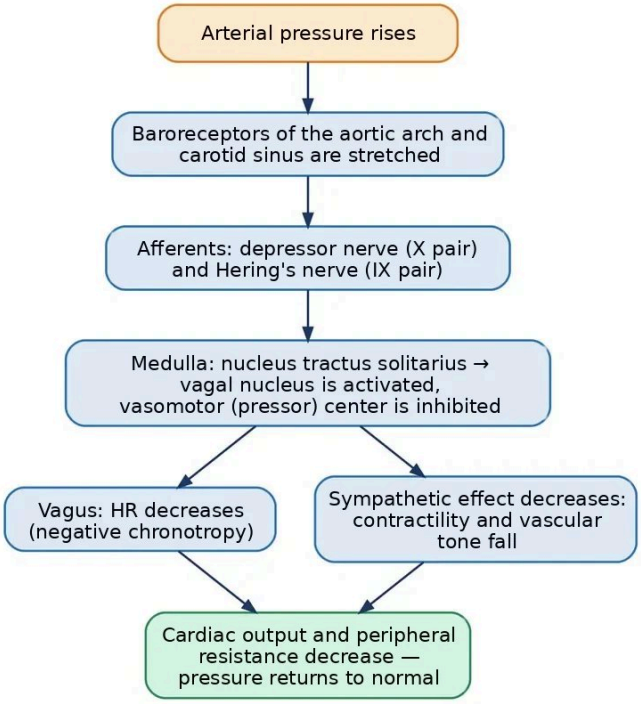


Figure 12.3. *Regulation of arterial pressure through the baroreceptor reflex*

CLINICAL SIGNIFICANCE

An attack of **paroxysmal supraventricular tachycardia** can sometimes be stopped by **vagal maneuvers**: the Valsalva maneuver (straining against a closed glottis), immersing the face in cold water, carotid sinus massage. They raise vagal tone and slow AV conduction. Pressure on the eyeballs (Dagnini–Aschner) is no longer recommended because of the risk of eye injury. In **carotid sinus hypersensitivity** a tight collar or a sharp turn of the head can cause bradycardia and fainting.

Influence of the cerebral cortex

Cardiac activity changes under the influence of emotions (fear, anger, excitement) and conditioned reflexes (an increase in HR in an athlete before the start). The hypothalamus and limbic system control the sympathetic and parasympathetic centers.

5. Humoral regulation of cardiac activity

Table 12.4. *Effects of humoral factors on the heart*

Factor	Effect
Adrenaline, noradrenaline	Positive chrono-, ino-, dromo- and bathmotropic effects via β_1
Thyroid hormones (T_3, T_4)	Increase the number of β-adrenergic receptors, increase HR and force of contraction (tachycardia in thyrotoxicosis)
Glucagon	Positive inotropic (via cAMP)
Corticosteroids	Enhance the effect of catecholamines
Acetylcholine	Negative effects via M_2

Factor	Effect
Natriuretic peptide (atrial)	Released when the atria are stretched; increases diuresis and natriuresis, dilates vessels
Increase in K^+ (hyperkalemia)	Excitability and conduction decrease; at high concentration the heart stops in diastole
Decrease in K^+ (hypokalemia)	Excitability increases, arrhythmias
Increase in Ca^{2+}	Force of contraction increases; at high concentration the heart stops in systole
Hypoxia, acidosis	Force of contraction decreases
Rise in temperature	HR increases (≈ 10 /min per 1 °C)

REMEMBER

Potassium stops the heart in **diastole**, and **calcium** in **systole**. That is why cardioplegic solutions (used to stop the heart temporarily during heart surgery) are high in potassium.

6. Laboratory work

Exercise 1: effect of vagus nerve stimulation on the frog heart

1. Immobilize the frog, expose the heart and isolate the vagosympathetic trunk in the neck.
2. Record the heart contractions with a kymograph.
3. Stimulate the vagosympathetic trunk with a rhythmic electric current (weak and strong) for 10–20 s.
4. Continue the stimulation and observe the “escape from the vagal effect”. **Result and analysis:** with weak stimulation the heart rate slows and the force of contraction decreases (negative chrono- and inotropic effects); with strong stimulation the heart stops in diastole. With prolonged stimulation the heart begins to contract again (escape). After stimulation is stopped, the rhythm is restored.

Exercise 2: the Goltz reflex (in the frog)

1. Open the frog's abdominal cavity and count the heart contractions.
2. Tap the abdominal organs (stomach, intestine) lightly several times with the handle of forceps.
3. Record the change in HR. **Result and analysis:** the heart slows or stops temporarily. Afferent impulses from the abdominal organs (celiac plexus) go to the medulla oblongata and excite the vagal center. After the vagus nerve is cut or atropine is given, the reflex disappears.

Exercise 3: the Dagnini–Aschner oculocardiac reflex in humans

1. The subject rests lying down for 5 minutes; the pulse is counted for 15 s and converted to 1 minute.
2. With the eyes closed, light pressure is applied to the eyeballs from both sides for 20–30 s (the pressure should not cause pain).
3. The pulse is counted again during the pressure (the last 15 s). **Interpretation:** a decrease in pulse of 4–12 /min — normal; more than 12 — high parasympathetic tone (vagotonia); no decrease or an increase — predominance of sympathetic tone (sympathicotonia).

Exercise 4: the orthostatic test

1. The subject's pulse is counted while lying for 5 minutes.
2. The subject stands up quickly; the pulse is counted after 1 minute.

Interpretation: an increase in pulse of 10–16 /min — normal (in the upright position venous return decreases, impulses from the baroreceptors decrease and the sympathetic influence increases). An increase of more than 20 /min — high sympathetic tone or low orthostatic stability.

7. Practical skills

Skill: evaluating autonomic tone

1. Determine the HR and blood pressure at rest.
2. Perform the Dagnini–Aschner reflex.
3. Perform the orthostatic test.
4. Evaluate respiratory sinus arrhythmia (the difference in HR during deep inspiration and expiration).
5. Summarize the results and determine the type of autonomic tone (eutonia, vagotonia, sympathicotonia).

Skill: detecting respiratory sinus arrhythmia on the ECG

1. Record the ECG with deep inspiration and expiration.
2. Measure the RR intervals during inspiration and expiration.
3. If the RR difference is more than 0.15 s and the P waves are unchanged — respiratory sinus arrhythmia (physiological).

Review questions

1. At what levels is cardiac activity regulated?
2. State the Frank–Starling law and explain its mechanism.
3. What are the Anrep phenomenon and the Bowditch staircase?
4. How are the peripheral intracardiac reflexes carried out?
5. List the types of effect of the vagus nerve on the heart.
6. What is the cellular mechanism of the vagal effect?
7. How do the sympathetic nerves affect the heart?
8. What is vagal tone, and how is it proved?
9. What is the cause of respiratory sinus arrhythmia?
10. List the intrinsic and conjugate reflexes of the heart.
11. How is the baroreceptor reflex carried out?
12. Explain the Goltz and Dagnini–Aschner reflexes.
13. Which hormones enhance cardiac activity?
14. How do K^+ and Ca^{2+} ions affect cardiac activity?
15. How is the orthostatic test performed and evaluated?

Test questions

1. According to the Frank–Starling law, what does the force of contraction depend on?
   A) On the heart rate
   B) On the degree of stretching of the myocardium in diastole
   C) On the pressure in the aorta
   D) On vagal tone
   E) On blood glucose

2. What are the mediator and receptor of the vagus nerve in the heart?
   A) Noradrenaline, β_1
   B) Adrenaline, α_1
   C) Acetylcholine, N-cholinergic receptor
   D) Acetylcholine, M_2-cholinergic receptor
   E) Serotonin, 5-HT

3. What does a positive chronotropic effect mean?
   A) An increase in the force of contraction
   B) An increase in conduction
   C) An increase in heart rate
   D) An increase in excitability
   E) An increase in tone

4. How does HR change when atropine is given, and why?
   A) It decreases — the sympathetic influence is blocked
   B) It does not change
   C) It decreases — the vagus is enhanced
   D) It increases — vagal tone is blocked
   E) The heart stops

5. What change does the vagus nerve cause in sinus node cells?
   A) Opening of K^+ channels, hyperpolarization and slowing of diastolic depolarization
   B) Increased Ca^{2+} entry
   C) An increase in cAMP
   D) Opening of Na^+ channels
   E) Depolarization

6. Through which receptors do the sympathetic nerves act on the heart?
   A) α_2
   B) β_1
   C) M_2
   D) N
   E) β_3

7. What is the gradual increase in the force of ventricular contraction when aortic pressure rises called?
   A) The Frank–Starling law
   B) The Bowditch staircase
   C) The Anrep phenomenon
   D) Bainbridge reflex
   E) Goltz reflex

8. What response does the baroreceptor reflex produce when arterial pressure rises?
   A) HR increases
   B) HR decreases and vessels dilate
   C) Vessels constrict
   D) Force of contraction increases
   E) There is no response

9. How does the pulse normally change when the eyeballs are pressed in the Dagnini–Aschner reflex?
   A) It increases by 20–30 /min
   B) It does not change
   C) It decreases by more than 30 /min
   D) It increases by 10–16 /min
   E) It decreases by 4–12 /min

10. What is the Bainbridge reflex?
    A) An increase in HR when the right atrium and venae cavae are stretched
    B) A decrease in HR when the eyeballs are pressed
    C) Cardiac arrest after a blow to the abdomen
    D) A decrease in HR when pressure rises
    E) An increase in HR in hypoxia

11. How does the heart stop when blood K^+ rises sharply?
    A) In systole
    B) With fibrillation
    C) It does not stop
    D) In a state of tetanus
    E) In diastole

12. In respiratory sinus arrhythmia, why does HR increase on inspiration?
    A) The sympathetic nerves are stimulated
    B) Adrenaline is released
    C) The tone of the vagal center decreases
    D) Blood K^+ decreases
    E) Hypoxia develops

13. What is tachycardia in thyrotoxicosis associated with?
    A) An increase in vagal tone
    B) Hypocalcemia
    C) A decrease in blood cortisol
    D) An increase in acetylcholine
    E) Thyroid hormones increasing the number of β-adrenergic receptors and enhancing metabolism

14. How does the pulse normally change in the orthostatic test?
    A) It increases by 10–16 /min
    B) It decreases by 4–12 /min
    C) It increases by more than 30 /min
    D) It does not change
    E) It decreases by half

15. Through which nerve is the Goltz reflex carried out?
    A) Sympathetic nerve
    B) The phrenic nerve
    C) The trigeminal nerve
    D) Vagus nerve
    E) The glossopharyngeal nerve

Clinical cases

Case 1

After heart transplantation, a patient's resting HR is 95–100 /min. During physical load the HR increases slowly, but the stroke volume increases well.

Questions: Why is the HR high at rest? By what mechanisms does the heart increase its work during exercise?

Answer: The transplanted heart is denervated — the vagus and sympathetic nerves do not reach it. Normally at rest vagal tone acts on the heart and slows the intrinsic rhythm of the sinus node (≈ 100 /min) to 60–80 /min; after denervation this influence is absent, so the HR is close to the intrinsic rhythm. During exercise the increase in cardiac work is provided by intracardiac mechanisms (the Frank–Starling law: venous return increases, the ventricles are stretched more and contract more strongly; intracardiac reflexes) and by humoral factors (adrenaline released from the adrenal glands, which acts more slowly).

Case 2

A 20-year-old young man was examined because his heart was “beating fast”: at rest HR 72 /min, 88 on deep inspiration, 64 /min on expiration. On the ECG every complex is preceded by a P wave of unchanged shape.

Questions: What is the condition? Explain its mechanism. Is it a pathology?

Answer: Respiratory sinus arrhythmia is a physiological condition, especially well expressed in young people. During inspiration the stretch receptors of the lungs and the activity of the respiratory center lower the tone of the vagal centers; in addition, venous return increases and the Bainbridge reflex is triggered — HR increases. During expiration vagal tone is restored and HR decreases. The rhythm comes from the sinus node (the P waves are identical) — it is not a pathology.

Case 3

A nurse mistakenly gave a rapid intravenous injection of potassium chloride solution to a patient with kidney failure. The patient developed bradycardia, followed by cardiac arrest.

Questions: Explain the mechanism of the effect of potassium on the heart. In which phase does the heart stop?

Answer: When blood K^+ rises sharply, the resting potential of the cardiomyocyte membrane decreases (depolarization) and the fast Na^+ channels are inactivated — excitability and conduction decrease and action potentials can no longer be generated. Excess K^+ also slows diastolic depolarization in the sinus node. As a result, bradycardia, conduction blocks and cardiac arrest in diastole develop. That is why potassium preparations are given intravenously only slowly, diluted and under ECG monitoring.

Quick review

Table 12.5. *Key concepts of topic 12*

Concept	Brief definition
The Frank–Starling law	Greater filling — stronger contraction; equalizes the work of the two ventricles
The Anrep phenomenon	Increase in contractile force when resistance rises
The Bowditch staircase	Increase in contractile force when the rate rises
Intracardiac nervous system	Metasympathetic; preserved after transplantation
Vagus nerve	Acetylcholine, M_2; all effects negative
Sympathetic nerve	Noradrenaline, β_1, cAMP; all effects positive
Chronotropic / inotropic	HR / force of contraction
Dromotropic / bathmotropic	Conduction / excitability
Vagal tone	Slows the HR at rest; proved with atropine
Respiratory arrhythmia	HR increases on inspiration; physiological
Baroreceptor reflex	Pressure ↑ → HR ↓, vessels dilate
Goltz reflex	Blow to the abdomen → cardiac arrest (vagus)
Dagnini–Aschner	Pressure on the eyes → pulse decreases by 4–12 /min
K^+ / Ca^{2+}	Stop the heart in diastole / systole
Orthostatic test	On standing the pulse rises by 10–16 /min

TOPIC 13

Arterial blood pressure and peripheral pulse. Microcirculation. Blood flow in the veins. Regulation of vascular tone

Learning objectives

To teach the student the basic laws of hemodynamics, the functional groups of blood vessels, arterial blood pressure and the factors that determine it, the arterial pulse and its characteristics, blood flow and exchange of substances in the microcirculatory bed, the mechanisms of blood flow in the veins, and the local, nervous and humoral regulation of vascular tone; to master measuring arterial pressure by the Korotkov method and examining the pulse.

By the end of the class the student should know:

- the basic laws of hemodynamics (Poiseuille's law, continuity of flow);
- the functional classification of blood vessels;
- the types and norms of arterial pressure and the factors that determine it;
- the characteristics of the arterial pulse and the sphygmogram;
- the structure of the microcirculatory bed, filtration and reabsorption in capillaries (Starling equilibrium);
- the factors that ensure venous return;
- the basal, myogenic, nervous and humoral regulation of vascular tone;
- the vasomotor center and the reflex regulation of blood pressure.

The student should be able to:

- measure arterial pressure by the Korotkov method;
- calculate the pulse pressure and mean arterial pressure;
- examine the pulse on the radial artery and evaluate its characteristics.

1. Basic laws of hemodynamics

The main cause of blood movement in the vessels is the **pressure difference** between the initial (aorta) and final (venae cavae) parts of the vascular system. Blood flow (Q) is directly proportional to the pressure difference (ΔP) and inversely proportional to vascular resistance (R): **Q = ΔP / R**.

According to **Poiseuille's law**, resistance $R = 8\eta l / \pi r^4$ — it is directly proportional to viscosity (η) and vessel length (l) and inversely proportional to **the fourth power of the radius**. Therefore, if the radius of a vessel decreases 2 times, the resistance increases 16 times. By changing the radius of its vessels the body regulates blood flow very effectively.

Continuity of flow: the same volume of blood passes through every cross-section of the vascular system per unit time. The **linear velocity** is inversely proportional to the **total cross-sectional area** of the vessels: 50 cm/s in the aorta, 0.05 cm/s in the capillaries (total cross-section 500–800 times larger), and it rises again in the veins (20–25 cm/s in the venae cavae). The circulation time (one complete circuit) is ≈ 20–23 s.

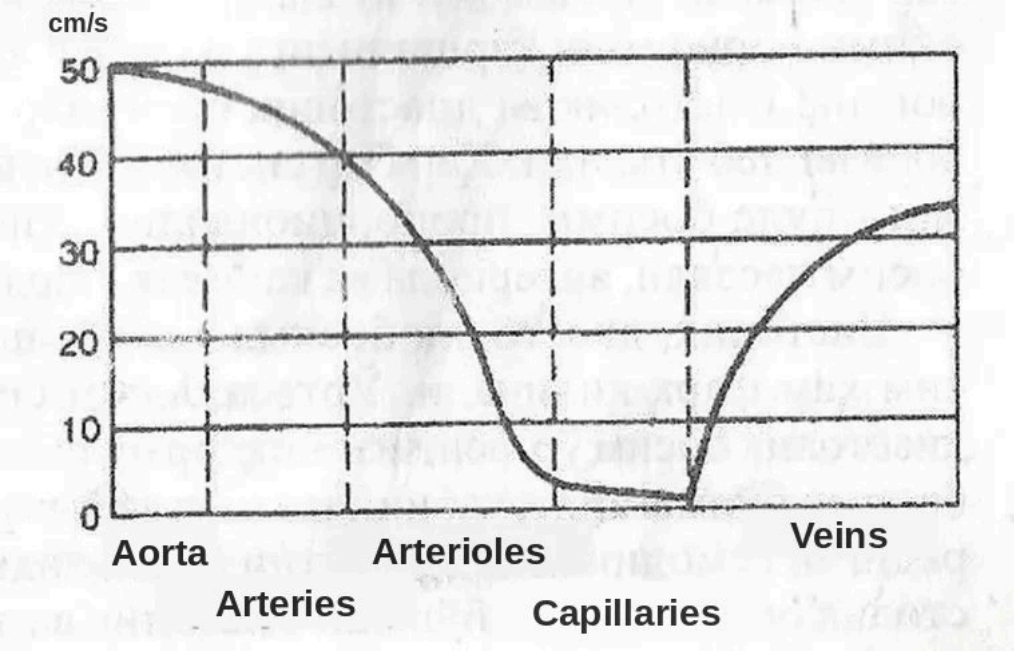


Figure 13.1. *Linear velocity of blood flow in different parts of the vascular system (Alavia O.T., 2018)*

Table 13.1. *Functional classification of blood vessels (B. Folkow)*

Group	Vessels	Function
Shock-absorbing (elastic)	Aorta, pulmonary artery, large arteries	Stretch in systole and recoil in diastole, turning intermittent ejection into continuous flow (“Windkessel”)
Resistance (resistive)	Small arteries, arterioles	≈ 50% of peripheral resistance; distribute blood among organs
Sphincters	Precapillary sphincters	Regulate the number of open capillaries
Exchange	Capillaries	Exchange of substances between blood and tissue
Capacitance (collecting)	Veins	Hold 70–80% of the blood; regulate venous return
Shunting	Arteriovenous anastomoses	Carry blood past the capillaries (skin, thermoregulation)

2. Arterial blood pressure

Arterial blood pressure is the pressure that blood exerts on the walls of the arteries.

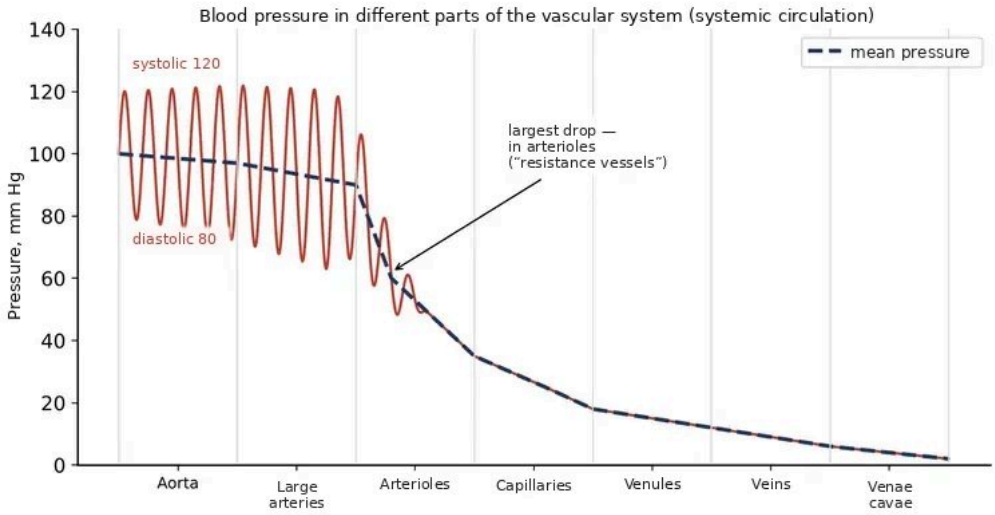


Figure 13.2. *Blood pressure in different parts of the vascular system*

Table 13.2. *Types of arterial pressure (brachial artery, adults)*

Type	Definition	Normal value
Systolic (maximum)	The highest pressure during ventricular systole	110–129 mm Hg
Diastolic (minimum)	The lowest pressure at the end of diastole	70–84 mm Hg
Pulse pressure	Systolic − diastolic	40–50 mm Hg
Mean dynamic pressure	Diastolic + 1/3 of pulse pressure	85–95 mm Hg

CAUTION

The criteria for arterial hypertension differ between international guidelines. According to WHO and European guidelines, an office blood pressure of **140/90 mm Hg or higher** is arterial hypertension; 120–139/80–89 is high-normal or elevated pressure. In the American (ACC/AHA, 2017) guidelines the threshold for hypertension is 130/80. Arterial hypotension is below 90/60.

Factors that determine arterial pressure:

1. **Minute volume (cardiac work)** — affects mainly systolic pressure.
2. **Total peripheral resistance** (arteriolar tone) — affects mainly diastolic pressure.
3. **Circulating blood volume**.
4. **Blood viscosity**.
5. **Elasticity of the walls of the large arteries** — in the elderly elasticity decreases, systolic pressure rises and the pulse pressure widens. **Mean arterial pressure = minute volume × total peripheral resistance.**

Physiological changes: it increases with age; rises during physical work, excitement and pain; falls during sleep; changes slightly on standing; fluctuates over the day (circadian rhythm).

Methods of measuring arterial pressure

- **Direct (invasive)** method — a cannula is inserted into an artery and connected to a manometer (S. Hales, 1733); today it is used as invasive monitoring in intensive care.
- **Indirect (non-invasive)** methods: the **Riva-Rocci** palpatory method (systolic pressure only) and **N.S. Korotkov's auscultatory method** (1905) — systolic and diastolic pressure; also the oscillometric method (automatic blood pressure monitors).

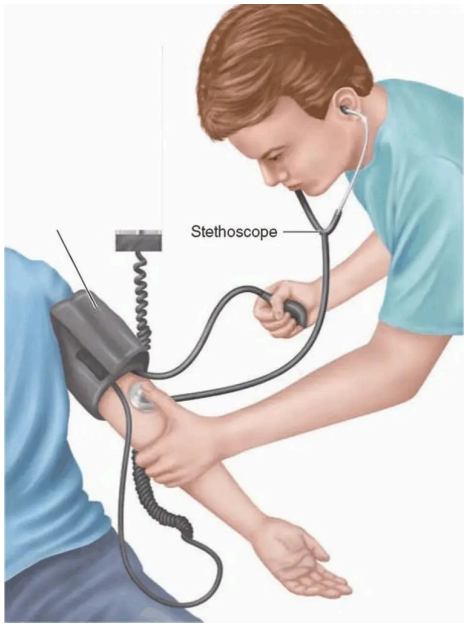


Figure 13.3. *Measuring arterial pressure by the Korotkov method (Alavia O.T., 2018)*

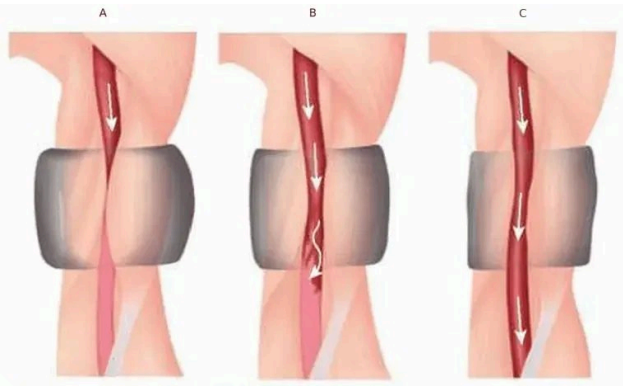


Figure 13.4. *Mechanism of the Korotkov sounds: A — cuff pressure above systolic, the artery is compressed; B — pressure between systolic and diastolic, blood flows turbulently (sounds are heard); C — pressure below diastolic, laminar flow (sounds disappear) (Alavia O.T., 2018)*

3. The arterial pulse

The **arterial pulse** is the rhythmic oscillation of the arterial walls caused by the ejection of blood into the aorta during cardiac systole. The pulse wave travels much faster than the blood itself: 4–6 m/s in the aorta, 8–12 m/s in the peripheral arteries (the velocity increases when the wall stiffens).

The **sphygmogram** is a graphic recording of the pulse. It shows the **anacrotic limb** (the ascending part — ejection of blood), the **catacrotic limb** (the descending part) and on it the **dicrotic wave** (closure of the aortic valves and the rebound of blood from them; preceded by the incisura).

Table 13.3. *Characteristics of the pulse (examined on the radial artery)*

Characteristic	What it indicates	Normal value
Rate	Number of beats per minute	60–80 /min; more — tachycardia, fewer — bradycardia
Rhythm	Equality of the intervals between beats	Rhythmic
Tension	The pressure needed to stop the pulse (depends on arterial pressure)	Satisfactory; a “hard” pulse at high pressure
Filling	Filling of the artery with blood (depends on stroke volume)	Well filled
Size	Amplitude of the pulse wave	Medium
Shape (speed)	Rate of rise and fall of pressure	Normal; “quick and high” in aortic valve insufficiency
Symmetry on both sides	—	Equal

CLINICAL SIGNIFICANCE

Pulse deficit — the heart rate (on auscultation) is greater than the pulse rate; in atrial fibrillation some contractions are so weak that the pulse wave does not reach the periphery. A difference in the pulse between the two arms may indicate narrowing of the aorta or its branches.

4. Microcirculation

The **microcirculatory bed** is a functional unit consisting of arterioles, metarterioles, precapillary sphincters, capillaries, venules and arteriovenous anastomoses. At rest only 25–35% of capillaries are open; in a working organ their number increases.

The **capillary** wall consists of a single layer of endothelium; its length is 0.5–1 mm, its diameter 5–10 µm, and blood stays in it for 1–2 s. Types of capillaries: **somatic** (continuous — muscle, skin, lung; in the CNS the blood–brain barrier), **visceral** (fenestrated — renal glomeruli, intestine, endocrine glands) and **sinusoidal** (discontinuous — liver, spleen, bone marrow).

Mechanisms of exchange of substances: diffusion (the main one — O_2, CO_2, glucose), filtration and reabsorption (water and substances dissolved in it), micropinocytosis (large molecules).

Starling equilibrium

Fluid movement across the capillary wall is determined by the balance of four forces: the **hydrostatic pressure** in the capillary (pushes fluid out), the **oncotic pressure** of plasma (holds fluid in), and the hydrostatic and oncotic pressures of tissue fluid.

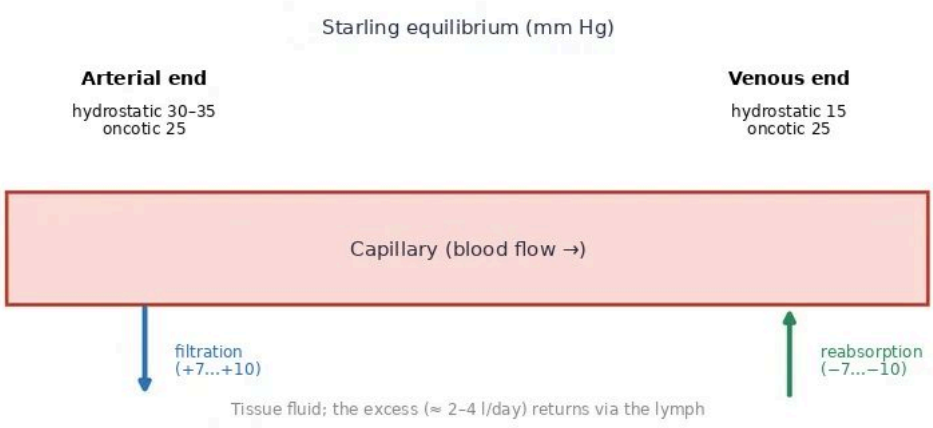


Figure 13.5. *Filtration and reabsorption in a capillary (Starling equilibrium)*

At the arterial end of the capillary the hydrostatic pressure exceeds the oncotic pressure — **filtration**; at the venous end the hydrostatic pressure has fallen — **reabsorption**. Filtration slightly exceeds reabsorption; the excess fluid and proteins return to the blood through the **lymphatic vessels** (2–4 l of lymph per day).

> **CLINICAL SIGNIFICANCE**
> Edema develops when the Starling equilibrium is disturbed: **increased hydrostatic pressure** (heart failure, venous thrombosis), **decreased oncotic pressure** (hypoalbuminemia — liver cirrhosis, nephrotic syndrome, starvation), **increased capillary permeability** (inflammation, allergy — histamine) and **impaired lymph flow** (lymphostasis).

5. Blood flow in the veins

Pressure in the veins is low (15–20 mm Hg in venules, 0–5 mm Hg in the venae cavae, negative during inspiration). Factors that ensure venous return:

1. **The residual force of the heart** (vis a tergo — the force pushing from behind) — the main factor.

2. **The suction effect of the chest** — during inspiration the negative pressure in the thoracic cavity increases, dilating the venae cavae and drawing in blood.
3. **Movement of the diaphragm** — during inspiration intra-abdominal pressure rises and squeezes blood out of the abdominal veins.
4. **The skeletal muscle "pump"** — when muscles contract the veins are compressed, and the valves allow blood to flow only toward the heart.
5. **Venous valves** (especially in the leg veins).
6. **The suction effect of the heart** — during ventricular systole the downward displacement of the AV boundary dilates the atria.
7. **Pulsation of the arteries** — compresses the adjacent veins.

> **CLINICAL SIGNIFICANCE**
> During prolonged standing and immobility blood pools in the leg veins; with valve insufficiency **varicose disease** develops. During prolonged bed rest and travel venous flow slows and the risk of **deep vein thrombosis** increases — therefore early mobilization, leg exercises and compression stockings are recommended.

6. Regulation of vascular tone

Vascular tone is the constant state of partial contraction of vascular smooth muscle. It consists of two parts:

- **Basal (myogenic) tone** — independent of innervation; the automaticity of smooth muscle and its response to stretch (the **Bayliss effect**: when pressure rises and the vessel is stretched, it contracts). This is the basis of **autoregulation**: in the kidney, brain and heart blood flow is kept constant even when pressure changes within 60–160 mm Hg.
- **Neurohumoral tone** — under the influence of autonomic nerves and humoral factors.

Local (metabolic) regulation

In working tissue a decrease in O_2, an increase in CO_2, H^+, lactate, adenosine and K^+, and a rise in temperature dilate the arterioles — **working (functional) hyperemia**. The endothelium produces **nitric oxide (NO)**, prostacyclin (vasodilators) and **endothelin** (a powerful vasoconstrictor). NO is released continuously under the action of the shear stress of blood flow.

Nervous regulation

- **Vasoconstrictors** — mainly **sympathetic** adrenergic fibers (noradrenaline, α_1-adrenergic receptors). They are tonically active: at rest they send impulses at 1–3 Hz. If the stimulation frequency increases, the vessels constrict; if it decreases, they dilate.
- **Vasodilators** — parasympathetic fibers (in the tongue, salivary glands, genital organs — acetylcholine, NO), sympathetic cholinergic fibers (in skeletal muscles), β_2-adrenergic receptors (in the vessels of skeletal muscle, heart and liver under the action of adrenaline), the axon reflex. The **vasomotor (circulatory) center** is in the **medulla oblongata** (F.V. Ovsyannikov, 1871): it has **pressor** (ventrolateral, exciting sympathetic neurons) and **depressor** parts. It is constantly tonically active; after transection above the medulla oblongata the pressure hardly changes, after transection below it the pressure falls sharply. Higher centers — the hypothalamus, limbic system and cerebral cortex (changes in pressure during emotions).

Reflex regulation of pressure

- **Depressor reflexes** — from the **baroreceptors** of the aortic arch and carotid sinus (topic 12): when pressure rises, sympathetic tone decreases and the vagus is activated — pressure falls. The baroreceptors provide short-term ("second-to-second") regulation.
- **Pressor reflexes** — from the carotid and aortic **chemoreceptors** (hypoxia, hypercapnia, acidosis) — pressure rises.

Humoral regulation

Table 13.4. *Humoral factors affecting vascular tone*

Constrictors (vasoconstrictors)	Dilators (vasodilators)
Adrenaline (via α_1 — skin, internal organs), noradrenaline	Adrenaline (via β_2 — skeletal muscle, heart)
Angiotensin II (renin–angiotensin–aldosterone system)	Nitric oxide (NO)
Vasopressin (antidiuretic hormone)	Histamine, bradykinin
Endothelin	Prostacyclin, prostaglandin E
Serotonin (in some vessels), thromboxane A_2	Natriuretic peptide
—	Adenosine, CO_2, H^+, lactate, K^+

REMEMBER

By the speed of regulation of arterial pressure there are three groups of mechanisms: **fast** (seconds — baroreceptor and chemoreceptor reflexes), **intermediate** (minutes–hours — the renin–angiotensin system, fluid shifts in capillaries, stress relaxation of vessels) and **long-term** (days — regulation of blood volume through the kidneys: aldosterone, vasopressin, natriuretic peptide, pressure natriuresis).

7. Laboratory work

Exercise 1: measuring arterial pressure by the Korotkov method

1. The subject sits quietly for 5 minutes; the arm is placed on the table at heart level, palm up.
2. The cuff is wrapped around the upper arm 2–2.5 cm above the elbow crease, loosely enough for one finger to fit under it.
3. The pulsation of the brachial artery is found in the cubital fossa and the stethoscope is placed there.
4. After the radial pulse disappears, the cuff is inflated a further 20–30 mm Hg.
5. The air is released slowly (2–3 mm Hg/s); the pressure at which the **first clear sound** appears is the **systolic pressure**.
6. The pressure at which the sounds **disappear completely** is the **diastolic pressure**.
7. The measurement is repeated after 2–3 minutes and the mean value is recorded; the first time, it is measured on both arms. **Calculate:** pulse pressure = systolic − diastolic; mean dynamic pressure = diastolic + pulse pressure / 3.

CAUTION

Measurement errors: a cuff that is too narrow (falsely high pressure in obese people), an arm positioned below heart level (reads high), deflating too fast (incorrect diastolic pressure), smoking or drinking coffee before measurement. **“Auscultatory gap”** — in some people (in hypertension) the sounds disappear temporarily after the systolic pressure and then reappear; therefore the cuff is inflated sufficiently under palpation control.

Exercise 2: examining the pulse

1. Palpate the radial artery of both arms in the region of the styloid process of the radius with three fingers (index, middle, ring).
2. Compare the pulse on the two sides.
3. Count for 30 s and multiply by 2 (if the rhythm is irregular — a full 1 minute).
4. Evaluate its rhythm, tension and filling.
5. Also examine the pulse on the carotid, femoral and dorsalis pedis arteries.

Exercise 3: effect of physical load on arterial pressure and pulse

1. Measure the pressure and pulse at rest.
2. The subject does 20 squats in 30 s (Martinet–Kushelevsky test).
3. Measure the pressure and pulse again immediately and every minute (for 3–5 minutes). **Interpretation:** normally the pulse increases by 50–75%, systolic pressure rises by 15–30 mm Hg, diastolic pressure does not change or falls slightly (vessels dilate in the working muscles), and the parameters recover within 3 minutes.

8. Practical skills

Skill: calculating hemodynamic parameters

1. Take the systolic and diastolic pressures.
2. Calculate the pulse pressure (normal 40–50).
3. Calculate the mean dynamic pressure.
4. If the minute volume is known, find the total peripheral resistance: R = mean pressure / minute volume.
5. Evaluate the results: a large pulse pressure — aortic valve insufficiency or reduced arterial elasticity in the elderly; a small one — aortic stenosis, blood loss.

Skill: analyzing the mechanism of edema by the Starling forces

1. Has the capillary hydrostatic pressure changed (venous stasis, heart failure)?
2. Has the plasma oncotic pressure changed (albumin level)?
3. Has capillary permeability increased (inflammation, allergy)?
4. Is lymph flow impaired?
5. Identify the predominant factor and name the type of edema.

Review questions

1. What is the main cause of blood movement in the vessels?
2. Explain Poiseuille's law; why is the vessel radius the most important factor?

3. What is the linear velocity of blood flow in different parts of the vascular system?
4. List the functional groups of blood vessels.
5. What are the types and norms of arterial pressure?
6. What factors does arterial pressure depend on?
7. What is the essence of the Korotkov method? How are the Korotkov sounds generated?
8. Explain the arterial pulse and the elements of the sphygmogram.
9. Which characteristics of the pulse are evaluated?
10. How is the microcirculatory bed structured?
11. How do filtration and reabsorption take place in the capillaries?
12. List the factors that ensure venous return.
13. What are basal tone and the Bayliss effect?
14. Where is the vasomotor center located, and how does it work?
15. List the humoral vasoconstrictor and vasodilator factors.

Test questions

1. According to Poiseuille's law, which factor has the strongest effect on vascular resistance?
   A) Blood viscosity
   B) Vessel length
   C) Heart rate
   D) Blood volume
   E) Vessel radius (to the fourth power)

2. Where is the linear velocity of blood flow lowest?
   A) In the aorta
   B) In the arteries
   C) In the venae cavae
   D) In the capillaries
   E) In the veins

3. Which vessels create the main part of the peripheral resistance?
   A) The aorta and large arteries
   B) Capillaries
   C) The arterioles
   D) Veins
   E) Arteriovenous anastomoses

4. Which factor mainly affects diastolic pressure?
   A) Total peripheral resistance
   B) Stroke volume
   C) Aortic elasticity
   D) Respiratory rate
   E) Venous return

5. How is systolic pressure determined by the Korotkov method?
   A) When the sounds disappear completely
   B) When the sounds are loudest
   C) When the radial pulse appears
   D) When the first clear sound appears
   E) When the cuff is fully inflated

6. If the pressure is 130/80 mm Hg, what is the mean dynamic pressure?
   A) 105
   B) ≈ 97
   C) 50
   D) 210
   E) 80

7. What is the dicrotic wave on the sphygmogram associated with?
   A) Atrial systole
   B) Rapid ventricular ejection
   C) Closure of the aortic valves
   D) Venous return
   E) Breathing

8. What predominates at the arterial end of a capillary?
   A) Reabsorption
   B) Oncotic pressure
   C) Only diffusion
   D) Filtration
   E) There is no movement

9. What is the main factor of venous return?
   A) The suction effect of the chest
   B) The skeletal muscle pump
   C) Movement of the diaphragm
   D) Venous valves
   E) The residual force of the heart (vis a tergo)

10. What is the Bayliss effect?
    A) Vasoconstriction under the influence of sympathetic nerves
    B) Vasodilation under the influence of adrenaline
    C) Contraction of a vessel when it is stretched by increased pressure (myogenic response)
    D) Vasodilation under the influence of NO
    E) Increased permeability under the influence of histamine

11. Where is the vasomotor center located?
    A) In the medulla oblongata
    B) In the spinal cord
    C) In the hypothalamus
    D) In the cerebellum
    E) In the cerebral cortex

12. Which is a powerful endothelial vasodilator factor?
    A) Endothelin
    B) Nitric oxide (NO)
    C) Angiotensin II
    D) Thromboxane A_2
    E) Vasopressin

13. Through which receptors does adrenaline dilate the vessels of skeletal muscle?
    A) α_1
    B) α_2
    C) M_2
    D) β_1
    E) β_2

14. In hypoalbuminemia, edema develops due to which factor?
    A) A decrease in plasma oncotic pressure
    B) An increase in capillary hydrostatic pressure
    C) Increased lymph flow
    D) Decreased capillary permeability
    E) A rise in arterial pressure

15. What is the normal pulse pressure?
    A) 10–20 mm Hg
    B) 40–50 mm Hg
    C) 80–90 mm Hg
    D) 100–120 mm Hg
    E) 120–140 mm Hg

Clinical cases

Case 1

A 72-year-old patient has an arterial pressure of 170/70 mm Hg. In youth his pressure was 120/80.

Questions: Calculate the pulse pressure and mean pressure. How can the rise in systolic and fall in diastolic pressure in old age be explained?

Answer: Pulse pressure = 170 − 70 = 100 mm Hg (normal 40–50). Mean pressure = 70 + 100/3 ≈ 103 mm Hg. In old age the walls of the aorta and large arteries lose their elastic fibers and stiffen (atherosclerosis, collagen accumulation). The shock-absorbing function weakens: in systole the stiff aorta dilates little — systolic pressure rises sharply; in diastole the aorta cannot recoil and push the blood onward — diastolic pressure falls. This condition is called isolated systolic hypertension and increases cardiovascular risk.

Case 2

A patient in the surgical ward has heart failure: edema of the legs and distended neck veins. Blood albumin is normal.

Questions: What is the main mechanism of the edema? Explain it on the basis of the Starling forces.

Answer: In heart failure the heart cannot pump the venous blood adequately — blood accumulates in the veins and venous pressure rises (distended neck veins). The increased venous pressure raises the hydrostatic pressure at the venous end of the capillaries; reabsorption decreases and filtration

increases. The oncotic pressure is normal (albumin is normal), so the main factor is the increase in hydrostatic pressure (congestive edema). Because of gravity, the edema appears in the lower parts of the body — the legs. In addition, sodium and water retention in the kidneys (RAAS activation) increases blood volume and aggravates the edema.

Case 3

After standing motionless in formation for a long time, a student's vision darkened and he lost consciousness. After he was laid down and his legs were raised, he quickly came round.

Questions: Explain the mechanism of the fainting. Why did raising the legs help?

Answer: During prolonged motionless standing, gravity causes blood to pool in the leg veins; the skeletal muscle pump does not work — venous return decreases. According to the Frank–Starling mechanism, ventricular filling and stroke volume decrease and arterial pressure falls. When compensation by the baroreceptor reflex (sympathetic activation) is insufficient, cerebral blood flow decreases and fainting (orthostatic syncope) develops. In the horizontal position with the legs raised, venous return is restored, stroke volume and cerebral blood flow increase — the patient quickly comes round.

Quick review

Table 13.5. *Key concepts of topic 13*

Concept	Brief definition
$Q = \Delta P / R$	Blood flow is directly proportional to the pressure difference and inversely proportional to resistance
Poiseuille's law	R is inversely proportional to the fourth power of the radius
Linear velocity	50 cm/s in the aorta, 0.05 cm/s in the capillaries
Resistance vessels	The arterioles
Systolic / diastolic	110–129 / 70–84 mm Hg
Pulse pressure	40–50 mm Hg
Mean pressure	Diastolic + 1/3 of pulse pressure
Korotkov method	First sound — systolic; disappearance of the sounds — diastolic
Dicrotic wave	Closure of the aortic valves
Starling equilibrium	Filtration at the arterial end, reabsorption at the venous end
Venous return	Vis a tergo, thoracic suction, muscle pump, valves
Bayliss effect	Myogenic contraction in response to stretch; autoregulation
Vasomotor center	Medulla oblongata (Ovsyannikov)
Vasoconstrictors	Sympathetic (α_1), angiotensin II, vasopressin, endothelin
Vasodilators	NO, prostacyclin, bradykinin, histamine, metabolites, β_2

TOPIC 14

Study of metabolism and energy exchange. Thermoregulation. Physiology of the endocrine glands. The hypothalamo-pituitary system

Learning objectives

To teach the student energy exchange in the body and the methods of measuring it, basal metabolism and daily energy expenditure, the constancy of body temperature and the mechanisms of thermoregulation, the general principles of the endocrine system, the classification of hormones and their mechanisms of action, and the structure and hormones of the hypothalamo-pituitary system; to master calculating basal metabolism and measuring body temperature.

By the end of the class the student should know:

- the methods of direct and indirect calorimetry of energy exchange, the respiratory quotient and the caloric equivalent of oxygen;
- the concept of basal metabolism, its magnitude and the factors affecting it;
- the specific dynamic action of food and daily energy expenditure;
- the norms of body temperature and its daily variations;
- the mechanisms of chemical and physical thermoregulation and the thermoregulatory center;
- the general properties, chemical classification and mechanisms of action of hormones;
- the regulation of endocrine gland activity and the feedback principle;
- the hormones of the hypothalamus and pituitary and their functions.

The student should be able to:

- determine the predicted basal metabolism using tables and formulas;
- calculate the deviation of basal metabolism from the norm using Read's formula;
- calculate energy expenditure from the results of the Douglas–Haldane method;
- measure body temperature and evaluate the temperature curve.

1. Energy exchange and methods of measuring it

Metabolism in the body consists of two processes: **assimilation (anabolism)** — synthesis of complex substances, which consumes energy; and **dissimilation (catabolism)** — breakdown of substances, which releases energy (topic 21). Ultimately all the energy that enters the body is converted into **heat** (apart from external mechanical work). Therefore energy expenditure can be estimated from the amount of heat released.

Table 14.1. *Methods of determining energy expenditure*

Method	Essence	Features
Direct calorimetry	The person stays in a special insulated chamber (Atwater–Benedict calorimeter); the heat released is measured by the warming of water flowing through the chamber wall	Most accurate, but complex and expensive
Indirect calorimetry — complete gas analysis	The O_2 consumed and the CO_2 released are measured (Douglas–Haldane method: the expired air is collected in a Douglas bag)	Calculated from the respiratory quotient and caloric equivalent

Method	Essence	Features
Indirect — incomplete gas analysis	Only the O_2 consumed is measured (spirometabolograph); the respiratory quotient is taken as 0.85	Widely used in the clinic
Calculation methods	Tables and formulas based on sex, age, height and weight (Harris–Benedict)	Quick, approximate

The **respiratory quotient (RQ)** is the ratio of the volume of CO_2 released to the volume of O_2 consumed. It depends on which substance is being oxidized.

Table 14.2. *Respiratory quotient and caloric equivalent of oxygen*

Substance oxidized	RQ	Caloric equivalent of 1 l O_2, kcal	Energy value of 1 g of substance, kcal (kJ)
Carbohydrates	1,0	5,05	4,1 (17,2)
Fats	0,7	4,69	9,3 (38,9)
Proteins	0,8	4,80	4,1 (17,2)
Mixed food	0,85–0,9	≈ 4,86–4,9	—

The **caloric equivalent of oxygen** is the amount of heat released when 1 l of O_2 is consumed. Energy expenditure = volume of O_2 consumed × caloric equivalent.

REMEMBER

The RQ can exceed 1.0 — **when fat is synthesized from carbohydrates** (in obesity) or in hyperventilation (when CO_2 is "washed out"), and after heavy physical work. An RQ below 0.7 — in prolonged starvation and diabetes mellitus (fats and ketone bodies are oxidized).

2. Basal metabolism

Basal metabolism is the minimum energy expenditure of the body needed for life (cellular metabolism, breathing, the work of the heart, kidneys and CNS, maintenance of body temperature). It is determined **under standard conditions** : in the morning, after waking, on an empty stomach (12–14 hours after the last meal), at physical and mental rest, lying down, at a **"comfort temperature"** (18–20 °C).

Magnitude: on average **1 kcal/kg/hour** (4.19 kJ); in a 70 kg man ≈ **1700 kcal/day** (7100 kJ), 10% less in women. Relative to body surface area: ≈ 40 kcal/m^2/hour. **Rubner's surface law**: in warm-blooded animals of different mass, basal metabolism is proportional not to body mass but to **body surface area**.

Factors affecting basal metabolism: sex (higher in men), age (relatively high in children, low in the elderly), height and weight, **thyroid hormones** (the strongest — it increases by up to 50–100% in thyrotoxicosis and decreases by 30–40% in hypothyroidism), temperature (in fever it increases by ≈ 10–13% for every 1 °C), season, nutrition, pregnancy.

CLINICAL SIGNIFICANCE

Read's formula gives an approximate estimate of the deviation of basal metabolism from the norm (%) from the pulse and arterial pressure: BM (%) = 0.75 × (pulse + pulse pressure × 0.74) − 72. Up to ± 10% is normal. This formula was used for screening thyroid function; today, for an accurate diagnosis, TSH and thyroid hormones are measured in the blood.

Specific dynamic action of food

After food intake energy expenditure increases (for digestion, absorption and metabolic transformations). With protein intake it reaches 30%, with carbohydrates 4–6%, with fats 2–4%; with mixed food — 10–15%.

Daily energy expenditure

Daily energy expenditure = basal metabolism + specific dynamic action + work increment (physical activity).

Table 14.3. *Daily energy expenditure by work intensity (approximate)*

Group	Occupations	Daily expenditure, kcal
I — mainly mental work	Researchers, students, doctors	2000–2600
II — light physical work	Services, automated production	2500–2800
III — moderate work	Lathe operator, driver, surgeon	2700–3000
IV — heavy physical work	Construction workers, agriculture	3000–3600
V — very heavy	Miners, porters, athletes	3500–4500 and more

The **physical activity ratio** is the ratio of daily expenditure to basal metabolism (1.4 in group I; 2.4 in group V).

3. Thermoregulation

Humans are **homeothermic (warm-blooded)** organisms: the core body temperature is kept relatively constant regardless of the ambient temperature. The body **core** (internal organs, brain — ≈ 37 °C) and **shell** (skin, subcutaneous tissue, limbs — variable temperature) are distinguished.

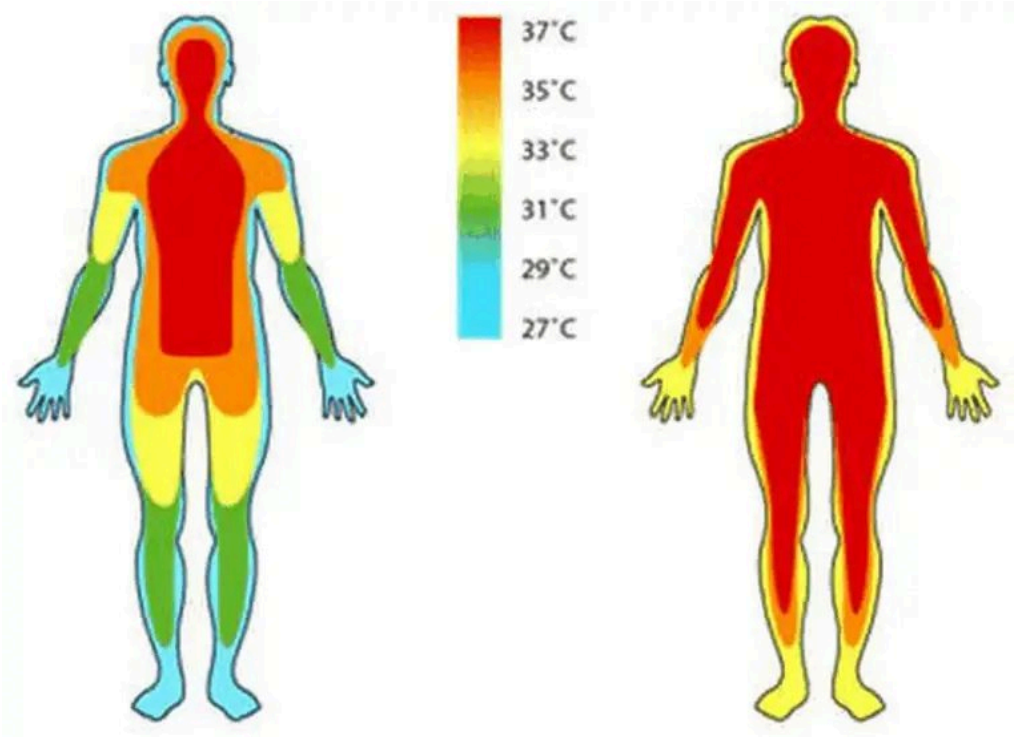


Figure 14.1. *Temperature of the body core and shell in a cold (left) and hot (right) environment (Alavia O.T., 2018)*

Table 14.4. *Normal values of body temperature*

Site of measurement	Norm, °C
Axilla	36,4–36,9
Oral cavity (sublingual)	36,8–37,3
Rectum	37,2–37,6
Brain, liver (core)	≈ 37,5–38

Daily variations of 0.5–1 °C: lowest at 3–6 a.m., highest in the evening at 4–7 p.m. In women after ovulation (under the influence of progesterone) the basal temperature rises by 0.3–0.6 °C.

Chemical thermoregulation (heat production)

The main sources of heat are the **skeletal muscles** (20% at rest, up to 90% during work), the **liver** and other internal organs. In the cold, heat production increases:

- **Shivering thermogenesis** — involuntary rhythmic contraction of skeletal muscles; it increases heat production 3–5 times.
- **Non-shivering thermogenesis** — intensification of metabolism under the influence of the sympathetic nervous system, noradrenaline and thyroid hormones; **brown adipose tissue** (abundant in infants; in its mitochondria thermogenin — the UCP-1 protein — uncouples oxidation from phosphorylation and converts energy directly into heat).

Physical thermoregulation (heat loss)

Table 14.5. *Routes of heat loss (at rest, at comfort temperature)*

Route	Essence	Share
Radiation	By infrared rays to colder objects	≈ 60 %
Evaporation	Evaporation of water from sweat and the respiratory tract (0.58 kcal per 1 g of water evaporated)	≈ 20 %
Convection	Through currents of air or water in contact with the skin	≈ 15 %
Conduction	To objects in direct contact	≈ 3 %

REMEMBER

When the ambient temperature is **higher** than body temperature, heat cannot be lost by radiation, convection or conduction — the body gains heat. The only route then is **evaporation of sweat**. If the air humidity is high, evaporation becomes difficult and the risk of heatstroke increases.

Heat loss is regulated through the **tone of skin vessels** (they constrict in the cold and dilate in the heat — the skin blood supply varies from 0.2 to 3 l/min), sweating, breathing, clothing and behavior (changing posture).

The thermoregulatory center

The **thermoregulatory center** is in the **hypothalamus**: its anterior part (the preoptic area) is mainly the center of heat loss; the posterior part is the center of heat production and conservation. The hypothalamus has a temperature “set point”; the center receives information from peripheral (skin) and central (hypothalamus, blood) thermoreceptors and keeps body temperature at this point.

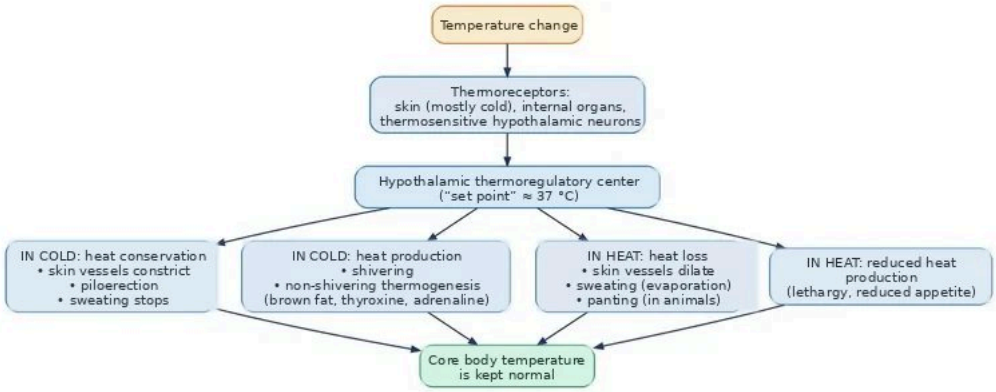


Figure 14.2. *Mechanisms of thermoregulation*

CLINICAL SIGNIFICANCE

Fever — under the action of pyrogens (bacterial toxins, IL-1, IL-6, TNF-α) prostaglandin E_2 is formed in the hypothalamus and the "set point" shifts upward; the body raises its temperature to the new level (chills, shivering). Antipyretics (paracetamol, ibuprofen) reduce prostaglandin synthesis. **Hyperthermia (heatstroke)**, in contrast, results from insufficient heat loss with an unchanged "set point" — antipyretics are ineffective and external cooling is needed.

4. General physiology of the endocrine system

The **endocrine system** is the set of endocrine glands and hormone-producing cells. **Hormones** are biologically active substances formed in specialized cells, released into the blood and acting at a distance on target cells.

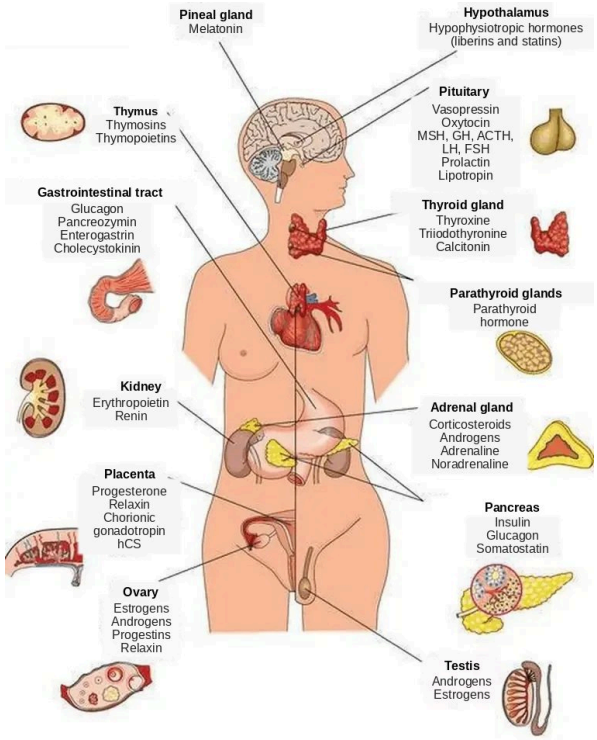


Figure 14.3. *The endocrine glands and their hormones (Alavia O.T., 2018)*

General properties of hormones: high biological activity (they act in very small concentrations — 10^{-12}–10^{-6} mol/l), specificity of action (only on target cells that have receptors), action at a distance, and a relatively short life span (inactivation in the liver and kidneys).

Table 14.6. *Chemical classification and mechanisms of action of hormones*

Group	Examples	Receptor location	Mechanism
Peptide and protein hormones	Insulin, glucagon, pituitary and hypothalamic hormones, PTH	Cell membrane	Second messengers (cAMP, Ca^{2+}, IP_3, DAG) or tyrosine kinase — rapid action
Amino acid derivatives — catecholamines	Adrenaline, noradrenaline, dopamine	Membrane	cAMP and other messengers
Amino acid derivatives — thyroid hormones	T_3, T_4	Inside the cell (nucleus)	Alter gene transcription — slow action
Steroid hormones	Cortisol, aldosterone, estrogens, testosterone, calcitriol (vitamin D)	Cytoplasm or nucleus	The hormone–receptor complex binds to DNA and alters protein synthesis

The activity of the endocrine glands is regulated mainly by the principle of **negative feedback**; the nervous system (directly or through the hypothalamus) and blood metabolites (glucose, Ca^{2+}) also have an influence.

Methods of studying endocrine gland function: removal of the gland (extirpation) followed by hormone administration; transplantation of the gland; determination of hormones and their metabolites in blood and urine (radioimmune and enzyme immunoassay methods); functional (stimulation and suppression) tests; radioisotope methods; imaging (ultrasound, CT, MRI).

5. The hypothalamo-pituitary system

The **hypothalamus** is part of the diencephalon; it is the highest center of autonomic, endocrine and behavioral functions. It converts nervous signals into hormonal signals. The **pituitary gland** is a small gland (0.5–0.6 g) located in the fossa of the sella turcica, the “conductor of the endocrine system”.

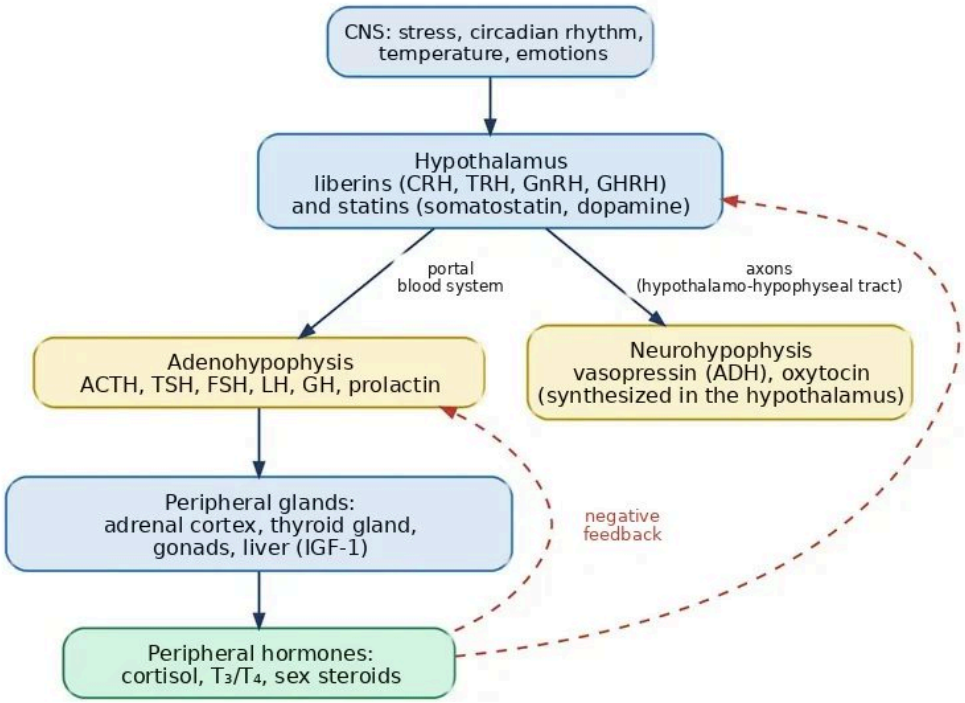


Figure 14.4. *The hypothalamo-pituitary system and negative feedback*

Neurosecretory cells of the hypothalamus:

- **Parvocellular nuclei** produce liberins and statins; they reach the adenohypophysis through the **hypophyseal portal blood system** and regulate the secretion of its hormones.
- **Magnocellular nuclei** (supraoptic and paraventricular) synthesize vasopressin and oxytocin; they are transported along the axons to the **neurohypophysis** and released from there into the blood.

Table 14.7. *Regulatory hormones of the hypothalamus*

Hormone	Effect
Corticoliberin (CRH)	Stimulates ACTH secretion
Thyroliberin (TRH)	Stimulates TSH and prolactin secretion
Gonadoliberin (GnRH)	Stimulates LH and FSH secretion (released in pulses)
Somatoliberin (GHRH)	Stimulates GH secretion
Somatostatin	Inhibits GH and TSH secretion
Dopamine (prolactostatin)	Inhibits prolactin secretion

Table 14.8. *Pituitary hormones*

Branch	Hormone	Main effect
Adenohypophysis	Somatotropin (GH, growth hormone)	Stimulates growth via IGF-1 from the liver; enhances protein synthesis, increases lipolysis, anti-insulin effect
Adenohypophysis	Thyrotropin (TSH)	Activity and growth of the thyroid gland
Adenohypophysis	Corticotropin (ACTH)	Adrenal cortex (mainly glucocorticoids)
Adenohypophysis	FSH	Follicle growth in women; spermatogenesis in men
Adenohypophysis	LH	Ovulation, corpus luteum; testosterone synthesis in men (Leydig cells)
Adenohypophysis	Prolactin	Development of the mammary glands and milk production
Intermediate lobe	Melanocyte-stimulating hormone (MSH)	Melanin synthesis (pigmentation)
Neurohypophysis	Vasopressin (antidiuretic hormone, ADH)	Water reabsorption in the renal collecting ducts (aquaporin-2); constricts vessels at high doses
Neurohypophysis	Oxytocin	Uterine contraction (childbirth), milk ejection (myoepithelium of the mammary glands)

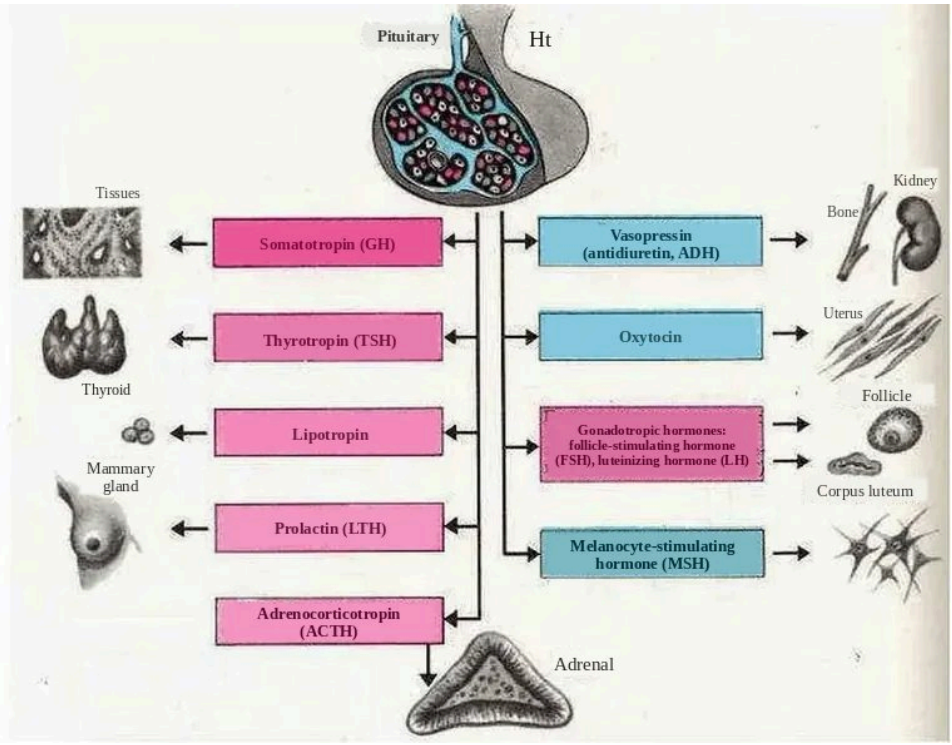


Figure 14.5. *Hormones of the adenohypophysis and neurohypophysis and their target organs (Alavia O.T., 2018)*

CLINICAL SIGNIFICANCE

GH excess: in childhood (growth plates not yet closed) — **gigantism**; in adults — **acromegaly** (enlargement of the hands and feet, nose, lower jaw and tongue). **GH deficiency** in childhood — **pituitary dwarfism** (short stature with preserved body proportions and intelligence). **ADH deficiency** — **diabetes insipidus**: 5–20 l of dilute urine per day, intense thirst.

MEMORY AID

Hormones of the adenohypophysis — **"FLAT PiG"**: **F**SH, **L**H, **A**CTH, **T**SH, **P**rolactin, (**i**), **G**rowth hormone (GH).

6. Laboratory work

Exercise 1: determining the predicted basal metabolism using tables and a formula

1. Determine the subject's sex, age, height (cm) and weight (kg).
2. Calculate the predicted basal metabolism by the Harris–Benedict formula: for men 66.5 + 13.75 × weight + 5.0 × height − 6.76 × age (kcal/day); for women 655.1 + 9.56 × weight + 1.85 × height − 4.68 × age.
3. Compare the result with the approximate estimate of 1 kcal/kg/hour.

Exercise 2: determining the deviation of basal metabolism by Read's formula

1. In the morning, on an empty stomach, after 10–15 minutes of lying rest, the subject's pulse and arterial pressure are measured three times; the lowest values are taken.
2. Calculate the pulse pressure.
3. BM (%) = 0.75 × (pulse + pulse pressure × 0.74) − 72.
4. **Interpretation:** ± 10% — normal; more than 10% — basal metabolism is increased (thyrotoxicosis suspected). **Example:** pulse 70, pressure 120/80 (pulse pressure 40) → 0.75 ×

(70 + 29.6) − 72 = 2.7% — normal.

Exercise 3: calculating energy expenditure from the results of the Douglas–Haldane method

1. Use the given data: for example, 40 l of air expired in 5 minutes, containing 16.6% O_2 and 3.6% CO_2 (atmospheric air contains 20.94% O_2 and 0.03% CO_2).
2. O_2 consumed: 40 × (20.94 − 16.6) / 100 = 1.74 l; CO_2 released: 40 × (3.6 − 0.03) / 100 = 1.43 l.
3. RQ = 1.43 / 1.74 = 0.82; from the table the caloric equivalent is ≈ 4.83 kcal.
4. Energy expenditure in 5 minutes: 1.74 × 4.83 = 8.4 kcal; in 1 hour: 8.4 × 12 = 101 kcal.

Exercise 4: measuring body temperature

1. Place the thermometer (if it is a mercury thermometer, shake it down below 35 °C) in the dry axilla; hold it with the arm pressed against the body for 10 minutes (with an electronic thermometer, until the signal).
2. Record the result; measure in the morning and evening and determine the daily difference.
3. Measure the sublingual temperature and compare it with the axillary temperature.

7. Practical skills

Skill: calculating daily energy requirement

1. Calculate basal metabolism (Harris–Benedict).
2. Choose the physical activity ratio according to the work group (1.4–2.4).
3. Daily requirement = basal metabolism × ratio.
4. Compare the result with calorie intake and evaluate the energy balance.

Skill: evaluating a disorder of the hypothalamo-pituitary system (scheme)

1. Determine the level of the peripheral hormone (for example, T_4).
2. Determine the corresponding tropic hormone (TSH).
3. If the peripheral hormone is low and the tropic hormone high — the gland itself is damaged (primary insufficiency).
4. If both are low — the pituitary or hypothalamus is damaged (secondary or tertiary).
5. Explain the result according to the feedback principle.

Review questions

1. What methods of measuring energy exchange are there?
2. What is the respiratory quotient, and what does it depend on?
3. What is the caloric equivalent of oxygen?
4. Under what conditions is basal metabolism determined, and what is its magnitude?
5. What factors affect basal metabolism?
6. What is the specific dynamic action of food?
7. What is the daily energy expenditure according to work intensity?
8. What are the norms and daily variations of body temperature?
9. Explain the mechanisms of chemical thermoregulation.

10. List the routes of heat loss.
11. Where is the thermoregulatory center, and how does it work?
12. What is the difference between fever and hyperthermia?
13. What are the general properties and chemical classification of hormones?
14. How do the mechanisms of action of peptide and steroid hormones differ?
15. What hormones does the pituitary gland have, and how are they regulated?

Test questions

1. What is the respiratory quotient when carbohydrates are oxidized?
   A) 0,7
   B) 0,8
   C) 0,85
   D) 0,5
   E) 1,0

2. What is the approximate basal metabolism of a 70 kg man?
   A) 500 kcal/day
   B) 1700 kcal/day
   C) 3500 kcal/day
   D) 5000 kcal/day
   E) 900 kcal/day

3. Which hormone increases basal metabolism most strongly?
   A) Thyroxine
   B) Insulin
   C) Aldosterone
   D) Prolactin
   E) Oxytocin

4. Which nutrient has the greatest specific dynamic action?
   A) Fats
   B) Carbohydrates
   C) Vitamins
   D) Water
   E) Proteins

5. At rest at comfort temperature, by which route is heat mainly lost?
   A) Radiation
   B) Evaporation
   C) Conductivity
   D) Convection
   E) With urine

6. When the ambient temperature is higher than body temperature, what is the only route of heat loss?
   A) Radiation
   B) Convection
   C) Evaporation of sweat
   D) Conductivity
   E) Constriction of skin vessels

7. Where is the thermoregulatory center located?
   A) In the medulla oblongata
   B) In the hypothalamus
   C) In the cerebellum
   D) In the spinal cord
   E) In the cerebral cortex

8. Which protein provides heat production in brown adipose tissue?
   A) Myosin
   B) Hemoglobin
   C) Albumin
   D) Insulin
   E) Thermogenin (UCP-1)

9. Where are the receptors for steroid hormones located?
   A) Only in the cell membrane
   B) Only in the mitochondria
   C) Outside the cell
   D) In the cytoplasm or nucleus
   E) In the ribosomes

10. Where are vasopressin and oxytocin synthesized?
    A) In the adenohypophysis
    B) In the neurohypophysis
    C) In the supraoptic and paraventricular nuclei of the hypothalamus
    D) In the adrenal gland
    E) In the pineal gland

11. What disease does GH excess cause in adults?
    A) Gigantism
    B) Dwarfism
    C) Diabetes insipidus
    D) Acromegaly
    E) Cretinism

12. Diabetes insipidus is associated with deficiency of which hormone?
    A) Vasopressin (ADH)
    B) Insulin
    C) Oxytocin
    D) ACTH
    E) Prolactin

13. Which substance inhibits prolactin secretion?
    A) Thyroliberin
    B) Somatoliberin
    C) Dopamine
    D) Corticoliberin
    E) Oxytocin

14. What is the main mechanism of the rise in temperature in fever?
    A) Increased sweating
    B) Increased heat loss
    C) A rise in ambient temperature
    D) An upward shift of the “set point” in the hypothalamus
    E) Muscle fatigue

15. What are the conditions for determining basal metabolism?
    A) Immediately after a meal
    B) In the morning, on an empty stomach, at rest, lying down, at comfort temperature
    C) After physical work
    D) In the evening, before going to bed
    E) In a cold room

Clinical cases

Case 1

A 30-year-old woman complains of weight loss, sweating, heat intolerance, palpitations and hand tremor. Pulse 110 /min, arterial pressure 140/70 mm Hg.

Questions: Calculate the deviation of basal metabolism by Read's formula. Which gland's function may be disturbed?

Answer: Pulse pressure = 140 − 70 = 70. BM = 0.75 × (110 + 70 × 0.74) − 72 = 0.75 × 161.8 − 72 ≈ 49%. Basal metabolism is increased by ≈ 50% above the norm. The clinical signs (weight loss, sweating, heat intolerance, tachycardia, tremor, wide pulse pressure) are typical of thyroid hyperfunction — thyrotoxicosis. Thyroid hormones increase oxidative processes and Na^+/K^+-ATPase activity in almost all cells, increase heat production and raise the sensitivity of β-adrenergic receptors. The diagnosis is confirmed by blood TSH (decreased) and free T_4, T_3 (increased).

Case 2

In summer, on a day with an air temperature of 40 °C and high humidity, a worker who had been working in the field for a long time suddenly developed headache, nausea and was close to losing consciousness. Body temperature 40.5 °C, skin dry and hot.

Questions: What is the condition? Explain its mechanism. Why does paracetamol not help in this condition?

Answer: Heatstroke (hyperthermia). When the ambient temperature is higher than body temperature, heat loss by radiation, convection and conduction is impossible; the only route is evaporation of sweat. Because the humidity was high, evaporation was ineffective; prolonged sweating led to dehydration and salt loss, and eventually sweating stopped (dry skin). Muscle work produced a lot of heat. The heat accumulated and the temperature rose. In this condition the “set point” in the hypothalamus has not changed — the thermoregulatory mechanisms are simply insufficient; therefore

antipyretics, which reduce prostaglandin synthesis, are ineffective. Help: move the person to shade and a cool place, external cooling (wet sheets, ice, a fan), and restoration of fluids and salts.

Case 3

An 8-year-old child is 105 cm tall (the norm for his age ≈ 127 cm), with normal body proportions and age-appropriate mental development. Blood GH and IGF-1 are low.

Questions: What is the condition? Explain the mechanism of action of GH.

Answer: Pituitary dwarfism (somatotropin deficiency). GH is formed in the adenohypophysis and stimulates the synthesis of insulin-like growth factor-1 (IGF-1, somatomedin) mainly in the liver and other tissues; IGF-1 stimulates the proliferation of chondrocytes in the growth plates of bones and ensures longitudinal growth. GH also enhances protein synthesis. In deficiency growth slows, but because thyroid hormones are normal, body proportions and mental development are preserved (the difference from cretinism). Treatment — recombinant GH until the growth plates close.

Quick review

Table 14.9. *Key concepts of topic 14*

Concept	Brief definition
Respiratory quotient	CO_2 / O_2; carbohydrate 1.0, fat 0.7, protein 0.8
Caloric equivalent	Heat per 1 l O_2 (≈ 4.8–5.0 kcal)
Basal metabolism	Minimum expenditure under standard conditions; ≈ 1 kcal/kg/hour
Rubner's law	Metabolism is proportional to body surface area
Specific dynamic action	Increased expenditure after a meal; up to 30% with protein
Read's formula	Deviation of BM from the pulse and pulse pressure
Body temperature	36.4–36.9 °C in the axilla
Chemical thermoregulation	Shivering and non-shivering thermogenesis (brown fat)
Physical thermoregulation	Radiation 60%, evaporation 20%, convection 15%
The thermoregulatory center	Hypothalamus
Fever	Shift of the “set point” (PGE_2)
Peptide hormones	Membrane receptors, second messengers
Steroid and thyroid hormones	Intracellular receptors, gene transcription
Adenohypophysis	GH, TSH, ACTH, FSH, LH, prolactin
Neurohypophysis	Vasopressin, oxytocin (synthesized in the hypothalamus)

TOPIC 15

Physiology of the thyroid, parathyroid, adrenal, pancreatic and sex glands

Learning objectives

To teach the student the formation, biological effects and regulation of secretion of the hormones of the peripheral endocrine glands — the thyroid, parathyroid and adrenal glands, the endocrine part of the pancreas and the sex glands — and the main manifestations of their dysfunction; to master evaluating hormonal status on the basis of clinical and laboratory parameters.

By the end of the class the student should know:

- the synthesis, effects and regulation of thyroid hormones and the signs of hypo- and hyperthyroidism;
- the hormonal regulation of calcium metabolism (parathyroid hormone, calcitonin, calcitriol);
- the effects of the hormones of the adrenal cortex (mineralocorticoids, glucocorticoids, androgens) and of the adrenal medulla;
- the hormones of the pancreatic islet apparatus and the regulation of blood glucose;
- the male and female sex hormones and the hormonal regulation of the menstrual cycle;
- stress and the general adaptation syndrome (H. Selye).

The student should be able to:

- interpret thyroid hormone, cortisol, glucose and calcium parameters;
- evaluate the result of a glucose tolerance test;
- explain the main signs of endocrine disorders from a physiological point of view.

1. The thyroid gland

The thyroid gland (20–30 g) lies on the anterior surface of the neck. Its structural and functional unit is the **follicle**: a vesicle lined by a single layer of thyrocytes and filled with **colloid** (thyroglobulin). Between the follicles lie the **parafollicular C cells**, which produce **calcitonin**.

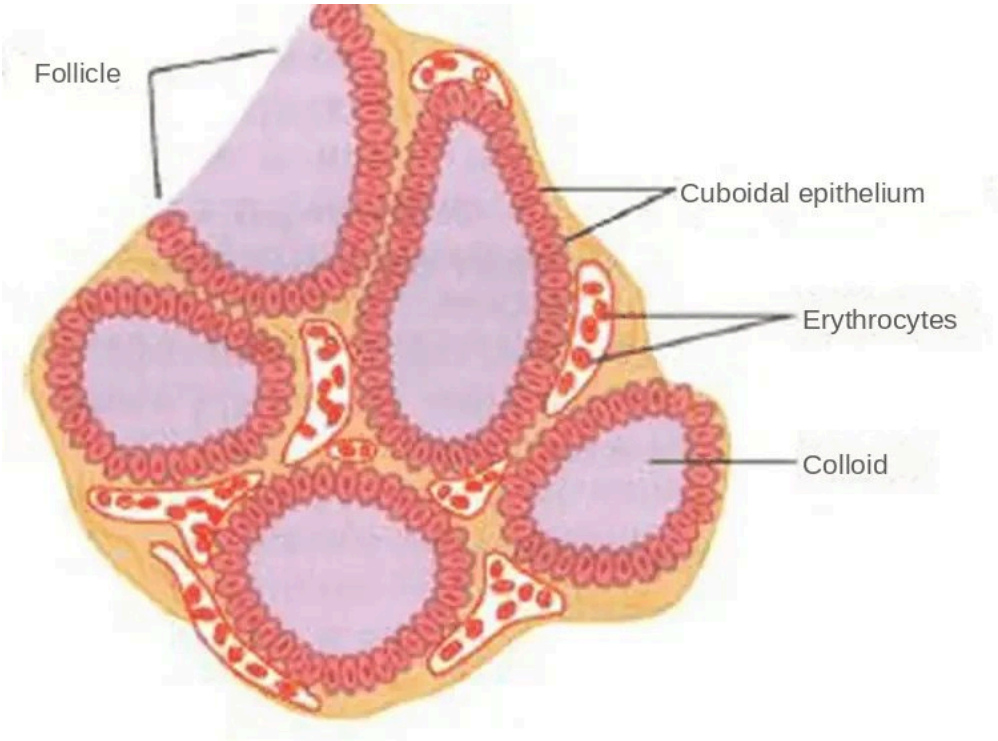


Figure 15.1. *Thyroid follicles (Alavia O.T., 2018)*

Synthesis of thyroid hormones

1. Thyrocytes actively take up **iodide** (I^-) from the blood through the Na^+/I^- symporter.
2. Thyroid peroxidase oxidizes iodide and attaches it to tyrosine residues of **thyroglobulin** (monoiodotyrosine, diiodotyrosine).
3. Iodotyrosines combine to form **thyroxine (T_4)** and **triiodothyronine (T_3)**; they are stored in the colloid.
4. Under the action of TSH thyroglobulin is taken back up by endocytosis and proteolyzed, and T_4 and T_3 are released into the blood. 90% of the secreted hormone is T_4, but the active form is **T_3** (3–5 times more potent); T_4 is converted to T_3 in peripheral tissues by deiodinase. In the blood the hormones are carried mainly bound to proteins. Adults need ≈ 150 µg of iodine per day.

Table 15.1. *Effects of thyroid hormones*

Effect	Content
Energy exchange	Increase basal metabolism, O_2 consumption and heat production (Na^+/K^+-ATPase, mitochondria)
Growth and development	Essential for the CNS (myelination, neuronal development) and bone growth; enhance the effect of GH
Metabolism	Enhance protein synthesis (at physiological doses), gluconeogenesis, lipolysis and cholesterol breakdown
Cardiovascular	Increase the number of β-adrenergic receptors — HR and force of contraction increase
Nervous system	Increase excitability, reflex speed and mental activity

Regulation: hypothalamus (TRH) → pituitary (TSH) → thyroid gland (T_4, T_3) — negative feedback. Cold stimulates TRH secretion.

Table 15.2. *Disorders of thyroid function*

Condition	Signs	Examples
Hyperthyroidism (thyrotoxicosis)	Weight loss, sweating, heat intolerance, tachycardia, hand tremor, nervousness, bulging eyes (exophthalmos)	Diffuse toxic goiter (Graves–Basedow disease)
Hypothyroidism (in adults)	Weight gain, cold sensitivity, bradycardia, drowsiness, memory impairment, dry and swollen skin (myxedema)	Autoimmune thyroiditis, after surgery
Hypothyroidism (congenital)	Cretinism — delayed mental and physical development, dwarfism	Congenital hypothyroidism, severe iodine deficiency in pregnancy
Endemic goiter	Enlargement of the gland (TSH rises in iodine deficiency)	Iodine-deficient regions

CLINICAL SIGNIFICANCE

If congenital hypothyroidism is not detected early, irreversible mental retardation (cretinism) develops. Therefore **TSH screening** is performed in all newborns, and if the disease is detected, treatment with thyroxine starts in the first weeks. **Iodized salt** is used to prevent iodine deficiency.

2. Regulation of calcium and phosphorus metabolism

Total blood calcium is **2.2–2.6 mmol/l**, ionized (active) calcium 1.1–1.3 mmol/l. Calcium is necessary for muscle contraction, mediator release, blood coagulation and excitability. It is regulated by three hormones.

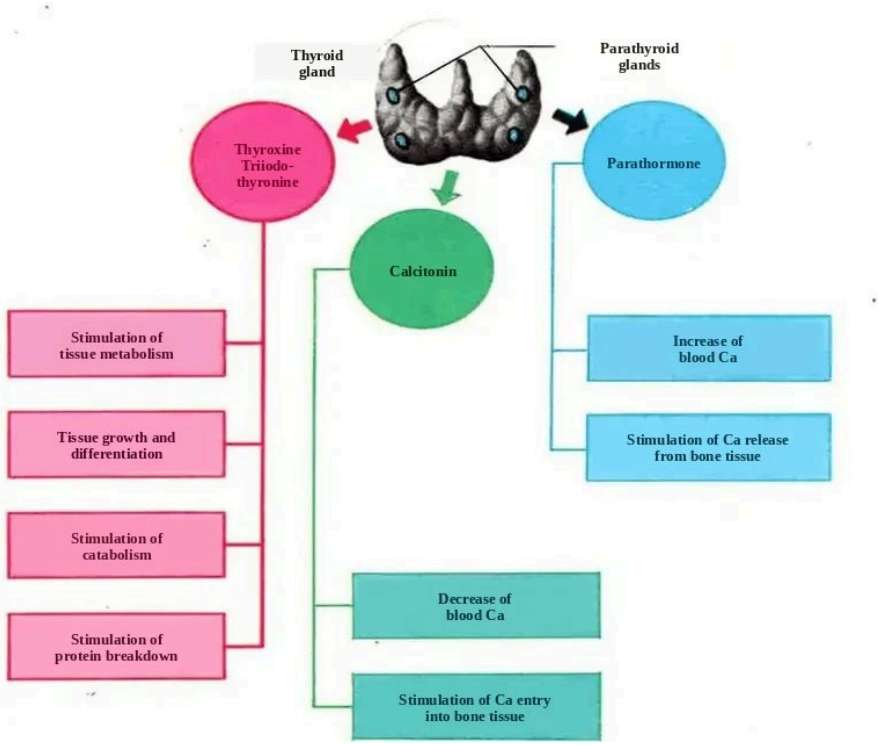


Figure 15.2. *Effects of the hormones of the thyroid and parathyroid glands (Alavia O.T., 2018)*

Table 15.3. *Hormones regulating calcium metabolism*

Hormone	Source	Effect	Blood Ca^{2+}
Parathyroid hormone (PTH)	Parathyroid glands (4)	Enhances release of Ca^{2+} from bone (osteoclasts) and Ca^{2+} reabsorption in the kidney, increases phosphate excretion; activates calcitriol synthesis in the kidney	Increases
Calcitriol (active form of vitamin D)	Skin (ultraviolet) → liver → kidney	Enhances absorption of Ca^{2+} and phosphate in the intestine; bone mineralization	Increases
Calcitonin	Thyroid C cells	Inhibits osteoclasts, enhances the transfer of Ca^{2+} into bone	Decreases

PTH secretion is stimulated by a **decrease in ionized Ca^{2+}** in the blood (through calcium-sensing receptors on the gland cells).

CLINICAL SIGNIFICANCE

Hypoparathyroidism (most often when the parathyroid glands are accidentally removed during thyroid surgery) — hypocalcemia: the excitability of nerves and muscles increases, **tetany** (cramps, “obstetrician's hand”, Chvostek and Trousseau signs), and in severe cases laryngospasm. In **hyperparathyroidism** — leaching of calcium from bone (osteoporosis, fractures), kidney stones, peptic ulcer. **Vitamin D deficiency** leads to rickets in children and osteomalacia in adults.

3. The adrenal glands

The adrenal gland consists of two independent parts: the **cortex** (mesodermal, steroid hormones) and the **medulla** (neuroectodermal, catecholamines).

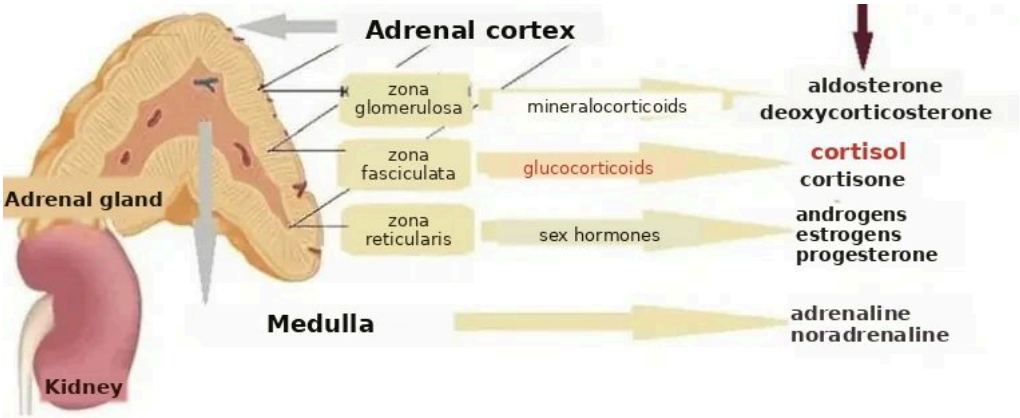


Figure 15.3. *Layers and hormones of the adrenal gland (Alavia O.T., 2018)*

Table 15.4. *Hormones of the adrenal cortex*

Zone	Hormones	Main effect	Control
Glomerulosa	Mineralocorticoids — aldosterone	Enhances reabsorption of Na^+ and water and secretion of K^+ and H^+ in the distal tubules and collecting ducts of the kidney; increases blood volume and pressure	Angiotensin II, increased blood K^+ (main); ACTH (minor)

Zone	Hormones	Main effect	Control
Fasciculata	Glucocorticoids — cortisol	Gluconeogenesis (hyperglycemia), protein catabolism, redistribution of fat; suppress inflammation and immunity; enhance the effect of catecholamines (permissive effect); adaptation to stress	ACTH (CRH); circadian rhythm — highest in the morning
Reticularis	Androgens (dehydroepiandrosterone) and small amounts of estrogens	Secondary sexual characteristics (hair growth in women), anabolic effect	ACTH

REMEMBER

Aldosterone is essential for life: without it the body loses sodium and water and potassium accumulates — in acute adrenal insufficiency blood pressure falls and hyperkalemia can stop the heart. **Cortisol** is the main “stress hormone”: its secretion increases several times during stress.

Table 15.5. *Disorders of adrenal cortex function*

Condition	Cause	Signs
Itsenko–Cushing syndrome (hypercortisolism)	Cortisol excess (pituitary tumor, adrenal tumor, long-term glucocorticoid therapy)	“Moon face”, obesity of the upper body, red-purple abdominal stripes (striae), arterial hypertension, hyperglycemia, osteoporosis, muscle weakness
Addison's disease (primary insufficiency)	Autoimmune or tuberculous damage to the cortex	Weakness, weight loss, arterial hypotension, skin hyperpigmentation (ACTH and MSH increase), hypoglycemia, hyperkalemia
Primary hyperaldosteronism (Conn syndrome)	Tumor of the zona glomerulosa	Arterial hypertension, hypokalemia, muscle weakness

The medulla

The chromaffin cells of the medulla are modified postganglionic sympathetic neurons; they are innervated by preganglionic sympathetic fibers (acetylcholine, N-cholinergic receptors). They secrete **adrenaline** (80%) and **noradrenaline** (20%). The effects of catecholamines resemble those of the sympathetic nervous system (the “fight or flight” reaction) and are discussed in detail in topic 26: increased HR and pressure, bronchodilation, glycogenolysis and lipolysis (hyperglycemia), pupil dilation, and redistribution of blood flow to the muscles.

Stress and the general adaptation syndrome

Stress (H. Selye, 1936) is the body's nonspecific adaptive reaction to strong stimuli (stressors). The **general adaptation syndrome** consists of three stages: the **alarm stage** (sharp activation of the sympathoadrenal system and the hypothalamus–pituitary–adrenal system), the **stage of resistance** and the **stage of exhaustion** (if the stressor acts for a long time, the adaptive capacities are exhausted and diseases develop). The main hormones are **catecholamines and cortisol**.

4. Endocrine function of the pancreas

The endocrine part of the pancreas is the **islets of Langerhans** (1–2% of the gland mass, 1–2 million islets).

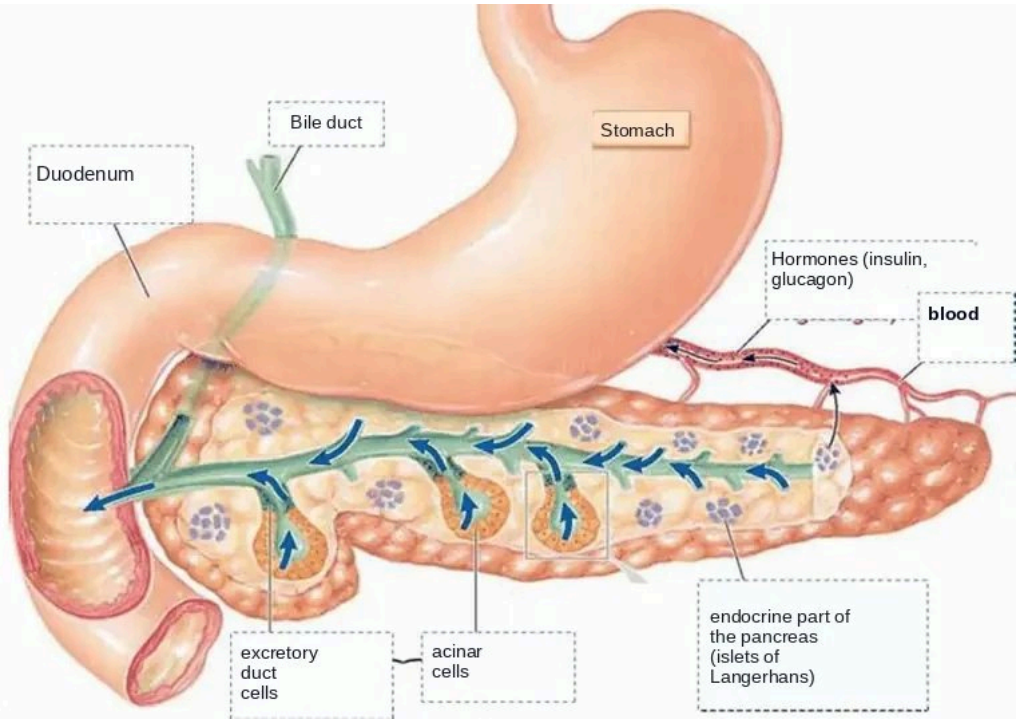


Figure 15.4. *The exocrine (acinar) and endocrine (islets of Langerhans) parts of the pancreas (Alavia O.T., 2018)*

Table 15.6. *Cells and hormones of the islets of Langerhans*

Cell	Share	Hormone	Main effect
β cells	60–70 %	Insulin	Lowers blood glucose; an anabolic hormone
α cells	20–25 %	Glucagon	Glycogenolysis and gluconeogenesis — raises glucose; lipolysis
δ cells	5–10 %	Somatostatin	Inhibits insulin and glucagon secretion
PP cells	< 5 %	Pancreatic polypeptide	Inhibits exocrine pancreatic secretion

Insulin is the body's only hypoglycemic hormone. Its effects: it enhances the entry of glucose into muscle and fat cells (by bringing GLUT4 transporters to the membrane); in the liver and muscle it increases glycogen synthesis and inhibits gluconeogenesis; it enhances lipogenesis, inhibits lipolysis and the formation of ketone bodies; it enhances protein synthesis; it increases the entry of K^+ into cells.

Regulation of insulin secretion: the main stimulus is a **rise in blood glucose** (glucose metabolism in β cells → ATP ↑ → ATP-sensitive K^+ channels close → depolarization → Ca^{2+} entry → exocytosis of insulin). Amino acids, intestinal hormones — **incretins** (GLP-1, GIP; that is why oral glucose releases more insulin than intravenous glucose) and the vagus nerve also enhance it; the sympathetic nerves (α_2), adrenaline and somatostatin inhibit it.

REMEMBER

There are many hormones that raise blood glucose (**counter-insulin** hormones): glucagon, adrenaline, cortisol, GH, thyroid hormones. Only one lowers it — **insulin**. That is why the body is better protected against hypoglycemia.

CLINICAL SIGNIFICANCE

Diabetes mellitus is chronic hyperglycemia resulting from insulin deficiency (type 1, autoimmune destruction of β cells) or insulin resistance with relative deficiency (type 2, often with obesity). Classic signs: **polyuria** (glucosuria — osmotic diuresis), **polydipsia** (thirst), **polyphagia**, weight loss. Diagnostic criteria: fasting venous plasma glucose ≥ 7.0 mmol/l, ≥ 11.1 mmol/l 2 hours into a glucose tolerance test, HbA_{1c} ≥ 6.5%.

5. The sex glands

Male sex glands (testes)

In the testes, **Leydig cells** produce **testosterone** under the influence of LH; **Sertoli cells** support spermatogenesis under the influence of FSH and testosterone and secrete **inhibin** (which inhibits FSH).

Effects of testosterone: formation of the male genital organs in the fetal period; primary and secondary sexual characteristics at puberty (deepening of the voice, growth of hair and beard, muscle mass); spermatogenesis; a strong **anabolic** effect (protein synthesis, muscle and bone mass); stimulation of erythropoiesis; sexual drive and behavior. In many tissues testosterone is converted into a more active form — **dihydrotestosterone**.

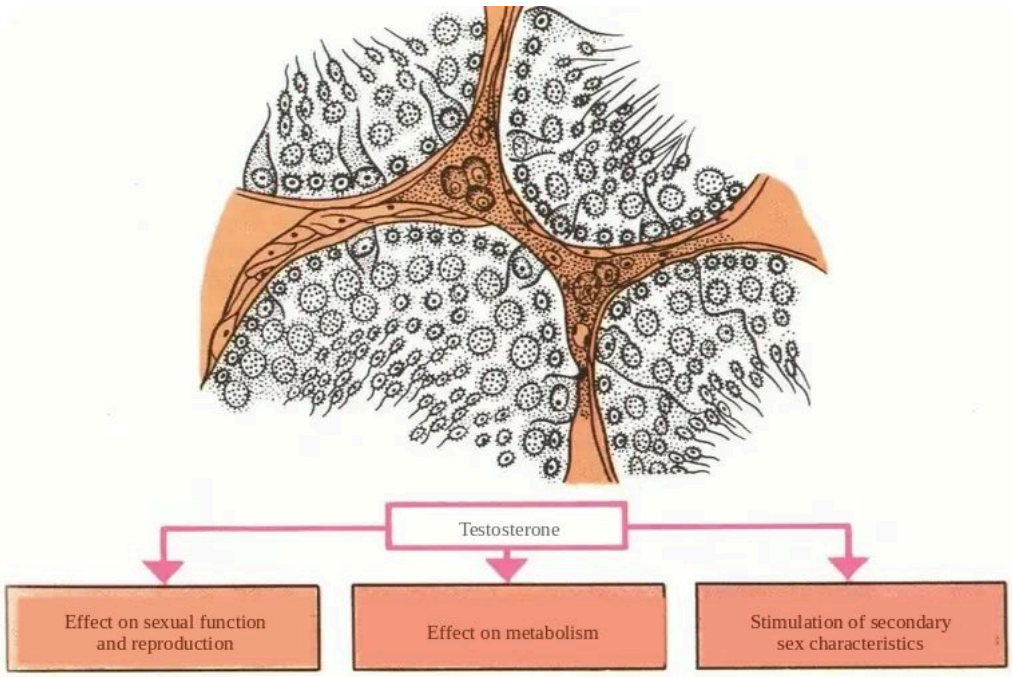


Figure 15.5. *Structure of the testis and effects of testosterone (Alavia O.T., 2018)*

Female sex glands (ovaries) and the menstrual cycle

The ovaries produce **estrogens** (estradiol — in follicle cells), **progesterone** (in the corpus luteum) and small amounts of androgens and inhibin.

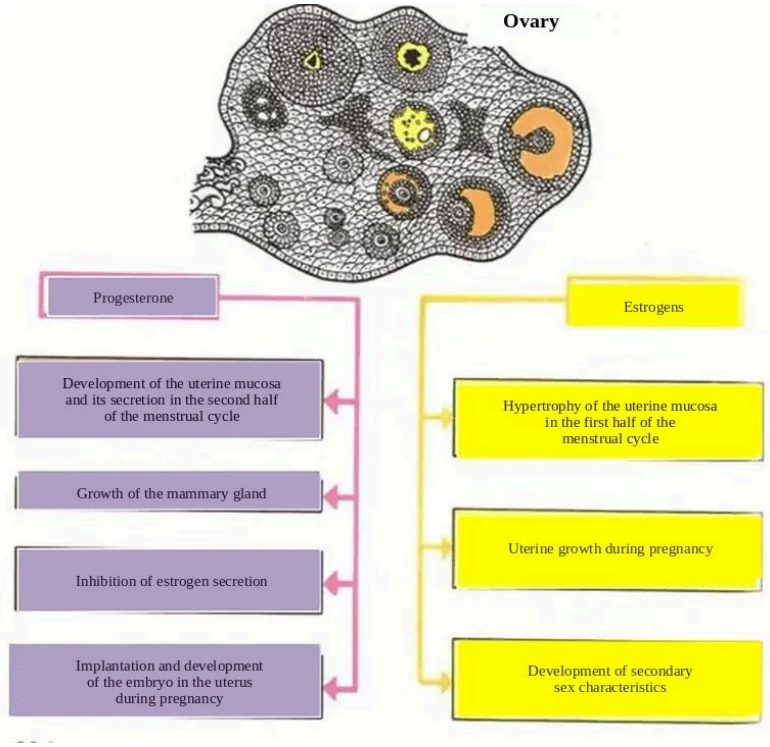


Figure 15.6. *Structure of the ovary and effects of progesterone and estrogens (Alavia O.T., 2018)*

Table 15.7. *Phases of the menstrual cycle (28-day cycle)*

Phase	Days	Hormones	In the ovary	In the uterine mucosa
Menstruation (desquamation)	1–5	Estrogen and progesterone low	New follicles begin to grow	The functional layer is shed
Follicular (proliferative)	5–14	FSH → follicle grows → estrogens rise	The dominant follicle matures	The endometrium thickens (proliferation)
Ovulation	≈ 14	The estrogen peak triggers the LH surge by positive feedback	The follicle ruptures and the egg is released	—
Luteal (secretory)	15–28	LH → corpus luteum → progesterone (and estrogens)	Corpus luteum	Glandular secretion — preparation for implantation

If there is no pregnancy, the corpus luteum regresses, progesterone falls and menstruation begins. In pregnancy **chorionic gonadotropin** (detected by pregnancy tests) maintains the corpus luteum; later the placenta produces progesterone and estrogens.

MEMORY AID

“Estrogen builds, progesterone preserves”: estrogen makes the endometrium grow (proliferation), and progesterone prepares it for pregnancy and maintains it (“pro-gestation” — for pregnancy).

6. Laboratory work

Exercise 1: evaluating the results of a glucose tolerance test

1. From the given results take the fasting glycemia and the glycemia 2 hours after drinking 75 g of glucose.
2. Plot the glycemic curve (0, 30, 60, 90, 120 minutes) on graph paper.

3. Interpretation (venous plasma): fasting < 6.1 and at 2 hours < 7.8 mmol/l — normal; at 2 hours 7.8–11.0 — impaired glucose tolerance; ≥ 11.1 — diabetes mellitus.
4. Analyze the shape of the curve (time and height of the peak, return to the baseline level).

Exercise 2: interpreting thyroid status parameters

1. Take the results of TSH (normal 0.4–4.0 mIU/l) and free T_4 (10–23 pmol/l).
2. TSH low, T_4 high — primary hyperthyroidism.
3. TSH high, T_4 low — primary hypothyroidism.
4. Both TSH and T_4 low — secondary (pituitary) hypothyroidism.
5. Compare the result with the clinical signs.

Exercise 3: testing the Chvostek and Trousseau signs (demonstration)

1. **Chvostek sign**: contraction of the facial muscles (corner of the mouth) when the exit point of the facial nerve in front of the ear is tapped lightly with a hammer.
2. **Trousseau sign**: spasm of the hand muscles (“obstetrician's hand”) when a blood pressure cuff is placed on the upper arm and kept above systolic pressure for 3 minutes.
3. If the signs are positive, neuromuscular excitability is increased (hypocalcemia).

7. Practical skills

Skill: suspecting an endocrine disorder from clinical signs

1. Evaluate the signs of energy exchange (weight, tolerance of heat/cold, HR) — thyroid gland.
2. Evaluate blood pressure and electrolytes (K^+, Na^+) — aldosterone.
3. Evaluate glycemia and glucosuria — insulin, cortisol.
4. Evaluate the calcium level and signs of tetany — parathyroid hormone.
5. Evaluate skin pigmentation, fat distribution and striae — cortisol and ACTH.
6. Identify the hormone–tropic hormone pair and determine the level of the disorder (primary or secondary).

Skill: determining the phase of the menstrual cycle from hormones

1. Determine which day of the cycle it is.
2. Evaluate the levels of estrogen, progesterone, LH and FSH.
3. Analyze the basal temperature chart: a rise of 0.3–0.6 °C after ovulation — the effect of progesterone.
4. Determine the phase and whether ovulation has occurred.

Review questions

1. Explain the structural unit of the thyroid gland and the stages of thyroid hormone synthesis.
2. List the effects of thyroid hormones on the body.
3. What are the main signs of hyperthyroidism and hypothyroidism?
4. Which hormones regulate calcium metabolism?
5. What are the signs of parathyroid hormone deficiency?
6. List the zones of the adrenal cortex and their hormones.

7. What are the effects of aldosterone, and how is its secretion regulated?
8. What are the main effects of cortisol?
9. Compare the signs of Itsenko–Cushing syndrome and Addison's disease.
10. Explain stress and the stages of the general adaptation syndrome.
11. List the cells and hormones of the islets of Langerhans.
12. How is insulin secretion regulated, and what are its effects?
13. List the counter-insulin hormones.
14. What are the effects of testosterone?
15. Explain the phases of the menstrual cycle and their hormonal regulation.

Test questions

1. Which is the active form of thyroid hormones?
   A) Thyroglobulin
   B) Triiodothyronine (T_3)
   C) Monoiodotyrosine
   D) Calcitonin
   E) TSH

2. Which hormone lowers blood calcium?
   A) Parathyroid hormone
   B) Calcitriol
   C) Cortisol
   D) Calcitonin
   E) Aldosterone

3. What is the complication of congenital hypothyroidism?
   A) Acromegaly
   B) Gigantism
   C) Exophthalmos
   D) Tetany
   E) Cretinism

4. What mainly stimulates aldosterone secretion?
   A) ACTH
   B) Insulin
   C) Angiotensin II and increased blood K^+
   D) Vasopressin
   E) Calcitonin

5. What effect does aldosterone have in the kidney?
   A) It enhances Na^+ excretion
   B) It increases glucose reabsorption
   C) It decreases calcium reabsorption
   D) It decreases water reabsorption
   E) It enhances Na^+ and water reabsorption and K^+ secretion

6. In which zone of the adrenal cortex is cortisol formed?
    A) Glomerulosa
    B) Fasciculata
    C) Reticularis
    D) In the medulla
    E) Equally in all zones

7. What is the main sign of hypoparathyroidism?
    A) Tetany (cramps)
    B) Polyuria
    C) Obesity
    D) Skin hyperpigmentation
    E) Exophthalmos

8. In which cells is insulin formed?
    A) In α cells
    B) In β cells
    C) In δ cells
    D) In PP cells
    E) In acinar cells

9. Through which transporter does insulin enhance glucose entry into muscle and fat cells?
    A) GLUT2
    B) SGLT1
    C) GLUT1
    D) GLUT4
    E) GLUT5

10. Which hormone is not counter-insulin?
    A) Glucagon
    B) Adrenaline
    C) Cortisol
    D) Somatotropin
    E) Insulin

11. What is the cause of skin hyperpigmentation in Addison's disease?
    A) A decrease in melatonin
    B) An increase in cortisol
    C) An increase in ACTH and MSH
    D) An increase in aldosterone
    E) A decrease in thyroxine

12. The surge of which hormone directly triggers ovulation?
    A) LH
    B) FSH
    C) Progesterone
    D) Prolactin
    E) Oxytocin

13. Which hormone does the corpus luteum mainly produce?
    A) Estradiol
    B) Testosterone
    C) FSH
    D) Progesterone
    E) Prolactin

14. Which cells of the testis produce testosterone?
    A) Leydig cells
    B) Sertoli cells
    C) Spermatogonia
    D) Spermatocytes
    E) Myoid cells

15. Which hormones mainly increase during stress?
    A) Insulin and calcitonin
    B) Prolactin and oxytocin
    C) Catecholamines and cortisol
    D) Aldosterone and vasopressin only
    E) Melatonin and estrogen

Clinical cases

Case 1

A 45-year-old woman complains of weakness, weight loss, dizziness and darkening of the skin. Arterial pressure 85/55 mm Hg, blood Na^+ 128, K^+ 5.8 mmol/l, glucose 3.4 mmol/l. Blood cortisol is low, ACTH high.

Questions: What is the condition? Explain each sign by linking it to the hormone deficiency.

Answer: Primary adrenal cortical insufficiency (Addison's disease). Because of aldosterone deficiency, Na^+ and water are lost in the kidney and K^+ is retained — hyponatremia, hyperkalemia, decreased blood volume and arterial hypotension. Because of cortisol deficiency, gluconeogenesis weakens — hypoglycemia, weakness; the effect of catecholamines on the vessels (permissive effect) decreases and the hypotension worsens. When cortisol falls, the pituitary releases a lot of ACTH by negative feedback; ACTH and MSH are formed from a common precursor (proopiomelanocortin) and stimulate melanocytes — the skin darkens.

Case 2

On the second day after thyroid resection, the patient developed numbness of the lips and fingertips, followed by painful cramps of the hand muscles. The Chvostek and Trousseau signs are positive.

Questions: What complication has developed? Explain its mechanism.

Answer: During the operation the parathyroid glands were removed or damaged — hypoparathyroidism. Because of parathyroid hormone deficiency, the release of calcium from bone and calcium reabsorption in the kidney decrease, calcitriol synthesis falls and intestinal calcium absorption decreases — hypocalcemia develops. When extracellular Ca^{2+} falls, the opening threshold of Na^+ channels decreases (membrane stability decreases) — the excitability of nerves and muscles increases and spontaneous impulses arise: paresthesias, tetany. Treatment — intravenous calcium preparations, then calcium and active vitamin D.

Case 3

A patient treated with prednisolone (a glucocorticoid) for a long time has a rounded face, fat accumulation in the upper body, purple stripes on the abdomen, arterial pressure 160/100 mm Hg and blood glucose 8.2 mmol/l.

Questions: What is the condition? Explain the mechanism of the signs. Why is it dangerous to stop the treatment abruptly?

Answer: Drug-induced (iatrogenic) Itsenko–Cushing syndrome. Glucocorticoids enhance gluconeogenesis and reduce glucose uptake by tissues — hyperglycemia; protein catabolism thins the skin and collagen — striae, muscle weakness, osteoporosis; they redistribute fat — “moon face”, obesity of the upper body; through mineralocorticoid action and increased vascular sensitivity to catecholamines — hypertension. During long-term treatment, exogenous glucocorticoids suppress CRH and ACTH by negative feedback, and the patient's own adrenal glands atrophy. If treatment is stopped abruptly, acute adrenal insufficiency may develop; therefore the dose is reduced gradually.

Quick review

Table 15.8. *Key concepts of topic 15*

Concept	Brief definition
Follicle	Unit of the thyroid gland; colloid — thyroglobulin
T_4 / T_3	T_3 is active; increases basal metabolism
Cretinism	Congenital hypothyroidism — mental and physical retardation
Parathyroid hormone	Raises Ca^{2+}; deficiency causes tetany
Calcitonin	Lowers Ca^{2+} (C cells)
Calcitriol	Intestinal absorption of Ca^{2+}
Aldosterone	Zona glomerulosa; retains Na^{+}, excretes K^{+}
Cortisol	Zona fasciculata; gluconeogenesis, anti-inflammatory, stress
Addison's disease	Cortical insufficiency; hypotension, hyperpigmentation
Itsenko–Cushing	Cortisol excess
Medulla	Adrenaline 80%, noradrenaline 20%
Insulin	β cells; the only hypoglycemic hormone
Glucagon	α cells; raises glucose
Testosterone	Leydig cells; anabolic, secondary sexual characteristics
Menstrual cycle	FSH → estrogen → LH surge (ovulation) → progesterone

TOPIC 16

Respiratory physiology. Mechanism of breathing. Alveolar-capillary gas exchange. Regulation of breathing

Learning objectives

To teach the student the stages of respiration, the biomechanics of inspiration and expiration, intrapleural pressure and its significance, lung compliance and the role of surfactant, the laws of gas exchange in the alveoli, the structure of the respiratory center and the reflex and humoral regulation of breathing; in laboratory work, to master recording respiratory movements and analyzing their changes under the influence of various factors.

By the end of the class the student should know:

- the stages of respiration (external respiration, transport of gases in the blood, tissue respiration);
- the inspiratory and expiratory muscles and the biomechanics of the respiratory cycle;
- intrapleural pressure and its origin, pneumothorax;
- the elastic recoil of the lungs and the function of surfactant;
- the composition of gases in air and in the alveoli, partial pressures;
- the diffusion of gases across the alveolar-capillary membrane;
- the structure of the respiratory center and the rhythm generator;
- the central and peripheral chemoreceptors and the Hering–Breuer reflex.

The student should be able to:

- demonstrate the mechanism of breathing with the Donders model;
- record respiratory movements by pneumography and analyze the effects of hyperventilation, breath-holding and exercise;
- perform and evaluate the Stange and Genchi tests.

1. Stages of respiration

Respiration is the set of processes that ensure gas exchange between the body and the external environment (uptake of O_2 and elimination of CO_2).

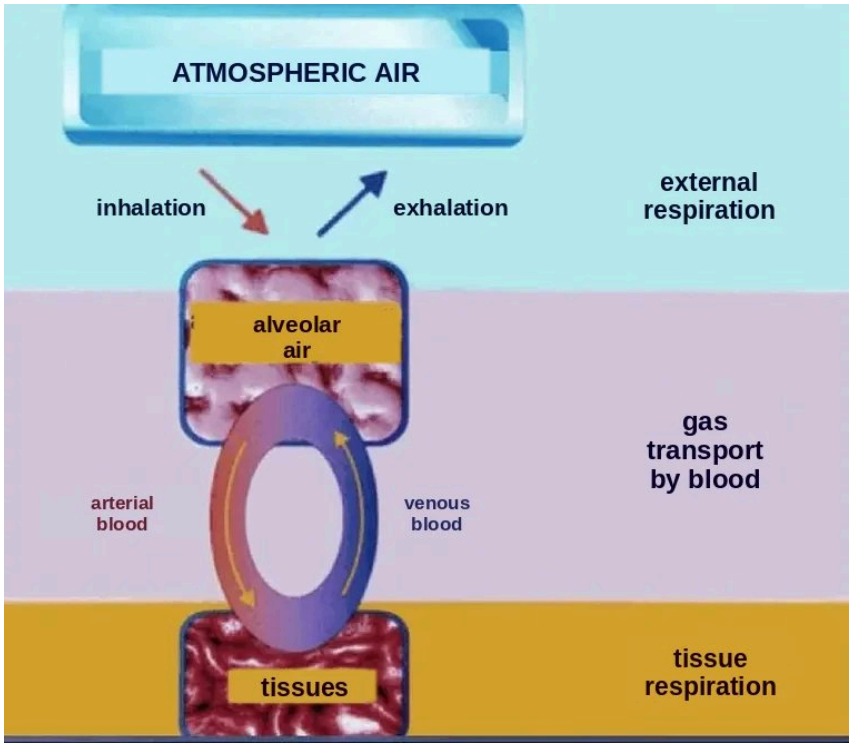


Figure 16.1. *Stages of respiration: external respiration, transport of gases in the blood, tissue respiration (Alavia O.T., 2018)*

1. **External respiration (pulmonary ventilation)** — exchange of air between the atmosphere and the alveoli.
2. **Gas exchange in the lungs** — between alveolar air and the blood of the pulmonary capillaries.
3. **Transport of gases in the blood** (topic 17).
4. **Gas exchange in the tissues** — between blood and tissues.
5. **Cellular (tissue) respiration** — the use of O_2 in oxidative processes in the mitochondria.

2. Mechanism of inspiration and expiration

Because the lung has no muscle tissue of its own, it cannot expand actively; it passively follows the movements of the chest. Between the lung and the chest wall there is a sealed **pleural cavity** (a thin layer of fluid).

Inspiration — an active process

Main inspiratory muscles: the diaphragm (70–80% of the volume change in quiet breathing; its dome descends 1.5–2 cm, up to 10 cm in deep breathing) and the **external intercostal muscles** (they raise the ribs and expand the chest anteroposteriorly and laterally). **Accessory muscles** (in deep breathing): sternocleidomastoid, scalene, pectoralis major and minor.

Sequence: impulse from the respiratory center → the diaphragm and external intercostal muscles contract → chest volume increases → intrapleural pressure becomes more negative → the lungs expand → alveolar pressure falls 1–2 cm H_2O below atmospheric → air enters the lungs.

Expiration

In quiet breathing it is **passive**: the respiratory muscles relax, and the **elastic recoil** of the lungs and the weight of the chest return it to the initial position; alveolar pressure exceeds atmospheric and air

leaves. In forced expiration (physical work, coughing, bronchial asthma) it is **active**: the **internal intercostal muscles** and the **abdominal muscles** contract.

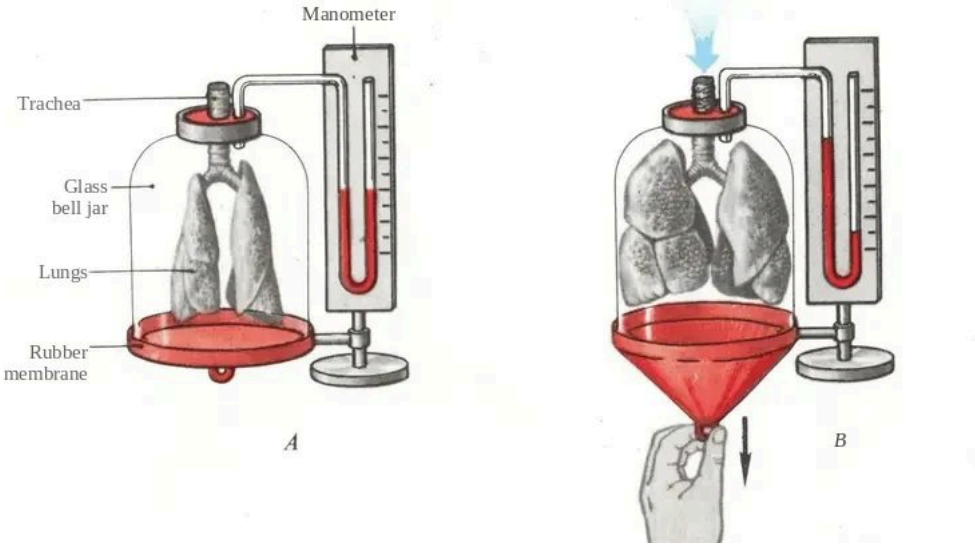


Figure 16.2. *Donders model: A — resting state; B — when the "diaphragm" (rubber) is pulled down, the lungs expand (Alavia O.T., 2018)*

Intrapleural pressure

Intrapleural pressure is **lower** than atmospheric ("negative"): −3…−5 cm H_2O (−2…−4 mm Hg) at the end of quiet expiration, −6…−8 during quiet inspiration, and down to −20 cm H_2O during deep inspiration.

Origin: the **elastic recoil** of the lungs (which tends to shrink the lung toward its root) and the outward pull of the chest wall act in opposite directions; the pleural layers cannot separate from each other and a negative pressure arises in the cavity. In the postnatal period the faster growth of the chest relative to the lungs also contributes.

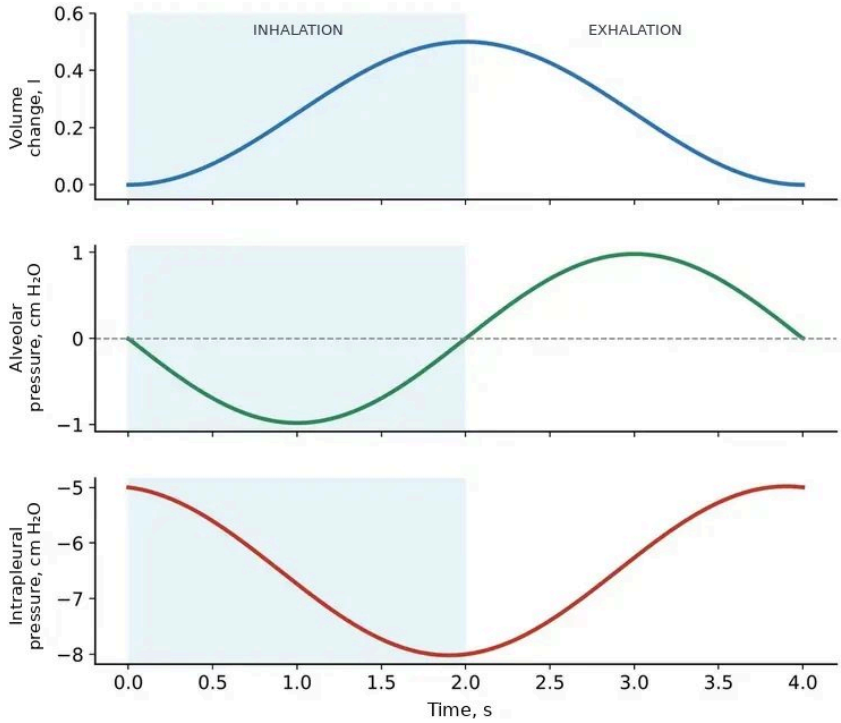


Figure 16.3. *Changes in lung volume and alveolar and intrapleural pressure during the respiratory cycle (drawn after Costanzo L.S., 2018)*

CLINICAL SIGNIFICANCE

Pneumothorax is the entry of air into the pleural cavity (chest injury, rupture of a bulla). The negative pressure disappears and the lung collapses due to its elastic recoil — it no longer takes part in breathing.

Valvular (tension) pneumothorax is life-threatening: with each inspiration air enters but does not leave, pressure rises and the mediastinum shifts. Emergency help — puncture and drainage of the pleural cavity.

Elastic recoil of the lungs and surfactant

The elastic recoil of the lungs is due to two factors: the **elastic fibers** of lung tissue (≈ 1/3) and the **surface tension** of the fluid layer on the inner surface of the alveoli (≈ 2/3).

Surfactant is a complex of phospholipids (mainly dipalmitoylphosphatidylcholine) and proteins produced by type II alveolocytes. Its functions: it **reduces surface tension** and increases lung compliance (reducing the work of breathing); it **prevents the collapse of small alveoli (atelectasis)** — according to Laplace's law the pressure in a small alveolus should be greater, but surfactant is denser in it and reduces the tension more; it keeps the alveoli dry (prevents pulmonary edema); and it has a protective function.

CLINICAL SIGNIFICANCE

Surfactant begins to be synthesized in the fetus at 24–28 weeks and becomes sufficient by 34–35 weeks. In premature infants surfactant deficiency leads to **respiratory distress syndrome** (hyaline membrane disease): the alveoli collapse and breathing becomes difficult. Prevention — glucocorticoids for the mother when preterm birth is threatened (they accelerate surfactant synthesis); treatment — endotracheal administration of surfactant.

Lung compliance is the change in lung volume per 1 cm H_2O change in pressure; normally ≈ 200 ml/cm H_2O (lungs and chest together — 100). It decreases in fibrosis and increases in emphysema.

Airway resistance is mainly in the medium-sized bronchi; it rises sharply when the bronchi narrow (asthma, bronchitis) (Poiseuille's law: the fourth power of the radius).

3. Gas exchange in the lungs

Table 16.1. *Composition of inspired, expired and alveolar air (%)*

Air	O_2	CO_2	N_2 and inert gases
Atmospheric (inspired) air	20,93	0,03	79,04
Expired air	16,0	4,0	80,0
Alveolar air	14,0	5,5	80,5

Because expired air is a mixture of alveolar air and “dead space” air, it contains more O_2 and less CO_2.

The **partial pressure of a gas** is the pressure attributable to that gas in a gas mixture: Pgas = Ptotal × gas fraction (%) / 100. In the alveoli the water vapor pressure (47 mm Hg) is taken into account: pO_2 alv = (760 − 47) × 0.14 ≈ 100 mm Hg.

Table 16.2. *Partial pressures and tensions of gases (mm Hg)*

Medium	pO_2	pCO_2
Atmospheric air (at sea level)	159	0,2
Alveolar air	100–104	40

Medium	pO_2	pCO_2
Arterial blood	95–100	40
Venous blood (mixed)	40	46
Tissues (interstitial fluid)	20–40	46–60

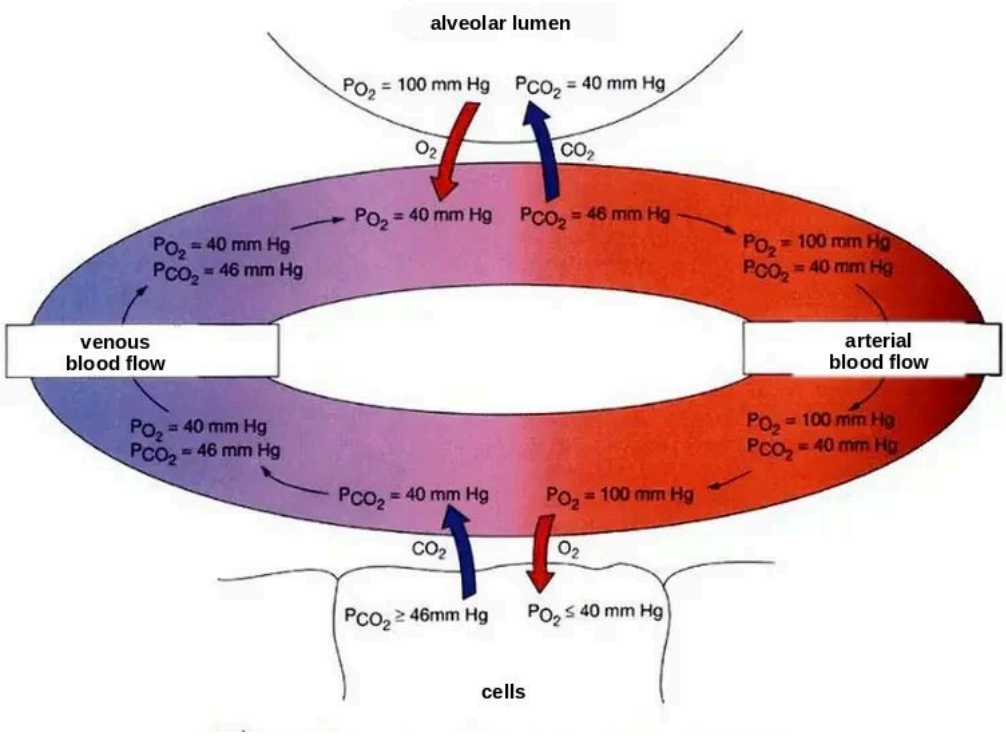


Figure 16.4. *Partial pressures of gases in the pulmonary and tissue capillaries (Alavia O.T., 2018)*

Gases pass from the alveoli into the blood and from the blood into the tissues only by **diffusion**, down the partial pressure gradient. In the lungs the gradient for O_2 is 100 − 40 = **60 mm Hg**, and for CO_2 46 − 40 = **6 mm Hg**. Nevertheless CO_2 diffuses well: its solubility is 20–25 times higher than that of O_2.

The alveolar-capillary membrane

The air–blood barrier is only **0.3–0.5 µm** thick: the surfactant layer, the alveolar epithelium (type I alveolocytes), the common basement membrane and the capillary endothelium; plus a layer of plasma and the erythrocyte membrane. The total diffusion surface is **70–100 m^2**. An erythrocyte passes through a capillary in ≈ 0.75 s, while gas equilibrium is reached in 0.25 s — there is a large reserve.

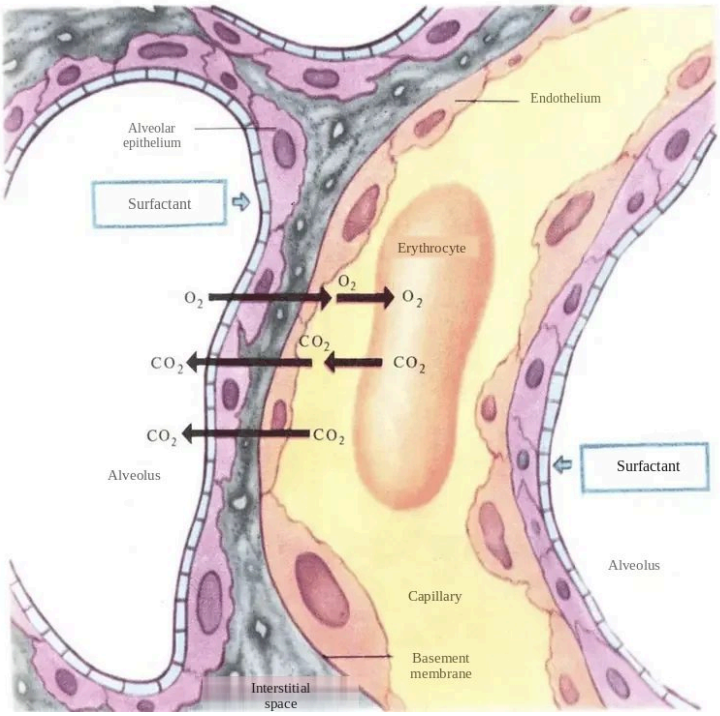


Figure 16.5. *The alveolar-capillary membrane and the diffusion of gases (Alavia O.T., 2018)*

According to **Fick's law**, the rate of diffusion is directly proportional to the partial pressure difference, the membrane area and the solubility of the gas, and inversely proportional to membrane thickness. **The diffusing capacity of the lung** is the volume of gas that diffuses in 1 minute at a gradient of 1 mm Hg; for O_2 at rest it is ≈ 25 ml/min/mm Hg.

The ventilation-perfusion ratio

For effective gas exchange, alveolar ventilation (≈ 4.2 l/min) and pulmonary blood flow (≈ 5 l/min) must be matched: **V/Q ≈ 0.8–1.0**. At the lung apices V/Q is high (ventilation predominates), at the base it is low (blood flow predominates). **Hypoxic vasoconstriction** — when O_2 falls in a poorly ventilated area, the vessels there constrict and blood is directed to well-ventilated areas.

> **CAUTION**
> Two extreme states of V/Q mismatch: **dead space** (ventilation present, no blood flow — pulmonary embolism) and **shunt** (blood flow present, no ventilation — pneumonia, atelectasis). In a shunt, even giving 100% oxygen does not fully correct the hypoxemia.

4. Regulation of breathing

The respiratory center

The **respiratory center** is the set of CNS neurons that generate the respiratory rhythm and adapt it to the body's needs. Its main part is in the **medulla oblongata** (N.A. Mislavsky, 1885):

- the **dorsal respiratory group** (in the region of the nucleus of the solitary tract) — mainly inspiratory neurons; it receives afferent information;
- the **ventral respiratory group** — inspiratory and expiratory neurons; within it the **pre-Bötzinger complex** is the respiratory **rhythm generator** (neurons with pacemaker properties);
- the **pneumotaxic center in the pons** — terminates inspiration on time and ensures the switch to expiration; its damage makes breathing deep and slow. After transection between the medulla

oblongata and the spinal cord breathing stops; after transection above the pons breathing is preserved.

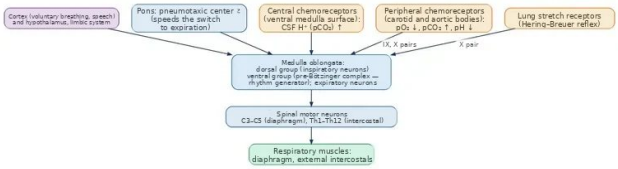


Figure 16.6. *The system of respiratory control*

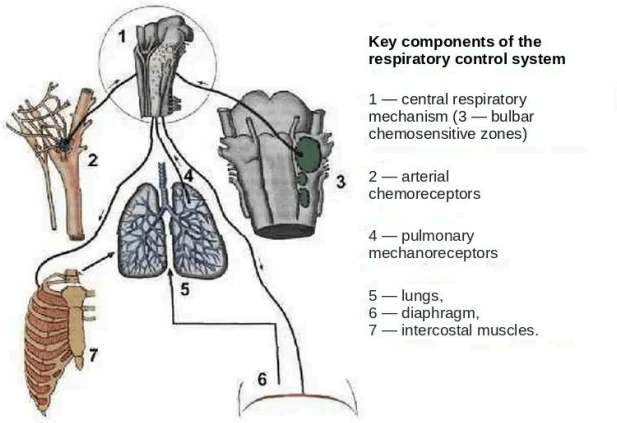


Figure 16.7. *Main links of the system of respiratory control (Alavia O.T., 2018)*

Humoral regulation — chemoreceptors

Table 16.3. *Chemoreceptors regulating breathing*

Receptors	Location	What they sense	Significance
Central	Ventral surface of the medulla oblongata	An increase in H^+ in the cerebrospinal fluid (mainly due to a rise in pCO_2 — CO_2 crosses the blood–brain barrier easily)	≈ 70–80% of the control of quiet breathing; sensitive to CO_2
Peripheral	The carotid bodies at the bifurcation of the carotid artery (cranial nerve IX — Hering's nerve) and the aortic bodies (cranial nerve X)	A fall in arterial pO_2 (mainly below 60 mm Hg), a rise in pCO_2, a fall in pH	The only receptors that respond to hypoxia

REMEMBER

The main humoral stimulus of breathing is **CO_2** (more precisely, H^+): a rise in pCO_2 of 1 mm Hg increases ventilation by 2–3 l/min. Hypoxia strongly stimulates breathing only when pO_2 falls below 60 mm Hg. **Fredericq's experiment** (cross-circulation of the heads of two dogs) and Haldane's experiments proved the humoral regulation of breathing.

CLINICAL SIGNIFICANCE

In patients with chronic obstructive pulmonary disease, because pCO_2 is constantly high, the central chemoreceptors "adapt" to it and breathing is driven mainly by **hypoxia** (the peripheral chemoreceptors). If such a patient is given high-concentration oxygen, the hypoxic drive disappears and breathing may weaken and hypercapnia worsen. Therefore oxygen is given to them under control, at a low flow.

Reflex regulation

- **Hering–Breuer reflex** — during inspiration the stretch receptors of the lungs are excited, and impulses travel along the vagus nerve to the respiratory center and stop inspiration (the switch from inspiration to expiration). In adults it is of little importance in quiet breathing and is activated in deep breathing (volume > 1–1.5 l); it is well expressed in newborns.
- **Protective reflexes** — coughing (irritant receptors of the larynx, trachea and bronchi), sneezing (nasal mucosa), reflex cessation of breathing (pungent substances, diving into water).

- **Proprioceptors** (muscles and joints) — rapidly increase ventilation at the start of physical work.
- **The cerebral cortex** — voluntary control of breathing (speech, singing, breath-holding), conditioned reflexes (increased breathing before the start).

5. Laboratory work

Exercise 1: demonstrating the mechanism of breathing with the Donders model

1. Examine the model: the lungs of an animal are suspended by the trachea inside a glass bell jar (its lower part closed by a rubber membrane); the cavity of the jar is connected to a manometer.
2. Pull the rubber membrane down (imitating contraction of the diaphragm).
3. Observe the change in pressure inside the jar and in lung volume.
4. Release the membrane. **Result and analysis:** when the membrane is pulled, the pressure inside the jar falls (the manometer shows negative pressure) and the lungs expand and air enters them — inspiration. When it is released, the lungs shrink because of their elastic recoil — expiration. The model shows that the lungs passively follow changes in chest volume.

Exercise 2: pneumography — recording respiratory movements

1. Tie the pneumograph cuff (or sensor) around the subject's chest and connect it to a kymograph or computer system.
2. Record quiet breathing for 1 minute and count the respiratory rate (normal 12–18 /min).
3. The subject breathes rapidly and deeply for 30 s (hyperventilation), then returns to normal breathing — continue recording.
4. Record breathing during voluntary breath-holding, speaking, swallowing and after 20 squats. **Expected result and analysis:** after hyperventilation a short **apnea** or weakening of breathing is observed — CO_2 has been “washed out” (hypocapnia), and the main stimulus of the respiratory center has decreased. After breath-holding breathing becomes deeper and faster — CO_2 has accumulated. During swallowing breathing stops reflexly. During physical load the rate and depth of breathing increase.

Exercise 3: the Stange and Genchi tests

1. **Stange test**: the seated subject takes a deep (not maximal) breath, pinches the nose and holds the breath; the time is measured with a stopwatch.
2. **Genchi test**: the breath-holding time after a quiet expiration is measured.
3. Repeat the Stange test after 1 minute of hyperventilation. **Interpretation:** the Stange test is 40–60 s in healthy adults (60–120 s in athletes), the Genchi test 20–40 s. After hyperventilation the breath-holding time increases — blood CO_2 decreases, and it takes longer to reach the “threshold” that excites the respiratory center.

6. Practical skills

Skill: calculating the partial pressure of a gas in alveolar air

1. Determine the atmospheric pressure (760 mm Hg at sea level).
2. Subtract the water vapor pressure in the alveoli (47 mm Hg).
3. Multiply by the percentage of the gas: $pO_2 = (760 - 47) \times 14 / 100 \approx 100$ mm Hg.

4. Repeat the calculation for the lower atmospheric pressure at altitude and assess the risk of hypoxia.

Skill: evaluating the type and rate of breathing

1. Without the subject noticing (as if counting the pulse), count the respiratory movements for 1 minute.
2. Determine the type of breathing: thoracic (mainly in women), abdominal (diaphragmatic — mainly in men), mixed.
3. Evaluate the rhythm and depth.
4. Compare the result with the age norm (12–18 /min in adults; 40–60 in newborns).

Review questions

1. What stages does respiration consist of?
2. List the inspiratory and expiratory muscles.
3. Explain the mechanism of inspiration step by step.
4. Why is quiet expiration passive?
5. What is intrapleural pressure, and how does it arise?
6. What happens in pneumothorax?
7. What factors does the elastic recoil of the lungs depend on?
8. What are the functions of surfactant and the consequences of its deficiency?
9. What is the composition of atmospheric, expired and alveolar air?
10. How is the partial pressure of a gas calculated?
11. What is the structure of the alveolar-capillary membrane, and what factors affect diffusion?
12. What is the ventilation-perfusion ratio?
13. What is the structure of the respiratory center, and where is the rhythm generator?
14. What is the difference between central and peripheral chemoreceptors?
15. Explain the Hering–Breuer reflex and the protective respiratory reflexes.

Test questions

1. Which muscle plays the main role in quiet inspiration?
   A) The internal intercostal muscles
   B) The abdominal muscles
   C) The sternocleidomastoid muscle
   D) The pectoralis major
   E) The diaphragm

2. What kind of process is quiet expiration?
   A) Active — the internal intercostal muscles contract
   B) Active — the abdominal muscles contract
   C) Passive — due to the elastic recoil of the lungs
   D) Active — the diaphragm contracts
   E) Only due to the accessory muscles

3. What is the intrapleural pressure at the end of quiet expiration?
   A) −3…−5 cm H_2O (below atmospheric)
   B) Equal to atmospheric pressure
   C) +5 cm H_2O
   D) −40 cm H_2O
   E) +20 cm H_2O

4. Which cells produce surfactant?
   A) Type I alveolocytes
   B) Type II alveolocytes
   C) Alveolar macrophages
   D) Ciliated epithelium
   E) Goblet cells

5. What is the main function of surfactant?
   A) To reduce the diffusion of gases
   B) To constrict the bronchi
   C) To reduce surface tension in the alveoli and prevent their collapse
   D) To bind oxygen
   E) Mucociliary clearance

6. What is the pO_2 in alveolar air?
   A) 159 mm Hg
   B) ≈ 100 mm Hg
   C) 40 mm Hg
   D) 760 mm Hg
   E) 20 mm Hg

7. What is the partial pressure gradient of CO_2 in the lungs?
   A) ≈ 6 mm Hg
   B) ≈ 60 mm Hg
   C) ≈ 100 mm Hg
   D) ≈ 40 mm Hg
   E) There is no gradient

8. By what mechanism do gases pass from the alveoli into the blood?
   A) Active transport
   B) Filtration
   C) Pinocytosis
   D) Diffusion
   E) Osmosis

9. Where is the respiratory rhythm generator located?
   A) In the ventral group of the medulla oblongata (pre-Bötzinger complex)
   B) In the cerebral cortex
   C) In the cerebellum
   D) In the hypothalamus
   E) In the spinal cord

10. Where are the receptors that sense a fall in arterial pO_2 located?
    A) On the surface of the medulla oblongata
    B) In the pulmonary alveoli
    C) In the tracheal mucosa
    D) In the hypothalamus
    E) In the carotid and aortic bodies

11. What is the main humoral stimulus of breathing at rest?
    A) A decrease in O_2
    B) Lactate
    C) Adrenaline
    D) An increase in CO_2 (H^+)
    E) Glucose

12. What causes the short cessation of breathing (apnea) after hyperventilation?
    A) An increase in O_2
    B) A decrease in blood CO_2 (hypocapnia)
    C) Fatigue of the respiratory muscles
    D) Acidosis
    E) The Hering–Breuer reflex

13. What is the Hering–Breuer reflex?
    A) The cough reflex
    B) The sneeze reflex
    C) Increased breathing in hypoxia
    D) Cessation of breathing when diving into water
    E) Termination of inspiration when the lungs are stretched (via the vagus nerve)

14. Why does the lung collapse in pneumothorax?
    A) The diaphragm is paralyzed
    B) The bronchi narrow
    C) The negative pressure in the pleural cavity disappears and the lung shrinks due to its elastic recoil
    D) Surfactant increases
    E) Alveolar pressure becomes negative

15. What is the normal Stange test result in healthy adults?
    A) 5–10 s
    B) 10–20 s
    C) 3–5 minutes
    D) 40–60 s
    E) Less than 1 s

Clinical cases

Case 1

A tall, thin 20-year-old young man suddenly developed sharp pain in the right side of the chest and shortness of breath. No breath sounds are heard on the right. On X-ray the right lung has collapsed toward its root, with air in the pleural cavity.

Questions: What is the condition? Why did the lung collapse? Why does this lung not expand during inspiration?

Answer: Spontaneous pneumothorax (a bulla on the lung surface has ruptured). Normally the pressure in the pleural cavity is lower than atmospheric — it keeps the lung "stuck" to the chest wall. When air enters, the pressure equalizes with atmospheric and the lung shrinks toward its root because of its own elastic recoil (elastic fibers and alveolar surface tension). Even when the chest expands during inspiration, no negative pressure is created in the pleural cavity — air simply enters the pleural cavity, and the lung does not expand passively. Treatment — removal of the air (puncture, drainage).

Case 2

A baby born at 29 weeks of pregnancy developed rapid breathing a few hours after birth; the soft parts of the chest are drawn inward, there is grunting on expiration, and cyanosis is increasing.

Questions: What is the condition? Explain its mechanism on the basis of the function of surfactant.

Answer: Respiratory distress syndrome of the newborn. Surfactant is synthesized in sufficient amounts after 34–35 weeks; a premature baby lacks it. As a result the surface tension in the alveoli remains high: according to Laplace's law the small alveoli collapse during expiration (atelectasis), lung compliance falls sharply and each inspiration requires great effort — chest retraction, shortness of breath. No gas exchange takes place in the collapsed alveoli (shunt) — hypoxemia and cyanosis. Treatment — administration of surfactant preparations into the trachea and respiratory support.

Case 3

A student who was nervous before an exam breathed rapidly and deeply. He developed dizziness and numbness of the fingers and around the lips. He was advised to breathe in and out of a paper bag.

Questions: Explain the mechanism of the signs. Why does breathing into a bag help?

Answer: Hyperventilation syndrome. Because of excessive ventilation a lot of CO_2 is removed from the blood — hypocapnia and respiratory alkalosis. The decrease in CO_2 constricts the cerebral vessels — cerebral blood flow decreases and dizziness occurs. Alkalosis enhances the binding of ionized calcium to proteins — free Ca^{2+} decreases and nerve excitability increases — paresthesias (numbness), and in severe cases cramps. When breathing into a bag, the expired CO_2 is re-inhaled, blood pCO_2 is restored and the signs disappear (but this method is used only briefly and only when it is clear that the cause is hyperventilation).

Quick review

Table 16.4. *Key concepts of topic 16*

Concept	Brief definition
Stages of respiration	External respiration, exchange in the lungs, transport, exchange in the tissues, cellular respiration
The diaphragm	Main inspiratory muscle (70–80%)
Quiet expiration	Passive — elastic recoil
Intrapleural pressure	−3…−5 (on expiration), −6…−8 cm H_2O (on inspiration)
Pneumothorax	Air enters the pleura — the lung collapses
Surfactant	Type II alveolocytes; reduces surface tension
Compliance	Lung compliance ≈ 200 ml/cm H_2O

Concept	Brief definition
Alveolar air	O_2 14%, CO_2 5.5%
Alveolar pO_2 / pCO_2	100 / 40 mm Hg
Gradients	O_2 — 60, CO_2 — 6 mm Hg
Air–blood barrier	0.3–0.5 μm; area 70–100 m^2
V/Q	≈ 0,8–1,0
The respiratory center	Medulla oblongata (pre-Bötzinger — rhythm); pons — pneumotaxic
Central chemoreceptors	H^+ in the CSF (CO_2)
Peripheral chemoreceptors	Carotid and aortic bodies — pO_2 ↓

TOPIC 17

Methods of functional examination of the respiratory system. Lung volumes. Spirography. Transport of gases in the blood. Tissue respiration. Pulse oximetry

Learning objectives

To teach the student lung volumes and capacities, the parameters of pulmonary ventilation, the methods of spirometry and spirography, the mechanisms of transport of oxygen and carbon dioxide in the blood, the oxyhemoglobin dissociation curve, gas exchange in the tissues and tissue respiration, and the principles of pulse oximetry; to master performing spirometry and evaluating the results.

By the end of the class the student should know:

- lung volumes and capacities and their normal values;
- minute ventilation, alveolar ventilation and dead space;
- spirographic parameters (FEV_1, Tiffeneau index, maximal voluntary ventilation);
- the forms of oxygen transport in the blood and the oxygen capacity;
- the oxyhemoglobin dissociation curve, its shape and shifts (Bohr effect);
- the forms of CO_2 transport in the blood, carbonic anhydrase and the chloride shift (Hamburger);
- gas exchange in the tissues and tissue respiration;
- the principle, norm and limitations of pulse oximetry.

The student should be able to:

- measure lung volumes with a spirometer and compare the VC with the predicted value;
- analyze a spirogram and distinguish obstructive from restrictive disorders;
- perform pulse oximetry and evaluate the result.

1. Lung volumes and capacities

Table 17.1. *Lung volumes (in a middle-aged man, approximate)*

Volume	Definition	Magnitude, ml
Tidal volume (TV)	Air entering (leaving) during quiet inspiration	400–600 (≈ 500)
Inspiratory reserve volume (IRV)	The maximum volume that can additionally be inhaled after a quiet inspiration	1500–3000
Expiratory reserve volume (ERV)	The volume that can additionally be exhaled after a quiet expiration	1000–1500
Residual volume (RV)	Air that remains in the lungs even after maximal expiration	1000–1500

Table 17.2. *Lung capacities (sums of volumes)*

Capacity	Composition	Magnitude, ml
Vital capacity (VC)	TV + IRV + ERV	3500–5000 in men, 3000–4000 in women
Inspiratory capacity	TV + IRV	≈ 3500
Functional residual capacity (FRC)	ERV + RV	≈ 2400

Capacity	Composition	Magnitude, ml
Total lung capacity (TLC)	VC + RV	4500–6000

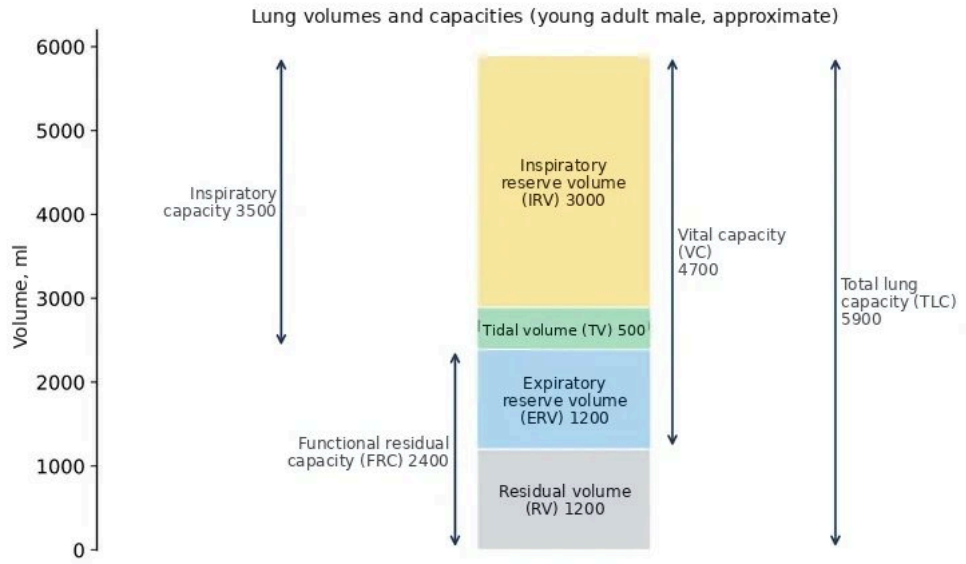


Figure 17.1. *Lung volumes and capacities*

The **functional residual capacity** is important: it dilutes the air entering with each breath (500 ml) and keeps the composition of alveolar air stable throughout the respiratory cycle — gas concentrations do not fluctuate sharply. The **residual volume** and FRC cannot be measured with an ordinary spirometer (helium dilution or body plethysmography is required).

> **CLINICAL SIGNIFICANCE**
> **In determining the cause of death**, whether the lung sinks in water is important: the lungs of a baby who breathed after birth contain residual (minimal) air, and a piece of lung does not sink in water; the lungs of a stillborn fetus sink (a forensic test).

2. Pulmonary ventilation

Minute ventilation (MV) = tidal volume × respiratory rate = 500 ml × 14–16 = **6–8 l/min**. During heavy physical work it increases to 100–150 l/min.

Dead space is the volume of the airways that does not take part in gas exchange:

- **Anatomical dead space** — from the nasopharynx to the terminal bronchioles: ≈ **150 ml** (≈ 2 ml/kg).
- **Physiological (functional) dead space** — anatomical plus alveoli that are ventilated but not perfused; in a healthy person it is almost equal to the anatomical dead space. **Alveolar ventilation** = (TV − dead space) × respiratory rate = (500 − 150) × 14 ≈ **4.2–5 l/min**. It is alveolar ventilation that determines gas exchange.

> **REMEMBER**
> **Rapid shallow breathing is ineffective**: for example, with a TV of 200 ml and a respiratory rate of 40 /min the MV is 8 l/min (close to normal), but alveolar ventilation is (200 − 150) × 40 = 2 l/min — hypoventilation. Therefore **deep, slow breathing** is more effective.

3. Spirometry and spirography

Spirometry is the measurement of lung volumes; **spirography** is the graphic recording of respiratory movements and volume changes over time. Modern computer spirometers also record the **"flow–volume" curve**.

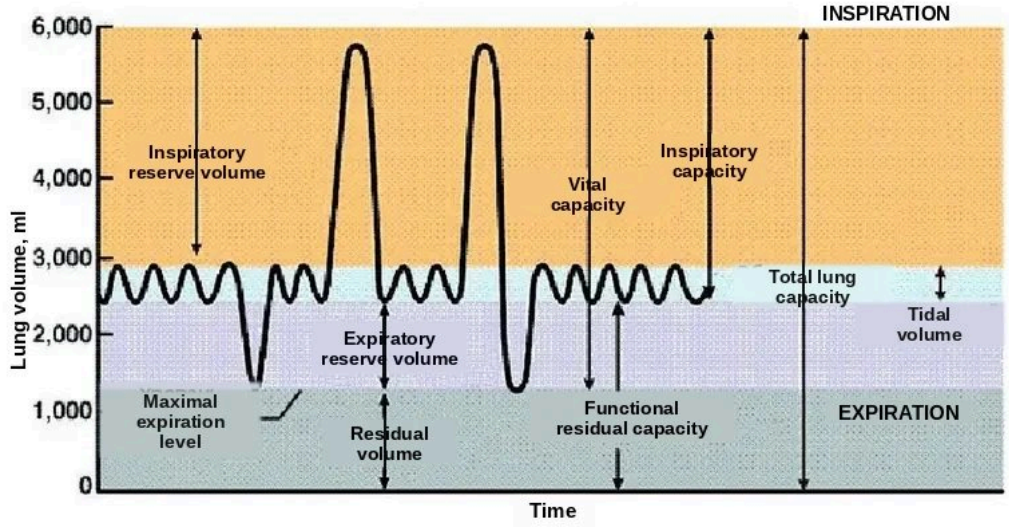


Figure 17.2. *Spirogram: lung volumes and capacities (Alavia O.T., 2018)*

Table 17.3. *Main dynamic parameters of spirography*

Parameter	Definition	Normal value
FVC (forced VC)	The volume of air exhaled as quickly and completely as possible after a maximal inspiration	≥ 80% of the predicted value
FEV_1	The volume in the first second of forced expiration	≥ 80% of the predicted value
Tiffeneau index (FEV_1 / VC or FEV_1 / FVC)	Indicator of bronchial patency	≥ 70% (75–85% in young people)
Peak expiratory flow (PEF)	The maximum flow rate during forced expiration	Depends on age, sex and height; also measured with a peak flow meter
Maximal voluntary ventilation (MVV)	Ventilation during 10–15 s of the fastest and deepest breathing, converted to 1 minute	80–150 l/min
Breathing reserve	MVV − MV	More than 85% of MVV

Table 17.4. *Types of ventilatory disorders*

Type	Essence	VC	FEV_1	Tiffeneau index	Examples
Obstructive	Airway patency is reduced	Normal or slightly reduced	Sharply reduced	< 70 %	Bronchial asthma, chronic obstructive pulmonary disease
Restrictive	The ability of the lungs to expand is reduced	Reduced	Proportionally reduced	Normal or increased	Pulmonary fibrosis, pleural adhesions, chest deformity, obesity
Mixed	Both	Reduced	Reduced	Reduced	Late stages of disease

CLINICAL SIGNIFICANCE

Bronchodilator test — an increase in FEV_1 of ≥ 12% and ≥ 200 ml 15–20 minutes after salbutamol inhalation shows that the obstruction is reversible and is typical of **bronchial asthma**; in chronic obstructive pulmonary disease the obstruction is not fully reversible.

The **predicted VC** is calculated by a formula (Baldwin, Cournand, Richards): for men VC (l) = (27.63 − 0.112 × age) × height (cm) / 1000; for women (21.78 − 0.101 × age) × height / 1000. If the actual VC is within ± 15–20% of the predicted value, it is normal.

4. Transport of oxygen in the blood

Oxygen is carried in the blood in two forms:

1. **Physically dissolved** — a small amount (0.3 ml per 100 ml of blood at a pO_2 of 100 mm Hg). But it is the dissolved O_2 that creates the partial pressure (tension) and ensures diffusion.
2. **Bound to hemoglobin (oxyhemoglobin)** — 98–99%. The **oxygen capacity of blood** is the maximum amount of O_2 that 1 l of blood can bind: Hb 150 g/l × 1.34 ml ≈ **200 ml O_2/l** (20 vol%). Arterial blood contains ≈ 190–200 ml/l of O_2 (saturation 96–98%), venous blood ≈ 140–150 ml/l (saturation ≈ 75%).

Arteriovenous difference — the difference in O_2 content between arterial and venous blood: ≈ 50 ml/l at rest. **Oxygen utilization coefficient** of the tissues = arteriovenous difference / O_2 in arterial blood ≈ 25–30%; up to 50–60% during physical work, and 60–70% in heart muscle even at rest.

The oxyhemoglobin dissociation curve

The relationship between the saturation of hemoglobin with O_2 (saturation, SO_2) and pO_2 is expressed by an **S-shaped** curve. The S shape results from the **cooperative effect**: once one heme binds O_2, the affinity of the others increases.

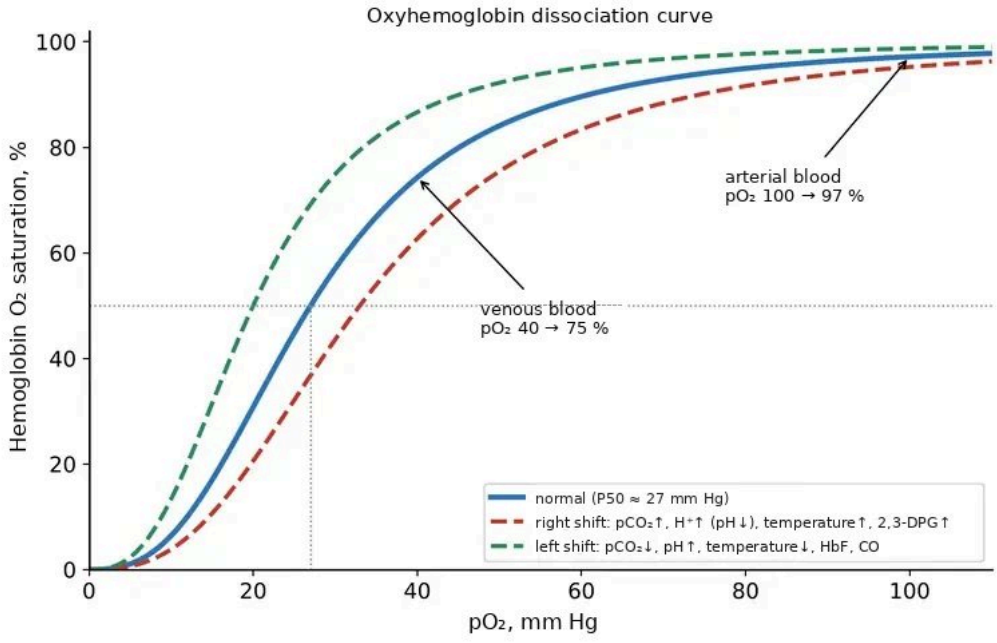


Figure 17.3. *The oxyhemoglobin dissociation curve and its shifts (drawn after Costanzo L.S., 2018)*

Physiological significance of the curve:

- **The upper flat part** (pO_2 70–100 mm Hg) — in the lungs: even if pO_2 falls slightly (altitude, lung diseases), the saturation remains high — oxygen supply is **reliable**.
- **The steep lower part** (pO_2 10–40) — in the tissues: a small fall in pO_2 releases a lot of O_2 — the tissues **easily obtain** O_2. **P_{50}** is the pO_2 at which hemoglobin is 50% saturated; normally ≈ **26–27 mm Hg.** An increase in P_{50} (a right shift of the curve) means that the affinity of Hb for O_2 has decreased and the delivery of O_2 to the tissues is facilitated.

Table 17.5. *Factors affecting the shift of the curve*

Right shift (O_2 delivery facilitated)	Left shift (O_2 retention enhanced)
Rise in pCO_2 and fall in pH (Bohr effect)	Fall in pCO_2, rise in pH (alkalosis)
Rise in temperature	Fall in temperature
Increase in 2,3-diphosphoglycerate (2,3-DPG) (hypoxia, high altitude, anemia)	Decrease in 2,3-DPG (stored preserved blood)
Physical work (all factors in working muscle)	Fetal hemoglobin (HbF), carboxyhemoglobin, methemoglobin

MEMORY AID

Shift to the right — **“CADET, face Right!”**: **C**O_2, **A**cid (H^+), **D**PG (2,3-DPG), **E**xercise (physical work), **T**emperature — when these increase the curve shifts **to the right** and the tissues receive more oxygen.

5. Transport of carbon dioxide in the blood

Table 17.6. *Forms of CO_2 transport in the blood*

Form	Share	Note
Bicarbonates (HCO_3^-)	≈ 70 %	Sodium bicarbonate in plasma, potassium bicarbonate in erythrocytes
Carbhemoglobin (carbamino compounds)	≈ 20–23 %	CO_2 binds to the amino groups of globin
Physically dissolved	≈ 7–10 %	—

Mechanism (in tissue capillaries):

1. CO_2 diffuses from the tissue into the plasma and then into the erythrocyte.
2. In the erythrocyte the enzyme **carbonic anhydrase** accelerates the reaction $CO_2 + H_2O \rightarrow H_2CO_3$ 13,000-fold.
3. H_2CO_3 dissociates into $H^+ + HCO_3^-$. H^+ is bound by deoxyhemoglobin (buffer).
4. HCO_3^- leaves the erythrocyte for the plasma and Cl^- enters the erythrocyte in its place — the **chloride shift (Hamburger phenomenon)**.
5. In the pulmonary capillaries all processes go in the reverse direction and CO_2 is released into the alveoli.

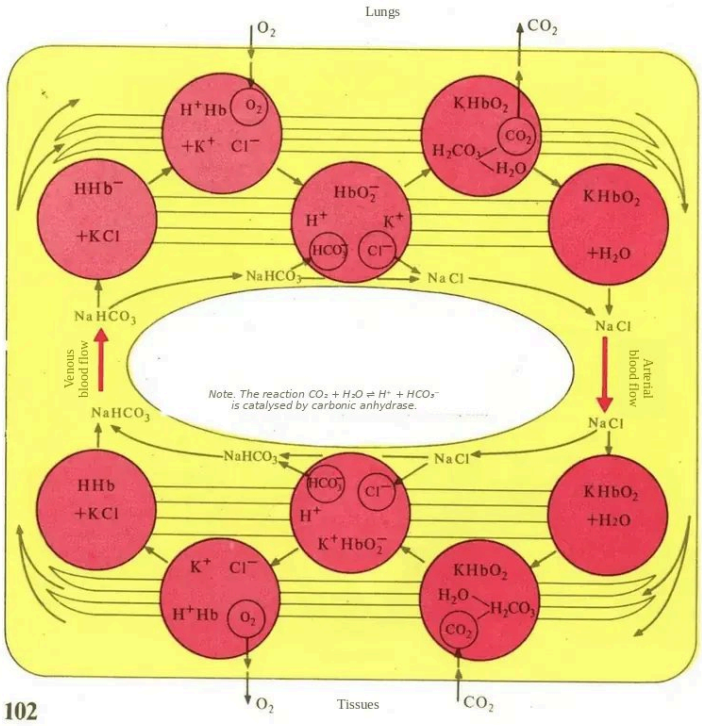


Figure 17.4. *Transport of CO_2 in the blood: processes inside the erythrocyte in the tissue and pulmonary capillaries (Alavia O.T., 2018)*

REMEMBER

The transport of O_2 and CO_2 is interrelated: **Bohr effect** — an increase in CO_2 and H^+ facilitates the release of O_2 by Hb (in the tissues); **Haldane effect** — deoxyhemoglobin binds CO_2 and H^+ better than oxyhemoglobin, so in the tissues the release of O_2 facilitates the uptake of CO_2, and in the lungs oxygenation facilitates the release of CO_2.

6. Gas exchange in the tissues and tissue respiration

In the tissue capillaries O_2 diffuses from the blood (pO_2 95–100) into the tissue (pO_2 20–40, 5–10 mm Hg in working muscle), and CO_2 from the tissue (pCO_2 46–60) into the blood (40). Diffusion is affected by capillary density, the number of open capillaries, the distance to the cells and the tissue's O_2 consumption. In muscles **myoglobin** stores O_2 and carries it inside the cell (its affinity for O_2 is higher than that of hemoglobin — $P_{50} \approx 1–5$ mm Hg).

Tissue (cellular) respiration is the use of O_2 as the final electron acceptor in the mitochondrial respiratory chain (cytochromes, cytochrome oxidase) and the synthesis of ATP by oxidative phosphorylation. For normal mitochondrial function an intracellular pO_2 of only 1–3 mm Hg is sufficient (**critical pO_2**).

CLINICAL SIGNIFICANCE

Cyanides (bitter almonds, cyanide salts) block **cytochrome oxidase** — even if there is enough oxygen in the blood, the cells cannot use it (**tissue hypoxia**). Venous blood remains saturated with oxygen (the arteriovenous difference narrows) and the skin is pink.

7. Pulse oximetry

Pulse oximetry is a non-invasive, continuous method of determining arterial oxygen saturation (**SpO_2**). The sensor (usually placed on a finger) transmits light of two wavelengths — **red (660 nm)** and **infrared (940 nm)**. Oxyhemoglobin absorbs more infrared light, deoxyhemoglobin more red

light. The device analyzes only the **pulsatile** (arterial) component, calculates SpO_2 from the ratio of absorption of the two lights and at the same time shows the pulse rate.

Table 17.7. *Evaluation of SpO_2*

SpO_2	Evaluation
95–100 %	Normal value
90–94 %	Mild hypoxemia (in chronic lung diseases the target level may be 88–92%)
85–89 %	Moderate hypoxemia — oxygen therapy
< 85 %	Severe hypoxemia

CAUTION

Limitations of pulse oximetry: in **carboxyhemoglobinemia** (carbon monoxide poisoning) and **methemoglobinemia** the result is wrong (falsely high or about ≈ 85%); peripheral vasospasm, cold hands, shock, movement, nail polish, bright ambient light and dark skin reduce accuracy. SpO_2 shows only oxygenation — **it does not evaluate ventilation (CO_2)**: in a patient receiving oxygen hypercapnia may be present even when SpO_2 is normal.

8. Laboratory work

Exercise 1: determining lung volumes with a water (or dry) spirometer

1. Wipe the mouthpiece with alcohol; the subject closes the nose with a clip.
2. **Tidal volume**: after a quiet inspiration, exhale quietly into the spirometer 3–5 times and calculate the mean.
3. **ERV**: after a quiet expiration, exhale maximally into the spirometer.
4. **VC**: take a maximal breath and exhale maximally into the spirometer; repeat three times and take the largest result.
5. **IRV** = VC − (TV + ERV).
6. Calculate the predicted VC by the formula and compare it with the actual VC.

Exercise 2: recording and analyzing a spirogram

1. Record quiet breathing (1 minute) on a computer spirograph — determine TV, respiratory rate and MV.
2. Perform the VC maneuver.
3. Forced expiration maneuver: take a maximal breath and exhale as quickly and completely as possible — determine FVC, FEV_1 and PEF.
4. Calculate the Tiffeneau index.
5. Compare the results with the predicted values and determine the type of ventilatory disorder.

Exercise 3: pulse oximetry

1. The subject's finger should be warm and the nail free of polish; put the sensor on the finger.
2. Wait for the signal (plethysmogram wave) to stabilize (10–20 s); record the SpO_2 and pulse readings.
3. Repeat the measurement after 30 s of breath-holding and after 20 squats.

4. Squeeze the finger tightly (reducing blood flow) and observe how the signal quality changes. **Expected result:** in a healthy person SpO_2 is 96–99%; brief breath-holding and light exercise hardly change it (because of the upper flat part of the dissociation curve and the O_2 reserve). When peripheral blood flow decreases, the signal deteriorates.

9. Practical skills

Skill: interpreting a spirometry result

1. Evaluate the FVC (VC) as a percentage of the predicted value (≥ 80% — normal).
2. Evaluate the FEV_1 relative to the predicted value.
3. Calculate the Tiffeneau index: < 70% — obstruction.
4. Index normal but FVC reduced — restriction is likely.
5. In obstruction, assess reversibility from the result of the bronchodilator test.

Skill: calculating blood oxygen parameters

1. Oxygen capacity = Hb (g/l) × 1.34.
2. O_2 in arterial blood = oxygen capacity × SaO_2 + 0.003 × pO_2 (ml/l).
3. Arteriovenous difference = arterial O_2 − venous O_2.
4. Utilization coefficient = arteriovenous difference / arterial O_2 × 100%.

Review questions

1. List the lung volumes and give their normal values.
2. Which volumes make up the VC, FRC and TLC?
3. What is the physiological significance of the functional residual capacity?
4. How are minute ventilation and alveolar ventilation calculated?
5. What are anatomical and physiological dead space?
6. Why is rapid shallow breathing ineffective?
7. What are the main spirographic parameters and the Tiffeneau index?
8. How are obstructive and restrictive disorders distinguished?
9. In what forms is oxygen carried in the blood? What is the oxygen capacity?
10. Explain the shape and physiological significance of the oxyhemoglobin dissociation curve.
11. List the factors that shift the curve to the right and to the left.
12. In what forms is CO_2 carried in the blood?
13. What is the role of carbonic anhydrase and the chloride shift?
14. Explain the Bohr and Haldane effects.
15. What are the principle and limitations of pulse oximetry?

Test questions

1. What is the tidal volume in quiet breathing?
   A) 150 ml
   B) 1500 ml
   C) ≈ 500 ml
   D) 3000 ml
   E) 5000 ml

2. Which volumes make up the vital capacity?
   A) Tidal volume + inspiratory reserve volume + expiratory reserve volume
   B) Tidal volume + residual volume
   C) Residual volume + expiratory reserve volume
   D) Only the inspiratory reserve volume
   E) TLC + residual volume

3. Which volume cannot be measured with an ordinary spirometer?
   A) Tidal volume
   B) VC
   C) IRV
   D) ERV
   E) Residual volume

4. What is the volume of the anatomical dead space?
   A) 500 ml
   B) ≈ 150 ml
   C) 1200 ml
   D) 50 ml
   E) 2400 ml

5. In which disorder does the Tiffeneau index decrease?
   A) In a restrictive disorder
   B) In obesity
   C) In pleural adhesions
   D) In an obstructive disorder (bronchial asthma)
   E) In pulmonary fibrosis

6. In what form is most of the oxygen carried in the blood?
   A) Physically dissolved
   B) As oxyhemoglobin
   C) As bicarbonate
   D) As carbhemoglobin
   E) With myoglobin

7. Which factor shifts the oxyhemoglobin dissociation curve to the right?
   A) Fall in temperature
   B) A rise in pH
   C) Fetal hemoglobin
   D) A fall in pCO_2
   E) An increase in 2,3-DPG

8. What is the Bohr effect?
   A) A decrease in the affinity of Hb for O_2 when CO_2 and H^+ increase
   B) The good binding of CO_2 by deoxyhemoglobin
   C) The exchange of HCO_3^- and Cl^-
   D) The binding of CO to Hb
   E) The storage of O_2 by myoglobin

9. In what form is most of the CO_2 carried?
   A) Physically dissolved
   B) Bicarbonates (HCO_3^-)
   C) Carbaminohemoglobin
   D) Carboxyhemoglobin
   E) Oxyhemoglobin

10. Which enzyme accelerates the hydration of CO_2 in the erythrocyte?
   A) Cytochrome oxidase
   B) Catalase
   C) Carbonic anhydrase
   D) Peroxidase
   E) Lactate dehydrogenase

11. What is the chloride shift (Hamburger phenomenon)?
   A) The entry of Cl^- into the erythrocyte in place of HCO_3^- leaving it
   B) The reabsorption of Cl^- in the kidney
   C) The exchange of Na^+ and K^+
   D) The binding of H^+ to hemoglobin
   E) The release of O_2 into the plasma

12. What is the normal P_{50}?
   A) 5 mm Hg
   B) 100 mm Hg
   C) 60 mm Hg
   D) 26–27 mm Hg
   E) 40 mm Hg

13. What is the mechanism of hypoxia in cyanide poisoning?
   A) Hb does not bind O_2
   B) Pulmonary ventilation is impaired
   C) Cytochrome oxidase is blocked and the cells cannot use O_2
   D) Blood flow stops
   E) Erythrocytes break down

14. In which situation does a pulse oximeter give a wrong result?
   A) In a healthy person at rest
   B) During light physical exercise
   C) On a finger without a nail
   D) When the respiratory rate is 16
   E) In carbon monoxide poisoning (carboxyhemoglobin)

15. If TV is 500 ml, the respiratory rate 15 /min and the dead space 150 ml, what is the alveolar ventilation?
    A) 7.5 l/min
    B) 2.25 l/min
    C) 9.75 l/min
    D) 5.25 l/min
    E) 0.35 l/min

Clinical cases

Case 1

A 25-year-old woman complains of cough and nocturnal attacks of breathlessness. Spirometry: FVC — 88% of predicted, FEV_1 — 58%, Tiffeneau index 62%. 20 minutes after salbutamol inhalation the FEV_1 increased by 22% (350 ml).

Questions: What type of ventilatory disorder is this? What does the result of the bronchodilator test show?

Answer: An obstructive disorder: the FEV_1 and Tiffeneau index (< 70%) are reduced, while the FVC is almost normal. When the bronchi narrow, airway resistance increases (according to Poiseuille's law, inversely to the fourth power of the radius), and rapid expiration of air becomes difficult. The bronchodilator test is positive (≥ 12% and ≥ 200 ml) — the obstruction is reversible; the spasm of the bronchial smooth muscle relaxed under the action of a β_2-adrenomimetic. This is typical of bronchial asthma.

Case 2

A mountaineer climbed to an altitude of 5000 m, where the atmospheric pressure is ≈ 405 mm Hg. A few days later his blood 2,3-DPG was found to have increased.

Questions: Calculate the approximate alveolar pO_2 at this altitude. What is the adaptive significance of the increase in 2,3-DPG?

Answer: In inspired air pO_2 = (405 − 47) × 0.21 ≈ 75 mm Hg; taking alveolar CO_2 and ventilation into account, alveolar pO_2 falls to ≈ 45–50 mm Hg (hyperventilation raises it slightly). At this pO_2 the saturation is about 80–85%. Hypoxia enhances 2,3-DPG synthesis in erythrocytes; 2,3-DPG binds to deoxyhemoglobin and reduces its affinity for O_2 — the dissociation curve shifts to the right. As a result, although O_2 binding in the lungs decreases slightly, O_2 release in the tissues is greatly facilitated — the oxygen supply of the tissues improves.

Case 3

A patient with severe pneumonia is breathing rapidly and shallowly: respiratory rate 40 /min, tidal volume 200 ml. SpO_2 86%.

Questions: Calculate the MV and alveolar ventilation. Why is such breathing ineffective?

Answer: MV = 200 × 40 = 8 l/min (it appears close to normal). Alveolar ventilation = (200 − 150) × 40 = 2 l/min — more than two times lower than normal (4.2–5 l/min). A large part of each breath (150 ml) ventilates the dead space and does not reach gas exchange. The result is alveolar hypoventilation, hypoxemia (SpO_2 86%) and a risk of CO_2 accumulation. In pneumonia the shunt in the inflamed alveoli also aggravates the hypoxemia.

Quick review

Table 17.8. *Key concepts of topic 17*

Concept	Brief definition
Tidal volume	≈ 500 ml
VC	TV + IRV + ERV; 3500–5000 ml in men
Residual volume	1000–1500 ml; not measured with a spirometer
FRC	ERV + RV; stabilizes the composition of alveolar air
MV	TV × respiratory rate = 6–8 l/min
Dead space	≈ 150 ml
Alveolar ventilation	(TV − dead space) × rate ≈ 4.2–5 l/min
Tiffeneau index	FEV_1 / VC ≥ 70%; reduced in obstruction
Oxygen capacity	Hb × 1.34 ≈ 200 ml/l
Dissociation curve	S-shaped; P_{50} ≈ 27 mm Hg
Shift to the right	CO_2, H^+, 2,3-DPG, temperature, exercise (Bohr effect)
CO_2 transport	HCO_3^- 70%, carbhemoglobin 20%, dissolved 10%
Carbonic anhydrase	$CO_2 + H_2O \rightleftarrows H_2CO_3$
Chloride shift	$HCO_3^- \leftrightarrow Cl^-$ exchange (Hamburger)
Pulse oximetry	Normal SpO_2 95–100%; inaccurate in CO poisoning

TOPIC 18

Methods of studying digestive function. Digestion in the oral cavity and stomach. Interrelation of the maxillofacial organs with the body's systems

Learning objectives

To teach the student the functions of the digestive system and the types of digestion, the methods of studying digestive function (I.P. Pavlov's chronic experimental methods and modern clinical methods), digestion in the oral cavity — the mechanisms of salivation, chewing and swallowing, digestion in the stomach — the composition of gastric juice and the phases and regulation of secretion, and the relationship of the maxillofacial organs with the other systems of the body.

By the end of the class the student should know:

- the secretory, motor, absorptive, excretory, endocrine and protective functions of the digestive system;
- the types of digestion (intrinsic, symbiotic, autolytic; cavity and membrane digestion);
- I.P. Pavlov's fistula methods and modern methods of examination;
- the composition and functions of saliva and the regulation of salivation;
- the mechanisms of chewing and swallowing, masticatiography;
- the composition of gastric juice and the significance of hydrochloric acid and pepsin;
- the phases of gastric secretion and its neurohumoral regulation;
- the relationship of the maxillofacial organs with digestion, respiration, speech and other systems.

The student should be able to:

- test the effect of salivary amylase on starch;
- demonstrate the effect of pepsin on protein and the role of HCl in it;
- evaluate the acidity parameters of gastric juice (total, free HCl);
- evaluate chewing efficiency.

1. Functions of the digestive system and types of digestion

Digestion is the mechanical and chemical processing of the complex substances of food (proteins, fats, carbohydrates) and their breakdown into simple compounds that have lost their species specificity and can be absorbed (amino acids, monosaccharides, fatty acids and monoglycerides).

Table 18.1. *Functions of the digestive system*

Function	Content
Secretory	Production of digestive juices (8–9 l per day)
Motor	Chewing, swallowing, mixing and propulsion of food, defecation
Absorption	Passage of digestion products, water, salts and vitamins into the blood and lymph
Excretory	Excretion of metabolic products, drugs and heavy metals
Endocrine	Gut hormones (gastrin, secretin, cholecystokinin, GIP, motilin, ghrelin)
Protective	Mucus, HCl, lysozyme, IgA, lymphoid tissue (gut-associated lymphoid tissue)
Metabolic and homeostatic	Participation in water-salt metabolism and acid–base balance

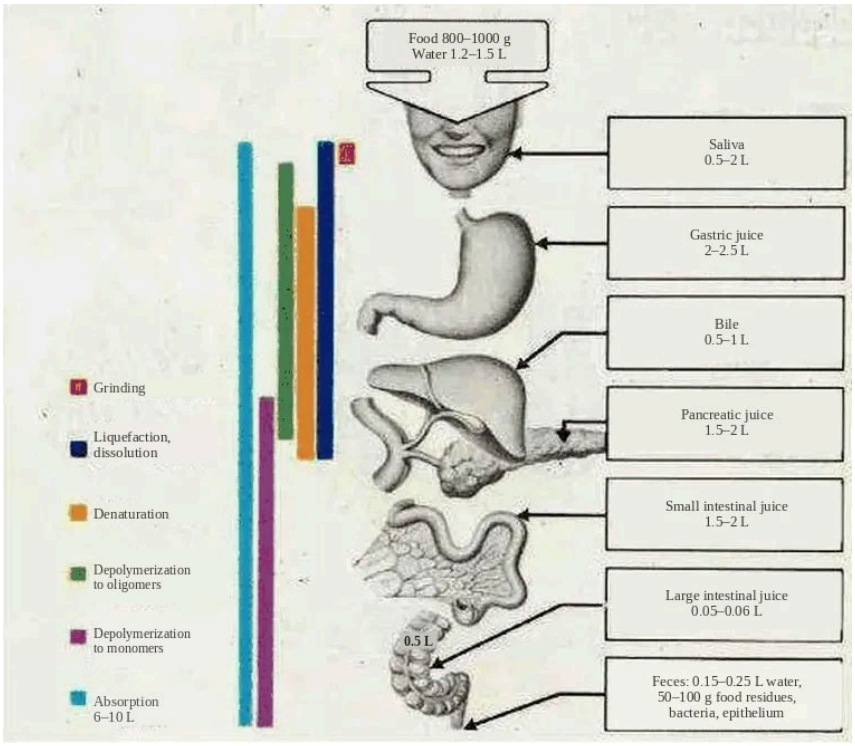


Figure 18.1. *Volume of digestive juices and the processes performed in different parts of the digestive system (Alavia O.T., 2018)*

Types of digestion (by the source of enzymes): **intrinsic** (the body's own enzymes), **symbiotic** (enzymes of the intestinal microflora — breakdown of fiber) and **autolytic** (the food's own enzymes, for example in breast milk). By location: **intracellular** (phagocytosis, pinocytosis) and **extracellular** — **cavity (distant)** digestion and **membrane (parietal)** digestion (A.M. Ugolev, topic 20).

The conveyor principle of digestion: each part of the digestive system processes the food in turn and prepares the next part; the activity of the preceding part influences that of the next (for example, the acid chyme formed in the stomach stimulates pancreatic secretion in the duodenum).

2. Methods of studying digestive function

I.P. Pavlov's chronic experimental methods

Before Pavlov, digestion was studied in acute experiments. I.P. Pavlov developed **chronic experiment** methods, in which a surgical operation is performed on the animal beforehand, allowing long-term observation under natural conditions (Nobel Prize, 1904).

Table 18.2. *Methods of studying the physiology of digestion*

Method	Essence	What it allows to be studied
Salivary duct fistula (D.D. Glinsky, Pavlov)	The orifice of the gland duct is brought out onto the skin of the cheek	Amount and composition of saliva, conditioned and unconditioned reflexes
Gastric fistula (V.A. Basov, 1842)	A tube is inserted into the stomach cavity	Gastric juice (mixed with food)
Sham feeding (Pavlov, 1889)	Gastric fistula + the esophagus is cut and its ends are brought out onto the neck (esophagotomy)	Pure gastric juice; the complex reflex phase — juice is secreted even though food does not enter the stomach
Small (isolated) stomach pouch — Heidenhain (1878)	Part of the stomach is isolated and its nerves are cut	Only humoral regulation

Method	Essence	What it allows to be studied
Small stomach pouch — Pavlov (1894)	Isolated with the vagal fibers preserved	Nervous and humoral regulation; reflects the processes in the main stomach
Pancreatic and bile duct fistulas	The ducts are brought out onto the skin	Pancreatic juice and bile secretion
Thiry–Vella method	An isolated segment of intestine is brought out onto the skin	Intestinal juice

Modern methods used in humans

- **Gastric juice sounding** (fractional examination with a thin tube) — basal and stimulated secretion, acidity.
- **Intragastric pH-metry** (24-hour) — evaluation of the acid-forming function.
- **Fibroesophagogastroduodenoscopy (FEGDS)**, colonoscopy, capsule endoscopy.
- **Contrast radiography** (barium), ultrasound, CT, MRI.
- **Electrogastrography**, manometry — motor activity.
- **Sialometry** — measurement of salivary flow rate; **masticatiography**, **electromyography** — activity of the masticatory muscles.
- **Radioisotope** and **breath tests** (for example, the urease breath test for Helicobacter pylori).

3. Digestion in the oral cavity

Food stays in the oral cavity for 15–20 s. Here food is processed **mechanically** (chewing), **moistened with saliva**, the **food bolus** is formed, partial chemical breakdown of carbohydrates begins and the taste of the food is evaluated.

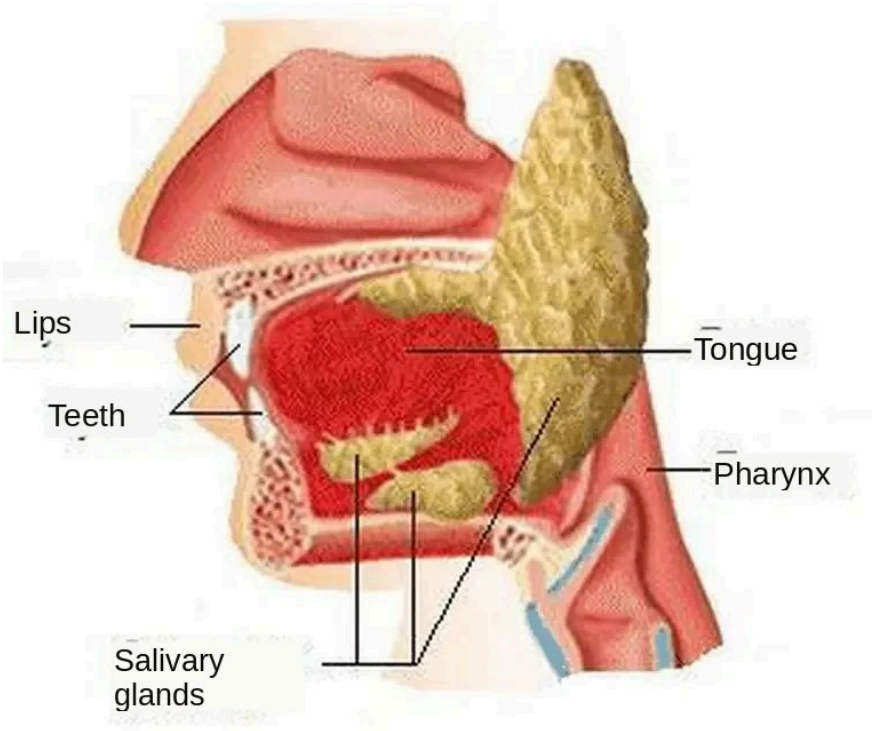


Figure 18.2. *The oral cavity and the salivary glands (Alavia O.T., 2018)*

Saliva

Saliva is produced by three pairs of major glands — the **parotid** (serous, watery, rich in enzymes), **submandibular** (mixed) and **sublingual** (mixed, rich in mucus) — and by many minor glands of the oral mucosa. **0.5–2 l** of saliva is secreted per day; 0.3–0.5 ml/min at rest and up to 3–4 ml/min when stimulated.

Table 18.3. *Composition and functions of saliva*

Component	Function
Water (99.4–99.5%)	Moistening and dissolving food (necessary for taste), forming the bolus
α-amylase (ptyalin)	Breaks down starch and glycogen to dextrins and maltose (optimal pH 6.8–7)
Maltase	Breaks down maltose to glucose
Mucin	Coats the bolus with mucus, facilitating swallowing; protects the mucosa
Lysozyme, lactoferrin, secretory IgA, thiocyanate	Antibacterial protection
Bicarbonates, phosphates	Buffer — protects tooth enamel from acids; pH 5.8–7.4
Calcium, phosphate	Remineralization of enamel
Active substances (kallikrein, growth factors, nerve growth factor)	Regeneration of the mucosa, wound healing

Regulation of salivation is mainly **reflex** (nervous): unconditioned reflexes (receptors of the oral cavity — taste, mechanical, temperature) and conditioned reflexes (the sight and smell of food, mealtime, kitchen sounds). The center is the **salivary nuclei of the medulla oblongata**.

- **Parasympathetic** influence (superior and inferior salivary nuclei; via cranial nerves VII and IX) — a large amount of watery, enzyme-rich saliva; vessels dilate.
- **Sympathetic** influence (Th_2–Th_4, superior cervical ganglion) — a small amount of thick, mucin-rich saliva (dry mouth during excitement).

REMEMBER

Both autonomic divisions **stimulate** salivation — unlike in other organs (synergism), but the composition of saliva differs: parasympathetic — “watery”, sympathetic — “thick” saliva.

CLINICAL SIGNIFICANCE

Xerostomia (dry mouth) — in Sjögren syndrome, radiotherapy of the head and neck, many drugs (antidepressants, antihistamines, M-cholinergic blockers), dehydration. Because the protective and remineralizing functions of saliva are lost, **caries increases sharply**, and candidiasis and disorders of taste and swallowing occur. **Sialorrhea** (excessive drooling) — in Parkinson's disease, cerebral palsy, and during teething.

Chewing

Chewing is the grinding of food, mixing it with saliva and forming the bolus. The chewing center is in the **medulla oblongata** and pons (the chewing rhythm generator); chewing can also be controlled voluntarily (cortex). The masticatory muscles are innervated by the motor fibers of the **trigeminal nerve** (cranial nerve V). Receptors — the oral mucosa, the tooth periodontium (pressure receptors —

regulate the biting force), and proprioceptors of the masticatory muscles and the temporomandibular joint.

Chewing pressure is 10–15 kg on the incisors and up to 50–80 kg on the molars (limited by the endurance of the periodontium).

Masticatiography is the graphic recording of the movements of the lower jaw. The masticatiogram shows the phases of one chewing period: I — rest; II — taking food into the mouth; III — the beginning of chewing (adaptation); IV — the main (rhythmic) chewing phase; V — formation of the bolus and swallowing.

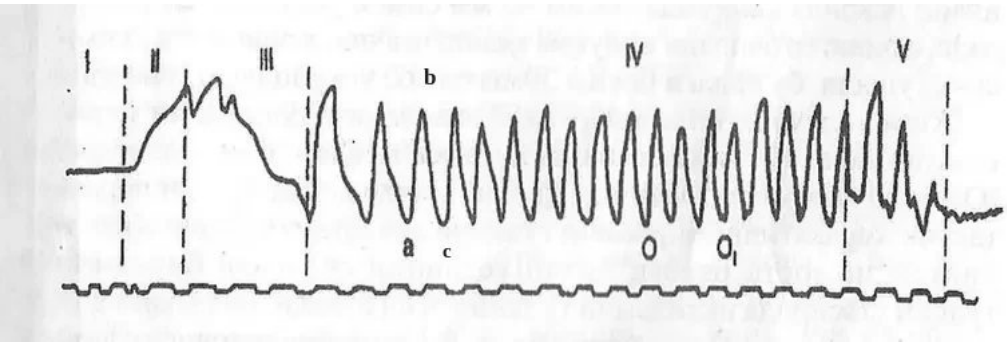


Figure 18.3. *Masticatiogram: phases of the chewing period (Alavia O.T., 2018)*

Swallowing

Swallowing is a complex reflex act that ensures the passage of the bolus from the oral cavity to the stomach (F. Magendie). It consists of three phases:

1. **Oral (voluntary) phase** — the tongue lifts the bolus toward the palate and pushes it into the pharynx.
2. **Pharyngeal (involuntary, rapid) phase** — the soft palate rises and closes the nasopharynx; the larynx rises and the **epiglottis** closes the laryngeal inlet; breathing stops reflexly (swallowing apnea); the upper esophageal sphincter relaxes. The center is in the medulla oblongata.
3. **Esophageal (involuntary, slow) phase** — a peristaltic wave carries the bolus to the stomach in 2–9 s (liquid 1–2 s); the lower esophageal sphincter relaxes reflexly.

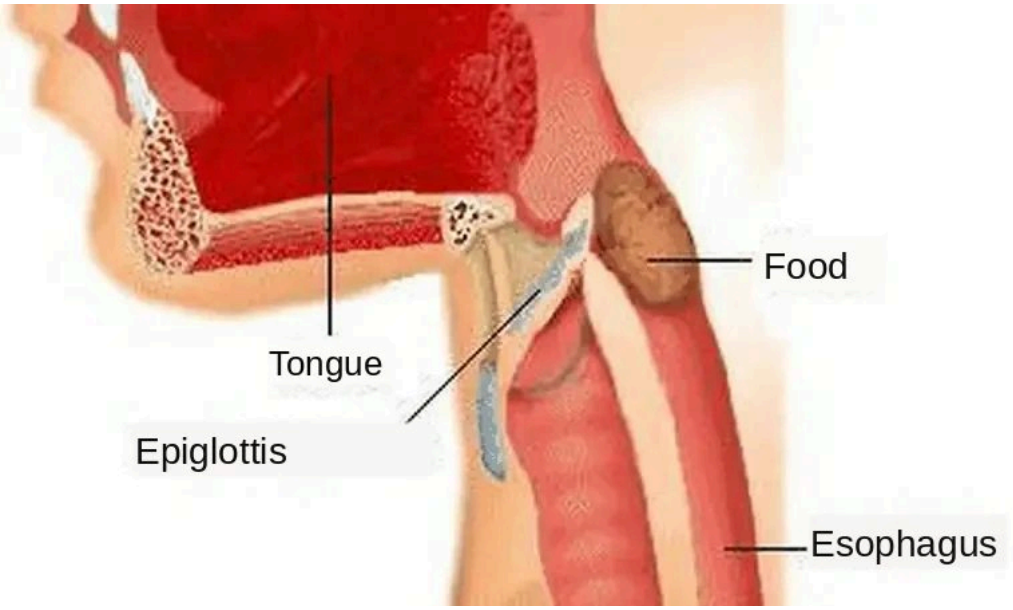


Figure 18.4. *The pharyngeal phase of swallowing: the epiglottis closes the airway (Alavia O.T., 2018)*

CLINICAL SIGNIFICANCE

When the swallowing center is damaged (stroke, bulbar palsy), **dysphagia** and entry of food into the airways (aspiration, aspiration pneumonia) are observed. In insufficiency of the lower esophageal sphincter the gastric contents return to the esophagus — **gastroesophageal reflux disease** (heartburn).

4. Interrelation of the maxillofacial organs with the body's systems

The maxillofacial region (oral cavity, teeth, jaws, temporomandibular joint, masticatory and facial muscles, salivary glands, tongue) is not only the initial part of digestion; it is closely linked with many systems of the body.

Table 18.4. *Relationship of the maxillofacial organs with other systems*

System	Nature of the relationship
Digestive system	Reflexes from the receptors of the oral cavity start gastric and pancreatic secretion and bile release (the complex reflex phase). Poor chewing (missing teeth, faulty dentures) burdens the function of the stomach and intestine
Respiratory system	The mouth and pharynx are a common pathway; the oral cavity takes part in breathing, and breathing stops reflexly during swallowing; when nasal breathing is impaired, mouth breathing affects the development of the teeth and occlusion
Speech	The tongue, lips, teeth and palate take part in the articulation of sounds; missing teeth and occlusal defects impair speech
Sensory systems	Taste, smell (retronasal olfaction), pain, temperature, tactile and proprioceptive sensation; evaluation of food quality
Cardiovascular system	Chronic foci of infection in the oral cavity (periodontitis, caries) contribute through bacteremia to infective endocarditis and atherosclerosis; reflex reactions during dental procedures (bradycardia, fainting)
Endocrine system	Periodontitis and xerostomia in diabetes mellitus, while periodontitis worsens glycemic control; gingivitis during hormonal changes (puberty, pregnancy)
Immune system	The tonsils (Waldeyer's ring), IgA and lysozyme in saliva — the first line of defense
Nervous system	Very extensive sensory innervation by the trigeminal nerve; toothache radiates to other areas; chewing has a toning effect on the central nervous system
Water-salt and acid–base balance	The sensation of thirst (dry mouth), loss of water and bicarbonates with saliva
Thermoregulation	In animals (dogs) heat loss through evaporation of saliva and panting

REMEMBER

The oral cavity is **the most important reflexogenic zone**: impulses from its receptors “start up” the whole digestive system. That is why chewing food well and perceiving its taste are also important for the subsequent stages of digestion.

5. Digestion in the stomach

Functions of the stomach: **storage** of food (up to 1.5–2.5 l), mechanical and chemical processing, formation of **chyme** and its passage in portions into the intestine; protective (bactericidal), endocrine and excretory (urea, drugs) functions, and production of **Castle's intrinsic factor**.

Gastric juice

2–2.5 l of gastric juice is secreted per day; it is a colorless acidic fluid (pH **0.8–1.5**; 1.5–3.5 when mixed with food in the stomach cavity).

Table 18.5. *Cells of the gastric glands and their products*

Cell	Location	Product
Chief (principal) cells	Glands of the fundus and body	Pepsinogens, gastric lipase
Parietal (oxyntic) cells	Glands of the fundus and body	Hydrochloric acid (HCl), Castle's intrinsic factor
Accessory (mucous) cells and surface epithelium	All parts	Mucin (mucus), bicarbonates
G cells	Antral (pyloric) part	Gastrin
ECL cells	Fundus and body	Histamine
D cells	All parts	Somatostatin

Functions of hydrochloric acid: converts pepsinogens into pepsin and creates the optimal pH for its action; denatures and swells proteins; bactericidal action; regulates the passage of food from the stomach to the intestine (pyloric reflex); stimulates the formation of secretin in the duodenum; aids iron absorption ($Fe^{3+} \rightarrow Fe^{2+}$).

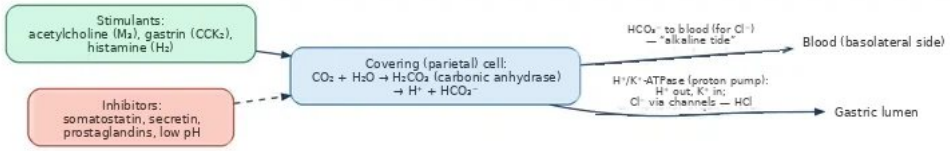


Figure 18.5. *Formation of hydrochloric acid in the parietal cell and its regulation*

Pepsins break down proteins into large polypeptides (optimal pH 1.5–2.0 and 3.2–3.5); they also break down collagen. **Gastric lipase** acts only on emulsified fats (milk fat); it is important in infants. In the stomach salivary amylase continues to work inside the bolus until the acid penetrates it.

The **gastric mucus-bicarbonate barrier** protects the stomach wall from self-digestion: the mucus gel layer, the bicarbonates beneath it (pH ≈ 7), the rapid renewal of epithelial cells, and prostaglandins (which enhance mucus and bicarbonate secretion and improve blood flow).

CLINICAL SIGNIFICANCE

Helicobacter pylori infection, **NSAIDs** (aspirin, ibuprofen — they reduce prostaglandin synthesis), alcohol, smoking and stress weaken the protective barrier — gastritis and **gastroduodenal ulcer** develop. Treatment uses **proton pump inhibitors** (omeprazole — blocks H^+/K^+-ATPase), H_2-histamine blockers (famotidine) and eradication of H. pylori.

Phases and regulation of gastric secretion

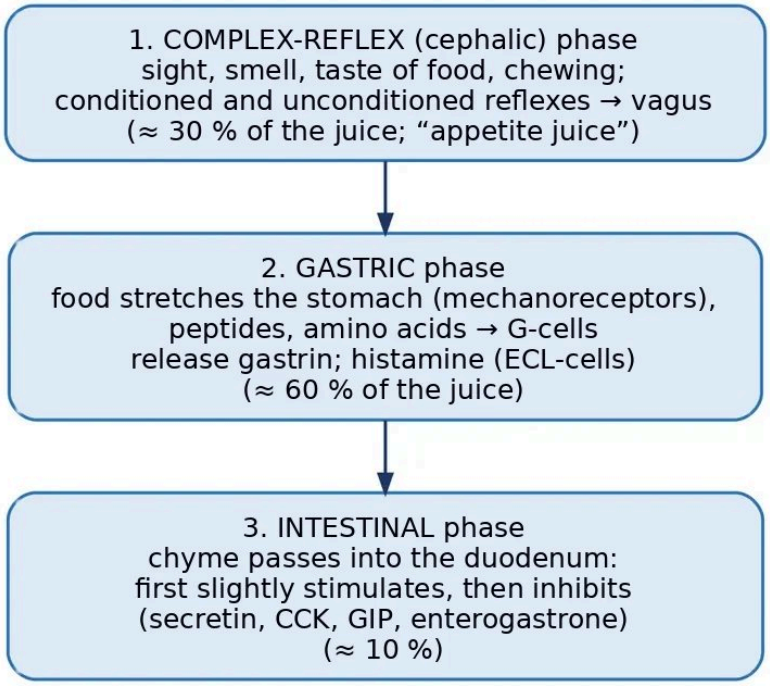


Figure 18.6. *Phases of gastric secretion*

Main stimulants: acetylcholine (vagus nerve; M_3-cholinergic receptors), **gastrin** (G cells; released under the influence of peptides, amino acids, gastric distension and the vagus) and **histamine** (ECL cells; H_2 receptors — the most powerful stimulant). **Inhibitors:** somatostatin, secretin, cholecystokinin, GIP, prostaglandins, a fall of pH in the stomach cavity below 1.5–2 (negative feedback), sympathetic influence.

Adaptation to the type of food (Pavlov): bread — the smallest amount of juice but the richest in enzymes; meat — the largest amount and the most acidic juice; milk — a small amount of juice poor in enzymes (fat has an inhibitory effect).

Motor activity of the stomach

When food enters, the proximal part of the stomach relaxes reflexly — **receptive relaxation** (via the vagus) — so food accumulates without a rise in pressure. In the distal part **peristaltic contractions** (≈ 3 /min; pacemaker — the Cajal cells on the greater curvature) mix the chyme and pass it in small portions through the pylorus into the duodenum. Food remains in the stomach for 3–10 hours: carbohydrate food is evacuated quickly, protein food more slowly, and fatty food most slowly. **Evacuation** is inhibited by acid, fats and hypertonic solutions in the duodenum (the **enterogastric reflex**, secretin, CCK).

Vomiting is a protective reflex; its center is in the medulla oblongata (the vomiting center and the trigger zone on the floor of the fourth ventricle, which is sensitive to toxins in the blood).

6. Laboratory work

Exercise 1: effect of salivary amylase on starch

1. Rinse the mouth with water, collect 2 ml of saliva and dilute it 5 times with water.
2. Pour 2 ml of 1% starch solution into each of three test tubes.

3. Add diluted saliva to tube 1, boiled saliva to tube 2 and water to tube 3 (control).
4. Keep the tubes at 37 °C for 10–15 minutes.
5. Take a sample from each tube and test it with iodine solution (for starch) and the Trommer (Fehling) test (for reducing sugars). **Result and analysis:** in tube 1 there is no blue color with iodine (the starch has been broken down) and the Trommer test is positive — maltose has formed. In tube 2 (boiled saliva — the enzyme is denatured) and tube 3 the starch is preserved (blue color). This proves that salivary amylase breaks down starch and is an enzyme of protein nature.

Exercise 2: effect of pepsin on protein and the role of HCl

1. Put identical pieces of fibrin (or egg white) into four test tubes.
2. Add to tube 1 — gastric juice; tube 2 — boiled gastric juice; tube 3 — neutralized gastric juice (with $NaHCO_3$); tube 4 — 0.5% HCl.
3. Keep at 37 °C for 30–40 minutes and compare the state of the protein. **Result and analysis:** the protein is digested (dissolves or becomes transparent) only in tube 1. Boiling inactivates pepsin; pepsin does not work in a neutral medium; HCl itself only swells the protein but does not break it down. Therefore protein digestion requires **pepsin and an acidic medium** together.

Exercise 3: evaluating the acidity of gastric juice (from results)

By the titration method, the acidity of gastric juice is expressed as the number of ml of 0.1 N NaOH needed to neutralize 100 ml of juice (titration units).

Table 18.6. *Normal values of gastric juice acidity (in titration units)*

Parameter	Basal secretion (fasting)	Stimulated secretion
Total acidity	40–60	80–100
Free HCl	20–40	65–85

Analyze the results: above normal — a hyperacid state (duodenal ulcer); low — a hypoacid state (atrophic gastritis); absence of free HCl — achlorhydria.

Exercise 4: evaluating chewing efficiency (Gelman test)

1. The subject chews 5 g of almonds (or standard nuts) for 50 s.
2. The chewed mass is passed through a sieve (2.4 mm mesh).
3. The percentage that passed through the sieve is calculated. **Interpretation:** with a healthy dentition chewing efficiency is close to 100%; when teeth are missing and the denture is unfit, it decreases sharply.

7. Practical skills

Skill: analyzing the factors regulating digestive secretion by phases

1. Indicate which receptors and which nerve take part in the complex reflex phase.
2. For the gastric phase, write down the mechanical (distension) and chemical (peptides) factors and the role of gastrin and histamine.
3. For the intestinal phase, indicate the stimulating and inhibiting hormones.
4. Explain the feedback (low pH → somatostatin → gastrin ↓).

Skill: measuring the salivary flow rate (sialometry)

1. In the morning, on an empty stomach, the subject spits saliva into a test tube for 5 minutes without swallowing (unstimulated saliva).
2. Measure the volume and calculate it in ml/min (normal 0.3–0.5 ml/min).
3. Stimulate with 2% citric acid solution and repeat the measurement (normal 1–3 ml/min).
4. Evaluate the result: unstimulated saliva < 0.1 ml/min — hyposalivation.

Review questions

1. List the functions of the digestive system.
2. What are intrinsic, symbiotic and autolytic digestion?
3. Explain I.P. Pavlov's chronic experimental methods.
4. What does the sham feeding experiment prove?
5. What is the difference between the Heidenhain and Pavlov small stomach pouches?
6. What are the composition and functions of saliva?
7. How is salivation regulated?
8. How is chewing carried out, and what phases does the masticatiogram show?
9. Explain the phases of swallowing.
10. Explain with examples the relationship of the maxillofacial organs with the other systems of the body.
11. List the cells of the gastric glands and their products.
12. What are the functions of hydrochloric acid?
13. How is the stomach wall protected from self-digestion?
14. Explain the phases of gastric secretion.
15. Which factors enhance and inhibit gastric secretion?

Test questions

1. Which enzyme in saliva breaks down starch?
   A) α-amylase (ptyalin)
   B) Pepsin
   C) Lipase
   D) Trypsin
   E) Lactase
2. What kind of saliva is secreted under the influence of the parasympathetic nerves?
   A) A small amount, thick, rich in mucin
   B) Salivation stops
   C) Only mucin
   D) Only lysozyme
   E) A large amount, watery, rich in enzymes

3. What did the sham feeding experiment prove?
   A) That gastric juice is evoked only by gastrin
   B) That gastric juice is secreted reflexly even when food does not enter the stomach
   C) The bactericidal action of saliva
   D) The composition of intestinal juice
   E) Bile secretion

4. In which type of small stomach pouch are the vagal fibers preserved?
   A) In the Heidenhain pouch
   B) In the Basov fistula
   C) In the Thiry–Vella method
   D) In the Pavlov pouch
   E) In none of them

5. Which cells of the stomach produce hydrochloric acid?
   A) Parietal (oxyntic) cells
   B) Chief cells
   C) Accessory cells
   D) G cells
   E) D cells

6. Which factor converts pepsinogen into pepsin?
   A) Enterokinase
   B) Trypsin
   C) Hydrochloric acid
   D) Bile acids
   E) Gastrin

7. Which is the most powerful stimulant of gastric secretion?
   A) Secretin
   B) Somatostatin
   C) Cholecystokinin
   D) Prostaglandins
   E) Histamine

8. In which cells is gastrin formed?
   A) In ECL cells
   B) In parietal cells
   C) In chief cells
   D) In the G cells of the antrum
   E) In D cells

9. Which structure do proton pump inhibitors (omeprazole) block?
   A) H_2-histamine receptors
   B) M_3-cholinergic receptors
   C) Gastrin receptors
   D) Carbonic anhydrase
   E) H^+/K^+-ATPase

10. What is the pH of gastric juice?
    A) 0,8–1,5
    B) 5,8–7,4
    C) 7,8–8,5
    D) 4,0–5,0
    E) 9,0–10,0

11. What closes the airways in the pharyngeal phase of swallowing?
    A) The tongue
    B) The soft palate and the epiglottis
    C) The esophageal sphincter
    D) Only the vocal cords
    E) The tonsils

12. Which nerve innervates the masticatory muscles?
    A) The facial nerve (VII)
    B) The hypoglossal nerve (XII)
    C) The vagus nerve (X)
    D) The trigeminal nerve (V)
    E) The glossopharyngeal nerve (IX)

13. In which phase of gastric secretion is the largest part of the juice secreted?
    A) Complex reflex
    B) Intestinal
    C) Gastric phase
    D) Equally in all phases
    E) Before a meal

14. What is the main danger of xerostomia for the teeth?
    A) Tooth growth accelerates
    B) Caries increases (the protective and remineralizing functions of saliva are lost)
    C) The enamel thickens
    D) Tartar decreases
    E) There is no danger

15. Which food is evacuated from the stomach most slowly?
    A) Carbohydrate food
    B) Liquid
    C) Fatty food
    D) Protein food
    E) Water

Clinical cases

Case 1

After radiotherapy for a head and neck tumor, a 55-year-old patient developed dry mouth, difficulty swallowing food and altered taste. Within six months caries developed in many teeth.

Questions: What are these changes associated with? Which functions of saliva are disturbed?

Answer: The radiotherapy damaged the salivary gland tissue — xerostomia (hyposalivation) developed. The water of saliva moistens food and dissolves taste substances — when it decreases,

taste is impaired; mucin lubricates the bolus — when it is lacking, swallowing becomes difficult. The salivary buffer systems (bicarbonate, phosphate) neutralize the acids formed by bacteria in dental plaque, calcium and phosphate remineralize the enamel, and lysozyme, lactoferrin and IgA provide antibacterial protection — because these functions were lost, caries developed rapidly. Help: saliva substitutes, fluoride preparations, oral hygiene.

Case 2

A 40-year-old patient has epigastric pain on an empty stomach and at night; the pain decreases after eating. FEGDS revealed a duodenal ulcer and H. pylori. Gastric juice acidity is increased.

Questions: Explain the physiological mechanism of ulcer development on the basis of the balance of protective and aggressive factors. What is the effect of a proton pump inhibitor?

Answer: An ulcer develops when the balance between aggressive factors (HCl, pepsin, H. pylori) and protective factors (the mucus-bicarbonate barrier, prostaglandins, blood flow, epithelial regeneration) is disturbed. H. pylori reduces the somatostatin-producing D cells in the antrum — gastrin release and HCl secretion increase; the excess acid enters the duodenum and damages the mucosa. Food buffers the acid — the pain decreases. Proton pump inhibitors irreversibly block the H^+/K^+-ATPase in the parietal cell — the final stage of HCl secretion stops (the response to any stimulant decreases) and the ulcer heals; H. pylori is eliminated with antibiotics.

Case 3

An elderly patient who has lost many teeth and does not wear a denture swallows food without chewing it well. He complains of heaviness in the stomach, bloating and constipation.

Questions: How does impaired chewing affect the subsequent stages of digestion?

Answer: Chewing grinds food, increases the surface on which digestive enzymes act and mixes it with saliva. Poorly chewed large pieces are digested slowly in the stomach and stay there long (a feeling of heaviness). The reflex influence from the receptors of the oral cavity (the complex reflex phase) weakens — the “start-up” of gastric and pancreatic secretion and bile release is reduced. Undigested residues enhance fermentation and putrefaction in the large intestine (bloating). That is why dental treatment and prosthetics matter for the whole digestive system.

Quick review

Table 18.7. *Key concepts of topic 18*

Concept	Brief definition
Types of digestion	Intrinsic, symbiotic, autolytic; cavity and membrane digestion
Pavlov's methods	Fistulas, sham feeding, small stomach pouch (chronic experiment)
Saliva	0.5–2 l/day; amylase, mucin, lysozyme, IgA, bicarbonate
Salivary center	Medulla oblongata; parasympathetic — watery, sympathetic — thick saliva
Chewing	Cranial nerve V; phases of the masticatiogram
Swallowing	Oral (voluntary), pharyngeal and esophageal (involuntary) phases
Oral cavity	Main reflexogenic zone; linked with digestion, respiration, speech and the sensory systems
Gastric juice	2–2.5 l/day; pH 0.8–1.5
Parietal cells	HCl and Castle's intrinsic factor; H^+/K^+-ATPase

Concept	Brief definition
Chief cells	Pepsinogens
Stimulants	Acetylcholine, gastrin, histamine (the strongest)
Inhibitors	Somatostatin, secretin, CCK, low pH
Phases of secretion	Complex reflex (30%), gastric (60%), intestinal (10%)
Protective barrier	Mucus + bicarbonate + prostaglandins
Evacuation	Fatty food is slowest; the enterogastric reflex inhibits it

TOPIC 19

Motor activity of the gastrointestinal tract. The role of the liver in digestion. Bile secretion and its properties. Secretory activity of the pancreas

Learning objectives

To teach the student the properties of the smooth muscles of the gastrointestinal tract and the types of movement, their nervous (enteric nervous system and autonomic nerves) and humoral regulation, the digestive and non-digestive functions of the liver, the formation, composition, properties and secretion of bile, and the composition of pancreatic juice and the regulation of its secretion.

By the end of the class the student should know:

- the electrical activity of the gastrointestinal smooth muscles (slow waves, Cajal cells);
- the structure and function of the enteric (metasympathetic) nervous system;
- the types of movement of the small and large intestine, the migrating motor complex;
- the nervous and humoral factors regulating motility;
- the functions of the liver;
- the composition of bile, the function of bile acids and their enterohepatic circulation;
- the regulation of bile formation and secretion;
- the composition of pancreatic juice, its enzymes and their activation, and the phases and regulation of secretion.

The student should be able to:

- demonstrate experimentally the role of bile in emulsifying fats;
- prove the importance of bile for the action of pancreatic lipase;
- explain the effect of adrenaline and acetylcholine on intestinal motility.

1. Motor activity of the gastrointestinal tract

Properties of smooth muscles

The wall of the gastrointestinal tract (except the upper esophagus and the external anal sphincter) contains smooth muscles: an inner **circular** and an outer **longitudinal** layer. They work as a **functional syncytium** (gap junctions).

Smooth muscles show rhythmic oscillations of the membrane potential — **slow waves** (the basic electrical rhythm); they are generated by the **interstitial cells of Cajal** (pacemakers). Slow waves do not cause contraction by themselves; if the membrane reaches threshold at their peak, **action potentials (spikes)** appear and the muscle contracts. Frequency: ≈ 3 /min in the stomach, 11–12 /min in the duodenum, 8–9 /min in the ileum.

The enteric nervous system

The **enteric (metasympathetic) nervous system** is an independent network of ≈ 100 million neurons in the gastrointestinal wall (the “second brain”):

- **The myenteric (Auerbach) plexus** — between the circular and longitudinal layers; mainly regulates **motor** activity.

- **The submucosal (Meissner) plexus** — regulates **secretion**, absorption and local blood flow.

The enteric nervous system carries out peristaltic and secretory reflexes even without the central nervous system; the autonomic nerves only modify its activity.

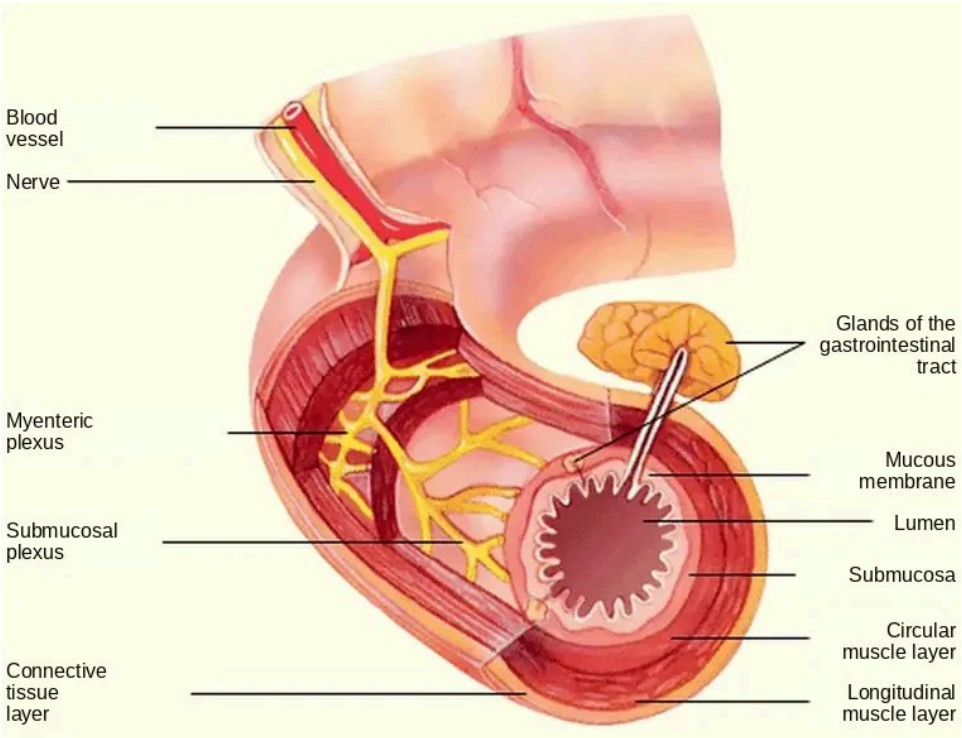


Figure 19.1. *Layers of the intestinal wall and nerve plexuses (Alavia O.T., 2018)*

Movements of the small intestine

Table 19.1. *Types of movement of the small intestine*

Type	Essence	Significance
Rhythmic segmentation	Simultaneous contraction of the circular muscles at many points; the intestine is divided into segments	Mixing the chyme, increasing its contact with the mucosa
Pendular	The longitudinal muscles contract and a part of the intestine moves back and forth	Mixing
Peristaltic	The circular muscles contract above and relax below — the wave spreads in the distal direction (the “law of the intestine”, Bayliss–Starling)	Propelling the chyme
Tonic contractions	Long-lasting contraction	Sphincters, tone
Migrating motor complex (MMC)	A wave of strong contractions spreading from the stomach to the ileum every 90–120 minutes on an empty stomach (motilin)	The “intestinal housekeeper” — sweeps out residues and bacteria

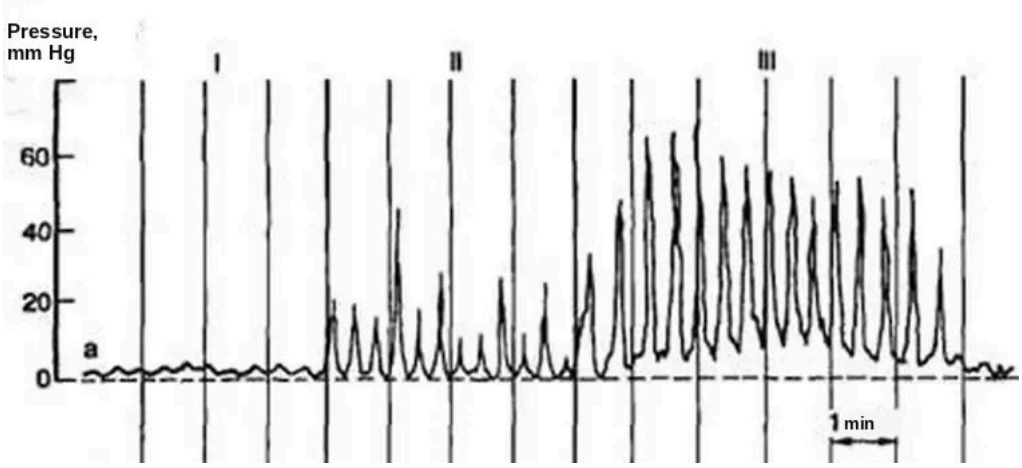


Figure 19.2. *Phases of the migrating motor complex in the small intestine on an empty stomach (I — quiescence, II — irregular contractions, III — strong rhythmic contractions) (Alavia O.T., 2018)*

The **ileocecal sphincter (Bauhin's valve)** regulates the passage of chyme from the ileum into the cecum in portions and prevents the reflux of colonic contents; it opens reflexly when food enters the stomach (the **gastroileal reflex**).

Regulation of motility

Table 19.2. *Factors affecting gastrointestinal motility*

Stimulating	Inhibiting
Parasympathetic (vagus and pelvic nerves, acetylcholine)	Sympathetic nerves (noradrenaline)
Distension of the intestinal wall (local reflex)	Adrenaline
Serotonin, gastrin, motilin, cholecystokinin, insulin	Secretin, GIP, glucagon, VIP
Coarse food (fiber), bile acids	Sharp distension of the intestine (irritation of the peritoneum)

Gastrointestinal reflexes: gastroileal, **gastrocolic** (when food enters the stomach, colonic motility and the urge to defecate increase), **enterogastric** (when the duodenum is distended, gastric motility and secretion are inhibited), **intestino-intestinal** (when one part of the intestine is strongly distended, the other parts are inhibited).

> **CLINICAL SIGNIFICANCE**
>
> **Paralytic ileus** is reflex inhibition of intestinal peristalsis (predominance of sympathetic influence) after abdominal surgery, in peritonitis and hypokalemia. The abdomen is distended and no peristalsis is heard. Treatment includes restoring electrolytes and AChE inhibitors (proserin/neostigmine).

2. The role of the liver in digestion

The liver is the largest gland in the body (1.5 kg), the “central biochemical laboratory”. It is the first to receive all the blood flowing from the intestine (through the portal vein).

Table 19.3. *Functions of the liver*

Function	Content
Digestive (secretory)	Bile formation — for the digestion and absorption of fats

Function	Content
Carbohydrate metabolism	Synthesis and breakdown of glycogen, gluconeogenesis — maintaining blood glucose
Protein metabolism	Synthesis of plasma proteins (albumin, fibrinogen, prothrombin and other clotting factors); deamination of amino acids; synthesis of urea (detoxification of ammonia)
Lipid metabolism	Synthesis of lipoproteins, cholesterol and phospholipids; β-oxidation of fatty acids; formation of ketone bodies
Detoxification (barrier)	Detoxification of toxins from the intestine (indole, skatole, phenol), drugs and alcohol by the cytochrome P450 system and conjugation
Excretory	Excretion of bilirubin, cholesterol, drugs and heavy metals with bile
Inactivation of hormones	Insulin, steroid hormones, thyroid hormones
Depot	Glycogen, iron (ferritin), vitamins (A, D, B_{12}), blood depot
Hematopoiesis	In the fetal period
Protective	Kupffer cells — phagocytosis
Thermoregulation	Produces a lot of heat

CLINICAL SIGNIFICANCE

In liver failure ammonia is not detoxified, accumulates in the blood and affects the brain (**hepatic encephalopathy**); synthesis of albumin and clotting factors decreases — edema, ascites and a tendency to bleed; bilirubin metabolism is disturbed and jaundice develops; the metabolism of drugs and hormones slows down (doses must be reduced).

3. Bile formation and secretion

Bile is formed **continuously** in liver cells (hepatocytes) (**0.5–1.2 l** per day), but it is released into the duodenum only during **digestion**. Between meals bile accumulates **in the gallbladder** and is concentrated 5–10 times (water and electrolytes are reabsorbed).

Table 19.4. *Composition of hepatic and gallbladder bile*

Parameter	Hepatic bile	Gallbladder bile
Color	Pale yellow-brown	Dark brown-green
pH	7,3–8,0	6,0–7,0
Dry matter	2,5 %	10–20 %
Bile acids	Low	High (5–10 times)

Composition of bile: water, **bile acids and their salts** (the main part), **bile pigments** (conjugated bilirubin — gives the color), cholesterol, phospholipids (lecithin), electrolytes (much bicarbonate), mucus.

Functions of bile:

1. **Emulsifies fats** — by reducing surface tension it breaks large fat droplets into small ones; the surface on which lipase acts increases thousands of times.
2. **Activates pancreatic and intestinal lipase**.
3. Forms **micelles** with fatty acids, monoglycerides, cholesterol and **fat-soluble vitamins** (A, D, E, K), ensuring their absorption.

4. **Neutralizes** the acid chyme coming from the stomach (bicarbonate) and stops the action of pepsin.
5. **Enhances intestinal peristalsis**.
6. **Bacteriostatic** action — reduces putrefactive processes.
7. **Excretory** — excretes bilirubin, cholesterol and drugs.
8. Stimulates bile formation and pancreatic secretion.

Enterohepatic circulation of bile acids

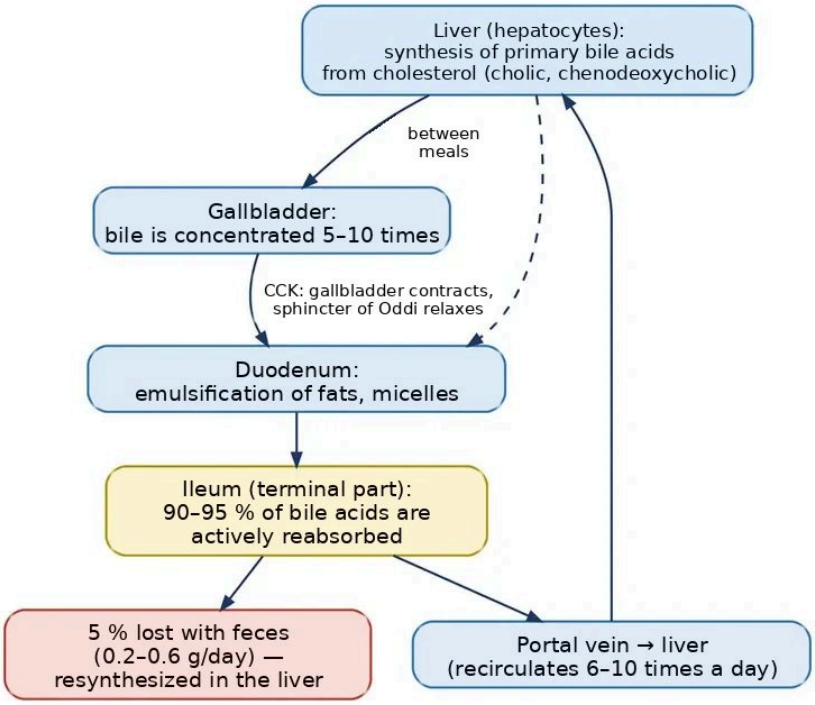


Figure 19.3. *Enterohepatic circulation of bile acids*

Regulation of bile formation and secretion

- **Bile formation (choleresis)** is enhanced by: bile acids returning to the blood (the strongest stimulus), **secretin** (water and bicarbonate), the vagus nerve, gastrin, and protein and fatty food.
- **Bile release (cholekinesis)** is enhanced by: **cholecystokinin (CCK)** — contracts the gallbladder and relaxes the **sphincter of Oddi** (the main factor); the vagus nerve. The strongest stimulants are egg yolk, milk, meat and fats. The sympathetic nerves relax the gallbladder and contract the sphincter.

CLINICAL SIGNIFICANCE

Gallstone disease — if the amount of cholesterol in bile increases relative to bile acids and phospholipids (lithogenic bile), cholesterol crystallizes and forms stones. After fatty food CCK contracts the gallbladder, and when a stone blocks the neck of the gallbladder, pain appears under the right costal margin (biliary colic). When the duct is completely blocked, bile does not enter the intestine: fats are not digested (**steatorrhea**), the stool is discolored (acholic stool), vitamin K is not absorbed (tendency to bleed), and **obstructive jaundice** develops.

4. Secretory activity of the pancreas

The exocrine part of the pancreas (acini and ducts) produces **1.5–2.0 l** of alkaline (pH **7.8–8.4**) juice per day. It is the most important digestive juice — it contains enzymes that break down all nutrients.

Table 19.5. *Enzymes of pancreatic juice*

Group	Enzyme (inactive form)	Activator	Effect
Proteolytic	Trypsinogen → trypsin	Enterokinase (enteropeptidase; duodenal mucosa), then trypsin itself (autocatalysis)	Breaks down proteins and peptides at internal bonds; activates the other proenzymes
Proteolytic	Chymotrypsinogen → chymotrypsin	Trypsin	Breaks down proteins
Proteolytic	Procarboxypeptidases → carboxypeptidase A, B	Trypsin	Breaks down peptides from the C-terminus
Proteolytic	Proelastase → elastase	Trypsin	Breaks down elastin and collagen
Amylolytic	α-amylase (secreted in active form)	—	Starch and glycogen to maltose and oligosaccharides
Lipolytic	Lipase (active; requires colipase and bile)	Bile acids, colipase (activated by trypsin)	Triglycerides to monoglycerides and fatty acids
Lipolytic	Phospholipase A_2 (inactive)	Trypsin	Breaks down phospholipids
Nucleolytic	Ribonuclease, deoxyribonuclease	—	Breaks down nucleic acids

The gland is protected from self-digestion: the proteolytic enzymes are secreted as **inactive proenzymes**; the acinar cells contain a **trypsin inhibitor** (SPINK1); the enzymes are activated only in the intestine.

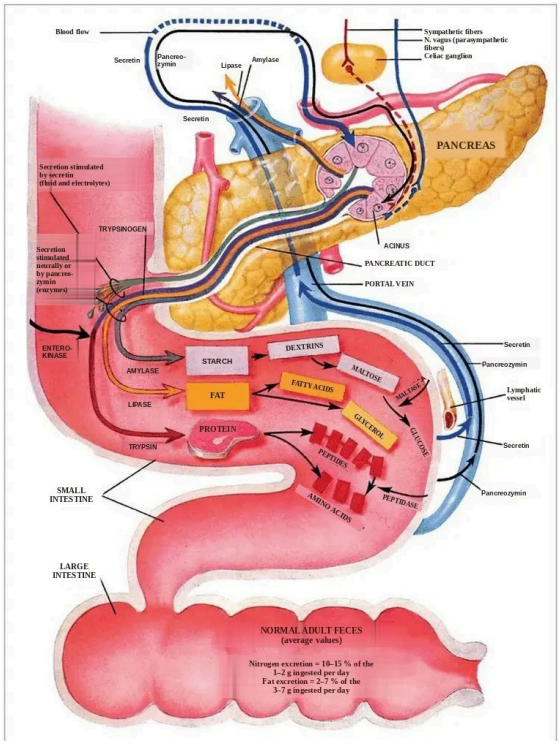


Figure 19.4. *Pancreatic juice, the regulation of its secretion and the breakdown of nutrients in the small intestine (Alavia O.T., 2018)*

Regulation of secretion

Pancreatic secretion also proceeds in three phases: **complex reflex** (vagus nerve, ≈ 20%), **gastric** (gastric distension — vagovagal reflex, gastrin; ≈ 10%) and **intestinal** (the main one — ≈ 70%).

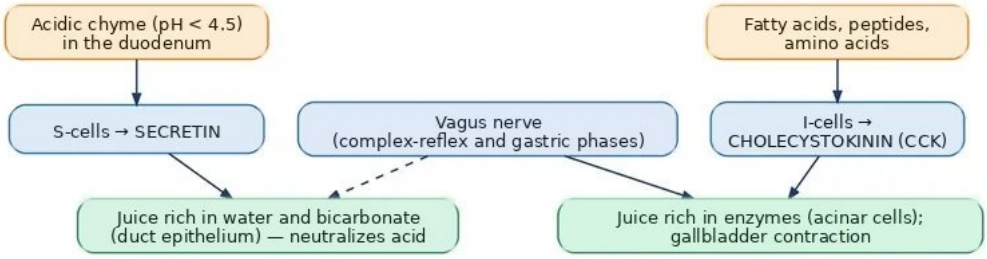


Figure 19.5. *Humoral regulation of pancreatic secretion*

REMEMBER

Secretin (the first hormone discovered — W. Bayliss and E. Starling, 1902) mainly evokes juice rich in **water and bicarbonates** — it neutralizes the acid chyme. **Cholecystokinin** mainly evokes juice rich in **enzymes** and contracts the gallbladder. Together they enhance each other's effect (potentiation).

Adaptation to the type of food: bread — much amylase; meat — much trypsin; milk and fatty food — much lipase.

CLINICAL SIGNIFICANCE

Acute pancreatitis — the enzymes are activated inside the gland and "digest" it (alcohol, blockage of the duct by a gallstone). **Amylase and lipase** rise sharply in the blood and urine (a diagnostic criterion). In **chronic pancreatitis** and **cystic fibrosis** exocrine insufficiency develops: fats are not digested (steatorrhea), protein digestion is impaired (creatorrhea), weight loss; treatment — enzyme preparations (pancreatin).

5. Laboratory work

Exercise 1: emulsification of fats by bile

1. Pour 2 ml of water into each of two test tubes.
2. Add 1 ml of bile (or a solution of bile salts) to tube 1 and 1 ml of water to tube 2.
3. Add 3–5 drops of vegetable oil to both, shake vigorously and let stand for 5 minutes.
4. Put a drop from each on a slide and examine under the microscope. **Result and analysis:** in the tube with bile the oil remains broken into small droplets for a long time (a stable emulsion), and small uniform droplets are seen under the microscope. In the tube with water the oil quickly separates to the surface. Bile acids reduce surface tension and emulsify fats.

Exercise 2: the importance of bile for the action of pancreatic lipase

1. Put 5 ml of milk (or fat emulsion) and phenolphthalein (pink in an alkaline medium) into each of three test tubes; bring them to a pale pink color with a weak alkali.
2. Add to tube 1 — pancreatin (lipase) solution + bile; tube 2 — pancreatin + water; tube 3 — water only.
3. Keep at 37 °C and observe the time of color change. **Result and analysis:** the pink color disappears fastest (fatty acids are formed and the medium becomes acidic) in tube 1, more

slowly in tube 2; in tube 3 the color does not change. Bile activates lipase and, by emulsifying fats, increases the surface on which the enzyme acts.

Exercise 3: effect of mediators on the motility of an isolated intestinal segment (demonstration or video)

1. An isolated segment of rabbit small intestine is suspended in a bath filled with Ringer–Tyrode solution at 37 °C and aerated with air, and its contractions are recorded.
2. Acetylcholine solution is added to the bath; recording continues.
3. After washing, adrenaline is added. **Result and analysis:** the isolated intestine contracts rhythmically on its own — automaticity (Cajal cells and the enteric nervous system). Acetylcholine increases the amplitude of contractions and the tone (parasympathetic effect), while adrenaline reduces them (sympathetic effect).

6. Practical skills

Skill: distinguishing types of jaundice by bile pigment metabolism (schematic)

1. Determine which bilirubin fraction is increased in the blood: unconjugated (indirect) or conjugated (direct).
2. Evaluate the presence of bilirubin (only conjugated bilirubin passes) and urobilin in the urine.
3. Evaluate the color of the stool (stercobilin).
4. Unconjugated bilirubin increased, stool dark, much urobilin in the urine — hemolytic (prehepatic) jaundice.
5. Conjugated bilirubin increased, stool discolored, bilirubin in the urine, no urobilin — obstructive (posthepatic) jaundice.
6. Both fractions increased — parenchymal (hepatic) jaundice.

Skill: recognizing exocrine pancreatic insufficiency from the signs

1. Evaluate whether the stool is abundant, fatty, shiny and foul-smelling (steatorrhea).
2. Identify weight loss and signs of fat-soluble vitamin deficiency.
3. Evaluate fecal elastase-1 (normal > 200 µg/g).
4. Explain the result by linking it to the function of the pancreatic enzymes.

Review questions

1. What is the electrical activity of the gastrointestinal smooth muscles?
2. What is the role of the interstitial cells of Cajal?
3. How is the enteric nervous system structured, and what functions does it perform?
4. List the types of movement of the small intestine.
5. What is the migrating motor complex, and what is its significance?
6. Which nervous and humoral factors regulate intestinal motility?
7. List the gastrointestinal reflexes.
8. List the main functions of the liver.
9. What are the composition and functions of bile?
10. What is the difference between hepatic and gallbladder bile?

11. Explain the enterohepatic circulation of bile acids.
12. How are bile formation and secretion regulated?
13. List the composition and enzymes of pancreatic juice.
14. How are the proteolytic enzymes activated, and how is the gland protected from self-digestion?
15. What is the effect of secretin and cholecystokinin on pancreatic secretion?

Test questions

1. Which cells generate slow waves in the gastrointestinal smooth muscles?
   A) Chief cells
   B) G cells
   C) Goblet cells
   D) Paneth cells
   E) Interstitial cells of Cajal

2. Which nerve plexus mainly regulates intestinal motility?
   A) The myenteric (Auerbach) plexus
   B) The submucosal (Meissner) plexus
   C) The solar plexus
   D) The brachial plexus
   E) The lumbar plexus

3. What is the wave of contractions repeated every 90–120 minutes on an empty stomach called?
   A) Segmentation
   B) Pendular movement
   C) Antiperistalsis
   D) Tonic contraction
   E) The migrating motor complex

4. Which factor inhibits intestinal motility?
   A) Acetylcholine
   B) Serotonin
   C) Adrenaline (sympathetic influence)
   D) Motilin
   E) Distension of the intestinal wall

5. Where does urea synthesis take place?
   A) In the kidney
   B) In the pancreas
   C) In the intestine
   D) In the liver
   E) In the spleen

6. What is the main digestive function of bile?
   A) Breaking down proteins
   B) Breaking down starch
   C) Emulsifying fats and activating lipase
   D) Activating pepsin
   E) Breaking down lactose

7. Which hormone contracts the gallbladder and relaxes the sphincter of Oddi?
   A) Secretin
   B) Gastrin
   C) Somatostatin
   D) Insulin
   E) Cholecystokinin

8. Where is the main part of the bile acids reabsorbed?
   A) In the duodenum
   B) In the terminal ileum
   C) In the large intestine
   D) In the stomach
   E) In the jejunum

9. Which enzyme converts trypsinogen into trypsin?
   A) Enterokinase (enteropeptidase)
   B) Pepsin
   C) Amylase
   D) Lipase
   E) Hydrochloric acid

10. What is the pH of pancreatic juice?
   A) 1,0–2,0
   B) 5,0–6,0
   C) 6,5–7,0
   D) 7,8–8,4
   E) 9,5–10,5

11. What kind of pancreatic juice does secretin stimulate?
   A) Rich in enzymes, small in volume
   B) Rich in water and bicarbonates
   C) Rich only in trypsin
   D) Rich in mucus
   E) Acidic

12. Which pancreatic enzyme is secreted in active form?
   A) Trypsinogen
   B) Chymotrypsinogen
   C) Proelastase
   D) α-amylase
   E) Procarboxypeptidase

13. What is observed when a duct is blocked by a gallstone?
   A) Steatorrhea and discoloration of the stool
   B) Darkening of the stool
   C) Only an increase in unconjugated bilirubin in the blood
   D) Protein digestion is enhanced
   E) Urobilin increases in the urine

14. Which enzymes are measured in the blood to diagnose acute pancreatitis?
    A) ALT and AST
    B) Creatine kinase
    C) Amylase and lipase
    D) Alkaline phosphatase
    E) Troponin

15. How much bile is formed per day?
    A) 50–100 ml
    B) 0.5–1.2 l
    C) 3–4 l
    D) 8–10 l
    E) 10 ml

Clinical cases

Case 1

After fatty food a 45-year-old obese woman developed severe pain under the right costal margin, followed by yellowing of the skin and sclerae, dark urine and pale stools. Blood conjugated bilirubin is increased.

Questions: What is the condition? Explain the signs on the basis of the physiology of bile.

Answer: Gallstone disease with blockage of the common bile duct by a stone — obstructive jaundice. Fatty food enhances the release of CCK in the duodenum; the gallbladder contracts, the stone moves and blocks the duct — biliary colic. Bile does not enter the intestine: the conjugated bilirubin in the liver returns to the blood (jaundice), and because it is water-soluble it passes into the urine (dark urine); no stercobilin is formed in the intestine (pale stools). Because bile acids are lacking, fats are not emulsified or digested (steatorrhea), and fat-soluble vitamins (vitamin K — blood coagulation) are not absorbed.

Case 2

A patient who has consumed alcohol for many years has weight loss, abundant shiny, fatty, foul-smelling stools and bloating. Fecal elastase-1 is reduced.

Questions: What is the condition? Explain the mechanism of the signs.

Answer: Chronic pancreatitis and exocrine pancreatic insufficiency. The acinar tissue has been replaced by fibrosis, and the production of lipase, proteases and amylase has decreased. Because of lipase deficiency triglycerides are not broken down or absorbed — steatorrhea (fatty, shiny stools); protein digestion is impaired — creatorrhea; undigested substances ferment and putrefy in the large intestine under the action of bacteria — bloating, foul smell. Because nutrients are not absorbed, weight loss develops. Treatment: stopping alcohol and enzyme preparations (pancreatin) with meals.

Case 3

On the second day after abdominal surgery the patient's abdomen is distended, no intestinal peristalsis is heard and no gas is passed. Blood K^+ 3.1 mmol/l.

Questions: Explain the mechanisms of the arrest of intestinal motility.

Answer: Postoperative paralytic ileus. When the peritoneum and intestine are irritated, the sympathetic nervous system is reflexly activated — noradrenaline inhibits the intestinal smooth

muscle and enteric neurons; pain and stress (catecholamines), anesthesia and opioid analgesics also weaken peristalsis. Hypokalemia hyperpolarizes the smooth muscle membrane and reduces excitability — motility weakens further. Treatment: restoration of potassium, early mobilization and, if needed, AChE inhibitors (proserin/neostigmine), which enhance the parasympathetic influence.

Quick review

Table 19.6. *Key concepts of topic 19*

Concept	Brief definition
Slow waves	Cajal cells; 3 /min in the stomach, 12 /min in the duodenum
Auerbach plexus	Motility
Meissner plexus	Secretion, absorption, blood flow
Segmentation	Mixing
Peristalsis	Propulsion (“law of the intestine”)
MMC	Fasting “intestinal housekeeper”; motilin
Gastrocolic reflex	The colon is activated when food enters the stomach
Liver functions	Bile, metabolism, urea, detoxification, protein synthesis
Bile	0.5–1.2 l/day; emulsifies, activates lipase, micelles
Enterohepatic circulation	90–95% of bile acids are reabsorbed in the ileum
Cholecystokinin	Gallbladder contraction, enzyme-rich juice
Secretin	Bicarbonate-rich juice (Bayliss–Starling, 1902)
Pancreatic juice	1.5–2 l/day; pH 7.8–8.4
Enterokinase	Activates trypsinogen
Proenzymes	Protect the gland from self-digestion

TOPIC 20

Digestion in the small and large intestine. Ugolev's experiment

Learning objectives

To teach the student digestion in the small intestine — the composition and secretion of intestinal juice, A.M. Ugolev's doctrine of cavity and membrane (parietal) digestion, the mechanisms of absorption of nutrients, water and electrolytes, the processes in the large intestine, the role of the microflora and the mechanism of defecation; in laboratory work, to master demonstrating the efficiency of membrane digestion compared with cavity digestion (Ugolev's experiment).

By the end of the class the student should know:

- the structural features of the small intestinal mucosa and the absorptive surface;
- the composition of intestinal juice and the regulation of its secretion;
- cavity, membrane and intracellular digestion;
- the final digestion and absorption of carbohydrates, proteins and fats;
- the absorption of water, electrolytes, vitamins and iron;
- water absorption in the large intestine, the microflora and its functions;
- the defecation reflex and its regulation.

The student should be able to:

- perform Ugolev's experiment and analyze the result;
- demonstrate the action of intestinal juice (disaccharidases);
- explain the signs of malabsorption from a physiological point of view.

1. Structural features of the small intestine

The small intestine (duodenum, jejunum and ileum) is 3–5 m long in a living person. Here digestion is completed and almost all nutrients are absorbed. Structures that increase the absorptive surface:

Table 20.1. *Structures that increase the absorptive surface of the small intestine*

Structure	Increase in surface
Circular folds (Kerckring's folds)	3 times
Villi (20–40 per 1 mm^2)	10 times
Microvilli (1500–3000 on each enterocyte — the “brush border”)	20 times
Total	≈ 600 times; total surface ≈ 200 m^2

In the center of each villus there is an arteriole, a capillary network, a venule and a lymphatic capillary (the **lacteal**); smooth muscle fibers contract the villus rhythmically, enhancing lymph and blood flow (the “villus pump”; its hormone is villikinin).

On the surface of the microvilli there is the **glycocalyx** — a mesh layer of glycoproteins and mucopolysaccharides; enzymes are adsorbed in it, and it acts as a barrier to the intestinal microflora.

2. Intestinal juice

Intestinal juice is produced by the **Brunner glands** (in the duodenum — rich in mucus and bicarbonate) and the **crypts of Lieberkühn** (throughout the intestine). **1.5–2.0 l** is secreted per day;

pH **7.2–8.6**.

The juice consists of two parts: a **liquid part** (water, electrolytes, bicarbonates, mucus) and a **dense part** — shed enterocytes (they disintegrate and release their enzymes into the intestinal lumen). The intestinal epithelium is completely renewed in 3–5 days.

Enzymes of intestinal juice (more than 20): **enterokinase** (enteropeptidase — activates trypsinogen), peptidases (aminopeptidases, dipeptidases), disaccharidases (**maltase, saccharase (sucrase), lactase**, isomaltase), intestinal lipase, phospholipases, alkaline phosphatase, nucleases.

Regulation of secretion is mainly **local**: mechanical and chemical stimulation of the mucosa by chyme (enteric nervous system reflexes); humoral factors — VIP, gastrin, secretin, CCK and motilin enhance it; somatostatin inhibits it.

3. Cavity and membrane digestion (A.M. Ugolev's doctrine)

A.M. Ugolev (1958) identified two main types of digestion in the small intestine:

- **Cavity (distant) digestion** — in the intestinal lumen, mainly under the action of pancreatic enzymes; large molecules (polymers) undergo initial breakdown (oligomers are formed).
- **Membrane (parietal) digestion** — on the surface of the microvilli, under the action of enzymes located in the glycocalyx and membrane; the intermediate products are broken down to the final stage (monomers). Membrane digestion is **the final stage of digestion** and is directly coupled with **absorption**.

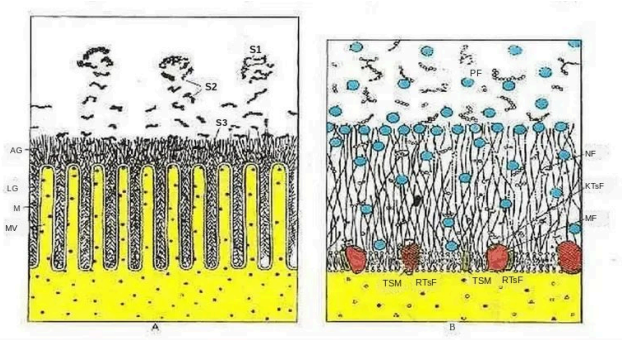


Figure 20.1. *Membrane digestion: A — microvilli and glycocalyx, substrate molecules reaching the membrane through the glycocalyx; B — location of the cavity and membrane enzymes and the transport systems (Alavia O.T., 2018)*

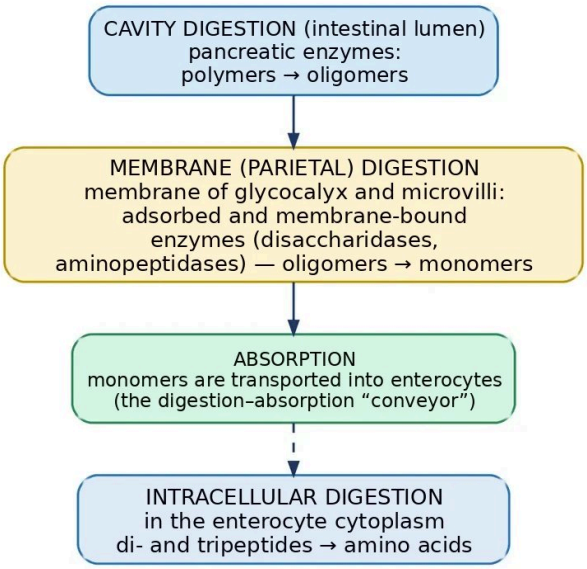


Figure 20.2. *Stages of digestion in the small intestine*

Table 20.2. *Comparative characteristics of cavity and membrane digestion*

Feature	Cavity digestion	Membrane digestion
Site	Intestinal lumen	Surface of the microvilli, glycocalyx
Enzymes	Pancreatic and intestinal juice enzymes (free)	Adsorbed and membrane-bound (enterocyte's own) enzymes
Substrates	Large molecules (polymers)	Intermediate products (oligomers, dimers)
Product	Oligomers	Monomers — absorbed immediately

Feature	Cavity digestion	Membrane digestion
Sterile conditions	Absent	Yes — the spaces between microvilli are too narrow for bacteria (glycocalyx barrier)
Efficiency	Relatively low	Very high (enzymes arranged in order, a “digestion-absorption conveyor”)

REMEMBER

Advantages of membrane digestion: 1) **high speed** — the enzymes are arranged in order over a large surface and the substrate is close to them; 2) **coupling of digestion and absorption** — the monomers formed are passed directly to the transport systems; 3) **sterility** — microorganisms cannot use the monomers.

CLINICAL SIGNIFICANCE

Lactase deficiency is the most common disorder of membrane digestion (widespread in adults). Lactose passes undigested into the large intestine: it draws water osmotically and bacteria ferment it — bloating, pain and diarrhea after milk. In **celiac disease** (an immune response to gluten) the villi atrophy — membrane digestion and the absorptive surface decrease sharply (malabsorption syndrome).

4. Absorption

Absorption is the passage of substances from the lumen of the digestive system into the blood and lymph. The main site is the **small intestine** (most substances in its proximal part, B_{12} and bile acids in the ileum). Little is absorbed in the oral cavity (nitroglycerin); water, alcohol and some drugs are absorbed in the stomach, and water and electrolytes in the large intestine.

Mechanisms of absorption: simple and facilitated diffusion, osmosis, filtration, primary and **secondary active transport** (driven by the Na^+ gradient), endocytosis (pinocytosis — antibodies from breast milk in infants).

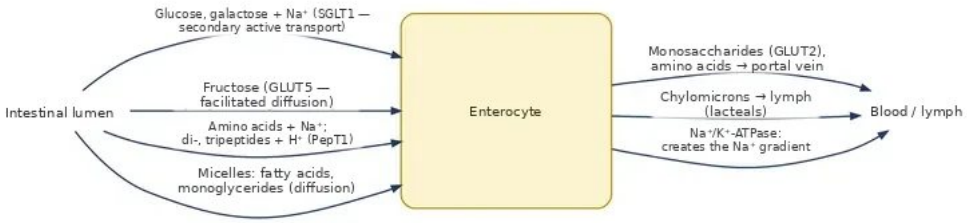


Figure 20.3. *Mechanisms of absorption of the main substances in the small intestine*

Table 20.3. *Absorption of nutrients*

Substance	Final digestion product	Mechanism of absorption	Where to
Carbohydrates	Glucose, galactose	Secondary active transport with Na^+ (SGLT1), then into the blood via GLUT2	Portal vein
Carbohydrates	Fructose	Facilitated diffusion (GLUT5)	Portal vein
Proteins	Amino acids, di- and tripeptides	Secondary active transport with Na^+ (amino acids) and H^+ (peptides, PepT1); peptides are broken down inside the cell	Portal vein

Substance	Final digestion product	Mechanism of absorption	Where to
Fats	Fatty acids, monoglycerides, cholesterol	They reach the membrane in bile micelles and enter by diffusion; in the enterocyte triglycerides are resynthesized and chylomicrons are formed	Lymph (lacteals) — into the blood via the thoracic duct
Short-chain fatty acids	—	Diffusion	Portal vein

Absorption of water, electrolytes, vitamins and iron

- **Water** — 8–9 l of fluid enters the intestine per day (2 l with food + 7 l of digestive juices); ≈ 7–8 l is absorbed in the small intestine and 1–1.5 l in the large intestine; 100–150 ml leaves with the feces. Water is absorbed passively along the **osmotic** gradient, together with dissolved substances (Na^+, glucose).
- **Na^+** — actively (Na^+/K^+-ATPase); together with glucose and amino acids; aldosterone enhances it in the large intestine.
- **Ca^{2+}** — absorbed actively in the duodenum under the action of calcitriol (vitamin D).
- **Iron** — in the duodenum; it is well absorbed as Fe^{2+} (ascorbic acid and HCl reduce Fe^{3+}); heme iron by a separate route; regulated by hepcidin according to the body's needs.
- **Vitamin B_{12}** — forms a complex with Castle's intrinsic factor and is absorbed **in the terminal ileum**.
- **Fat-soluble vitamins (A, D, E, K)** — in micelles, with the participation of bile acids.

> **CLINICAL SIGNIFICANCE**
> **Oral rehydration** (in dehydration, for example in cholera) is based on the co-transport of glucose and sodium (SGLT1): when a glucose-salt solution is drunk, Na^+ is absorbed together with glucose and water follows — even in severe secretory diarrhea. This discovery has saved the lives of millions of children.

5. Digestion in the large intestine

0.5–1.5 l of chyme enters the large intestine per day. Its main functions are **absorption of water and electrolytes** (formation of feces), microbial digestion, and storage and elimination of feces.

Juice of the large intestine — small in amount (50–60 ml per day), rich in mucus, alkaline (pH 8.5–9.0), with few enzymes. The mucus protects the intestinal wall and binds the fecal masses together.

The intestinal microflora

The large intestine contains 10^{11}–10^{12} /g bacteria (more than 500 species), mainly anaerobes: bifidobacteria, lactobacilli, bacteroides, and also Escherichia coli and others. Their total mass is 1–1.5 kg.

Table 20.4. *Functions of the intestinal microflora*

Function	Content
Digestion	Ferments fiber (cellulose, pectin) to form short-chain fatty acids — an energy source for colonocytes
Vitamin synthesis	Vitamin K, B vitamins (B_1, B_2, B_6, B_{12}, folate, biotin)

Function	Content
Protection (colonization resistance)	Prevents the multiplication of pathogenic microorganisms
Development of the immune system	“Training” local immunity, IgA synthesis
Metabolic	Metabolism of bile acids, bilirubin (urobilinogen, stercobilin) and cholesterol
Putrefaction and fermentation	Indole, skatole and phenol from protein residues (detoxified in the liver); gases

CLINICAL SIGNIFICANCE

Long-term antibiotic treatment disturbs the balance of the microflora (**dysbacteriosis**): diarrhea, vitamin K deficiency, candidiasis and Clostridioides difficile infection (pseudomembranous colitis) may develop. Fiber and fermented milk products support the beneficial microflora.

Motility of the large intestine and defecation

Movements in the large intestine are slow: **pendular**, **haustral** contractions (mixing, aiding water absorption), **peristaltic**, and 3–4 times a day, often after meals (**gastrocolic reflex**) — **strong propulsive contractions (mass movements)** that push the contents into the rectum. Food residues remain in the large intestine for 12–24 hours.

Defecation is a complex reflex act. When feces enter the rectum and stretch its wall (pressure 40–50 cm H_2O), the mechanoreceptors are excited. **The center of the defecation reflex** is in the **sacral** (S_2–S_4) segments of the spinal cord; through the pelvic (parasympathetic) nerves it ensures contraction of the rectum and relaxation of the **internal anal sphincter** (smooth muscle, involuntary). The **external anal sphincter** is striated muscle (pudendal nerve) and is controlled **voluntarily**: the cerebral cortex can delay or enhance defecation (this control develops by the age of two). Defecation is aided by contraction of the abdominal muscles and descent of the diaphragm (straining).

CAUTION

When the spinal cord is damaged above the sacral segments, the defecation reflex is preserved but voluntary control is lost (reflex automatic emptying). When the sacral segments themselves are damaged, the reflex is lost — fecal retention and incontinence are observed.

6. Laboratory work

Exercise 1: Ugolev's experiment — proving membrane digestion

Aim: to show the efficiency of digestion associated with the intestinal mucosa (membrane digestion) compared with cavity digestion.

1. Take the small intestine of a freshly killed rat (or another animal), wash it with Ringer's solution and divide it into three pieces of equal length.
2. **Sample 1** (membrane + cavity digestion): a piece of intestine is cut lengthwise and placed with the mucosa intact in a substrate solution (starch or maltose, 37 °C).
3. **Sample 2** (cavity digestion only): a piece of intestine is shaken several times in the same volume of solution and the “washings” are collected (non-adsorbed enzymes) — the substrate is added to these washings.
4. **Sample 3** (control): substrate and solution only.
5. Incubate all tubes at 37 °C for 20–30 minutes.

6. Determine the amount of glucose formed by the glucose oxidase method (or test strips, the Trommer test) and compare the results. **Expected result and analysis:** the most glucose is formed in the sample with the mucosa (1); considerably less in the washings (2); and none in the control (3). Therefore the main part of the enzymes is not free in the intestinal lumen but **bound to the membrane and glycocalyx of the enterocytes**, and the final stage of digestion proceeds efficiently precisely there. This experiment proves Ugolev's doctrine of membrane digestion.

Exercise 2: effect of intestinal juice on disaccharides

1. Pour 2 ml of 1% maltose, sucrose and lactose solution into three test tubes respectively.
2. Add intestinal mucosa homogenate (or intestinal juice) to each.
3. Keep at 37 °C for 20 minutes and determine glucose. **Result and analysis:** glucose is formed in all tubes — intestinal juice contains maltase, sucrase and lactase, which break down disaccharides to monosaccharides. Lactase activity is often lower in adults.

7. Practical skills

Skill: physiological analysis of the signs of malabsorption

1. Evaluate the character of the stool: liquid and abundant (osmotic or secretory diarrhea), fatty (steatorrhea — impaired fat absorption).
2. Identify signs of deficiency: anemia (iron, B_{12}, folate), tendency to bleed (vitamin K), bone fragility (Ca^{2+}, vitamin D), edema (protein).
3. Determine the level of the disorder: cavity digestion (pancreas, bile), membrane digestion (lactase), absorption (celiac disease, ileal resection), transport (lymph).
4. Suggest diagnostic tests: fecal elastase, hydrogen breath test (lactose), celiac antibodies, endoscopy and biopsy.

Skill: physiological justification of the composition of an oral rehydration solution

1. Show that the solution contains glucose and sodium in approximately equimolar proportion.
2. Explain the principle of operation of the SGLT1 transporter.
3. Explain how water is absorbed along the osmotic gradient following sodium and glucose.
4. State the importance of adding potassium and bicarbonate (citrate).

Review questions

1. Which structures increase the absorptive surface of the small intestine?
2. What are the structure and function of a villus?
3. List the composition and enzymes of intestinal juice.
4. How is intestinal secretion regulated?
5. What is cavity digestion?
6. What is membrane digestion, and who discovered it?
7. Explain the advantages of membrane digestion.
8. How is Ugolev's experiment performed, and what does it prove?
9. How are carbohydrates absorbed?
10. How are the products of protein digestion absorbed?
11. What is the role of bile and chylomicrons in fat absorption?

12. Where and how are iron, calcium and vitamin B_{12} absorbed?
13. What processes take place in the large intestine?
14. List the functions of the intestinal microflora.
15. What is the mechanism of the defecation reflex, and where is its center?

Test questions

1. What is the total absorptive surface of the small intestinal mucosa?
   A) 2 m^2
   B) 20 m^2
   C) ≈ 200 m^2
   D) 2000 m^2
   E) 0.5 m^2
2. Which intestinal enzyme activates trypsinogen?
   A) Maltase
   B) Lactase
   C) Alkaline phosphatase
   D) Enterokinase
   E) Sucrase
3. Who is the author of the doctrine of membrane digestion?
   A) I.P. Pavlov
   B) W. Bayliss
   C) E. Starling
   D) I.M. Sechenov
   E) A.M. Ugolev
4. Where does membrane digestion take place?
   A) In the stomach cavity
   B) On the surface of the microvilli and in the glycocalyx
   C) In the center of the intestinal lumen
   D) In the liver
   E) In the lumen of the large intestine
5. What is the advantage of membrane digestion?
   A) It breaks down only large molecules
   B) It takes place with the participation of microorganisms
   C) Digestion and absorption are coupled; high speed and sterility
   D) It takes place without pancreatic enzymes
   E) It takes place only in the large intestine
6. By what mechanism does glucose enter the enterocyte?
   A) Simple diffusion
   B) Pinocytosis
   C) Filtration
   D) Through GLUT5
   E) Secondary active transport with Na^+ (SGLT1)

7. Where are the products of fat digestion mainly absorbed?
   A) Into the portal vein
   B) Into the lymph (as chylomicrons)
   C) Into the gastric veins
   D) They return to the intestinal lumen
   E) Into the splenic vein

8. Where is vitamin B_{12} absorbed?
   A) In the terminal ileum
   B) In the stomach
   C) In the duodenum
   D) In the large intestine
   E) In the oral cavity

9. Which vitamin does the intestinal microflora synthesize?
   A) Vitamin C
   B) Vitamin K
   C) Vitamin A
   D) Vitamin D
   E) Vitamin E

10. In Ugolev's experiment, in which sample is the most glucose formed?
    A) Substrate only (control)
    B) In the intestinal washings
    C) In boiled intestine
    D) In gastric juice
    E) In the intestinal segment with intact mucosa

11. Where is the center of the defecation reflex?
    A) In the medulla oblongata
    B) In the hypothalamus
    C) In the lumbar segments of the spinal cord
    D) In the sacral segments of the spinal cord (S_2–S_4)
    E) In the cerebellum

12. What is the mechanism of diarrhea after milk in lactase deficiency?
    A) The osmotic effect and fermentation of undigested lactose
    B) An increase in secretin
    C) An increase in gastric juice
    D) Weakening of intestinal motility
    E) An allergic reaction

13. How much intestinal juice is secreted per day?
    A) 50 ml
    B) 10 l
    C) 1.5–2.0 l
    D) 0.2 l
    E) 5–6 l

14. What kind of muscle is the internal anal sphincter, and how is it controlled?
    A) Smooth muscle, involuntary
    B) Striated, voluntary
    C) Striated, involuntary
    D) Smooth muscle, voluntary
    E) Cardiac muscle

15. What mechanism is oral rehydration based on?
    A) Absorption of fructose through GLUT5
    B) Active transport of water
    C) Active secretion of potassium
    D) Co-transport of glucose and sodium (SGLT1)
    E) Absorption of fats with micelles

Clinical cases

Case 1

A 22-year-old young man complains of bloating, rumbling in the abdomen and watery diarrhea 1–2 hours after drinking milk. He tolerates yogurt and cheese well.

Questions: What condition can be suspected? Which type of digestion is disturbed? Why can he tolerate yogurt?

Answer: Lactase deficiency (lactose intolerance). Lactase is an enzyme bound to the membrane of the microvilli, i.e. membrane digestion is disturbed. Undigested lactose, as an osmotically active substance, draws water into the intestine, and in the large intestine bacteria ferment it, forming gases (H_2, CO_2) and organic acids — bloating, rumbling and osmotic diarrhea. In yogurt and cheese most of the lactose has been broken down by lactic acid bacteria, so they are well tolerated. The diagnosis is confirmed by the hydrogen breath test.

Case 2

A year after resection of the terminal ileum (1 m) because of Crohn's disease, a patient developed weakness, macrocytic anemia and fatty stools.

Questions: Explain these signs by linking them to the functions of the ileum.

Answer: Two important processes take place in the terminal ileum: vitamin B_{12} is absorbed as a complex with Castle's intrinsic factor, and 90–95% of the bile acids are reabsorbed (enterohepatic circulation). After resection B_{12} is not absorbed — when the stores are exhausted (months to years), DNA synthesis is disturbed and megaloblastic (macrocytic) anemia develops. Bile acids are lost and the liver cannot replace them sufficiently — micelles are not formed, fat absorption is impaired (steatorrhea) and fat-soluble vitamins are also deficient. Treatment: B_{12} injections, fat restriction.

Case 3

A patient with cholera is losing 10–15 l of watery stool per day. In a setting where intravenous fluids were not available, drinking a glucose-salt solution was recommended.

Questions: What is the mechanism of diarrhea in cholera? Why is the glucose-salt solution absorbed by the intestine?

Answer: Cholera toxin permanently activates adenylate cyclase in enterocytes — cAMP increases, and Cl^-, followed by Na^+ and water, is secreted into the intestinal lumen through CFTR chloride

channels (secretory diarrhea); the active absorption of NaCl is also inhibited. However, the glucose–sodium cotransporter (SGLT1) is not damaged by the toxin. When the glucose-salt solution is drunk, Na^+ is actively absorbed together with glucose, and water is absorbed along the resulting osmotic gradient — the fluid loss is compensated. This is the physiological basis of oral rehydration.

Quick review

Table 20.5. *Key concepts of topic 20*

Concept	Brief definition
Absorptive surface	Folds × villi × microvilli ≈ 200 m^2
Glycocalyx	A mesh on the surface of the microvilli; adsorption of enzymes, barrier
Intestinal juice	1.5–2 l/day; enterokinase, disaccharidases, peptidases
Cavity digestion	In the intestinal lumen; polymers → oligomers
Membrane digestion	On the microvilli; oligomers → monomers (Ugolev)
Ugolev's experiment	Digestion in an intestinal segment with mucosa is more efficient than in the washings
Glucose absorption	SGLT1 (with Na^+) → GLUT2
Fructose	GLUT5
Fats	Micelles → enterocyte → chylomicrons → lymph
B_{12}	With Castle's factor, in the terminal ileum
Iron, calcium	In the duodenum
Large intestine	Absorption of water and electrolytes, microflora
Microflora	Fermentation of fiber, vitamins K and B, protection
Defecation center	Spinal cord S_2–S_4; external sphincter voluntary
Oral rehydration	Glucose + Na^+ cotransport

TOPIC 21

Metabolism. Metabolic bases of physiological functions. The plastic and energetic role of nutrients. Diet and nutritional norms

Learning objectives

To teach the student the essence of metabolism, the processes of assimilation and dissimilation, the plastic and energetic significance of proteins, fats, carbohydrates, water, minerals and vitamins in the body, the regulation of metabolism, and the principles of rational nutrition, diet and nutritional norms; to master compiling and evaluating a daily ration.

By the end of the class the student should know:

- the concepts of metabolism, anabolism (assimilation) and catabolism (dissimilation);
- the plastic and energetic functions of nutrients;
- protein metabolism, nitrogen balance, essential amino acids, the protein minimum and optimum;
- lipid and carbohydrate metabolism and their regulation;
- the significance of water, minerals and vitamins;
- the neurohumoral regulation of metabolism;
- the principles of rational nutrition and the norms for nutrients and energy.

The student should be able to:

- calculate and evaluate the nitrogen balance;
- calculate the energy value and nutrient ratio of a daily ration;
- calculate and evaluate the body mass index.

1. General concept of metabolism

Metabolism is the exchange of substances and energy between the body and the external environment; it is the main sign of life. It consists of two interrelated processes:

- **Assimilation (anabolism)** — the synthesis of complex compounds specific to the body from simple substances coming from outside; **energy is consumed** (ATP). This process ensures cell growth, renewal and the accumulation of reserves.
- **Dissimilation (catabolism)** — the breakdown of complex organic substances to simple ones (CO_2, H_2O, urea); **energy is released** — it is stored as ATP and partly converted into heat.

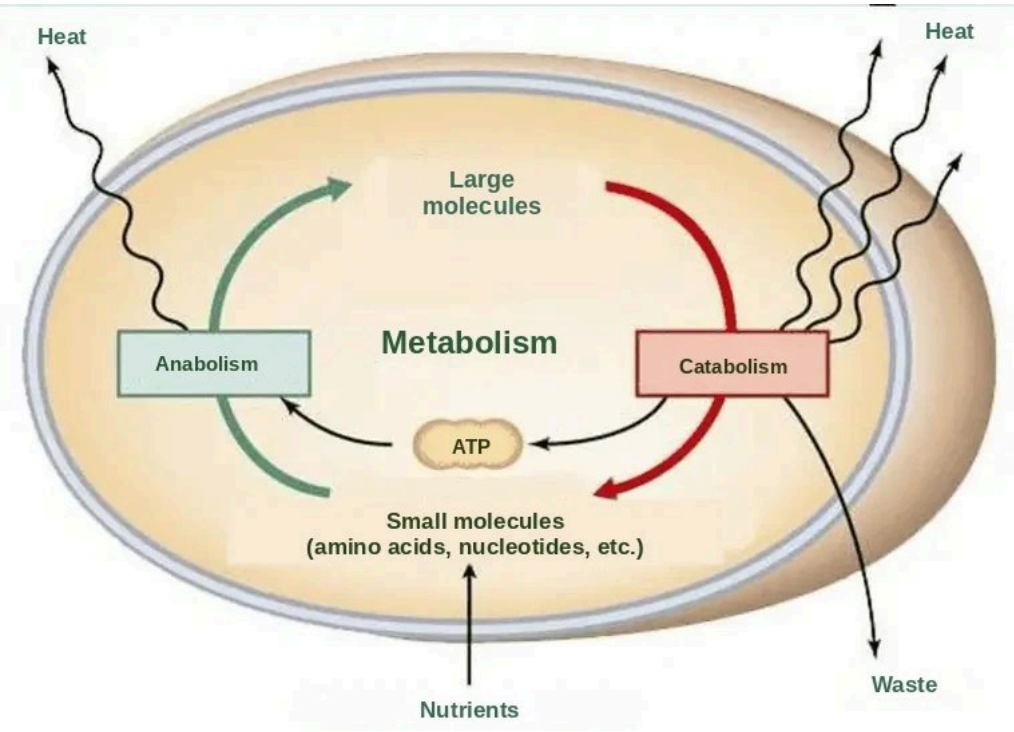


Figure 21.1. *Metabolism: anabolism and catabolism, formation of ATP and heat (Alavia O.T., 2018)*

In childhood and during growth assimilation predominates; in adults the two processes are in balance; in old age and in diseases (fever, tumors, starvation) dissimilation predominates.

Metabolic bases of physiological functions: every function — muscle contraction, the nerve impulse, secretion, active transport, synthesis — is carried out with the energy of ATP. ATP is formed in the cell in three ways: **creatine phosphate** (seconds — fast but short), **anaerobic glycolysis** (minutes — lactate is formed) and **oxidative phosphorylation** (in mitochondria — the most efficient: ≈ 30–32 ATP from 1 molecule of glucose).

2. The plastic and energetic role of nutrients

Table 21.1. *Functions of nutrients*

Substance	Plastic role	Energetic role	Energy value
Proteins	The main plastic material: cell structures, enzymes, hormones, antibodies, hemoglobin, transport proteins	Secondary (10–15% of daily energy)	4.1 kcal/g (17.2 kJ)
Fats (lipids)	Cell membranes (phospholipids, cholesterol), steroid hormones, myelin, mechanical protection and thermal insulation	Most energy, the main reserve (25–30%)	9.3 kcal/g (38.9 kJ)
Carbohydrates	Glycoproteins, nucleic acids (ribose), glycosaminoglycans	The main fast energy source (55–60%); almost the only source for the CNS	4.1 kcal/g (17.2 kJ)

Protein metabolism

Proteins are not stored in reserve in the body and are constantly renewed (≈ 300–400 g of protein is renewed per day, mostly from recycled amino acids). When proteins are broken down, most of the nitrogen is excreted in the urine as **urea**.

Nitrogen balance is the ratio between the amount of nitrogen taken in with food and the amount excreted from the body (protein contains on average 16% nitrogen: 1 g nitrogen = 6.25 g protein).

- **Nitrogen equilibrium** — intake = output (healthy adults).
- **Positive nitrogen balance** — intake > output (growth, pregnancy, recovery from illness, anabolic hormones — GH, testosterone, insulin).
- **Negative nitrogen balance** — intake < output (starvation, protein deficiency, severe diseases, burns, fever, glucocorticoid excess, old age). **Essential amino acids** are not synthesized in the body and must come with food: **valine, leucine, isoleucine, lysine, methionine, threonine, tryptophan, phenylalanine** (in children also **histidine**, and according to some sources arginine). Proteins that contain all the essential amino acids in adequate proportions are **complete** (meat, fish, eggs, milk); proteins lacking any of them are **incomplete** (many plant proteins; for example, maize protein is low in lysine and tryptophan).

MEMORY AID

Essential amino acids: **"Val Leu Ile Lys Met Thr Trp Phe"** — **Va**line, **Le**ucine, **I**soleucine, **Ly**sine, **Met**hionine, **Th**reonine, **Tr**yptophan, **Phe**nylalanine.

The protein minimum (Rubner) is the smallest amount of protein needed to maintain nitrogen equilibrium (≈ 30–45 g/day; 0.6 g/kg). **The protein optimum** is the amount needed for full health and high working capacity: **0.8–1 g/kg** in adults (≈ 70–100 g/day), at least half of which should be animal protein; more is needed for children, pregnant and lactating women and athletes (1.2–2 g/kg).

CLINICAL SIGNIFICANCE

Protein-energy malnutrition in children: **kwashiorkor** (mainly protein deficiency — edema, enlarged abdomen, fatty liver, changes in skin and hair; the edema is due to the fall in oncotic pressure from hypoalbuminemia) and **marasmus** (general calorie deficiency — severe wasting without edema).

Lipid metabolism

Lipids are the energy reserve (10–15 kg in adipose tissue — energy for several weeks), a component of membranes, and a source of hormones and fat-soluble vitamins. **Essential fatty acids** are the polyunsaturated **linoleic** (ω-6) and **linolenic** (ω-3) acids; from them arachidonic acid and eicosanoids (prostaglandins, thromboxanes, leukotrienes) are formed. Sources — vegetable oils, fish oil.

Regulation: lipogenesis is enhanced by insulin; lipolysis is enhanced by adrenaline, noradrenaline, glucagon, GH, cortisol and thyroid hormones. Lipids are transported in the blood in **lipoproteins** (chylomicrons, VLDL, LDL, HDL).

Carbohydrate metabolism

Carbohydrates are the main fast energy source. Glucose is stored in the **liver** and **muscles as glycogen** (100–120 g in the liver, 300–400 g in the muscles). Liver glycogen is used to maintain blood glucose, while muscle glycogen is used only by the muscle itself. When glycogen stores are exhausted (on days 1–2 of starvation), **gluconeogenesis** (from amino acids, lactate and glycerol) is activated.

The blood glucose level (**3.3–5.5 mmol/l**) is regulated by insulin and the counter-insulin hormones (glucagon, adrenaline, cortisol, GH) (topic 15). The **brain** obtains almost all its energy from glucose (≈ 120 g per day); in prolonged starvation it partly uses ketone bodies.

Fiber (dietary fiber) — indigestible carbohydrates (cellulose, pectin): enhances intestinal peristalsis, is a substrate for the beneficial microflora, and slows the absorption of cholesterol and glucose (25–30 g per day is recommended).

3. Water and mineral metabolism

Water makes up ≈ 60% of body mass; it is the substance most essential for life (without water a person lives 3–7 days, without food 30–50 days). The daily requirement is **2–2.5 l** (≈ 30–35 ml/kg): 1.2–1.5 l from drinks, ≈ 0.8–1 l from food, ≈ 0.3 l of metabolic (oxidation) water. Excretion: 1–1.5 l with urine, 0.5–0.7 l through the skin (sweat and insensible evaporation), 0.3–0.4 l with breath, 0.1–0.2 l with feces. Water metabolism is regulated by **thirst** (hypothalamus), **vasopressin**, **aldosterone** and natriuretic peptide.

Table 21.2. *Main minerals*

Element	Daily requirement (adults)	Main function	Sources
Sodium (as table salt)	NaCl up to 5 g (WHO)	Extracellular osmotic pressure, excitability	Table salt
Potassium	3.5–4.7 g	Intracellular cation, membrane potential	Fruit, vegetables, dried fruit
Calcium	1000–1200 mg	Bones and teeth, blood coagulation, muscle contraction	Dairy products
Phosphorus	700–1200 mg	Bone, ATP, nucleic acids, phospholipids	Meat, fish, milk
Magnesium	300–400 mg	Enzyme cofactor, neuromuscular excitability	Grains, nuts, legumes
Iron	Men 10, women 18 mg	Hemoglobin, myoglobin, cytochromes	Meat, liver
Iodine	150 µg	Thyroid hormones	Iodized salt, seafood
Fluorine	1–4 mg	Resistance of tooth enamel	Water, tea
Zinc	8–12 mg	Enzymes, immunity, wound healing	Meat, grains

4. Vitamins

Vitamins are organic substances that are not synthesized (or are insufficiently synthesized) in the body, are needed in very small amounts and work mainly as enzyme cofactors. Their deficiency is **hypovitaminosis** (complete absence — avitaminosis), their excess **hypervitaminosis** (mainly the fat-soluble ones: A, D).

Table 21.3. *Main vitamins*

Vitamin	Function	Signs of deficiency
A (retinol)	Visual pigment (rhodopsin), epithelial differentiation	“Night blindness” (nyctalopia), xerophthalmia, dry skin
D (calciferol)	Absorption of calcium and phosphorus, bone mineralization	Rickets (children), osteomalacia (adults)
E (tocopherol)	Antioxidant, protection of membranes	Hemolysis, neurological disorders
K (phylloquinone)	Synthesis of factors II, VII, IX, X	Tendency to bleed

Vitamin	Function	Signs of deficiency
B_1 (thiamine)	Carbohydrate metabolism (pyruvate dehydrogenase)	Beriberi, Wernicke–Korsakoff syndrome (in alcoholism)
B_2 (riboflavin)	FAD, FMN — oxidation-reduction	Cheilitis (cracks at the corners of the mouth), glossitis
PP (niacin)	NAD, NADP	Pellagra (“3 D”: dermatitis, diarrhea, dementia)
B_6 (pyridoxine)	Amino acid metabolism	Seizures, anemia, dermatitis
B_9 (folic acid)	DNA synthesis	Megaloblastic anemia; neural tube defects in the fetus
B_{12} (cobalamin)	DNA synthesis, myelin	Megaloblastic anemia, neurological disorders
C (ascorbic acid)	Collagen synthesis, antioxidant, iron absorption	Scurvy: bleeding gums, loss of teeth, slow wound healing

REMEMBER

Fat-soluble vitamins — A, D, E, K (“ADEK”): absorbed with the participation of bile and fats and accumulate in the body (liver, adipose tissue) — there is a risk of hypervitaminosis. **Water-soluble vitamins** (B group, C) do not accumulate (except B_{12} — the liver stores enough for years), and the excess is excreted in the urine — they must be consumed regularly.

5. Regulation of metabolism

Metabolism is regulated by **neurohumoral** means. The highest center is the **hypothalamus**: it contains the **hunger** (lateral nuclei) and **satiety** (ventromedial nuclei, arcuate nucleus) centers; the centers of thermoregulation, thirst and the endocrine system are also located there. The hypothalamus influences metabolism through the endocrine glands (insulin, glucagon, thyroid hormones, cortisol, GH, sex hormones) and through the autonomic nervous system.

Regulation of appetite and food intake: ghrelin (secreted in the stomach on an empty stomach — enhances the sensation of hunger); **leptin** (from adipose tissue — a signal about fat stores, reduces appetite); **insulin**, **CCK**, **GLP-1**, **PYY** (satiety signals); gastric distension (via the vagus).

6. Principles and norms of rational nutrition

Principles of **rational (healthy) nutrition**:

1. **Energy balance** — the energy value of food should match the daily energy expenditure (topic 14).
2. **Balanced nutrition** — the optimal ratio of nutrients: proteins : fats : carbohydrates ≈ **1 : 1 : 4** (by mass); by energy, respectively 10–15%, 25–30%, 55–60%.
3. **Meal pattern** — at least 3–4 meals a day, at the same times; the last meal 2–3 hours before sleep.
4. **Variety** — obtaining sufficient amounts of all essential substances, vitamins and minerals.
5. **Food safety** and proper culinary processing.

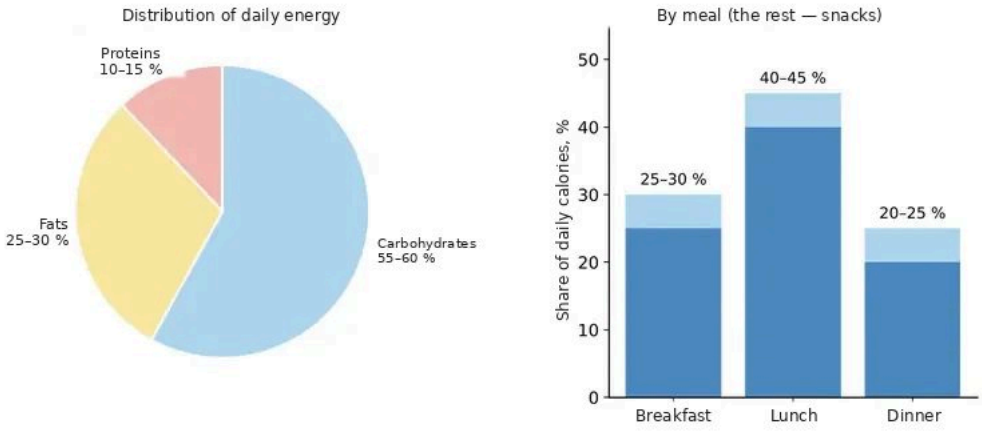


Figure 21.2. *Distribution of daily energy by nutrients and meals*

Table 21.4. *Main recommendations for adults (based on WHO and national norms, approximate)*

Parameter	Recommendation
Proteins	0.8–1 g/kg (55% animal protein)
Fats	25–30% of energy; saturated fats < 10%; trans fats < 1%
Carbohydrates	55–60% of energy; “free” sugars < 10% (preferably < 5%)
Fiber	25–30 g/day
Table salt	< 5 g/day
Fruit and vegetables	At least 400 g/day
Water	30–35 ml/kg

CAUTION

Norms differ according to age, sex, work intensity, climate, physiological state (pregnancy, lactation) and health; the figures in the table are for average adults. In children the protein and energy requirements per 1 kg of body mass are 2–3 times higher than in adults.

Body mass index (BMI) = body mass (kg) / height $(m)^2$. Evaluation (WHO): < 18.5 — underweight; 18.5–24.9 — normal; 25–29.9 — overweight; ≥ 30 — obesity.

CLINICAL SIGNIFICANCE

Obesity results from energy intake exceeding expenditure over a long period; it increases the risk of type 2 diabetes mellitus, arterial hypertension, atherosclerosis, coronary heart disease, some tumors and joint diseases. Abdominal (visceral) obesity is especially dangerous: the risk increases when the waist circumference is > 94 cm in men and > 80 cm in women.

7. Laboratory work

Exercise 1: calculating the nitrogen balance

Given: 90 g of protein was taken with the daily food; 13 g of nitrogen was excreted with urine and 1.5 g with feces.

1. Calculate the nitrogen intake: 90 g / 6.25 = 14.4 g.
2. Nitrogen excreted: 13 + 1.5 = 14.5 g.
3. Nitrogen balance: 14.4 − 14.5 = −0.1 g ≈ 0 — **nitrogen equilibrium**.

4. In the same way calculate the other variants given by the instructor (a growing child, a burn patient) and determine the type of balance.

Exercise 2: evaluating a daily ration

1. Write down all the products consumed in the last 24 hours and their mass.
2. Using food composition tables, calculate the amount of protein, fat and carbohydrate in each product.
3. Calculate the energy value: protein × 4.1 + fat × 9.3 + carbohydrate × 4.1 (kcal).
4. Compare the result with your own daily energy expenditure (topic 14).
5. Compare the nutrient ratio and the distribution of energy among meals with the recommendations; write down the shortcomings.

Exercise 3: determining the body mass index

1. Measure height (cm) and body mass (kg).
2. Calculate the BMI.
3. Measure the waist circumference at the level of the navel.
4. Evaluate the results by the WHO criteria.

8. Practical skills

Skill: compiling a daily ration for a student

1. Determine the daily energy requirement (for example, 2500 kcal — group I).
2. Allocate 12% to proteins (300 kcal ≈ 73 g), 28% to fats (700 kcal ≈ 75 g) and 60% to carbohydrates (1500 kcal ≈ 366 g).
3. Distribute the energy among meals (breakfast 25–30%, lunch 40–45%, dinner 20–25%).
4. Compile a menu including animal and plant proteins, fruit and vegetables and dairy products.
5. Control the amounts of salt, sugar and saturated fats.

Skill: suspecting hypovitaminosis from clinical signs

1. Vision (poorer vision in the dark) — vitamin A.
2. Bleeding gums, slow wound healing — vitamin C.
3. Cracks at the corners of the mouth, glossitis — B_2, PP.
4. Megaloblastic anemia — B_{12} or folate.
5. Bone deformities and sweating in children — vitamin D.
6. Tendency to bleed — vitamin K.

Review questions

1. What are metabolism, assimilation and dissimilation?
2. In what ways is ATP formed in the cell?
3. What are the plastic and energetic roles of proteins, fats and carbohydrates?
4. What is nitrogen balance? Explain its types with examples.
5. List the essential amino acids.
6. How do complete and incomplete proteins differ?

7. What are the protein minimum and optimum?
8. Name the essential fatty acids and their significance.
9. How are carbohydrates stored in the body?
10. What is the daily water balance?
11. What is the significance of the main minerals?
12. What is the difference between fat-soluble and water-soluble vitamins?
13. How are hunger and satiety regulated?
14. List the principles of rational nutrition.
15. How is the body mass index calculated and evaluated?

Test questions

1. Which feature is characteristic of assimilation?
   A) Energy is released
   B) Complex substances are broken down
   C) CO_2 and water are formed
   D) Urea is formed
   E) Complex substances are synthesized and energy is consumed

2. How much energy is released when 1 g of fat is oxidized?
   A) 4.1 kcal
   B) 17.2 kcal
   C) 9.3 kcal
   D) 1.5 kcal
   E) 2.0 kcal

3. What is the nitrogen balance in a growing child?
   A) Positive
   B) Negative
   C) In equilibrium
   D) Zero and unchanging
   E) Always negative

4. Which is an essential amino acid?
   A) Glycine
   B) Lysine
   C) Alanine
   D) Glutamic acid
   E) Serine

5. What is the protein optimum for adults?
   A) 0.1–0.2 g/kg
   B) 3–4 g/kg
   C) 0.8–1 g/kg
   D) 5 g/kg
   E) 10 g/kg

6. Which are the essential fatty acids?
   A) Linoleic and linolenic acids
   B) Palmitic and stearic acids
   C) Butyric acid
   D) Acetic acid
   E) Oleic acid

7. What is the main function of liver glycogen?
   A) To supply energy to the liver only
   B) For muscle contraction
   C) Bile formation
   D) Maintaining the blood glucose level
   E) Protein synthesis

8. Which vitamin is fat-soluble?
   A) Vitamin D
   B) Vitamin C
   C) Vitamin B_1
   D) Vitamin B_{12}
   E) Vitamin PP

9. Deficiency of which vitamin causes scurvy?
   A) A
   B) D
   C) C
   D) K
   E) B_1

10. Which gastric hormone enhances the sensation of hunger?
    A) Leptin
    B) Insulin
    C) Cholecystokinin
    D) Ghrelin
    E) Secretin

11. In what form is nitrogen mainly excreted when protein is oxidized?
    A) Ammonia
    B) Urea
    C) Creatinine
    D) Uric acid
    E) Amino acids

12. What is the protein : fat : carbohydrate ratio (by mass) in rational nutrition?
    A) 4 : 1 : 1
    B) 1 : 4 : 1
    C) 1 : 1 : 1
    D) 2 : 1 : 1
    E) 1 : 1 : 4

13. What causes edema in kwashiorkor?
    A) Heart failure
    B) Hypoalbuminemia and a fall in oncotic pressure
    C) Excessive salt intake
    D) Kidney failure
    E) Allergy

14. What does a BMI of 27 kg/m^2 indicate?
    A) Underweight
    B) Normal value
    C) Grade II obesity
    D) Overweight
    E) Morbid obesity

15. Pellagra (“3 D”) is a deficiency of which vitamin?
    A) B_1 (thiamine)
    B) A (retinol)
    C) E (tocopherol)
    D) B_6 (pyridoxine)
    E) PP (niacin)

Clinical cases

Case 1

A week after extensive burns, a patient is excreting 25 g of nitrogen per day in the urine while taking 75 g of protein with food. The patient is losing weight and the wounds are healing slowly.

Questions: Calculate and evaluate the nitrogen balance. Why has this condition developed, and what measures are needed?

Answer: Nitrogen intake = 75 / 6.25 = 12 g; nitrogen output ≥ 25 g; balance = 12 − 25 = −13 g/day — a sharply negative nitrogen balance (catabolism predominates). In burns, stress hormones (cortisol, catecholamines, glucagon) and inflammatory cytokines enhance protein breakdown (mainly in muscles); amino acids are used for gluconeogenesis and wound healing; protein is also lost from the burn surface. As a result muscle mass decreases and immunity and regeneration weaken. Measures: increase protein to 1.5–2 g/kg, provide sufficient energy (otherwise protein is used for energy), vitamins (C, A) and trace elements (zinc).

Case 2

A student eats mainly bread, pasta and sweet tea every day, with little meat, milk, fruit or vegetables. His daily energy intake is 2800 kcal (expenditure 2400 kcal). In recent months his weight has increased, but he complains of fatigue and bleeding gums.

Questions: Evaluate the shortcomings of the ration from a physiological point of view.

Answer: The ration is excessive in energy (positive energy balance — weight gain) and far too rich in carbohydrates, especially rapidly absorbed (“free”) sugar. At the same time it is unbalanced: complete (animal) proteins, essential amino acids, calcium (milk) and iron (meat) are lacking; there are few fruit and vegetables — vitamin C deficiency (bleeding gums — impaired collagen synthesis) and little fiber. The fatigue may be related to deficiency of iron and B vitamins. Recommendation:

match energy to expenditure, reduce sugar, add protein sources, dairy products and at least 400 g of fruit and vegetables a day.

Case 3

A patient who has consumed alcohol for a long time and eaten poorly has disturbed gait, impaired eye movements and a sharp decline in memory. Thiamine was prescribed before giving him glucose solution.

Questions: Deficiency of which vitamin is this? Why is thiamine given before glucose?

Answer: Vitamin B_1 (thiamine) deficiency — Wernicke's encephalopathy (with Korsakoff syndrome). Thiamine pyrophosphate is a cofactor of the enzymes pyruvate dehydrogenase and α-ketoglutarate dehydrogenase — it is necessary for the oxidation of carbohydrates. The brain obtains its energy mainly from glucose, so in thiamine deficiency the neurons suffer from an energy shortage. If glucose is given without thiamine, glucose metabolism increases and uses up the little thiamine that remains, and the neurological condition may worsen. In alcoholism thiamine intake is low, it is poorly absorbed and little is stored in the liver.

Quick review

Table 21.5. *Key concepts of topic 21*

Concept	Brief definition
Assimilation	Synthesis; energy is consumed
Dissimilation	Breakdown; energy is released
Energy value	Protein 4.1, fat 9.3, carbohydrate 4.1 kcal/g
Nitrogen balance	1 g nitrogen = 6.25 g protein; positive in growth, negative in starvation
Essential amino acids	Valine, leucine, isoleucine, lysine, methionine, threonine, tryptophan, phenylalanine
Protein optimum	0.8–1 g/kg
Essential fatty acids	Linoleic (ω-6), linolenic (ω-3)
Glycogen	In the liver — for blood glucose; in muscle — for the muscle
Water requirement	2–2.5 l (30–35 ml/kg)
ADEK	Fat-soluble vitamins
Ghrelin / leptin	Hunger / satiety signals
Nutrient ratio	Protein : fat : carbohydrate = 1 : 1 : 4
Energy distribution	10–15 % / 25–30 % / 55–60 %
BMI	Kg / m^2; normal 18.5–24.9
Table salt	< 5 g/day

TOPIC 22

Physiology of excretion. Physiology and regulation of the kidneys. Regulation of electrolytes, pH and extracellular fluid volume by the kidneys

Learning objectives

To teach the student the excretory organs and their functions, the structure of the nephron and the features of renal blood flow, the processes of urine formation — glomerular filtration, tubular reabsorption and secretion, the mechanism of urine concentration, the neurohumoral regulation of kidney function and the homeostatic functions of the kidneys (water-salt balance, osmotic pressure, blood volume, acid–base balance); to master evaluating clearance and urinalysis.

By the end of the class the student should know:

- the excretory organs and the non-excretory functions of the kidneys;
- the structure of the nephron, the types of nephron, renal blood flow and its autoregulation;
- the mechanism of glomerular filtration, the effective filtration pressure and the GFR;
- reabsorption and secretion in the proximal tubules, the loop of Henle, the distal tubule and the collecting ducts;
- the renal threshold for glucose;
- the countercurrent system and urine concentration;
- the effects of vasopressin, aldosterone, the renin–angiotensin system, natriuretic peptide and parathyroid hormone on the kidney;
- the role of the kidneys in regulating acid–base balance;
- the amount, composition and properties of urine and the micturition reflex.

The student should be able to:

- calculate the GFR from creatinine clearance;
- evaluate the Zimnitsky test;
- interpret the results of a routine urinalysis from a physiological point of view.

1. Excretory organs and functions of the kidneys

Excretion is the removal from the body of the end products of metabolism, excess water, salts and foreign substances. Excretory organs: the **kidneys** (the main ones — products of nitrogen metabolism, water, salts), the **lungs** (CO_2, water vapor, volatile substances), the **skin** (sweat glands — water, salts, urea), the **digestive system** (bile pigments, cholesterol, heavy metals, undigested residues), the salivary glands.

Table 22.1. *Functions of the kidneys*

Function	Content
Excretory	Excretion of urea, creatinine, uric acid, drugs and toxins
Homeostatic	Water balance, osmotic pressure (osmoregulation), ionic composition (Na^+, K^+, Ca^{2+}, phosphate), blood volume
Acid–base balance	Secretion of H^+, reabsorption of HCO_3^- and formation of new bicarbonate

Function	Content
Endocrine	Renin (arterial pressure), erythropoietin (erythropoiesis), calcitriol (the active form of vitamin D), prostaglandins, kinins
Metabolic	Gluconeogenesis (in prolonged starvation), breakdown of proteins and peptide hormones
Regulation of arterial pressure	Through blood volume (Na^+ and water) and the renin–angiotensin system

2. The nephron and renal blood flow

The **nephron** is the structural and functional unit of the kidney; each kidney has ≈ 1 million nephrons. The nephron consists of the **renal corpuscle** (the network of glomerular capillaries and the surrounding **Shumlyansky–Bowman capsule**) and the system of **tubules**: **proximal tubule → loop of Henle** (descending and ascending limbs) **→ distal tubule → collecting duct**.

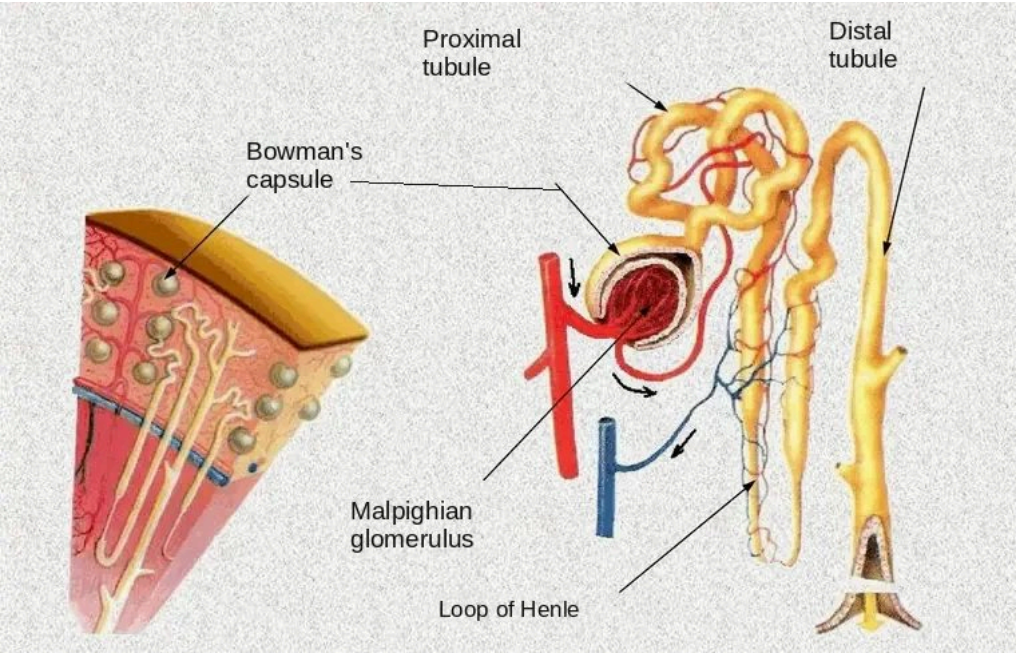


Figure 22.1. *Structure of the nephron (Alavia O.T., 2018)*

Types of nephron: cortical nephrons — 80–85%, with a short loop of Henle; **juxtamedullary** nephrons — 15–20%, with a long loop extending deep into the medulla; they play the main role in concentrating urine.

The **juxtaglomerular apparatus (JGA)** consists of the **juxtaglomerular cells** in the wall of the afferent arteriole (they secrete renin), the **macula densa** of the distal tubule (it senses NaCl in the tubule) and the mesangial cells.

Renal blood flow

Although the kidneys make up 0.5% of body mass, they receive **20–25%** of the minute volume (**≈ 1.2 l/min**) — not because of their own needs, but because of their function of forming urine. Features:

- **two capillary networks**: the glomerular capillaries (high pressure — for filtration) and the peritubular capillaries (low pressure — for reabsorption);
- the efferent arteriole has a smaller diameter than the afferent one — the pressure in the glomerulus is high;

- **autoregulation**: when arterial pressure changes within **70–180 mm Hg**, renal blood flow and GFR hardly change (the myogenic mechanism — the Bayliss effect — and **tubuloglomerular feedback** via the macula densa).

3. Mechanism of urine formation

Urine formation consists of three processes: **glomerular filtration**, **tubular reabsorption** and **tubular secretion**. Amount excreted = filtered − reabsorbed + secreted.

Glomerular filtration

Blood plasma, with almost all its components except proteins, is filtered from the glomerular capillaries into the capsular space — **primary urine** (ultrafiltrate) is formed. **The filtration barrier**: the fenestrated endothelium, the (negatively charged) basement membrane and the slit diaphragms between the foot processes of the podocytes. Substances with a molecular mass greater than 70,000 Da and negatively charged substances (albumin) hardly pass.

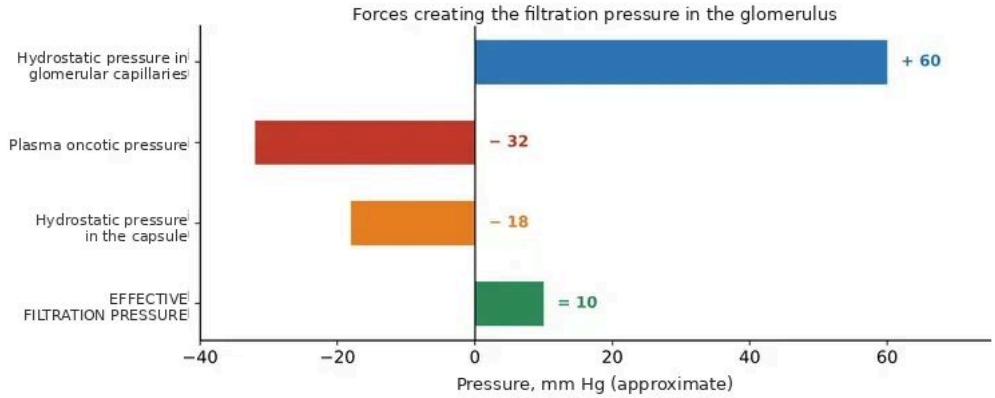


Figure 22.2. *Forces creating the filtration pressure in the glomerulus (drawn after Costanzo L.S., 2018)*

Effective filtration pressure = hydrostatic pressure in the glomerular capillaries − (plasma oncotic pressure + hydrostatic pressure in the capsule) ≈ 60 − (32 + 18) ≈ **10 mm Hg**.

Glomerular filtration rate (GFR) is the volume of primary urine formed by both kidneys per unit time: **≈ 125 ml/min** (≈ **180 l/day**). Of this, **99%** is reabsorbed and about **1.5 l** of final (secondary) urine is excreted. The **filtration fraction** is the ratio of GFR to renal plasma flow (≈ 20%).

Factors affecting GFR: a sharp fall in arterial pressure (< 70 mm Hg — filtration decreases; it stops at < 40–50), the tone of the afferent and efferent arterioles, plasma oncotic pressure (GFR increases in hypoproteinemia), the pressure in the capsule (GFR decreases when the urinary tract is obstructed), the number of active glomeruli and the filtration surface.

CLINICAL SIGNIFICANCE

GFR is the main indicator of kidney function; it is evaluated by **creatinine clearance** or by formulas (CKD-EPI) based on blood creatinine, age and sex. If GFR < 60 ml/min/1.73 m^2 persists for more than 3 months — **chronic kidney disease**; < 15 — end-stage renal failure (dialysis or transplantation).

Tubular reabsorption

Table 22.2. *Reabsorption in different parts of the nephron*

Part	What and how much is reabsorbed	Features
Proximal tubule	Na^+ ≈ 65%, water ≈ 65%, K^+, Cl^-, HCO_3^- ≈ 80–90%; glucose and amino acids ≈ 100%; phosphate, uric acid; filtered small proteins (pinocytosis)	Obligatory reabsorption; iso-osmotic; microvilli, many mitochondria
Descending limb of the loop of Henle	Water (aquaporin-1)	Impermeable to salts — the filtrate becomes concentrated
Thick ascending limb of the loop of Henle	Na^+, K^+, $2Cl^-$ (NKCC2) ≈ 25%; Ca^{2+}, Mg^{2+}	Impermeable to water — the “diluting segment”; the filtrate becomes hypotonic
Distal tubule	Na^+, Cl^- (NCC) ≈ 5%; Ca^{2+} (under the action of PTH)	Poorly permeable to water
Collecting duct	Na^+ (aldosterone), water (vasopressin — aquaporin-2), urea	Facultative reabsorption — regulated by hormones

Glucose reabsorption takes place in the proximal tubule together with Na^+ through the **SGLT2** (mainly) and SGLT1 transporters and is limited by the number of transporters (the **transport maximum**). When blood glucose exceeds **8.9–10 mmol/l** (the **renal threshold**), the transporters are saturated and glucose appears in the urine — **glucosuria**.

> **CLINICAL SIGNIFICANCE**
> **Diuretics** block reabsorption in different parts of the nephron: **furosemide** — NKCC2 in the ascending limb of the loop of Henle (the strongest, “loop diuretics”); **hydrochlorothiazide** — NCC in the distal tubule; **spironolactone** — the aldosterone receptors in the collecting duct (potassium-sparing); **SGLT2 inhibitors** (empagliflozin) — glucose reabsorption in the proximal tubule (in the treatment of diabetes).

Tubular secretion

The tubular epithelium **actively secretes** substances from the blood into the tubular lumen: **H^+** (proximal and distal tubules, intercalated cells of the collecting ducts), **K^+** (principal cells of the distal tubule and collecting ducts, under the action of aldosterone), **NH_3/NH_4^+**, and organic acids and bases (para-aminohippuric acid, penicillin, many drugs, partly creatinine) — in the proximal tubule.

4. Mechanism of urine concentration

The human kidney can concentrate urine **up to 4 times** relative to plasma (up to 1200 mOsm/kg) or dilute it (down to 50 mOsm/kg). This is achieved by the **countercurrent multiplier system**:

1. In the descending and ascending limbs of the loop of Henle fluid flows in opposite directions.
2. The **ascending limb**, being impermeable to water, actively pumps NaCl into the interstitium — the interstitium becomes hypertonic.
3. The **descending limb** is permeable to water — water moves out into the hypertonic interstitium and the filtrate becomes concentrated.
4. Repeated along the loop, this process creates an **osmotic gradient** in the medulla from the cortex (300 mOsm/kg) to the tips of the papillae (1200 mOsm/kg). The return of **urea** from the collecting ducts to the interstitium (urea recycling) also contributes greatly.
5. The **vasa recta**, acting as a countercurrent exchanger, also do not wash out the gradient.

6. When **vasopressin** is present, **aquaporin-2** channels are inserted into the walls of the collecting ducts and water is reabsorbed into the hypertonic interstitium — **concentrated urine**; without vasopressin — **dilute urine**.

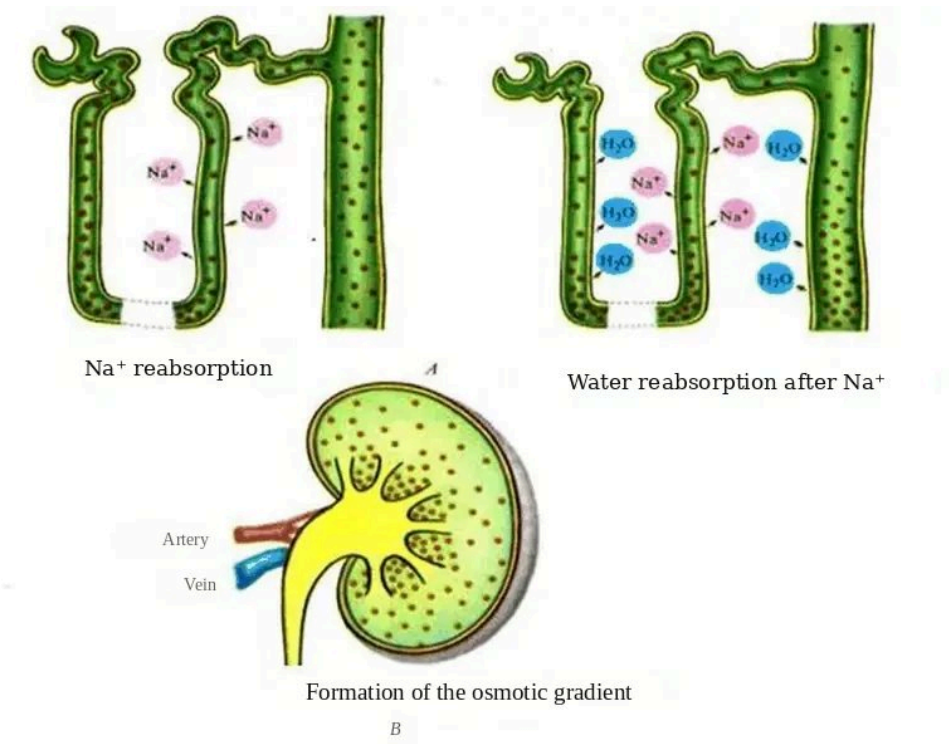


Figure 22.3. *Reabsorption of Na^+ and water in the proximal tubule (A) and formation of the osmotic gradient in the renal medulla (B) (Alavia O.T., 2018)*

5. Regulation of kidney function

Humoral regulation

Table 22.3. *Hormones regulating kidney function*

Hormone	Source	Stimulus	Effect on the kidney
Vasopressin (ADH)	Hypothalamus → neurohypophysis	Increase in plasma osmolarity (osmoreceptors; even 1–2% is enough), decrease in blood volume	Water reabsorption through aquaporin-2 in the collecting ducts ↑ — urine decreases and becomes concentrated
Aldosterone	Adrenal cortex	Angiotensin II, hyperkalemia	Na^+ reabsorption ↑, K^+ and H^+ secretion ↑ — blood volume increases
Angiotensin II	Renin–angiotensin system	Renin release	Constricts the efferent arteriole and maintains GFR; Na^+ reabsorption in the proximal tubule ↑; release of aldosterone and ADH, thirst
Natriuretic peptide	Cardiac atria	Stretching of the atria (increased blood volume)	GFR ↑, Na^+ reabsorption ↓ — natriuresis and diuresis; inhibits renin and aldosterone
Parathyroid hormone	Parathyroid glands	Decrease in Ca^{2+}	Ca^{2+} reabsorption in the distal tubule ↑, phosphate reabsorption ↓; calcitriol synthesis ↑

Renin release is enhanced by: a fall in pressure in the afferent arteriole (baroreceptor mechanism), a reduced NaCl supply to the macula densa, and the sympathetic nerves (β_1).

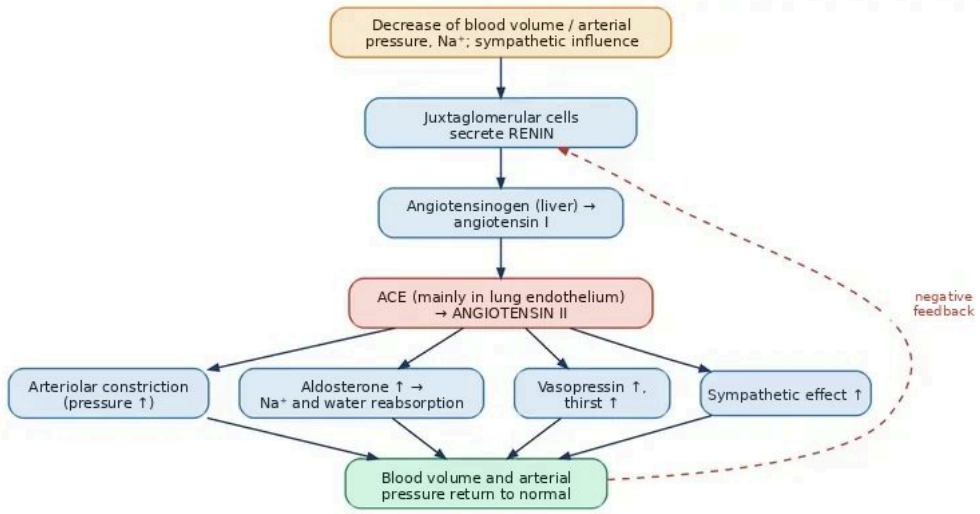


Figure 22.4. *The renin–angiotensin–aldosterone system*

Nervous regulation

The kidney is innervated mainly by **sympathetic** nerves: they constrict the arterioles (with strong influence GFR decreases — in pain and stress a **reflex decrease in diuresis** — pain anuria) and enhance renin release and Na^+ reabsorption. The conditioned-reflex increase in diuresis with heavy drinking shows the influence of the cerebral cortex on the kidney (K.M. Bykov).

Regulation of acid–base balance by the kidneys

The kidneys excrete 50–100 mmol of **non-volatile acids** (sulfuric and phosphoric acids — from protein metabolism) from the body per day and conserve bicarbonate:

1. **Reabsorption of filtered bicarbonate** (mainly in the proximal tubule, with the participation of carbonic anhydrase).
2. **Secretion of H^+** and its binding by the **phosphate buffer** (**titratable acidity**: $HPO_4^{2-} + H^+ \rightarrow H_2PO_4^-$).
3. **Ammoniagenesis**: NH_3 is formed from glutamine in the tubular cells, combines with H^+ and is excreted as **NH_4^+**; in acidosis this route increases sharply. For each H^+ excreted, one **new HCO_3^-** is returned to the blood. Urine pH can vary between **4.5 and 8.0** (on average 5.5–6.5).

6. Urine and micturition

Table 22.4. *Main parameters of urine (adults)*

Parameter	Normal value
Daily amount (diuresis)	1–1.5 l (≈ 70–80% of the fluid drunk)
Color	Straw-yellow (urobilin, urochrome)
Clarity	Clear
Specific gravity	1.010–1.025 (varies within 1.003–1.035 during the day)
pH	5.0–7.0 (alkaline with plant food, acidic with meat food)
Protein	None or traces (< 0.033 g/l; < 150 mg per day)

Parameter	Normal value
Glucose	Absent
Ketone bodies	Absent
Urea	20–35 g/day
Creatinine	1–2 g/day

Disorders of diuresis: polyuria (> 2 l/day — diabetes mellitus and insipidus, diuretics), **oliguria** (< 500 ml/day — dehydration, shock, acute kidney failure), **anuria** (< 100 ml/day), **nocturia** (nighttime diuresis exceeding daytime).

Micturition

Urine passes from the renal pelvis through the ureters into the bladder by peristaltic contractions. When 250–300 ml has accumulated in the bladder (pressure above 15–20 cm H_2O), the stretch receptors are excited and the urge to urinate appears. **The center of the micturition reflex** is in the **sacral segments of the spinal cord (S_2–S_4)**: the parasympathetic (pelvic) nerves contract the bladder muscle (detrusor) and relax the internal sphincter; the sympathetic nerves (Th_{11}–L_2), conversely, help to store urine. The **external sphincter** (striated, pudendal nerve) is controlled voluntarily; the **pontine micturition center** (Barrington's center) and the cerebral cortex coordinate the reflex and provide voluntary control (developed by 2–3 years of age).

7. Laboratory work

Exercise 1: calculating GFR from creatinine clearance

Clearance is the volume of plasma completely cleared of a given substance by the kidneys per unit time: C = (U × V) / P, where U is the concentration of the substance in urine, V the minute diuresis (ml/min) and P its concentration in plasma. The clearance of a substance that is only filtered and neither reabsorbed nor secreted (inulin, in practice creatinine) equals the GFR.

Given: daily urine 1440 ml (V = 1 ml/min); urine creatinine 9 mmol/l; plasma creatinine 0.08 mmol/l.

1. C = 9 × 1 / 0.08 = 112.5 ml/min.
2. **Interpretation:** normal 90–130 ml/min (decreases with age) — GFR is normal.
3. Calculate and evaluate the other variants given by the instructor.

> **CAUTION**
> Creatinine is secreted in small amounts in the tubules, so creatinine clearance is 10–20% higher than the true GFR. In the clinic, GFR estimated from blood creatinine by formulas (CKD-EPI) is widely used.

Exercise 2: evaluating the Zimnitsky test

1. Eating as usual, the patient collects urine in a separate container every 3 hours over 24 hours (8 portions).
2. The volume and specific gravity (with a urometer) of each portion are determined.
3. Determine the daytime (6–18) and nighttime (18–6) diuresis, the daily diuresis and the maximum and minimum specific gravity.

Interpretation: normally the daily diuresis is 70–80% of the fluid drunk, daytime diuresis is 2–3 times greater than nighttime; the specific gravity is ≥ 1.020 in at least one portion, and the difference

between portions is ≥ 0.010–0.015. **Hyposthenuria** (specific gravity always < 1.012) or **isosthenuria** (specific gravity hardly changes, ≈ 1.010) — the concentrating ability of the kidney is impaired (chronic kidney failure, diabetes insipidus).

Exercise 3: determining the physicochemical properties of urine

1. Evaluate the color and clarity.
2. Measure the specific gravity with a urometer (fill the cylinder with urine and lower the urometer so that it does not touch the wall).
3. Determine the pH with indicator paper.
4. Test for protein, glucose, ketone bodies and blood with test strips.

8. Practical skills

Skill: physiological interpretation of a routine urinalysis

1. Compare the volume, color, clarity, specific gravity and pH with the norm.
2. If there is glucosuria, compare it with blood glucose: ≥ 10 mmol/l in the blood — hyperglycemic; normal — renal glucosuria.
3. Proteinuria — damage to the glomerular filtration barrier or impaired tubular reabsorption.
4. Specific gravity always low — impaired concentrating ability (vasopressin deficiency or kidney disease).
5. Ketonuria — carbohydrate deficiency (starvation, decompensated diabetes).

Skill: distinguishing the effects of vasopressin and aldosterone in clinical situations

1. Evaluate the urine volume and specific gravity: abundant and dilute — vasopressin is lacking; scanty and concentrated — much vasopressin (dehydration).
2. Evaluate blood Na^+ and K^+: Na^+ retention and K^+ loss — much aldosterone; the reverse — aldosterone is lacking.
3. Also evaluate arterial pressure and blood volume.

Review questions

1. Which organs perform the excretory function?
2. List the non-excretory functions of the kidneys.
3. Explain the structure of the nephron and the types of nephron.
4. What are the features of renal blood flow and its autoregulation?
5. How does glomerular filtration take place, and what is the effective filtration pressure?
6. What is GFR, and what factors affect it?
7. Which substances are reabsorbed in the proximal tubule?
8. What is the difference between obligatory and facultative reabsorption?
9. What is the renal threshold for glucose?
10. Which substances are secreted in the tubules?
11. Explain the mechanism of urine concentration.
12. What are the effects of vasopressin and aldosterone on the kidney?
13. Explain the renin–angiotensin–aldosterone system.

14. How do the kidneys regulate acid–base balance?
15. How is the micturition reflex carried out?

Test questions

1. What proportion of the minute volume do the kidneys receive?
   A) 5 %
   B) 50 %
   C) 1–2 %
   D) 20–25 %
   E) 70 %

2. How much primary urine is formed per day?
   A) 1.5 l
   B) 15 l
   C) ≈ 180 l
   D) 500 l
   E) 50 l

3. What is the effective filtration pressure?
   A) 60 mm Hg
   B) 32 mm Hg
   C) 120 mm Hg
   D) ≈ 10 mm Hg
   E) 0 mm Hg

4. Where is glucose completely reabsorbed?
   A) In the proximal tubule
   B) In the loop of Henle
   C) In the distal tubule
   D) In the collecting duct
   E) In the bladder

5. What is the renal threshold for glucose?
   A) 3.3–5.5 mmol/l
   B) 8.9–10 mmol/l
   C) 15–20 mmol/l
   D) 1–2 mmol/l
   E) 25 mmol/l

6. What effect does vasopressin have in the collecting ducts?
   A) It decreases Na^+ reabsorption
   B) It enhances K^+ secretion
   C) It increases glucose reabsorption
   D) It increases GFR
   E) It enhances water reabsorption through aquaporin-2

7. Where is renin formed?
   A) In the liver
   B) In the juxtaglomerular cells
   C) In the lungs
   D) In the adrenal gland
   E) In the pituitary gland

8. What is the feature of the thick ascending limb of the loop of Henle?
   A) It is highly permeable to water
   B) It reabsorbs glucose
   C) It is sensitive to vasopressin
   D) It only secretes
   E) It actively reabsorbs NaCl and is impermeable to water

9. What is the effect of natriuretic peptide on the kidney?
   A) It enhances Na^+ reabsorption
   B) It increases renin release
   C) It enhances natriuresis and diuresis
   D) It concentrates urine
   E) It increases aldosterone release

10. In which part of the nephron does furosemide act?
   A) In the proximal tubule
   B) In the ascending limb of the loop of Henle (NKCC2)
   C) In the distal tubule (NCC)
   D) In the collecting duct
   E) In the glomerulus

11. Which substance do the kidneys form to bind H^+ when excreting acids?
   A) Ammonia (excreted as NH_4^+)
   B) Urea
   C) Creatinine
   D) Glucose
   E) Albumin

12. What is the normal creatinine clearance?
   A) 10–20 ml/min
   B) 30–40 ml/min
   C) 200–300 ml/min
   D) 90–130 ml/min
   E) 1.5 l/min

13. Where is the center of the micturition reflex?
   A) In the sacral segments of the spinal cord (S_2–S_4)
   B) In the medulla oblongata
   C) In the hypothalamus
   D) In the cerebellum
   E) In the cervical segments of the spinal cord

14. What does isosthenuria in the Zimnitsky test indicate?
    A) Kidney function is normal
    B) A lot of fluid has been drunk
    C) There is only glucosuria
    D) Blood glucose is increased
    E) The concentrating ability of the kidney is impaired

15. Which is not an effect of angiotensin II?
    A) It constricts arterioles
    B) It enhances aldosterone release
    C) It enhances natriuresis
    D) It causes thirst
    E) It enhances vasopressin release

Clinical cases

Case 1

After a head injury a patient began to excrete 8–10 l of urine per day with a specific gravity of 1.001–1.004, and intense thirst. Blood glucose is normal, sodium slightly increased.

Questions: What is the condition? Explain its mechanism.

Answer: Central diabetes insipidus. The injury has damaged the hypothalamus–neurohypophysis system and vasopressin release has decreased. Without vasopressin, aquaporin-2 channels are not inserted into the membrane of the collecting ducts and water is not reabsorbed into the hypertonic medulla — a large volume of dilute urine is excreted (polyuria, hyposthenuria). The loss of water raises plasma osmolarity and Na^+ — the osmoreceptors and the thirst center are excited (polydipsia). Glucose is normal — it is not diabetes mellitus. Treatment — a synthetic vasopressin analogue (desmopressin).

Case 2

After a car accident with a blood loss of 1.5 l, a patient has an arterial pressure of 80/50 mm Hg and hourly urine output of 15 ml (normal ≥ 0.5 ml/kg/hour).

Questions: Why has urine output decreased sharply? What compensatory mechanisms have been activated?

Answer: Arterial pressure has fallen below the limit of renal autoregulation (70 mm Hg), and the sympathetic nerves have constricted the afferent arterioles — the hydrostatic pressure in the glomerular capillaries and the effective filtration pressure have decreased sharply — GFR has fallen (oliguria). Compensatory mechanisms: sympathetic activation (tachycardia, vasoconstriction); the renin–angiotensin–aldosterone system (angiotensin II constricts vessels and, by constricting the efferent arteriole, partly maintains GFR; aldosterone retains Na^+ and water); vasopressin release (through the baroreceptors in response to reduced blood volume) — water reabsorption increases, and urine decreases and becomes concentrated. These mechanisms are aimed at preserving blood volume; prolonged hypotension can lead to acute kidney injury.

Case 3

A patient with diabetes mellitus has blood glucose of 14 mmol/l, urine glucose of 2%, daily diuresis of 3 l and thirst.

Questions: Explain the mechanism of the glucosuria and polyuria.

Answer: In the proximal tubule glucose is reabsorbed through the SGLT2 and SGLT1 transporters, whose number is limited (the transport maximum). When blood glucose exceeds the renal threshold (8.9–10 mmol/l), the amount of filtered glucose exceeds the capacity of the transporters and the excess passes into the urine — hyperglycemic glucosuria. The glucose remaining in the tubules, as an osmotically active substance, retains water and reduces its reabsorption — osmotic diuresis (polyuria). The loss of water causes thirst (polydipsia).

Quick review

Table 22.5. *Key concepts of topic 22*

Concept	Brief definition
Nephron	Renal corpuscle + tubules; ≈ 1 million in each kidney
Renal blood flow	≈ 1.2 l/min (20–25% of the minute volume)
Autoregulation	Blood flow and GFR constant at pressures of 70–180 mm Hg
Effective filtration pressure	60 − (32 + 18) ≈ 10 mm Hg
GFR	≈ 125 ml/min, 180 l/day
Proximal tubule	65% of Na^+ and water; glucose and amino acids 100%
Ascending limb of Henle	NaCl reabsorption (NKCC2), impermeable to water; furosemide
Collecting duct	Facultative reabsorption: vasopressin (water), aldosterone (Na^+)
Renal threshold	8.9–10 mmol/l for glucose
Concentration	Countercurrent system; up to 1200 mOsm/kg
RAAS	Renin → angiotensin II → aldosterone; raises pressure and volume
Natriuretic peptide	From the atria; natriuresis and diuresis
Acid excretion	H^+ secretion, titratable acidity, NH_4^+
Diuresis	1–1.5 l/day; specific gravity 1.010–1.025
Clearance	C = U × V / P; GFR by creatinine 90–130 ml/min

TOPIC 23

Properties of nerve centers. The spinal cord: spinal reflexes, conducting pathways, methods of examination

Learning objectives

To teach the student the reflex activity of the central nervous system, the concept of the nerve center and its properties, the transmission of excitation in central synapses, the types of inhibition in the CNS and the principles of reflex coordination, the reflex and conducting functions of the spinal cord, and spinal shock and the consequences of spinal cord injury; to master examining tendon reflexes in humans.

By the end of the class the student should know:

- the reflex and the reflex arc, monosynaptic and polysynaptic reflexes;
- the properties of the nerve center (one-way conduction, delay, summation, transformation, aftereffect, tone, fatigue, sensitivity to chemicals, plasticity);
- excitatory and inhibitory postsynaptic potentials;
- central inhibition (Sechenov's experiment), postsynaptic and presynaptic inhibition;
- the principles of reflex coordination (convergence, divergence, irradiation, reciprocal innervation, dominant, final common path, occlusion, facilitation);
- the segmental structure, reflexes and conducting pathways of the spinal cord;
- spinal shock and Brown-Séquard syndrome.

The student should be able to:

- determine the reflex time in a spinal frog and show the importance of an intact reflex arc;
- perform Sechenov's experiment;
- examine and evaluate the knee, Achilles, biceps, triceps and abdominal reflexes in humans.

1. Reflex activity of the central nervous system

A **reflex** is the body's response to stimulation of receptors with the participation of the CNS. The material basis of the reflex is the **reflex arc**: receptor → afferent (sensory) neuron → nerve center (interneurons) → efferent (motor) neuron → effector.

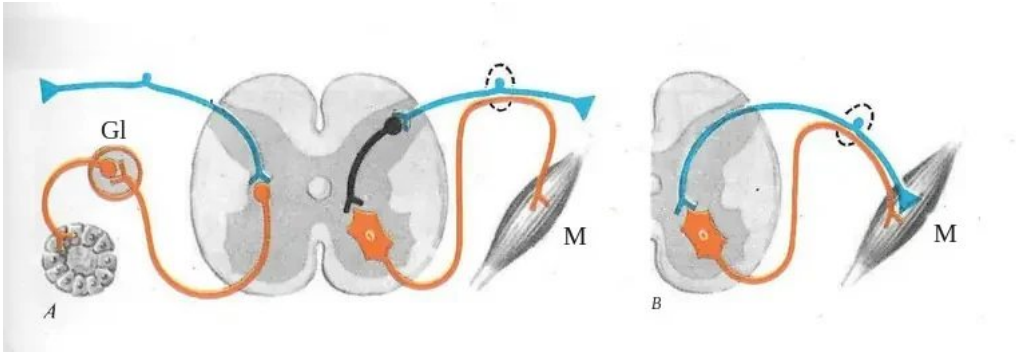


Figure 23.1. *Reflex arcs: A — polysynaptic (through an interneuron); B — monosynaptic (Alavia O.T., 2018)*

- **Monosynaptic** reflex arc — the afferent neuron ends directly on the motoneuron (one central synapse): **tendon (myotatic) reflexes** — knee, Achilles.

- **Polysynaptic** reflex arc — through one or more interneurons: flexor (protective), skin and autonomic reflexes. **Reflex time** is the time from the stimulus to the beginning of the response. It consists of the excitation of the receptors, the conduction of excitation along the afferent and efferent fibers, and the **central reflex time** (the delay at the synapses of the center). When the stimulus strength increases, the reflex time shortens (summation).

A **nerve center** is the set of neurons that carry out a particular reflex or function. Functionally, the center of a complex function is located at different levels of the CNS (for example, the respiratory center — spinal cord, medulla oblongata, pons, hypothalamus, cortex).

2. Central synapses: EPSP and IPSP

There are thousands of synapses on the surface of a central neuron. Depending on the type of mediator and receptor:

- **Excitatory postsynaptic potential (EPSP)** — **depolarization** of the postsynaptic membrane (Na^+ entry); the main mediator is **glutamate** (AMPA, NMDA receptors), and also acetylcholine. A single EPSP is small (1–2 mV); for a neuron to be excited, many EPSPs must summate.
- **Inhibitory postsynaptic potential (IPSP)** — **hyperpolarization** (Cl^- entry or K^+ efflux); mediators — **GABA** (the main one in the brain) and **glycine** (in the spinal cord and brainstem). The neuron integrates all EPSPs and IPSPs at the **axon hillock**: if the result reaches the critical level, an action potential is generated.

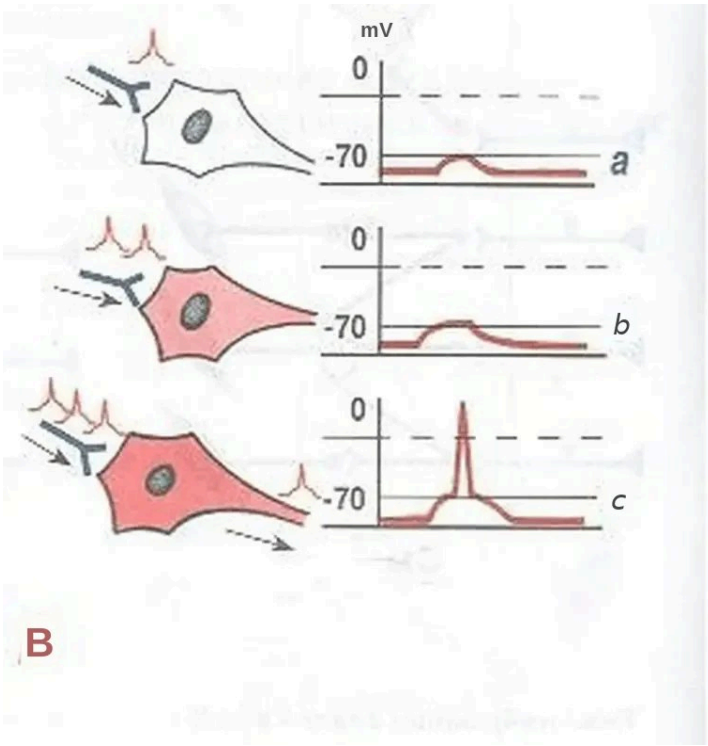


Figure 23.2. *Summation of postsynaptic potentials: a, b — subthreshold EPSPs with one and two impulses; c — appearance of an action potential with summation (Alavia O.T., 2018)*

3. Properties of nerve centers

Table 23.1. *Main properties of nerve centers*

Property	Content	Cause
One-way conduction	Excitation passes from the afferent to the efferent neuron	A property of chemical synapses

Property	Content	Cause
Synaptic (central) delay	Conduction of excitation slows in the center; 0.3–0.5 ms at each synapse	Release and diffusion of the mediator
Summation — temporal (successive)	Frequent subthreshold impulses from one afferent fiber add up	Addition of EPSPs
Summation — spatial	Impulses arriving simultaneously from several afferent fibers add up	Addition of EPSPs at different synapses
Transformation of rhythm	The frequency of impulses leaving the center differs from that entering it	Neuronal properties, afterpotentials
Aftereffect (reflex aftereffect)	The response continues after the stimulus has stopped	Afterdepolarization, circulation of impulses in loops (reverberation) — the basis of short-term memory
Tone	Constant weak activity even at rest	Continuous afferent input from receptors, humoral factors
Rapid fatigue	With prolonged stimulation the reflex weakens	Depletion of mediator at the synapses
High sensitivity to chemicals and hypoxia	Narcotics, poisons and lack of O_2 act rapidly	High metabolism of synapses and neurons (the brain consumes 20% of O_2)
Plasticity	Ability to reorganize and restore functions	New synaptic connections, changes in synaptic efficacy

CLINICAL SIGNIFICANCE

The **sensitivity of neurons to hypoxia** is very high: if blood supply to the brain stops, consciousness is lost within 10 s, and irreversible changes begin in cortical neurons within **4–6 minutes**. That is why resuscitation must start immediately after cardiac arrest. **Plasticity**, in turn, is the basis of partial recovery of functions after a stroke and of rehabilitation.

4. Inhibition in the central nervous system

Inhibition is an active nervous process that prevents the appearance or spread of excitation. It arises as a result of excitation and does not itself spread.

Central inhibition was discovered by I.M. Sechenov (1862): when the frog brain was transected at the level of the optic thalami and a crystal of NaCl was placed on the cut surface, the time of the withdrawal reflex on immersing the foot in sulfuric acid was sharply prolonged. Therefore the higher parts of the brain can inhibit spinal reflexes.

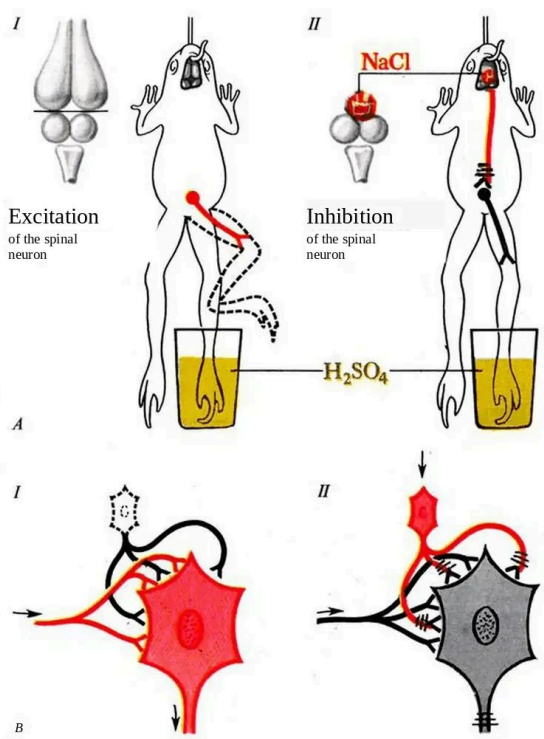


Figure 23.3. *A — Sechenov's experiment: when the optic thalami are stimulated with NaCl, the spinal reflex is inhibited; B — synaptic connections involving inhibitory neurons (Alavia O.T., 2018)*

Table 23.2. *Types of central inhibition*

Type	Mechanism	Example
Postsynaptic	An inhibitory interneuron evokes an IPSP (hyperpolarization) on the postsynaptic membrane; mediator GABA or glycine	Reciprocal inhibition, recurrent inhibition through Renshaw cells
Presynaptic	An axo-axonal synapse partially depolarizes the presynaptic terminal — mediator release decreases; mediator GABA	Filtering the flow of sensory information (in the dorsal horns of the spinal cord)
Recurrent	A collateral of the motoneuron axon excites a Renshaw cell, which inhibits the same motoneuron	Limiting the activity of motoneurons
Reciprocal	Inhibition of the center of the antagonist muscle	When the flexor contracts, the extensor relaxes
Pessimal	Prolonged depolarization of the neuron under very frequent impulses	Vvedensky's pessimum
Post-excitatory	Trace hyperpolarization after strong excitation	—

CLINICAL SIGNIFICANCE

Strychnine blocks glycine receptors and **tetanus toxin** blocks the release of glycine and GABA — postsynaptic inhibition in the spinal cord is lost: any weak stimulus causes generalized convulsions. Conversely, **benzodiazepines** and **barbiturates** enhance the action of $GABA_A$ receptors and have sedative and anticonvulsant effects.

5. Principles of reflex coordination

Table 23.3. *Principles of coordination*

Principle	Content
Convergence	Impulses arriving at one neuron from many neurons
Divergence	The axon of one neuron contacts many neurons — spread of excitation
Irradiation	Wide spread of excitation in the CNS with a strong stimulus (many muscles are recruited into the response)
Reciprocal (interrelated) innervation	When one center is excited, the antagonist center is inhibited (C. Sherrington, N.Ye. Vvedensky)
Final common path (Sherrington's “funnel”)	Afferent pathways outnumber efferent ones; different reflexes are carried out through the same motoneurons — there is competition for them
Facilitation (relief)	When two afferent inputs are stimulated simultaneously, the response is greater than the sum of their separate responses (subliminal fringe neurons)
Occlusion	When two inputs are stimulated simultaneously, the response is smaller than the sum (the common neurons are excited by both)
Dominant (A.A. Ukhtomsky)	A temporarily dominant focus of excitation in the CNS: it draws the excitation of other centers to itself, becomes stronger and inhibits competing reflexes
Reverse afferentation	Correction based on information about the result of the movement
Subordination	The higher divisions control the lower ones

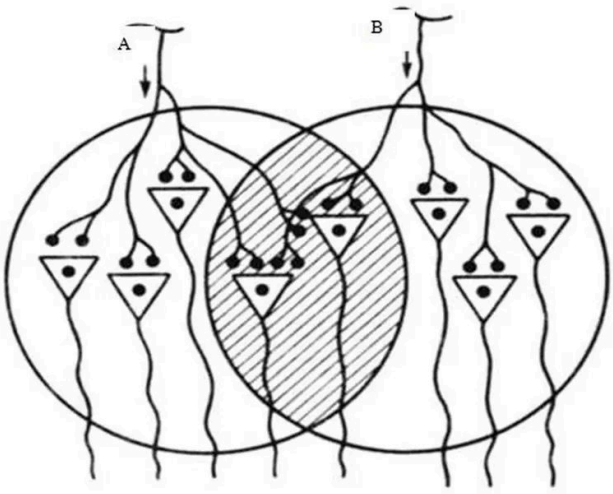


Figure 23.4. *Occlusion and facilitation: common neurons of inputs A and B (hatched area) (Alavia O.T., 2018)*

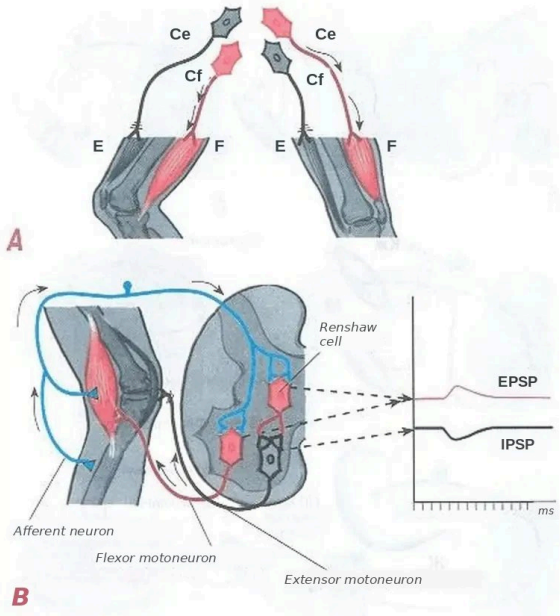


Figure 23.5. *A — reciprocal innervation of flexor and extensor muscles; B — reciprocal inhibition: EPSP in the extensor motoneuron and IPSP in the flexor motoneuron in the myotatic reflex (Alavia O.T., 2018)*

REMEMBER

Features of the dominant (Ukhtomsky): high excitability; persistence of excitation; summation of excitations (“drawing them to itself”); inertia (it persists after the stimulus has stopped). The dominant is the basis of attention, motivation (hunger, thirst) and behavior.

6. Structure and functions of the spinal cord

The spinal cord lies in the vertebral canal and consists of 31 segments (8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal). A pair of spinal nerves leaves each segment.

Bell–Magendie law: the dorsal roots consist of **sensory** (afferent) fibers, and the ventral roots of **motor** (efferent) fibers.

Gray matter: dorsal horns — sensory (inter)neurons; **ventral horns** — **α-motoneurons** (extrafusal fibers of skeletal muscles) and **γ-motoneurons** (muscle spindles); **lateral horns** (Th_1–L_2 — sympathetic, S_2–S_4 — parasympathetic preganglionic neurons). **White matter** — conducting pathways.

A **dermatome** is the area of skin receiving sensory innervation from one spinal segment; a **myotome** is the muscles innervated by one segment. One muscle usually receives innervation from several segments (the principle of reliability).

Reflexes of the spinal cord

Table 23.4. *Spinal reflexes*

Group	Examples	Features
Myotatic (stretch, tendon) reflexes	Knee, Achilles, biceps, triceps	Monosynaptic; the muscle spindles (Ia afferents) are stimulated; the basis of muscle tone and posture
Inverse myotatic (Golgi) reflex	Relaxation of a muscle when its tension becomes excessive	Golgi tendon organs (Ib afferents) — protect the muscle from rupture
Flexor (protective, nociceptive) reflexes	Withdrawal of the limb from a painful stimulus	Polysynaptic
Crossed extensor reflex	When one leg flexes, the opposite leg extends	Maintaining balance
Rhythmic reflexes	Stepping, scratching	Locomotor generators
Postural reflexes	Tonic neck reflexes (partly)	—
Autonomic reflexes	Micturition, defecation, erection, ejaculation, vascular and sweating reflexes	Lateral horns

Muscle spindles are intrafusal fibers inside the muscle that sense muscle stretch; their sensitivity is controlled by **γ-motoneurons** (the **γ-loop**). **Golgi tendon organs** sense muscle tension.

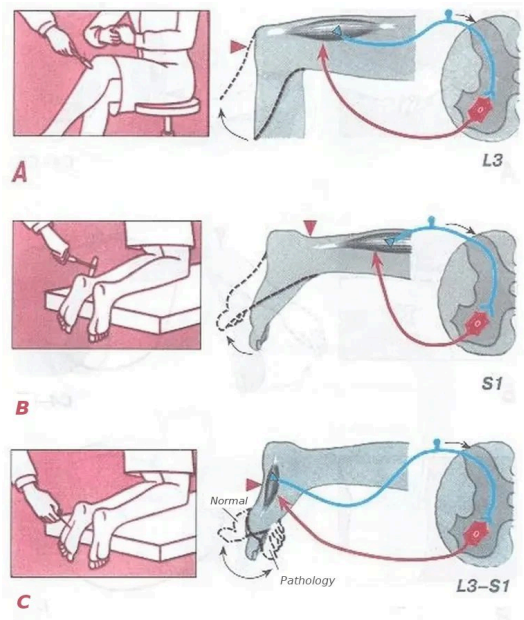


Figure 23.6. *Examination of tendon reflexes: A — knee (L_3–L_4), B — Achilles (S_1–S_2), C — plantar (L_5–S_1) reflexes (Alavia O.T., 2018)*

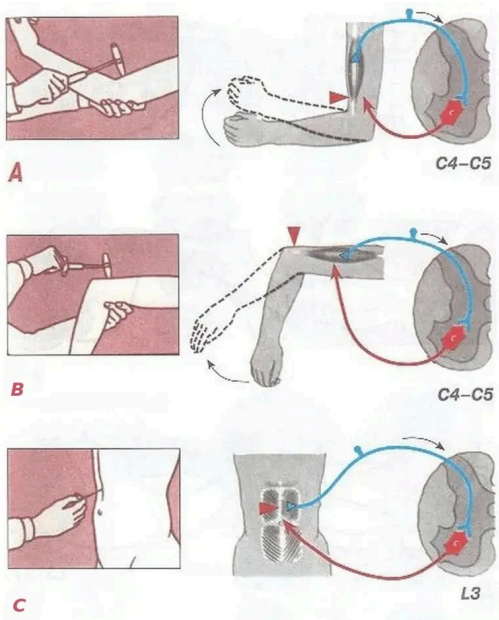


Figure 23.7. *A — biceps (C_5–C_6), B — triceps (C_7–C_8), C — abdominal (Th_7–Th_{12}) reflexes (Alavia O.T., 2018)*

CAUTION

Textbooks give the reflex segments with slight differences (for example, the knee reflex L_2–L_4 or L_3–L_4). The values most commonly used in clinical neurology are: biceps — C_5–C_6, triceps — C_7–C_8, abdominal — Th_7–Th_{12}, knee — L_3–L_4, Achilles — S_1–S_2.

Conducting pathways

Table 23.5. *Main conducting pathways of the spinal cord*

Route	Direction	What it conducts	Site of crossing
Fasciculus gracilis (Goll) and fasciculus cuneatus (Burdach) (dorsal columns)	Ascending	Proprioceptive (muscle-joint) sensation, fine touch, vibration	In the medulla oblongata (medial lemniscus)
Spinothalamic (lateral and anterior)	Ascending	Pain, temperature, crude touch	In the spinal cord, at the level of the entering segment (1–2 segments higher)
Spinocerebellar (Flechsig and Gowers)	Ascending	Unconscious proprioceptive information to the cerebellum	Partly
Corticospinal (pyramidal) — lateral and anterior	Descending	Voluntary movements (especially fine movements of the fingers)	80–90% in the medulla oblongata (pyramidal decussation)
Extrapyramidal (rubrospinal, reticulospinal, vestibulospinal, tectospinal)	Descending	Muscle tone, posture, automatic movements	Various

Spinal shock

Spinal shock is the temporary loss of all reflexes below the level of transection (or injury) of the spinal cord. Its cause is the sudden interruption of the activating influence of the higher centers (brainstem, cortex). Its duration depends on the evolutionary level of the animal: minutes in the frog,

days in the dog, weeks in the monkey, and **weeks–months** in humans. Later the reflexes recover and even become exaggerated (hyperreflexia), while voluntary movements and sensation do not recover.

> **CLINICAL SIGNIFICANCE**
>
> **Brown-Séquard syndrome** — in transverse damage to one half of the spinal cord: on the injured side — central paralysis (pyramidal tract) and loss of deep (proprioceptive) sensation (dorsal columns); on the opposite side — loss of pain and temperature sensation (the spinothalamic tract crosses in the spinal cord). It is a classic example for understanding where these pathways cross.

7. Laboratory work

Exercise 1: determining the reflex time in a spinal frog (Türck method)

1. Suspend a spinal frog (with the brain removed) by the jaw on a stand; wait 5–10 minutes for spinal shock to pass.
2. Dip the tip of the hind foot into 0.5% sulfuric acid solution and measure the time until the leg is withdrawn with a metronome or stopwatch.
3. Wash the foot with water after each measurement.
4. Repeat the experiment with 0.3% and 1% acid. **Result and analysis:** as the acid concentration increases, the reflex time shortens — a strong stimulus excites more receptors, and summation in the center reaches threshold faster.

Exercise 2: the importance of an intact reflex arc

1. In a spinal frog make a circular cut in the skin of the leg and remove the skin from the foot (the receptors are removed) — there is no reflex on dipping into acid.
2. Cut the sciatic nerve in the other leg — the reflex disappears.
3. Destroy the spinal cord (with a probe) — all reflexes disappear. **Result and analysis:** when any part of the reflex arc (receptor, afferent pathway, center, efferent pathway) is damaged, the reflex is not carried out.

Exercise 3: Sechenov's experiment

1. Transect the frog brain in front of the optic thalami (removing the cerebral hemispheres) (a “thalamic frog”).
2. Determine the reflex time by the Türck method.
3. Place a NaCl crystal on the cut surface of the optic thalami and measure the reflex time again.
4. Remove the crystal, wash the surface with Ringer's solution and measure again. **Result and analysis:** when the NaCl crystal is applied, the reflex time is sharply prolonged or the reflex disappears; after washing it recovers. Excitation of neurons in the region of the optic thalami inhibits spinal reflexes — proof of **central inhibition**.

Exercise 4: examining tendon reflexes in humans

1. **Knee reflex**: the subject sits and crosses one leg over the other (or lets the legs hang freely); the tendon below the kneecap is struck with a neurological hammer — contraction of the quadriceps and extension of the lower leg.
2. **Achilles reflex**: the subject kneels on a chair with the feet hanging freely; the Achilles tendon is struck — plantar flexion of the foot.

3. **Biceps reflex**: the arm is half-flexed at the elbow; the examiner places a thumb on the biceps tendon and strikes it — flexion at the elbow.
4. **Triceps reflex**: the arm is flexed at the elbow at a right angle; the triceps tendon is struck above the olecranon — extension.
5. **Abdominal reflexes**: with the subject lying down, the abdominal skin is stroked quickly from the side toward the center (upper, middle, lower) — contraction of the abdominal muscles.
6. Compare the reflexes on the two sides (symmetry); **the Jendrassik maneuver** — the subject hooks the hands together and pulls, which enhances the knee reflex (distraction of attention and activation of γ-motoneurons). **Interpretation:** normally the reflexes are moderately brisk and symmetrical. Reduction or absence — damage to the reflex arc (peripheral nerve, root, segment); exaggeration (hyperreflexia, clonus) — loss of the inhibitory influence of the higher centers (upper motor neuron lesion).

8. Practical skills

Skill: determining the level of spinal cord injury from the reflexes

1. Examine the biceps, triceps, abdominal, knee and Achilles reflexes.
2. Write down which reflexes are absent, reduced or exaggerated.
3. Determine the segment at which the reflexes are lost (the level of injury).
4. Exaggeration of the reflexes below (after spinal shock) shows that the influence of the higher centers has been interrupted.
5. Compare the boundary of the sensory disturbance with the dermatomes.

Skill: drawing a reflex arc

1. Draw the receptor (muscle spindle), the afferent neuron (in the spinal ganglion), the center (ventral horn), the efferent neuron (α-motoneuron) and the effector.
2. Distinguish monosynaptic and polysynaptic arcs.
3. Add the inhibitory interneuron going to the antagonist muscle (reciprocal inhibition).

Review questions

1. What are a reflex and a reflex arc?
2. What is the difference between monosynaptic and polysynaptic reflexes?
3. What does the reflex time consist of, and what affects it?
4. How are EPSPs and IPSPs generated?
5. List and explain the properties of nerve centers.
6. What are temporal and spatial summation?
7. Explain Sechenov's experiment.
8. What is the difference between postsynaptic and presynaptic inhibition?
9. Explain Renshaw cells and recurrent inhibition.
10. What is reciprocal innervation?
11. Explain the principles of the final common path, occlusion and facilitation.
12. Name the dominant and its features.
13. List the reflexes of the spinal cord.

14. What are the main conducting pathways of the spinal cord?
15. What are spinal shock and Brown-Séquard syndrome?

Test questions

1. Which is an example of a monosynaptic reflex?
   A) The flexor reflex
   B) Crossed extensor reflex
   C) The abdominal reflex
   D) The micturition reflex
   E) The knee reflex

2. How is an IPSP generated?
   A) Na^+ entry and depolarization
   B) Ca^{2+} entry
   C) Cl^- entry (or K^+ efflux) and hyperpolarization
   D) Under the action of glutamate
   E) Under the action of acetylcholine on N-cholinergic receptors

3. Which is the main inhibitory mediator in the spinal cord?
   A) Glutamate
   B) Glycine
   C) Acetylcholine
   D) Norepinephrine
   E) Dopamine

4. Who discovered central inhibition?
   A) C. Sherrington
   B) A.A. Ukhtomsky
   C) I.P. Pavlov
   D) N.E. Vvedensky
   E) I.M. Sechenov

5. What is the addition of frequent subthreshold impulses from one afferent fiber called?
   A) Spatial summation
   B) Occlusion
   C) Irradiation
   D) Temporal (successive) summation
   E) Dominant

6. What is the mechanism of presynaptic inhibition?
   A) Hyperpolarization of the postsynaptic membrane
   B) Depolarization of the presynaptic terminal through an axo-axonal synapse and a decrease in mediator release
   C) An increase in mediator
   D) An increase in the number of receptors
   E) Opening of Na^+ channels

7. What kind of inhibition do Renshaw cells provide?
   A) Presynaptic
   B) Pessimal
   C) Recurrent postsynaptic
   D) Conditioned
   E) External

8. What is the inhibition of the extensor center when the flexor center is excited called?
   A) Occlusion
   B) Dominant
   C) Convergence
   D) Reciprocal innervation
   E) Irradiation

9. Who is the author of the doctrine of the dominant?
   A) A.A. Ukhtomsky
   B) I.M. Sechenov
   C) C. Sherrington
   D) P.K. Anokhin
   E) I.P. Pavlov

10. According to the Bell–Magendie law, what fibers do the dorsal roots consist of?
    A) Motor
    B) Autonomic efferent
    C) Mixed, in equal proportion
    D) Sensory (afferent)
    E) Only sympathetic

11. Which pathway conducts pain and temperature sensation?
    A) The fasciculi of Goll and Burdach
    B) The spinothalamic tract
    C) The corticospinal tract
    D) The spinocerebellar tract
    E) The rubrospinal tract

12. Through which segments does the arc of the Achilles reflex close?
    A) S_1–S_2
    B) C_5–C_6
    C) L_3–L_4
    D) Th_7–Th_{12}
    E) C_7–C_8

13. What is the cause of spinal shock?
    A) Increased blood flow in the spinal cord
    B) Paralysis of the muscles
    C) Sudden interruption of the activating influence of the higher centers
    D) Damage to the receptors
    E) An increase in mediators

14. What is the cause of convulsions in strychnine poisoning?
    A) Blockade of glutamate receptors
    B) An increase in acetylcholine
    C) Blockade of Na^+ channels
    D) Enhanced GABA synthesis
    E) Blockade of glycine receptors and loss of postsynaptic inhibition

15. What is observed on the injured side in Brown-Séquard syndrome?
    A) Motor paralysis and loss of deep sensation
    B) Only loss of pain sensation
    C) Only loss of temperature sensation
    D) No change
    E) Complete bilateral paralysis

Clinical cases

Case 1

After a motorcycle accident a patient has a complete spinal cord injury in the thoracic region (Th_{10}). In the first weeks there were no movements, sensation or reflexes in the legs, and urinary retention was observed. After 2 months the knee and Achilles reflexes returned and even became exaggerated, but voluntary movements and sensation did not recover.

Questions: What is the condition in the first weeks called? Why did the reflexes later return and become exaggerated?

Answer: In the first weeks spinal shock developed — the spinal neurons below the injury were suddenly deprived of the constant activating impulses from the higher centers (brainstem, cortex), and their excitability fell sharply. Over time the excitability of the lower neurons recovers (increased receptor sensitivity, new synaptic connections — plasticity); the arcs of the knee (L_3–L_4) and Achilles (S_1–S_2) reflexes lie below the injury and are intact — the reflexes recover. Because the inhibitory influence of the higher centers is absent, they become exaggerated (hyperreflexia). Voluntary movement and sensation do not recover because the descending and ascending pathways needed for them have been severed.

Case 2

After a stab wound a patient was found to have paralysis and loss of joint-position sensation in the right leg, and loss of pain and temperature sensation in the left leg.

Questions: Where and how is the spinal cord damaged? Explain the signs on the basis of the conducting pathways.

Answer: The right half of the spinal cord has been damaged transversely (Brown-Séquard syndrome). The corticospinal (pyramidal) tract crosses in the medulla oblongata and descends in the spinal cord on its own side — paralysis on the injured (right) side. The dorsal columns (Goll and Burdach), which conduct proprioceptive sensation, do not cross in the spinal cord — deep sensation is lost on the right side. Pain and temperature sensation are conducted by the spinothalamic tract, which crosses in the spinal cord at the level of the entering segment — therefore pain and temperature sensation are lost on the opposite (left) side.

Case 3

During a neurological examination the patient's knee reflex was very weak on both sides. The doctor asked the patient to hook his hands together in front of him and pull hard, and the reflex became clearly visible.

Questions: What is this maneuver called? Explain its mechanism.

Answer: The Jendrassik maneuver. While pulling the hands hard, the patient's attention is distracted from the leg being tested (the voluntary inhibitory influence of the cortex decreases); in addition, excitation irradiates in the CNS, and spinal motoneurons and γ-motoneurons are activated — the sensitivity of the muscle spindles increases. As a result the knee reflex (a monosynaptic myotatic reflex) is enhanced. The maneuver helps to determine whether the reflex is truly reduced or only appeared weak because of the examination conditions.

Quick review

Table 23.6. *Key concepts of topic 23*

Concept	Brief definition
Reflex arc	Receptor → afferent → center → efferent → effector
Monosynaptic reflex	Tendon (myotatic) reflexes
EPSP	Depolarization; glutamate
IPSP	Hyperpolarization; GABA, glycine
Properties of nerve centers	One-way conduction, delay, summation, aftereffect, tone, fatigue
Sechenov's experiment	NaCl on the optic thalami — the spinal reflex is inhibited
Presynaptic inhibition	Axo-axonal synapse; mediator release decreases
Renshaw cells	Recurrent inhibition
Reciprocal innervation	The antagonist center is inhibited
Dominant	Dominant focus of excitation (Ukhtomsky)
Bell–Magendie law	Dorsal root — sensory, ventral — motor
The spinothalamic tract	Pain and temperature; crosses in the spinal cord
The corticospinal tract	Voluntary movement; crosses in the medulla oblongata
Reflex levels	Biceps C_5–C_6, triceps C_7–C_8, knee L_3–L_4, Achilles S_1–S_2
Spinal shock	Temporary loss of reflexes below the level of transection

TOPIC 24

Activity of the different parts of the brain: the medulla oblongata, the pons, the midbrain

Learning objectives

To teach the student the structural features, reflex and conducting functions and vital centers of the brainstem (medulla oblongata, pons, midbrain), its role in regulating muscle tone and posture (decerebrate rigidity, static and statokinetic reflexes), and the descending and ascending influences of the reticular formation; in laboratory work, to master examining brainstem reflexes and cranial nerve function.

By the end of the class the student should know:

- the composition of the brainstem and the location of the cranial nerve nuclei;
- the vital centers and reflexes of the medulla oblongata;
- the functions of the pons;
- the functions of the midbrain nuclei (corpora quadrigemina, red nucleus, substantia nigra, nuclei of cranial nerves III–IV);
- decerebrate rigidity and its mechanism;
- static (postural and righting) and statokinetic reflexes (R. Magnus);
- the structure of the reticular formation and its activating and inhibitory influences.

The student should be able to:

- observe brainstem reflexes in the frog and rat (video);
- examine the pupillary, oculocardiac, swallowing and blink (corneal) reflexes and cranial nerve function in humans;
- demonstrate statokinetic reflexes (nystagmus).

1. General description of the brainstem

The **brainstem** comprises the medulla oblongata, the pons and the midbrain (some authors also include the diencephalon). It connects the spinal cord with the higher parts of the brain and performs the following functions:

1. **Conducting** — all ascending and descending pathways pass through it.
2. **Reflex** — the nuclei of cranial nerves III–XII are located here; centers of vital reflexes.
3. **Regulation of muscle tone and posture** — through the vestibular, reticular and red nucleus systems.
4. **Integrative** — regulation of the general activity of the CNS through the reticular formation (sleep–wakefulness).

2. The medulla oblongata

The medulla oblongata contains the nuclei of cranial nerves **IX–XII** (glossopharyngeal, vagus, accessory, hypoglossal), as well as part of the nuclei of nerves V and VIII.

Table 24.1. *Centers and reflexes of the medulla oblongata*

Group	Center or reflex
Vital centers	The respiratory center (dorsal and ventral groups), the circulatory (vasomotor) center, the center regulating cardiac activity (vagal nuclei)
Digestive reflexes	Sucking, chewing, swallowing, salivation, gastric and pancreatic secretion, vomiting
Protective reflexes	Coughing, sneezing, lacrimation, blinking (corneal — often together with the pons), vomiting
Postural reflexes	Tonic neck and labyrinthine reflexes (vestibular nuclei)
Others	Sweating, vascular tone reflexes

REMEMBER

Damage to the medulla oblongata (injury, stroke, herniation of the brain into the foramen magnum) is **extremely dangerous to life**: the respiratory and circulatory centers are damaged — breathing stops and arterial pressure falls sharply. That is why the medulla oblongata is called the **"vital knot"** (Flourens).

Conducting function: in the medulla oblongata 80–90% of the **pyramidal tracts** cross (the pyramidal decussation); in the gracile and cuneate nuclei the proprioceptive pathways synapse on the second neuron and cross as the **medial lemniscus**.

CLINICAL SIGNIFICANCE

Bulbar syndrome — when the nuclei or nerves of cranial nerves IX, X and XII are damaged: impaired swallowing (dysphagia, return of liquid food through the nose), hoarseness (dysphonia), speech disturbance (dysarthria), atrophy and fibrillation of the tongue, loss of the pharyngeal and soft palate reflexes. The risk of aspiration pneumonia is high.

3. The pons

The pons contains the nuclei of cranial nerves **V–VIII** (trigeminal, abducens, facial, vestibulocochlear).

Functions:

- **Conducting** — pathways from the cortex to the cerebellum (the **cortico-ponto-cerebellar pathway**; pontine nuclei), ascending and descending pathways.
- **Reflex** — chewing (nerve V), facial expression (nerve VII), turning the eyeball outward (nerve VI), the corneal reflex (afferent V, efferent VII), primary processing of auditory and vestibular information (nuclei of nerve VIII).
- **Regulation of breathing** — the **pneumotaxic center** (controls the switch from inspiration to expiration).
- **Regulation of sleep** — the locus coeruleus (noradrenergic neurons) and the mechanisms of paradoxical (REM) sleep.

4. The midbrain

The midbrain consists of the corpora quadrigemina (tectum), the cerebral peduncles, the red nucleus, the substantia nigra, the periaqueductal gray matter around the aqueduct of Sylvius and the nuclei of cranial nerves III–IV.

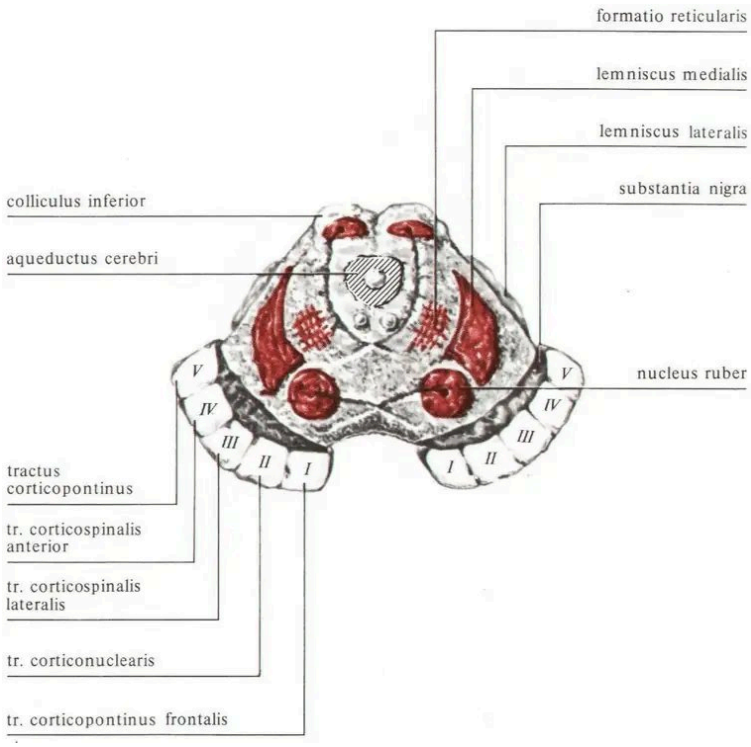


Figure 24.1. *Cross-section of the midbrain: superior and inferior colliculi, red nucleus, substantia nigra, reticular formation, conducting pathways (Alavia O.T., 2018)*

Table 24.2. *Functions of the midbrain structures*

Structure	Function
Superior colliculi of the corpora quadrigemina	The primary visual center: turning the eyes and head toward light and moving objects (the orienting reflex — the “what is it?” reflex), coordination of eye movements
Inferior colliculi	The primary auditory center: turning the head and ears toward a sound (the auditory orienting reflex), the startle reflex
Red nucleus	Regulation of muscle tone — inhibits extensor tone and enhances flexor tone (rubrospinal tract); righting reflexes
Substantia nigra	Dopaminergic neurons — control of movement through the basal ganglia; coordination of chewing and swallowing movements
Nuclei of cranial nerves III and IV	Movement of the eyeball, constriction of the pupil (the Yakubovich–Edinger–Westphal nucleus — parasympathetic), accommodation
Periaqueductal gray matter	Pain control (the antinociceptive system — endogenous opioids), defensive behavior

CLINICAL SIGNIFICANCE

Parkinson's disease is the death of the dopaminergic neurons of the substantia nigra: slowness of movement (bradykinesia), muscle rigidity (the “cogwheel” phenomenon), resting tremor and postural instability. Treatment — the dopamine precursor **levodopa** (dopamine itself does not cross the blood–brain barrier), dopamine agonists.

The pupillary reflex: when light enters the eye, the pupil constricts (direct reaction), and the pupil of the other eye constricts as well (**crossed (consensual) reaction**). Pathway: retina → optic nerve → pretectal nucleus (midbrain) → Yakubovich–Edinger–Westphal nuclei on both sides → nerve III (parasympathetic) → ciliary ganglion → pupillary constrictor muscle.

5. The role of the brainstem in regulating muscle tone

Decerebrate rigidity

When an animal's brainstem is transected **between the midbrain (red nucleus) and the vestibular nuclei** (between the colliculi), **decerebrate rigidity** develops (C. Sherrington): the **tone of all the extensor (antigravity) muscles rises sharply** — the legs are stretched straight, the head is thrown back and the tail raised; if the animal is set on its feet it stands like a “wooden” figure.

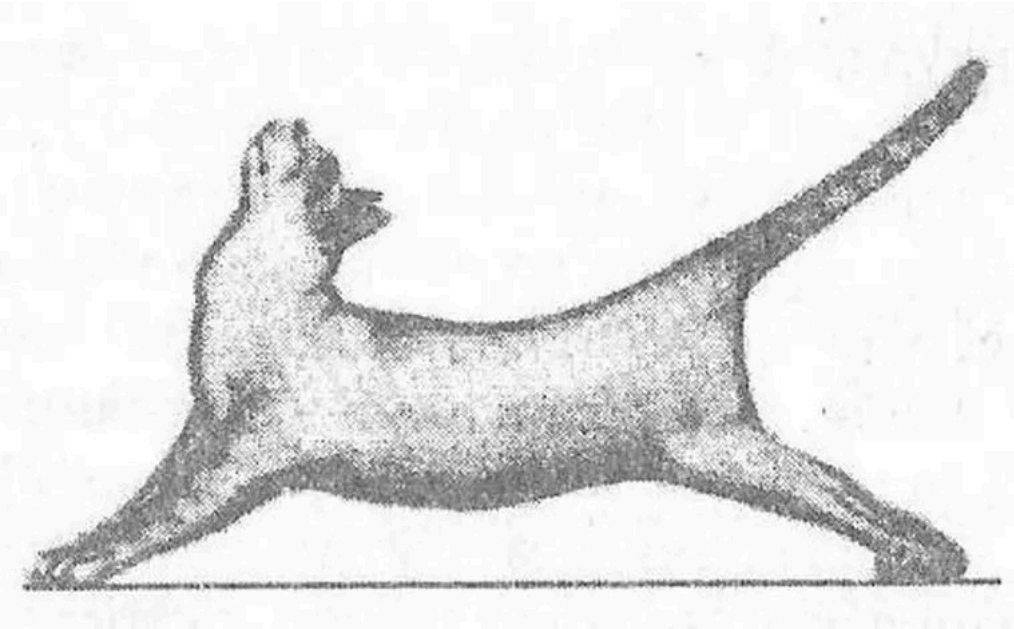


Figure 24.2. *Decerebrate rigidity — a sharp increase in extensor muscle tone in a cat (Alavia O.T., 2018)*

Mechanism: the transection interrupts the **inhibitory** influence of the red nucleus and the higher centers (cortex, basal ganglia) on the extensor muscles, while the **activating** influence of the vestibular nuclei (**Deiters' nucleus**) and the pontine reticular formation on the extensor motoneurons is preserved and becomes dominant. The rigidity is mediated mainly by activation of **γ-motoneurons** (γ-rigidity): if the dorsal roots are cut (afferent input from the muscle spindles is lost), the rigidity disappears. If the transection is made below the vestibular nuclei, rigidity does not develop.

> **CLINICAL SIGNIFICANCE**
> In humans, in severe brain injuries, stroke and brainstem compression, a **decerebrate posture** (arms and legs extended, forearms rotated inward) and a **decorticate posture** (arms flexed, legs extended — injury above the red nucleus) are observed. The decerebrate posture is considered a prognostically worse sign (the motor response is scored on the Glasgow scale).

Static and statokinetic reflexes (R. Magnus)

Table 24.3. *Postural reflexes of the brainstem*

Group	Reflex	Receptors	Center	Content
Static — postural reflexes	Tonic neck reflexes	Proprioceptors of the neck muscles and joints	Medulla oblongata	When the head is thrown back, the arms extend and the legs flex; when it is bent forward — the reverse; when the head is turned to one side, the limbs on that side extend

Group	Reflex	Receptors	Center	Content
Static — postural reflexes	Tonic labyrinthine reflexes	Vestibular (otolith) apparatus	Medulla oblongata (vestibular nuclei)	Change in extensor muscle tone depending on the position of the head in space
Static — righting reflexes	Righting of the head and trunk	Labyrinthine, skin, visual and neck proprioceptors	Midbrain (red nucleus)	When the animal is laid on its side, first its head and then its trunk return to the upright position
Statokinetic reflexes	Elevator reflex	Otolith apparatus (linear acceleration)	Midbrain	On descending the legs extend, on ascending they flex
Statokinetic reflexes	Response to rotation, eye and head nystagmus	Semicircular canals (angular acceleration)	Brainstem	During rotation, slow opposite movement of the head and eyes followed by a quick return
Statokinetic reflexes	Righting during a fall (a cat landing on its feet)	Labyrinth, vision	Midbrain	Turning in the air and landing on the feet

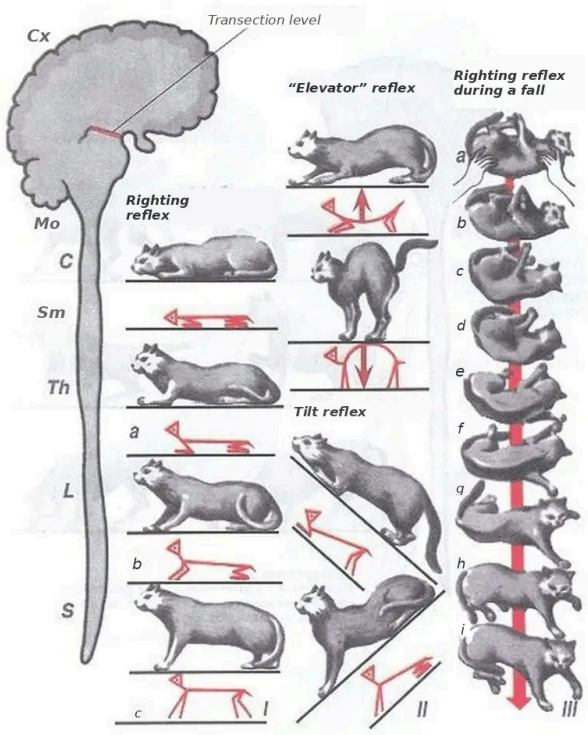


Figure 24.3. *Righting reflexes in the rat (I), the "elevator" reflex and postural reflex on a slope (II), the righting reflex during a fall (III) (Alavia O.T., 2018)*

MEMORY AID

Remembering the tonic neck reflexes: **"Look up at the sky — the arms extend"** (head back — the arms extend and the legs flex: a cat looking up at a shelf), **"Look down — the arms flex"** (a cat looking down at a mouse hole).

6. The reticular formation

The **reticular formation (RF)** is a diffuse network of neurons with highly branched dendrites and axons located throughout the brainstem (from the upper spinal cord to the diencephalon).

Its neurons receive information from all sensory systems (via collaterals), but this information is **nonspecific** (it reflects the strength rather than the type of stimulus).

The ascending activating system (H. Magoun, G. Moruzzi, 1949): the RF diffusely activates the cerebral cortex — it maintains **wakefulness**, attention and consciousness. When the RF is stimulated, a sleeping animal wakes up and on the EEG the slow rhythms are replaced by fast ones (desynchronization); when the RF is damaged (or the brain is transected below the midbrain), the animal falls into a state resembling permanent sleep (coma).

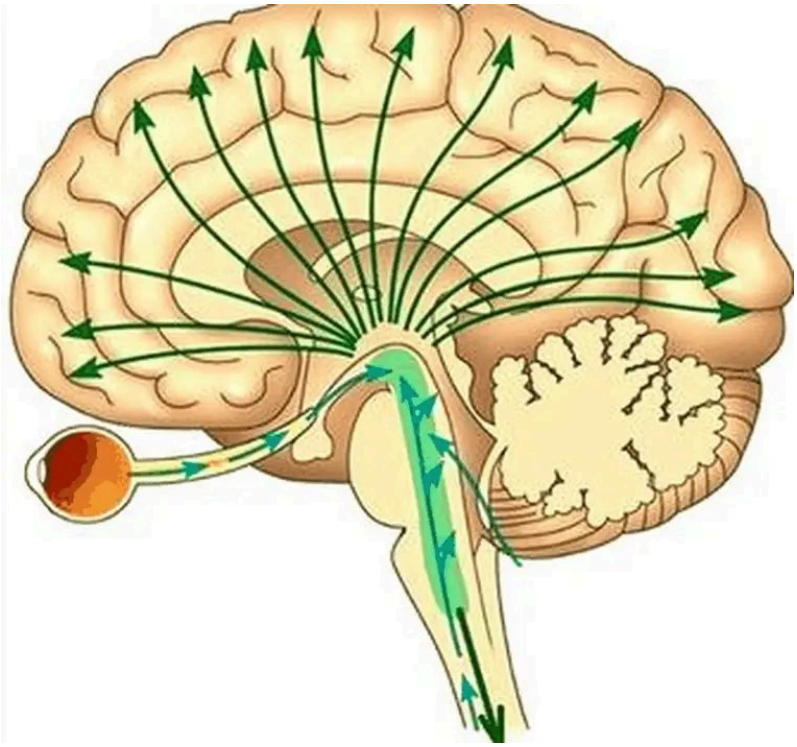


Figure 24.4. *The ascending activating influence of the reticular formation: collaterals of the sensory pathways to the RF, and diffuse impulses from the RF to the cortex (Alavia O.T., 2018)*

Descending influences (Magoun): the **medial (medullary) part** of the RF **inhibits** the activity of spinal motoneurons, while the **lateral (pontine and midbrain) part facilitates** (activates) it. These influences act mainly through the γ-motoneurons and regulate muscle tone and reflexes.

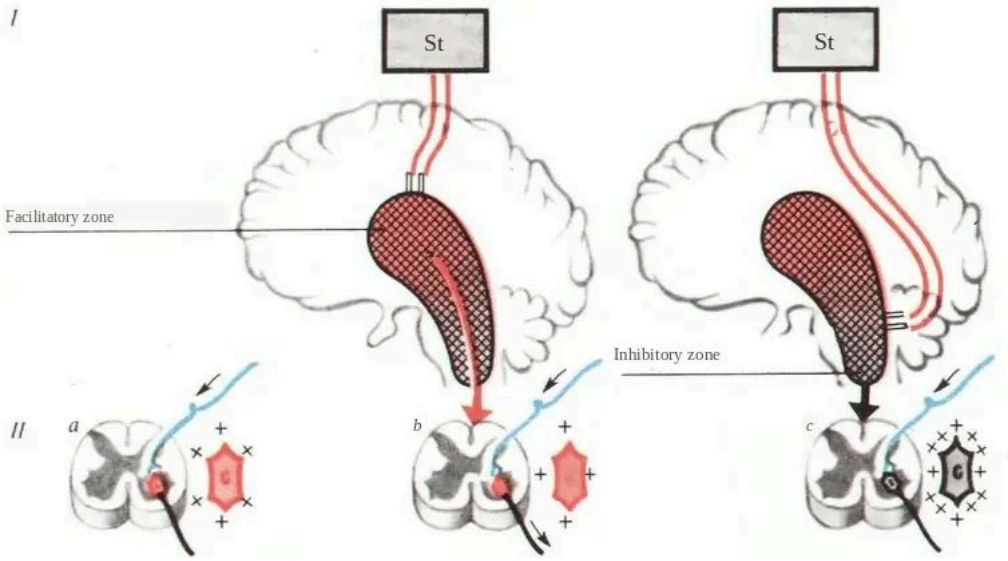


Figure 24.5. *Descending facilitatory (b) and inhibitory (c) influences of the reticular formation: St — stimulator; a — initial state (Alavia O.T., 2018)*

Other functions of the RF: participation in the respiratory and circulatory centers; regulation of autonomic functions; regulation of the sleep–wake cycle (noradrenergic — locus coeruleus, serotonergic — raphe nuclei); modulation of pain; attention and orienting reactions.

> **CLINICAL SIGNIFICANCE**
> **Barbiturates** and general anesthetics cause sleep and narcosis mainly by inhibiting the ascending activating system of the RF. Damage to the brainstem RF (injury, stroke, tumors) leads to impaired consciousness — **stupor and coma**.

7. Laboratory work

Exercise 1: observing brainstem reflexes in the frog

1. Observe a normal frog: its sitting posture at rest, jumping, swimming, righting when turned on its back, and the “croaking” reflex when the abdomen is lightly pressed.
2. Test the same reflexes in a frog with the cerebral hemispheres removed (the midbrain and lower parts preserved).
3. Test them in a frog transected above the medulla oblongata.
4. Test them in a spinal frog. **Result and analysis:** in the frog with the midbrain preserved, the postural and righting reflexes are retained (the center of these reflexes is in the midbrain); in the medullary frog the righting reflexes are lost, but breathing and some tonic reflexes are preserved; in the spinal frog only segmental reflexes remain, with no posture or breathing.

Exercise 2: examining brainstem reflexes and cranial nerves in humans

1. **Pupillary reflex** (nerve III, midbrain): cover the eye with the palm for a few seconds, then uncover it and observe the constriction of the pupil; check the crossed reaction in the other eye.
2. **Corneal (blink) reflex** (nerves V–VII, pons): when the cornea is touched carefully with the tip of a cotton wisp, the eye closes (instructor demonstration).
3. **Swallowing and pharyngeal reflex** (nerves IX–X, medulla oblongata): touching the posterior pharyngeal wall with a spatula — a gag movement.
4. **Oculocardiac (Dagnini–Aschner) reflex** (nerves V–X): light pressure on the eyeballs — slowing of the pulse (topic 12).
5. **Eye movements** (nerves III, IV, VI): the subject follows a hammer in all directions with the eyes.
6. **Facial muscles** (nerve VII): wrinkling the forehead, closing the eyes tightly, showing the teeth; **masticatory muscles** (nerve V): clenching the jaws.
7. **Protruding the tongue** (nerve XII): the tongue should be in the midline. **Interpretation:** the reflexes should be equal and brisk on both sides; asymmetry or absence indicates damage to the corresponding cranial nerve or the brainstem.

Exercise 3: demonstrating statokinetic reflexes (Bárány chair)

1. The subject sits in a rotating chair with the head bent 30° forward and the eyes closed.
2. He is rotated evenly 10 times over 20 s and then stopped suddenly.
3. The subject opens his eyes and looks ahead — rhythmic movements of the eyeballs (**post-rotational nystagmus**) are observed; their duration is measured with a stopwatch.

Interpretation: normally post-rotational nystagmus lasts 25–35 s. Prolonged nystagmus or nausea and pallor indicate increased excitability of the vestibular apparatus (in occupational selection — pilots, sailors).

8. Practical skills

Skill: evaluating the pupillary reaction

1. Darken the room slightly.
2. Direct a flashlight beam at each eye from the side.
3. Observe the direct and crossed reactions.
4. Assess the equality of pupil size (absence of anisocoria).
5. If the reaction is absent or sluggish, consider damage to nerve III or the midbrain, or potent drugs (opioids — the pupil constricts; atropine — it dilates).

Skill: estimating the level of brainstem damage from the syndromes

1. Disorders of breathing and circulation, impaired swallowing and speech — the medulla oblongata.
2. Paralysis of the facial muscles, inability to turn the eye outward, impaired chewing — the pons.
3. Absence of the pupillary reaction, inability to turn the eye up and in — the midbrain.
4. Impaired consciousness — the reticular formation.
5. If the cranial nerve disturbance is on one side and the paralysis on the opposite side — an **alternating (crossed) syndrome** (unilateral damage to the brainstem).

Review questions

1. Which parts make up the brainstem, and what are its functions?
2. Which cranial nerve nuclei are located in the medulla oblongata?
3. List the vital centers of the medulla oblongata.
4. Name the protective and digestive reflexes of the medulla oblongata.
5. What is bulbar syndrome?
6. Explain the functions of the pons.
7. What functions do the superior and inferior colliculi of the midbrain perform?
8. What are the functions of the red nucleus and the substantia nigra?
9. Explain the pathway of the pupillary reflex.
10. How does decerebrate rigidity arise, and what is its mechanism?
11. Explain the tonic neck and labyrinthine reflexes.
12. Where is the center of the righting reflexes?
13. Give examples of statokinetic reflexes.
14. What is the ascending activating influence of the reticular formation?
15. What are the descending influences of the reticular formation?

Test questions

1. Which part is called the “vital knot”?
   A) Midbrain
   B) The cerebellum
   C) The diencephalon
   D) The pons
   E) Medulla oblongata

2. Which center is located in the medulla oblongata?
   A) The respiratory center
   B) The thermoregulatory center
   C) The hunger center
   D) The cortical visual center
   E) The speech center

3. Where is the pneumotaxic center located?
   A) In the medulla oblongata
   B) In the hypothalamus
   C) In the cerebellum
   D) In the spinal cord
   E) In the pons

4. What function do the superior colliculi of the midbrain perform?
   A) Primary auditory center
   B) Regulation of breathing
   C) Primary visual center — visual orienting reflexes
   D) Regulation of chewing
   E) Regulation of swallowing

5. In Parkinson's disease, the neurons of which structure die?
   A) Red nucleus
   B) The locus coeruleus
   C) The vestibular nuclei
   D) The substantia nigra (dopaminergic neurons)
   E) Inferior colliculi

6. With which transection does decerebrate rigidity develop?
   A) Between the spinal cord and the medulla oblongata
   B) Between the midbrain (red nucleus) and the vestibular nuclei
   C) Above the diencephalon
   D) On removal of the cerebral hemispheres
   E) On removal of the cerebellum

7. In decerebrate rigidity, the tone of which muscles rises sharply?
   A) Flexor muscles
   B) Only the arm muscles
   C) Only the respiratory muscles
   D) Extensor (antigravity) muscles
   E) Facial muscles

8. How does the red nucleus affect muscle tone?
   A) It enhances extensor tone
   B) It inhibits extensor tone and enhances flexor tone
   C) They have no effect
   D) It acts only on the eye muscles
   E) It raises the tone of all muscles

9. Where is the parasympathetic nucleus that constricts the pupil located?
   A) In the midbrain (the Yakubovich–Edinger–Westphal nucleus)
   B) In the medulla oblongata
   C) In the thoracic segments of the spinal cord
   D) In the pons
   E) In the hypothalamus

10. Which reflex is the righting, first of the head and then of the trunk, when an animal is laid on its side?
    A) Elevator reflex
    B) Tonic neck reflex
    C) Righting reflex
    D) Nystagmus
    E) Myotatic reflex

11. What does the ascending influence of the reticular formation provide?
    A) Activation of the cortex, wakefulness and attention
    B) It induces sleep
    C) Only muscle tone
    D) Only pain sensation
    E) Only vision

12. Stimulation of which receptors is post-rotational nystagmus associated with?
    A) The otolith apparatus
    B) Neck proprioceptors
    C) Visual receptors
    D) Skin receptors
    E) The semicircular canals

13. What are the afferent and efferent nerves of the corneal reflex?
    A) II and III
    B) V and VII
    C) IX and X
    D) VIII and VI
    E) XI and XII

14. Which receptors are excited in the elevator reflex?
    A) The semicircular canals
    B) The organ of Corti
    C) Otolith apparatus (linear acceleration)
    D) Visual receptors
    E) Muscle spindles

15. Which sign is characteristic of bulbar syndrome?
    A) Absence of the pupillary reaction
    B) Inability to turn the eye outward
    C) Increased muscle tone
    D) Impaired swallowing, voice and speech
    E) Loss of hearing

Clinical cases

Case 1

After a stroke a 65-year-old patient has difficulty swallowing, liquid comes back through the nose, the voice is hoarse and speech is unintelligible. When protruded, the tongue deviates to the left, and fibrillations are seen in the tongue.

Questions: Which part of the brain is damaged? Which cranial nerves are affected? What is the main danger for the patient?

Answer: The medulla oblongata is damaged — bulbar syndrome. Impaired swallowing and elevation of the soft palate (return of liquid through the nose) and hoarseness — damage to the nuclei of nerves IX (glossopharyngeal) and X (vagus); deviation and fibrillation of the tongue — damage to the nucleus of nerve XII (hypoglossal) (the tongue deviates toward the damaged side). The main dangers: entry of food and saliva into the airways (aspiration, aspiration pneumonia) and, if the process spreads, damage to the nearby respiratory and circulatory centers.

Case 2

A patient with a severe brain injury is unconscious; his arms and legs are stretched straight, the forearms rotated inward and the head thrown back, and this posture intensifies with painful stimuli.

Questions: What is this condition? Explain its mechanism.

Answer: Decerebrate posture (decerebrate rigidity). The injury is below the red nucleus and above the vestibular nuclei (at the midbrain–pons level). The inhibitory influence of the red nucleus and higher centers on the extensor muscles has been interrupted, while the activating influence of the vestibular (Deiters') nuclei and the pontine reticular formation is preserved and dominant — extensor muscle tone has risen sharply (mainly through the γ-motoneurons). This posture indicates serious brainstem damage and a poor prognosis; the unconsciousness is due to damage to the ascending activating system of the reticular formation.

Case 3

In a patient under general anesthesia, the anesthesiologist regularly checks the state of the pupils and their reaction to light.

Questions: What is the pathway of the pupillary reflex? What might its loss indicate?

Answer: Light excites the photoreceptors of the retina; impulses travel along the optic nerve to the pretectal nucleus in the midbrain and from there to the Yakubovich–Edinger–Westphal nuclei on both sides; parasympathetic fibers in cranial nerve III go to the ciliary ganglion and from there to the pupillary constrictor muscle — the pupil constricts (direct and crossed reaction). Because the reflex center is in the midbrain, its loss and dilation of the pupils may indicate deep depression of brainstem function (excessively deep anesthesia), severe hypoxia or damage to the midbrain. That is why the state of the pupils is an important indicator of the depth of anesthesia and the condition of the brain.

Quick review

Table 24.4. *Key concepts of topic 24*

Concept	Brief definition
Brainstem	Medulla oblongata, pons, midbrain
Medulla oblongata	Nuclei of nerves IX–XII; respiratory and circulatory centers — the “vital knot”
Protective reflexes	Coughing, sneezing, vomiting, lacrimation
The pons	Nuclei of nerves V–VIII; pneumotaxic center; cortico-ponto-cerebellar pathway
Superior colliculi	Visual orienting reflexes
Inferior colliculi	Auditory orienting reflexes
Red nucleus	Inhibits extensor tone; righting reflexes
Substantia nigra	Dopamine; Parkinson's disease
Pupillary reflex	Pretectal nucleus → Yakubovich–Edinger–Westphal → nerve III
Decerebrate rigidity	Transection between the red nucleus and Deiters' nucleus — extensor tone ↑
Tonic neck and labyrinthine reflexes	Postural reflexes; medulla oblongata
Righting reflexes	Midbrain (Magnus)
Statokinetic reflexes	Elevator reflex, nystagmus
Ascending influence of the RF	Activates the cortex — wakefulness (Magoun, Moruzzi)
Descending influence of the RF	Medial — inhibitory, lateral — facilitatory

TOPIC 25

Activity of the different parts of the brain: the cerebellum, diencephalon, basal ganglia and cerebral cortex

Learning objectives

To teach the student the structure and connections of the cerebellum and its role in coordinating movements, the functions of the diencephalon (thalamus and hypothalamus), the importance of the basal (subcortical) ganglia in controlling movement, the sensory, motor and associative areas of the cerebral cortex, the limbic system and electroencephalography; to master performing tests of cerebellar function in humans.

By the end of the class the student should know:

- the functional divisions of the cerebellum (archicerebellum, paleocerebellum, neocerebellum) and their connections;
- the signs of cerebellar damage (Luciani's triad, ataxia, asynergia, dysmetria, intention tremor, nystagmus);
- the types and functions of the thalamic nuclei;
- the autonomic, endocrine and motivational centers of the hypothalamus;
- the structure and connections of the basal ganglia and the syndromes of their damage;
- the cytoarchitectonics and functional areas of the cortex;
- the concept of the somatosensory and motor homunculus;
- the functions of the limbic system;
- the EEG rhythms and their significance.

The student should be able to:

- perform and evaluate the Romberg test, the finger-to-nose and heel-to-knee tests and the adiadochokinesis test;
- distinguish the main rhythms on an EEG recording.

1. The cerebellum

The cerebellum lies behind the pons and medulla oblongata; it consists of two hemispheres and the vermis. Its cortex has three layers (molecular, Purkinje cell and granular), and inside there are nuclei: the **fastigial**, **globose**, **emboliform** and **dentate** nuclei.

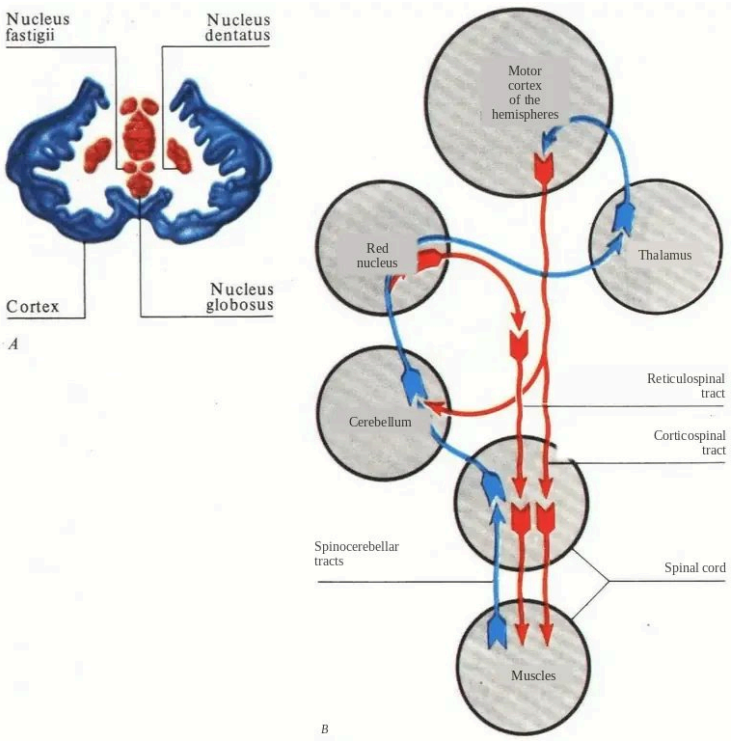


Figure 25.1. *Cerebellar nuclei (A) and the connections of the cerebellum in the control of movement (B) (Alavia O.T., 2018)*

The only output neurons of the cerebellar cortex are the Purkinje cells; they exert an **inhibitory** (GABA) influence on the cerebellar nuclei. Information reaches the cerebellum through **mossy** fibers (mainly from the spinal cord and pontine nuclei) and **climbing** fibers (from the inferior olivary nucleus).

Table 25.1. *Functional divisions of the cerebellum*

Branch	Structures	Main connections	Function
Vestibulocerebellum (archicerebellum)	Flocculonodular lobe	The vestibular nuclei	Balance, control of eye movements
Spinocerebellum (paleocerebellum)	The vermis and the paravermal part of the hemispheres	Spinal cord (spinocerebellar tracts), red nucleus, reticular formation	Muscle tone, posture and correction of ongoing movements
Cerebrocerebellum (neocerebellum)	The lateral part of the hemispheres, the dentate nucleus	Cortex (via the pons) → thalamus → motor cortex	Planning and programming of complex voluntary movements, motor memory, timing

Functions of the cerebellum: maintaining balance and posture; regulating muscle tone; coordinating voluntary movements (adjusting accuracy, smoothness, speed and force); motor learning (memory of motor skills); influence on autonomic functions.

When the cerebellum is removed (L. Luciani's experiments), three classic signs develop — **Luciani's triad**: **atonia** (reduced muscle tone), **asthenia** (reduced muscle strength and rapid fatigue), **astasia** (tremor-like muscle contractions — impaired standing). Later **ataxia** (incoordination of movements), **dysmetria**, **asynergia**, **intention tremor**, **dysarthria** (scanning speech) and **nystagmus** are added.

Table 25.2. *Signs of cerebellar damage*

Sign	Content
Static ataxia	Inability to keep balance when standing (swaying in the Romberg position, whether the eyes are open or closed)
Dynamic ataxia	Gait disturbance — walking with the legs wide apart, swaying (a “drunken gait”)
Dysmetria	Misjudging the amplitude of a movement (hyper- and hypometria)
Intention tremor	Tremor that increases as the target is approached (in the finger-to-nose test)
Adiadochokinesis	Inability to alternate opposite movements rapidly
Asynergia	Disturbance of the cooperative work of muscles
Scanning speech (dysarthria)	Speech broken into syllables, slow and explosive
Nystagmus	Rhythmic oscillation of the eyeballs

CLINICAL SIGNIFICANCE

In alcohol intoxication cerebellar function is disturbed first: swaying gait, impaired speech, inaccurate movements — that is why the finger-to-nose test and walking along a straight line are used to examine an intoxicated person. In chronic alcoholism the cerebellar vermis atrophies.

2. The diencephalon

The thalamus (optic thalamus)

The **thalamus** is the main “relay station” on the way of all sensory pathways (except smell) to the cortex and an integrative center (the “gateway to the cortex”).

Table 25.3. *Thalamic nuclei*

Group	Examples	Function
Specific (relay)	Ventral posterolateral and posteromedial (somatosensory), lateral geniculate body (vision), medial geniculate body (hearing)	Transmit particular sensory information to the corresponding projection area of the cortex
Motor nuclei	Ventral lateral and ventral anterior	From the cerebellum and basal ganglia to the motor cortex
Associative	Pulvinar, mediodorsal nucleus	Connected with the associative areas of the cortex; complex integration
Nonspecific	Intralaminar and reticular nuclei	From the reticular formation; diffusely activate the cortex and regulate its rhythm

CLINICAL SIGNIFICANCE

Thalamic syndrome (Dejerine–Roussy) — in damage to the thalamus (stroke): reduction of all types of sensation on the opposite side, but severe, agonizing, hard-to-treat pain (thalamic pain), hyperpathia and emotional changes.

Hypothalamus

The **hypothalamus** is the lower part of the diencephalon (≈ 4 g), the highest integrative center of **autonomic, endocrine and emotional-motivational** functions. It has more than 30 nuclei; it has a

very rich blood supply and the blood–brain barrier is weak here — it senses changes in the blood directly.

Table 25.4. *Functions of the hypothalamus*

Function	Centers and mechanisms
Autonomic regulation	Anterior part — parasympathetic effects; posterior part — sympathetic effects
Endocrine regulation	Liberins and statins (adenohypophysis), vasopressin and oxytocin (neurohypophysis) — topic 14
Thermoregulation	Anterior part — heat loss; posterior — heat production
Food intake	Lateral nuclei — hunger center; ventromedial nuclei — satiety center; arcuate nucleus (leptin and ghrelin signals)
Drinking	Thirst center (osmoreceptors)
Circadian rhythms	Suprachiasmatic nucleus — the “biological clock” (light information from the retina; melatonin via the pineal gland)
Sleep and wakefulness	Ventrolateral preoptic nucleus (sleep), lateral hypothalamus (orexin — wakefulness)
Emotions and motivation	Together with the limbic system: anger, fear, sexual and parental behavior

CLINICAL SIGNIFICANCE

When the ventromedial nuclei of the hypothalamus are damaged (tumors, injury), appetite increases sharply and **hypothalamic obesity** develops; damage to the lateral nuclei leads to refusal of food and wasting. Loss of orexin neurons causes **narcolepsy** (sudden falling asleep during the day).

3. The basal (subcortical) ganglia

The **basal ganglia** are clusters of gray matter located within the white matter of the cerebral hemispheres: the **caudate nucleus** and the **putamen** — together the **striatum**; the **globus pallidus (pallidum)**; and, functionally related to them, the **subthalamic nucleus** and the **substantia nigra** in the midbrain.

Connections: cortex (almost all areas) → **striatum** → globus pallidus / substantia nigra → **thalamus** → cortex (motor and prefrontal). This **cortex–basal ganglia–thalamus–cortex loop** works through two pathways:

- **The direct pathway enhances** the influence of the thalamus on the cortex — it “starts up” the required movements;
- **The indirect pathway** (through the subthalamic nucleus) **inhibits** the thalamus — it suppresses unwanted movements. **Dopamine** from the **substantia nigra** activates the direct pathway (D_1 receptors) and inhibits the indirect pathway (D_2) — as a result it increases motor activity.

Functions of the basal ganglia: initiating and stopping movements; selecting motor programs; automatic movements (arm swing during walking, facial expression); regulating muscle tone; setting the amplitude and speed of movement; motor learning; cognitive and motivational functions.

Table 25.5. *Syndromes of basal ganglia damage*

Syndrome	Lesion	Signs
Akinetic-rigid (parkinsonism)	Dopaminergic neurons of the substantia nigra	Hypokinesia, bradykinesia, rigidity, resting tremor, poor facial expression (mask-like face), shuffling small-step gait

Syndrome	Lesion	Signs
Hyperkinetic — chorea	Striatum (Huntington's disease)	Involuntary rapid, irregular, dance-like movements, reduced muscle tone
Athetosis	Striatum	Slow, worm-like involuntary movements (in the fingers)
Ballism	Subthalamic nucleus	Sharp, wide-amplitude flinging movements of the limbs (on one side — hemiballism)

REMEMBER

Damage to the basal ganglia has **two kinds** of result: **reduced movement** (hypokinesia — Parkinson's disease; dopamine deficiency) or **excessive involuntary movements** (hyperkinesias — chorea, ballism). The difference from cerebellar damage: in basal ganglia disease the **initiation and amplitude** of movements are disturbed, whereas in the cerebellum their **accuracy and coordination** are disturbed.

4. The cerebral cortex

The cortex is a layer of gray matter 1.5–4.5 mm thick, with a surface area of ≈ 2200 cm^2 and 14–16 billion neurons. Phylogenetically, the **ancient cortex (archicortex)**, the **old cortex (paleocortex)** and the **new cortex (neocortex — 96%)** are distinguished. The neocortex consists of **6 layers**; layer IV (internal granular) mainly receives afferent inputs (from the thalamus), and layer V (internal pyramidal — with the giant Betz pyramidal cells in the motor cortex) is the main efferent output. K. Brodmann divided the cortex cytoarchitectonically into **52 areas**.

Functional organization: neurons are grouped into **vertical columns** — each column processes a particular type of stimulus (V. Mountcastle).

Table 25.6. *Functional areas of the cortex*

Area	Location (Brodmann areas)	Function	Consequence of damage
Primary somatosensory	Postcentral gyrus (1, 2, 3)	Skin and muscle-joint sensation; somatotopically organized	Sensory disturbance on the opposite side
Primary motor	Precentral gyrus (4)	Voluntary movements; somatotopic	Central paralysis on the opposite side
Premotor and supplementary motor	Area 6	Planning of movements, complex movement sequences	Apraxia (inability to perform purposeful movements)
Primary visual	Occipital lobe, calcarine sulcus (17)	Reception of visual information	Loss of visual fields, cortical blindness
Primary auditory	Temporal lobe, Heschl's gyri (41, 42)	Reception of sounds	Cortical deafness with bilateral damage
Taste and smell	Insula, parietal operculum (43); smell — paleocortex (parahippocampal gyrus, uncus)	Perception of taste and smell	—
Motor speech center (Broca)	Left inferior frontal gyrus (44, 45)	The motor program of speech	Motor aphasia — understands but cannot speak

Area	Location (Brodmann areas)	Function	Consequence of damage
Sensory speech center (Wernicke)	Left superior temporal gyrus (22)	Comprehension of spoken language	Sensory aphasia — fluent but meaningless speech; does not understand what is said
Prefrontal associative	Anterior part of the frontal lobe	Planning, decision-making, control of behavior, working memory	Behavioral disturbance, lack of initiative, loss of self-criticism
Parietal associative	Parietal lobe	Spatial orientation, body schema	Agnosia, neglect of the opposite side

Somatotopic organization — the homunculus

By stimulating the cortex electrically in patients during surgery, W. Penfield found that the sensory and motor areas are **somatotopically** organized: each part of the body is represented in a particular place of the cortex (on the opposite side). The projection area of a body part depends not on its size but on its **functional significance** (receptor density, precision of movements) — the hand, fingers, lips and tongue occupy a very large area.

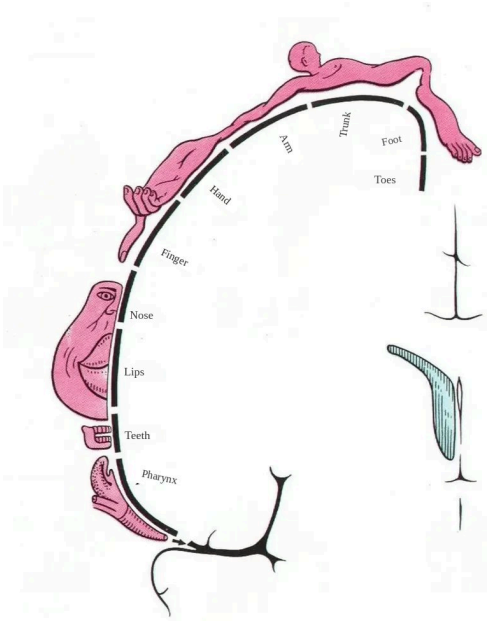


Figure 25.2. *The sensory homunculus — projection of the body on the postcentral gyrus (Alavia O.T., 2018)*

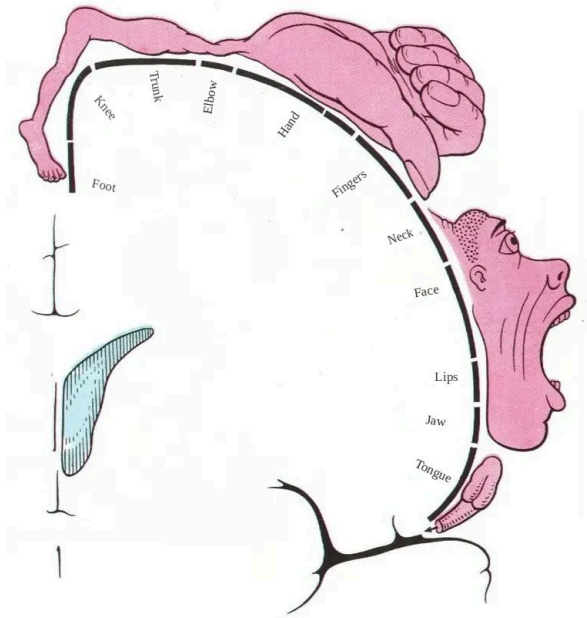


Figure 25.3. *The motor homunculus — projection of the muscles on the precentral gyrus (Alavia O.T., 2018)*

5. The limbic system

The **limbic system** (the “visceral brain”, P. MacLean) is a set of cortical and subcortical structures: the **hippocampus**, the cingulate gyrus, the **amygdala**, the septum, the hypothalamus (mammillary bodies), the anterior thalamic nuclei and others. They are connected through the **Papez circuit** (hippocampus → fornix → mammillary bodies → anterior thalamic nuclei → cingulate gyrus → hippocampus).

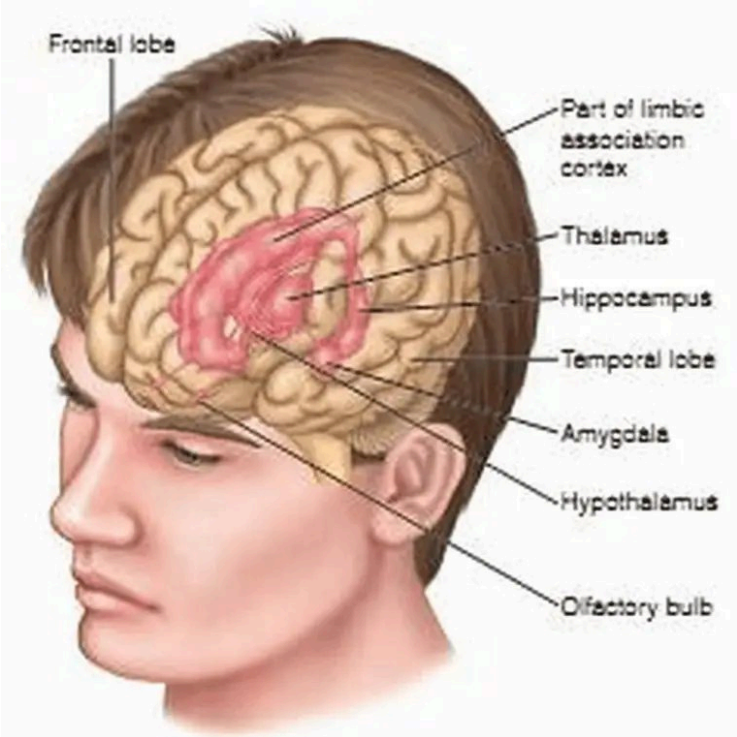


Figure 25.4. *Structures of the limbic system (Alavia O.T., 2018)*

Functions: emotions (fear, anger, pleasure — amygdala); **motivation** and instinctive behavior (feeding, defensive, sexual); **memory** — the hippocampus transfers short-term memory into long-term memory (consolidation); coordination of autonomic and endocrine reactions with emotions.

> **CLINICAL SIGNIFICANCE**
> In bilateral damage to the **hippocampus** (the famous patient H.M. after epilepsy surgery), the ability to remember new events is lost (anterograde amnesia), while old memories and motor skills are preserved. **Alzheimer's disease** begins with the early death of hippocampal neurons.

6. Electroencephalography

Electroencephalography (EEG) is a method of recording the biopotentials of the brain from the scalp (H. Berger, 1929). EEG waves mainly reflect the sum of the **postsynaptic potentials** of cortical pyramidal neurons.

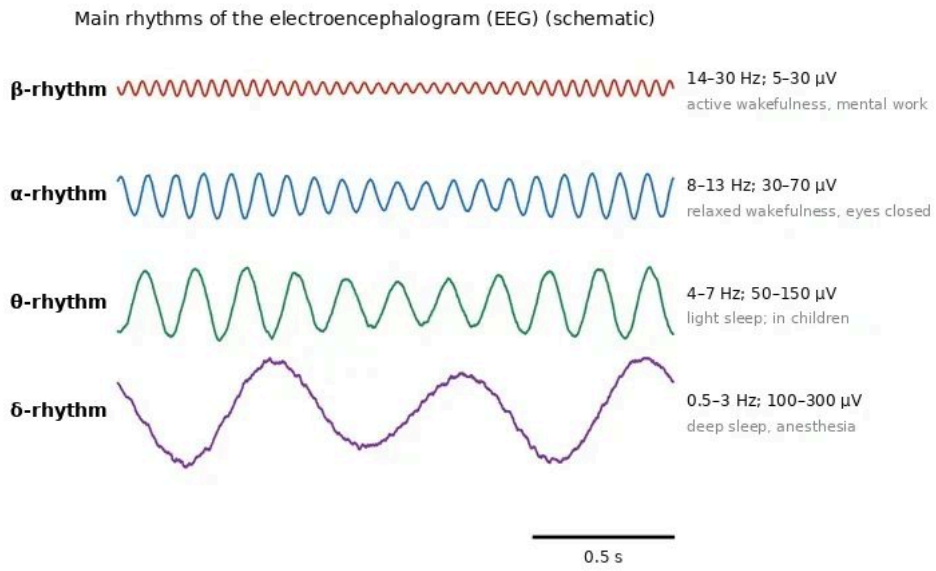


Figure 25.5. *Main EEG rhythms*

Table 25.7. *EEG rhythms*

Rhythm	Frequency, Hz	Amplitude, μV	When observed
α (alpha)	8–13	30–70	Quiet wakefulness with the eyes closed; well expressed over the occipital region
β (beta)	14–30	5–30	Active wakefulness, mental activity, attention
γ (gamma)	30–100	< 10	High cognitive activity, binding of information
θ (theta)	4–7	50–150	Light sleep, normal in children; pathological in awake adults
δ (delta)	0,5–3	100–300	Deep (slow-wave) sleep, anesthesia, infants; severe pathology in awake adults

α-rhythm blockade (desynchronization) — when the eyes are opened or a new stimulus appears, the α-rhythm is replaced by the β-rhythm; this reflects the activation of the cortex by the reticular formation and the thalamus.

CLINICAL SIGNIFICANCE

EEG is the main method in the diagnosis of **epilepsy** — seizure activity appears as “spike-and-wave” complexes. EEG is also used to evaluate sleep disorders (polysomnography), the depth of anesthesia and the degree of coma, and to confirm **brain death** (electrical silence).

7. Laboratory work

Exercise 1: examining cerebellar function in humans

1. **Romberg test**: the subject stands with the feet together, the arms stretched forward and the eyes closed (30–60 s). Observe swaying or a tendency to fall. A more difficult variant — standing with the feet in one line (heel to toe).
2. **Finger-to-nose test**: stretch the arm out to the side, then, with the eyes closed, touch the tip of the nose with the index finger (with each hand). Evaluate accuracy and tremor.

3. **Heel-to-knee test**: lying down, place the heel of one leg on the knee of the other and slide it down along the shin.
4. **Adiadochokinesis test**: rapidly turning the palms up and down alternately (supination–pronation).
5. **Gait**: walking along a straight line, heel-to-toe walking. **Interpretation:** normally stability is maintained in the Romberg position and the tests are performed accurately, without tremor and smoothly. In cerebellar damage — swaying, tremor on approaching the target (intention tremor), "overshooting" (dysmetria), slow and incorrect alternation of movements.

> **CAUTION**
> If swaying in the Romberg position increases when the eyes are **closed**, this is most often a sign of **sensory (proprioceptive) ataxia** (damage to the dorsal columns or peripheral nerves — visual control compensates). In cerebellar ataxia the swaying persists even with the eyes open.

Exercise 2: analyzing an EEG recording (demonstration or a prepared recording)

1. The subject is at rest with the eyes closed — find the α-rhythm in the occipital leads and determine its frequency from the number of waves per 1 s.
2. The subject opens the eyes — observe the α-rhythm blockade (its replacement by the β-rhythm).
3. A mental load (calculation) is given — record the enhancement of the β-rhythm.
4. 3 minutes of hyperventilation — observe the appearance of slow waves (a provocative test).

8. Practical skills

Skill: determining the level of a motor disorder

1. Evaluate muscle strength and tone: weakness, spasticity and hyperreflexia — the pyramidal tract (upper motor neuron).
2. Rigidity ("cogwheel"), bradykinesia, resting tremor — the basal ganglia (extrapyramidal system).
3. Reduced muscle tone, ataxia, intention tremor, dysmetria — the cerebellum.
4. Atrophy, fibrillations, loss of reflexes — the lower motor neuron.
5. Explain the result by linking it to the function of the corresponding structures.

Skill: localizing a speech disorder

1. Evaluate whether the patient's speech is fluent, whether he understands what is said and whether he can repeat.
2. Speech not fluent, sparse, but comprehension preserved — Broca's area (motor aphasia).
3. Speech fluent but meaningless, comprehension impaired — Wernicke's area (sensory aphasia).
4. Only articulation impaired (dysarthria), language structure preserved — brainstem or cerebellum.

Review questions

1. Explain the functional divisions of the cerebellum and their connections.
2. What is the function of the Purkinje cells?
3. What is Luciani's triad?
4. List the main signs of cerebellar damage.

5. Into which groups are the thalamic nuclei divided?
6. List the functions of the hypothalamus.
7. Where are the hunger and satiety centers located?
8. Which structures make up the basal ganglia, and what connections do they have?
9. What is the role of the direct and indirect pathways, and what is the effect of dopamine?
10. What is the difference between parkinsonism and chorea?
11. What are the cytoarchitectonics and functional areas of the cortex?
12. What is the homunculus? Why do the hand and face occupy a large area?
13. What is observed when Broca's and Wernicke's speech centers are damaged?
14. What are the composition and functions of the limbic system?
15. What are the main EEG rhythms and their significance?

Test questions

1. Which are the only output neurons of the cerebellar cortex?
   A) Granule cells
   B) Basket cells
   C) Golgi cells
   D) Purkinje cells
   E) Stellate cells

2. What does Luciani's triad include?
   A) Rigidity, tremor, bradykinesia
   B) Aphasia, agnosia, apraxia
   C) Atonia, asthenia, astasia
   D) Ptosis, miosis, enophthalmos
   E) Hyperreflexia, clonus, spasticity

3. Tremor that increases as the target is approached is characteristic of damage to which structure?
   A) Substantia nigra
   B) The thalamus
   C) Hypothalamus
   D) The cerebellum
   E) The spinal cord

4. Which thalamic nucleus relays visual information to the cortex?
   A) The medial geniculate body
   B) The ventral posterolateral nucleus
   C) The intralaminar nuclei
   D) The ventral lateral nucleus
   E) The lateral geniculate body

5. Where is the satiety center located?
   A) In the lateral nuclei of the hypothalamus
   B) In the thalamus
   C) In the ventromedial nuclei of the hypothalamus
   D) In the cerebellum
   E) In the medulla oblongata

6. Which hypothalamic nucleus acts as the “biological clock”?
   A) The suprachiasmatic nucleus
   B) The supraoptic nucleus
   C) The paraventricular nucleus
   D) The ventromedial nucleus
   E) The arcuate nucleus

7. Which is one of the main signs of Parkinson's disease?
   A) Intention tremor
   B) Rigidity and bradykinesia
   C) Dance-like involuntary movements
   D) Scanning speech
   E) Reduced muscle tone

8. Damage to which structure causes chorea?
   A) Striatum
   B) The cerebellum
   C) The thalamus
   D) The hippocampus
   E) The pons

9. Where is the primary motor cortex located?
   A) In the postcentral gyrus
   B) In the occipital lobe
   C) In the temporal lobe
   D) In the precentral gyrus (area 4)
   E) In the insula

10. What disorder is observed when Wernicke's area is damaged?
    A) Motor aphasia
    B) Cortical blindness
    C) Paralysis on the opposite side
    D) Ataxia
    E) Sensory aphasia — failure to understand speech

11. What does the size of the projection area in the homunculus depend on?
    A) On the size of the organ
    B) On the functional significance of the organ (receptor density, precision of movements)
    C) On the mass of the organ
    D) On its blood supply
    E) On age

12. Which structure plays the main role in transferring short-term memory into long-term memory?
    A) The cerebellum
    B) The thalamus
    C) The hippocampus
    D) Substantia nigra
    E) Medulla oblongata

13. Which EEG rhythm predominates in quiet wakefulness with the eyes closed?
    A) α-rhythm
    B) β-rhythm
    C) δ-rhythm
    D) θ-rhythm
    E) γ-rhythm

14. What does the recording of a δ-rhythm in an awake adult indicate?
    A) Normal value
    B) Active mental work
    C) Quiet wakefulness
    D) Excitement
    E) Severe pathology (brain damage)

15. What condition develops when the ventromedial nuclei of the hypothalamus are damaged?
    A) Wasting
    B) Hypothalamic obesity (a sharp increase in appetite)
    C) Diabetes insipidus
    D) Hypothermia
    E) Insomnia

Clinical cases

Case 1

A 45-year-old patient sways when walking, walks with the legs wide apart, and his speech is broken into syllables and slow. In the finger-to-nose test the tremor increases as the target is approached and the finger “overshoots” the nose. In the Romberg position he sways even with the eyes open.

Questions: Which structure is damaged? Explain each sign.

Answer: The cerebellum is damaged (cerebellar ataxia). The cerebellum ensures the accuracy, smoothness and coordination of movements: it compares the planned movement with information about the ongoing movement (from proprioceptors) and corrects errors. When it is damaged: dynamic ataxia — a swaying, wide-based gait; intention tremor — on approaching the target the corrective movements overshoot and oscillation appears; dysmetria — misjudging the amplitude of a movement (“overshooting”); scanning speech — incoordination of the movements of the speech muscles; static ataxia persists regardless of visual control (the difference from sensory ataxia).

Case 2

For two years a 68-year-old man's movements have been slowing; he has a “pill-rolling” tremor of the hands at rest, a mask-like face, walks stooped with small steps, and uneven resistance is felt on passive flexion of his arms.

Questions: What is the disease? Explain its pathophysiological basis through the connections of the basal ganglia.

Answer: Parkinson's disease (akinetic-rigid syndrome). The dopaminergic neurons of the substantia nigra die and the supply of dopamine to the striatum decreases. Normally dopamine activates the direct pathway (D_1) and inhibits the indirect pathway (D_2), facilitating the passage of movement-initiating signals through the thalamus to the motor cortex. In dopamine deficiency the indirect (inhibitory) pathway predominates — the activation of the cortex by the thalamus decreases:

initiating movements becomes difficult (akinesia, bradykinesia), automatic movements disappear (mask-like face, no arm swing), muscle tone increases (“cogwheel” rigidity) and a resting tremor appears. Treatment — levodopa, dopamine agonists.

Case 3

After a left-sided stroke the patient understands what is said and follows instructions, but speaking is very difficult for him: few words, short, broken phrases; there is weakness in the right arm.

Questions: Which area of the cortex is damaged? Why is there weakness in the right arm?

Answer: Broca's area (areas 44–45) in the inferior frontal gyrus of the left hemisphere is damaged — motor (expressive) aphasia: comprehension of speech (Wernicke's area) is preserved, but the construction of the motor program of speech is impaired. Broca's area lies close to the primary motor cortex (the lower part of the precentral gyrus — the projection of the face and arm); because this too is damaged and the pyramidal tracts cross in the medulla oblongata, weakness (central paresis) has developed in the opposite (right) arm.

Quick review

Table 25.8. *Key concepts of topic 25*

Concept	Brief definition
Purkinje cells	The only output of the cerebellar cortex; inhibitory
Function of the cerebellum	Balance, tone, accuracy and coordination of movements
Luciani's triad	Atonia, asthenia, astasia
Cerebellar signs	Ataxia, dysmetria, intention tremor, adiadochokinesis, nystagmus
The thalamus	The “gateway” for sensory pathways (except smell)
Hypothalamus	Autonomic, endocrine, thermoregulation, hunger–satiety, circadian rhythm
The suprachiasmatic nucleus	Biological clock
Basal ganglia	Striatum, globus pallidus, subthalamic nucleus, substantia nigra
Parkinsonism	Dopamine deficiency: rigidity, bradykinesia, resting tremor
Chorea, ballism	Hyperkinesias (striatum, subthalamic nucleus)
Homunculus	Somatotopic projection; hand and face large (Penfield)
Broca / Wernicke	Motor / sensory speech centers (left hemisphere)
Limbic system	Emotions, motivation, memory (hippocampus, amygdala)
α-rhythm	8–13 Hz; quiet wakefulness, eyes closed
β / θ / δ rhythms	Activity / light sleep / deep sleep

TOPIC 26

The autonomic nervous system: sympathetic, parasympathetic and metasympathetic divisions. The adrenal medulla

Learning objectives

To teach the student the structural and functional features of the autonomic nervous system, its sympathetic, parasympathetic and metasympathetic divisions, the mediators and receptors in autonomic ganglia and synapses, the dual innervation of organs, autonomic reflexes and centers, and the effects of the hormones of the adrenal medulla; to master tests evaluating autonomic tone in humans.

By the end of the class the student should know:

- the differences between the autonomic and somatic nervous systems;
- the structure of the sympathetic, parasympathetic and metasympathetic divisions and the location of their centers;
- the functions of autonomic ganglia and the axon reflex;
- the mediators and receptors of autonomic synapses (N- and M-cholinergic receptors, α- and β-adrenergic receptors);
- the effects of the sympathetic and parasympathetic divisions on organs;
- the types of autonomic reflexes and the centers regulating autonomic functions;
- the hormones of the adrenal medulla and their effects, the sympathoadrenal system.

The student should be able to:

- examine and evaluate dermographism;
- perform the clino-orthostatic test and calculate the Kérdö index;
- determine the type of autonomic tone.

1. General description of the autonomic nervous system

The **autonomic nervous system (ANS)** is the part of the nervous system that regulates the activity of the internal organs, blood vessels, glands, smooth muscles and heart, and metabolism. It maintains homeostasis and ensures the body's adaptation to different conditions of activity (the **adaptive-trophic** function — L.A. Orbeli). Its activity is usually not subject to consciousness (“autonomic”).

Table 26.1. *Differences between the somatic and autonomic nervous systems*

Feature	Somatic nervous system	Autonomic nervous system
Effectors	Skeletal muscles	Smooth muscles, heart, glands, adipose tissue
Outflow from the centers	Along the whole spinal cord and brainstem	Focal (from particular segments)
Efferent pathway	One neuron (motoneuron)	Two neurons: preganglionic and postganglionic, with an autonomic ganglion between them
Fibers	Thick myelinated (Aα), fast	Preganglionic — B (thinly myelinated), postganglionic — C (unmyelinated), slow
Voluntary control	Present	Almost none

Feature	Somatic nervous system	Autonomic nervous system
Mediator at the effector	Acetylcholine (N-cholinergic receptor)	Acetylcholine (M) or noradrenaline (α, β) and others

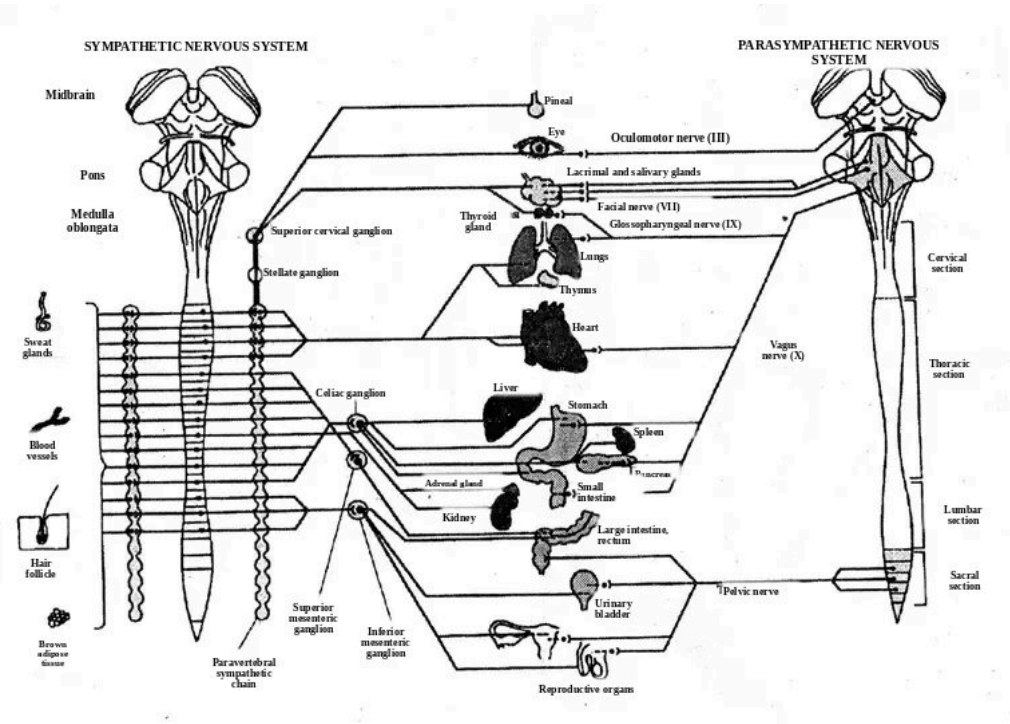


Figure 26.1. *The somatic and autonomic nervous systems: centers, ganglia and innervated organs (Alavia O.T., 2018)*

2. Divisions of the autonomic nervous system

The sympathetic division

Its centers are in the lateral horns of the **thoracic and lumbar segments (Th_1–L_2 (L_3))** of the spinal cord (the **thoracolumbar division**). The preganglionic fibers are **short**, and the ganglia lie close to the center: in the **paravertebral** (sympathetic trunk) and **prevertebral** (celiac, superior and inferior mesenteric plexus) ganglia. The postganglionic fibers are **long**. One preganglionic fiber connects with many (≈ 1 : 10–30) postganglionic neurons — a **widespread (generalized)** effect.

The parasympathetic division

Its centers are in the **brainstem** (nuclei of cranial nerves III, VII, IX, X) and in the **sacral segments (S_2–S_4)** of the spinal cord (the **craniosacral division**). The preganglionic fibers are **long**, and the ganglia lie next to or inside the organ (intramural); the postganglionic fibers are **short**. The effect is **local**. ≈ 75% of the parasympathetic fibers run in the **vagus nerve**.

Table 26.2. *Parasympathetic fibers*

Nerve	Center	Organs innervated
III (oculomotor)	Midbrain (Yakubovich–Edinger–Westphal nucleus)	Pupillary constrictor and ciliary muscle
VII (facial)	Pons (superior salivary nucleus)	Lacrimal, submandibular and sublingual salivary glands
IX (glossopharyngeal)	Medulla oblongata (inferior salivary nucleus)	Parotid salivary gland

Nerve	Center	Organs innervated
X (vagus)	Medulla oblongata (dorsal nucleus, nucleus ambiguus)	Heart, bronchi, stomach, intestine (up to the left colic flexure), liver, pancreas, kidneys
Pelvic nerves	Spinal cord S_2–S_4	Distal part of the large intestine, rectum, bladder, genital organs

CAUTION

Some organs have **only sympathetic** innervation: most blood vessels, the sweat glands, the adrenal medulla, the smooth muscles of the skin (piloerectors) and the splenic capsule. Their activity is regulated by an **increase or decrease** in sympathetic tone.

The metasympathetic division

The **metasympathetic nervous system** (A.D. Nozdrachev) is the set of **intramural** nerve plexuses in the walls of hollow organs: the gastrointestinal tract (Auerbach and Meissner plexuses — the enteric nervous system), heart, bladder, uterus and bronchi. Its features:

- it has its own sensory, inter- and efferent neurons — it **carries out local reflexes without the CNS** (for example, intestinal peristalsis);
- **automaticity** (it has its own rhythm generators);
- it innervates only internal organs and does not form part of somatic reflex arcs;
- it connects with the sympathetic and parasympathetic fibers, “relays” and modulates their influence;
- its mediators are diverse: acetylcholine, noradrenaline, serotonin, ATP (purinergic), VIP, NO, substance P.

Autonomic ganglia and the axon reflex

In addition to transmitting impulses, autonomic ganglia also function as **peripheral nerve centers**: convergence, multiplication, summation and rhythm transformation of impulses take place in them; **peripheral reflexes** can close through the ganglia.

The **axon reflex** is a reflex carried out through the branching of a sensory neuron's axon, without the participation of the CNS or even a ganglion: when the skin is stimulated, the impulse passes from one branch to another and dilates nearby vessels (reddening of the skin — dermographism, the “red flare” in inflammation).

3. Mediators and receptors of autonomic synapses

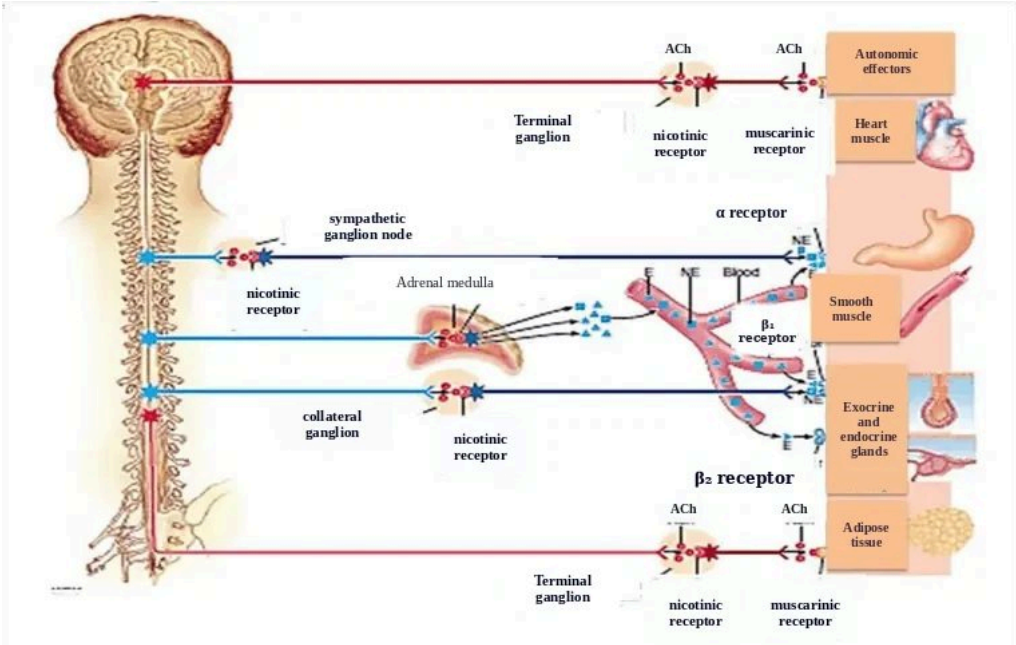


Figure 26.2. *Synapses, mediators and receptors of the autonomic nervous system: parasympathetic (upper and lower rows) and sympathetic (middle) pathways, the adrenal medulla (Alavia O.T., 2018)*

Table 26.3. *Mediators and receptors of the autonomic nervous system*

Synapse	Mediator	Receptor
In all autonomic ganglia (preganglionic → postganglionic)	Acetylcholine	N_N-cholinergic receptor (neuronal nicotinic receptor)
In the adrenal medulla	Acetylcholine	N_N-cholinergic receptor
Parasympathetic postganglionic → effector	Acetylcholine	M-cholinergic receptors (M_1–M_5)
Sympathetic postganglionic → effector (most)	Norepinephrine	α_1, α_2, β_1, β_2, β_3-adrenergic receptors
Sympathetic postganglionic → sweat glands, some skeletal muscle vessels	Acetylcholine	M-cholinergic receptors (sympathetic cholinergic fibers)
Sympathetic postganglionic → renal vessels	Dopamine	D_1

Table 26.4. *Adrenergic receptors and their effects*

Receptor	Location	Effect
α_1	Vessels (skin, internal organs), pupillary dilator muscle, sphincters	Smooth muscle contraction: vasoconstriction, pupil dilation
α_2	Presynaptic terminals, pancreatic β cells	Inhibition of noradrenaline release (feedback), reduced insulin secretion
β_1	Heart, juxtaglomerular cells	Increase in HR and force of contraction, renin release
β_2	Bronchi, skeletal muscle vessels, uterus, liver	Smooth muscle relaxation: bronchodilation and vasodilation; glycogenolysis
β_3	Adipose tissue, bladder	Lipolysis, relaxation of the detrusor

MEMORY AID

Remembering the mediators: **“In the ganglion — always acetylcholine (N)”**; **“Parasympathetic — acetylcholine at the end too (M)”**; **“Sympathetic — noradrenaline at the end (α, β)”**, the exception being the **sweat glands** (sympathetic, but acetylcholine, M).

CLINICAL SIGNIFICANCE

Many drugs act on autonomic receptors: **atropine** (M-cholinergic blocker) — dilates the pupil, reduces salivation and sweating, increases HR, dilates the bronchi; **salbutamol** (β_2-agonist) — dilates the bronchi in bronchial asthma; **metoprolol** (β_1-blocker) — reduces HR and arterial pressure; **prazosin** (α_1-blocker) — dilates vessels; **pilocarpine** (M-cholinomimetic) — constricts the pupil (in glaucoma). In poisoning with **organophosphorus compounds**, acetylcholine accumulates and a strong parasympathetic effect develops (pupil constriction, profuse salivation and sweating, bronchospasm, bradycardia, diarrhea).

4. Effects of the sympathetic and parasympathetic divisions on organs

Most organs have **dual innervation**; the divisions usually have **opposite (antagonistic)** effects, but their effects are **complementary (functional synergism)** and together finely adjust the organ's activity. Sometimes both divisions act in the same direction (salivary glands — topic 18).

Table 26.5. *Effects of the sympathetic and parasympathetic divisions on organs*

Organ	Sympathetic influence	Parasympathetic effect
Pupil	Dilates (mydriasis; α_1 — radial muscle)	Constricts (miosis; M — sphincter)
Ciliary muscle	Relaxes (far vision)	Contracts (near vision — accommodation)
Salivary glands	Scanty, thick saliva	Abundant, watery saliva
Heart	HR, force of contraction and conduction increase (β_1)	HR decreases, AV conduction slows (M_2)
Bronchi	Dilate (β_2)	Constrict, mucus secretion increases (M_3)
Vessels of skin and internal organs	Constrict (α_1)	No effect on most
Skeletal muscle vessels	Dilate (β_2, cholinergic)	—
Gastrointestinal motility and secretion	Inhibited; sphincters contract	Enhanced; sphincters relax
Liver	Glycogenolysis, gluconeogenesis (β_2, α_1)	Glycogen synthesis
Adipose tissue	Lipolysis (β_3)	—
Bladder	Detrusor relaxes, sphincter contracts — urine retention	Detrusor contracts, sphincter relaxes — micturition
Sweat glands	Sweating (cholinergic)	—
Adrenal medulla	Release of adrenaline and noradrenaline	—
Genital organs	Ejaculation	Erection

REMEMBER

The sympathetic division — the “fight or flight” reaction: mobilization of energy, increased blood supply to the heart and skeletal muscles, inhibition of digestion. **The parasympathetic division** — “rest and digest”: saving and restoring energy, digestion, excretion.

5. Autonomic reflexes and centers

Table 26.6. *Types of autonomic reflexes*

Type	Receptors → effector	Example
Viscero-visceral	Internal organ → internal organ	Baroreceptor reflex; Goltz reflex
Viscero-dermal	Internal organ → skin	Pain and hyperesthesia in the Zakharyin–Head zones
Viscero-motor	Internal organ → skeletal muscles	Protective tension of the abdominal muscles in appendicitis
Dermo-visceral	Skin → internal organ	Reduction of intestinal spasm when a hot-water bottle is applied; the basis of physiotherapy
Somato-visceral	Skeletal muscles → internal organ	Increased HR and breathing during physical work

CLINICAL SIGNIFICANCE

Zakharyin–Head zones — when an internal organ is diseased, pain is **referred** to the skin area innervated from the same segment (referred pain): in cardiac ischemia — to the left arm and under the scapula, in acute cholecystitis — to the right scapula and shoulder, in renal colic — to the groin. The cause is the **convergence** of visceral and skin afferents onto the same neurons of the spinal cord.

Centers regulating autonomic functions:

- **The spinal cord** — centers of micturition, defecation, erection, sweating and vascular reflexes (segmental level).
- **The medulla oblongata and pons** — the centers of breathing, circulation, the heart, salivation, swallowing, vomiting and coughing.
- **The midbrain** — the pupillary reflex, accommodation.
- **The hypothalamus** — the **highest autonomic center**: its anterior part — parasympathetic effects, its posterior part — sympathetic effects; thermoregulation, hunger, thirst; it integrates autonomic, endocrine and behavioral reactions.
- **The limbic system, cerebellum, basal ganglia and cerebral cortex** — autonomic reactions associated with emotions and behavior (palpitations with excitement, pallor with fear), conditioned autonomic reflexes.

6. The adrenal medulla

The adrenal medulla is embryologically and functionally a **modified sympathetic ganglion**: its chromaffin cells resemble postganglionic neurons, but instead of an axon they release hormones directly into the blood. They are innervated by **preganglionic sympathetic fibers** (acetylcholine, N_N receptors).

Hormones secreted: **adrenaline (≈ 80%)** and **noradrenaline (≈ 20%)**. Their synthesis: tyrosine → DOPA → dopamine → noradrenaline → adrenaline (the enzyme phenylethanolamine-N-

methyltransferase — activated by cortisol; the blood flowing from the adrenal cortex helps this). Inactivation — by the enzymes MAO and COMT; the metabolite (vanillylmandelic acid) is excreted in the urine.

Table 26.7. *Comparative effects of adrenaline and noradrenaline*

Parameter	Adrenaline	Norepinephrine
Receptor affinity	α and β (strong for β_2)	Mainly α and β_1 (weak for β_2)
HR	Increases	May not change or may decrease reflexly
Systolic pressure	Increases	Increases
Diastolic pressure	Does not change or decreases (muscle vessels dilate)	Increases (general vasoconstriction)
Bronchi	Dilate	Weak effect
Metabolic effect (glycogenolysis, lipolysis, hyperglycemia)	Strong	Weaker

The **sympathoadrenal system** is the single functional union of the sympathetic nervous system and the adrenal medulla (W. Cannon). It is activated in stress, pain, fear, physical work, cold, blood loss and hypoglycemia and adapts the body to an emergency: cardiac work and arterial pressure increase, blood is redistributed to the muscles, heart and brain, the bronchi dilate, blood glucose and fatty acids increase, blood coagulation accelerates and fatigue decreases.

CLINICAL SIGNIFICANCE

Pheochromocytoma is a catecholamine-producing tumor of the adrenal medulla: paroxysmal severe arterial hypertension, headache, sweating, palpitations, pallor, hyperglycemia. Diagnosis — metanephrines in blood and urine. **Adrenaline** is the first-line drug in anaphylactic shock (α — constricts vessels, β_1 — strengthens the heart, β_2 — dilates the bronchi and reduces the release of mediators from mast cells).

7. Laboratory work

Exercise 1: examining dermographism

1. Draw a line on the skin of the subject's chest or forearm with a blunt spatula using moderate pressure.
2. After 10–20 s observe the color, width and persistence of the line that appears. **Interpretation: white dermographism** (a white line that disappears quickly) — predominance of sympathetic tone (constriction of skin vessels); **red dermographism** (a wide red line persisting for a long time — more than 1–3 minutes) — predominance of parasympathetic tone; **raised (urticarial)** dermographism — increased vascular permeability. Normally — a narrow pink line that disappears in 1–2 minutes. The axon reflex takes part in the mechanism of the red line.

Exercise 2: the clino-orthostatic test

1. **Orthostatic test:** measure the pulse and pressure after 5 minutes of lying down; measure them again 1 minute after standing up. Normally the pulse increases by 10–16 /min (sympathetic reactivity).
2. **Clinostatic test:** the pulse is measured after lying down from the standing position. Normally the pulse decreases by 4–12 /min (parasympathetic reactivity). **Interpretation:** an increase in pulse

of > 16 in the orthostatic test — increased sympathetic reactivity; a decrease of > 12 in the clinostatic test — increased parasympathetic reactivity.

Exercise 3: calculating the Kérdö autonomic index

1. Measure the diastolic pressure (D) and pulse (P) at rest.
2. Kérdö index = (1 − D / P) × 100. **Interpretation:** close to 0 — autonomic balance (eutonia); positive — predominance of sympathetic tone; negative — predominance of parasympathetic tone.

Example: D = 80, P = 70 → (1 − 80/70) × 100 = −14 — parasympathetic predominance.

Exercise 4: reaction of the pupil to light and drugs (demonstration)

1. Compare the pupil size in light and in darkness.
2. Observe dilation of the pupil under the influence of fear or pain (for example, when the skin is pinched) (a sympathetic reflex).
3. Analyze the effect of atropine (dilates) and pilocarpine (constricts) in ophthalmology using videos or pictures.

8. Practical skills

Skill: comprehensive evaluation of the type of autonomic tone

1. Determine the pulse, pressure and respiratory rate at rest.
2. Evaluate skin color, moisture and dermographism.
3. Calculate the Kérdö index.
4. Perform the orthostatic and clinostatic tests.
5. Add the Dagnini–Aschner reflex (topic 12).
6. Summarize the results and determine the type of autonomic tone: sympathicotonia (tachycardia, white dermographism, dry skin, wide pupils, positive Kérdö index), vagotonia (bradycardia, red dermographism, moist skin, negative Kérdö index), eutonia.

Skill: predicting a drug's effect from the receptors

1. Determine which receptor the drug acts on (agonist or blocker).
2. Write down which organs the receptor is located in.
3. Determine the physiological effect of the receptor.
4. An agonist enhances that effect; with a blocker, the effect of the opposite division predominates.
5. List the expected and side effects (for example, a β-blocker may cause bronchospasm in asthma).

Review questions

1. What are the differences between the autonomic and somatic nervous systems?
2. Where are the centers and ganglia of the sympathetic division located?
3. List the centers and fibers of the parasympathetic division.
4. What are the features of the metasympathetic nervous system?
5. What functions do autonomic ganglia perform? What is the axon reflex?
6. Which mediator and receptors operate in the autonomic ganglia?

7. What are the mediators of the parasympathetic and sympathetic postganglionic synapses?
8. What are the types of adrenergic receptors and their effects?
9. Compare the effects of the sympathetic and parasympathetic divisions on the heart, bronchi, gastrointestinal tract and eye.
10. Which organs have only sympathetic innervation?
11. Explain the types of autonomic reflexes with examples.
12. What are the Zakharyin–Head zones?
13. Where are the centers regulating autonomic functions?
14. What hormones does the adrenal medulla produce, and how do they differ?
15. What is the sympathoadrenal system, and when is it activated?

Test questions

1. Which mediator operates in all autonomic ganglia?
   A) Acetylcholine
   B) Norepinephrine
   C) Adrenaline
   D) Dopamine
   E) Serotonin

2. Where are the centers of the sympathetic division located?
   A) In the brainstem
   B) In the sacral segments of the spinal cord
   C) Only in the hypothalamus
   D) In the lateral horns of the thoracic and lumbar segments (Th_1–L_2) of the spinal cord
   E) In the cerebellum

3. In which nerve do most of the parasympathetic fibers run?
   A) The facial nerve
   B) The glossopharyngeal nerve
   C) Vagus nerve
   D) The oculomotor nerve
   E) Pelvic nerves

4. How are the sweat glands innervated?
   A) Parasympathetic, acetylcholine
   B) Sympathetic, but the mediator is acetylcholine (M receptors)
   C) Sympathetic, noradrenaline
   D) Somatic nerves
   E) They have no innervation

5. Which adrenergic receptor dilates the bronchi?
   A) β_2
   B) α_1
   C) α_2
   D) β_1
   E) β_3

6. How does the pupil change under parasympathetic influence?
   A) Dilate
   B) It does not change
   C) It dilates in one eye only
   D) It constricts (miosis)
   E) It first dilates, then constricts

7. Which is a feature of the metasympathetic nervous system?
   A) It innervates only skeletal muscles
   B) Its centers are in the spinal cord
   C) It uses only noradrenaline
   D) It is completely subordinate to the CNS
   E) It lies in the walls of hollow organs and carries out local reflexes without the CNS

8. What is the cause of pain in the Zakharyin–Head zones?
   A) Convergence of visceral and skin afferents onto the same neurons of the spinal cord
   B) Inflammation of the skin
   C) Constriction of skin vessels
   D) Passage of mediators into the blood
   E) Damage to somatic nerves

9. What effect does atropine have?
   A) It constricts the pupil
   B) It dilates the pupil, reduces salivation and increases HR
   C) It constricts the bronchi
   D) It enhances intestinal motility
   E) It enhances sweating

10. Which hormone does the adrenal medulla mainly secrete?
    A) Cortisol
    B) Aldosterone
    C) Norepinephrine
    D) Dopamine
    E) Adrenaline

11. How does the bladder change under sympathetic influence?
    A) The detrusor contracts and the sphincter relaxes
    B) Micturition increases
    C) No effect
    D) The detrusor relaxes and the sphincter contracts — urine is retained
    E) The bladder ruptures

12. What does white dermographism indicate?
    A) Predominance of parasympathetic tone
    B) The norm
    C) Predominance of sympathetic tone
    D) Allergy
    E) Impaired blood coagulation

13. If the diastolic pressure is 70 and the pulse 84, what is the Kérdö index?
    A) −14
    B) 0
    C) +50
    D) −20
    E) ≈ +17

14. Which structure is the highest autonomic center?
    A) Medulla oblongata
    B) Hypothalamus
    C) The spinal cord
    D) The cerebellum
    E) The pons

15. What is the main sign of pheochromocytoma?
    A) Arterial hypotension
    B) Bradycardia
    C) Paroxysmal arterial hypertension, sweating, palpitations
    D) Hypoglycemia
    E) Profuse salivation

Clinical cases

Case 1

A farmer who had been working with an organophosphorus insecticide has sharply constricted pupils, profuse salivation and sweating, shortness of breath (bronchospasm), bradycardia, abdominal pain and diarrhea, and muscle twitching.

Questions: Explain the mechanism of poisoning. Which drug is used as an antidote, and why?

Answer: Organophosphorus compounds irreversibly inhibit acetylcholinesterase — acetylcholine accumulates in the synapses without being broken down. The M-cholinergic receptors are excessively stimulated (muscarinic effects): miosis, profuse salivation and sweating, bronchoconstriction and increased mucus, bradycardia, increased intestinal motility. Stimulation of N-cholinergic receptors (nicotinic effects) — fasciculations in the skeletal muscles, then paralysis. The antidote is atropine (an M-cholinergic blocker): it eliminates the muscarinic effects; AChE reactivators (pralidoxime) are also used. In respiratory failure — artificial ventilation.

Case 2

Before an exam a student's heart is beating fast, his palms are sweating, his mouth is dry, his pupils are dilated and he has no appetite.

Questions: Which system is activated? Explain each sign in terms of mediators and receptors.

Answer: Because of stress the sympathoadrenal system has been activated (an emotional reaction — via the limbic system and hypothalamus). Tachycardia — the action of noradrenaline and adrenaline on β_1 receptors in the heart. Sweating of the palms — the sympathetic cholinergic innervation of the sweat glands (acetylcholine, M receptors). Dry mouth — under sympathetic influence little thick saliva is secreted and the parasympathetic influence decreases. Pupil dilation — α_1 receptors in the radial muscle. Lack of appetite and inhibition of digestion — the sympathetic division inhibits gastrointestinal motility and secretion. These are elements of the “fight or flight” reaction.

Case 3

A patient with bronchial asthma was prescribed a non-selective β-blocker (propranolol) for hypertension, after which his attacks of breathlessness increased.

Questions: Why did the asthma worsen? Which drug would have been a better choice?

Answer: The smooth muscles of the bronchi contain β_2-adrenergic receptors, which dilate the bronchi under the action of adrenaline. A non-selective β-blocker blocks the β_2 receptors together with β_1 (heart) — the bronchodilating sympathetic influence is lost and the parasympathetic (M_3 — bronchoconstricting) influence predominates — bronchospasm increases. Non-selective β-blockers are contraindicated in asthmatic patients; if necessary, cardioselective β_1-blockers are chosen with caution, or antihypertensive drugs of other groups (calcium channel blockers, ACE inhibitors).

Quick review

Table 26.8. *Key concepts of topic 26*

Concept	Brief definition
ANS	Regulates internal organs, vessels, glands and the heart; two-neuron efferent pathway
Sympathetic centers	Lateral horns of Th_1–L_2; ganglia next to the spine
Parasympathetic centers	Nuclei of nerves III, VII, IX, X and S_2–S_4; ganglia next to the organ
Metasympathetic system	Intramural plexuses; local reflexes, automaticity
Axon reflex	Through branches of a sensory axon (without the CNS)
In the ganglia	Acetylcholine, N_N-cholinergic receptor
Parasympathetic effector	Acetylcholine, M-cholinergic receptor
Sympathetic effector	Noradrenaline, α and β; sweat glands — acetylcholine
α_1 / β_1 / β_2	Vasoconstriction / heart / bronchodilation
Sympathetic influence	“Fight or flight”: tachycardia, mydriasis, bronchodilation, inhibition of digestion
Parasympathetic effect	“Rest and digest”: bradycardia, miosis, enhanced digestion
Zakharyin–Head zones	Referral of visceral pain to the skin
Hypothalamus	Highest autonomic center
Medulla	Adrenaline 80%, noradrenaline 20%; a modified sympathetic ganglion
Kérdö index	(1 − D/P) × 100; positive — sympathetic predominance

TOPIC 27

Physiology of the sensory systems. The visual system. The optical and receptor systems of the eye

Learning objectives

To teach the student the general doctrine of analyzers (sensory systems), the classification and properties of receptors, the structure of the visual system, the optical system of the eye, accommodation and refractive errors, the photoreceptor apparatus of the retina, phototransduction, theories of color vision, and the visual pathways and centers; in laboratory work, to master testing visual acuity, the visual field, color vision and the pupillary reflexes.

By the end of the class the student should know:

- the concept of the analyzer (I.P. Pavlov) and its divisions;
- the classification and properties of receptors (adaptation, sensitivity, receptor potential);
- the psychophysical laws of sensation (Weber–Fechner);
- the optical system of the eye, refractive power and the diopter;
- the mechanism of accommodation and refractive errors;
- the properties of rods and cones and phototransduction;
- the theory of color vision and color blindness;
- dark and light adaptation;
- the visual pathways and centers and disorders of the visual fields.

The student should be able to:

- determine visual acuity with the Sivtsev chart;
- determine the visual field with a perimeter;
- test color vision with the Rabkin plates;
- detect the blind spot with Mariotte's experiment;
- test the reaction of the pupil to light and accommodation.

1. General concept of the sensory systems

An **analyzer** (I.P. Pavlov) is the set of nervous structures that receive, process and analyze stimuli from the external and internal environment. It consists of three divisions:

1. **The peripheral (receptor) division** — converts the energy of the stimulus into a nerve impulse.
2. **The conducting division** — afferent pathways and intermediate centers (brainstem, thalamus).
3. **The central (cortical) division** — the corresponding area of the cerebral cortex; here the **sensation** is formed. In modern physiology the term "**sensory system**" is also used.

Receptors

Table 27.1. *Classification of receptors*

Feature	Types
By location	Exteroceptors (external environment), interoceptors (internal organs), proprioceptors (muscles, tendons, joints)

Feature	Types
By distance of the stimulus	Distant (vision, hearing, smell) and contact (taste, touch)
By nature of the stimulus	Mechanoreceptors, thermoreceptors, chemoreceptors, photoreceptors, nociceptors (pain)
By structure	Primary sensory (the receptor is the sensory neuron itself: touch, smell) and secondary sensory (a specialized receptor cell transmits to a neuron through a synapse: vision, hearing, taste, vestibular)
By rate of adaptation	Rapidly adapting (phasic — touch, smell) and slowly adapting (tonic — pain, proprioceptors, baroreceptors)

Properties of receptors:

- **High sensitivity to the adequate stimulus** (the eye can detect a few photons of light).
- **Transformation** — the energy of the stimulus is converted into a **receptor potential** (a graded, local response) and then into action potentials; the strength of the stimulus is coded by the **frequency** of impulses.
- **Adaptation** — a decrease in sensitivity to a constant stimulus.
- **Rhythmic response** (spontaneous activity). **Weber–Fechner law:** the intensity of sensation is proportional to the **logarithm** of the stimulus strength. Weber's law: the smallest noticeable change in stimulus strength is a constant fraction of the initial strength (for example, ≈ 1/30 for a weight).

2. Structure and optical system of the eye

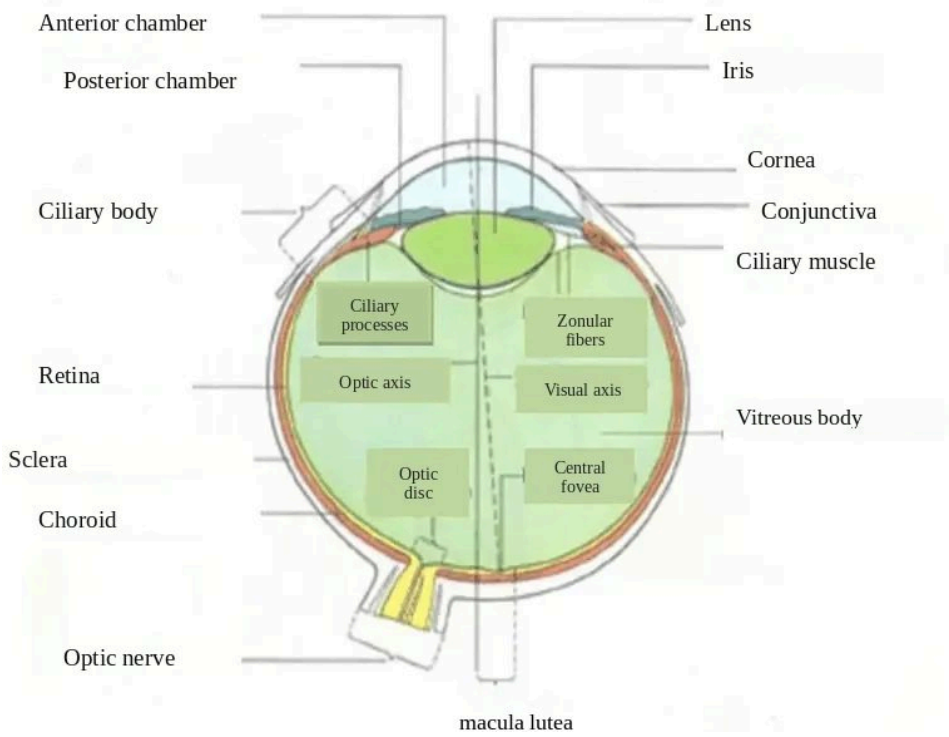


Figure 27.1. *Structure of the eyeball (Alavia O.T., 2018)*

The **optical (refracting) system** of the eye: the **cornea**, the aqueous humor of the anterior chamber, the **lens** and the **vitreous body**. Refractive power is measured in **diopters (D)**: D = 1 / focal length (m).

Table 27.2. *Refractive power of the optical system of the eye*

Structure	Refractive power, D
Cornea	≈ 43 (the main one)
Lens — looking at a distance	≈ 19
Lens — looking at the nearest point	≈ 33
The whole eye (looking at a distance)	≈ 59–60

A **real, reduced and inverted** image is formed in the eye; the cortex perceives it in the upright position.

Accommodation

Accommodation is the adaptation of the eye to see objects at different distances clearly; it is achieved by changing the curvature of the lens.

- **Looking at a distance**: the ciliary muscle is relaxed, the **zonules of Zinn** are taut — the lens is flattened (refracts less).
- **Looking at a near object**: the **ciliary muscle contracts** (parasympathetic — nerve III, M-cholinergic receptors), the zonules of Zinn relax — the lens becomes **convex** owing to its own elasticity (refracts more). At the same time the pupil constricts and the eyes converge (the **near triad**).

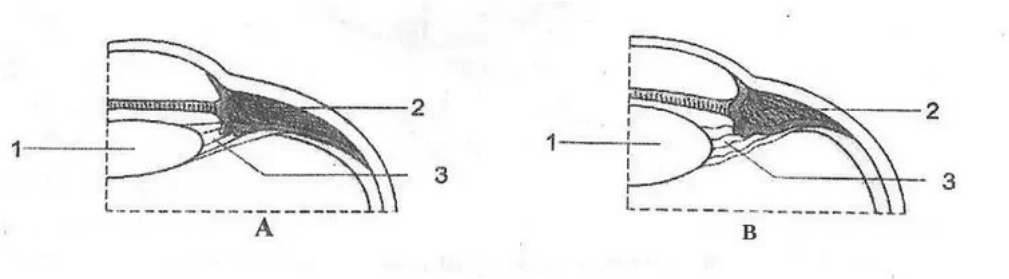


Figure 27.2. *Mechanism of accommodation: A — looking at a distance; B — looking near (1 — lens, 2 — ciliary muscle, 3 — zonules of Zinn) (Alavia O.T., 2018)*

The **near point of clear vision** recedes with age: ≈ 7 cm at 10 years, 10 cm at 20, 20–25 cm at 40, 80–100 cm at 60.

CLINICAL SIGNIFICANCE

Presbyopia (age-related farsightedness) — after 40–45 years the lens loses its elasticity and the ability to accommodate decreases: reading nearby text becomes difficult and it has to be held farther away. Correction — reading glasses with convex (converging) lenses.

Refractive errors

Table 27.3. *Types of refraction*

Type	Essence	Focus	Correction
Emmetropia (normal)	Eye length and refractive power match	On the retina	—
Myopia (nearsightedness)	The eyeball is long or the refractive power is large	In front of the retina	Concave (diverging, “−”) lens
Hypermetropia (farsightedness)	The eyeball is short or the refractive power is small	Behind the retina	Convex (converging, “+”) lens

Type	Essence	Focus	Correction
Astigmatism	The cornea (or lens) has different curvature in different meridians	Several foci	Cylindrical lens

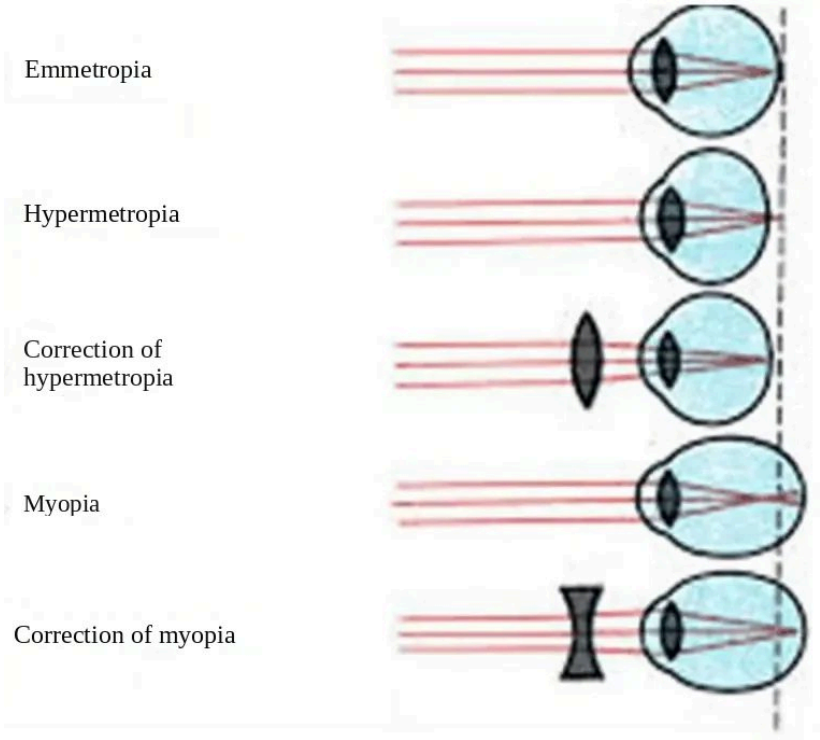


Figure 27.3. *Types of refraction and their correction (Alavia O.T., 2018)*

MEMORY AID
“Myope — Minus”: nearsightedness is corrected with a “−” (concave) lens; **“Hypermetrope — Plus”** — farsightedness with a “+” (convex) lens.

The pupil and its reflexes

The pupil acts as the diaphragm of a camera: it regulates the amount of light entering the eye, reduces spherical aberration and increases the depth of focus. Its diameter is 1.5–8 mm. The **constrictor** muscle is parasympathetic (nerve III), the **dilator** muscle sympathetic (α_1). The pupil constricts in light (direct and crossed reaction) and when looking near; it dilates in darkness, fear and pain.

3. The receptor apparatus of the eye

The **retina** is a 10-layer structure; the photoreceptors are located in its outermost layer (next to the pigment epithelium), and light passes through the other layers of the retina to reach them. Signal pathway: **photoreceptors → bipolar cells → ganglion cells** (whose axons form the optic nerve); horizontal and amacrine cells provide **lateral inhibition** (contrast enhancement).

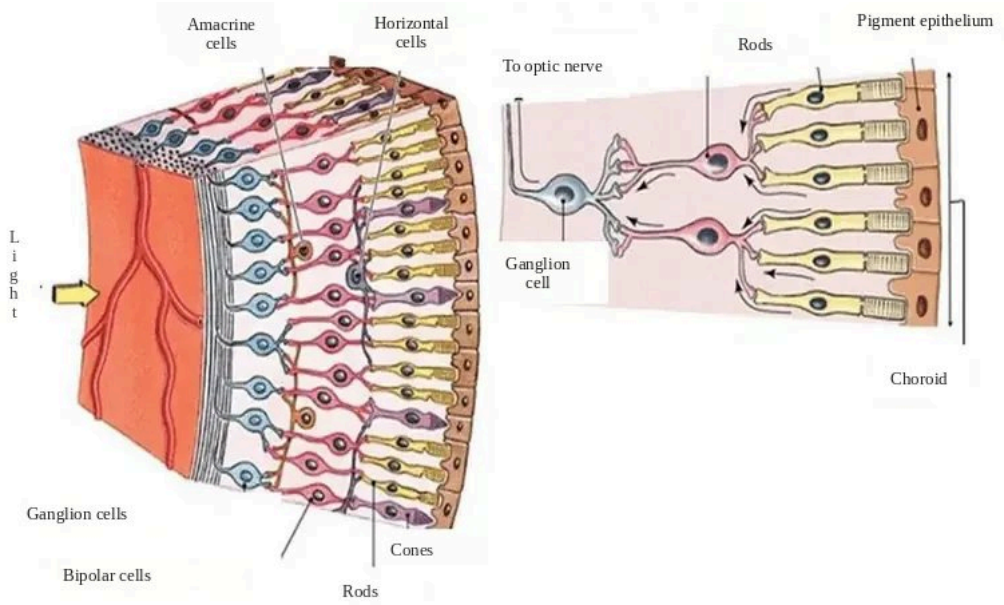


Figure 27.4. *Structure of the retina: photoreceptors (rods and cones), bipolar, ganglion, horizontal and amacrine cells (Alavia O.T., 2018)*

Table 27.4. *Comparative characteristics of rods and cones*

Feature	Rods	Cones
Number	≈ 120 million	≈ 6–7 million
Location	Mainly at the periphery of the retina; absent in the fovea	Mainly in the fovea (macula lutea)
Pigment	Rhodopsin (visual purple)	Iodopsins (3 kinds: blue, green, red)
Sensitivity to light	Very high — vision in darkness (twilight)	Low — daylight vision
Color discrimination	None (black and white)	Present
Visual acuity	Low (many rods to one ganglion cell — convergence)	High (in the fovea 1 cone → 1 bipolar → 1 ganglion cell)
Adaptation	Slow (30–40 minutes to darkness)	Fast

The **blind spot** is the exit point of the optic nerve (the disc); there are no photoreceptors there. The **fovea of the macula lutea** is the site of highest visual acuity.

Phototransduction

1. In darkness the photoreceptors contain much **cGMP**, and the cGMP-gated Na^+ channels are open — the **"dark current"** flows; the membrane is relatively depolarized (−40 mV) and **glutamate** is released continuously.
2. Light converts **11-cis-retinal into all-trans-retinal** in **rhodopsin**; rhodopsin is activated.
3. Activated rhodopsin activates the G protein **transducin → phosphodiesterase** breaks down cGMP.
4. cGMP decreases and the Na^+ channels close — the photoreceptor **hyperpolarizes** (−70 mV) and glutamate release decreases.
5. This change is processed in the bipolar and ganglion cells, and action potentials are generated in the ganglion cells.

REMEMBER

Unlike other receptors, photoreceptors respond to light by **hyperpolarization**. Retinal is formed from **vitamin A** — that is why vitamin A deficiency causes **night blindness** (hemeralopia, "chicken blindness").

Dark and light adaptation

- **Dark adaptation** — on passing from light to darkness, sensitivity increases thousands of times; first through the cones (5–10 minutes), then through the rods (up to 30–40 minutes) — resynthesis of rhodopsin, pupil dilation, neuronal convergence.
- **Light adaptation** is fast (a few minutes): breakdown of rhodopsin, pupil constriction.

Color vision

The three-component theory (T. Young, H. Helmholtz): the retina has three kinds of cones, sensitive to **blue (short-wave, ≈ 420 nm)**, **green (medium, ≈ 530 nm)** and **red (long-wave, ≈ 560 nm)** light. Any color is perceived as a combination of their excitation to different degrees. **The opponent theory** (E. Hering) explains the processing of signals in the ganglion cells and thalamus in opposing pairs (red–green, blue–yellow, white–black); the two theories complement each other.

Table 27.5. *Disorders of color vision*

Disorder	Missing or weak cone	Frequency
Protanopia (protanomaly)	Red	≈ 1–2% of men
Deuteranopia (deuteranomaly)	Green — Daltonism (the most common)	≈ 5–6% of men
Tritanopia	Blue	Very few
Achromasia	All cones	Very rare — only black and white

Red-green color blindness is inherited as an **X-linked recessive** trait — ≈ 8% of men and ≈ 0.5% of women.

CLINICAL SIGNIFICANCE

Disorders of color vision are tested with the **Rabkin polychromatic plates** (or Ishihara plates) in occupational selection for drivers, pilots, railway workers and some medical specialties.

4. Visual pathways and centers

The axons of the ganglion cells form the **optic nerve**. In the **optic chiasm** the fibers from the **nasal (medial) halves** of the retinas cross, while those from the temporal (lateral) halves do not. Thus each **optic tract** carries information about the opposite half of the visual field of both eyes. The fibers go to the **lateral geniculate body** (thalamus) and partly to the superior colliculi of the midbrain (orienting reflexes) and the pretectal area (pupillary reflex). From the lateral geniculate body the **optic radiation** (Gratiolet's bundle) goes to the **primary visual cortex** (area 17, V_1) around the calcarine sulcus of the **occipital lobe**; areas 18–19 are the secondary (associative) visual areas.

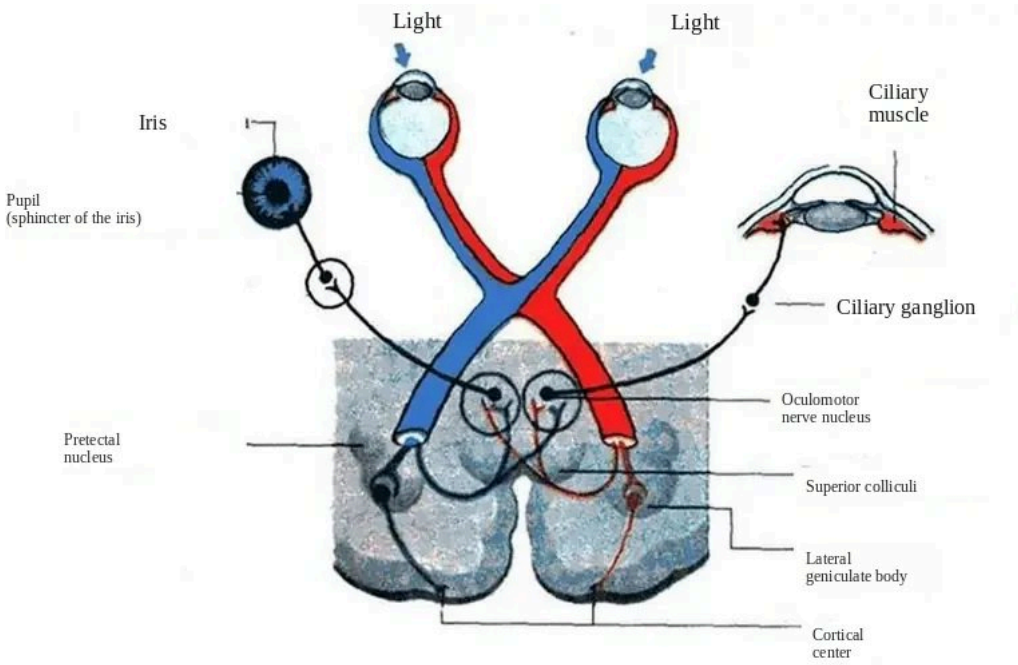


Figure 27.5. *Visual pathways: optic nerve, chiasm, lateral geniculate body, midbrain and cortical centers, pathway of the pupillary reflex (Alavia O.T., 2018)*

Table 27.6. *Level of damage to the visual pathway and the visual field defect*

Site of damage	Result
One optic nerve	Blindness of that eye (amaurosis)
Center of the chiasm (pituitary tumor)	Bitemporal hemianopia — the temporal fields of both eyes are lost
Optic tract, radiation or cortex (on one side)	Homonymous hemianopia — the half of the field on the opposite side is lost in both eyes

In the cortex visual information is processed in two parallel streams: the **dorsal stream** (to the parietal lobe — “where?”: movement, spatial location) and the **ventral stream** (to the temporal lobe — “what?”: shape, color, face recognition).

Binocular vision is vision with both eyes: depth (stereoscopic) perception and judgment of distance; the images of the two eyes are fused in the cortex.

5. Laboratory work

Exercise 1: determining visual acuity (Sivtsev chart)

1. The subject sits **5 m** from a well-lit chart; one eye is covered with an occluder.
2. The examiner points with a pointer to the letters on the chart from top to bottom; the subject reads them.
3. Write down the V number beside the lowest line the subject can read correctly; the second eye is also tested.

Interpretation: Snellen formula: V = d / D, where d is the test distance (5 m) and D the distance at which a normal eye should read that line. If line 10 is read from 5 m, V = 1.0 (normal). Visual acuity is the reciprocal of the smallest angle needed to see two points separately (normally 1 minute of arc).

Exercise 2: determining the visual field (perimetry)

1. The subject rests the chin on the support in front of the perimeter, with one eye closed, and looks at the central mark with the other.
2. A white (then red, green, blue) object is moved slowly along the perimeter arc from the periphery toward the center.
3. The angle at which the object is first seen is marked on the form; the arc is rotated every 15–30° and the test is repeated. **Interpretation:** normal limits for white: 90° outward, 60° inward, 50–55° upward, 65–70° downward. The field for colored objects is narrower (blue > red > green), because the cones are mainly in the center of the retina.

Exercise 3: detecting the blind spot (Mariotte's experiment)

1. Draw a cross on the left of a sheet of paper and a black circle 8–10 cm to its right.
2. Close the left eye and look at the cross with the right eye.
3. Slowly bring the paper closer to the eye and then move it away. **Result and analysis:** at a certain distance (≈ 20–25 cm) the circle disappears — its image falls on the optic disc (the blind spot), where there are no photoreceptors. In everyday life the blind spot is not noticed because of binocular vision, the constant small movements of the eyes and the “filling in” of the image by the cortex.

Exercise 4: testing color vision (Rabkin plates)

1. The subject sits 1 m from the plates in daylight.
2. Show each plate for 5–7 s; the subject names the number or shape on it.
3. Compare the results with the key in the instructions. **Interpretation:** if all plates are read correctly — normal trichromacy; errors on certain plates identify the type of protanopia or deuteranopia (anomaly).

Exercise 5: testing the pupillary reaction and accommodation

1. Test the direct and crossed reaction of the pupil to light (topic 24).
2. **Reaction to accommodation and convergence**: the subject looks into the distance, then at a finger 10–15 cm away — the pupils constrict and the eyes turn inward.
3. Determine the **near point of clear vision**: slowly bring small print toward the eye and measure the distance at which the letters become blurred.

6. Practical skills

Skill: physiological explanation of a refractive error

1. Determine at which distance the subject sees poorly (far or near).
2. Show in a drawing whether the focus is in front of the retina (myopia) or behind it (hypermetropia).
3. Justify the type of lens needed (concave or convex).
4. If the person is over 40, assess the likelihood of presbyopia.

Skill: determining the site of damage from the visual field defect

1. Determine whether the visual field defect is in one eye or both.
2. Complete blindness of one eye — the optic nerve.

3. The temporal fields of both eyes — the chiasm (pituitary tumor).
4. The field on one side in both eyes — the tract, radiation or occipital cortex on the opposite side.

Review questions

1. What is an analyzer, and what divisions does it consist of?
2. How are receptors classified?
3. What is the difference between primary and secondary sensory receptors?
4. Explain receptor adaptation and the Weber–Fechner law.
5. Describe the optical system of the eye and its refractive power.
6. How is accommodation carried out?
7. What is presbyopia?
8. How are myopia, hypermetropia and astigmatism corrected?
9. What functions does the pupil perform, and which muscles control it?
10. List the differences between rods and cones.
11. Explain the stages of phototransduction.
12. How does dark adaptation take place?
13. Explain the theories of color vision. What is Daltonism?
14. Describe the visual pathways and centers.
15. How does the visual field change when different parts of the visual pathway are damaged?

Test questions

1. Which is the main refracting medium of the eye?
   A) Cornea
   B) The lens
   C) The vitreous body
   D) The aqueous humor
   E) The retina

2. How does the lens change when looking at a near object?
   A) It flattens
   B) It does not change
   C) It becomes convex (the ciliary muscle contracts, the zonules of Zinn relax)
   D) It moves forward
   E) It becomes cloudy

3. Where is the focus formed in myopia, and how is it corrected?
   A) Behind the retina; convex lens
   B) In front of the retina; concave lens
   C) On the retina; no correction needed
   D) Several foci; cylindrical lens
   E) In the lens; convex lens

4. Which is the visual pigment of rods?
    A) Iodopsin
    B) Melanin
    C) Hemoglobin
    D) Myoglobin
    E) Rhodopsin

5. Where in the retina is visual acuity highest?
    A) In the blind spot
    B) At the periphery of the retina
    C) In the ciliary body
    D) In the fovea of the macula lutea
    E) In the iris

6. How do photoreceptors respond to light?
    A) Depolarization
    B) Action potential
    C) Hyperpolarization
    D) They do not respond
    E) Ca^{2+} entry

7. Deficiency of which vitamin causes night blindness (hemeralopia)?
    A) Vitamin A
    B) Vitamin C
    C) Vitamin D
    D) Vitamin B_{12}
    E) Vitamin K

8. What is the blind spot?
    A) The macula lutea
    B) The center of the iris
    C) A cloudy area of the lens
    D) The exit point of the optic nerve (no photoreceptors)
    E) The location of the ciliary muscle

9. In Daltonism (deuteranopia), the function of which cones is impaired?
    A) Red
    B) Blue
    C) Green
    D) All cones
    E) Rods

10. What visual field defect is observed when a pituitary tumor compresses the chiasm?
    A) Blindness of one eye
    B) Homonymous hemianopia
    C) Loss of central vision only
    D) Loss of color vision
    E) Bitemporal hemianopia

11. Where is the primary visual cortex located?
    A) In the temporal lobe
    B) In the occipital lobe (area 17)
    C) In the parietal lobe
    D) In the frontal lobe
    E) In the insula

12. With which chart is visual acuity determined?
    A) The Rabkin chart
    B) A perimeter
    C) The Ishihara chart
    D) The Sivtsev chart
    E) The Bárány chair

13. By which nerve does the ciliary muscle contract in accommodation?
    A) Sympathetic nerves (α_1)
    B) Parasympathetic fibers of the oculomotor nerve (III)
    C) The optic nerve (II)
    D) The trigeminal nerve (V)
    E) The abducens nerve (VI)

14. In which division of the analyzer is sensation formed?
    A) In the cortical (central) division
    B) In the receptors
    C) In the afferent nerves
    D) In the spinal cord
    E) Only in the thalamus

15. How long does complete dark adaptation take?
    A) 1–2 seconds
    B) 1 minute
    C) 5 hours
    D) 24 hours
    E) 30–40 minutes

Clinical cases

Case 1

A 46-year-old teacher complains that recently he has had to read text in books and on his phone with his arm outstretched, holding it farther away. He sees distant objects as well as before.

Questions: What is the condition? Explain its mechanism and name the method of correction.

Answer: Presbyopia — an age-related decline in the ability to accommodate. To see a near object the ciliary muscle must contract, the zonules of Zinn relax and the lens become convex owing to its own elasticity. With age the lens becomes denser and loses its elasticity — it cannot become sufficiently convex, and the near point of clear vision recedes (more than 20–25 cm at 40 years). Distance vision (without accommodation) does not change. Correction — reading glasses with convex (converging, “+”) lenses.

Case 2

A 35-year-old woman has headaches, menstrual irregularity and milk discharge from the breasts. She says that on the road she fails to see cars coming from the side. On perimetry the temporal fields of both eyes are lost.

Questions: What is the type of visual field defect, and what is its cause? Explain on the basis of the structure of the visual pathways.

Answer: Bitemporal hemianopia. Here the central part of the optic chiasm is damaged — where the fibers from the nasal (medial) halves of the retinas cross. The nasal half of the retina receives the temporal (lateral) part of the visual field, so the temporal fields of both eyes are lost. Because the chiasm lies directly above the pituitary gland, the most common cause is a pituitary tumor; in this patient it may be a prolactin-producing tumor (prolactinoma): hyperprolactinemia leads to milk discharge (galactorrhea) and menstrual disturbance.

Case 3

A young man undergoing a medical examination for a driving license could not see many of the numbers on the Rabkin plates, especially those made of red and green dots.

Questions: What is the condition? Explain its cause and inheritance.

Answer: A red-green color vision defect (protanopia/protanomaly or deuteranopia/deuteranomaly — Daltonism). According to the three-component theory, colors are perceived through the excitation to different degrees of three kinds of cones (blue, green, red); if the red or green cone pigment is absent or altered, red and green cannot be distinguished. The genes for the green and red cone pigments are located on the X chromosome, and the defect is inherited as an X-linked recessive trait: it is common in men (who have one X chromosome) (≈ 8%), while women are most often carriers. Such people distinguish traffic lights not by color but by position and brightness; this is taken into account in occupational selection.

Quick review

Table 27.7. *Key concepts of topic 27*

Concept	Brief definition
Analyzer	Receptor, conducting and cortical divisions (Pavlov)
Weber–Fechner law	Sensation is proportional to the logarithm of the stimulus
Refractive power	Eye ≈ 60 D; cornea ≈ 43 D
Accommodation	Near — the ciliary muscle contracts, the lens becomes convex
Presbyopia	Age-related decline in accommodation
Myopia / hypermetropia	Focus in front of the retina (“−” lens) / behind it (“+” lens)
Rods	120 million; rhodopsin; vision in darkness
Cones	6–7 million; in the macula lutea; color and sharp vision
Phototransduction	Isomerization of retinal → transducin → cGMP ↓ → hyperpolarization
Vitamin A	Source of retinal; deficiency causes night blindness
Three-component theory	Blue, green and red cones (Young–Helmholtz)
Daltonism	Red-green blindness; X-linked recessive

Concept	Brief definition
Chiasm	The nasal-half fibers cross; its damage — bitemporal hemianopia
Visual cortex	Occipital lobe, area 17
The Sivtsev chart	Visual acuity at 5 m; normal V = 1.0

TOPIC 28

Physiology of the auditory system and the vestibular apparatus

Learning objectives

To teach the student the structure of the auditory system, the conduction of sound in the outer, middle and inner ear, mechanoelectrical transduction in the organ of Corti, the theories of coding pitch and loudness, the auditory pathways and centers, binaural hearing, and the structure and receptors of the vestibular apparatus, vestibular reflexes and testing of vestibular excitability; in laboratory work, to master testing hearing acuity and bone and air conduction (Rinne and Weber tests).

By the end of the class the student should know:

- the structure and functions of the outer, middle and inner ear;
- the sound-amplifying and protective mechanisms of the middle ear;
- the structure of the cochlea, the composition of endolymph and perilymph, the endocochlear potential;
- mechanoelectrical transduction in hair cells;
- the theories of coding pitch (frequency) and loudness;
- the limits of hearing, the auditory pathways and centers, sound localization;
- the structure of the vestibular apparatus (otolith organs and semicircular canals) and their adequate stimuli;
- vestibular reflexes and vestibulo-autonomic reactions.

The student should be able to:

- determine hearing acuity with whispered speech;
- perform and evaluate the Rinne and Weber tests with a tuning fork;
- read an audiogram;
- evaluate the excitability of the vestibular apparatus with tests.

1. Structure of the auditory system

Sound is mechanical vibration propagating in air (or another medium). Its physical characteristics: **frequency** (Hz) — perceived as pitch (low or high); **amplitude (intensity)** — perceived as loudness, measured in **decibels (dB)**; **timbre** — the composition of harmonics.

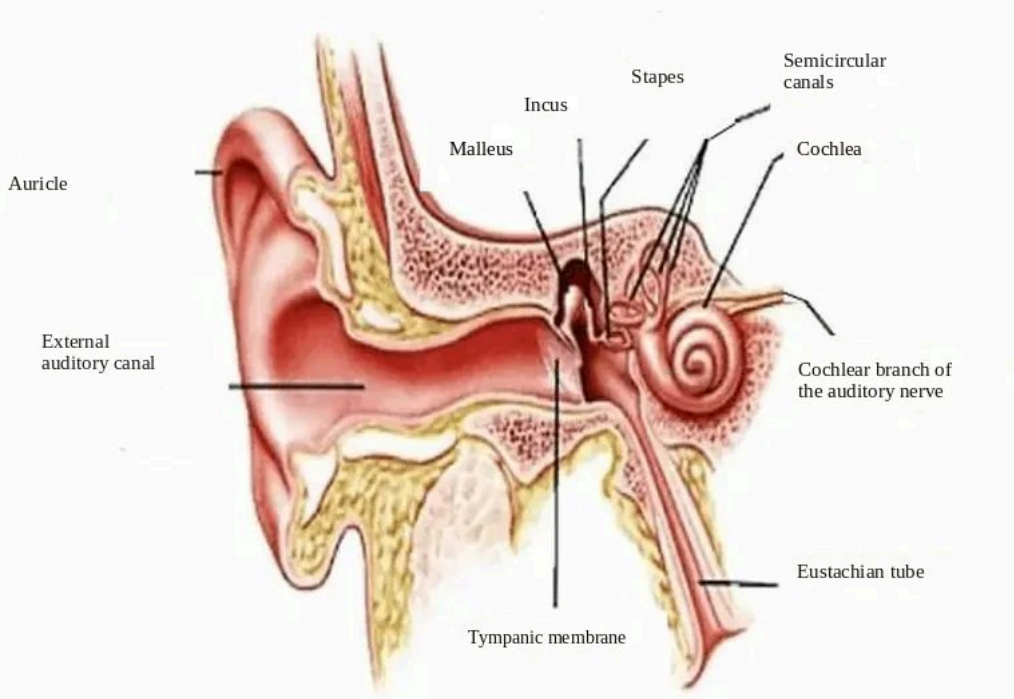


Figure 28.1. *Structure of the ear: outer, middle and inner ear (Alavia O.T., 2018)*

Table 28.1. *Parts of the ear and their functions*

Branch	Structures	Function
Outer ear	Auricle, external auditory canal	Collecting and directing sound, helping to localize the sound source, resonance (2–4 kHz), protection (earwax)
Middle ear	Tympanic membrane, auditory ossicles (malleus, incus, stapes), muscles, Eustachian tube	Transmitting sound from air to fluid and amplifying it; protection
Inner ear	Cochlea (hearing), vestibule and semicircular canals (vestibular)	Receptor apparatus

Functions of the middle ear

1. **Amplification of sound pressure (impedance matching)** — when sound passes from air into fluid, 99.9% of its energy is reflected. The middle ear overcomes this: the area of the tympanic membrane (≈ 55 mm^2) is ≈ 17 times larger than that of the oval window (≈ 3.2 mm^2), and with the lever system of the ossicles (≈ 1.3 times) the pressure increases **≈ 20–22 times**.
2. **Protection** — with loud sound (> 70–80 dB) the **tensor tympani** (nerve V) and the **stapedius muscle** (nerve VII) contract reflexly and limit the movement of the ossicles (the **acoustic reflex**; latency 10–40 ms — it does not protect against sudden explosions).
3. **Pressure equalization** — the **Eustachian (auditory) tube** opens during swallowing and yawning and equalizes the pressure in the middle ear with atmospheric pressure.

CLINICAL SIGNIFICANCE

When a plane is landing or when diving, the external pressure changes; if the Eustachian tube does not open (cold, inflammation), the tympanic membrane is pressed and pain and hearing loss occur (barotrauma). Swallowing, yawning or the Valsalva maneuver help. In children the Eustachian tube is short and wide, so pharyngeal infections quickly spread to the middle ear (**otitis**).

Bone conduction

Sound reaches the inner ear **through the air** (the main route) and **through vibration of the skull bones** (bone conduction). The fact that we hear our own voice differently on a recording is due to the contribution of bone conduction.

2. The inner ear: the cochlea and the organ of Corti

The **cochlea** is a bony canal coiled 2.5 times; it contains three cavities: the **scala vestibuli** (connected with the oval window) and the **scala tympani** (with the round window) — filled with **perilymph** (rich in Na^+, similar to extracellular fluid); and the **scala media (cochlear duct)** — filled with **endolymph** (**rich in K^+**, ≈ 150 mmol/l). The endolymph has a positive charge of **+80 mV** relative to the perilymph (the **endocochlear potential**; generated by the stria vascularis).

The **organ of Corti (spiral organ)** lies on the basilar membrane: **inner hair cells** (≈ 3500, one row — the main sensory receptors; 90–95% of the afferent fibers come to them) and **outer hair cells** (≈ 12,000, 3–4 rows — the **cochlear amplifier**: by contracting they enhance the vibration of the basilar membrane and increase sensitivity). Above the stereocilia of the hair cells lies the **tectorial membrane**.

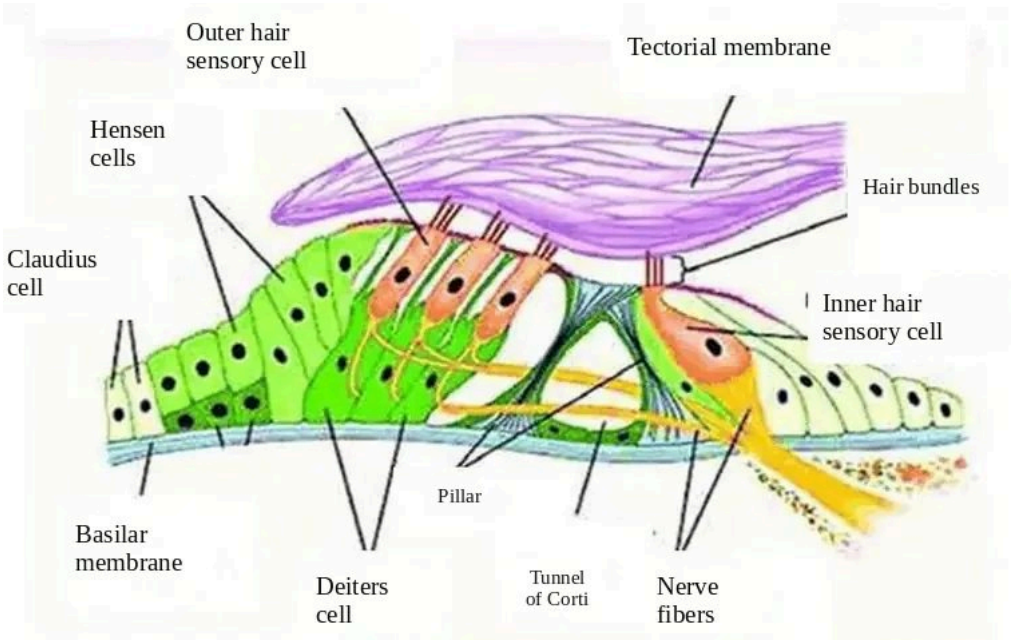


Figure 28.2. *Structure of the organ of Corti (Alavia O.T., 2018)*

Mechanoelectrical transduction

When the basilar membrane vibrates, the stereocilia of the hair cells shift relative to the tectorial membrane. When the stereocilia bend **toward the tallest stereocilium**, the tip links at their ends open the mechanosensitive channels and **K^+ enters the cell from the endolymph** (because of the high K^+ and the large electrical gradient) — the cell **depolarizes** (receptor potential), Ca^{2+} channels open, the mediator (**glutamate**) is released and action potentials arise in the neurons of the spiral ganglion. Bending in the opposite direction causes hyperpolarization.

> **REMEMBER**
> In hair cells depolarization is caused not by Na^+, as in other cells, but by **K^+ entry** — because the stereocilia are immersed in K^+-rich endolymph. This is a special feature of the auditory and vestibular receptors.

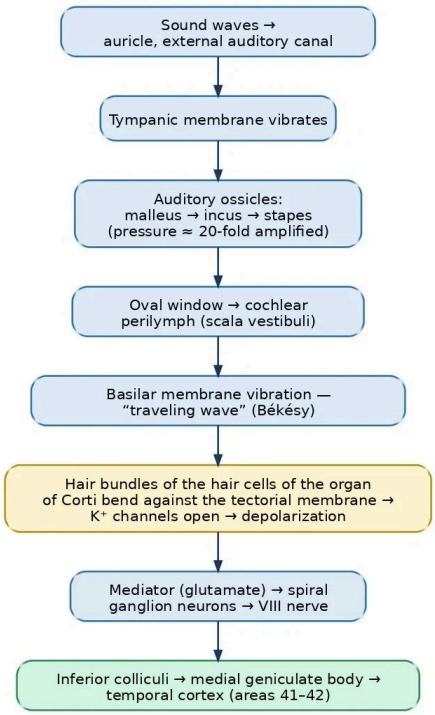


Figure 28.3. *Stages of the conduction and reception of sound*

Coding of pitch and loudness

- **The place (resonance) theory** (H. Helmholtz; the modern **"traveling wave"** theory — G. von Békésy, Nobel Prize 1961): the width and stiffness of the basilar membrane change from its base to its apex. **High-frequency** sounds produce maximal vibration at the **base** of the membrane (next to the oval window, the narrow and stiff part), and **low-frequency** sounds at the **apex** (the wide and soft part). Each frequency excites a particular group of hair cells — **tonotopy** (preserved in the auditory pathways and cortex).
- **The frequency (telephone, volley) theory** — at low frequencies (< 1000–4000 Hz) neurons fire impulses at a rhythm matching the frequency of the sound (several neurons in turn — the "volley" principle).
- **Loudness** is coded by the number of excited neurons and the frequency of impulses.

Limits of hearing

Humans hear sounds **from 16–20 Hz to 20,000 Hz** (the upper limit falls with age — 10–12 kHz at 60). The highest sensitivity is in the **1000–4000 Hz** range (speech frequencies 300–4000 Hz). Below 16 Hz — **infrasound**; above 20 kHz — **ultrasound**.

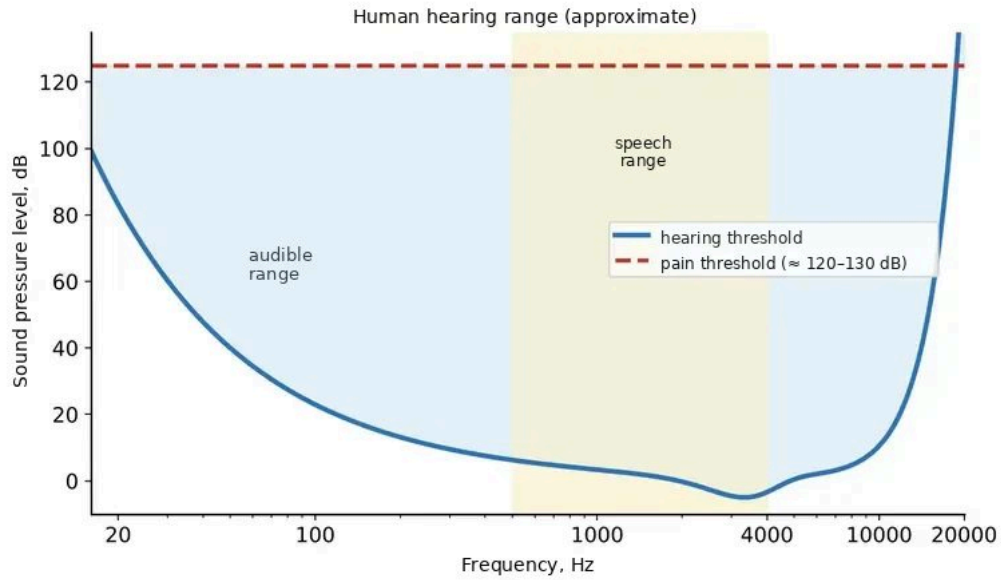


Figure 28.4. *The human hearing area: thresholds of hearing and pain (approximate)*

Table 28.2. *Sound intensity levels (approximate)*

Sound	dB
Threshold of hearing	0
Whisper	20–30
Ordinary conversation	50–60
Street traffic	70–80
Rock concert, loud music in headphones	100–110
Threshold of pain, near a jet engine	120–140

CLINICAL SIGNIFICANCE

Noise-induced hearing loss — prolonged exposure to noise above 85 dB irreversibly damages the outer hair cells (first around 4000 Hz). Listening to loud music in headphones is increasing this problem among young people. Prevention — protective equipment, limiting volume. Aminoglycoside antibiotics (gentamicin) also have an **ototoxic** effect on hair cells.

3. Auditory pathways and centers

Axons of the spiral ganglion neurons (the cochlear part of nerve VIII) → **cochlear nuclei** in the medulla oblongata → **superior olivary complex** (pons; information from the two ears is compared for the first time) → lateral lemniscus → **inferior colliculi** (midbrain; auditory orienting reflexes) → **medial geniculate body** (thalamus) → **primary auditory cortex** (temporal lobe, Heschl's gyri, **areas 41–42**). The pathways partly cross, so each hemisphere receives information **from both ears** — unilateral cortical damage does not cause deafness.

Binaural hearing and sound localization — determining the direction of a sound source. Mechanisms: the **difference in arrival time** of the sound at the two ears (for low frequencies; differences down to 10 µs are detected) and the **difference in intensity** (for high frequencies — the head creates a “shadow”); these differences are analyzed by the superior olivary complex.

4. The vestibular apparatus

The **vestibular apparatus** is the organ that senses the position of the head in space and its movement (acceleration). It consists of the **vestibule** (otolith organs — the **utricle** and **saccule**) and three **semicircular canals** located in mutually perpendicular planes; all are filled with endolymph.

Table 28.3. *Receptors of the vestibular apparatus*

Structure	Receptor structure	Adequate stimulus
Utricle — horizontal	Macula: an otolith membrane (calcium carbonate crystals — otoliths) above the hair cells	Tilting of the head (gravity) and horizontal linear acceleration
Saccule — vertical	Macula and otoliths	Vertical linear acceleration (elevator)
Semicircular canals (3)	The cupula (a gelatinous partition) in the ampullae and the hair cells in it	Angular (rotational) acceleration — the start and end of head rotation

Hair cells have several stereocilia and one long **kinocilium**. When the stereocilia bend toward the kinocilium — depolarization (impulses increase); in the opposite direction — hyperpolarization (impulses decrease). Even at rest there is constant spontaneous firing.

How the semicircular canals work: when rotation starts, the endolymph lags behind the canal wall because of inertia and bends the cupula in the direction **opposite** to the rotation; during uniform rotation the endolymph moves together with the canal — the cupula returns to its initial position; on a sudden stop the endolymph continues to move and bends the cupula in the **opposite** direction — **post-rotational sensations** (a feeling of rotating the other way) and **nystagmus** appear.

Vestibular pathways: vestibular ganglion neurons → **vestibular nuclei** (medulla oblongata and pons) → the cerebellum (flocculonodular lobe), the spinal cord (vestibulospinal tract — extensor muscle tone), the nuclei of the oculomotor nerves (medial longitudinal fasciculus), the thalamus → cortex (parietal and temporal lobes), autonomic centers, the reticular formation.

Table 28.4. *Vestibular reactions*

Reaction	Content
Vestibulo-ocular reflex	When the head moves, the eyes move in the opposite direction, keeping the image stable on the retina; nystagmus
Vestibulospinal (static and statokinetic) reflexes	Maintaining balance and posture by redistributing muscle tone (topic 24)
Vestibulo-autonomic reactions	With strong or prolonged stimulation — nausea, vomiting, pallor, sweating, changes in HR and pressure — motion sickness (sea, air, car sickness)
Vestibulo-sensory reactions	Dizziness, sensation of movement

CLINICAL SIGNIFICANCE

Motion sickness (kinetosis) develops when there is a mismatch between vestibular, visual and proprioceptive information (for example, inside a ship — the eye sees a motionless room while the vestibular apparatus senses rocking). **Benign positional vertigo** (the most common form of dizziness) — otoliths falling into a semicircular canal; brief severe dizziness when the head position changes; treated with special maneuvers (the Epley maneuver).

5. Laboratory work

Exercise 1: determining hearing acuity with speech

1. The subject stands sideways to the examiner, closes the ear not being tested with a finger and closes the eyes (so as not to lip-read).
2. After expiration, using the residual air, the examiner **whispers** two-digit numbers (for example, 21, 47); starting from 6 m, he comes closer if the subject cannot repeat them.
3. Write down the greatest distance at which the subject repeated correctly; the other ear is also tested. **Interpretation:** normally whispered speech is heard from **6 m** or more, and ordinary conversational speech up to 20 m.

Exercise 2: the Rinne test

1. Place the stem of a vibrating tuning fork (128 or 512 Hz) on the mastoid process (bone conduction).
2. As soon as the subject stops hearing the sound, bring the tuning fork close to the ear canal (1–2 cm away) (air conduction).
3. Ask whether the sound is heard again. **Interpretation:** normally air conduction lasts longer than bone conduction (≈ 2 times) — **Rinne positive**. When the **sound-conducting** apparatus (outer and middle ear) is damaged, bone conduction is longer — **Rinne negative**. When the **sound-perceiving** apparatus (inner ear, nerve) is damaged, both are shortened but the ratio is preserved — Rinne positive.

Exercise 3: the Weber test (lateralization)

1. Place a vibrating tuning fork on the top of the head (in the midline).
2. Ask in which ear the sound is heard louder.
3. Close one ear with a finger and repeat the test. **Result and analysis:** normally the sound is heard equally in both ears (in the middle of the head). When the ear is closed or **conduction** is impaired (otitis, earwax plug), the sound lateralizes **toward the affected (closed) ear** — external noise does not act there and the sound coming through the bone is heard better. When **perception** is impaired, the sound lateralizes **toward the healthy ear**.

Exercise 4: testing the excitability of the vestibular apparatus

1. **Bárány rotation test** — as in topic 24: after 10 rotations in 20 s, the duration of nystagmus (normal 25–35 s), autonomic reactions and loss of balance are evaluated.
2. **Yashchenko (Voyachek) test** — after rotation, standing with the head bent forward and then walking along a straight line with the eyes open; the degree of deviation is evaluated.
3. Stability **in the Romberg position** is evaluated. **Interpretation:** prolonged nystagmus, severe nausea, pallor and sweating — high vestibular excitability (predisposition to motion sickness); important in occupational selection (pilots, cosmonauts, sailors).

6. Practical skills

Skill: determining the type of hearing loss from tuning fork tests

1. Perform the Rinne test on both ears.
2. Perform the Weber test.

3. Rinne negative + Weber toward the affected ear — **conductive** hearing loss (outer or middle ear).
4. Rinne positive (both shortened) + Weber toward the healthy ear — **sensorineural** hearing loss (inner ear or auditory nerve).
5. State that the result needs to be confirmed by audiometry.

Skill: reading an audiogram

1. Note that the horizontal axis shows frequency (125–8000 Hz) and the vertical axis the hearing threshold (dB, increasing from top to bottom).
2. Find the air conduction (o — right, x — left) and bone conduction lines.
3. Thresholds of 0–20 dB — normal; 20–40 — mild, 40–70 — moderate, 70–90 — severe, > 90 dB — profound hearing loss.
4. If there is a gap between the air and bone lines — conductive; if both are lowered together — a sensorineural disorder.

Review questions

1. What are the physical characteristics of sound and how are they perceived?
2. What are the functions of the outer ear?
3. How does the middle ear amplify sound?
4. What is the acoustic reflex, and what is its significance?
5. What is the function of the Eustachian tube?
6. Describe the structure of the cochlea and the composition of endolymph and perilymph.
7. What are the functions of the inner and outer hair cells?
8. How does mechanoelectrical transduction take place in hair cells?
9. Explain the place (traveling wave) theory of pitch coding.
10. What are the limits of human hearing?
11. Describe the auditory pathways and centers.
12. How is the direction of a sound source determined?
13. How is the vestibular apparatus structured, and what stimuli do its parts sense?
14. How do the semicircular canals sense rotation?
15. Explain the vestibular reflexes and the mechanism of motion sickness.

Test questions

1. Approximately how many times does the middle ear amplify sound pressure?
   A) 2 times
   B) 20–22 times
   C) 100 times
   D) 1000 times
   E) It does not amplify

2. What is special about the composition of endolymph?
   A) Much Na^+, little K^+
   B) Rich in proteins
   C) Very much Ca^{2+}
   D) Much K^+ (≈ 150 mmol/l)
   E) The same as extracellular fluid

3. Which ion causes depolarization in hair cells?
   A) Na^+
   B) Cl^-
   C) K^+ (entering from the endolymph)
   D) Mg^{2+}
   E) HCO_3^-

4. In which part of the basilar membrane do high-frequency sounds produce maximal vibration?
   A) At the base (next to the oval window)
   B) At the apex
   C) In the middle
   D) Equally over the whole membrane
   E) In the tectorial membrane

5. Which are the main sensory receptors?
   A) Outer hair cells
   B) Supporting cells
   C) Vestibular cells
   D) Spiral ganglion neurons
   E) Inner hair cells

6. What frequency range of sounds does a human hear?
   A) 1–100 Hz
   B) 100–1000 Hz
   C) From 16–20 Hz to 20,000 Hz
   D) 20,000–100,000 Hz
   E) 5–50 Hz

7. Where is the primary auditory cortex located?
   A) In the occipital lobe
   B) In the frontal lobe
   C) In the parietal lobe
   D) In the insula
   E) In the temporal lobe (areas 41–42)

8. Which nerve innervates the tensor tympani muscle?
   A) The facial nerve
   B) The trigeminal nerve
   C) Vagus nerve
   D) The vestibulocochlear nerve
   E) The glossopharyngeal nerve

9. What is the adequate stimulus of the semicircular canals?
   A) Angular (rotational) acceleration
   B) Gravity
   C) Vertical linear acceleration
   D) Sound
   E) Temperature

10. What stimulus does the saccule sense?
   A) Rotation
   B) Sound
   C) Vertical linear acceleration
   D) Light
   E) Temperature

11. What does a negative Rinne test indicate?
   A) Normal value
   B) Damage to the inner ear
   C) Damage to the auditory nerve
   D) Impairment of the sound-conducting apparatus (outer or middle ear)
   E) Damage to the auditory cortex

12. If in the Weber test the sound lateralizes toward the healthy ear, what does this indicate?
   A) Normal value
   B) An earwax plug
   C) Middle ear inflammation
   D) Perforation of the tympanic membrane
   E) A sensorineural (sound-perceiving) disorder

13. Why does the acoustic reflex not protect against a sudden explosion?
   A) The muscles are too strong
   B) Its latency (10–40 ms) is long
   C) It responds only to low sounds
   D) It works in only one ear
   E) The reflex does not exist

14. In determining the direction of a sound source, where is information from the two ears compared for the first time?
   A) In the superior olivary complex
   B) In the cochlear nuclei
   C) In the spiral ganglion
   D) Only in the auditory cortex
   E) In the cerebellum

15. From what distance should whispered speech normally be heard?
   A) 0.5 m
   B) 1 m
   C) 2 m
   D) 6 m or more
   E) 50 m

Clinical cases

Case 1

After a cold a child's right ear began to hurt, his hearing decreased and his temperature rose. The Rinne test is negative on the right and positive on the left; in the Weber test the sound is heard louder in the right ear.

Questions: What is the type and cause of the hearing loss? Explain the tuning fork tests.

Answer: Acute right-sided otitis media — conductive hearing loss. In children the Eustachian tube is short and wide, so infection easily passes from the nasopharynx to the middle ear; fluid accumulates in the middle ear and restricts the movement of the tympanic membrane and ossicles — air conduction of sound is impaired. Rinne negative: bone conduction (directly to the inner ear) is heard longer than air conduction. Weber lateralizes toward the affected ear: external noise does not enter the affected ear, and the sound coming through the bone is perceived as relatively louder; the inner ear is healthy.

Case 2

A patient treated with gentamicin for a long time developed ringing in the ears, bilateral hearing loss and unsteadiness when walking. Rinne is positive on both sides, and there is no Weber lateralization.

Questions: What disorder has developed? Explain its mechanism.

Answer: The ototoxic (and vestibulotoxic) effect of aminoglycosides — bilateral sensorineural hearing loss. Gentamicin accumulates in the endolymph and damages the hair cells (first the outer hair cells at the base of the cochlea, which receive high frequencies) and the hair cells of the vestibular apparatus. Because the sound-conducting apparatus is healthy, Rinne is positive (air and bone conduction are shortened equally); because the damage is symmetrical, there is no lateralization in the Weber test. Damage to the vestibular hair cells leads to loss of balance. That is why doses and blood concentrations are monitored during treatment with aminoglycosides.

Case 3

A passenger traveling by ship for the first time developed nausea, vomiting, pallor and cold sweat while sitting in the cabin. When he went out on deck and looked at the horizon, he felt better.

Questions: What is the condition? Explain its mechanism and why looking at the horizon helps.

Answer: Motion sickness (seasickness). When the ship rocks, the vestibular apparatus (otolith organs and semicircular canals) continuously registers changing accelerations, while in the cabin the eye sees motionless surroundings — a mismatch arises between vestibular, visual and proprioceptive information. Through the connections of the vestibular nuclei with the autonomic centers (the vomiting center, vagal nuclei), vestibulo-autonomic reactions develop: nausea, vomiting, pallor, sweating. When looking at the horizon, visual information matches vestibular information (the eye also sees the rocking) — the mismatch decreases and the symptoms ease.

Quick review

Table 28.5. *Key concepts of topic 28*

Concept	Brief definition
Middle ear	Amplifies pressure 20–22 times; acoustic reflex; Eustachian tube
Endolymph	Much K^+; +80 mV endocochlear potential

Concept	Brief definition
Inner hair cells	Main receptors
Outer hair cells	Cochlear amplifier
Transduction	Bending of stereocilia → K^+ entry → depolarization → glutamate
Place theory	High frequency — at the base, low — at the apex (Békésy)
Limits of hearing	16–20 Hz – 20 kHz; most sensitive at 1–4 kHz
Auditory cortex	Temporal lobe, areas 41–42
Sound localization	Time and intensity differences; superior olivary complex
Utricle and saccule	Otoliths; gravity and linear acceleration
The semicircular canals	Cupula; angular acceleration
Nystagmus	Vestibulo-ocular reflex; 25–35 s in the Bárány test
Rinne test	Air > bone — normal; negative — conduction disorder
Weber test	Toward the affected ear — conductive; toward the healthy side — sensorineural
Motion sickness	Mismatch of sensory information; vestibulo-autonomic reaction

TOPIC 29

The taste and olfactory systems. The pain sensory system. Determining the pain threshold — algometry

Learning objectives

To teach the student the structure, receptors, transduction mechanisms, conducting pathways and centers of the taste and olfactory systems and the features of taste and smell, the types of pain, the nociceptive and antinociceptive (anti-pain) systems, and the theories of pain control (the gate theory, endogenous opioids); in laboratory work, to master determining the taste threshold, the taste map and the pain threshold (algometry).

By the end of the class the student should know:

- the structure of the taste papillae and buds, the basic taste qualities and their transduction mechanisms;
- the taste pathways and centers and the adaptation and changes of taste sensation;
- the olfactory epithelium, receptors, transduction and pathways;
- the definition, biological significance and types of pain;
- the types of nociceptors and the pain-conducting fibers and pathways;
- pain mediators and hyperalgesia;
- the antinociceptive system and endogenous opioids, the “gate theory”;
- referred and phantom pain.

The student should be able to:

- determine the taste threshold;
- test taste sensitivity in different regions of the tongue (gustometry);
- determine and evaluate the pain threshold with an algometer;
- evaluate pain intensity with scales.

1. The taste system

Taste receptors

Taste buds are onion-shaped structures located in the papillae of the tongue (fungiform, circumvallate, foliate) and in the mucosa of the soft palate, pharynx and epiglottis. Adults have ≈ 5000–10,000 buds. Each bud consists of **50–150 taste receptor cells** (renewed every 10–14 days), supporting cells and basal cells; at its tip there is a **taste pore** into which the microvilli of the receptor cells project. Taste receptors are **secondary sensory** receptors.

For taste to be sensed, substances must **dissolve in saliva**.

Table 29.1. *Basic taste qualities and transduction mechanisms*

Taste	Acting substances	Transduction mechanism	Biological significance
Salty	Na^+ ions	Direct entry of Na^+ through ENaC channels — depolarization	Electrolyte balance
Sour	H^+ ions (acids)	Entry of H^+ through proton channels (OTOP1)	Avoidance of spoiled, fermented food

Taste	Acting substances	Transduction mechanism	Biological significance
Sweet	Sugars, some amino acids, artificial sweeteners	G protein–coupled receptors (T1R2/T1R3) → gustducin → second messengers	Energy-rich food
Bitter	Alkaloids (quinine, caffeine), many poisons	G protein–coupled receptors (T2R family, ≈ 25 kinds)	Protection from poisonous substances (lowest threshold)
Umami	Glutamate (meat, cheese, tomatoes), nucleotides	G protein–coupled receptors (T1R1/T1R3)	Protein-rich food

REMEMBER

The taste threshold is lowest for **bitter** substances (quinine ≈ 0.000008 M) and higher for salty and sweet ones (table salt ≈ 0.01 M, sucrose ≈ 0.01 M). This is an evolutionary defense: most poisons taste bitter.

CAUTION

The **"taste map"** of older textbooks (sweet at the tip of the tongue, sour and salty at the edges, bitter at the root) is, according to modern data, a **simplified** idea: all taste qualities are sensed by the buds in all regions of the tongue, with only slight differences in sensitivity (for example, the root of the tongue is slightly more sensitive to bitter). Testing sensitivity by region in laboratory work shows this.

Taste pathways and centers

Impulses from the anterior 2/3 of the tongue travel via the **facial nerve** (VII, chorda tympani), from the posterior 1/3 via the **glossopharyngeal nerve** (IX), and from the pharynx and epiglottis via the **vagus nerve** (X) to the **nucleus of the solitary tract** in the medulla oblongata → the ventral posteromedial nucleus of the thalamus → the taste area of the cortex (**insula and parietal operculum**, area 43). The nucleus of the solitary tract also has connections with the hypothalamus and limbic system (emotional evaluation of taste, appetite) and with the centers of salivation and digestive reflexes.

Features of taste: adaptation (especially rapid to sweet and salty); **contrast** (sour is felt more strongly after sweet); **mixed taste** (taste together with smell, temperature, texture and pungency — capsaicin sensations — forms "flavor"); sensitivity decreases with age.

CLINICAL SIGNIFICANCE

Disorders of taste: ageusia (loss), hypogeusia (reduction), dysgeusia (alteration — a metallic taste). Causes: facial nerve palsy (taste is lost in the anterior 2/3 of the tongue), dry mouth, some drugs (metronidazole, ACE inhibitors), zinc deficiency, radiation and chemotherapy, smoking, viral infections (loss of smell and taste in COVID-19).

2. The olfactory system

The **olfactory epithelium** lies in the upper part of the nasal cavity (the superior nasal concha and the upper part of the septum), over an area of ≈ 5 cm^2. It contains ≈ 10–20 million **olfactory receptor cells** — these are **primary sensory** receptors (bipolar neurons whose axons form the olfactory nerve) and the only neurons that are renewed, every 30–60 days. The **cilia** at the tip of their dendrite lie in the mucus.

Transduction: the odorant dissolves in the mucus and binds to **olfactory receptor proteins** on the cilia (≈ 350–400 functional genes in humans) → G protein (G_{olf}) → adenylate cyclase → **cAMP** ↑ → cation channels open (Na^+, Ca^{2+} enter) → Ca^{2+}-dependent Cl^- channels open and Cl^- leaves →

depolarization → action potentials. Each receptor cell expresses **one** type of receptor protein; but one odor excites many receptors to different degrees — odors are distinguished by a **combinatorial code** (R. Axel and L. Buck, Nobel Prize 2004). Humans can distinguish several thousand (according to some data more than a million) odors.

Olfactory pathways: the fibers of the olfactory nerve (I) pass through the openings of the ethmoid bone to the **olfactory bulb**; there, in the **glomeruli**, they synapse with mitral cells (all cells with the same receptor converge on one glomerulus) → **olfactory tract** → **primary olfactory cortex** (piriform cortex, amygdala, entorhinal cortex) — **bypassing the thalamus** (the only sense to do so). Then to the orbitofrontal cortex (also via the thalamus) — conscious discrimination of odors.

REMEMBER

Smell is the only sense that reaches the cortex **bypassing the thalamus**, and it is directly connected with the **limbic system** (amygdala, hippocampus). That is why smells evoke strong emotions and memories (the “Proust effect”).

Features of smell: very high sensitivity (mercaptans are detected at 10^{-11} g per 1 l of air); **rapid adaptation** (ceasing to notice a constant smell within a few minutes); sensitivity is slightly higher in women; it decreases with age. A large part of food “flavor” is actually **retronasal smell** — that is why food seems “tasteless” during a cold.

CLINICAL SIGNIFICANCE

Anosmia (loss of smell) — in colds, polyps, head injury (the olfactory nerve fibers are torn at the ethmoid bone) and viral infections; it is also observed as an early sign **in Parkinson's and Alzheimer's diseases**. Anosmia is dangerous: the person does not smell a gas leak, smoke or spoiled food. For safety, a strong-smelling substance (mercaptan) is added to household gas.

3. The pain sensory system

Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage (International Association for the Study of Pain, IASP). Pain is a **protective signal**: it forces the body to escape the harmful factor and encourages it to protect the damaged organ. However, chronic pain may lose its protective significance and become a disease in its own right.

Components of pain: sensory-discriminative (location, intensity, character), emotional-affective (unpleasantness, fear), autonomic (tachycardia, rise in pressure, sweating, pupil dilation), motor (protective reflexes) and cognitive-evaluative components.

Table 29.2. *Types of pain*

Feature	Types
Location	Superficial somatic (skin), deep somatic (muscle, joint, bone), visceral (internal organs)
Duration	Acute (signal, protective) and chronic (more than 3 months)
Origin	Nociceptive (stimulation of receptors), neuropathic (damage to the nervous system — diabetic neuropathy, neuralgia), nociplastic (central change in sensitivity)
Quality	First (sharp, well localized — Aδ fibers) and second (aching, burning, poorly localized — C fibers)

Nociceptors and pain pathways

Nociceptors are **free nerve endings** (primary sensory receptors); they are present in almost all tissues (except brain tissue). Types: **mechanical**, **thermal** (> 45 °C or < 15 °C; TRPV1 — the

capsaicin receptor) and the most widespread, **polymodal** (responding to mechanical, thermal and chemical stimuli). Nociceptors **adapt slowly** or not at all — pain persists as long as the stimulus lasts.

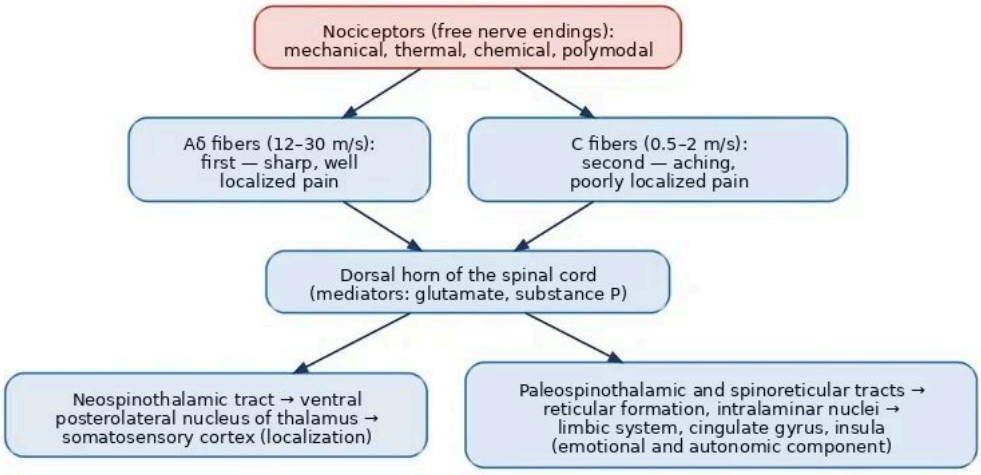


Figure 29.1. *Pain-conducting fibers and pathways*

Pain mediators (algogens) are formed in damaged tissue and excite or sensitize nociceptors: **bradykinin**, **prostaglandins** (E_2 — sensitizes nociceptors), **histamine**, **serotonin**, **K^+ and H^+ ions**, ATP, **substance P** and calcitonin gene-related peptide (released from nerve endings, causing neurogenic inflammation).

Hyperalgesia is increased pain sensitivity: **primary** (in the damaged area — sensitization of nociceptors by mediators) and **secondary** (in surrounding areas — **central sensitization** of spinal neurons). **Allodynia** — pain evoked by a normally painless stimulus (touch) (skin after a burn).

> **CLINICAL SIGNIFICANCE**
> **Nonsteroidal anti-inflammatory drugs** (NSAIDs — ibuprofen, diclofenac) inhibit cyclooxygenase and reduce the synthesis of **prostaglandins** — the sensitization of nociceptors decreases (a peripheral effect). **Local anesthetics** block Na^+ channels in nerve fibers and stop the conduction of impulses.

4. The antinociceptive system and pain control

The gate theory

The "gate theory" (R. Melzack and P. Wall, 1965): in the dorsal horns of the spinal cord there is a "gate" for the transmission of pain signals; it is controlled by **inhibitory interneurons**. When the thick tactile **Aβ fibers** are activated, the inhibitory neurons are excited and the transmission of pain signals to the brain decreases ("the gate closes"). That is why rubbing a bruised spot reduces pain. This theory is the basis of **transcutaneous electrical nerve stimulation (TENS)**, massage and acupuncture.

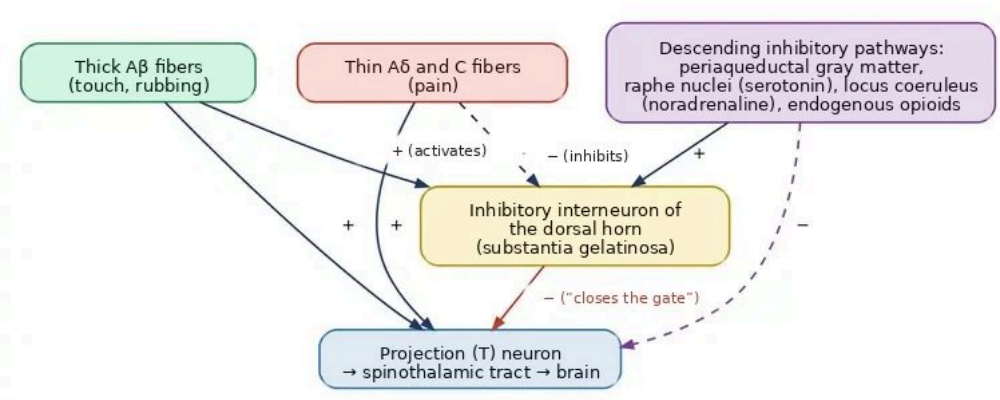


Figure 29.2. *The "gate theory" of pain and descending inhibitory influences*

The endogenous antinociceptive system

The **antinociceptive system** is the set of structures that limit and regulate pain sensation:

- **Descending inhibitory pathways** from the **periaqueductal gray matter** (midbrain), the **raphe nuclei** (serotonin) and the **locus coeruleus** (noradrenaline) to the dorsal horns of the spinal cord; electrical stimulation of these areas causes strong analgesia (stimulation-produced analgesia).
- **Endogenous opioid peptides** — **endorphins** (β-endorphin), **enkephalins** and **dynorphins**: they bind to opioid receptors (μ, δ, κ) and inhibit the transmission of pain signals presynaptically and postsynaptically.
- **Non-opioid mechanisms** — serotonin, noradrenaline, GABA, cannabinoids.

REMEMBER

Reduced perception of pain during stress, heavy physical exertion or injury (for example, in battle or in a sports competition) is **stress analgesia**: endorphins and the descending inhibitory system are activated. The **placebo** effect is also partly due to the release of endogenous opioids (naloxone, an opioid antagonist, reduces it).

CLINICAL SIGNIFICANCE

Morphine and other opioid analgesics bind to the μ receptors on which endogenous opioids act and produce strong analgesia; their side effects are depression of the respiratory center, pupil constriction, constipation and dependence. The antagonist in opioid poisoning is **naloxone**.

Referred and phantom pain

- **Referred pain** — when an internal organ is diseased, pain is felt in the skin area innervated from the same segment (**Zakharyin–Head zones**): heart — left arm and scapula, diaphragm — shoulder, duodenum — epigastrium (topic 26). The cause is the **convergence** of visceral and somatic afferents onto the same neurons of the spinal cord.
- **Phantom pain** is the sensation of pain in an amputated limb: it is associated with the damaged nerve endings and with reorganization of the somatotopic "map" in the spinal cord and cortex.

5. Laboratory work

Exercise 1: determining the taste threshold

1. Prepare solutions of sucrose (sweet), table salt (salty), citric acid (sour) and quinine (bitter) in decreasing concentrations.

2. The subject rinses the mouth with water; starting with the lowest concentration, 1–2 drops are applied to the tongue with a pipette.
3. Write down the lowest concentration at which the subject identifies the taste — this is the **taste threshold**.
4. After each solution the mouth is rinsed with water and there is a 1–2-minute break. **Interpretation:** compare the thresholds of the different substances — the threshold for bitter is the lowest.

Exercise 2: testing taste sensitivity in different regions of the tongue (gustometry)

1. The subject keeps the tongue protruded.
2. One of the taste solutions is applied with a brush to the tip, lateral edges and root of the tongue in turn.
3. Without drawing the tongue into the mouth, the subject indicates the taste with a hand signal or a card.
4. Mark the results on a diagram. **Result and analysis:** all taste qualities are perceived in most regions of the tongue, with slight differences in sensitivity; in the central back part of the tongue there are few buds — sensitivity is low.

Exercise 3: determining olfactory acuity and adaptation

1. Vials of different odorous substances (vinegar, mint, vanillin, coffee) are brought in turn to one nostril (the other is closed), and the subject identifies the smell with the eyes closed.
2. Adaptation: the subject smells one odor continuously and the time until it is no longer perceived is recorded; then another odor is presented. **Result and analysis:** adaptation to a constant odor occurs in 1–3 minutes, but another odor is perceived immediately — adaptation is **selective** (only to that odor).

Exercise 4: determining the pain threshold — algometry

1. **Mechanical algometry** (tensoalgometer): the tip of the instrument is placed perpendicular to the skin (for example, the inner surface of the forearm or the palm) and the pressure is increased gradually.
2. The subject says when the pressure turns into a **sensation of pain** — this is the **pain threshold** (kg/cm^2 or N).
3. Increasing the pressure further, the subject indicates the unbearable level — the **pain tolerance threshold** (under the instructor's supervision, with caution).
4. Repeat the measurement in different areas of the body (fingertip, palm, forearm, shoulder).
5. Measure the threshold again after local rubbing or applying cold. **Interpretation:** the pain threshold differs in different areas of the body (depending on nociceptor density); individual differences are large. An increase in the threshold after rubbing confirms the “gate theory”. Other types of algometry: **thermal** (a heat beam) and **electrical** algometry.

6. Practical skills

Skill: evaluating pain intensity

1. **Visual analogue scale (VAS)**: the subject marks his pain on a 10 cm line (0 — no pain, 10 — the worst pain imaginable).

2. **Numerical rating scale (0–10)**: rating pain with a number.
3. In children the **faces scale** is used.
4. Ask about the location, character and duration of the pain and the factors that worsen and relieve it.
5. Evaluate the autonomic signs (pulse, pressure, sweating).

Skill: distinguishing the type of pain by mechanism

1. Well localized, sharp, immediately after the stimulus — first (Aδ) pain.
2. Aching, poorly localized, delayed — second (C) pain.
3. Pain in a skin area in disease of an internal organ — referred pain.
4. Burning, “like an electric shock”, with allodynia — neuropathic pain.
5. Justify the appropriate analgesic approach for each type (NSAIDs, opioids, anticonvulsants).

Review questions

1. How are taste buds structured, and where are they located?
2. Explain the basic taste qualities and their transduction mechanisms.
3. Why is the taste threshold lowest for bitter substances?
4. What are the taste pathways and centers?
5. Why is the idea of the “taste map” simplified?
6. How are the olfactory receptors structured, and what feature do they have?
7. How does transduction take place in olfaction?
8. What is special about the olfactory pathways?
9. What is pain, and what is its biological significance?
10. Describe the types of nociceptors and the pain-conducting fibers.
11. List the pain mediators. What are hyperalgesia and allodynia?
12. Explain the “gate theory”.
13. What is the role of the antinociceptive system and endogenous opioids?
14. What is the mechanism of referred and phantom pain?
15. How is algometry performed?

Test questions

1. To which type of receptors do taste receptors belong?
   A) Primary sensory
   B) Mechanoreceptors
   C) Thermoreceptors
   D) Proprioceptors
   E) Secondary sensory

2. For which taste is the threshold lowest?
   A) Sweet
   B) Salty
   C) Sour
   D) Bitter
   E) Umami

3. Which nerve conducts taste sensation from the anterior 2/3 of the tongue?
   A) The glossopharyngeal nerve
   B) The facial nerve (chorda tympani)
   C) Vagus nerve
   D) The hypoglossal nerve
   E) The motor fibers of the trigeminal nerve

4. Which substance evokes the umami taste?
   A) Quinine
   B) Sucrose
   C) Citric acid
   D) Glutamate
   E) Table salt

5. Which is a special feature of the olfactory receptors?
   A) They are secondary sensory receptors
   B) They are never renewed
   C) They do not adapt
   D) They go to the cortex through the thalamus
   E) They are primary sensory bipolar neurons that are regularly renewed

6. Which sense reaches the cortex bypassing the thalamus?
   A) Vision
   B) Hearing
   C) Smell
   D) Taste
   E) Pain

7. Which fibers conduct first, sharp, well-localized pain?
   A) Aδ fibers
   B) C fibers
   C) Aβ fibers
   D) Aα fibers
   E) B fibers

8. What is the role of prostaglandins in pain?
   A) They stop pain directly
   B) They enhance endorphin synthesis
   C) They only constrict vessels
   D) They block Na^+ channels
   E) They sensitize nociceptors (hyperalgesia)

9. Who are the authors of the “gate theory”?
   A) Pavlov and Sechenov
   B) Hodgkin and Huxley
   C) Axel and Buck
   D) Melzack and Wall
   E) Young and Helmholtz

10. Why does rubbing a bruised spot reduce pain?
   A) Prostaglandins increase
   B) Aβ fibers activate inhibitory neurons in the dorsal horn and “close the gate”
   C) Nociceptors are destroyed
   D) Blood flow stops
   E) C fibers are activated

11. Which is an endogenous opioid peptide?
   A) β-endorphin
   B) Bradykinin
   C) Substance P
   D) Histamine
   E) Glutamate

12. What is the spread of pain to the left arm in cardiac ischemia called?
   A) Phantom pain
   B) Allodynia
   C) Referred pain
   D) Neuropathic pain
   E) First pain

13. From which structure do the descending pathways that counteract pain originate?
   A) From the cerebellum
   B) From the periaqueductal gray matter
   C) From the olfactory bulb
   D) From the hippocampus
   E) From the ventral horns of the spinal cord

14. How do NSAIDs (ibuprofen) reduce pain?
   A) They activate opioid receptors
   B) They block Na^+ channels
   C) They inhibit cyclooxygenase and reduce prostaglandin synthesis
   D) They block glutamate receptors
   E) They enhance endorphin release

15. What does algometry determine?
   A) The pain threshold and pain tolerance
   B) The taste threshold
   C) Visual acuity
   D) Olfactory acuity
   E) The limits of hearing

Clinical cases

Case 1

Neighbors noticed a household gas leak in a patient's home, but the patient himself smelled nothing. A year earlier he had injured the back of his head in a car accident, and since then food has seemed “tasteless”.

Questions: What disorder has developed? Explain its mechanism. Why does food seem “tasteless”?

Answer: Post-traumatic anosmia. When the back of the head is struck, the brain shifts back and forth and the delicate olfactory nerve fibers passing through the openings of the ethmoid bone are torn — impulses from the olfactory receptors no longer reach the bulb. The “flavor” of food consists mainly of retronasal smell (odorants passing from the mouth through the pharynx into the nasal cavity); the taste receptors distinguish only the five basic tastes (sweet, salty, sour, bitter, umami). When smell is lost, the complex flavor of food is not perceived. Anosmia is dangerous — the patient does not smell gas, smoke or spoiled food; installing gas and smoke detectors at home is recommended.

Case 2

A footballer injured his leg during a match, but felt almost no pain until the end of the game. After the game the pain increased.

Questions: How can this phenomenon be explained?

Answer: Stress analgesia. The strong emotional and physical strain during competition activates the antinociceptive system: β-endorphin and enkephalins are released in the pituitary and CNS, and descending inhibitory pathways from the periaqueductal gray matter, the raphe nuclei (serotonin) and the locus coeruleus (noradrenaline) to the dorsal horns of the spinal cord stop the transmission of pain signals. Activation of the sympathoadrenal system also contributes. When the stress passes these influences decrease, while in the damaged tissue inflammatory mediators (prostaglandins, bradykinin) have sensitized the nociceptors — the pain increases.

Case 3

A 50-year-old man complains of pressing pain behind the sternum during physical exertion, spreading to the left arm and under the scapula.

Questions: How can the spread of the pain to the left arm be explained?

Answer: This is referred pain — typical of angina pectoris. Visceral afferent fibers from the heart enter the Th_1–Th_5 segments of the spinal cord; somatic afferents from the skin of the inner surface of the left arm and the chest wall enter the same segments. They converge on the same projection neurons in the dorsal horn. Based on usual experience, the brain “interprets” the signals from these neurons as coming from a skin (somatic) source — the pain is felt in the left arm and scapular region (Zakharyin–Head zones). This sign is important in diagnosing myocardial ischemia.

Quick review

Table 29.3. *Key concepts of topic 29*

Concept	Brief definition
Taste bud	50–150 secondary sensory cells; renewed in 10–14 days
Five tastes	Sweet, salty, sour, bitter, umami

Concept	Brief definition
Bitter taste	Lowest threshold; T2R receptors; protection
Taste nerves	VII (anterior 2/3), IX (posterior 1/3), X; nucleus of the solitary tract
Taste cortex	Insula and parietal operculum
Olfactory receptors	Primary sensory neurons; ≈ 400 kinds of receptor proteins; cAMP
Olfactory pathway	Bulb → piriform cortex; without the thalamus; linked with the limbic system
Pain	A sensory and emotional experience associated with damage
Nociceptors	Free nerve endings; mechanical, thermal, polymodal; little adaptation
Aδ / C fibers	First sharp / second aching pain
Algogens	Bradykinin, prostaglandins, histamine, serotonin, K^+, H^+
The gate theory	Aβ fibers inhibit the pain signal in the dorsal horn (Melzack–Wall)
Endogenous opioids	Endorphins, enkephalins, dynorphins
Referred pain	Convergence of visceral and somatic afferents
Algometry	Measuring the pain threshold and the tolerance threshold

TOPIC 30

Higher nervous activity. Conditioned reflexes. Types of higher nervous activity. Signal systems. Functional asymmetry of the cerebral hemispheres

Learning objectives

To teach the student I.P. Pavlov's doctrine of higher nervous activity (HNA), innate and acquired forms of behavior, the conditions, mechanisms and types of formation of conditioned reflexes, the inhibition of conditioned reflexes, the types of HNA, the first and second signal systems, memory, sleep and the functional asymmetry of the cerebral hemispheres; to master methods of forming a conditioned reflex and determining the type of HNA in humans.

By the end of the class the student should know:

- the concepts of higher and lower nervous activity;
- the differences between unconditioned reflexes and instincts and conditioned reflexes;
- the conditions and mechanism of formation of a conditioned reflex (temporary connection);
- the classification of conditioned reflexes;
- the external and internal inhibition of conditioned reflexes;
- the types of HNA and their relationship to temperaments;
- the first and second signal systems and the specifically human types of HNA;
- the types and mechanisms of memory and the phases of sleep;
- the functional asymmetry of the cerebral hemispheres.

The student should be able to:

- explain the formation of a pupillary or blink conditioned reflex in humans;
- determine the capacity of short-term memory;
- determine the type of HNA and functional asymmetry (dominant hand, eye, ear) with tests.

1. The concept of higher nervous activity

Higher nervous activity (HNA) (I.P. Pavlov) is the joint activity of the cerebral cortex and the subcortical structures; it ensures the most perfect individual adaptation (behavior) of animals and humans to a changing external environment. Its main mechanism is **conditioned reflexes**. **Lower nervous activity**, by contrast, coordinates the activity of the internal organs through the spinal cord and brainstem (mainly unconditioned reflexes).

The conceptual basis of Pavlov's doctrine was created by **I.M. Sechenov** in his work “Reflexes of the Brain” (1863): mental activity is also reflex in nature.

Innate and acquired forms of behavior

Table 30.1. *Differences between unconditioned and conditioned reflexes*

Feature	Unconditioned reflexes	Conditioned reflexes
Origin	Innate, hereditary	Acquired during life
Species specificity	Common to the species (a species trait)	Individual

Feature	Unconditioned reflexes	Conditioned reflexes
Stability	Constant, preserved throughout life	Variable: formed and extinguished
Stimulus	Adequate (to a particular receptive field)	Any (indifferent) stimulus can become a signal
Reflex arc	Ready-made, permanent (at the level of the spinal cord and brainstem)	Formed through a temporary connection (with the participation of the cortex)
Biological significance	Adaptation to constant conditions	Adaptation to changing conditions, anticipation of events

Instincts are complex innate programs of behavior (chains of unconditioned reflexes): feeding, defensive, sexual, parental and social instincts. They are triggered by **motivation** (a need — hunger, thirst).

2. Conditioned reflexes

Conditions for the formation of a conditioned reflex

1. **Coincidence in time of two stimuli**: an indifferent (neutral) stimulus (for example, light, a bell) must be given **before** the **unconditioned stimulus** (food) and applied **together with it several times** (reinforcement).
2. The conditioned signal must begin **1–5 s before** the unconditioned stimulus (if it is given after the unconditioned stimulus, the reflex is not formed).
3. The unconditioned stimulus must be biologically **stronger** (the animal must be hungry).
4. **An active, alert state of the cortex** and the absence of extraneous stimuli.
5. The indifferent stimulus must be of moderate strength.

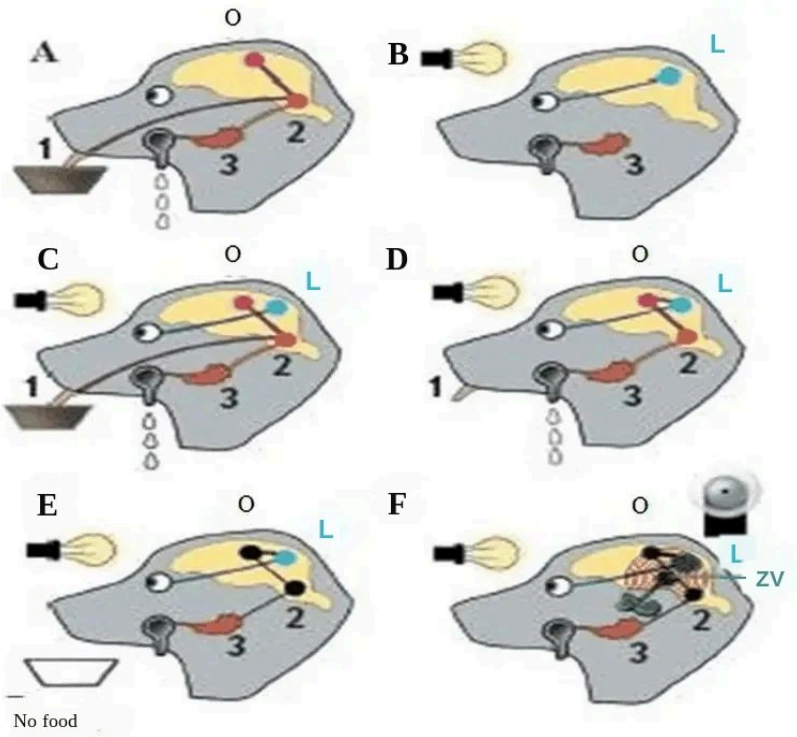


Figure 30.1. *Formation of a conditioned salivary reflex: A — unconditioned reflex (food → saliva); B — orienting to light; C — combining light and food; D — conditioned reflex (saliva to light alone); E — extinction (without reinforcement); F — second-order reflex (bell) (O — food center, L — light center, ZV — sound center, 1 — food, 2 — salivary center, 3 — salivary gland) (Alavia O.T., 2018)*

Mechanism of formation of a conditioned reflex

According to Pavlov, when a conditioned reflex is formed, a **temporary connection** arises in the cortex between the **center of the conditioned signal** (for example, the visual center) and the **cortical representation of the unconditioned reflex** (the food center). The strong (dominant) unconditioned center “draws to itself” the excitation in the weaker center of the conditioned signal (Ukhtomsky's dominant). With repeated pairing this path is “worn in” — synaptic efficacy increases.

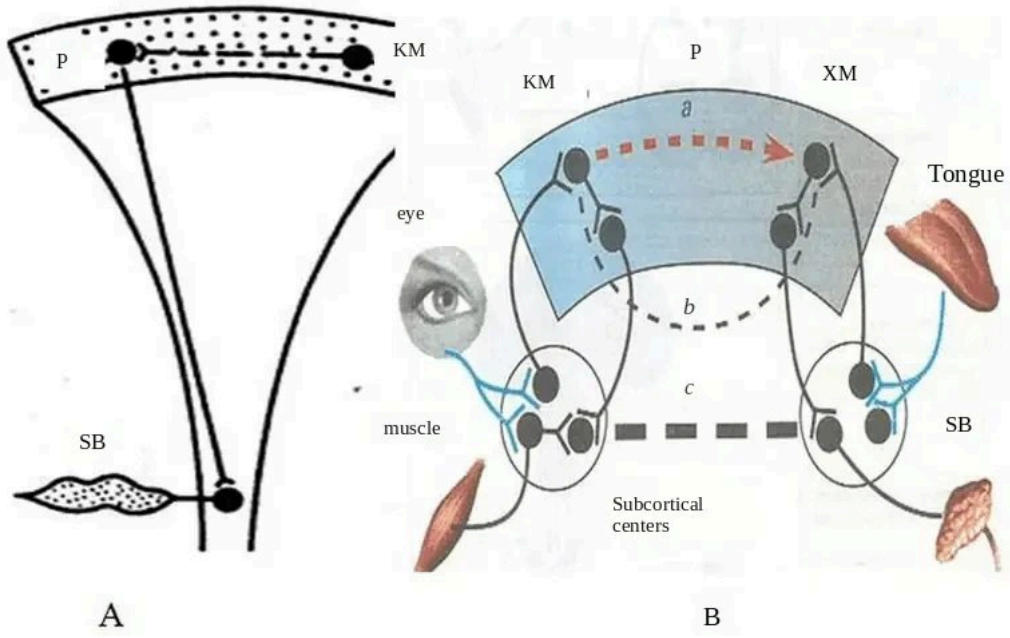


Figure 30.2. *Scheme of the temporary connection: connections between the center of the conditioned signal (KM — visual center) and the center of the unconditioned reflex (XM — taste center) in the cortex (a) and through subcortical structures (b, c); SB — salivary gland (Alavia O.T., 2018)*

According to modern ideas, the temporary connection is formed not only in the cortex but in a **multilevel** system involving **cortical and subcortical** structures (thalamus, hippocampus, amygdala, reticular formation, cerebellum). Its cellular basis is **synaptic plasticity**: long-term potentiation (in the hippocampus — NMDA glutamate receptors, Ca^{2+}), synthesis of new proteins and formation of new synapses (E. Kandel, Nobel Prize 2000).

Classification of conditioned reflexes

Table 30.2. *Types of conditioned reflexes*

Feature	Types
By type of unconditioned reflex	Feeding, defensive, sexual, orienting, motor, autonomic (cardiac, respiratory, urinary)
By nature of the signal	Natural (a natural feature of the unconditioned stimulus — the smell or sight of food) and artificial (bell, light)
By receptors	Exteroceptive, interoceptive, proprioceptive
By the time between signal and reinforcement	Coincident (0.5–1 s), delayed (1–3 minutes), trace (reinforced after the signal ends), temporal (at regular time intervals — the “biological clock”)
By order	First-order (on an unconditioned basis) and higher-order (second, third — on the basis of a previous conditioned reflex)
By complexity	Simple and complex (to a set of stimuli); the dynamic stereotype — a stable system of responses to a particular sequence of stimuli (habits, daily routine)

Feature	Types
Instrumental (operant) reflexes	The animal's own action is reinforced (J. Konorski; B. Skinner)

CLINICAL SIGNIFICANCE

Conditioned reflexes are of great importance in medicine: nausea in a patient on entering the chemotherapy room (conditioned nausea), “white coat” hypertension (a rise in pressure at the sight of the doctor), part of the placebo effect, and the daily routine (regular times of eating and sleeping — a dynamic stereotype) are explained on the basis of conditioned reflexes. That is why the patient's environment and regimen are taken into account in treatment.

3. Inhibition of conditioned reflexes

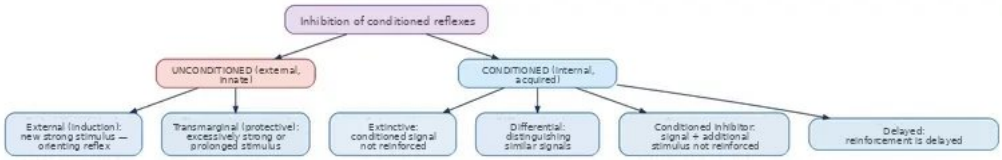


Figure 30.3. *Types of inhibition of conditioned reflexes*

Table 30.3. *Types of inhibition of conditioned reflexes*

Type	Mechanism and example	Significance
External (inductive) — unconditioned	A new strong stimulus appearing during a conditioned reflex creates a new dominant focus of excitation in the cortex (the orienting reflex — “what is it?”) and inhibits the conditioned reflex	Shifting attention to a more important new signal
Transmarginal (protective) — unconditioned	An excessively strong or prolonged stimulus reduces the response	Protection of neurons from exhaustion and damage
Extinctive — conditioned (internal)	If the conditioned signal is not reinforced, the reflex gradually fades; it may recover spontaneously after a while	Getting rid of reflexes that have lost their biological significance
Differential — conditioned	The response to a signal similar to the reinforced one but not reinforced is inhibited (for example, a metronome at 100 beats/min is reinforced, at 60 — not)	Fine discrimination of stimuli
Conditioned inhibitor — conditioned	If the conditioned signal is given together with an additional stimulus and not reinforced, the response to this combination is inhibited	—
Delayed — conditioned	When reinforcement is delayed, there is no response at the beginning of the signal; it appears close to the time of reinforcement	Estimating time, patience

REMEMBER

External inhibition is **innate** (unconditioned) and develops immediately; internal inhibition is **acquired** (develops with practice) and arises in the cortex within the arc of the conditioned signal. Internal inhibition is weaker in youth and old age and in fatigue.

4. Types of higher nervous activity

The **type of HNA** is the set of innate and acquired individual properties of the nervous system. Pavlov distinguished the types of HNA on the basis of three properties of the processes of **excitation and inhibition**: **strength**, **balance** and **mobility**. These types correspond to the temperaments of Hippocrates.

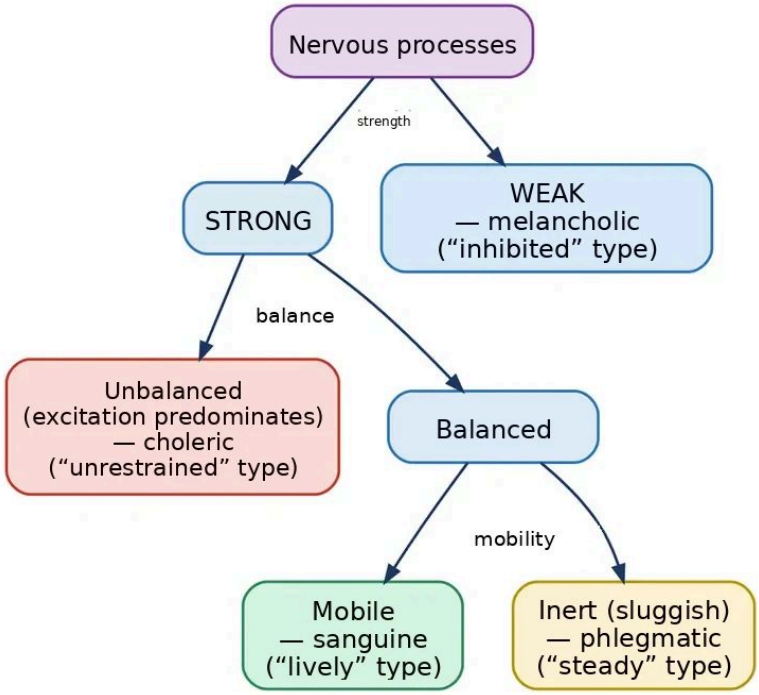


Figure 30.4. *Types of higher nervous activity (I.P. Pavlov) and temperaments*

Table 30.4. *Types of HNA and temperaments*

Type of HNA	Nervous processes	Temperament (Hippocrates)	Features
Strong, balanced, mobile (“lively”)	Strong, balanced, mobile	Sanguine	Active, adapts quickly, hardworking, quickly settles into new conditions
Strong, balanced, inert (“calm”)	Strong, balanced, slow	Phlegmatic	Calm, patient, steady; acquires new skills slowly but firmly
Strong, unbalanced (“unrestrained”)	Strong, excitation predominates over inhibition	Choleric	Energetic, quick-tempered, easily excited, finds it hard to control himself
Weak (“inhibited”)	Both processes weak	Melancholic	Sensitive, tires quickly, cautious, does not tolerate strong stimuli

Specifically human types of HNA (according to the ratio of the signal systems): the **artistic type** (the first signal system predominates — imaginative, emotional thinking), the **thinking type** (the second signal system predominates — abstract, logical thinking) and the **intermediate (mixed) type**.

CAUTION

“Pure” types of HNA are rare; most people have mixed features. The type is shaped not only by heredity but also by upbringing and living conditions (the phenotype). Therefore the type of HNA is not a diagnosis or a “label”, but a guide for an individual approach.

5. Signal systems

- **The first signal system** is the perception of environmental stimuli in the form of concrete images through **direct sensations** (vision, hearing, smell, taste, skin). It is common to animals and humans.
- **The second signal system** is signaling through **speech (the word)**; the word is the "signal of signals" (Pavlov). It is specific to humans and is the basis of **abstract thinking**, the formation of concepts, generalization, upbringing and education, and the transmission of social experience to the next generations. A word can evoke a conditioned reflex just like first-signal stimuli (for example, the word "lemon" increases salivation). The two signal systems are closely interconnected; the second signal system develops on the basis of the first and controls it.

Speech centers: Broca's (motor) and Wernicke's (sensory) areas — topic 25; for reading and writing, the angular gyrus and the corresponding areas of the frontal lobe.

6. Memory and sleep

Memory

Memory is the ability to receive, store and retrieve information.

Table 30.5. *Types of memory*

Type	Duration	Mechanism
Sensory (iconic, echoic)	0.25–2 s	Trace processes in receptors and the primary cortex
Short-term (working)	Seconds–minutes; capacity 7 ± 2 items	Circulation of impulses in neuronal loops (reverberation), prefrontal cortex
Long-term	Hours–years	Synaptic changes, protein synthesis, new synapses; the hippocampus in consolidation

Long-term memory is divided into **declarative** (explicit — facts and events; hippocampus and temporal lobe) and **nondeclarative** (implicit — skills, habits, conditioned reflexes; basal ganglia, cerebellum). **Consolidation** is the transfer of short-term memory into long-term memory; sleep (especially deep and paradoxical sleep) plays an important role in it.

Sleep

Sleep is a special active state, different from wakefulness — a periodic reorganization of brain activity; it is controlled by the hypothalamus (ventrolateral preoptic nucleus), the brainstem reticular formation and the suprachiasmatic nucleus (the "biological clock"); the pineal hormone **melatonin** is released in darkness and increases sleepiness.

Table 30.6. *Phases of sleep*

Phase	EEG	Features
Slow-wave (NREM) sleep — stages 1–2	θ-rhythm, sleep spindles, K-complexes	Light sleep
Slow-wave — stage 3 (deep)	δ-rhythm	Difficult to awaken; GH is secreted; restoration
Rapid (paradoxical, REM) sleep	Desynchronization resembling wakefulness	Rapid eye movements, sharp fall in muscle tone, dreaming, memory consolidation

4–6 cycles (each 90–100 minutes) alternate during the night; toward morning the proportion of REM sleep increases. Adults need 7–9 hours of sleep.

7. Functional asymmetry of the cerebral hemispheres

Although the cerebral hemispheres are similar in structure, they differ in function — **functional asymmetry**. This was clearly demonstrated by R. Sperry (Nobel Prize 1981) in studies of **"split-brain"** patients (with the corpus callosum severed).

Table 30.7. *Functional specialization of the hemispheres (in right-handed people)*

Left hemisphere	Right hemisphere
Speech (in 95% of people), reading, writing	Spatial perception, maps, drawings
Logical, analytical, sequential thinking	Imaginative, holistic (synthetic), intuitive thinking
Calculation, mathematical operations	Face recognition, perception of emotions
Perception of time, sequence	Music (melody), speech intonation
Control of the right side of the body	Control of the left side of the body

The hemispheres are connected through the **corpus callosum** and work together. Asymmetry also appears in the **dominant hand** (≈ 90% of people are right-handed), and the dominant eye, ear and leg.

> **CLINICAL SIGNIFICANCE**
> When the left hemisphere is damaged (stroke) — right-sided paralysis and **aphasia**; when the right hemisphere is damaged — left-sided paralysis, neglect of the left side, impaired spatial orientation and face recognition, and sometimes denial of one's own illness (anosognosia). In children the speech centers are plastic: if the left hemisphere is damaged at an early age, speech may "move" to the right hemisphere.

8. Laboratory work

Exercise 1: forming a conditioned blink reflex in humans (demonstration)

1. The subject sits in front of a device that blows a puff of air into the eye; the blinking movement is recorded.
2. An indifferent stimulus (a sound signal or light) is given, and it is checked that it does not cause blinking.
3. 0.5–1 s after the sound signal a puff of air is blown into the eye (the unconditioned stimulus — the blink reflex); this is repeated 10–20 times.
4. Then the sound signal alone is given.

Result and analysis: after several pairings, blinking appears to the sound alone — a **conditioned defensive reflex** has been formed. Presenting the sound several times without reinforcement leads to **extinction** of the reflex (internal inhibition).

Exercise 2: determining the capacity of short-term memory

1. Twelve unrelated two-digit numbers (or words) are read to the subject at 1 s intervals.
2. The subject immediately writes down those he remembers, in any order.
3. The number written correctly is counted. **Interpretation:** normally **7 ± 2** items (Miller's number). Material linked by meaning (grouping) is easier to remember.

Exercise 3: determining functional asymmetry

1. **Dominant hand**: clapping the hands (which hand is on top), crossing the arms on the chest, interlocking the fingers (which thumb is on top).
2. **Dominant eye**: look at a distant object through a hole in a sheet of paper with both eyes, then close one eye at a time — the eye with which the object does not “shift” is dominant.
3. **Dominant leg**: kicking a ball, hopping on one leg.
4. Summarize the results: right-handed, left-handed or ambidextrous (using both hands equally).

Exercise 4: determining the type of HNA (based on questionnaires and observation)

1. Fill in the Eysenck or another temperament questionnaire (extraversion–introversion and emotional stability scales).
2. Place the result in the Eysenck circle and determine the temperament type (stable extravert — sanguine, unstable extravert — choleric, stable introvert — phlegmatic, unstable introvert — melancholic).
3. Perform a reaction speed (sensorimotor reaction time) test and compare the results.

9. Practical skills

Skill: drawing up a plan for forming a conditioned reflex

1. Choose an unconditioned reflex (for example, food — salivation).
2. Choose an indifferent signal and check that it does not evoke a response by itself.
3. Give the signal 1–5 s before the unconditioned stimulus and repeat the pairing.
4. Ensure an alert state of the cortex, motivation (hunger) and the absence of extraneous stimuli.
5. Test with the signal alone to check that the reflex has formed; if you present it without reinforcement, observe its extinction.

Skill: taking HNA features into account when working with patients

1. Observe the patient's temperament and emotional reactions.
2. Give information to a patient of the melancholic type step by step, calmly and in a supportive tone.
3. Speak patiently, clearly and briefly with a patient of the choleric type.
4. Help turn the treatment regimen (times of taking drugs, meals) into a dynamic stereotype.
5. Take into account the effect of the word (the second signal system): the doctor's word can both improve and worsen the patient's condition (iatrogenesis).

Review questions

1. What is higher nervous activity, and who studied it?
2. List the differences between unconditioned and conditioned reflexes.
3. What is an instinct?
4. What conditions are necessary for the formation of a conditioned reflex?
5. How is a temporary connection formed? What are the modern ideas?
6. How are conditioned reflexes classified?
7. What is a dynamic stereotype?

8. Explain the types of external inhibition of conditioned reflexes.
9. List the types of internal inhibition and give their significance.
10. Describe the types of HNA according to Pavlov and the temperaments.
11. What is the difference between the first and second signal systems?
12. What are the specifically human types of HNA?
13. What are the types and mechanisms of memory?
14. What are the phases of sleep and their significance?
15. Explain the functional asymmetry of the cerebral hemispheres.

Test questions

1. Who is the author of the doctrine of higher nervous activity?
   A) I.M. Sechenov
   B) C. Sherrington
   C) A.A. Ukhtomsky
   D) P.K. Anokhin
   E) I.P. Pavlov

2. Which feature is characteristic of conditioned reflexes?
   A) Innate
   B) Common to the species
   C) Acquired during life, individual
   D) Constant, preserved throughout life
   E) Carried out only through the spinal cord

3. When should the indifferent signal be given to form a conditioned reflex?
   A) After the unconditioned stimulus
   B) 1 hour before the unconditioned stimulus
   C) Never together with the unconditioned stimulus
   D) Only during sleep
   E) Shortly (1–5 s) before the unconditioned stimulus

4. What is the gradual disappearance of a reflex when the conditioned signal is not reinforced called?
   A) External inhibition
   B) Transmarginal inhibition
   C) Extinctive inhibition
   D) Differentiation
   E) Delayed inhibition

5. What is the inhibition of a reflex when a new strong stimulus appears during a conditioned reflex called?
   A) Extinctive
   B) Differentiation
   C) Delayed
   D) External (inductive)
   E) Conditioned inhibitor

6. Which temperament corresponds to the strong, balanced, mobile type of HNA?
   A) Sanguine
   B) Phlegmatic
   C) Choleric
   D) Melancholic
   E) None

7. Which temperament corresponds to the strong, unbalanced (excitation predominates) type of HNA?
   A) Sanguine
   B) Choleric
   C) Phlegmatic
   D) Melancholic
   E) The artistic type

8. What is the basis of the second signal system?
   A) Speech (the word)
   B) Visual images
   C) Taste sensations
   D) Unconditioned reflexes
   E) Instincts

9. Which type of HNA is specific only to humans?
   A) Sanguine
   B) Phlegmatic
   C) Choleric
   D) The thinking type
   E) Melancholic

10. What is the capacity of short-term memory?
    A) 7 ± 2 items
    B) 1–2 items
    C) 50–100 items
    D) Unlimited
    E) 20 ± 5 items

11. Which inhibition is based on discriminating similar signals?
    A) Extinctive
    B) Differentiation
    C) External
    D) Transmarginal
    E) Delayed

12. Which feature is characteristic of rapid (paradoxical) sleep?
    A) Predominance of the δ-rhythm on the EEG
    B) Increased muscle tone
    C) High secretion of GH
    D) Rapid eye movements and dreaming
    E) Very difficult awakening

13. In right-handed people, what is the main function of the left hemisphere?
    A) Spatial perception
    B) Face recognition
    C) Speech and logical thinking
    D) Perception of musical melody
    E) Control of the left side of the body

14. What is a stable system of responses to a particular sequence of stimuli called?
    A) An instinct
    B) A dynamic stereotype
    C) Dominant
    D) The orienting reflex
    E) Imprinting

15. What is the cellular basis of long-term memory?
    A) Loss of mediators
    B) Death of neurons
    C) Reduced excitability
    D) Refractoriness
    E) Synaptic plasticity (long-term potentiation, protein synthesis, new synapses)

Clinical cases

Case 1

After several courses of chemotherapy, a patient began to experience nausea and vomiting as soon as he entered the hospital building or saw the nurse (before the drug had been given).

Questions: Explain this phenomenon from the standpoint of HNA physiology. How can it be reduced?

Answer: This is a conditioned reflex (conditioned nausea). The chemotherapy drug is an unconditioned stimulus that excites the vomiting center; indifferent stimuli such as the hospital environment, the sight of the nurse and the smell of the room acted several times before and together with the drug administration — a temporary connection was formed in the cortical and subcortical structures. Now these signals by themselves excite the vomiting center. Ways to reduce it: effective use of antiemetic drugs from the first course (preventing the unconditioned response — the reflex is not formed), changing the environment, relaxation and psychotherapy (presenting the signal without reinforcement to develop extinctive inhibition).

Case 2

A student wakes up every morning at 7, has breakfast and goes to class at 8. Even on vacation he wakes at 7 without an alarm clock, and his appetite appears at that time. When he moved to a new country (a 5-hour time difference), he had daytime sleepiness and nighttime insomnia for a week.

Questions: How can these phenomena be explained?

Answer: On the basis of a regular daily routine, temporal conditioned reflexes and a dynamic stereotype have formed: waking at a certain time and the appearance of appetite (secretion of digestive juices) have become conditioned reflexes. Circadian rhythms (the suprachiasmatic nucleus of the hypothalamus — the “biological clock”, melatonin) also underlie this. When the time zone changes abruptly, the internal biological clock and the system of conditioned reflexes no longer

match external time — desynchronosis (jet lag): the sleep–wake cycle is disturbed. Readjustment of the new stereotype and the biological clock (roughly 1 day for every hour of difference) takes several days; morning light and a regular new regimen speed up adaptation.

Case 3

After a stroke a patient cannot move his right arm and leg and cannot speak, but he can hum tunes and songs and recognizes familiar people by their faces.

Questions: Which hemisphere is damaged? Why are some functions preserved?

Answer: The left hemisphere is damaged. Paralysis of the right arm and leg — damage to the motor cortex or pyramidal tract of the left hemisphere (the pathways cross). Loss of speech (aphasia) — in 95% of right-handed people the speech centers are in the left hemisphere. Humming the melody of tunes and songs and face recognition are mainly functions of the right hemisphere, which is not damaged. This demonstrates functional asymmetry; teaching patients to speak through singing (melodic intonation therapy) in speech rehabilitation is based on using the capacities of the right hemisphere.

Quick review

Table 30.8. *Key concepts of topic 30*

Concept	Brief definition
HNA	The behavior-providing activity of the cortex and subcortex (Pavlov)
Unconditioned reflex	Innate, species-specific, permanent
Conditioned reflex	Acquired, individual, through a temporary connection
Condition of formation	Indifferent signal before the unconditioned stimulus, repeatedly paired
Temporary connection	Between the signal center and the unconditioned center; synaptic plasticity
A dynamic stereotype	A stable system of responses to a sequence of stimuli
External inhibition	A new strong stimulus (orienting reflex) — innate
Internal inhibition	Extinctive, differentiation, conditioned inhibitor, delayed
Sanguine	Strong, balanced, mobile
Phlegmatic	Strong, balanced, inert
Choleric	Strong, unbalanced
Melancholic	Weak
Second signal system	Speech — the “signal of signals”; only in humans
Memory	Short-term 7 ± 2; consolidation — hippocampus
Functional asymmetry	Left — speech, logic; right — spatial, imaginative, music

Answers to the test questions

Topic 1: 1-B; 2-A; 3-E; 4-C; 5-D; 6-A; 7-B; 8-C; 9-D; 10-A; 11-B; 12-E; 13-C; 14-E; 15-D.

Topic 2: 1-C; 2-B; 3-D; 4-A; 5-C; 6-E; 7-B; 8-A; 9-C; 10-D; 11-B; 12-E; 13-A; 14-E; 15-D.

Topic 3: 1-D; 2-A; 3-B; 4-C; 5-B; 6-E; 7-C; 8-A; 9-D; 10-E; 11-D; 12-E; 13-C; 14-A; 15-B.

Topic 4: 1-C; 2-A; 3-B; 4-C; 5-D; 6-B; 7-D; 8-E; 9-A; 10-E; 11-A; 12-D; 13-E; 14-B; 15-C.

Topic 5: 1-A; 2-B; 3-A; 4-C; 5-E; 6-D; 7-B; 8-E; 9-C; 10-B; 11-A; 12-D; 13-C; 14-D; 15-E.

Topic 6: 1-E; 2-D; 3-A; 4-C; 5-D; 6-B; 7-C; 8-E; 9-D; 10-E; 11-A; 12-B; 13-A; 14-C; 15-B.

Topic 7: 1-C; 2-A; 3-E; 4-D; 5-B; 6-D; 7-A; 8-D; 9-E; 10-A; 11-C; 12-B; 13-E; 14-B; 15-C.

Topic 8: 1-D; 2-E; 3-C; 4-A; 5-B; 6-E; 7-B; 8-E; 9-A; 10-C; 11-D; 12-B; 13-C; 14-D; 15-A.

Topic 9: 1-B; 2-E; 3-A; 4-D; 5-C; 6-A; 7-E; 8-B; 9-C; 10-D; 11-B; 12-C; 13-E; 14-A; 15-D.

Topic 10: 1-D; 2-C; 3-E; 4-B; 5-D; 6-E; 7-C; 8-E; 9-A; 10-B; 11-D; 12-A; 13-B; 14-A; 15-C.

Topic 11: 1-B; 2-C; 3-E; 4-D; 5-C; 6-A; 7-B; 8-C; 9-B; 10-D; 11-A; 12-D; 13-E; 14-A; 15-E.

Topic 12: 1-B; 2-D; 3-C; 4-D; 5-A; 6-B; 7-C; 8-B; 9-E; 10-A; 11-E; 12-C; 13-E; 14-A; 15-D.

Topic 13: 1-E; 2-D; 3-C; 4-A; 5-D; 6-B; 7-C; 8-D; 9-E; 10-C; 11-A; 12-B; 13-E; 14-A; 15-B.

Topic 14: 1-E; 2-B; 3-A; 4-E; 5-A; 6-C; 7-B; 8-E; 9-D; 10-C; 11-D; 12-A; 13-C; 14-D; 15-B.

Topic 15: 1-B; 2-D; 3-E; 4-C; 5-E; 6-B; 7-A; 8-B; 9-D; 10-E; 11-C; 12-A; 13-D; 14-A; 15-C.

Topic 16: 1-E; 2-C; 3-A; 4-B; 5-C; 6-B; 7-A; 8-D; 9-A; 10-E; 11-D; 12-B; 13-E; 14-C; 15-D.

Topic 17: 1-C; 2-A; 3-E; 4-B; 5-D; 6-B; 7-E; 8-A; 9-B; 10-C; 11-A; 12-D; 13-C; 14-E; 15-D.

Topic 18: 1-A; 2-E; 3-B; 4-D; 5-A; 6-C; 7-E; 8-D; 9-E; 10-A; 11-B; 12-D; 13-C; 14-B; 15-C.

Topic 19: 1-E; 2-A; 3-E; 4-C; 5-D; 6-C; 7-E; 8-B; 9-A; 10-D; 11-B; 12-D; 13-A; 14-C; 15-B.

Topic 20: 1-C; 2-D; 3-E; 4-B; 5-C; 6-E; 7-B; 8-A; 9-B; 10-E; 11-D; 12-A; 13-C; 14-A; 15-D.

Topic 21: 1-E; 2-C; 3-A; 4-B; 5-C; 6-A; 7-D; 8-A; 9-C; 10-D; 11-B; 12-E; 13-B; 14-D; 15-E.

Topic 22: 1-D; 2-C; 3-D; 4-A; 5-B; 6-E; 7-B; 8-E; 9-C; 10-B; 11-A; 12-D; 13-A; 14-E; 15-C.

Topic 23: 1-E; 2-C; 3-B; 4-E; 5-D; 6-B; 7-C; 8-D; 9-A; 10-D; 11-B; 12-A; 13-C; 14-E; 15-A.

Topic 24: 1-E; 2-A; 3-E; 4-C; 5-D; 6-B; 7-D; 8-B; 9-A; 10-C; 11-A; 12-E; 13-B; 14-C; 15-D.

Topic 25: 1-D; 2-C; 3-D; 4-E; 5-C; 6-A; 7-B; 8-A; 9-D; 10-E; 11-B; 12-C; 13-A; 14-E; 15-B.

Topic 26: 1-A; 2-D; 3-C; 4-B; 5-A; 6-D; 7-E; 8-A; 9-B; 10-E; 11-D; 12-C; 13-E; 14-B; 15-C.

Topic 27: 1-A; 2-C; 3-B; 4-E; 5-D; 6-C; 7-A; 8-D; 9-C; 10-E; 11-B; 12-D; 13-B; 14-A; 15-E.

Topic 28: 1-B; 2-D; 3-C; 4-A; 5-E; 6-C; 7-E; 8-B; 9-A; 10-C; 11-D; 12-E; 13-B; 14-A; 15-D.

Topic 29: 1-E; 2-D; 3-B; 4-D; 5-E; 6-C; 7-A; 8-E; 9-D; 10-B; 11-A; 12-C; 13-B; 14-C; 15-A.

Topic 30: 1-E; 2-C; 3-E; 4-C; 5-D; 6-A; 7-B; 8-A; 9-D; 10-A; 11-B; 12-D; 13-C; 14-B; 15-E.

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